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OF

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A. R. FORSYTH.

19 May, 1897.

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C. XII. *b*

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CLASSIFICATION.

799.

ON CURVILINEAR COORDINATES.

[*From the Quarterly Journal of Pure and Applied Mathematics*, vol. XIX. (1883), pp. 1—22.]

THE present memoir is based upon Mr Warren's "Exercises in Curvilinear and Normal Coordinates," *Camb. Phil. Trans.* t. XII. (1877), pp. 455—502, and has for a principal object the establishment of the six differential equations of the second order corresponding to his six equations for normal coordinates: but the notation is different; the results are more general, inasmuch as I use throughout general curvilinear coordinates instead of his normal coordinates; and as regards my six equations for general curvilinear coordinates, the terms containing differential coefficients of the first order are presented under a different form.

If the position of a point in space is determined by the rectangular coordinates x, y, z; then p, q, r being each of them a given function of x, y, z, we have conversely x, y, z, each of them a given function of p, q, r, which are thus in effect coordinates serving to determine the position of the point, and are called curvilinear coordinates.

But it is not in the first instance necessary to regard x, y, z as rectangular coordinates, or even as Cartesian coordinates at all; we are simply concerned with the two sets of variables x, y, z and p, q, r, each variable of the one set being a given function of the variables of the other set; and, in particular, the x, y, z are regarded as being each of them a given function of the p, q, r.

Except as regards the symbols ξ, η, ζ presently mentioned, the suffixes 1, 2, 3 refer to the variables p, q, r respectively, and are used to denote differentiations in regard to these variables, viz.

$$x_1,\ x_2,\ x_3,\ x_{11},\ x_{22},\ x_{33},\ \ldots$$

are written to denote

$$\frac{dx}{dp},\ \frac{dx}{dq},\ \frac{dx}{dr},\ \frac{d^2x}{dp^2},\ \frac{d^2x}{dq^2},\ \frac{d^2x}{dr^2},\ \&c.,$$

and so in other cases; in particular,

$$\begin{array}{lll} x_1,\ x_2,\ x_3 & \text{denote} & \dfrac{dx}{dp},\ \dfrac{dx}{dq},\ \dfrac{dx}{dr}, \\ y_1,\ y_2,\ y_3 & \text{,,} & \dfrac{dy}{dp},\ \dfrac{dy}{dq},\ \dfrac{dy}{dr}, \\ z_1,\ z_2,\ z_3 & \text{,,} & \dfrac{dz}{dp},\ \dfrac{dz}{dq},\ \dfrac{dz}{dr}. \end{array}$$

The minors formed with these differential coefficients are denoted by suffixed letters ξ, η, ζ, thus

$$\begin{array}{lll} \xi_1,\ \xi_2,\ \xi_3 & \text{denote} & y_2z_3 - y_3z_2,\ y_3z_1 - y_1z_3,\ y_1z_2 - y_2z_1, \\ \eta_1,\ \eta_2,\ \eta_3 & \text{,,} & z_2x_3 - z_3x_2,\ z_3x_1 - z_1x_3,\ z_1x_2 - z_2x_1, \\ \zeta_1,\ \zeta_2,\ \zeta_3 & \text{,,} & x_2y_3 - x_3y_2,\ x_3y_1 - x_1y_3,\ x_1y_2 - x_2y_1, \end{array}$$

so that, as regards these letters ξ, η, ζ, the suffixes do not denote differentiations.

The determinant

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \text{ is put } = L;$$

and the symbols (a, b, c, f, g, h), (A, B, C, F, G, H) are defined as follows:

$$\begin{array}{ll} \mathrm{a} = x_1^2 + y_1^2 + z_1^2, & A = \xi_1^2 + \eta_1^2 + \zeta_1^2, \\ \mathrm{b} = x_2^2 + y_2^2 + z_2^2, & B = \xi_2^2 + \eta_2^2 + \zeta_2^2, \\ \mathrm{c} = x_3^2 + y_3^2 + z_3^2, & C = \xi_3^2 + \eta_3^2 + \zeta_3^2, \\ \mathrm{f} = x_2x_3 + y_2y_3 + z_2z_3, & F = \xi_2\xi_3 + \eta_2\eta_3 + \zeta_2\zeta_3, \\ \mathrm{g} = x_3x_1 + y_3y_1 + z_3z_1, & G = \xi_3\xi_1 + \eta_3\eta_1 + \zeta_3\zeta_1, \\ \mathrm{h} = x_1x_2 + y_1y_2 + z_1z_2, & H = \xi_1\xi_2 + \eta_1\eta_2 + \zeta_1\zeta_2. \end{array}$$

We have then, further,

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = L, \qquad \begin{vmatrix} \mathrm{a} & \mathrm{h} & \mathrm{g} \\ \mathrm{h} & \mathrm{b} & \mathrm{f} \\ \mathrm{g} & \mathrm{f} & \mathrm{c} \end{vmatrix} = L^2,$$

$$\begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix} = L^2, \qquad \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = \frac{1}{L^2};$$

$$(A,\ B,\ C,\ F,\ G,\ H) = (\mathrm{bc} - \mathrm{f}^2,\ \mathrm{ca} - \mathrm{g}^2,\ \mathrm{ab} - \mathrm{h}^2,\ \mathrm{gh} - \mathrm{af},\ \mathrm{hf} - \mathrm{bg},\ \mathrm{fg} - \mathrm{ch}),$$

$$L^2\,(\mathrm{a},\ \mathrm{b},\ \mathrm{c},\ \mathrm{f},\ \mathrm{g},\ \mathrm{h}) = (BC{-}F^2,\ CA{-}G^2,\ AB{-}H^2,\ GH{-}AF,\ HF{-}BG,\ FG{-}CH),$$

which equations are at once proved, and are fundamental ones in the theory.

It is convenient to add that we have

$$\begin{vmatrix} \dfrac{dp}{dx} & \dfrac{dp}{dy} & \dfrac{dp}{dz} \\ \dfrac{dq}{dx} & \dfrac{dq}{dy} & \dfrac{dq}{dz} \\ \dfrac{dr}{dx} & \dfrac{dr}{dy} & \dfrac{dr}{dz} \end{vmatrix} = \frac{1}{L} \begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix},$$

that is, $\dfrac{dp}{dx} = \dfrac{1}{L}\,\xi_1$, &c.; and, further,

$$dx^2 + dy^2 + dz^2 = (\mathrm{a},\ \mathrm{b},\ \mathrm{c},\ \mathrm{f},\ \mathrm{g},\ \mathrm{h} \mid dp,\ dq,\ dr)^2,$$

$$dp^2 + dq^2 + dr^2 = \frac{1}{L^2}\,(A,\ B,\ C,\ F,\ G,\ H \mid dx,\ dy,\ dz)^2.$$

Differentiating the values of a, b, c, f, g, h, we have $\tfrac{1}{2}\mathrm{a}_1 = x_1x_{11} + y_1y_{11} + z_1z_{11}$, which may be written in the abbreviated form $\tfrac{1}{2}\mathrm{a}_1 = 1.11$; similarly

$$x_1x_{12} + x_2x_{11} + y_1y_{12} + y_2y_{11} + z_1z_{12} + z_2z_{11},$$

which in like manner may be written $\mathrm{f}_1 = 2.13 + 3.12$, and so in other cases; observe that, in the duad part of any symbol, the order of the numbers is immaterial, $2.13 = 2.31$. The whole system of equations is

$$\begin{array}{lll}
\tfrac{1}{2}\mathrm{a}_1 = 1.11, & \tfrac{1}{2}\mathrm{a}_2 = 1.12, & \tfrac{1}{2}\mathrm{a}_3 = 1.13, \\
\tfrac{1}{2}\mathrm{b}_1 = 2.12, & \tfrac{1}{2}\mathrm{b}_2 = 2.22, & \tfrac{1}{2}\mathrm{b}_3 = 2.23, \\
\tfrac{1}{2}\mathrm{c}_1 = 3.13, & \tfrac{1}{2}\mathrm{c}_2 = 3.23, & \tfrac{1}{2}\mathrm{c}_3 = 3.33, \\
\mathrm{f}_1 = 2.13 + 3.12, & \mathrm{f}_2 = 2.23 + 3.22, & \mathrm{f}_3 = 2.33 + 3.23, \\
\mathrm{g}_1 = 3.11 + 1.13, & \mathrm{g}_2 = 3.12 + 1.23, & \mathrm{g}_3 = 3.13 + 1.33, \\
\mathrm{h}_1 = 1.12 + 2.11, & \mathrm{h}_2 = 1.22 + 2.12, & \mathrm{h}_3 = 1.23 + 2.13.
\end{array}$$

These may also be written

$$\begin{array}{lll}
1.11 = \tfrac{1}{2}\mathrm{a}_1, & 1.22 = \mathrm{h}_2 - \tfrac{1}{2}\mathrm{b}_1, & 1.33 = \mathrm{g}_3 - \tfrac{1}{2}\mathrm{c}_1, \\
2.11 = \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2, & 2.22 = \tfrac{1}{2}\mathrm{b}_2, & 2.33 = \mathrm{f}_3 - \tfrac{1}{2}\mathrm{c}_2, \\
3.11 = \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3, & 3.22 = \mathrm{f}_2 - \tfrac{1}{2}\mathrm{b}_3, & 3.33 = \tfrac{1}{2}\mathrm{c}_3, \\
1.23 = \tfrac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3), & 1.31 = \tfrac{1}{2}\mathrm{a}_3, & 1.12 = \tfrac{1}{2}\mathrm{a}_2, \\
2.23 = \tfrac{1}{2}\mathrm{b}_3, & 2.31 = \tfrac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3), & 2.12 = \tfrac{1}{2}\mathrm{b}_1, \\
3.23 = \tfrac{1}{2}\mathrm{c}_2, & 3.31 = \tfrac{1}{2}\mathrm{c}_1, & 3.12 = \tfrac{1}{2}(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3).
\end{array}$$

It is to be observed that we can, from each system of three equations, express a set of second differential coefficients of the x, y, z in terms of the first differential

1—2

coefficients of the a, b, c, f, g, h; thus the three equations containing 11, written at length, are

$$\begin{aligned} x_1x_{11} + y_1y_{11} + z_1z_{11} &= \tfrac{1}{2}\mathrm{a}_1, \\ x_2 \quad + y_2 \quad + z_2 \quad &= \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2, \\ x_3 \quad + y_3 \quad + z_3 \quad &= \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3, \end{aligned}$$

three linear equations for the determination of x_{11}, y_{11}, z_{11}; hence for x_{11} we have

$$x_{11}\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \tfrac{1}{2}\mathrm{a}_1(y_2z_3 - y_3z_2) + (\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2)(y_3z_1 - y_1z_3) + (\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)(y_1z_2 - y_2z_1),$$

or, what is the same thing,

$$L\,x_{11} = \tfrac{1}{2}\mathrm{a}_1\,\xi_1 + (\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2)\,\xi_2 + (\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)\,\xi_3,$$

and so for y_{11}, z_{11}; or if (as in the sequel) we desire the value of a linear function $\alpha x_{11} + \beta y_{11} + \gamma z_{11}$, calling this for a moment $\Box$, we join to the foregoing a new equation

$$\alpha x_1 + \beta y_1 + \gamma z_1 = \Box,$$

and then, eliminating the three quantities, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \Box \\ x_1 & y_1 & z_1 & \tfrac{1}{2}\mathrm{a}_1 \\ x_2 & y_2 & z_2 & \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2 \\ x_3 & y_3 & z_3 & \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3 \end{vmatrix} = 0,$$

giving $L\,\Box$ as a linear function of $\tfrac{1}{2}\mathrm{a}_1$, $\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2$, $\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3$.

We can in like manner form the expressions for the second differential coefficients of the a, b, c, f, g, h: these will of course contain third differential coefficients of the x, y, z.

Writing down only what is wanted, we have

$$\begin{aligned} \tfrac{1}{2}\mathrm{a}_{22} &= 12.12 + 1.122, \\ \tfrac{1}{2}\mathrm{b}_{11} &= 12.12 + 2.112, \\ \mathrm{h}_{12} &= 12.12 + 11.22 + 1.122 + 2.112, \end{aligned}$$

where of course 12.12 denotes $x_{12}^2 + y_{12}^2 + z_{12}^2$, 1.122 denotes $x_1.x_{122} + y_1.y_{122} + z_1.z_{122}$, and so in other cases: it follows that

$$\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) = 12.12 - 11.22,$$

so that the third differential coefficients of x, y, z, which enter into the expression of $\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}$ destroy each other, and this combination contains really only second differential coefficients of x, y, z.

Similarly

$$\begin{aligned}
\mathrm{f}_{12} &= 31.23 + 33.12 + 3.123 + 2.133,\\
\mathrm{g}_{31} &= 31.23 + 33.12 + 3.123 + 1.233,\\
\mathrm{c}_{12} &= 2\cdot 31.23 + 2\cdot 3.123,\\
\mathrm{h}_{33} &= 2\cdot 31.23 + 1.233 + 2.133,
\end{aligned}$$

and thence

$$\mathrm{f}_{12} + \mathrm{g}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33} = -2\,(31.23 - 33.12),$$

so that here again we have a combination containing only second differential coefficients of x, y, z.

There are thus, in all, the six combinations

$$\begin{aligned}
\mathrm{b}_{33} + \mathrm{c}_{22} - 2\mathrm{f}_{23} &\quad \dotfill (\mathfrak{A}),\\
\mathrm{c}_{11} + \mathrm{a}_{33} - 2\mathrm{g}_{31} &\quad \dotfill (\mathfrak{B}),\\
\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12} &\quad \dotfill (\mathfrak{C}),\\
\mathrm{g}_{23} + \mathrm{h}_{32} - \mathrm{a}_{23} - \mathrm{f}_{11} &\quad \dotfill (\mathfrak{f}),\\
\mathrm{h}_{31} + \mathrm{f}_{13} - \mathrm{b}_{31} - \mathrm{g}_{22} &\quad \dotfill (\mathfrak{g}),\\
\mathrm{f}_{12} + \mathrm{g}_{21} - \mathrm{c}_{12} - \mathrm{h}_{33} &\quad \dotfill (\mathfrak{h}),
\end{aligned}$$

each really containing only second differential coefficients of x, y, z; and we thus understand how each of these combinations may be expressible in terms of the first differential coefficients of $(\mathrm{a},\mathrm{b},\mathrm{c},\mathrm{f},\mathrm{g},\mathrm{h})$. We have, in fact, for thus expressing these combinations the six equations called $(\mathfrak{A})$, $(\mathfrak{B})$, $(\mathfrak{C})$, $(\mathfrak{f})$, $(\mathfrak{g})$, $(\mathfrak{h})$ about to be obtained, and which are the generalisations of Warren's six equations for normal coordinates.

I consider the several determinants of the form

$$\begin{vmatrix} x_{\alpha\beta} & y_{\alpha\beta} & z_{\alpha\beta}\\ x_2 & y_2 & z_2\\ x_3 & y_3 & z_3 \end{vmatrix},$$

in all 18, since the suffixes for the top row may be 11, 22, 33, 23, 31, 12, and those for the second and third rows 2, 3; 3, 1; or 1, 2. Each determinant is a linear function of second differential coefficients of x, y, z (thus the determinant written down is $\xi_1 x_{11} + \eta_1 y_{11} + \zeta_1 z_{11}$, and as such it can, by what precedes, be expressed by means of the first differential coefficients of the a, b, c, f, g, h. Thus, if the determinant above written down be called D, writing ξ_3, η_3, ζ_3, for α, β, γ, we have

$$\begin{vmatrix} \xi_3 & \eta_3 & \zeta_3 & \square\\ x_1 & y_1 & z_1 & \tfrac12\mathrm{a}_1\\ x_2 & y_2 & z_2 & \mathrm{h}_1 - \tfrac12\mathrm{a}_2\\ x_3 & y_3 & z_3 & \mathrm{g}_1 - \tfrac12\mathrm{a}_3 \end{vmatrix} = 0,$$

that is,

$$L\,\square = -\begin{vmatrix} & \xi_3 & \eta_3 & \zeta_3\\ \tfrac12\mathrm{a}_1 & x_1 & y_1 & z_1\\ \mathrm{h}_1 - \tfrac12\mathrm{a}_2 & x_2 & y_2 & z_2\\ \mathrm{g}_1 - \tfrac12\mathrm{a}_3 & x_3 & y_3 & z_3 \end{vmatrix},$$

where the right-hand side is

$$\begin{aligned}&= \tfrac{1}{2}\mathrm{a}_1 \cdot \xi_3 + \eta_3 \cdot \zeta\zeta\\ &= (\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2)(-(\tfrac{1}{2}\xi_2 + \eta_2 \cdot \zeta\zeta))\\ &\quad + (\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)(\xi_1^2 + \eta_1^2 + \zeta_1^2),\end{aligned}$$

which is

$$= G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

or, as this might be written,

$$= (G,\ F,\ C \,\langle\, \tfrac{1}{2}\mathrm{a}_1,\ \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2,\ \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3\rangle).$$

Retaining the former form, the result is

$$L\square = G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

and it is now very easy to write down the complete system of the 18 equations; viz. if the determinant above written down be called 11.1.2, and so in other cases, then we have

	2.3	3.1	1.2
	A	H	G
	H	B	F
	G	F	C
$L.11$	$\frac{1}{2}\mathrm{a}_1$	$\mathrm{h}_1 - \frac{1}{2}\mathrm{a}_2$	$\mathrm{g}_1 - \frac{1}{2}\mathrm{a}_3$
$.\,22$	$\mathrm{h}_2 - \frac{1}{2}\mathrm{b}_1$	$\frac{1}{2}\mathrm{b}_2$	$\mathrm{f}_2 - \frac{1}{2}\mathrm{b}_3$
$.\,33$	$\mathrm{g}_3 - \frac{1}{2}\mathrm{c}_1$	$\mathrm{f}_3 - \frac{1}{2}\mathrm{c}_2$	$\frac{1}{2}\mathrm{c}_3$
$.\,23$	$-\frac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3)$	$\frac{1}{2}\mathrm{c}_1$	$\frac{1}{2}\mathrm{b}_3$
$.\,31$	$\frac{1}{2}\mathrm{c}_2$	$\frac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3)$	$\frac{1}{2}\mathrm{a}_3$
$.\,12$	$\frac{1}{2}\mathrm{b}_3$	$\frac{1}{2}\mathrm{a}_2$	$\frac{1}{2}(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)$

read for instance

$$L.11.1.2 = G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

the equation obtained above.

There are eighteen functions as shown by the following diagram:

	2.3 3.1 1.2
$(22)(33) - (23)^2$	2.3
$(33)(11) - (31)^2$	3.1
$(11)(22) - (12)^2$	1.2
$(31)(12) - (11)(23)$	2.3
$(12)(23) - (22)(31)$	3.1
$(23)(31) - (33)(12)$	1.2,

viz. in any line of the diagram the bracketed duads may belong to any one at pleasure, but all to the same, of the three pairs 2.3, 3.1, 1.2; thus the first line might be

$$(22.3.1)(33.3.1) - (23.3.1)^2,$$

or, instead of the 3.1, we might have 2.3 or 1.2. But of the 18 functions I distinguish 6, viz. those in which for the six lines respectively the pairs are 2.3, 3.1, 1.2, 2.3, 3.1, 1.2, as shown in the diagram. Each of these six functions can be obtained under two different forms, and by equating these we have the equations ($\mathfrak{A}$), ($\mathfrak{B}$), ($\mathfrak{C}$), ($\mathfrak{f}$), ($\mathfrak{g}$), ($\mathfrak{h}$), before referred to; thus ($\mathfrak{C}$) is obtained by equating two different forms of the function $(11.1.2)(22.1.2) - (12.1.2)^2$, and ($\mathfrak{f}$) is obtained by equating two different forms of the function $(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$.

The determinants 11.1.2, &c., may be denoted by accented letters a, b, c, f, g, h, as follows:

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}, \ldots = \mathrm{a}', \mathrm{b}', \mathrm{c}', \mathrm{f}', \mathrm{g}', \mathrm{h}',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_3 & y_3 & z_3 \\ x_1 & y_1 & z_1 \end{vmatrix}, \ldots = \mathrm{a}'', \mathrm{b}'', \mathrm{c}'', \mathrm{f}'', \mathrm{g}'', \mathrm{h}'',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}, \ldots = \mathrm{a}''', \mathrm{b}''', \mathrm{c}''', \mathrm{f}''', \mathrm{g}''', \mathrm{h}''';$$

viz.

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \mathrm{a}', \quad \&\mathrm{c}.$$

In this notation, ($\mathfrak{C}$) is obtained by equating two values of $\mathrm{a}''\mathrm{b}'' - (\mathrm{h}'')^2$, and ($\mathfrak{f}$) by equating two values of $\mathrm{f}'\mathrm{f}' - \mathrm{g}''\mathrm{h}''$.

The forms which I call the second are those given by the immediate substitution of the foregoing values of (11.1.2), &c.; to obtain the first forms, I proceed to calculate those of ($\mathfrak{C}$) and ($\mathfrak{f}$).

Forming by the ordinary rule the product of the determinants (11.1.2) and (22.1.2), which are

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} x_{22} & y_{22} & z_{22} \\ x_1 & y_1 & z_1 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

this is

$$\begin{vmatrix} 11.22 & 1.11 & 2.11 \\ 1.22 & 1.1 & 1.2 \\ 2.22 & 1.2 & 2.2 \end{vmatrix},$$

where 11.22 denotes $x_{11}x_{22}+y_{11}y_{22}+z_{11}z_{22}$, and the like for the other symbols. In like manner, the square of the determinant (12.1.2), that is, of

$$\begin{vmatrix} x_{12} & y_{12} & z_{12} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

is

$$\begin{vmatrix} 12.12 & 1.12 & 2.12 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 2.1 & 2.2 \end{vmatrix},$$

or, observing that in the two resulting determinants the terms 11.22 and 12.12 are multiplied by the same factor, the expression for the difference gives

$$\begin{aligned} &(11.1.2)(22.1.2) - (12.1.2)^2 \\ &= (11.22 - 12.12)\begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} & 1.11 & 2.11 \\ 1.22 & 1.1 & 1.2 \\ 2.22 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} & 1.12 & 2.12 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

containing $11.22 - 12.12$, which, by what precedes, is

$$= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12});$$

the other terms are also known, viz. the whole value is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\begin{vmatrix} 0 & \frac{1}{2}\mathrm{b}_1 & \frac{1}{2}(\mathrm{b}_2 - \mathrm{h}_1) \\ \frac{1}{2}\mathrm{a}_2 - \mathrm{h}_1 & \mathrm{a} & \mathrm{h} \\ \frac{1}{2}\mathrm{b}_2 - \mathrm{h}_1 & \mathrm{h} & \mathrm{b} \end{vmatrix} - \mathrm{h}\begin{vmatrix} 0 & \frac{1}{2}\mathrm{a}_1 & \frac{1}{2}\mathrm{a}_2 \\ \frac{1}{2}\mathrm{b}_1 & \mathrm{a} & \mathrm{h} \\ \frac{1}{2}\mathrm{b}_2 - \mathrm{h}_1 & \mathrm{h} & \mathrm{b} \end{vmatrix}, \end{aligned}$$

which is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\big(\tfrac{1}{2}\mathrm{b}_1^2 - \tfrac{1}{2}\mathrm{b}_2\mathrm{h}_1 + \tfrac{1}{2}\mathrm{a}_2\mathrm{b}_2\big) \\ &\quad + \mathrm{b}\big(-\tfrac{1}{2}\mathrm{a}_2^2 - \tfrac{1}{2}\mathrm{h}_1^2 + \tfrac{1}{2}\mathrm{a}_1\mathrm{h}_2\big) \\ &\quad + \mathrm{h}\big[\tfrac{1}{2}(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_1\mathrm{b}_2) + \mathrm{h}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{b}_1\mathrm{h}_1\big] \\ &= (\mathrm{ab} - \mathrm{h}^2)k, \quad \text{where } k \text{ is the measure of curvature.} \end{aligned}$$

In the same way, we have

$$\begin{aligned} &(23.1.2)(31.1.2) - (33.1.2)(12.1.2) \\ &= \begin{vmatrix} 23.31 & 1.31 & 2.31 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 12.33 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

which is

$$= (23.31 - 12.33)\begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} 0 & 1.13 & 2.13 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 0 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix},$$

containing $23.31 - 12.33$, which by what precedes is

$$= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33});$$

the other terms are also known, and the value is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\begin{vmatrix} 0 & \tfrac{1}{2}\mathrm{a}_3 & \tfrac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3) \\ \tfrac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3) & \mathrm{a} & \mathrm{h} \\ \tfrac{1}{2}\mathrm{b}_3 & \mathrm{h} & \mathrm{b} \end{vmatrix} \\ &\quad - \mathrm{h}\begin{vmatrix} 0 & \tfrac{1}{2}\mathrm{c}_1 & \tfrac{1}{2}\mathrm{c}_2 \\ \tfrac{1}{2}\mathrm{a}_2 & \mathrm{a} & \mathrm{h} \\ \tfrac{1}{2}\mathrm{b}_3 & \mathrm{h} & \mathrm{b} \end{vmatrix}, \end{aligned}$$

or finally this is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\big(\tfrac{1}{2}\mathrm{b}_3\mathrm{f}_1 - \tfrac{1}{2}\mathrm{b}_3\mathrm{g}_2 - \tfrac{1}{2}\mathrm{b}_3\mathrm{h}_3 + \tfrac{1}{2}\mathrm{a}_3\mathrm{b}_2 - \mathrm{b}_3\mathrm{h}_3\big) \\ &\quad + \mathrm{b}\big(\tfrac{1}{2}\mathrm{a}_3\mathrm{g}_2 + \dots\big) \\ &\quad + \mathrm{h}\big(\dots\big). \end{aligned}$$

The remaining four of the six functions can of course be obtained from the two just found by a cyclical interchange of the letters and suffix-numbers 1, 2, 3, and it is not worth while to write down the values.

The two values of $(11.1.2)(22.1.2) - (12.1.2)^2$ are

$$\begin{aligned} &-\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) + \mathrm{a}(\tfrac{1}{2}\mathrm{b}_1^2 - \tfrac{1}{2}\mathrm{b}_2\mathrm{h}_1 + \tfrac{1}{2}\mathrm{a}_2\mathrm{b}_2) \\ &\quad + \mathrm{b}(-\tfrac{1}{2}\mathrm{a}_2^2 - \tfrac{1}{2}\mathrm{h}_1^2 + \tfrac{1}{2}\mathrm{a}_1\mathrm{h}_2) \\ &\quad + \mathrm{h}\big[\tfrac{1}{2}(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_1\mathrm{b}_2) + \mathrm{h}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{b}_1\mathrm{h}_1\big], \end{aligned}$$

and

$$\{G\cdot\tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)\}\cdot\{G(\mathrm{h}_2 - \tfrac{1}{2}\mathrm{b}_1) + F\cdot\tfrac{1}{2}\mathrm{b}_2 + C(\mathrm{f}_2 - \tfrac{1}{2}\mathrm{b}_3)\},$$

where, in the first value, for a, b, h, $\mathrm{ab} - \mathrm{h}^2$ we must write $L^{-2}(BC - F^2)$, $L^{-2}(CA - G^2)$, $L^{-2}(FG - CH)$, and $L^{-2}C^2$; making this change, multiplying by 4 and equating, we obtain

$$\begin{aligned} &-2L^2C(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\ &\quad + (BC - F^2)(\mathrm{b}_1^2 - 2\mathrm{b}_2\mathrm{h}_1 + \mathrm{a}_2\mathrm{b}_2) \\ &\quad + (CA - G^2)(\mathrm{a}_2^2 - 2\mathrm{a}_1\mathrm{h}_2 + \mathrm{a}_1\mathrm{b}_1) \\ &\quad + (FG - CH)(\mathrm{a}_1\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_1 + 4\mathrm{h}_1\mathrm{h}_2 - 2\mathrm{b}_1\mathrm{h}_1 - 2\mathrm{a}_2\mathrm{h}_2) \\ &\quad - \{G\cdot\mathrm{a}_1 + F(2\mathrm{h}_1 - \mathrm{a}_2) + C(2\mathrm{g}_1 - \mathrm{a}_3)\}\{G(2\mathrm{h}_2 - \mathrm{b}_1) + F\mathrm{b}_2 + C(2\mathrm{f}_2 - \mathrm{b}_3)\} \\ &\quad + \{G\mathrm{b}_1 + F\mathrm{b}_2 + C(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)\}^2 \;=\; 0. \end{aligned}$$

C. XII. 2

Developing the fourth and fifth lines, it appears that in this expression the coefficients of F^2, G^2 and FG each of them vanish; the whole equation is thus divisible by C, and omitting this factor throughout, the equation becomes

$$\begin{aligned}
0 = {} & -2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\
& + A(\mathrm{a}_2^2 - 2\mathrm{a}_2\mathrm{h}_1 + \mathrm{a}_1\mathrm{b}_1) \\
& + B(\mathrm{b}_1^2 - 2\mathrm{b}_1\mathrm{h}_2 + \mathrm{a}_2\mathrm{b}_2) \\
& + C\big\{ -\mathrm{a}_3\mathrm{b}_3 + 2\mathrm{a}_2\mathrm{f}_3 + 2\mathrm{b}_1\mathrm{g}_3 - 4\mathrm{f}_3\mathrm{g}_3 + (\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)^2\big\} \\
& + F\big\{(\mathrm{a}_3\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3) + 2(\mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_2\mathrm{g}_1) + 2(\mathrm{b}_1\mathrm{h}_3 - \mathrm{b}_3\mathrm{h}_1) + 2(\mathrm{a}_2\mathrm{f}_2 + \mathrm{b}_1\mathrm{f}_2 - 2\mathrm{h}_1\mathrm{f}_2)\big\} \\
& + G\big\{\mathrm{a}_3\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3 + 2(\mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) + 2(\mathrm{a}_2\mathrm{h}_3 - \mathrm{a}_3\mathrm{h}_2) + 2(\mathrm{a}_2\mathrm{g}_2 + \mathrm{b}_1\mathrm{g}_1 - 2\mathrm{g}_1\mathrm{h}_2)\big\} \\
& + H\big\{\ 2(\mathrm{a}_2\mathrm{h}_2 + \mathrm{b}_1\mathrm{h}_1 - 2\mathrm{h}_1\mathrm{h}_2)\big\};
\end{aligned}$$

this is substantially the required equation ($\mathfrak{C}$), but the form of it may be greatly simplified.

Forming the identity,

$$\begin{aligned}
0 &= 2L[(\mathrm{a}_2 - \mathrm{h}_1)L_1 + (\mathrm{b}_1 - \mathrm{h}_2)L_2] \\
&= -A(\mathrm{a}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{a}_1\mathrm{h}_1 + \mathrm{a}_2^2) \\
&\quad - B(\mathrm{b}_1^2 - \mathrm{b}_1\mathrm{h}_2 - \mathrm{b}_2\mathrm{h}_1 + \mathrm{a}_2\mathrm{b}_2) \\
&\quad - C(\mathrm{b}_1\mathrm{c}_2 + \mathrm{a}_2\mathrm{c}_1 - \mathrm{c}_1\mathrm{h}_2 - \mathrm{c}_2\mathrm{h}_1) \\
&\quad - 2F(\mathrm{b}_1\mathrm{f}_2 + \mathrm{a}_2\mathrm{f}_1 - \mathrm{f}_1\mathrm{h}_2 - \mathrm{f}_2\mathrm{h}_1) \\
&\quad - 2G(\mathrm{b}_1\mathrm{g}_2 + \mathrm{a}_2\mathrm{g}_1 - \mathrm{g}_1\mathrm{h}_2 - \mathrm{g}_2\mathrm{h}_1) \\
&\quad - 2H(\mathrm{a}_2\mathrm{h}_2 + \mathrm{b}_1\mathrm{h}_1 - 2\mathrm{h}_1\mathrm{h}_2),
\end{aligned}$$

we add hereto the last preceding equation; the coefficients of A, B, F, G, H thus assume new and simple forms, but the coefficient of C requires a further transformation.

Assume

$$\begin{aligned}
\Omega = {} & (\mathrm{b}_1\mathrm{c}_1 - \mathrm{c}_2\mathrm{h}_1) + (\mathrm{c}_2\mathrm{a}_2 - \mathrm{c}_1\mathrm{g}_2) + (\mathrm{a}_3\mathrm{b}_3 - \mathrm{h}_3^2) \\
& + (\mathrm{g}_2\mathrm{h}_3 + \mathrm{g}_3\mathrm{h}_2 - \mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) + (\mathrm{h}_3\mathrm{f}_1 + \mathrm{h}_1\mathrm{f}_3 - \mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_3\mathrm{g}_1) + (\mathrm{f}_1\mathrm{g}_2 + \mathrm{f}_2\mathrm{g}_1 - \mathrm{c}_2\mathrm{h}_1 - \mathrm{c}_1\mathrm{h}_2);
\end{aligned}$$

then, if we add to the equation C multiplied by this value and subtract $C\Omega$, the coefficient of C takes its proper form, and the equation is

$$\begin{aligned}
0 = {} & -2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\
& + 2L[(\mathrm{b}_1 - \mathrm{h}_2)L_2 + (\mathrm{a}_2 - \mathrm{h}_1)L_1] - C\Omega \\
& + A\{\ (\mathrm{a}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_1)\ \} \\
& + B\{\ (\mathrm{b}_1\mathrm{h}_2 - \mathrm{b}_2\mathrm{h}_1)\ \} \\
& + C\{-(\mathrm{a}_2\mathrm{f}_2 - \mathrm{a}_1\mathrm{f}_1) + (\mathrm{b}_1\mathrm{g}_1 - \mathrm{b}_2\mathrm{g}_2) - (\mathrm{g}_2\mathrm{h}_1 - \mathrm{g}_1\mathrm{h}_2) - (\mathrm{h}_1\mathrm{f}_2 - \mathrm{h}_2\mathrm{f}_1) + 3(\mathrm{f}_1\mathrm{g}_2 - \mathrm{f}_2\mathrm{g}_1)\} \\
& + F\{-(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3) + 2(\mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_3\mathrm{g}_1) + 2(\mathrm{b}_1\mathrm{h}_3 - \mathrm{b}_3\mathrm{h}_1) - 2(\mathrm{h}_1\mathrm{f}_3 - \mathrm{h}_3\mathrm{f}_1)\} \\
& + G\{-(\mathrm{a}_2\mathrm{b}_3 - \mathrm{a}_3\mathrm{b}_2) - 2(\mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) - 2(\mathrm{a}_2\mathrm{h}_3 - \mathrm{a}_3\mathrm{h}_2) - 2(\mathrm{g}_1\mathrm{h}_3 - \mathrm{g}_3\mathrm{h}_1)\} \\
& + H\{\ \cdots\ \} \\
= {} & 0;
\end{aligned}$$

where, as a verification, observe that the letters (a,b), (f,g) may be interchanged if at the same time we interchange the suffixes 1 and 2.

The two values of (23.1.2)(31.1.2) − (33.1.2)(12.1.2) are

$$\begin{aligned}
&-\tfrac{1}{2}(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})(\mathrm{ab}-\mathrm{h}^2)\\
&\quad+\mathrm{a}\bigl(\tfrac{1}{2}\mathrm{b}_1\mathrm{f}_3-\tfrac{1}{2}\mathrm{b}_3\mathrm{f}_1-\tfrac{1}{2}\mathrm{b}_1\mathrm{c}_2+\tfrac{1}{2}\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3\bigr)\\
&\quad+\mathrm{b}\bigl(\tfrac{1}{2}\mathrm{a}_3\mathrm{g}_2+\ldots-\mathrm{a}_3\mathrm{h}_3\bigr)+\mathrm{h}\bigl[\ldots-\tfrac{1}{2}(\mathrm{f}_1+\mathrm{g}_2)^2\bigr],
\end{aligned}$$

and

$$\{G\cdot\tfrac{1}{2}\mathrm{a}_3+F(\mathrm{f}_2-\tfrac{1}{2}\mathrm{c}_1)+C\cdot\tfrac{1}{2}\mathrm{c}_3\}\cdot\{G\cdot\tfrac{1}{2}\mathrm{a}_2+F\cdot\tfrac{1}{2}\mathrm{b}_3+C\cdot\tfrac{1}{2}(\mathrm{f}_1+\mathrm{g}_2-\mathrm{h}_3)\},$$

and here for a, b, h, $\mathrm{ab}-\mathrm{h}^2$ substituting their values, multiplying by 4, and equating, we have

$$\begin{aligned}
&-2L^2C(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})\\
&\quad+(BC-F^2)(2\mathrm{b}_1\mathrm{f}_3-\mathrm{b}_3\mathrm{f}_1-\mathrm{b}_1\mathrm{c}_2+\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3)\\
&\quad+(CA-G^2)(2\mathrm{a}_2\mathrm{g}_3-\mathrm{a}_3\mathrm{g}_2-\mathrm{a}_2\mathrm{c}_1+\mathrm{a}_1\mathrm{f}_3-\mathrm{a}_3\mathrm{h}_3)\\
&\quad+(FG-CH)(\mathrm{a}_3\mathrm{b}_3+\mathrm{b}_1\mathrm{c}_1+\mathrm{a}_2\mathrm{c}_2-2\mathrm{b}_1\mathrm{g}_3-2\mathrm{a}_2\mathrm{f}_3+\mathrm{h}_3^2-\mathrm{f}_2\mathrm{g}_1+2\mathrm{f}_1\mathrm{g}_2-\mathrm{g}_1^2\mathrm{a}_2^2)\\
&\quad-\{G(-\mathrm{f}_1+\mathrm{g}_2+\mathrm{h}_3)+F\mathrm{b}_3+C\mathrm{c}_2\}\{G\mathrm{a}_3+F(\mathrm{f}_2-\mathrm{g}_1+\mathrm{h}_2)+C\mathrm{c}_1\}\\
&\quad+\{G(2\mathrm{g}_3-\mathrm{c}_2)+F(2\mathrm{f}_3-\mathrm{c}_2)+C\mathrm{c}_3\}\{G\mathrm{a}_2+F\mathrm{b}_2+C(\mathrm{f}_1+\mathrm{g}_2-\mathrm{h}_3)\}=0.
\end{aligned}$$

Here again the terms in F^2, FG, G^2 all vanish, hence the whole equation divides by C; throwing this factor out, we have

$$\begin{aligned}
0=&-2L(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})\\
&+A(\;2\mathrm{a}_2\mathrm{g}_3-\mathrm{a}_3\mathrm{g}_2-\mathrm{a}_2\mathrm{c}_1+\mathrm{a}_1\mathrm{f}_3-\mathrm{a}_3\mathrm{h}_3\;)\\
&+B(\;2\mathrm{b}_1\mathrm{f}_3-\mathrm{b}_3\mathrm{f}_1-\mathrm{b}_1\mathrm{c}_2+\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3\;)\\
&+C(\;-\mathrm{c}_1\mathrm{c}_2+\mathrm{c}_3\mathrm{f}_1+\mathrm{c}_3\mathrm{g}_2-\mathrm{c}_3\mathrm{h}_3\;)\\
&+F\{\;-(\mathrm{b}_3\mathrm{c}_1-\mathrm{b}_1\mathrm{c}_3)-2\mathrm{c}_2\mathrm{f}_1+2\mathrm{f}_1\mathrm{f}_3+2\mathrm{g}_2\mathrm{f}_3-2\mathrm{f}_3\mathrm{h}_3\;\}\\
&+G(\;\mathrm{a}_2\mathrm{c}_3-\mathrm{a}_3\mathrm{c}_2-2\mathrm{c}_1\mathrm{g}_3+2\mathrm{f}_2\mathrm{g}_3+2\mathrm{g}_2\mathrm{g}_3-2\mathrm{g}_3\mathrm{h}_3\;)\\
&+H(\;-\mathrm{a}_3\mathrm{b}_3-\mathrm{b}_1\mathrm{c}_1-\mathrm{a}_2\mathrm{c}_2+2\mathrm{b}_1\mathrm{g}_3+2\mathrm{a}_2\mathrm{f}_3-\mathrm{h}_3^2+\mathrm{f}_2\mathrm{g}_1-2\mathrm{f}_1\mathrm{g}_2+\mathrm{g}_1^2\mathrm{a}_2^2\;)=0,
\end{aligned}$$

which is substantially the required equation ($\mathfrak{f}$); but the form has to be altered. First, multiply by 2, then forming the expression

$$\begin{aligned}
&2L[(\mathrm{f}_3-\mathrm{c}_2)L_1+(\mathrm{g}_3-\mathrm{c}_1)L_2+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)L_3]\\
&\qquad=(\mathrm{f}_3-\mathrm{c}_2)(A\mathrm{a}_1+B\mathrm{b}_1+C\mathrm{c}_1+2F\mathrm{f}_1+\ldots)\\
&\qquad\quad+(\mathrm{g}_3-\mathrm{c}_1)(A\mathrm{a}_2+B\mathrm{b}_2+C\mathrm{c}_2+2F\mathrm{f}_2+\ldots)\\
&\qquad\quad+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)(A\mathrm{a}_3+B\mathrm{b}_3+C\mathrm{c}_3+2F\mathrm{f}_3+2G\mathrm{g}_3+\ldots),
\end{aligned}$$

2—2

this is

$$\begin{aligned}
&= A(-a_1c_2 - a_2c_1 + a_1f_1 + a_2f_1 + a_2g_2 + a_3g_3 - 2a_3h_3)\\
&\quad + B(-b_1c_2 - b_2c_1 + b_1f_1 + b_1f_3 + b_2g_3 + b_3g_2 - 2b_3h_3)\\
&\quad + C(-2c_1c_2 + c_1f_1 + c_3f_1 + c_2g_2 + c_3g_2 - 2c_3h_3)\\
&\quad + 2F(-c_1f_1 - c_2f_2 + 2f_1f_3 + f_2g_3 + f_3g_2 - 2f_3h_3)\\
&\quad + 2G(-c_1g_2 - c_2g_1 + f_1g_2 + f_2g_1 + 2g_2g_3 - 2g_3h_3)\\
&\quad + 2H(-c_1h_1 - c_2h_2 + g_1h_2 + g_2h_1 + h_1f_2 + h_2f_1 - 2h_3^2);
\end{aligned}$$

then adding this expression

$$(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3$$

and subtracting its foregoing value, we have an equation with new coefficients of A, B, C, F, G, H, all of which, except that of H, are in the proper form, and for the coefficient of H, recurring to the foregoing value of Ω, we must add to the equation $2H$ multiplied by the value of Ω and subtract $2H\Omega$. The final result is

$$\begin{aligned}
0 = &- 4L(g_{23} + f_{31} - c_{12} - h_{33})\\
&+ 2L[(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3] - 2H\Omega\\
&+ A\{\ 3(a_2g_3 - a_3g_2) - (c_1a_2 - c_2a_1) + (a_1f_3 - a_3f_1)\ \}\\
&+ B\{\ -3(b_3f_1 - b_1f_3) - (b_1c_2 - b_2c_1) - (b_3g_2 - b_2g_3)\ \}\\
&+ C\{\ (c_2f_1 - c_1f_2) - (c_1g_2 - c_2g_1)\ \}\\
&+ 2F\{\ -(b_3c_1 - b_1c_3) + (c_1f_2 - c_2f_1) - (f_2g_3 - f_3g_2)\ \}\\
&+ 2G\{\ -(c_3a_2 - c_2a_3) - (c_1g_2 - c_2g_1) - (f_1g_3 - f_3g_1)\ \}\\
&+ 2H\{\ -(b_3g_1 - b_1g_3) + (a_2f_3 - a_3f_2)\ \} = 0,
\end{aligned}$$

which, it will be observed, remains unaltered by the interchange of a and b, f and g, A and B, F and G, the suffix numbers 1 and 2 being at the same time interchanged.

We have thus the required six equations, in which

$$\mathfrak{K} = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

$$\begin{aligned}
\Omega = &\ b_1c_1 - c_2h_1 + c_2a_2 - c_1g_2 + a_3b_3 - h_3^2\\
&+ (g_2h_3 + g_3h_2 - a_2f_3 - a_3f_2) + (h_3f_1 + h_1f_3 - b_1g_3 - b_3g_1) + (f_1g_2 + f_2g_1 - c_2h_1 - c_1h_2),
\end{aligned}$$

and where I write also, for shortness, ab|2 for $a_1b_2 - a_2b_1$, &c.; viz. the equations are

$$\begin{aligned}
0 = &- 2L(b_{33} + c_{22} - 2f_{23})\\
&+ 2L[(c_3 - f_2)L_2 + (b_2 - f_3)L_3] - A\Omega\\
&+ A\{\ -\mathrm{bg}|31 + \mathrm{ch}|2 + 3\cdot \mathrm{gh}|23 - \mathrm{hf}|31 - \mathrm{fg}|12\ \}\\
&+ B\{\ -\mathrm{bf}|23\ \}\\
&+ C\{\ +\mathrm{cf}|23\ \}\\
&+ F\{\ -\mathrm{bc}|23\ \}\\
&+ G\{\ -\mathrm{bc}|31 + 2\cdot \mathrm{cf}|12 + 2\cdot \mathrm{ch}|23 - 2\cdot \mathrm{fg}|23\ \}\\
&+ H\{\ -2\cdot \mathrm{bg}|23 - 2\cdot \mathrm{hf}|23\ \} \qquad (\mathfrak{A}).
\end{aligned}$$

$$
\begin{aligned}
0 = &- 2L(\mathrm{c}_{11} + \mathrm{a}_{33} - 2\mathrm{g}_{31}) \\
&+ 2L[(\mathrm{c}_3 - \mathrm{g}_1)L_1 + (\mathrm{a}_1 - \mathrm{g}_3)L_3] - B\Omega \\
&+ A\{\ \mathrm{ag}|31\ \} \\
&+ B\{\ \mathrm{af}|23 - \mathrm{ch}|12 - \mathrm{gh}|23 + 3\cdot\mathrm{hf}|31 - \mathrm{fg}|31\ \} \\
&+ C\{\ -\mathrm{cg}|31\ \} \\
&+ F\{\ -\mathrm{ca}|23 - 2\cdot\mathrm{cg}|12 - 2\cdot\mathrm{ch}|31 - 2\cdot\mathrm{fg}|31\ \} \\
&+ G\{\ -\mathrm{ca}|31 - 2\cdot\mathrm{gh}|31\ \} \\
&+ H\{\ \ldots\ \} \qquad (\mathfrak{B}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\
&+ 2L[(\mathrm{b}_1 - \mathrm{h}_2)L_2 + (\mathrm{a}_2 - \mathrm{h}_1)L_1] - C\Omega \\
&+ A\{\ -\mathrm{ah}|12\ \} \\
&+ B\{\ +\mathrm{bh}|12\ \} \\
&+ C\{\ -\mathrm{af}|23 + \mathrm{bg}|31 - \mathrm{gh}|23 - \mathrm{hf}|31 + 3\cdot\mathrm{fg}|12\ \} \\
&+ F\{\ -\mathrm{ab}|23 + 2\cdot\mathrm{bg}|12 + 2\cdot\mathrm{bh}|31 - 2\cdot\mathrm{hf}|12\ \} \\
&+ G\{\ -\mathrm{ab}|31 - 2\cdot\mathrm{af}|12 + 2\cdot\mathrm{ah}|23 - 2\cdot\mathrm{gh}|12\ \} \\
&+ H\{\ -\mathrm{ab}|12\ \} \qquad (\mathfrak{C}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 4L(\mathrm{g}_{12} + \mathrm{h}_{31} - \mathrm{a}_{23} - \mathrm{f}_{11}) \\
&+ 2L[(\mathrm{g}_2 + \mathrm{h}_3 - 2\mathrm{f}_1)L_1 + (\mathrm{g}_3 - \mathrm{a}_1)L_2 + (\mathrm{h}_1 - \mathrm{a}_1)L_3] - 2F\Omega \\
&+ A\{\ -\mathrm{ag}|12 - \mathrm{ah}|31\ \} \\
&+ B\{\ -\mathrm{ab}|23 + \mathrm{bg}|12 + 3\cdot\mathrm{bh}|31\ \} \\
&+ C\{\ -\mathrm{ca}|23 - 3\cdot\mathrm{cg}|12 - \mathrm{ch}|31\ \} \\
&+ 2F\{\ \ldots + \mathrm{bg}|31 - \mathrm{ch}|12\ \} \\
&+ 2G\{\ -\mathrm{ca}|12 + \mathrm{ag}|23 - \mathrm{gh}|31\ \} \\
&+ 2H\{\ -\mathrm{ab}|31 - \mathrm{ah}|23 - \mathrm{gh}|12\ \} \qquad (\mathfrak{f}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 4L(\mathrm{h}_{12} + \mathrm{f}_{31} - \mathrm{b}_{23} - \mathrm{g}_{22}) \\
&+ 2L[(\mathrm{f}_2 - \mathrm{b}_3)L_1 + (\mathrm{h}_1 + \mathrm{f}_3 - 2\mathrm{g}_2)L_2 + (\mathrm{h}_2 - \mathrm{b}_1)L_3] - 2G\Omega \\
&+ A\{\ -\mathrm{ab}|31 - 3\cdot\mathrm{ah}|23 - \mathrm{af}|12\ \} \\
&+ B\{\ \mathrm{bh}|23 - \mathrm{bf}|12\ \} \\
&+ C\{\ -\mathrm{bc}|31 + \mathrm{ch}|23 + 3\cdot\mathrm{cf}|12\ \} \\
&+ 2F\{\ -\mathrm{bc}|12 - \mathrm{bf}|31 - \mathrm{hf}|23\ \} \\
&+ 2G\{\ \ldots + \mathrm{ch}|12 - \mathrm{af}|23\ \} \\
&+ 2H\{\ -\mathrm{ab}|23 + \mathrm{bh}|31 - \mathrm{hf}|12\ \} \qquad (\mathfrak{g}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &-4L(\mathrm{f}_{12}+\mathrm{g}_{31}-\mathrm{c}_{23}-\mathrm{h}_{33})\\
&+2L[(\mathrm{f}_3-\mathrm{c}_2)L_1+(\mathrm{g}_3-\mathrm{c}_1)L_2+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)L_3]-2H\Omega\\
&+A\{\ -\mathrm{ca}|12+\mathrm{af}|31+3\cdot\mathrm{ag}|23\ \}\\
&+B\{\ -\mathrm{bc}|12-3\cdot\mathrm{bf}|31-\mathrm{bg}|23\ \}\\
&+C\{\ \mathrm{cf}|31-\mathrm{cg}|23\ \}\\
&+F\{\ -\mathrm{bc}|31+\mathrm{cf}|12-\mathrm{fg}|23\ \}\\
&+G\{\ -\mathrm{ca}|23-\mathrm{cg}|12-\mathrm{fg}|31\ \}\\
&+H\{\ \ldots+\mathrm{af}|23-\mathrm{bg}|31\ \}\qquad(\mathfrak{h}).
\end{aligned}
$$

It would be possible in these equations to introduce the symbols $AB|12$, &c., in place of ab|12, &c., and then writing $A=B=C=1$, all these symbols other than those where the letters are GH, HF or FG would vanish, and we should obtain Mr Warren's six equations for normal coordinates. But in the general case it would seem that there is not any advantage in the introduction of the new symbols $AB|12$, &c., and I retain by preference the equations in the form in which I have given them.

To the foregoing may be joined a symmetrical equation obtained (as by Mr Warren) by multiplying the several equations by a, b, c, f, g, h respectively, and adding; the result is in the first instance obtained in the form

$$\mathfrak{X}-\mathfrak{Y}+\mathfrak{Z}=0,$$

where

$$
\begin{aligned}
\mathfrak{X} = &\ \mathrm{a}(\mathrm{b}_{33}+\mathrm{c}_{22}-2\mathrm{f}_{23})\\
&+\mathrm{b}(\mathrm{c}_{11}+\mathrm{a}_{33}-2\mathrm{g}_{31})\\
&+\mathrm{c}(\mathrm{a}_{22}+\mathrm{b}_{11}-2\mathrm{h}_{12})\\
&+2\mathrm{f}(\mathrm{g}_{12}+\mathrm{h}_{31}-\mathrm{a}_{23}-\mathrm{f}_{11})\\
&+2\mathrm{g}(\mathrm{h}_{12}+\mathrm{f}_{31}-\mathrm{b}_{23}-\mathrm{g}_{22})\\
&+2\mathrm{h}(\mathrm{f}_{12}+\mathrm{g}_{31}-\mathrm{c}_{23}-\mathrm{h}_{33}).
\end{aligned}
$$

For Ψ, collecting the terms which contain L_1, L_2, L_3 respectively, and attending to the values of (A,B,C,F,G,H),

$$=(\mathrm{bc}-\mathrm{f}^2,\ \mathrm{ca}-\mathrm{g}^2,\ \mathrm{ab}-\mathrm{h}^2,\ \mathrm{gh}-\mathrm{af},\ \mathrm{hf}-\mathrm{bg},\ \mathrm{fg}-\mathrm{ch}),$$

this is easily reduced to the form

$$\Psi=(A_1+H_2+G_3)L_1+(H_1+B_2+F_3)L_2+(G_1+F_2+C_3)L_3.$$

The term in Ω is

$$=-(A\mathrm{a}+B\mathrm{b}+C\mathrm{c}+2F\mathrm{f}+2G\mathrm{g}+2H\mathrm{h})\Omega=-3L^2\Omega,\ \text{as above.}$$

For the calculation of $\Box$, collecting the terms, we have $\Box =$

23	A	B	C	F	G	H
bc				−a		
ca			−f	−b − 2h		
ab		−f	−c			−2g
af		b − c			−2g	2h
bf		−a				
cf			a			
ag	3h				2f	2b
bg		−h				−2a
cg			−h			
ah	−3g			−2c − 2f		
bh		g				
ch			g		2a	
gh	−3a − b − c					
hf				−2g		−2a
fg				−2h − 2a		

31	A	B	C	F	G	H
bc			−g	−2h	−a	
ca					−b	
ab	−g				−c	−2f
af	h					2b
bf		−3h		−2g		−2a
cf			h			
ag	b					
bg	−a		+c	2f		−2h
cg			−b			
ah	−f					
bh		3f		2c		2g
ch			−f − 2b			
gh				−2f		−2b
hf	−a	3b − c				
fg				−2b − 2h		

12	A	B	C	F	G	H
bc		−h		−2g		−a
ca	−h				−2f	−b
ab						−c
af	−g				−2c	
bf		−g				
cf			3g	2h	2a	
ag	f					
bg		f		2c		
cg			−3f − 2b	−2h		
ah	−c					
bh		c				
ch	a − b			−2f + 2g		
gh					−2c	−2f
hf				−2c		−2g
fg	−a − b + 3c					

Some however of the coefficients require reduction; for instance, that of bc31 is

$$= -(\mathrm{a}, \mathrm{h}, \mathrm{g} \rangle G, F, C) - \mathrm{h}F = -\mathrm{h}F;$$

and so $-\mathrm{h}B - 2\mathrm{g}F - \mathrm{a}H$ is

$$= -(\mathrm{a}, \mathrm{h}, \mathrm{g} \rangle H, B, F) - \mathrm{g}F = -\mathrm{g}F.$$

After these reductions, the value is found to be or, arranging the terms in a different order, this may be written

$$\square =$$

$$\begin{aligned}
&A\{\ \mathrm{a}(3\cdot \mathrm{gh}|23 - \mathrm{hf}|31 - \mathrm{fg}|12) + \mathrm{b}\cdot \mathrm{ag}|31 - \mathrm{c}\cdot \mathrm{ah}|12 - \mathrm{c}\cdot \mathrm{ah}|12 \\
&\qquad + \mathrm{f}(\mathrm{ag}|12 - \mathrm{ah}|31) + \mathrm{g}(2\cdot \mathrm{gh}|12 - \mathrm{ah}|23) + \mathrm{h}(2\cdot \mathrm{gh}|31 + \mathrm{ag}|23)\ \} \\
&+ B\{\ -\mathrm{a}\cdot \mathrm{bf}|23 + \mathrm{b}(-\mathrm{gh}|23 + 3\cdot \mathrm{hf}|31 - \mathrm{fg}|12) + \mathrm{c}\cdot \mathrm{bh}|12 \\
&\qquad + \mathrm{f}(2\cdot \mathrm{hf}|12 + \mathrm{bh}|31) + \mathrm{g}(\mathrm{bh}|23 - \mathrm{bf}|12) + \mathrm{h}(2\cdot \mathrm{hf}|23 - \mathrm{bf}|31)\ \} \\
&+ C\{\ \mathrm{a}\cdot \mathrm{cf}|23 - \mathrm{b}\cdot \mathrm{cg}|31 + \mathrm{c}(-\mathrm{gh}|23 - \mathrm{hf}|31 + 3\cdot \mathrm{fg}|12) \\
&\qquad + \mathrm{f}(2\cdot \mathrm{fg}|31 - \mathrm{cg}|12) + \mathrm{g}(2\cdot \mathrm{fg}|23 + \mathrm{cf}|12) + \mathrm{h}(\mathrm{cf}|31 - \mathrm{cg}|23)\ \} \\
&+ F\{\ -\mathrm{a}\cdot \mathrm{bc}|23 - \mathrm{b}\cdot \mathrm{ch}|31 + \mathrm{c}\cdot \mathrm{bg}|12 \\
&\qquad + \mathrm{f}(\mathrm{bg}|31 - \mathrm{ch}|12) + \mathrm{g}(\mathrm{bg}|23 - \mathrm{bc}|12) - \mathrm{h}(\mathrm{ch}|23 + \mathrm{bc}|31)\ \} \\
&+ G\{\ +\mathrm{a}\cdot \mathrm{ch}|23 - \mathrm{b}\cdot \mathrm{ca}|31 - \mathrm{c}\cdot \mathrm{af}|12 \\
&\qquad - \mathrm{f}(\mathrm{af}|31 + \mathrm{ca}|12) + \mathrm{g}(\mathrm{ch}|12 - \mathrm{af}|23) + \mathrm{h}(\mathrm{ch}|31 - \mathrm{ca}|23)\ \} \\
&+ H\{\ -\mathrm{a}\cdot \mathrm{bg}|23 + \mathrm{b}\cdot \mathrm{af}|31 - \mathrm{c}\cdot \mathrm{ab}|12 \\
&\qquad + \mathrm{f}(\mathrm{af}|12 - \mathrm{ab}|31) - \mathrm{g}(\mathrm{bg}|12 + \mathrm{ab}|23) + \mathrm{h}(\mathrm{af}|23 - \mathrm{bg}|31)\ \}.
\end{aligned}$$

Attending to the values of A, B, C, F, G, H, we have

$$\begin{aligned}
A_1 + B_2 + C_3 + 2F_4 + 2G_5 + 2H_6 \ &= \mathrm{bc}_{11} + \mathrm{cb}_{11} - 2\mathrm{ff}_{11} + 2(\mathrm{b}_1\mathrm{c}_1 - \mathrm{f}_1^2) \\
&+ \mathrm{ca}_{22} + \mathrm{ac}_{22} - 2\mathrm{gg}_{22} + 2(\mathrm{c}_2\mathrm{a}_2 - \mathrm{g}_2^2) \\
&+ \mathrm{ab}_{33} + \mathrm{ba}_{33} - 2\mathrm{hh}_{33} + 2(\mathrm{a}_3\mathrm{b}_3 - \mathrm{h}_3^2) \\
&+ 2\{\mathrm{gh}_{23} + \mathrm{hg}_{23} - \mathrm{af}_{23} - \mathrm{fa}_{23} + \mathrm{g}_2\mathrm{h}_3 + \mathrm{g}_3\mathrm{h}_2 - \mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2\} \\
&+ 2\{\mathrm{hf}_{31} + \mathrm{fh}_{31} - \mathrm{bg}_{31} - \mathrm{gb}_{31} + \mathrm{h}_3\mathrm{f}_1 + \mathrm{h}_1\mathrm{f}_3 - \mathrm{b}_3\mathrm{g}_1 - \mathrm{b}_1\mathrm{g}_3\} \\
&+ 2\{\mathrm{fg}_{12} + \mathrm{gf}_{12} - \mathrm{ch}_{12} - \mathrm{hc}_{12} + \mathrm{f}_1\mathrm{g}_2 + \mathrm{f}_2\mathrm{g}_1 - \mathrm{c}_1\mathrm{h}_2 - \mathrm{c}_2\mathrm{h}_1\},
\end{aligned}$$

which is, in fact, $= 2\mathfrak{X} + 2\Omega$.

Hence the foregoing equation $(\mathfrak{M})$ may also be written

$$-L^2\left(A_{11} + B_{22} + C_{33} + 2F_{23} + 2G_{31} + 2H_{12}\right) + 2L\Psi - L^2\Omega + \square = 0,$$

where Ψ, Ω, $\square$ have their before-mentioned values.

In the particular case where $\mathrm{f} = 0$, $\mathrm{g} = 0$, $\mathrm{h} = 0$, we have

$$A,\ B,\ C,\ F,\ G,\ H = \mathrm{bc},\ \mathrm{ca},\ \mathrm{ab},\ 0,\ 0,\ 0; \quad L^2 = \mathrm{abc};$$

the equation $(\mathfrak{X})$ becomes

$$-2\mathrm{abc}(\mathrm{b}_{33} + \mathrm{c}_{22}) + [\mathrm{c}(\mathrm{a}_1\mathrm{bc} + \mathrm{b}_1\mathrm{ca} + \mathrm{c}_1\mathrm{ab}) + \mathrm{b}(\mathrm{a}_1\mathrm{bc} + \mathrm{b}_1\mathrm{ca} + \mathrm{c}_1\mathrm{ab})] - \mathrm{bc}(\mathrm{b}_1\mathrm{c}_2 + \mathrm{c}_2\mathrm{a}_2 + \mathrm{a}_3\mathrm{b}_3) = 0,$$

that is,

$$-2\mathrm{abc}(\mathrm{b}_{33} + \mathrm{c}_{22}) - \mathrm{bc}\,\mathrm{b}_2\mathrm{c}_1 + \mathrm{ca}(\mathrm{b}_1\mathrm{c}_1 + \mathrm{b}_3^2) + \mathrm{ab}(\mathrm{b}_2\mathrm{c}_3 + \mathrm{c}_1^2) = 0,$$

and the equation ($\mathfrak{f}$) becomes

$$4\mathrm{abc}\cdot\mathrm{a}_{23}-[\mathrm{a}_3(\mathrm{a}_2\mathrm{bc}+\mathrm{b}_2\mathrm{ca}+\mathrm{c}_2\mathrm{ab})+\mathrm{a}_2(\mathrm{a}_3\mathrm{bc}+\mathrm{b}_3\mathrm{ca}+\mathrm{c}_3\mathrm{ab})]$$
$$-\mathrm{ca}(\mathrm{a}_2\mathrm{b}_3-\mathrm{a}_3\mathrm{b}_2)-\mathrm{ab}(\mathrm{c}_2\mathrm{a}_3-\mathrm{c}_3\mathrm{a}_2)=0,$$

that is,

$$4\mathrm{abc}\cdot\mathrm{a}_{23}-2\mathrm{bc}\cdot\mathrm{a}_2\mathrm{a}_3-2\mathrm{ca}\cdot\mathrm{a}_2\mathrm{b}_3-2\mathrm{ab}\cdot\mathrm{a}_3\mathrm{b}_2=0.$$

Dividing by 2abc and 4abc respectively, and completing the system, the six equations are

$$\mathrm{b}_{33}+\mathrm{c}_{22}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{c}_3^2}{2\mathrm{c}}-\frac{\mathrm{bc}_1}{2\mathrm{a}}+\frac{\mathrm{bc}_2}{2\mathrm{b}}+\frac{\mathrm{bc}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{A})$$
$$\mathrm{c}_{11}+\mathrm{a}_{33}-\frac{\mathrm{c}_3^2}{2\mathrm{c}}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{ca}_1}{2\mathrm{a}}+\frac{\mathrm{ca}_2}{2\mathrm{b}}+\frac{\mathrm{ca}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{B})$$
$$\mathrm{a}_{22}+\mathrm{b}_{11}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{ab}_1}{2\mathrm{a}}+\frac{\mathrm{ab}_2}{2\mathrm{b}}+\frac{\mathrm{ab}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{C})$$
$$\mathrm{a}_{23}-\frac{\mathrm{a}_2\mathrm{a}_3}{2\mathrm{a}}-\frac{\mathrm{a}_2\mathrm{b}_3}{2\mathrm{b}}-\frac{\mathrm{a}_3\mathrm{c}_2}{2\mathrm{c}}=0 \qquad (\mathfrak{f})$$
$$\mathrm{b}_{31}-\frac{\mathrm{b}_3\mathrm{b}_1}{2\mathrm{b}}-\frac{\mathrm{b}_3\mathrm{a}_1}{2\mathrm{a}}-\frac{\mathrm{b}_1\mathrm{c}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{g})$$
$$\mathrm{c}_{12}-\frac{\mathrm{c}_1\mathrm{c}_2}{2\mathrm{c}}-\frac{\mathrm{c}_1\mathrm{b}_2}{2\mathrm{b}}-\frac{\mathrm{c}_2\mathrm{a}_1}{2\mathrm{a}}=0. \qquad (\mathfrak{h})$$

These are in fact Lamé's equations, *Leçons sur les coordonnées curvilignes*, etc., Paris (1859), pp. 76, 78, viz. the first of the equations (8), p. 76, is

$$\frac{d^2H}{dp_2\,dp_3}=\frac{1}{H_2}\frac{dH_2}{dp_3}\frac{dH}{dp_2}+\frac{1}{H_3}\frac{dH_3}{dp_2}\frac{dH}{dp_3},$$

which, in the notation of the present paper, is

$$(\sqrt{\mathrm{a}})_{23}=\frac{1}{\sqrt{\mathrm{b}}}(\sqrt{\mathrm{b}})_3(\sqrt{\mathrm{a}})_2+\frac{1}{\sqrt{\mathrm{c}}}(\sqrt{\mathrm{c}})_2(\sqrt{\mathrm{a}})_3.$$

Here

$$(\sqrt{\mathrm{a}})_2=\tfrac{1}{2}\frac{\mathrm{a}_2}{\sqrt{\mathrm{a}}},$$
$$(\sqrt{\mathrm{a}})_{23}=\tfrac{1}{2}\left(\frac{\mathrm{a}_{23}}{\sqrt{\mathrm{a}}}-\tfrac{1}{2}\frac{\mathrm{a}_2\mathrm{a}_3}{\mathrm{a}\sqrt{\mathrm{a}}}\right);$$

and the equation therefore is

$$\frac{\mathrm{a}_{23}}{\sqrt{\mathrm{a}}}-\tfrac{1}{2}\frac{\mathrm{a}_2\mathrm{a}_3}{\mathrm{a}\sqrt{\mathrm{a}}}=\frac{1}{2}\frac{\mathrm{a}_2\mathrm{b}_3}{\sqrt{\mathrm{b}}\sqrt{\mathrm{a}}}+\tfrac{1}{2}\frac{\mathrm{a}_3\mathrm{c}_2}{\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}},$$

which, multiplying by $\sqrt{\mathrm{a}}$, gives the foregoing equation

$$\mathrm{a}_{23}-\frac{\mathrm{a}_2\mathrm{a}_3}{2\mathrm{a}}-\frac{\mathrm{a}_2\mathrm{b}_3}{2\mathrm{b}}-\frac{\mathrm{a}_3\mathrm{c}_2}{2\mathrm{c}}=0.$$

The first of the equations (9) p. 78 is

$$\frac{d}{dp_1}\left(\frac{1}{H_1}\frac{dH}{dp_1}\right)+\frac{d}{dp_2}\left(\frac{1}{H_2}\frac{dH_2}{dp_2}\right)=\frac{1}{H_2}\frac{dH_1}{dp_1}\frac{dH_1}{dp_2},$$

which, in the notation of the present paper, is

$$\left(\frac{(\sqrt{\mathrm{c}})_1}{\sqrt{\mathrm{a}}}\right)_1+\left(\frac{(\sqrt{\mathrm{c}})_2}{\sqrt{\mathrm{b}}}\right)_2=\frac{1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}}(\sqrt{\mathrm{a}})_2(\sqrt{\mathrm{b}})_1.$$

This gives first

$$\frac{(\sqrt{\mathrm{c}})_{11}}{\sqrt{\mathrm{a}}}-\tfrac{1}{2}\,\frac{(\sqrt{\mathrm{c}})_1\mathrm{a}_1}{\mathrm{a}\sqrt{\mathrm{a}}}+\frac{(\sqrt{\mathrm{c}})_{22}}{\sqrt{\mathrm{b}}}-\tfrac{1}{2}\,\frac{(\sqrt{\mathrm{c}})_2\mathrm{b}_2}{\mathrm{b}\sqrt{\mathrm{b}}}=\frac{1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}}(\sqrt{\mathrm{a}})_2(\sqrt{\mathrm{b}})_1,$$

and then

$$\tfrac{1}{2}\,\frac{\mathrm{c}_{11}}{\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}-\tfrac{1}{4}\,\frac{\mathrm{c}_1^2}{\mathrm{c}\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}-\tfrac{1}{4}\,\frac{\mathrm{c}_1\mathrm{a}_1}{\mathrm{a}\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}+\tfrac{1}{2}\,\frac{\mathrm{c}_{22}}{\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}-\tfrac{1}{4}\,\frac{\mathrm{c}_2^2}{\mathrm{c}\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}-\tfrac{1}{4}\,\frac{\mathrm{c}_2\mathrm{b}_2}{\mathrm{b}\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}=\tfrac{1}{4}\,\frac{\mathrm{a}_2\mathrm{b}_1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}},$$

which, on multiplying by $\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}$ and reducing, gives the foregoing equation

$$\mathrm{a}_{22}+\mathrm{b}_{11}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{ab}_1}{2\mathrm{a}}+\frac{\mathrm{ab}_2}{2\mathrm{b}}-\frac{\mathrm{a}_2\mathrm{b}_1+\mathrm{a}_1\mathrm{b}_2}{2\mathrm{c}}=0.$$

800.

NOTE ON THE STANDARD SOLUTIONS OF A SYSTEM OF LINEAR EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 38–40.]

To fix the ideas, the equations are assumed to be without constant terms. Supposing the system to be insufficient for the determination of the ratios of the unknown quantities, then regarding the unknown quantities as having a definite order of arrangement, there are certain solutions which may be regarded as standard solutions. Take the unknown quantities to be A, B, C, D, E, F, G, &c.; then assuming $A = 0$, or else A, B, each $= 0$, or else A, B, C, each $= 0$, as many equations as possible, the system as thus modified will have a definite solution; for instance, the assumed equations A, B, C, D, E, each $= 0$, may give a definite solution in which F is not $= 0$, and it may then for convenience be put $= 1$. We have thus a solution beginning with $F = 1$. This being so, there will be a solution or solutions with $F = 0$; we cannot then have A, B, C, D, E, each $= 0$, but we again assume $A = 0$, or else A, B, each $= 0$, as many equations as possible; suppose A, B, C, each $= 0$ give a definite solution, with D not $= 0$; and then taking it for convenience to be $= 1$, we have a solution beginning $D = 1$, and for which $F = 0$. Going on in this manner we obtain, it may be, a solution beginning $B = 1$, and for which $D = 0$, $F = 0$; and so on, the process stopping, if not sooner, with a solution beginning $A = 1$, and with the initial letters of the preceding solutions, each $= 0$. We have in this manner the system of standard solutions, of a form such as

A	B	C	D	E	F	G	
0	0	0	0	0	1	$*$	
0	0	0	1	$*$	0	$*$	3—2
0	1	$*$	0	$*$	0	$*$	
1	0	$*$	0	$*$	0	$*$	

where the $*$ denotes a value which is not in general $= 0$, but which may in any particular case happen to be so.

For instance, let it be required for the binary quartic $(a, b, c, d, e \rangle x, y)^4$, to find the asyzygetic seminvariants of the degree 4 and weight 6. Assuming for the seminvariant the value in the left-hand column of the diagram, the unknown coefficients being A, B, C, D, E, F, G, this must be reduced to zero by the operation

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e;$$

and we thus obtain as many equations as there are terms of the degree 4 and weight 5, as appearing by the second column

		A	B	C	D	E	F	G
a^2e^2	$ad^2b + 2\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
A	$= a^2ce$	1						
B	$= ab^2e$		1					
C	$= abcd$			1				
D	$= ac^3$				1			
E	$= b^3d$					1		
F	$= b^2c^2$						1	

[The table on book p. 20 records, for each of the seven monomials in the diagram, the corresponding column giving the result of Δ acting upon it; the resulting standard solutions $A = 1, B = -1, \ldots, F = 1$ enumerate the irreducible composites P and Q used on p. 21.]

As is known, there is no irreducible solution, but only the composite forms

$$\begin{aligned} \mathrm{I} \ &= \ a(ace - ad^2 - b^2e + 2bcd - c^3), \\ \mathrm{II} &= \ (ac - b^2)(ae - 4bd + 3c^2), \end{aligned}$$

the developed values of which are given above: II (as a form beginning $A = 1$ and with $B = 0$) can be nothing else than, and is in fact $= Q$: and so I (as a form beginning with $A = 1$, $B = -1$) can be nothing else than, and is in fact $= Q - P$; that is, we have

$$\begin{aligned} P &= -\mathrm{I} + \mathrm{II}, & \quad \text{or} \quad \mathrm{I} &= -P + Q, \\ Q &= \phantom{-\mathrm{I} +} \mathrm{II}, & \mathrm{II} &= Q; \end{aligned}$$

and so, in general, we have a standard set of values for the asyzygetic seminvariants of a given degree and weight; or, what is the same thing, for the covariants of a given deg-order.

801.

ON SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 131–138.]

The present paper is a somewhat fragmentary one, but it contains some results which seem to me to be worth putting on record.

I consider here not any binary quantic in particular, but the whole series $(a, b, c \,\rangle x, y)^2$, $(a, b, c, d \,\rangle x, y)^3$, &c.; or in a somewhat different point of view, I consider the indefinite series of coefficients $(a, b, c, d, e, \ldots)$; here, instead of covariants and invariants, we have only seminvariants; viz. a seminvariant is a function reduced to zero by the operator

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + 5e\partial_f + \ldots,$$

for instance, seminvariants are

$$a, \quad ac - b^2, \quad a^2d - 3abc + 2b^3, \quad a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2,$$

$$ae - 4bd + 3c^2, \quad ace - ad^2 - b^2e + 2bcd - c^3, \quad \&c.$$

A seminvariant is of a certain degree θ in the coefficients, and of a certain weight w (viz. the coefficients a, b, c, d, ... are reckoned as being of the weights 0, 1, 2, 3, ... respectively); it is, moreover, of a certain rank ρ; viz. according as the highest letter therein is a, c, d, e, ... (it is never b), the rank is taken to be 0, 2, 3, 4, ..., and we have $w = \frac{1}{2}\rho\theta$. The seminvariant... may be regarded as belonging to a quantic $(a, \ldots \rangle x, y)^n$, the order of which is n, equal to or greater than ρ; viz. in regard to such quantic the seminvariant, say A, is the leading coefficient of a covariant

$$(A, B, \ldots, K \,\rangle x, y)^{n\theta - 2w},$$

where the weights of the successive coefficients are w, $w+1$, ... up to $n\theta - w$; hence the number of terms less unity, that is, μ, is $= n\theta - 2w$; the least value of μ is thus $= \rho\theta - 2w$, which is either zero, or positive; in the former case, $w = \frac{1}{2}\rho\theta$, the seminvariant is an invariant of the quantic $(a, \ldots \rangle x, y)^\rho$, the order of which is equal to the rank of the seminvariant; but if $w < \frac{1}{2}\rho\theta$, then it is the leading coefficient of a covariant $(A, B, \ldots, E \,\rangle x, y)^{\rho\theta - 2w}$ of the same quantic $(a, \ldots \rangle x, y)^\rho$; and in every case, taking $n > \rho$, the seminvariant is the leading coefficient A of a covariant $(A, B, \ldots, K \,\rangle x, y)^{n\theta - 2w}$ of a quantic $(a, \ldots \rangle x, y)^n$.

Take A as belonging to the quantic $(a,\dots \rangle x,y)^n$; corresponding to such quantic, we have an operator Λ of the same rank n, viz.

$$\begin{aligned} \Lambda &= 2b\partial_a + c\partial_b && \text{for } n=2, \\ &= 3b\partial_a + 2c\partial_b + d\partial_c && n=3, \\ &= 4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d && n=4, \end{aligned}$$

Operating with Λ on A, we have a series of terms

$$A,\ \Lambda A,\ \Lambda^2 A,\ \dots,\ \Lambda^{n\theta-2w}A,$$

but the next term $\Lambda^{n\theta-2w+1}A$, and of course every succeeding term, is $=0$, and this being so, the coefficients of the covariant $(A,B,\dots,K\,\rangle x,y)^{n\theta-2w}$ are

$$\left(1,\ \frac{1}{1}\Lambda,\ \frac{1}{1\cdot 2}\Lambda^2,\ \dots\right)A,$$

or what is the same thing, each coefficient is obtained from the next preceding one by the formulae

$$B=\frac{1}{1}\Lambda A, \qquad C=\frac{1}{2}\Lambda B, \qquad D=\frac{1}{3}\Lambda C, \dots.$$

The coefficients A and K, B and J, … are derived one from the other by reversal of the order of the coefficients of $(a,b,\dots \rangle x,y)^n$, with or without a change of sign, and thus it is only necessary to calculate up to the middle coefficient, or pair of coefficients; and we obtain, moreover, a verification.

Calculating in this manner the covariant

$$(A,B,\dots,K\,\rangle x,y)^{n\theta-2w}$$

which belongs to the quantic $(a,\dots \rangle x,y)^n$, if we herein change a, b, c, … into $ax+by$, $bx+cy$, $cx+dy$, … we obtain the covariant belonging to the quantic $(a,\dots \rangle x,y)^{n+1}$; and in this covariant making the like change, or what is the same thing, in the first-mentioned covariant changing a, b, c, … into $(a,b,c\,\rangle x,y)^2$, $(b,c,d\,\rangle x,y)^2$, $(c,d,e\,\rangle x,y)^2$, … we have the covariant belonging to $(a,\dots \rangle x,y)^{n+2}$; and in like manner we obtain the covariant belonging to the quantic $(a,\dots \rangle x,y)^n$ of any given order n.

In particular, if $w=\frac{1}{2}\rho\theta$, that is, if the given seminvariant be an invariant of $(a,\dots \rangle x,y)^\rho$, then we obtain the series of covariants directly from A, by therein changing a, b, c, … into $ax+by$, $bx+cy$, $cx+dy$, … and in the result making the like change; or what is the same thing, in A changing a, b, c, … into $(a,b,c\,\rangle x,y)^2$, $(b,c,d\,\rangle x,y)^2$, $(c,d,e\,\rangle x,y)^2$, … and so on until we obtain the covariant for the quantic $(a,\dots \rangle x,y)^n$ of the given order n.

A seminvariant which cannot be expressed as a rational and integral function of lower seminvariants is said to be irreducible. The theory is distinct from that of the irreducible covariants of a quantic of a given order; for instance, as regards the cubic $(a,b,c,d\,\rangle x,y)^3$, we have the irreducible covariant (invariant)

$$a^2d^2+4ac^3+4b^3d-6abcd-3b^2c^2,$$

but this is not an irreducible seminvariant; it is

$$=(ac-b^2)(ae-4bd+3c^2)-a(ace-ad^2-b^2e+2bcd),$$

or, what is the same thing, there is not for the quartic $(a,b,c,d,e\,\rangle x,y)^4$ or for the higher quantics, any irreducible covariant having this for the leading coefficient.

We may consider the question to determine the number of asyzygetic seminvariants of a given degree and weight. For instance, taking the weights up to 12, so that the series of letters extends as far as m, then for the degrees 1, 2, 3 we have as follows:

$w =$	0	1	2	3	4	5	6	7	8	9	10	11	12
Deg. 1	a	b	c	d	e	f	g	h	i	j	k	l	m
Num.	1	1	1	1	1	1	1	1	1	1	1	1	1
Diff.	0	0	0	0	0	0	0	0	0	0	0	0	0
Deg. 2	a^2	ab	ac	ad	ae	af	ag	ah	ai	aj	ak	al	am
		b^2	bc	bd	be	bf	bg	bh	bi	bj	bk	bl	
			c^2	cd	ce	cf	cg	ch	ci	cj	ck	cl	
				d^2	de	df	dg	dh	di	dj	dk		
					e^2	ef	eg	eh	ei	ej	ek		
						f^2	fg	fh	fi	fj			
							g^2	gh	gi	gj			
								h^2	hi	ij			
									i^2	ij			
Num.	1	1	2	2	3	3	4	4	5	5	6	6	7
Diff.	1	0	1	0	1	0	1	0	1	0	1	0	1
Deg. 3	a^3	a^2b	a^2c	a^2d	a^2e	a^2f	a^2g	a^2h	a^2i	a^2j	a^2k	a^2l	a^2m
		ab^2	abc	abd	abe	abf	abg	abh	abi	abj	abk	abl	
			b^3	ac^2	acd	ace	acf	acg	ach	aci	acj	ack	
					b^2d	b^2e	ad^2	ade	adf	adg	adh	adi	
					bc^2	bcd	ae^2	b^2h	aef	aeg	aeh		
						c^3	bce	bch	b^2i	b^2j			
							bd^2	bdf	bcg	bdi			
							c^2d	bef	cd^2	cei			
								cde	ce^2	def			
								d^2e		d^2g			
Num.	1	1	2	3	4	5	7	8	10	12	14	16	19
Diff.	1	0	1	1	1	1	2	1	2	2	2	2	3

For the degree 1, the line of differences shows that the only seminvariant is ($W = 0$), the seminvariant a.

For the degree 2, the line of differences 0, 1, 1, 0, ..., shows that the number of seminvariants is $= 1$ for each even degree, $= 0$ for each odd degree; thus for the weight there is a seminvariant $= a^2$ which of course is not, irreducible; while for each of the other even weights we have a single irreducible seminvariant; as is well known, the forms are

$W =$	2	4	6	8	10	
	$ac - b^2$	$ae - 4bd + 3c^2$	$ag - 6bf + 15ce - 10d^2$	$ai - 8bh + 28cg - 56df + 35e^2$	$ak - \dots$	$am - 12bl + 6$

For degree 3, the line of differences shows that the

W	$= 0$	1	2	3	4	5	6	7	8	9	10	11	12
Nos. irr. $=$	0	1	1	1	1	1	2	1	2	2	2	2	3

but inasmuch as for each even weight there is a quadric seminvariant, which multiplied by a given a cubic seminvariant, to obtain the number of irreducible cubic seminvariants we subtract

1	0	1	0	1	0	1	0	1	0	1	0	1
0	0	0	1	0	1	1	1	1	2	1	2	2

or the numbers of irreducible cubic seminvariants are as in the line last written down.

C. XII. 4

There is a convenience however in giving, for each even weight, as well the rejected reducible covariant; and the entire series of results is found to be

$W =$	0	2	4	6	8	10	12	14	16	18	20	22	24
a^2+1	$ac+1$	a^2d+1	ad^2+1	ae^2+1	af^2+1	ag^2+1	ah^2+1	$\ldots$	$\ldots$	$\ldots$	$\ldots$	$\ldots$	
$\vdots$													

[Book p. 26 (PDF p. 46) gives an extensive multi-column tabulation, indexed by even weights from $W = 0$ up to $W = 12$ and beyond, of the asyzygetic and irreducible cubic seminvariants with the integer multipliers needed for their reductions. The verifications carried out by Cayley involve degrees 1, 2, 3 cubic seminvariants whose entries are sums of monomials in $a, b, c, \ldots$ with small integer coefficients, in the format shown above.]

The canonical form given for the quintic in my Tenth Memoir on Quantics [693] belongs to a series, viz. writing now the small roman letters (instead of the italic letters) for the series of coefficients, and using the italic letters a, c, d, e, f, ... to denote seminvariants, they are as follows:

$$\begin{array}{ll}
0 & a = \mathrm{a}, \\
2 & c = \mathrm{ac} - \mathrm{b}^2, \\
3 & d = \mathrm{a}^2\mathrm{d} - 3\mathrm{abc} + 2\mathrm{b}^3 \quad (= f \text{ in the tenth memoir}), \\
4 & e = \mathrm{ae} - 4\mathrm{bd} + 3\mathrm{c}^2 \quad (= b \text{ in the tenth memoir}), \\
5 & f = \mathrm{a}^2\mathrm{f} - 5\mathrm{abe} + 2\mathrm{acd} + 8\mathrm{b}^2\mathrm{d} - 6\mathrm{bc}^2, \\
6 & g = \mathrm{ag} - 6\mathrm{bf} + 15\mathrm{ce} - 10\mathrm{d}^2, \\
7 & h = \mathrm{a}^2\mathrm{h} - 7\mathrm{abg} + 9\mathrm{acf} - 5\mathrm{ade} + 12\mathrm{b}^2\mathrm{f} - 30\mathrm{bce} + 20\mathrm{bd}^2, \\
8 & i = \mathrm{ai} - 8\mathrm{bh} + 28\mathrm{cg} - 56\mathrm{df} + 35\mathrm{e}^2, \\
 & \&\mathrm{c}.
\end{array}$$

Writing also (instead of d in the tenth memoir)

$$\bar{e} = \mathrm{ace} - \mathrm{ad}^2 - \mathrm{b}^2\mathrm{e} + 2\mathrm{bcd} - \mathrm{c}^3,$$

so that the equation $a^2d^2 + a^2bc - 4c^3 \cdot f = 0$ of the tenth memoir is in the present notation $a^2\bar{e} - a^2ec + 4c^3 - d^2 = 0$, then the series of canonical forms is

$$\begin{array}{rl}
\text{Quadric} & (1, 0, c\,\rangle x, y)^2, \\
\text{Cubic} & (1, 0, c, d\,\rangle x, y)^3, \\
\text{Quartic} & (1, 0, c, d, a^2e - 3c^2\,\rangle x, y)^4, \\
\text{Quintic} & (1, 0, c, d, a^2e - 3c^2, a^2f - 2cd\,\rangle x, y)^5, \\
 & \&\mathrm{c}.
\end{array}$$

the series of coefficients being

$$\begin{array}{ccccccccc}
1, & 0, & c, & d, & a^2e+1, & a^2f+1, & a^2g+1, & a^2h+1, & a^2i+1, \\
 & & & & c^2-3, & cd-2, & a^2ce-15, & & \\
 & & & & c^3+45, & a^2de+5, & a^4e^2-35, & & \\
 & & & & d^2+10, & c^2d+3, & a^2c^2e+630, & & \\
 & & & & & & d^2f+56, & & \\
 & & & & & & c^4-1575, & & \\
 & & & & & & cd^2-392 & &
\end{array}$$

these values being, in fact, the expressions in terms of the seminvariants a, c, d, &c. of

$$1, \quad 0, \quad \mathrm{ac}, \quad \mathrm{a}^2\mathrm{d}, \quad \mathrm{a}^3\mathrm{e}, \quad \mathrm{a}^4\mathrm{f}, \quad \mathrm{a}^5\mathrm{g}, \quad \mathrm{a}^6\mathrm{h}, \quad \mathrm{a}^7\mathrm{i}.$$

I annex verifications of the foregoing values. The original page tabulates, for the canonical-form coefficients a^2e-3c^2, a^2f-2cd, $a^2g-15a^2ce+45c^3+10d^2$, $a^2h-28a^2cg+35a^2df-630ab^2f+112c^2e+\ldots$, the integer multipliers attached to each monomial a^2e, abd, ac^2, b^2c, a^2f, abe, $abcd$, ac^3, a^2g, abf, acf, ade, b^3, b^2d, b^2e, bce, bcf, bd^2, c^2d, c^2e, c^3, cd^2, d^2, …, displayed as a sequence of small tables side by side. The first such block (for a^2e-3c^2) is

	a^2e	$-3c^2$
a^2e	$+1$	
abd	-4	
ac^2	$+3$	-3
b^2c		$+8$

	a^2f	$-2cd$
a^2f	$+1$	
a^2cd		-3
abe	-5	
$abcd$	-2	
ac^3		$+10$

	a^2g	$-15a^2ce$	$+45c^3$	$+10d^2$
a^2g	$+1$			
abf	-6			
acf		-15		
ade		-10		
abe	$+5$			
b^3			$+45$	
b^2d			-100	$+10$
b^2e				
bce			$+45$	
bd^2			-30	$+5$
c^2e			-15	
c^3			$+45$	
cd^2			-30	$+5$
d^2				$+6$

The remaining block, for $a^2h-28a^2cg+35a^2df-630ab^2f+112c^2e+\ldots$, follows the same layout; the entries continue on book p. 29.

	a^2i	$-36a^2bh$	$-56a^2dg$	$+35a^2e^2$	$+630a^2cf$	$-1575c^4$
ai	$+1$					
abh	-8					
ach						
acg	$+28$	-36				
adg	-56		$+56$			
adf	-56					
ae^2	$+35$			$+35$		
b^2g		$+290$				
b^2h						
bcg		-460	$+1200$			
bdf		$+840$	-840			
be^2			-315	$+112$		
c^2g		$+1890$			$+630$	-1575
cd^2			$+1080$			
ce^2				$+672$	-315	
d^2f					$+6300$	-6275
de^2						$+1004$

It would be interesting to obtain the general law for the expressions of the canonical coefficients in terms of the seminvariants.

802.

NOTE ON CAPTAIN MACMAHON'S PAPER, "ON THE DIFFERENTIAL EQUATION $X^{-\frac{1}{3}}dx + Y^{-\frac{1}{3}}dy + Z^{-\frac{1}{3}}dz = 0$."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 182–184.]

In general, if $f = (x, y, 1)^3, = 0$ be the equation of a cubic curve, and if

$$d\omega = \frac{dx}{\dfrac{df}{dy}} = -\frac{dy}{\dfrac{df}{dx}},$$

then if 1, 2, 3 are the intersections of the curve by an arbitrary right line, the coordinates of these points being (x_1, y_1), (x_2, y_2), (x_3, y_3) respectively, we have, by Abel's theorem,

$$d\omega_1 + d\omega_2 + d\omega_3 = 0,$$

viz. this is the differential relation corresponding to the integral relation which expresses that the three points are the intersections of the cubic curve by a right line, or say to the integral equation

$$\nabla, = \begin{vmatrix} y_1, & y_2, & y_3 \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

in which equation y_1, y_2, y_3 are regarded as functions of x_1, x_2, x_3 respectively, given by means of the equations $f_1 = 0$, $f_2 = 0$, $f_3 = 0$, which express that the points are on the cubic curve. See my "Memoir on the Abelian and Theta Functions," [819].

In particular, if the equation of the curve is

$$f, = \tfrac{1}{3}y^3 - (A + 3Bx + 3Cx^2 + Dx^3), = \tfrac{1}{3}(y^3 - X), = 0,$$

then

$$d\omega = \frac{dx}{y^2}, \quad = \frac{dx}{X^{\frac{2}{3}}};$$

and corresponding to the differential relation

$$X_1^{-\frac{1}{3}}\, dx_1 + X_2^{-\frac{1}{3}}\, dx_2 + X_3^{-\frac{1}{3}}\, dx_3 = 0,$$

we have the integral relation

$$\nabla, = \begin{vmatrix} X_1^{\frac{1}{3}}, & X_2^{\frac{1}{3}}, & X_3^{\frac{1}{3}} \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

viz. this last equation, as containing no arbitrary constant, is a particular integral of the differential equation.

If instead of x_1, x_2, x_3 we write x, y, z, then we have

$$\nabla = \begin{vmatrix} X^{\frac{1}{3}}, & Y^{\frac{1}{3}}, & Z^{\frac{1}{3}} \\ x, & y, & z \\ 1, & 1, & 1 \end{vmatrix}, = 0,$$

as a particular integral of the differential equation

$$X^{-\frac{1}{3}}\, dx + Y^{-\frac{1}{3}}\, dy + Z^{-\frac{1}{3}}\, dz = 0.$$

To rationalize the integral equation, write

$$\alpha,\ \beta,\ \gamma = y - z,\ z - x,\ x - y \quad (\text{so that } \alpha + \beta + \gamma = 0),$$

the equation is

$$\alpha X^{\frac{1}{3}} + \beta Y^{\frac{1}{3}} + \gamma Z^{\frac{1}{3}} = 0;$$

and we thence have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma\, X^{\frac{1}{3}} Y^{\frac{1}{3}} Z^{\frac{1}{3}}.$$

The left-hand side is

$$\begin{aligned} &\alpha^3\,(A + 3Bx + 3Cx^2 + Dx^3) \\ &\quad + \beta^3\,(A + 3By + 3Cy^2 + Dy^3) \\ &\quad + \gamma^3\,(A + 3Bz + 3Cz^2 + Dz^3); \end{aligned}$$

or assuming

$$\alpha',\ \beta',\ \gamma' = x(y - z),\ y(z - x),\ z(x - y) \quad (\text{so that } \alpha' + \beta' + \gamma' = 0),$$

this is

$$= A(\alpha^3 + \beta^3 + \gamma^3) + 3B(\alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma') + 3C(\alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2) + D(\alpha'^3 + \beta'^3 + \gamma'^3).$$

But taking λ arbitrary, and

$$a,\ b,\ c = \alpha + \lambda\alpha',\ \beta + \lambda\beta',\ \gamma + \lambda\gamma',$$

then

$$a + b + c = 0,$$

whence

$$a^3 + b^3 + c^3 = 3abc;$$

or substituting for a, b, c their values, and comparing the coefficients of the several powers of λ,

$$\begin{aligned}
\alpha^3+\beta^3+\gamma^3 &= 3\alpha\beta\gamma,\\
\alpha^2\alpha'+\beta^2\beta'+\gamma^2\gamma' &= \alpha^2\beta'\gamma'+\beta^2\gamma'\alpha'+\gamma^2\alpha'\beta' = \alpha\beta\gamma\,(x+y+z),\\
\alpha\alpha'^2+\beta\beta'^2+\gamma\gamma'^2 &= \alpha\alpha'^2\beta\gamma+\beta\beta'^2\gamma\alpha+\gamma\gamma'^2\alpha\beta = \alpha\beta\gamma\,(yz+zx+xy),\\
\alpha'^3+\beta'^3+\gamma'^3 &= 3\alpha'\beta'\gamma' = 3\alpha\beta\gamma\cdot xyz.
\end{aligned}$$

Hence we have

$$\alpha^3X+\beta^3Y+\gamma^3Z = 3\alpha\beta\gamma\,[A+B(x+y+z)+C(yz+zx+xy)+Dxyz],$$

or the integral equation is

$$[A+B(x+y+z)+C(yz+zx+xy)+Dxyz]^3 = XYZ,$$

that is,

$$\sqrt[3]{\{A+B(x+y+z)+C(yz+zx+xy)+Dxyz\}^3} = \sqrt[3]{XYZ},$$

the elegant result given by Capt. MacMahon at the beginning of his paper.

The author in a letter to me, dated Jan. 13, 1883, remarks that the particular integral of the equation in question

$$X^{-\frac{1}{3}}\,dx+Y^{-\frac{1}{3}}\,dy+Z^{-\frac{1}{3}}\,dz=0,$$

is expressible as a determinant in a rational form as follows. Writing it $XYZ=P^3$, where

$$P = A+B(x+y+z)+C(yz+zx+xy)+Dxyz,$$

then the form is

$$\nabla, = \begin{vmatrix} 1, & \left(\frac{1}{3}P\dfrac{dX}{dx}-X\dfrac{dP}{dx}\right), & X\\ 1, & \left(\frac{1}{3}P\dfrac{dY}{dy}-Y\dfrac{dP}{dy}\right), & Y\\ 1, & \left(\frac{1}{3}P\dfrac{dZ}{dz}-Z\dfrac{dP}{dz}\right), & Z \end{vmatrix} = 0,$$

for, as shown by Captain MacMahon in his paper, each of the three terms such as

$$Z\left(\tfrac{1}{3}P\frac{dY}{dy}-Y\frac{dP}{dy}\right)-Y\left(\tfrac{1}{3}P\frac{dZ}{dz}-Z\frac{dP}{dz}\right),$$

which compose the determinant, is divisible by $XYZ-P^3$.

It may be added that we have identically

$$\tfrac{1}{3}P\frac{dX}{dx}-X\frac{dP}{dx} = (AC-B^2)(2x-y-z)+(AD-BC)(x^2-yz)+(BD-C^2)(x^2y+x^2z-2xyz),$$

and of course like values for the other two expressions in the determinant.

803.

ON MR ANGLIN'S FORMULA FOR THE SUCCESSIVE POWERS OF THE ROOT OF AN ALGEBRAICAL EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 223, 224.]

Suppose $x^m - px^{m-1} + qx^{m-2} - \ldots = 0$, then the successive powers x^m, x^{m+1}, x^{m+2}, &c. of x can be expressed in the form $Px^{m-1} - Qx^{m-2} + Rx^{m-3} -$ &c. Mr Anglin has obtained for this purpose a very elegant formula, with a demonstration which (it occurred to me) might be presented under a somewhat simplified form; and he has permitted me to draw up the present Note.

Take, for greater convenience, the equation to be

$$x^4 - px^3 + qx^2 - rx + s = 0,$$

and let h_1, h_2, h_3, ... be the sums of the homogeneous products of the roots, of the orders 1, 2, 3, &c. respectively; then, writing also $h_0 = 1$, we have

$$\begin{aligned}
h_1 &= h_0 p, \\
h_2 &= h_1 p - h_0 q, \\
h_3 &= h_2 p - h_1 q + h_0 r, \\
h_4 &= h_3 p - h_2 q + h_1 r - h_0 s, \\
h_5 &= h_4 p - h_3 q + h_2 r - h_1 s, \\
&\ldots
\end{aligned}$$

And this being so, starting from the equation

$$x^4 = px^3 - qx^2 + rx - s,$$

that is,

$$= h_1 x^3 - h_0 q x^2 + h_0 r x - h_0 s,$$

we obtain successively

$$
\begin{aligned}
x^5 &= h_1(px^3 - qx^2 + rx - s) \\
&\quad - h_0qx^3 + h_0rx^2 - h_0sx \\
&= h_2x^3 - (h_1q - h_0r)x^2 + (h_1r - h_0s)x - h_1s, \\
x^6 &= h_2(px^3 - qx^2 + rx - s) \\
&\quad - (h_1q - h_0r)x^3 + (h_1r - h_0s)x^2 - h_1sx \\
&= h_3x^3 - (h_2q - h_1r + h_0s)x^2 + (h_2r - h_1s)x - h_2s, \\
x^7 &= h_3(px^3 - qx^2 + rx - s) \\
&\quad - (h_2q - h_1r + h_0s)x^3 + (h_2r - h_1s)x^2 - h_2sx \\
&= h_4x^3 - (h_3q - h_2r + h_1s)x^2 + (h_3r - h_2s)x - h_3s,
\end{aligned}
$$

and so on, the characteristic feature being that by the introduction of the symbols h, the coefficient of x^3 presents itself at each step as a monomial, and the coefficients of the lower powers require no reduction. It is obvious that the process is a perfectly general one, and that for the equation

$$x^m - p_1x^{m-1} + p_2x^{m-2} - \ldots + (-)^m p_m = 0,$$

the formula is

$$
\begin{aligned}
x^{m+s} = \quad & h_{s+1}x^{m-1} \\
& - (h_sp_2 - h_{s-1}p_3 + \ldots)\, x^{m-2} \\
& + (h_sp_3 - h_{s-1}p_4 + \ldots)\, x^{m-3} \\
& \vdots \\
& + (-)^{m-1}(h_sp_{m-1} - h_{s-1}p_{m-2} + \ldots)\, x \\
& + (-)^{m-1}\, h_sp_m,
\end{aligned}
$$

where, as regards each power of x, the series forming the coefficient thereof is continued as far as possible, that is, up to the term which contains p_m or h_0 as the case may be.

804.

ON THE ELLIPTIC-FUNCTION SOLUTION OF THE EQUATION $x^3+y^3-1=0$.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv. (1883), pp. 106–109.]

I HAD occasion to find elliptic-function expressions for the coordinates $(x,\ y)$ of a point on the cubic curve $x^3+y^3=1$. These are derivable from the formulæ given, Legendre, *Fonctions Elliptiques*, t. I. pp. 185, 186, for the reduction to elliptic integrals of the integral $R=\int\frac{dz}{(1-z^3)^{\frac{1}{2}}}$. Legendre, writing

$$z=\frac{\sqrt{4y^3-1}-\sqrt{3}}{\sqrt{4y^3-1}+\sqrt{3}},$$

and then

$$m'^2=2 \quad \text{and} \quad m'^2y=1+x',$$

finds first

$$R=m\sqrt{3}\int\frac{dx'}{\sqrt{x'^3+3x'^2+3}},$$

and then writing $r=\sqrt[6]{3}$, $x=\tan\frac{1}{2}\phi$, and $c^2=\frac{1}{4}(2-r^2)$, finds

$$R=\tfrac{1}{2}mr\int\frac{d\phi}{\sqrt{1-c^2\sin^2\phi}};$$

we have therefore only to write $\sin\phi=\operatorname{sn}u$, to modulus

$$c,\ =\ \tfrac{1}{2}\sqrt{2-\sqrt[3]{3}},$$

and we thence obtain an expression for z in terms of the elliptic functions $\operatorname{sn}u$, $\operatorname{cn}u$, $\operatorname{dn}u$.

Writing x instead of z, and k for c, then

$$m=\sqrt[3]{2},\quad r=\sqrt[6]{3};\quad k=\tfrac{1}{2}\sqrt{2-r^2},\quad k'=\tfrac{1}{2}\sqrt{2+r^2}.$$

5–2

Working out the substitutions, the resulting formulæ are

$$x = \frac{2r\,\mathrm{sn}\,u\,\mathrm{dn}\,u - (1+\mathrm{cn}\,u)^2}{2r\,\mathrm{sn}\,u\,\mathrm{dn}\,u + (1+\mathrm{cn}\,u)^2},$$

$$y = \frac{m(1+\mathrm{cn}\,u)\,[1+r^2+(1-r^2)\,\mathrm{cn}\,u]}{2r\,\mathrm{sn}\,u\,\mathrm{dn}\,u + (1+\mathrm{cn}\,u)^2},$$

where the modulus is k as above; and these values give

$$x^3+y^3=1, \quad \frac{dx}{(1-x^3)^{\frac{2}{3}}}, = \frac{-\,dy}{(1-y^3)^{\frac{2}{3}}} = \tfrac{1}{2}mr\,du.$$

The verification is interesting enough; starting from the expression for x, and for shortness representing it by

$$x = \frac{A-B}{A+B},$$

we have

$$1-x^3 = \frac{2B(3A^2+B^2)}{(A+B)^3}, \quad = \frac{m^3(1+\mathrm{cn}\,u)^3(3A^2+B^2)}{[2r\,\mathrm{sn}\,u\,\mathrm{dn}\,u+(1+\mathrm{cn}\,u)^2]^3}.$$

We find

$$3A^2+B^2 = 12r^2\,\mathrm{cn}^2\,u\ \mathrm{dn}^2\,u + (1+\mathrm{cn}\,u)^4,$$

$$= (1+\mathrm{cn}\,u)\,[12r^2(1-\mathrm{cn}\,u)(k'^2+k^2\,\mathrm{cn}^2\,u) + (1+\mathrm{cn}\,u)^3],$$

where the term in { } is a perfect cube

$$= [1+\mathrm{cn}\,u + r^2(1-\mathrm{cn}\,u)]^3.$$

The last-mentioned expression is, in fact,

$$(1+\mathrm{cn}\,u)^3 + r^2(1-\mathrm{cn}\,u)\,[3(1+\mathrm{cn}\,u)^2 + 3r^2(1-\mathrm{cn}\,u)(1+\mathrm{cn}\,u) + r^4(1-\mathrm{cn}\,u)^2],$$

where the second term is

$$= 12r^2(1-\mathrm{cn}\,u)\,[\tfrac{1}{4}(1+\mathrm{cn}\,u)^2 + \tfrac{1}{4}r^2(1-\mathrm{cn}^2\,u)],$$

that is, it is

$$= 12r^2(1-\mathrm{cn}\,u)\,(k'^2+k^2\,\mathrm{cn}^2\,u).$$

We have consequently

$$1-x^3 = \frac{m^3(1+\mathrm{cn}\,u)^3\,[1+r^2+(1-r^2)\,\mathrm{cn}\,u]^3}{[2r\,\mathrm{sn}\,u\,\mathrm{dn}\,u+(1+\mathrm{cn}\,u)^2]^3},$$

or extracting the cube root $y, = \sqrt[3]{1-x^3}$, has its foregoing value: and the differential expressions are then verified.

Suppose $y=1$, we have

$$(m-1)(1+\mathrm{cn}\,u)^2 + mr^2(1-\mathrm{cn}^2\,u) = 2r\,\mathrm{sn}\,u\ \mathrm{dn}\,u,$$

that is,

$$(m-1)^2(1+\operatorname{cn}u)^4+2m(m-1)r^2(1+\operatorname{cn}u)^2(1-\operatorname{cn}u)$$
$$+3m^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)^2=r^2(1-\operatorname{cn}u)\{4-4k^2(1-\operatorname{cn}^2u)\},$$

or observing that the right-hand side is

$$=r^2(1-\operatorname{cn}u)\{(1+\operatorname{cn}u)^2+(1-\operatorname{cn}u)^2+r^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)\},$$

and multiplying by $\frac{1}{1+\operatorname{cn}u}$, the equation becomes

$$0=\tfrac{1}{2}(m-1)^2r^2(1+\operatorname{cn}u)^3+(2m^2-2m+1)(1+\operatorname{cn}u)^2(1-\operatorname{cn}u)$$
$$+(m^2-1)r^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)^2-(1-\operatorname{cn}u)^3;$$

viz. this is

$$0=\tfrac{1}{2}r^2(m^2-1)(1+\operatorname{cn}u)-(1-\operatorname{cn}u)^3,$$

as is immediately verified: hence writing $\frac{1}{2}r^2=\dfrac{1}{m^2}$, we have for the value in question, $y=1$,

$$(m^2-1)(1+\operatorname{cn}u)-r^2(1-\operatorname{cn}u)=0,$$

or say

$$m^2(1+\operatorname{cn}u)=(1+\operatorname{cn}u)+r^2(1-\operatorname{cn}u),$$

that is,

$$\operatorname{cn}u=\frac{r^2+1-m^2}{r^2-1+m^2},$$

which is one of the values of $\operatorname{cn}u$ derived from the equation $x=0$; but this equation $x=0$ gives, not the foregoing equation, but

$$m^2(1+\operatorname{cn}u)^3=[(1+\operatorname{cn}u)+r^2(1-\operatorname{cn}u)]^3,$$

viz. the three values of $\operatorname{cn}u$ are the foregoing value and the two values obtained therefrom by changing m into ωm and $\omega^2 m$ respectively, ω being an imaginary cube root of unity. In fact, the curve $x^3+y^3-1=0$ has at the point $x=0$, $y=1$ an inflexion, the tangent being $y=1$, so that this line meets the curve in the point counting three times; but the line $x=0$ meets the curve in the point, and besides in two imaginary points.

805.

NOTE ON ABEL'S THEOREM.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. IV. (1883), pp. 119–122.]

CONSIDERING Abel's theorem in so far as it relates to the first kind of integrals, and as a differential instead of an integral theorem, the theorem may be stated as follows:

We have a fixed curve $f(x,\ y,\ 1) = 0$ of the order m; this implies a relation $f'(x)\,dx + f'(y)\,dy = 0$, between the differentials dx, dy of the coordinates of a point on the curve; and we may therefore write

$$d\omega = \frac{dx}{f'(y)} = -\frac{dy}{f'(x)},$$

and, instead of dx or dy, use $d\omega$ to denote the displacement of a point $(x,\ y)$ on the curve.

Taking for greater simplicity the fixed curve to be a curve without nodes or cusps, and therefore of the deficiency $\frac{1}{2}(m-1)(m-2)$, we consider its intersections by a variable curve $\phi(x,\ y,\ 1) = 0$ of the order n. And then, if $(x,\ y,\ 1)^{m-3}$ denote an arbitrary rational and integral function of $(x,\ y)$ of the order $m-3$, the theorem is that we have between the displacements $d\omega_1$, $d\omega_2$, ..., $d\omega_{mn}$ of the mn points of intersection, the relation

$$\Sigma(x,\ y,\ 1)^{m-3}\,d\omega = 0,$$

where the left-hand side is the sum of the values of $(x,\ y,\ 1)^{m-3}\,d\omega$ belonging to the mn points of intersection respectively.

For the proof, observe that, varying in any manner the curve ϕ, we obtain

$$\frac{d\phi}{dx}\,dx + \frac{d\phi}{dy}\,dy + \delta\phi = 0,$$

where $\delta\phi$ is that part which depends on the variation of the coefficients, of the whole variation of ϕ; viz. if $\phi = ax^n + bx^{n-1}y + \ldots$, then $\delta\phi = x^n\, da + x^{n-1}y\, db + \ldots$; $\delta\phi$ is thus, in regard to the coordinates $(x,\ y)$, a rational and integral function of the order n.

Writing in this equation

$$dx,\ dy = \frac{df}{dy}\, d\omega,\ -\frac{df}{dx}\, d\omega,$$

the equation becomes

$$\left(\frac{d\phi}{dx}\frac{df}{dy} - \frac{d\phi}{dy}\frac{df}{dx}\right) d\omega + \delta\phi = 0,$$

or say

$$-J(f,\ \phi)\, d\omega + \delta\phi = 0,$$

that is,

$$d\omega = \frac{\delta\phi}{J(f,\ \phi)};$$

and then multiplying each side by the arbitrary function $(x,\ y,\ 1)^{m-3}$, we have

$$\Sigma(x,\ y,\ 1)^{m-3}\, d\omega = \Sigma\frac{(x,\ y,\ 1)^{m-3}}{J(f,\ \phi)}\,\delta\phi,$$

where $\delta\phi$ being of the order n in the variables, the numerator is a rational and integral function of $(x,\ y)$ of the order $m+n-3$: hence by a theorem contained in Jacobi's paper "Theoremata nova algebraica circa systema duarum æquationum inter duas variables," *Crelle*, t. xiv. (1835), pp. 281–288, [*Ges. Werke*, t. iii., pp. 285–294], the sum on the right-hand side is $= 0$: hence the required result $\Sigma(x,\ y,\ 1)^{m-3}\, d\omega = 0$.

Observing that $(x,\ y,\ 1)^{m-3}$ is an arbitrary function, the equation just obtained breaks up into the equations

$$\Sigma\, d\omega = 0,\quad \Sigma x\, d\omega = 0,\quad \Sigma y\, d\omega = 0,\ \ldots,\quad \Sigma x^{m-3}\, d\omega = 0,\ \ldots,\quad \Sigma y^{m-3}\, d\omega = 0,$$

viz. the number of equations is

$$1 + 2 + \ldots + (m-1),\ = \tfrac{1}{2}(m-1)\,(m-2),$$

which is $= p$, the deficiency of the curve.

Suppose the fixed curve $f(x,\ y,\ 1) = 0$ is a cubic, $m = 3$, and we have the single relation $\Sigma\, d\omega = 0$, where the summation refers to the $3n$ points of intersection of the cubic and of the variable curve of the order n, $\phi(x,\ y,\ 1) = 0$.

In particular, if this curve be a line, $n = 1$, and the equation is $d\omega_1 + d\omega_2 + d\omega_3 = 0$; here the two points $(x_1,\ y_1)$, $(x_2,\ y_2)$ taken at pleasure on the cubic, determine the line, and they consequently determine uniquely the third point of intersection $(x_3,\ y_3)$; there should thus be a single equation giving the displacement $d\omega_3$ in terms of the displacements $d\omega_1$, $d\omega_2$; viz. this is the equation just found

$$d\omega_1 + d\omega_2 + d\omega_3 = 0.$$

So if the variable curve be a conic, $n = 2$; and we have between the displacements of the six points the relation

$$d\omega_1 + d\omega_2 + \ldots + d\omega_6 = 0 :$$

here five of the points determine the conic, and they therefore determine uniquely the sixth point; and there should be between the displacements a single relation as just found.

If the variable curve be a cubic, $n = 3$, and we have between the displacements of the nine points the relation

$$d\omega_1 + d\omega_2 + \ldots + d\omega_9 = 0 :$$

here eight of the points do *not* determine the cubic ϕ, but they nevertheless determine the ninth point, viz. (reproducing the reasoning which establishes this well-known and fundamental theorem as to cubic curves) if $\phi_0 = 0$ be a particular cubic through the 8 points, then the general cubic is $\phi_0 + kf = 0$, and the intersections with $f = 0$ are given by the equations $\phi_0 = 0$, $f = 0$; whence the ninth point is independent of k, and is determined uniquely by the 8 points. There should thus be a single relation between the displacements, viz. this is the relation just found.

And so if the variable curve be a quartic, or curve of any higher order, it appears in like manner that there should be a single relation between the displacements; this relation being in fact the foregoing relation $\Sigma\, d\omega = 0$.

But take the fixed curve to be a quartic, $m = 4$: then we have between the displacements $d\omega$ the relation

$$\Sigma(x,\ y,\ 1)\, d\omega = 0,$$

that is, the three equations

$$\Sigma x\, d\omega = 0, \quad \Sigma y\, d\omega = 0, \quad \Sigma\, d\omega = 0.$$

If the variable curve is a conic, $n = 2$, then there are 8 points of intersection; 5 of these taken at pleasure determine the conic, and they consequently determine the remaining 3 points of intersection: hence there should be 3 equations. And so if the variable curve be a curve of any higher order, then by considerations similar to those made use of in the case where the first curve is a cubic it appears that the number of equations between the displacements $d\omega$ should always be $= 3$.

But if the variable curve be a line, $n = 1$, then the number of the points of intersection is $= 4$: 2 of these taken at pleasure determine the line, and they consequently determine the remaining 2 points of intersection; and the number of equations between the displacements $d\omega$ should thus be $= 2$. But by what precedes, we have the 3 equations

$$d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 = 0,$$

$$x_1 d\omega_1 + x_2 d\omega_2 + x_3 d\omega_3 + x_4 d\omega_4 = 0,$$

$$y_1 d\omega_1 + y_2 d\omega_2 + y_3 d\omega_3 + y_4 d\omega_4 = 0;$$

here the 4 points of intersection are on a line $y = ax + b$; we have therefore $y_1 = ax_1 + b, \ldots, y_4 = ax_4 + b$; the equations between the $d\omega$'s give

$$(y_1 - ax_1 - b)\, d\omega_1 + \ldots + (y_4 - ax_4 - b)\, d\omega_4 = 0,$$

that is, is a single relation $0 = 0$; or the 3 equations thus reduce themselves to 2 independent equations.

Again, if the fixed curve be a quintic, $m = 5$, there are here between the displacements the 6 equations

$$\Sigma x^2\, d\omega = 0, \quad \Sigma xy\, d\omega = 0, \quad \Sigma y^2\, d\omega = 0,$$

$$\Sigma x\, d\omega = 0, \quad \Sigma y\, d\omega = 0, \quad \Sigma\, d\omega = 0;$$

the two cases in which the number of independent equations is less than 6 are (i) when the variable curve is a line, and (ii) when the variable curve is a conic. For the line $n = 1$, and the number should be $= 3$. We have the above 6 equations; but the equation of the line is $ax+by+c = 0$, that is, we have $ax_1+by_1+c = 0$, &c.; we deduce the 3 identical equations

$$\Sigma x(ax + by + c) = 0, \quad \Sigma y(ax + by + c) = 0, \quad \Sigma(ax + by + c) = 0,$$

and the number of independent equations is thus $6 - 3, = 3$ as it should be.

So when the variable curve is a conic, $n = 2$; the number of independent equations should be $= 5$. The points of intersection lie on a conic $(a,\ b,\ c,\ f,\ g,\ h\langle x,\ y,\ 1\rangle)^2 = 0$; we have therefore the several equations $(a,\ b,\ c,\ f,\ g,\ h\langle x_1,\ y_1,\ 1\rangle)^2 = 0$, &c.: we have therefore the single identical equation

$$\Sigma(a,\ b,\ c,\ f,\ g,\ h\langle x,\ y,\ 1\rangle)^2\, d\omega = 0,$$

and the number of independent equations is $6 - 1, = 5$ as it should be.

Obviously the like considerations apply to the case where the fixed curve is a curve of any given order whatever.

806.

DETERMINATION OF THE ORDER OF A SURFACE.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 29–32.]

[On p. lxx of the Prolegomena to C. Taylor's *Introduction to the Geometry of Conics* (1881) occurs the following passage:

"*Proof and extension of Newton's Descriptio Organica.*

"Let two angles AOB and $A\omega B$ of given magnitudes turn about O and ω respectively, and let the intersection A trace a curve of the nth order. For a given position of the arm OB there are n positions of A and therefore n of B. When OB is in the position $O\omega$ the n B's coincide with ω, which is therefore an n-fold point on the locus of B, as is also the point O; and since any line through O (or ω) meets the locus of B in n other points, the locus is of the order $2n$. Its order is the same when $A\omega B$ is a zero-angle or straight line.

"Let a given trihedral angle $O(ABC)$—or a plane OBC and a line OA rigidly attached to it—turn about O, and let a variable plane through a fixed point ω meet OA in A and the plane OBC in BC; then if the line BC describes a ruled surface of the order n the point A describes a surface of the order $4n$."

And in a foot-note it is stated that the author is indebted to Professor Cayley for the determination of the order of this surface. The following paper contains Professor Cayley's determination, which was communicated by him to Mr Taylor.]*

LEMMA. Take $(a,\ b,\ c,\ f,\ g,\ h)$ the six coordinates of a line; these are connected by the equation

$$af + bg + ch = 0.$$

[* l.c., p. 29.]

In order that the line may belong to a ruled surface, we must have between the coordinates three more equations, say these are

$$\begin{aligned} F\,(a,\ b,\ c,\ f,\ g,\ h) &= 0, \\ G\,(\quad "\quad) &= 0, \\ H\,(\quad "\quad) &= 0, \end{aligned}$$

of the orders p, q, r respectively; then the scroll is of the order $n = 2pqr$.

For expressing that the line meets an arbitrary line $(a',\ b',\ c',\ f',\ g',\ h')$, we have the linear relation

$$f'a + g'b + h'c + a'f + b'g + c'h = 0;$$

the five relations determine the ratios $a : b : c : f : g : h$, and the number of systems of values is = product of orders $= 2 \cdot p \cdot q \cdot r \cdot 1$, $= 2pqr$, viz. this is the number of lines meeting the arbitrary line; or, what is the same thing, it is = order of ruled surface.

Consider now a trihedral angle $OABC$ rotating about a fixed point O which may be taken for the origin, and consider a fixed point ω. Let the lines OB, OC each meet a line L, and the plane ωL intersect OA in a point P; then it is to be shown that, if L is a line of a ruled surface of the order n, the locus of P is a surface of the order $4n$.

Observe that, for a given position of the line L, the position of the lines OB, OC is not determinate, but that the angle BOC has any position at pleasure in the plane OL, or say it rotates round the line ON which is the normal at O to the plane OL; the line OA therefore also rotates about ON, being always inclined to it at a determinate angle θ, or the locus of OA is a right cone axis ON and semi-aperture $= \theta$.

The points P are the intersections of the several lines of the cone with the fixed plane ωL, or say that (for the given position of L) the locus of P is the conic C which is the intersection of the cone in question by the plane ωL. And then varying the position of L, the required surface is the locus of the corresponding conics C.

Take now

$$Ax + By + Cz + D = 0, \quad A'x + B'y + C'z + D' = 0,$$

for the equations of a particular line L; then writing

$$AD' - A'D,\ BD' - B'D,\ CD' - C'D,\ BC' - B'C,\ CA' - C'A,\ AB' - A'B = (a,\ b,\ c,\ f,\ g,\ h)$$

respectively, these are the six coordinates of the line L.

The equation of the plane OL is

$$(AD' - A'D)\,x + (BD' - B'D)\,y + (CD' - C'D)\,z = 0,$$

that is,

$$ax + by + cz = 0; 6 - 2$$

and the equations of the normal ON are therefore

$$\frac{x}{a}=\frac{y}{b}=\frac{z}{c};$$

we have therefore for the cone

$$\frac{ax+by+cz}{\sqrt{(a^2+b^2+c^2)}\,\sqrt{(x^2+y^2+z^2)}}=\cos\theta;$$

or if, for convenience, $\cos^2\theta=k$, then the equation of the cone is

$$(ax+by+cz)^2=k\,(a^2+b^2+c^2)\,(x^2+y^2+z^2);$$

say this is

$$[a^2-k\,(a^2+b^2+c^2),\ b^2-k\,(a^2+b^2+c^2),\ c^2-k\,(a^2+b^2+c^2),\ bc,\ ca,\ ab\langle x,\ y,\ z\rangle]^2=0. \quad \ldots(1)$$

Taking then $(x_0,\ y_0,\ z_0)$ for the coordinates of the point ω, the equation of the plane ωL is

$$\frac{Ax+By+Cz+D}{Ax_0+By_0+Cz_0+D}=\frac{A'x+B'y+C'z+D'}{A'x_0+B'y_0+C'z_0+D'},$$

viz. it is

$$(BC'-B'C)(yz_0-y_0z)+\ldots+(AD'-A'D)(x-x_0)+\ldots=0,$$

that is,

$$f(yz_0-y_0z)+g(zx_0-z_0x)+h(xy_0-x_0y)+a(x-x_0)+b(y-y_0)+c(z-z_0)=0,$$

or say

$$x(hy_0-gz_0+a)+y(-hx_0+fz_0+b)+z(gx_0-fy_0+c)+(-ax_0-by_0-cz_0)=0, \quad \ldots(2)$$

viz. (1) and (2) are the equations of the conic C, and the coordinates $(a,\ b,\ c,\ f,\ g,\ h)$ satisfy of course the equation

$$af+bg+ch=0. \qquad \ldots\ldots(3)$$

Considering now the line L as belonging to a ruled surface, the coordinates $(a,\ \ldots)$ satisfy as before three equations

$$F\,(a,\ b,\ c,\ f,\ g,\ h)=0, \qquad \ldots\ldots(4)$$
$$G\,(\quad ” \quad)=0, \qquad \ldots\ldots(5)$$
$$H\,(\quad ” \quad)=0, \qquad \ldots\ldots(6)$$

of the orders p, q, r respectively, and we can from the six equations eliminate a, b, c, f, g, h. The resulting equation $\nabla=0$ contains the coefficients of (1) in the order $1\cdot2\cdot p\cdot q\cdot r=2pqr$ (which is the product of the orders of the other 5 equations), and the coefficients of (2) in the order $2\cdot2\cdot p\cdot q\cdot r$ (which is the product of the orders of the other 5 equations). But the coefficients of (1) being quadric functions of $(x,\ y,\ z)$, and those of (2) being linear functions of $(x,\ y,\ z)$, the aggregate order in $(x,\ y,\ z)$ is

$$2\cdot2pqr+1\cdot4pqr,\ =8pqr;$$

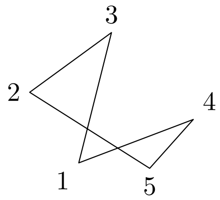
or, since the order of the ruled surface is $n,\ =2pqr$, the order of the locus is $4n$; which is the above-stated theorem.

807.

A PROOF OF WILSON'S THEOREM.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 41.]

LET n be a prime number; and imagine n points, the vertices of a regular polygon; any polygon which can be formed with these n points as vertices is either regular or else it is one of a set of n equal and similar polygons. For instance, $n = 5$, the polygon as shown in the figure is one of a set of 5 equal and similar



polygons; in fact, if the points taken in their cyclical order, but beginning at pleasure with any one of the 5 points are called 1, 2, 3, 4, 5, then we have 5 such polygons 13254; and so in general. The whole number of polygons is $\frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \ldots (n-1)$; and the number of the regular polygons is $\frac{1}{2}(n-1)$; hence the number of the remaining polygons is $= \frac{1}{2}(n-1)\,[1 \cdot 2 \ldots (n-2) - 1]$; and this number must therefore be divisible by n; that is, $1 \cdot 2 \ldots (n-1) - n + 1$ is divisible by n; or, what is the same thing, $1 \cdot 2 \ldots (n-1) + 1$ is divisible by n, which is the theorem in question.

808.

NOTE ON A FORM OF THE MODULAR EQUATION IN THE TRANSFORMATION OF THE THIRD ORDER.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 173, 174.]

In my *Treatise on Elliptic Functions*, pp. 214–216, writing only $\frac{1}{J}$, $\frac{1}{J'}$ instead of Ω, Ω', and α, β instead of α', β', I have shown as follows: viz. if k, λ denote as usual the original modulus, and the transformed modulus, and if

$$J = \frac{(k^4+14k^2+1)^3}{108k^4(1-k^2)^4}, \quad J' = \frac{(\lambda^4+14\lambda^2+1)^3}{108\lambda^4(1-\lambda^2)^4},$$

then the relation between J and J' can be found by the elimination of α, β from the equations

$$\alpha+\beta=1,$$

$$J = \frac{(1+8\alpha)^3}{64\alpha(1-\alpha)^3}, \quad J' = \frac{(1+8\beta)^3}{64\beta(1-\beta)^3}.$$

By a very slight change we obtain the result given by Prof. Klein in his paper, "Ueber die Transformation der elliptischen Functionen, &c.," *Math. Ann.* t. xiv. (1879), pp. 111–172; viz., see p. 143, the relation is to be obtained by the elimination of τ, τ' from the equation $\tau\tau'=1$, and the equations

$$J : J-1 : 1 = (\tau-1)(9\tau-1)^3 : (27\tau^2-18\tau-1)^2 : -64\tau;$$

$$J' : J'-1 : 1 = (\tau'-1)(9\tau'-1)^3 : (27\tau'^2-18\tau'-1)^2 : -64\tau';$$

these last equations being equivalent to two equations only in virtue of the identity

$$(\tau-1)(9\tau-1)^3+64\tau = (27\tau^2-18\tau-1)^2,$$

and the like identity in τ'.

In fact, writing $\alpha = \frac{\tau}{\tau-1}$, $\beta = \frac{\tau'}{\tau'-1}$, the equation $\alpha+\beta=1$ becomes $\tau\tau'=1$; and then for α, β substituting their values, we have

$$J = \frac{(9\tau-1)^3(\tau-1)}{-64\tau}, \quad J' = \frac{(9\tau'-1)^3(\tau'-1)}{-64\tau'},$$

which are the formulæ in question.

809.

SCHRÖTER'S CONSTRUCTION OF THE REGULAR PENTAGON.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 177.]

THE following construction of the regular pentagon, analogous to the more complicated one for the polygon of 17 sides, is given in the paper, H. Schröter, Zur v. Staudtschen "Construction des regulären Siebenzehnecks," *Crelle*, t. lxxv. (1873), pp. 13–24.

Take in a circle AB, CD diameters at right angles to each other, and at C, D draw in the directions A to B and B to A respectively, the lines Cc, $= 4$ radii, and Dd, $= 1$ radius; draw cd meeting the circle in the points E and F; draw CE and CF cutting AB in the points e and f respectively, and through these at right angles to AB draw the chords 34 and 25 respectively, then we have $A2345$ a regular pentagon.

810.

NOTE ON A SYSTEM OF EQUATIONS.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 191, 192.]

THE equations are

$$x^2 = ax + by, \quad xy = cx + dy, \quad y^2 = ex + fy,$$

where

$$\frac{b}{d} = \frac{a-d}{c-f} = \frac{c}{e};$$

or, what is the same thing, if a, b, c, d are given, then

$$e = \frac{cd}{b}, \quad f = c - \frac{d\,(a-d)}{b};$$

and this being so, the equations are equivalent to two independent equations; viz. starting from the first and the second equations, we have

$$dx^2 - bxy = (ad - bc)\,x,$$

that is,

$$dx - by = (ad - bc);$$

and thence

$$dxy - by^2 = (ad - bc)\,y,$$

or

$$d\,(ex + dy) - by^2 = (ad - bc)\,y;$$

which, attending to the values of e and f, is the third equation

$$y^2 = ex + fy.$$

We have

$$\frac{x}{y} = \frac{ax + by}{cx + dy}, \quad \frac{y}{x} = \frac{ex + fy}{cx + dy};$$

that is,

$$cx^2 - (a-d)\,xy - by^2 = 0,$$
$$ex^2 - (c-f)\,xy - dy^2 = 0,$$

and eliminating the $(x,\ y)$ from these equations we have an equation $\Omega = 0$ as the condition that the original three equations may have a single common root; the before-mentioned equations $\dfrac{b}{d} = \dfrac{a-d}{c-f} = \dfrac{c}{e}$ are the conditions in order that the three equations may have two common roots, that is, that there may be two systems of $(x,\ y)$ satisfying the three equations.

We have, moreover, $y\,(x-d) = ex$, $x\,(y-c) = dy$, and substituting these values, say $y = \dfrac{ex}{x-d}$ and $x = \dfrac{dy}{y-c}$, in the first and third equations respectively, they become

$$x - a = \frac{bc}{x-d}, \quad y - f = \frac{de}{y-c},$$

that is,

$$x^2 - (a+d)\,x + ad - bc = 0,$$
$$y^2 - (c+f)\,y + cf - de = 0,$$

which are quadric equations for x and y respectively; it is easy to express the second equation (like the first) in terms of $(a,\ b,\ c,\ d)$, and the first equation (like the second) in terms of $(c,\ d,\ e,\ f)$, but the forms are less simple.

Suppose $(a,\ b,\ c,\ d) = (-1,\ -1,\ 1,\ 0)$, then we have $(e,\ f) = (0,\ 1)$, the two equations in $x : y$ become $x^2 + xy + y^2 = 0$, and $0 = 0$ respectively; those in x and y become $x^2 + x + 1 = 0$, $y^2 - 2y + 1 = 0$ respectively; this is right, for the three equations are

$$x^2 = -x - y, \quad xy = x, \quad y^2 = y;$$

viz. from the third equation we have $y = 1$, a value satisfying the second equation, and then the first equation becomes $x^2 + x + 1 = 0$, or, if we please, $x^2 + xy + y^2 = 0$; the values in fact being $x = \omega$, an imaginary cube root of unity, and $y = 1$.

In the general case, the values $(x,\ y)$ may be regarded as units in a complex numerical theory, viz. if $(a,\ b,\ c,\ d,\ e,\ f)$ are integers, and $p,\ q,\ p',\ q',\ P,\ Q$ are also integers, then the product of the two complex integers $px + qy$ and $p'x + q'y$ will be a complex integer $Px + Qy$.

811.

ON THE LINEAR TRANSFORMATION OF THE THETA FUNCTIONS.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 54–60.]

THE functions referred to are the single Theta Functions; these may be defined as doubly infinite products, as was in fact done in my "Mémoire sur les fonctions doublement périodiques," *Liouv.* t. x. (1845), pp. 385–420, [25]; and it is interesting to consider from this point of view the theory of their linear transformation: this I propose to do in the present paper, adopting throughout the notation of Smith's* "Memoir on the Theta and Omega Functions."

The periods K, iK' are, in general, imaginary quantities

$$K = A + Bi,$$

$$iK' = C + Di,$$

where $AD - BC$ is positive; writing then $\omega = \dfrac{iK'}{K}$, and $q = e^{i\pi\omega}$, also for shortness

$$(q1) = 2q^{\frac{1}{8}} \prod_1^\infty (1 - q^{2n})^3,$$

where $q^{\frac{1}{8}}$ denotes $e^{\frac{i\pi\omega}{8}}$, the expression of the odd theta-function $\vartheta_1(x,\ \omega)$ as a doubly infinite product is

$$\vartheta_1(x,\ \omega) = (q1)\, x \prod{}' \left(1 + \frac{x}{m\pi + n\pi\omega}\right), \quad \left(\frac{\mu}{\nu} = \infty\right),$$

where $(m,\ n)$ have any positive or negative integer values (the combination $m = 0,\ n = 0$ excluded) from $m = -\mu$ to μ, and $n = -\nu$ to ν, μ and ν being each ultimately infinite but so that μ is infinite in comparison with ν; this condition in regard to the limits is indicated by $\mu/\nu = \infty$; and similarly $\nu/\mu = \infty$ would indicate that ν was infinite in comparison with μ.

[* Smith's *Collected Mathematical Papers*, vol. ii., pp. 415–621.]

The condition as to the limits might be that $(m,\ n)$ have any positive or negative values (excluding as before) such that the modulus of $m+n\omega$ does not exceed a positive value T, which is ultimately taken to be infinite; this condition may be indicated by $\mathrm{mod} = \infty$.

The values of the double product corresponding to the different conditions as to the limits are not equal, but they differ only by an exponential factor, the exponent being a multiple of x^2; we thus have

$$x\,\Pi\Pi\Big(1+\frac{x}{mK+niK'}\Big)\Big(\frac{\mu}{\nu}=\infty\Big) = \exp\left(\nabla x^2\right) x\,\Pi\Pi\Big(1+\frac{x}{mK+niK'}\Big)\ (\mathrm{mod}=\infty),$$

where ∇ is a determinate value, depending on K and K'; and similarly

$$x\,\Pi\Pi\Big(1+\frac{x}{m\Lambda+ni\Lambda'}\Big)\Big(\frac{\mu}{\nu}=\infty\Big) = \exp\left(\Box x^2\right) x\,\Pi\Pi\Big(1+\frac{x}{m\Lambda+ni\Lambda'}\Big)\ (\mathrm{mod}=\infty),$$

where $\Box$ is a determinate value depending in like manner on Λ, Λ'.

We have, then, as above

$$\begin{aligned}\vartheta_1(x,\ \omega) &= (q1)\,x\,\Pi\Pi\Big(1+\frac{x}{m\pi+n\pi\omega}\Big)\Big(\frac{\mu}{\nu}=\infty\Big)\\ &= (q1)\frac{\pi}{K}\cdot\frac{Kx}{\pi}\,\Pi\Pi\left(1+\frac{\frac{Kx}{\pi}}{mK+niK'}\right)\\ &= (q1)\frac{\pi}{K}\,\exp\Big(\nabla\frac{K^2x^2}{\pi^2}\Big)\,\frac{Kx}{\pi}\,\Pi\Pi\left(1+\frac{\frac{Kx}{\pi}}{mK+niK'}\right)\ (\mathrm{mod}=\infty),\end{aligned}$$

viz. we have thus defined $\vartheta_1(x,\ \omega)$ as a doubly infinite product with the limiting condition $(\mathrm{mod}=\infty)$; if for x we write $\dfrac{\pi x}{h}$, h arbitrary, we have

$$\vartheta_1\Big(\frac{\pi x}{h},\ \omega\Big) = (q1)\frac{\pi}{K}\,\exp\Big(\nabla\frac{K^2x^2}{h^2}\Big)\,\frac{Kx}{h}\,\Pi\Pi\left(1+\frac{\frac{Kx}{h}}{mK+niK'}\right)\ (\mathrm{mod}=\infty),$$

and similarly, if $\Omega=\dfrac{i\Lambda'}{\Lambda}$, $Q=e^{i\pi\Omega}$, then

$$\begin{aligned}\vartheta_1\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\} &= (Q1)\frac{\pi}{\Lambda}\,\exp\Big\{(a+b\Omega)^2\,\Box\,\Lambda^2\,\frac{x^2}{h^2}\Big\}\\ &\times(a+b\Omega)\frac{\Lambda x}{h}\,\Pi\Pi\left(1+\frac{(a+b\Omega)\frac{\Lambda x}{h}}{m\Lambda+ni\Lambda'}\right)\ (\mathrm{mod}=\infty).\end{aligned}$$

In the case of a linear transformation, we have

$$\omega=\begin{vmatrix}a, & b\\ c, & d\end{vmatrix}\Theta\,\Omega,\quad\text{that is,}\quad \omega=\frac{c+d\Omega}{a+b\Omega}, 7-2$$

where a, b, c, d are positive or negative integers such that $ad - bc = +1$; it is to be shown that the two infinite products are in this case identical; this being so, we have

$$\frac{\vartheta_1\Big\{(a+b\Omega)\dfrac{\pi x}{h},\ \Omega\Big\}}{\vartheta_1\Big\{\dfrac{\pi x}{h},\ \omega\Big\}} = \frac{(Q1)}{(q1)}\ \exp\big\{(a+b\Omega)^2\,\square\,\Lambda^2 - \nabla\,K^2\big\}\,\frac{x^2}{h^2},$$

viz. the two functions differ only by a constant factor and by an exponential factor, the exponent being a multiple of x^2: after all reductions, this factor is found to be

$$= \exp\left(-\,i\pi b\,(a+b\Omega)\,\frac{x^2}{h^2}\right).$$

We have

$$\omega = \frac{c+d\Omega}{a+b\Omega},$$

or since

$$\omega = \frac{iK'}{K}, \quad \Omega = \frac{i\Lambda'}{\Lambda},$$

this is

$$\frac{iK'}{K} = \frac{c\Lambda + di\Lambda'}{a\Lambda + bi\Lambda'},$$

or say

$$\frac{1}{M}K = a\Lambda + bi\Lambda',$$
$$\frac{1}{M}iK' = c\Lambda + di\Lambda',$$

either of which equations may be taken as a definition of the multiplier M. We have

$$\frac{K}{M\Lambda} = a + bi\Omega,$$

$$\begin{aligned}\frac{1}{M}(mK + niK') &= (ma+cn)\,\Lambda + (bm+dn)\,i\Lambda'\\ &= m'\Lambda + n'i\Lambda',\end{aligned}$$

if

$$m' = am + cn,$$
$$n' = bm + dn.$$

Here to any integer values of $(m,\ n)$ there correspond integer values of m', n'; and conversely, in virtue of the equation $ad - bc = 1$, to any integer values of m', n' there correspond integer values of m, n. The two products are

$$\Pi\Pi\left(1 + \frac{\dfrac{Kx}{h}}{m'\Lambda + n'i\Lambda'}\right),\ (\mathrm{mod} = \infty),$$

$$\Pi\Pi\left(1 + \frac{(a+b\Omega)\dfrac{\Lambda x}{h}}{m\Lambda + ni\Lambda'}\right),\ (\mathrm{mod} = \infty).$$

But, as above, we have $\dfrac{K}{M} = (a + b\Omega)\,\Lambda$: and then, observing that in the first of the two products we may for m', n' write m, n, it at once appears that the two products are identical.

The exponential factor, writing therein $(a + b\Omega)\,\Lambda = \dfrac{K}{M}$, becomes

$$\exp\left[\left(\frac{\square}{M^2} - \nabla\right)\frac{K^2x^2}{h^2}\right].$$

The values of ∇, $\square$ are at once obtained by means of a formula* given in my Memoir, viz. we have

$$\nabla = -\tfrac{1}{2}(B + \beta),$$

where

$$B = \frac{\pi(\omega\nu + \omega'\nu')}{\Pi\,\Upsilon\text{ mod }(\omega\nu - \omega'\nu)},$$

$$\beta = \frac{\pi i}{\Pi\,\Upsilon}\frac{(\omega\nu - \omega'\nu')}{\text{mod}(\omega\nu - \omega'\nu)}.$$

Comparing with the present notation

$$\Omega = \omega + \omega' i,\ = A + Bi = K,$$

$$\Upsilon = \nu + \nu' i,\ = C + Di = K'i,$$

so that Ω, Υ denote K, $K'i$, and ω, ω', ν, ν' denote A, B, C, D respectively: $\omega\nu - \omega'\nu$ is thus $= AD - BC$, which has been assumed to be positive; hence also $\text{mod}(\omega\nu - \omega'\nu) = AD - BC$, and the formula becomes

$$\nabla = -\tfrac{1}{2}\pi\left\{\frac{AC + BD}{i\,(AD - BC)} + 1\right\}\frac{1}{KK'}.$$

Now writing

$$\Lambda = A_1 + B_1 i,$$

$$i\Lambda' = C_1 + D_1 i,$$

then we have

$$\frac{1}{M}(A + Bi) = a\Lambda + bi\Lambda' = a\,(A_1 + B_1 i) + b\,(C_1 + D_1 i),$$

$$\frac{1}{M}(C + Di) = c\Lambda + di\Lambda' = c\,(A_1 + B_1 i) + d\,(C_1 + D_1 i);$$

consequently, if

$$M = \rho\,(\cos\theta + i\sin\theta),$$

[* *Collected Mathematical Papers*, t. i., p. 164. The denominator factor $i^2\Upsilon$ has been omitted (p. 165) by mistake.]

we have

$$
\begin{aligned}
\frac{1}{\rho}(A\cos\theta+B\sin\theta)&=aA_1+bC_1,\\
\frac{1}{\rho}(-A\sin\theta+B\cos\theta)&=aB_1+bD_1,\\
\frac{1}{\rho}(C\cos\theta+D\sin\theta)&=cA_1+dC_1,\\
\frac{1}{\rho}(-C\sin\theta+D\cos\theta)&=cB_1+dD_1,
\end{aligned}
$$

and thence

$$\frac{1}{\rho^2}(AD-BC)=(ad-bc)(A_1D_1-B_1C_1),\ =(A_1D_1-B_1C_1).$$

Hence $A_1D_1-B_1C_1$ is positive, and we have

$$\square=-\tfrac{1}{2}\pi\left\{\frac{A_1C_1+B_1D_1}{i\,(A_1D_1-B_1C_1)}+1\right\}\frac{1}{\Lambda\Lambda'}.$$

Take K_1 the conjugate of K, Λ_1 the conjugate of Λ, then

$$K_1=A-Bi,\quad \Lambda_1=A_1-B_1i,$$

$$iK'=C+Di,\quad i\Lambda'=C_1+D_1i.$$

We have

$$iK_1K'=AC+BD+i\,(AD-BC),$$

and therefore

$$\frac{K_1K'}{AD-BC}=\frac{AC+BD}{i\,(AD-BC)}+1,\quad \nabla=\frac{-\frac{1}{2}\pi}{AD-BC}\cdot\frac{K_1}{K},$$

and similarly

$$\square=\frac{-\frac{1}{2}\pi}{A_1D_1-B_1C_1}\cdot\frac{\Lambda_1}{\Lambda}.$$

The exponential is

$$\left(\frac{\square}{M^2}-\nabla\right)\frac{K^2x^2}{h^2},$$

and we have

$$\frac{\square}{M^2}-\nabla=\frac{-\frac{1}{2}\pi}{M^2(A_1D_1-B_1C_1)}\,\frac{\Lambda_1}{\Lambda}+\frac{\frac{1}{2}\pi}{AD-BC}\,\frac{K_1}{K},$$

which is

$$
\begin{aligned}
&=\frac{-\frac{1}{2}\pi}{M^2(A_1D_1-B_1C_1)}\,\frac{\Lambda_1}{\Lambda}+\frac{\frac{1}{2}\pi}{\rho^2(A_1D_1-B_1C_1)}\,\frac{K_1}{K}\\
&=\frac{-\frac{1}{2}\pi}{A_1D_1-B_1C_1}\left(\frac{1}{M^2}\,\frac{\Lambda_1}{\Lambda}-\frac{1}{\rho^2}\,\frac{K_1}{K}\right).
\end{aligned}
$$

But $\rho/M = \cos\theta - i\sin\theta$, or calling this for a moment P, then $1/M^2 = P^2/\rho^2$, and the formula may be written

$$\frac{\square}{M^2} - \nabla = \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\left(P\frac{\Lambda_1}{\Lambda} - P^{-1}\frac{K_1}{K}\right)$$

$$= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\frac{1}{K\Lambda}\{(\cos\theta - i\sin\theta)\,\Lambda_1 K - (\cos\theta + i\sin\theta)\,\Lambda K_1\}.$$

The term in { } is

$$\begin{aligned}
&(\cos\theta - i\sin\theta)\,(A+Bi)\,(A_1 - B_1 i) - (\cos\theta + i\sin\theta)\,(A - Bi)\,(A_1 + B_1 i)\\
&\quad = 2\cos\theta\,[-(AB_1 - A_1B)\,i] - 2i\sin\theta\,(AA_1 + BB_1),\\
&\quad = -2i\,\{(AB_1 - A_1B)\,\cos\theta + (AA_1 + BB_1)\,\sin\theta\},\\
&\quad = -2i\,\{B_1\,(A\cos\theta + B\sin\theta) - A_1\,(-A\sin\theta + B\cos\theta)\},\\
&\quad = -2i\rho\,\{B_1\,(aA_1 + bC_1) - A_1\,(aB_1 + bD_1)\},\\
&\quad = +2ib\rho\,(A_1D_1 - B_1C_1).
\end{aligned}$$

Hence

$$\begin{aligned}
\frac{\square}{M^2} - \nabla &= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\,2ib\rho\,(A_1D_1 - B_1C_1)\frac{1}{K\Lambda}\\
&= -\frac{i\pi bP}{\rho}\cdot\frac{1}{K\Lambda} = -\frac{i\pi b}{M}\cdot\frac{1}{K\Lambda},
\end{aligned}$$

and the exponential thus is

$$= \exp\left(-\frac{i\pi b}{M}\cdot\frac{1}{K\Lambda}\cdot\frac{K^2x^2}{h^2}\right) = \exp\left(-i\pi b\,\frac{K}{M\Lambda}\cdot\frac{x^2}{h^2}\right);$$

or, since $\dfrac{K}{M\Lambda} = (a + b\Omega)$, this is

$$= \exp\left(-i\pi b\,(a + b\Omega)\,\frac{x^2}{h^2}\right);$$

and we have thus the required formula

$$\frac{\vartheta_1\left\{(a+b\Omega)\dfrac{\pi x}{h},\ \Omega\right\}}{\vartheta_1\left\{\dfrac{\pi x}{h},\ \omega\right\}} = \frac{(Q1)}{(q1)}\,(a + b\Omega)\,\exp\left(-i\pi b\,(a + b\Omega)\,\frac{x^2}{h^2}\right).$$

usepackage[a4paper,margin=0.65in]geometry

812.

ON ARCHIMEDES' THEOREM FOR THE SURFACE OF A CYLINDER.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 107, 108.]

The measure of the surface of a cylinder was first obtained by Archimedes in his Treatise on the Sphere and Cylinder (Book i., Prop. xiv.), *Œuvres d'Archimède*, par F. Peyrard, 4º Paris, 1807, pp. 26–31; viz. Archimedes showed that the surface of the cylinder was equal to the area of a circle, having its radius a mean proportional between the height and the diameter of the circular base [$S = 2\pi ah, = \pi\,\{\sqrt{(2a\,.\,h)}\}^2$].

The following is in effect his demonstration:

He considers regular polygons (with the same number of sides) inscribed in and circumscribed about a circle; and, as regards the cylinder, the prisms standing on these polygons.

Say for the circular base of the cylinder we have

$S^\times$	surface of circumscribed prism,
S	cylinder,
$S_\times$	inscribed prism;

and for the circle, having its radius a mean proportional between the height and the diameter of the circular base,

$B^\times$	area of circumscribed polygon,
B	circle,
$B_\times$	inscribed polygon,

where the four polygons referred to by $S^\times$, $S_\times$, $B^\times$, $B_\times$ have all of them the same number of sides.

It is in the preceding propositions (by means of an axiom as to curve lines) shown that

$$S^{\times} > S > S_{\times}, \quad B^{\times} > B > B_{\times}\,;$$

and it is further shown that

$$S^{\times} = B^{\times}, \quad S_{\times} = B_{\times}.$$

It is moreover shown that, by taking the number of sides sufficiently large, the ratio $B^{\times} : B_{\times}$, or say the fraction $B^{\times}/B_{\times}$ (which is greater than 1) may be made less than any given quantity $1 + \varepsilon$.

It is then to be shown that $S = B$.

If not, then

either

$B < S$.

This being so, it is possible to make

$B_{\times}/B^{\times} < S/B$,

that is,

$S^{\times}/B^{\times} < S/B$,

or

$S^{\times}/S < B^{\times}/B$,

which is absurd, since

$S^{\times}/S > 1\,;\ B^{\times}/B < 1\,;$

or else

$B > S$.

This being so, it is possible to make

$B_{\times}/B^{\times} < B/S$,

that is,

$B^{\times}/S^{\times} < B/S$,

or

$B^{\times}/B < S^{\times}/S$,

which is absurd, since

$B^{\times}/B > 1\,;\ S^{\times}/S < 1\,;$

and consequently $S = B$, the theorem in question.

I take the opportunity of referring to two theorems by Archimedes, Lemmas, Prop. v. and vi., Peyrard, pp. 429–435, which relate to the contacts of circles. We have in each of them the figure which he calls the Arbelon, viz. if A, C, B are points in this order on the same straight line, then the figure consists of the three semicircles on the diameters AC, CB, and AB respectively, and the Arbelon is the space included between the three semi-circumferences.

In Prop. v., we have the common tangent at C to the two semicircles AC, CB; this divides the Arbelon into two mixtilinear triangles (each bounded by the common tangent, one of the smaller semicircles, and a portion of the larger semi-circle), and inscribing each of these a circle, the theorem is that the two inscribed circles are of equal magnitude.

In Prop. vi., the theorem is that the radii of the smaller semicircles being as 3 : 2, then the radius of the circle inscribed in the Arbelon is to the diameter of the larger semicircle as 6 to 19. But it is noticed that the demonstration would apply to any other value of the ratio; and, in fact, if the radii of the two smaller circles are as $a : b$, then the radius of the inscribed circle is to the diameter of the larger semicircle as ab to a^2+ab+b^2, which is the general form of the theorem.

813.

[NOTE ON MR GRIFFITHS' PAPER "ON A DEDUCTION FROM THE ELLIPTIC-INTEGRAL FORMULA $y = \sin(A + B + C + \ldots)$".]

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), p. 81.]

Consider, for instance,

the cubic transformation

$$y = \frac{x\,[1 + 2\alpha' - (1 + \alpha')^2\,x^2]}{1 - \alpha^2 x^2},$$

where $\alpha^2 + \alpha'^2 = 1$.

This implies

$$\sqrt{1 - y^2} = \frac{\sqrt{1 - x^2}\,[1 - (1 + \alpha')^2\,x^2]}{1 - \alpha^2 x^2},$$

viz., $\sqrt{1 - y^2}$ = a rational multiple of $\sqrt{1 - x^2}$.

Also the quadric transformation

$$z = \frac{1 - (1 + \beta'^2)\,x^2}{1 - \beta^2 x^2}\,;$$

where $\beta^2 + \beta'^2 = 1$.

This implies

$$\sqrt{1 - z^2} = \frac{\sqrt{1 - x^2}\,.\,2\beta' x}{1 - \beta^2 x^2},$$

viz., $\sqrt{1 - z^2}$ = a rational multiple of $\sqrt{1 - x^2}$.

Hence, assuming

$$u = yz - \sqrt{1 - y^2}\,\sqrt{1 - z^2},$$

which is a rational function

$$= \frac{x\,(a_0 - a_2 x^2 + a_4 x^4)}{1 - \alpha^2 x^2\,.\,1 - \beta^2 y^2},$$

we have

$$\sqrt{1 - u^2} = y\sqrt{1 - z^2} + z\sqrt{1 - y^2},$$

which is $= \sqrt{1 - x^2}$ multiplied by a like rational function.

That is, in defining the a_0, a_2, a_4 functions of the two arbitrary coefficients α, β, as above, we have in effect so determined them that $\sqrt{1-u^2}$ shall be $= \sqrt{1-x^2}$ multiplied by a rational function of x.

We can then further determine a_0, a_2, a_4 in such wise that the change of x into $\dfrac{1}{kx}$ shall change u into $\dfrac{1}{\lambda u}$; and, this being so, making the change in $\sqrt{1-x^2}$, we obtain $\sqrt{1-\lambda^2u^2}$ in the form, $\sqrt{1-k^2x^2}$ multiplied by a rational function of x; viz. u is a function of x such that

$$\frac{du}{\sqrt{1-u^2}\,\sqrt{1-\lambda^2u^2}} = \frac{M\,dx}{\sqrt{1-x^2}\,\sqrt{1-k^2x^2}}.$$

The theory is thus in effect Jacobi's—with the novelty of combining two lower transformations in such wise that the assumed expression for u as a rational function of x shall give

$$\sqrt{1-u^2} = \sqrt{1-x^2} \text{ multiplied by a rational function of } x.$$

It is not necessary that the equations

$$y = \text{rational function of } x \quad \text{and} \quad z = \text{rational function of } x$$

should be elliptic-function transformations. All that is required is that they should be such as to give $\sqrt{1-y^2}$ and $\sqrt{1-z^2}$ each $= \sqrt{1-x^2}$ multiplied by a rational function of x.

814.

ON DOUBLE ALGEBRA.

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), pp. 185–197. Read April 3, 1884.]

1. I consider the Double Algebra formed with the extraordinary symbols, or "extraordinaries" x, y, which are such that

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

or, as these equations may also be written,

$$\begin{array}{c|cc} & x & y \\ \hline x & (a,\ b) & (c,\ d) \\ y & (e,\ f) & (g,\ h) \end{array} ,$$

where a, b, c, d, e, f, g, h are ordinary symbols, or say coefficients; all coefficients being commutative and associative *inter se* and with the extraordinaries x, y.

The system depends in the first instance on the eight parameters a, b, c, d, e, f, g, h; but we may, instead of the extraordinaries x, y, consider the new extraordinaries connected therewith by the linear relations $\xi = \alpha x + \beta y$, $\eta = \gamma x + \delta y$, where the coefficients α, β, γ, δ may be determined so as to establish between the eight parameters any four relations at pleasure (or, what is the same thing, α, β, γ, δ are what I call "apocatastic" constants): and the number of parameters is thus properly $8 - 4, = 4$.

2. The extraordinaries here considered are not in general associative; differing herein from the imaginaries of Peirce's Memoir, "Linear Associative Algebra" (1870),

reprinted in the *American Mathematical Journal*, t. iv. (1881), pp. 97–227, which as appears by the title, refers only to associative imaginaries. I recall some definitions and results. The symbol x is said to be *idempotent* if $x^2 = x$, *nilpotent* if $x^2 = 0$; and the systems of associative symbols are expressed as much as may be by means of such idempotent and nilpotent symbols: thus the linear systems are $(a_1)\, x^2 = x$, $(b_1)\, x^2 = 0$. A double linear system composed of independent symbols, that is, symbols x, y each belonging to its own linear system, and moreover such that $xy = yx = 0$, is said to be "mixed"; thus the mixed double systems are

$$\begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & y \end{array}\,, \qquad \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & 0 \end{array}\,, \qquad \begin{array}{c|cc} & x & y \\ \hline x & 0 & 0 \\ y & 0 & 0 \end{array}\,.$$

But Peirce excludes those from consideration, attending only to the pure systems, which he finds to be

$$(a_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & y & 0 \end{array}\,, \qquad (b_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & 0 & 0 \end{array}\,, \qquad (c_2)\ \begin{array}{c|cc} & x & y \\ \hline x & 0 & x \\ y & 0 & 0 \end{array}\,.$$

To these, however, should be added the system

$$(d_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & y & 0 \end{array}\,;$$

see post, No. 19.

3. In the general theory, where the symbols are not in the first instance taken to be associative, we may of course establish between the coefficients such relations as will make the symbols associative; and the question presents itself to show how .in this case the system reduces itself to one of Peirce's systems. This I considered in my note "On Associative Imaginaries," *Johns Hopkins University Circular*, No. 15 (1882), p. 211 [822]; I there obtained, as the general form of the commutative and associative system,

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= \frac{cd}{b}\,x + \frac{d^2 + bc - cd}{b}\,y, \end{aligned}$$

the relation of which to Peirce's system was, as I there remarked, pointed out to me by Mr C. S. Peirce: this will be considered in the sequel, Nos. 13 to 19.

4. Starting now with the general equations

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

we may attempt to find an extraordinary; $\xi, = \alpha x + \beta y$ (α, β coefficients), such that $\xi^2 = K\xi$ (K a coefficient). In general, K is not $= 0$, and when it is not $= 0$, it may without loss of generality be taken to be $= 1$; we have then $\xi^2 = \xi$, the *idempotent* symbol. But K may be $= 0$; and then $\xi^2 = 0$, ξ a *nilpotent* symbol. To include the two cases, I retain K, it being understood that, when K is not $= 0$, it may be taken to be $= 1$. We have

$$\begin{aligned} \xi^2 &= \alpha^2\,(ax + by) + \alpha\beta\,(\overline{c+e}\,x + \overline{d+f}\,y) + \beta^2\,(gx + hy) \\ &= [\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2]\,x + [\alpha^2 b + (d+f)\,\alpha\beta + h\beta^2]\,y. \end{aligned}$$

Hence, when this is $= K\xi$, we have

$$\frac{\alpha}{\beta} \cdot K(\alpha x + \beta y),$$

$$\frac{\alpha}{\beta} = \frac{\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2}{\alpha^2 b + (d+f)\,\alpha\beta + h\beta^2},$$

a cubic equation for the determination of the ratio $\alpha : \beta$; and, for any particular value of the ratio, we can in general determine the absolute magnitudes, so that

$$K, = \frac{1}{\alpha}\,[\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2], = \frac{1}{\beta}\,[b\alpha^2 + (d+f)\,\alpha\beta + h\beta^2],$$

shall be $= 1$. If, however, for the given value of the ratio we have only

$$\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2 = 0, \quad b\alpha^2 + (d+f)\,\alpha\beta + h\beta^2 = 0,$$

one of these equations, of course, implying the other, then the value of K is $= 0$.

5. It follows that there are in general three idempotent symbols ξ, η, ζ, that is, extraordinaries such that $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$. The cubic equation may, however, have two equal roots, or three equal roots identically; in this last case, any extraordinary $\alpha x + \beta y$ is an idempotent symbol, and we have to consider in detail further on we may, instead of an idempotent symbol or

symbols, have a nilpotent symbol or symbols. It might be convenient to use the term Pierce's symbol which is in general idempotent, but which may be also nilpotent. Writing $\dfrac{\alpha}{\beta} = \dfrac{-y}{x}$, we obtain a cubic equation $\Omega = (x, y)^3 = 0$, where obviously the linear factors of Ω are the just-mentioned functions ξ, η, ζ; that is, we have ξ, η, ζ as the linear factors of the cubic function

$$\Omega = gx^3 + (h - c - e)\, x^2 y + (a - d - f)\, xy^2 + by^3;$$

each such factor, except in the case where it is nilpotent, being determined so that it shall be idempotent. The cubic function of course vanishes identically if

$$g = 0, \quad h - c - e = 0, \quad a - d - f = 0, \quad b = 0.$$

6. Two extraordinaries $\xi, = \alpha x + \beta y$; $\eta, = \gamma x + \delta y$ (α, β, γ, δ coefficients); may be such that $\xi\eta = 0$; this, of course, does not imply $\eta\xi = 0$; we have, in fact,

$$\begin{aligned}
\xi\eta &= (\alpha x + \beta y)\,(\gamma x + \delta y) \\
&= \alpha\gamma x^2 + \alpha\delta xy + \beta\gamma yx + \beta\delta y^2 \\
&= \alpha\gamma\,(ax + by) + \alpha\delta\,(cx + dy) + \beta\gamma\,(ex + fy) + \beta\delta\,(gx + hy) \\
&= (\alpha\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta)\,x + (b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta)\,y\,;
\end{aligned}$$

and the required condition is satisfied if

$$a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta = 0,$$

$$b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta = 0.$$

Writing these equations first under the form

$$\gamma\,(a\alpha + e\beta) + \delta\,(c\alpha + g\beta) = 0,$$

$$\gamma\,(b\alpha + f\beta) + \delta\,(d\alpha + h\beta) = 0,$$

and then under the form

$$\alpha\,(a\gamma + c\delta) + \beta\,(e\gamma + g\delta) = 0,$$

$$\alpha\,(b\gamma + d\delta) + \beta\,(f\gamma + h\delta) = 0,$$

we have

$$(a\alpha + e\beta)\,(d\alpha + h\beta) - (b\alpha + f\beta)\,(c\alpha + g\beta) = 0,$$

a quadric equation for the determination of $\alpha : \beta$; and similarly

$$(a\gamma + c\delta)\,(f\gamma + h\delta) - (b\gamma + d\delta)\,(e\gamma + g\delta) = 0,$$

a quadric equation for the determination of $\gamma : \delta$; that is, there are two values of the left-hand factor ξ, and two values of the right-hand factor η. But, of course, these correspond each to each, viz. either factor being given, the other factor is determined uniquely.

Writing successively $\dfrac{\alpha}{\beta} = \dfrac{-y}{x}$, and $\dfrac{\gamma}{\delta} = \dfrac{-y'}{x'}$, we have the quadric functions $(x, y)^2$,

$$\Phi = (ah - fg)\,x^2 + (-ah - cd + bg + cf)\,xy + (cd - be)\,y^2,$$

$$\Phi_1 = (ah - dg)\, x'^2 + (-ah - cf + bg + de)\, x'y' + (cf - be)\, y'^2,$$

where the linear factors of Φ are the two values ξ_1, ξ_2 of the left-hand factor ξ, and the linear factors of Φ_1 are the two values η_1, η_2 of the right-hand factor η.

7. In the commutative case, $c = e$, and $d = f$, we have

$$\Phi = \Phi_1 = (ah - dg)\, x^2 + (-ah + bg)\, xy + (cd - be)\, y^2\,;$$

here $\xi\eta = 0$, $\eta\xi = 0$, and the values may be taken to be $(\xi_1,\ \eta_1) = (\xi,\ \eta)$, $(\xi_2,\ \eta_2) = (\eta,\ \xi)$, so that $\xi_1\xi_2 = \xi\eta$, as above. The value of Ω is

$$\Omega = gx^3 + (h - 2c)\,x^2y + (a - 2d)\,xy^2 + by^3.$$

8. In the commutative and associative case, taking ξ, η, ζ to be the three idempotent symbols, $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$; we have

$$\xi\,(\eta - \xi\eta) = \xi\eta - \xi^2\eta = \xi\eta - \xi\eta = 0\,;$$

and in like manner we have the six equations

$$\xi\,(\eta{-}\xi\eta) = 0,\ \xi\,(\zeta{-}\xi\zeta) = 0\,;\ \eta\,(\xi{-}\eta\xi) = 0,\ \eta\,(\zeta{-}\eta\zeta) = 0\,;\ \zeta\,(\xi{-}\zeta\xi) = 0,\ \zeta\,(\eta{-}\zeta\eta) = 0\,;$$

viz. regarding the right-hand factor as being in each case expressed as a linear function of x, y, we have apparently six distinct factors $\eta - \xi\eta$, $\zeta - \xi\zeta$, &c. each of these products would be in any one case identically $= 0$, viz. the case of the second factor being $= 0$. But we know there are only two right-hand factors η_1, η_2, that is, $\eta - \xi\eta$ must be in some way or other identically a linear function of ξ; the same as $\zeta - \xi\zeta$, but here suppose (a, b coefficients, neither of them $= 0$); we thence have

$$\xi\,(a\xi + b\eta) = a\xi^2 + b\xi\eta,\ = a\xi + b\eta,\ = a\xi + b\eta,$$

that is, $(a^2 - a)\,\xi + (b^2 - b)\,\eta = 0$; whence $a = 1$, $b = 1$, and therefore $\xi\eta = \xi + \eta$; hence also $\xi\zeta$ and $\eta\zeta = \xi + \zeta$ and $\eta + \zeta$; $\xi\eta\,\eta\zeta$. The six products consequently are $\xi\eta$, $\eta\zeta$, $\eta\xi$, $\zeta\xi$, $\zeta\eta$, each $= 0$ or identically $= 0$.

9. In verification of the theorem that for the commutative and associative system the cubic function Ω contains the quadric function Φ as a factor, we may write, as above,

$$y = \frac{cd}{b}, \quad h = \frac{d^2 + bc - cd}{b},$$

values which give

$$\begin{aligned} b\Phi &= (bc^2 - acd)\,x^2 + (-ad^2 - abc + a\,d + bcd)\,xy + (bcd - b^2e)\,y^2 \\ &= -(ad - bc)\,[cx^2 + (d - a)\,xy - by^2], \\ b\Omega &= cdx^3 + (d^2 - bc)\,x^2y + (ab - 2bd)\,xy^2 + b^2y^3, \\ &= (dx - by)\,[cx^2 + (d - a)\,xy - by^2], \end{aligned}$$

which gives the theorem in question. And observe further that

$$\begin{aligned} (dx - by)\xi = d^2\,(ax + by)\,c - 2bd\,(cx + dy) + b^2\,\left(\frac{cd}{b}\,x + \frac{d^2 + bc - cd}{b}\,y\right), \\ = -(ad - bc)\,(dx - by), \end{aligned}$$

that is, disregarding coefficients, the two idempotent symbols ξ, η are the linear factors of $cx^2 + (d - a)\,xy - by^2$, and the third idempotent symbol ζ is $= dx - by$.

10. Introducing coefficients in order to make the symbols ξ, η, ζ idempotent, and writing accordingly

$$\xi = \frac{1}{K}\,[cx + \tfrac{1}{2}\,(d - a + \sqrt{\nabla}\,)\,y], \quad \nabla = (d-a)^2 + 4bc,$$

so that

$$\eta = \frac{1}{L}\,[cx + \tfrac{1}{2}\,(d - a - \sqrt{\nabla}\,)\,y], \quad \xi\eta = \frac{c}{KL}\,[cx^2 + (d-a)\,xy - by^2],$$

$$\zeta = \frac{1}{P}\,(dx - by),$$

we have to verify that it is possible to determine K, L, P so that $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$, $\xi\eta = \xi + \eta$. The last equation gives

$$\frac{d}{P} = \frac{c}{K} + \frac{c}{L},$$

$$-\frac{2b}{P} = \frac{d - a + \sqrt{\nabla}}{K} + \frac{d - a - \sqrt{\nabla}}{L},$$

and we thence have

$$\frac{d\,(d-a) + 2bc - d\sqrt{\nabla}}{P} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{d\,(d-a) + 2bc + d\sqrt{\nabla}}{P} = \frac{2c\sqrt{\nabla}}{L},$$

and we can from the equation $\xi^2 = \xi$ find P; viz. comparing the coefficients of x, we have

$$\frac{d}{P} = \frac{1}{P^2}\left(ad^2 - 2b\,dc + b\frac{cd}{b}\right), \ = \frac{d}{P^2}\,(ad - bc), \text{ that is, } P = (ad - bc),$$

or the values of K and L are

$$\frac{[d\,(d-a) + 2bc - d\sqrt{\nabla}\,]}{ad - bc} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{[d\,(d-a) + 2bc + d\sqrt{\nabla}\,]}{ad - bc} = +\frac{2c\sqrt{\nabla}}{L},$$

which should agree with the values of K and L found from the equations $\xi^2 = \xi$, $\eta^2 = \eta$, respectively. Comparing the coefficients of x, the first of these equations gives

$$\begin{aligned}
\frac{c}{K} &= \frac{1}{K^2}\left\{c^2 a + c\,(d - a + \sqrt{\nabla}\,)\,c + \tfrac{1}{4}\,(d - a + \sqrt{\nabla}\,)^2\,\frac{cd}{b}\right\} \\
&= \frac{c}{4bK^2}\left\{4abc + 4bc\,(d-a) + d\,[(d-a)^2 + \nabla\,] + 2\,[2bc + d\,(d-a)]\,\sqrt{\nabla}\right\} \\
&= \frac{c}{2bK^2}\left\{d\nabla + [d\,(d-a) + 2bc]\,\sqrt{\nabla}\right\},
\end{aligned}$$

that is,

$$K = \frac{1}{2b}\sqrt{\nabla}\,[d\,(d-a) + 2bc) + d\sqrt{\nabla}\,],$$

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and the equation for K becomes

$$\frac{d\,(d-a)+2bc-d\sqrt{\nabla}}{ad-bc}=\frac{-4bc}{d\,(d-a)+2bc+d\sqrt{\nabla}}\,;$$

that is,

$$[d\,(d-a)+2bc]^2-d^2\,[(d-a)^2+4bc]=-4bc\,(ad-bc),$$

which is right; and similarly the equation for L leads to this same equation.

11. We may now establish, on the principles appearing in No. 5, the different forms of the system. Using ideas and aid as abbreviations for idempotent and nilpotent respectively, there are in all 11 cases.

(i) 3 idems. Taking two of these to be x and y, the system is

$$x^2=x,\quad xy=ex+dy,\quad yx=ex+fy,\quad y^2=y.$$

Hence $\Omega=(1-c-e)\,x^2y+(1-d-f)\,xy^2=0$, so that the third factor is

$$(1-c-e)\,x+(1-d-f)\,y.$$

This must not reduce itself to x or y, for, if so, there would be a twofold idem.; viz. as negative conditions we must have $c+e\neq 1$, $d+f\neq 1$.

And we have

$$[(1-c-e)\,x+(1-d-f)\,y]^2=[1-(c+e)\,(d+f)]\,[(1-c-e)\,x+(1-d-f)\,y],$$

which must be an idem.; viz. we have the further negative condition $(c+e)\,(d+f)\neq 1$.

(ii) 2 idems and 1 nil. This arises from (i) by assuming therein

$$(c+e)\,(d+f)=1,\ \text{say}\ d+f=\frac{1}{c+e}\,;$$

for then, writing $z=-\,(c+e)\,x+y$, we have

$$z^2=(c+e)^2\,x^2-(c+e)\,[(c+e)\,x+(d+f)\,y]+y,$$
$$=[1-(c+e)\,(d+f)]\,y,\ =0\,;$$

viz. z is a nil. And, if in the equations instead of the idem y we introduce the nil z, then the equations assume the form

$$x^2=x,\quad xz=[(c+e)\,d-c]\,x+dz,\quad zx=[(c+e)\,f-c]\,x+fz,\quad z^2=0\,;$$

with the idem y, $=(c+e)\,x+z$, we have the negative conditions $c+e\neq 0$, implying $d+f\neq 0$.

But the equations are obtained in a more simple form by taking a for the idem and y for the nil; viz. we then have $x^2=x$, $xy=ax+dy$, $yx=ex+fy$, $y^2=0$: we must then have z, $=-(c+e)\,x+(1-d-f)\,y$, for an idem; this gives $z^2=-(c+e)\,(d+f)\,z$, and we have the negative conditions $c+e\neq 0$, $d+f\neq 0$ or 1.

(iii) 1 idem and 2 nils. This may be deduced from (ii) by writing therein $d + f = 0$; for then $z, = -(c + e)\,x + y$, is a nil. The equations are $x^2 = x$, $xy = ax - dy$, $y^2 = 0$: and if, instead of the idem y, we introduce therein z by the equation $z = -(c + e)\,x + y$, the equations become $x^2 = x$, $yz = [-(c+e)\,d+a]\,x+dz$, $zx = [-(c+e)\,f+c]\,x+cz$, $z^2 = 0$, with the negative condition $c + e \neq 0$.

But it is more simple to take x, y as the nils: the equations then are $x^2 = 0$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$. We must have $z, = ex + cy$, for an idem; this gives $z^2 = (c + e + f + d)\,xy$, becomes $= z$ and $= y$; so that x or y is a twofold nil. Or, what comes to the same thing, we have

$$\Omega = -(c + e)\,x^2y - (d + f)\,xy^2,$$

and Ω has as twofold factor if $c + e = 0$ or $d + f = 0$.

(iv) A twofold idem and a onefold idem. Taking x for the twofold idem and y for the onefold idem, Ω must reduce itself to $(1 - c - e)\,x^2y$, we must have $d + f = 1$, or say $f = 1 - d$. The equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + (1 - d)\,y$, $y^2 = y$; and we have the negative conditions $c + e \neq 0$, for otherwise Ω would vanish identically.

(v) A twofold idem and a onefold nil. Taking x for the twofold idem and y for the onefold nil, the equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$; and we have the negative conditions $c + e \neq 0$, $d + f \neq 0$.

(vi) A twofold nil and a onefold idem. Taking those to be x and y, then $d+f = 0$, and the equations are $x^2 = 0$, $xy = cx+dy$, $yx = ex-dy$, $y^2 = y$; we have the negative condition $c + e \neq 0$.

(vii) A twofold nil and a onefold nil. Taking those to be x and y, we have $d+f = 0$ and the equations are $x^2 = 0$, $xy = cx + dy$, $yx = ex - dy$, $y^2 = 0$; with the negative condition $c + e \neq 0$.

(viii) A threefold idem. Taking this to be x, then Ω must reduce itself to gx^3, viz. we must have $h = c + e$, $1 = d + f$; and the equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + (c + e)\,y$, $y^2 = gx + (c + e)\,y$; we have the negative condition $g \neq 0$, for otherwise Ω would vanish identically.

(ix) A threefold nil. Taking this to be x, then we must have $h = c + e$, $0 = d + f$; the equations are $x^2 = 0$, $xy = cx + dy$, $yx = ex - dy$, $y^2 = gx + (c + e)\,y$; and there is again the negative condition $g \neq 0$.

(x) $\Omega = 0$ identically; infinity of idems, 1 nil. Ω will vanish identically if $g = 0$, $h = c+e$, $a = d+f$, $b = 0$. If there is 1 idem, there will be no infinity of idems, and 1 nil. For, assume an idem x, $x^2 = x$; and, if possible, let there be no other idem; then there will be a nil y, $y^2 = 0$. We have $c+e = 0$, $d + f = 1$; and the equations

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are $x^2 = x$, $xy = cx + dy$, $yx = -cx + (1-d)\,y$, $y^2 = 0$; whence $xy + yx = y$. Taking α, β arbitrary coefficients, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta y, \ = \alpha\,(\alpha x + \beta y)\,;$$

hence for $\alpha x + \beta y$ to be an idem, except in the case $\alpha = 0$, when it is the original nil y.

If besides the idem x we have an idem y, then the conditions are $c+e = 1$, $d + f = 1$; the equations are

$$x^2 = x, \quad xy = cx + dy, \quad yx = (1-c)\,x + (1-d)\,y, \quad y^2 = y\,;$$

whence $xy + yx = x + y$. Considering the combination $\alpha x + \beta y$, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta\,(x + y) + \beta^2 y, \ = (\alpha + \beta)\,(\alpha x + \beta y).$$

This is an idem, except in the case $\alpha + \beta = 0$, when it is a nil; or, say we have the single nil $x - y$. We have thus again an infinity of idems, 1 nil.

(xi) $\Omega = 0$ identically; an infinity of nils. Taking the two nils x and y, the conditions are $c + e = 0$, $d + f = 0$, the equations are $x^2 = 0$, $xy = cx + dy$, $yx = -cx - dy$, $y^2 = 0$; whence $xy + yx = 0$, or there are an infinity of nils.

viz. $\alpha x + \beta y$ is a nil; or there are an infinity of nils.

12. The different cases may be grouped together as follows:—

A. 2 idems, (i), (ii), (iv), (x).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = y.$

B. 1 idem and 1 nil, (ii), (iii), (v), (vi), (x).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = 0.$

C. 2 nils, (iii), (vii), (xi).

Equations $x^2 = 0,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = 0.$

D. Threefold idem, (viii).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex - (1-d)\,y,\ y^2 = gx + (c+e)\,y.$

E. Threefold nil, (ix).

Equations $x^2 = 0,\ xy = cx + dy,\ yx = ex - dy,\ y^2 = gx + (c+e)\,y.$

The several cases of A, B, C respectively are distinguished by negative conditions which need not be here repeated.

13. I consider, as in my Note before referred to, the conditions in order that the system may be associative. We have the 8 products, x^3, x^2y, xyx, xy^2, yx^2, yxy, y^2x, y^3, giving rise to equations $x \,.\, x^2 = x^2 \,.\, x$, $x \,.\, xy = xy \,.\, x$, $\ldots$, $y \,.\, y^2 = y^2 \,.\, y$, which, on putting therein for x^2, xy, yx, y^2 their values, must be satisfied identically. We thus obtain in

the first instance 16 relations, but some of these are repeated, and we have actually only 12 relations; viz. the relations are

(twice) $b\,(c-e)=0,$

$b\,(f-d)=0,$

$g\,(c-e)=0,$

(twice) $g\,(f-d)=0,$

(twice) $bg-cd=0,$

(twice) $bg-ef=0,$

$c\,(e-h)+g\,(d-a)=0,$

$d\,(d-a)+b\,(c-b)=0,$

$e\,(e-h)+g\,(f-a)=0,$

$b\,(e-h)+f\,(f-a)=0,$

$a\,(e-c)-cf+de=0,$

$h\,(f-d)-cf+de=0.$

14. From the first four equations it appears that either $b=0$ and $g=0$, or else $c=e$ and $d=f$. I attend first to the latter case, viz. we here have the commutative system

$$x^2=ax+by,\quad xy=yx=cx+dy,\quad y^2=gx+hy.$$

In order that this may be associative, we must still have the relations

$$\begin{aligned} bg-cd&=0,\\ c\,(c-h)+g\,(d-a)&=0,\\ d\,(d-a)+b\,(c-b)&=0,\end{aligned}$$

or, as they may also be written,

$$\left|\begin{matrix} b, & -c, & d-a\\ -d, & g, & c-b\end{matrix}\right|=0.$$

These are satisfied by $g=\dfrac{cd}{b}$, $h=\dfrac{d^2+bc-cd}{b}$, and we have thus the commutative and associative system of the Note.

Every system is of the form A, B, C, D, or E; and it can be shown that the commutative and associative system is not of the form D. For, if D were commutative, we should have $c=e$, $d=\frac{1}{2}$, viz. the equations will be $x^2=x$, $xy=yx=cx+\frac{1}{2}y$, $y^2=gx+2cy$, that is,

$$a,\ b,\ c,\ d,\ y,\ h=1,\ 0,\ c,\ \tfrac{1}{2},\ 2c\,;$$

and the last of the three relations, viz. $d\,(d-a)+b\,(c-b)=0$, would thus be $\frac{1}{2}\,(\frac{1}{2}-1)=0$, which is not satisfied. Hence the commutative and associative system can only be of one of the forms A, B, C, and E.

15. First, of the form be A, B, or C ; there will be the two idem-or-nil symbols x and y, that is, we may assume $b = 0$, $g = 0$, and the associative conditions then become $cd = 0$, $c\,(c - h) = 0$, $d\,(d - a) = 0$, viz. for the forms A, B, C,

$$\begin{array}{lll} \text{A.} & x^2 = x, & y^2 = y, \\ \text{B.} & x^2 = x, & y^2 = 0, \\ \text{C.} & x^2 = 0, & y^2 = 0. \end{array}$$

these are $cd = 0$, $c\,(c - 1) = 0$, $d\,(d - 1) = 0$; $c = 0$ or 1, $d = 0$ or 1,

„ $cd = 0$, $c^2 = 0$, $d\,(d - 1) = 0$; $c = 0$, $d = 0$ or 1,

„ $cd = 0$, $c^2 = 0$, $d^2 = 0$; $c = 0$, $d = 0$.

But for the form A, if $c = 0$, $d = 1$, then is $xy = yx = y$, then, writing $z = x - y$, we have $z^2 = x$, $yz = zy = 0$, then, writing $x' = -y$, $z = z$, we have $x'^2 = -x'$, then $x' = -x$, that is, in each of these is reduced to the first case $c = 0$, $d = 0$, that is, $x^2 = x$, $xy = 0$, $y^2 = y$.

For the form B, if $c = 0$, $d = 1$, the system is $x^2 = x$, $xy = yx = y$, $y^2 = 0$; and this cannot be reduced to the first case $x^2 = x$, $xy = yx = 0$, $y^2 = 0$.

For the form C, the system is $x^2 = 0$, $xy = yx = 0$, $y^2 = 0$. In each of these is only the commutative, but not associative, conditions $b = 0$, $g = 0$, as above.

For the form E, here we have $a = 0$, $b = 0$, $c = e$, $d = f$, and the system may be commutative, $h = 2c$, viz. the equations must be $x^2 = 0$, $xy = yx = cx$, $y^2 = gx + 2cy$. The associative conditions then give $c = 0$; the equations are $x^2 = 0$, $xy = yx = 0$, $y^2 = gx$. Writing $\dfrac{x}{g}$ instead of x, and for convenience interchangingly x and y, the equations are

$$x^2 = y, \quad xy = yx = 0, \quad y^2 = 0.$$

16. The commutative associative system is thus seen to be reducible as follows:—

A system is $x^2 = x,\ xy = yx = 0,\ y^2 = y,$ first mixed system, *see* No. 2.

B. ,, $x^2 = x,\ xy = yx = y,\ y^2 = 0,$ Peirce's system (a_1).

or else

B. ,, $x^2 = x,\ xy = yx = 0,\ y^2 = 0,$ second mixed system.

C. ,, $x^2 = 0,\ xy = yx = 0,\ y^2 = 0,$ third mixed system.

E. ,, $x^2 = y,\ xy = yx = 0,\ y^2 = 0,$ Peirce's system (c_1).

I said, at the end of my Note before referred to, that it had been pointed out to me "that my system [the commutative associative system], in the general case $ad - bc$ not $= 0$, is expressible as a mixture of two algebras of the form (a_1), see *American Journal of Mathematics*, vol. iv., p. 120; whereas, if $ad - bc = 0$, it is reducible to the form (a), see p. 122 (l.c.)." The accurate conclusion is as above, that the commutative associative system is either a mixed system of one of the three forms, or else a system (a_1), or (c_1).

17. Considering next the non-commutative associative system, we have here, *vide* No. 14, $b = 0$, $g = 0$; and the relations which remain to be satisfied then are

$$cd = 0,\ ef = 0,\ c\,(c - h) = 0,\ e\,(e - h) = 0,\ d\,(d - a) = 0,\ f\,(f - a) = 0,$$

$$a\,(e - c) - cf + de = 0,\ h\,(f - d) - cf + de = 0.$$

The first equation gives $cd = 0$, viz. $c = 0$ or $d = 0$; but we may attend exclusively to the case $c = 0$, for the case $d = 0$ may be deduced from this by the interchange of x, y. We have then $ef = 0$; and it will be convenient to separate the cases

I. $c = 0,\ e \neq 0,\ x^2 = ax,\ xy = yx = 0,\ y^2 = hy$;

II. $c = 0,\ e \neq 0,\ f = 0,$;

III. $c = 0,\ e = 0,\ f \neq 0,$.

I. (a) $d = 0$; $x^2 = ax,\ xy = yx = 0,\ y^2 = hy$: commutative, and so included in what precedes.

I. (b) $d = a,\ h = 0$; $x^2 = ax,\ xy = ay,\ yx = 0,\ y^2 = 0$, which is Peirce's system (b_1).

I. (c) $d = a,\ h = a$; $x^2 = ax,\ xy = ay,\ yx = 0,\ y^2 = ay$: commutative, and so included in what pr

II. (d) $d = 0,\ f = a$: $x^2 = ax,\ xy = 0,\ yx = ay,\ y^2 = 0$, writing as we may do $a = 1$, this is $x^2 =$

II. (e) $d = a,\ e = h$: $x^2 = ax,\ xy = ay,\ yx = hx,\ y^2 = hy$, writing as we may do $a = 1,\ h = 1$, thi

Thus x, z form the system $x^2 = x$, $xz = z$, $zx = 0$, $z^2 = 0$ (or, what is the same thing, y, z form a system $y^2 = y$, $yz = 0$, $zy = z$, $z^2 = 0$); each of these is Peirce's system (b_1).

The conclusion is that every non-commutative associative system is either Peirce's system (b_1), or else the omitted system (d_1). Hence, disregarding the mixed systems, every associative system is either (a_1), (b_1), (c_1), or (d_1).

19. It may be proper to show that the systems (b_1), $x^2 = x$, $xy = y$, $yx = 0$, $y^2 = 0$, and (d_1), $x^2 = x$, $xy = 0$, $yx = y$, $y^2 = 0$, or say

	a	b	c	d	e	f	g	h
(b_1)	1	0	0	1	0	0	0	0
(d_1)	1	0	0	0	0	1	0	0

are really distinct from each other. Observe that they each belong to the case (x), $\Omega = 0$, an infinity of idoms and 1 nil; viz. in each of them writing $z = x + \beta y$, β an arbitrary coefficient, we have $z^2 = x^2 + \beta\,(xy + yx) + \beta^2 y^2$, $= x + \beta y$, $= z$, we have z an idom, and y is the only nil. And, this being so, in the second system, $xy = 0$, $yx = y$; viz. the system is $x^2 = x$, $xy = 0$, $yx = y$, $y^2 = 0$, retaining, when we write therein z for x, its original form. And similarly, in the second system, $xy = 0$, $yx = y$; viz. the system is $x^2 = x$, $xy = y$, $yx = 0$, $y^2 = 0$, retaining, when we write therein z for x, its original form. The two are thus distinct systems, in no wise transformable the one into the other.

815.

THE BINOMIAL EQUATION $x^p - 1 = 0$; QUINQUISECTION. SECOND PART.

[From the *Proceedings of the London Mathematical Society*, vol. xvi. (1885), pp. 61–63.]

In the paper, "The binomial equation $x^p - 1 = 0$; quinquisection," *Proc. Lond. Math. Soc.*, t. xii. (1881), pp. 15, 16, [744], I considered for an exponent $p = 5n + 1$, the five periods X, Y, Z, W, T connected by the equations

$$\begin{array}{rcccccc} & & X, & Y, & Z, & W, & T \\ X^2 & = & a, & b, & c, & d, & e \\ XY & = & f, & g, & h, & i, & j \\ XZ & = & k, & l, & m, & n, & o, \end{array}$$

and the equations deduced from these by cyclical permutations of the periods and of the coefficients of each set; but I did not obtain completely even the linear relations connecting the coefficients. I since found, by induction from the examples given in the Table 1, that the coefficients could be expressed linearly in terms of the linearly independent integer numbers a, β, f, k as follows: viz. introducing for convenience the new number θ, such that

$$a + \beta + \theta = \tfrac{1}{5}(p - 1),$$

then the expressions in question are

$$\begin{array}{rcccccc} a,\, b,\, c,\, d,\, e & = & -1 - 2\theta + a + \beta, & -\theta - a + \beta + f, & -\theta - a + \beta + k, & -a - 2\beta - k, & -2a - \beta - \\ f,\, g,\, h,\, i,\, j & = & f, & \theta - a - f, & a, & \beta, & a, \\ k,\, l,\, m,\, n,\, o & = & k, & a, & \theta - \beta - k, & \beta, & a, \end{array}$$

and I found further that, substituting these values of the coefficients in the 20 quadric relations referred to in the former paper, the 20 relations reduced themselves to two equations only, viz. these were

$$\theta\,(-2\alpha + \beta + k) + 3a^2 - \beta^2 + \alpha\,(f - k - 1) - \beta f + f^2 - 2fk = 0,$$

$$-\theta^2 + \theta\,(3\beta + 2k + f) + a^2 - a\beta - 3\beta - \alpha k + \beta\,(1 - f - k) - k^2 - 2fk = 0.$$

The final result thus is that the coefficients are expressed as functions of the five numbers a, β, f, k, θ, connected by the linear equation $\alpha+\beta+\theta=\frac{1}{5}(p-1)$, and the two quadric equations. I remark that formulae equivalent to these were obtained and proved by Mr F. S. Carey in his Trinity Fellowship Dissertation, 1884; viz. writing $n=\frac{1}{5}(p-1)$, his formulae were

$$\begin{array}{rcc} a,\ b,\ c,\ d,\ e & = & n-a,\ \beta-n,\ \gamma-n,\ \delta-n,\ \varepsilon-n, \\ f,\ g,\ h,\ i,\ j & = & f,\ \theta-a-f,\ \alpha,\ \beta,\ \rho, \\ k,\ l,\ m,\ n,\ o & = & \gamma,\ \alpha,\ \delta,\ \beta,\ \sigma,\ \alpha, \end{array}$$

with the three linear relations

$$\begin{gathered} \alpha+\beta+\gamma+\delta+\varepsilon=n-1, \\ \beta+\varepsilon+2\rho+\sigma=n, \\ \gamma+\delta+\rho+2\sigma=n, \end{gathered}$$

and the two quadric relations

$$\begin{aligned} \delta^2+\gamma^2+2\sigma a+(\rho-\sigma)(\delta+\gamma)-2\rho(\rho+\sigma)=(\delta-\gamma)(\beta-\varepsilon), \\ \beta^2+\varepsilon^2+2\rho a+(\sigma-\rho)(\beta+\varepsilon)-2\sigma(\rho+\sigma)=(\gamma-\delta)(\beta-\varepsilon), \end{aligned}$$

the coefficients being thus expressed in terms of the seven numbers a, β, γ, δ, ε, ρ, σ connected by five equations. The equivalence of the two sets of formulae may be shown without difficulty.

To the Table 2 of the Quintic Equations, given in the paper, may be added the following result from Legendre's *Théorie des Nombres*, Ed. 3, t. ii., p. 213,

p	q^6	q^5	q^4	q^3	q^2	q	1
641	1	+1	−256	−564	+3258	−3120	

$= 0$,

calculated by him for the isolated case $p=641$.

$$\frac{x}{1-\beta y}+\frac{y}{1-\gamma z}+\frac{z}{1-\alpha\beta}=0 \quad \text{and} \quad \frac{\xi}{a\,(\gamma-\beta)}+\frac{\eta}{\beta\,(\gamma-a)}+\frac{\zeta}{\gamma\,(\beta-a)}=0$$

respectively, we have 6+6 equations of like forms; but these two systems each of 6 equations are equivalent to each other, so that instead of $6+16+3+3=28$, we have $6+16+6\,(=6)=28$ equations for the 28 bitangents[1]. I make another slight change of notation by introducing the single letters f, g, h to denote the reciprocals $\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$; and I consider the whole question as follows. The theory is based on the equations

$$\begin{aligned}
ax+by+cz+f\xi+g\eta+h\zeta&=0,\\
a_1x+b_1y+c_1z+f_1\xi+g_1\eta+h_1\zeta&=0,\\
a_2x+b_2y+c_2z+f_2\xi+g_2\eta+h_2\zeta&=0,\\
a_3x+b_3y+c_3z+f_3\xi+g_3\eta+h_3\zeta&=0,
\end{aligned}$$

where $af=bg=ch=a_1f_1$ &c., $=a_3h_3$, $=1$: and the coefficients a, b, c, a_1, b_1, c_1, a_2, b_2, c_2 are arbitrary. The equations $x=0$, $y=0$, $z=0$ represent any three given lines: and

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816.

ON THE BITANGENTS OF A PLANE QUARTIC.

[From *Crelle's Journal der Mathem.*, t. xciv. (1883), pp. 93–115; *Camb. Phil. Soc. Proc.*, t. iv. (188

Riemann in the paper "Zur Theorie der Abelschen Functionen für den Fall $p=3$," *Werke*, pp. 456–479, has given a remarkably elegant solution of the problem of the bitangents of a plane quartic. But his formulae may be improved by a slight change; viz. we may in his first equation $x+y+z+f\xi+g\eta+h\zeta=0$ introduce coefficients so as to bring this into the same form as the other three equations. It thus appears that, instead of his $3+3$ equations of the forms

$$\frac{x}{1-\beta y}+\frac{y}{1-\gamma z}+\frac{z}{1-\alpha\beta}=0 \quad \text{and} \quad \frac{\xi}{a\,(\gamma-\beta)}+\frac{\eta}{\beta\,(\gamma-a)}+\frac{\zeta}{\gamma\,(\beta-a)}=0$$

[1] It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

[1] It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

considering ξ, η, ζ to be determined as linear functions of x, y, z by means of the first three equations, then (the coefficients a, b, c, a_1, b_1, c_1, a_2, b_2, c_2 being determined accordingly) the equations $\xi = 0$, $\eta = 0$, $\zeta = 0$ will also represent any three given lines. But observe that, if the equation of the first of these lines is $\alpha x + my + nz = 0$, ξ is not $=$ an arbitrary multiple $\theta\,(\alpha + my + nz)$ of the linear function $\alpha + my + nz$, but is $= G\,(\alpha + my + nz)$, G being a completely determinate value: and the like as regards η and ζ respectively.

The coefficients a_3, b_3, c_3 of the fourth equation are such that this fourth equation is a mere consequence of the other three: viz. we must have

$$\begin{aligned}
a_3 &= \lambda a + \lambda_1 a_1 + \lambda_2 a_2,\\
b_3 &= \lambda b + \lambda_1 b_1 + \lambda_2 b_2,\\
c_3 &= \lambda c + \lambda_1 c_1 + \lambda_2 c_2,\\
f_3 &= \lambda f + \lambda_1 f_1 + \lambda_2 f_2,\\
g_3 &= \lambda g + \lambda_1 g_1 + \lambda_2 g_2,\\
h_3 &= \lambda h + \lambda_1 h_1 + \lambda_2 h_2,
\end{aligned}$$

or, what is the same thing, we must have

$$\left\|\begin{matrix} a, & b, & c, & f, & g, & h\\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1\\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2\\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \end{matrix}\right\| = 0,$$

viz. each of the determinants formed with four columns out of this matrix is equal to 0.

Using the equations in λ, λ_1, λ_2, and writing for shortness

$$\begin{aligned}
a_3 f_3 = a_0 f_0, &\quad a_3 g_3 + a_0 f_1 = a f_1 + a_1 f_0 = F,\ F_1,\ F_2,\\
b_3 g_3 = b_0 g_0, &\quad b_3 g_3 + b_0 g_1 = b g_1 + b_1 g_0 = G,\ G_1,\ G_2,\\
c_3 h_3 = c_0 h_0, &\quad c_3 h_3 + c_0 h_1 = c h_1 + c_1 h_0 = H,\ H_1,\ H_2\,;
\end{aligned}$$

then, forming the products $a_3 f_3$, $b_3 g_3$, $c_3 h_3$, we find

$$\begin{aligned}
1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 &= P\lambda\lambda_1 + F_1\lambda\lambda_2 + F_2\lambda_1\lambda_2,\\
,\,, &= G\lambda\lambda_1 + G_1\lambda\lambda_2 + G_2\lambda_1\lambda_2,\\
,\,, &= H\lambda\lambda_1 + H_1\lambda\lambda_2 + H_2\lambda_1\lambda_2,
\end{aligned}$$

or, as these equations may be written,

$$\lambda\lambda_1 : \lambda\lambda_2 : \lambda_1\lambda_2 = \begin{vmatrix} 1, & F, & F_2\\ 1, & G, & G_2\\ 1, & H, & H_2 \end{vmatrix} : \begin{vmatrix} 1, & F, & F_1\\ 1, & G, & G_1\\ 1, & H, & H_1 \end{vmatrix} : \begin{vmatrix} 1, & F_1, & F_2\\ 1, & G_1, & G_2\\ 1, & H_1, & H_2 \end{vmatrix}.$$

10–2

say for shortness

$$= \Pi : \Lambda : \Lambda_1 : \Lambda_2.$$

We have thus

$$\frac{\lambda^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda_1\Lambda_2}{\Pi\Lambda},$$

$$\frac{\lambda_1^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda\Lambda_2}{\Pi\Lambda_1},$$

$$\frac{\lambda_2^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda\Lambda_1}{\Pi\Lambda_2},$$

and thence

$$\frac{1}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{1}{\Pi}\left(\frac{\Lambda_1\Lambda_2}{\Lambda} + \frac{\Lambda\Lambda_2}{\Lambda_1} + \frac{\Lambda\Lambda_1}{\Lambda_2}\right),$$

equations which give λ, λ_1, λ_2, and consequently a_3, b_3, c_3, f_3, g_3, h_3 in terms of known quantities, the several expressions depending on the single radical

$$\sqrt{\frac{1}{\Pi}\left(\frac{\Lambda_1\Lambda_2}{\Lambda} + \frac{\Lambda\Lambda_2}{\Lambda_1} + \frac{\Lambda\Lambda_1}{\Lambda_2}\right)}.$$

Observe that we have rationally

$$\lambda : \lambda_1 : \lambda_2 = \frac{1}{\Lambda} : \frac{1}{\Lambda_1} : \frac{1}{\Lambda_2}.$$

I consider now the quartic curve

$$\sqrt{x\xi} + \sqrt{y\eta} + \sqrt{z\zeta} = 0,$$

and I write down the equations of the 28 bitangents, each with its triple-theta characteristic as given by Riemann, and also with its dual symbol, derived from Hesse's method. I assume that these characteristics and dual symbols are properly attached to the several lines: and I insert also the current nos. 1 to 28. The equations are:

Current No.	Dual Symbol	Charac-teristic	
1	18	111 111	$x = 0,$
2	28	001 011	$y = 0,$
3	38	011 001	$z = 0,$
4	23	010 010	$\xi = 0,$
5	15	100 110	$\eta = 0,$
6	12	110 100	$\zeta = 0,$

Current No.	Dual Symbol	Charac-teristic	
7	49	101 100	$ax + by + cz = 0,\ f\xi + g\eta + h\zeta = 0,$
8	14	010 011	$f_1\xi + b_1y + c_1z = 0,\ a_1x + g_1\eta + h_1\zeta = 0,$
9	24	100 011	$a_2x + g_2\eta + c_2z = 0,\ f_2\xi + b_2y + h_2\zeta = 0,$
10	34	100 101	$a_3x + b_3y + h_3\zeta = 0,\ f_3\xi + g_3\eta + c_3z = 0,$
11	58	100 010	$a_1x + f_1\xi + c_1z = 0,$
12	16	001 010	$f_1\xi + b_1y + h_1\zeta = 0,$
13	25	101 010	$a_2x + g_2\eta + h_2\zeta = 0,$
14	35	111 110	$a_3x + g_3\eta + h_3\zeta = 0,$
15	68	110 010	$a_1x + b_1y + h_1\zeta = 0,$
16	16	001 101	$f_1\xi + g_1\eta + h_1\zeta = 0,$
17	26	111 101	$a_2x + b_2y + h_2\zeta = 0,$
18	36	101 011	$a_3x + b_3y + h_3\zeta = 0,$
19	78	010 110	$a_1x + b_1y + c_1z = 0,$
20	17	101 110	$f_1\xi + b_1y + c_1z = 0,$
21	27	011 110	$a_2x + f_2\xi + c_2z = 0,$
22	37	000 111	$a_3x + b_3y + h_3\zeta = 0,$
23	67	100 100	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0,\ \frac{1}{c_2h_2-c_3h_3} + \frac{1}{b_2f_2-b_3f_3} + \frac{1}{f_2g_2-f_3g_3} = 0,$ $\frac{x}{bc-b_2c_2} + \frac{y}{ca-c_2a_2} + \frac{z}{ab-a_2b_2} = 0,\ \frac{1}{c_1h_1-c_3h_3} + \frac{1}{a_1h_1-a_3h_3} + \frac{1}{f_1g_1-f_3g_3} = 0,$
24	57	110 001	$\frac{x}{bc-b_3c_3} + \frac{y}{ca-c_3a_3} + \frac{z}{ab-a_3b_3} = 0,$
25	56	010 001	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0,$
26	45	000 011	$\frac{x}{b_2f_2-b_3f_3} = \frac{y}{c_2g_2-c_3g_3} = \frac{z}{a_2h_2-a_3h_3},$
27	46	001 110	$\frac{x}{b_1f_1-b_3f_3} = \frac{y}{c_1g_1-c_3g_3} = \frac{z}{a_1h_1-a_3h_3},$
28	47	111 010	$\frac{x}{b_1f_1-b_2f_2} = \frac{y}{c_1g_1-c_2g_2} = \frac{z}{a_1h_1-a_2h_2},$

As regards each of the bitangents 7 to 22, the equivalence of the two forms of equation is at once seen by means of the fundamental equations $ax+by+cz+f\xi+g\eta+h\zeta=0$: as regards the remaining bitangents 23 to 28, the equation of each of these is expressible in eight different forms, which are written down in full for the bitangent 23; the equivalence of the eight forms of equation must of course be proved.

I proceed to show that each of the 28 lines is, in fact, a bitangent to the quartic curve

$$\sqrt{x\xi}+\sqrt{y\eta}+\sqrt{z\zeta}=0,$$

or, what is the same thing, that writing this equation in the form

$$\Omega=x^2\xi^2+y^2\eta^2+z^2\zeta^2-2yz\eta\zeta-2zx\zeta\xi-2xy\xi\eta=0,$$

then that by means of any one of these equations Ω becomes a perfect square.

This is obviously the case for each of the equations 1, 2, 3, 4, 5, 6.

For the equations 7, 8, 9, 10, observe that the equation of the quartic may be written:

$$\sqrt{axf\xi}+\sqrt{byg\eta}+\sqrt{czh\zeta}=0,$$

or, what is the same thing,

$$\Omega=ax^2\xi^2,f^2\xi^2+\ldots=0.$$

Hence writing x, y, z, ξ, η, ζ in place of ax, by, cz, $f\xi$, $g\eta$, $h\zeta$, so that now

$$x+y+z+\xi+\eta+\zeta=0,$$

the equation of the line 7 is expressed in the two equivalent forms $x+y+z=0$, $\xi+\eta+\zeta=0$. We have for Ω its original value

$$\Omega=x^2\xi^2+y^2\eta^2+z^2\zeta^2-2yz\eta\zeta-2zx\zeta\xi-2xy\xi\eta,$$

$=(x\xi-y\eta-z\zeta)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $x=-x-y$, $\xi=-\xi-y\eta$, we have $x\xi-y\eta-z\zeta=xy+y\xi$ and consequently $\Omega=(x\eta+y\xi)^2=(\xi y-y\xi)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $z=-x-y$, $\zeta=-\xi-\eta$, we have to apply this result to the equations 11, 12, 13, 14; 15, 16, 17, 18; 19, 20, 21, 22.

We come now to the equation of the line 23, say this is in the first instance taken to be

$$\frac{x}{bc-b_1c_1}+\frac{y}{ca-c_1a_1}+\frac{z}{ab-a_1b_1}=0;$$

if from this equation, by means of the two fundamental equations

$$ax+by+cz+f\xi+g\eta+h\zeta=0,$$
$$a_1x+b_1y+c_1z+f_1\xi+g_1\eta+h_1\zeta=0,$$

we eliminate first (y, z), secondly (z, x), and thirdly (x, y), it is found that ξ, η, ζ will also disappear in the three cases respectively, and that the resulting equations are

$$\begin{aligned}
\frac{x}{bc - b_1c_1} + \frac{\eta}{bb_1\,(ca - c_1a_1)} + \frac{\zeta}{cc_1\,(ab - a_1b_1)} &= 0,\\
-\frac{y}{aa_1\,(bc - b_1c_1)} + \frac{\eta}{ca - c_1a_1} + \frac{\zeta}{cc_1\,(ab - a_1b_1)} &= 0,\\
\frac{\xi}{aa_1\,(bc - b_1c_1)} - \frac{\eta}{bb_1\,(ca - c_1a_1)} + \frac{z}{ab - a_1b_1} &= 0,
\end{aligned}$$

so that each of these is, in fact, equivalent to the original equation in (x, y, z): and observe that, adding together the three equations, we reproduce the original equation in (x, y, z).

Write for shortness

$$\begin{aligned}
P &= bc - b_1c_1, & \mathfrak{a} &= bc - b_1c_1,\\
Q &= ca - c_1a_1, & \mathfrak{B} &= ca - c_1a_1,\\
R &= ab - a_1b_1, & \gamma &= ab - a_1b_1,
\end{aligned}$$

then the equation in (x, y, z) is

$$QRx + RPy + PQz = 0,$$

so that, eliminating y and z, we find

$$\begin{vmatrix} b, & c, & ax + f\xi + g\eta + h\zeta\\ b_1, & c_1, & a_1x + f_1\xi + g_1\eta + h_1\zeta\\ RP, & PQ, & QRx \end{vmatrix} = 0.$$

In this equation, the coefficient of x is

$$\begin{aligned}
(bc_1 - b_1c)\,QR &+ (cm_1 - c_1a)\,RP + (ab_1 - a_1b)\,PQ,\\
&= (bc_1 - b_1c)\,[aPb + a_1b_1c_1 - aa_1\,(bc_1 - b_1c)]\\
&\quad +(cm_1 - c_1a)\,[bP + a_1b_1c_1 - bb_1\,(ca_1 - c_1a)]\\
&\quad +(ab_1 - a_1b)\,[cP + a_1b_1c_1 - cc_1\,(ab_1 - a_1b)],\\
= -aa_1\,(bc_1 - b_1c)^2 &- bb_1\,(ca_1 - c_1a)^2 - cc_1\,(ab_1 - a_1b)^2,\\
&= -aa_1\gamma.
\end{aligned}$$

The coefficient of ξ is

$$= RP\,(cf_1 - c_1f) - PQ\,(bf_1 - b_1f),$$

which (observing that $cf_1 - c_1f = \dfrac{c}{a_1} - \dfrac{c_1}{a} = \dfrac{1}{aa_1}\,Q$ and $bf_1 - b_1f = \dfrac{b}{a_1} - \dfrac{b_1}{a} = \dfrac{1}{aa_1}\,R$) is $= 0$.

The coefficient of η is

$$RP\,(cg_1 - c_1g) - PQ\,(bg_1 - b_1g),$$

which is

$$
\begin{aligned}
&= RP\,\frac{1}{bb_1}\,(bc_1 - b_1c_1) - PQ\,\frac{1}{bb_1}\,(b^2 - b_1^2),\\
&= \frac{P}{bb_1}\,[(bc_1 - b_1c_1)\,(ab - a_1b_1) - (b^2 - b_1^2)\,(ca - c_1a_1)],\\
&= \frac{P}{bb_1}\,[b_1^2ca + b^2c_1a_1 - bb_1\,(ac_1 + a_1c)],\\
&= -\,\frac{P}{bb_1}\,\gamma\mathfrak{a}\,;
\end{aligned}
$$

and similarly the coefficient of ζ is

$$= -\,\frac{P}{cc_1}\,\mathfrak{a}\beta.$$

We have thus in x, η, ζ the equation

$$-aa_1\gamma x - \frac{P}{bb_1}\,\gamma\mathfrak{a}\,\eta + \frac{P}{cc_1}\,\mathfrak{a}\beta\,\zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{bb_1\beta} + \frac{\zeta}{cc_1\gamma} = 0\,;$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$-\frac{\xi}{aa_1\mathfrak{a}} + \frac{y}{Q} + \frac{\zeta}{cc_1\gamma} = 0,$$

$$\frac{\xi}{aa_1\mathfrak{a}} + \frac{\eta}{bb_1\beta} + \frac{z}{R} = 0,$$

which are the required three equations.

We have next to show that the line is a bitangent—write for a moment $aa_1\mathfrak{a}$, $bb_1\beta$, $cc_1\gamma = \lambda$, μ, ν, then the three equations give

$$x,\ y,\ z = P\Big(-\frac{\eta}{\mu} + \frac{\zeta}{\nu}\Big), \quad Q\Big(-\frac{\zeta}{\nu} + \frac{\xi}{\lambda}\Big), \quad R\Big(-\frac{\xi}{\lambda} + \frac{\eta}{\mu}\Big),$$

and we thence have

$$\begin{aligned}\Omega = P^2&\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)^2\\ &+Q^2\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right)^2\\ &+R^2\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)^2\\ &-2QR\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right)\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)\\ &-2RP\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)\\ &-2PQ\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right)\ ;\end{aligned}$$

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Contents

819. On Two Cases of the Quadric Transformation between Two Planes (concluded)

[p. 101] Writing for convenience $(a,\ b,\ c)$ for the coordinates of the fixed point, and $(x_1,\ y_1,\ z_1)$, $(x_2,\ y_2,\ z_2)$ for those of the other two points, the formulæ with A, B, C give thus the correspondence

$$x_2 : y_2 : z_2 \;=\; bcx_1^2 - a^2y_1z_1 \;:\; ca\,y_1^2 - b^2z_1x_1 \;:\; ab\,z_1^2 - c^2x_1y_1,$$

which is the first of the two cases in question. These equations give reciprocally

$$x_1 : y_1 : z_1 \;=\; bc\,x_2^2 - a^2y_2z_2 \;:\; ca\,y_2^2 - b^2z_2x_2 \;:\; ab\,z_2^2 - c^2x_2y_2,$$

or the correspondence is a (1, 1) quadric correspondence.

The formulæ with P, Q, R give in like manner

$$x_2 : y_2 : z_2 = a(ax_1^2 + by_1^2 + cz_1^2) - a_1(a''\,y_1z_1 + b'\,z_1x_1 + c'\,x_1y_1),\ \text{\&c.},$$

or if for shortness

$$\Omega_1 = ax_1^2 + by_1^2 + cz_1^2, \qquad \Theta_1 = a''\,y_1z_1 + b'\,z_1x_1 + c'\,x_1y_1,$$

then

$$x_2 : y_2 : z_2 \;=\; a\Omega_1 + a_1\Theta_1 \;:\; b\Omega_1 + b_1\Theta_1 \;:\; c\Omega_1 + c_1\Theta_1,$$

which is the second of the two cases. We have reciprocally

$$x_1 : y_1 : z_1 \;=\; a\Omega_2 + a_1\Theta_2 \;:\; b\Omega_2 + b_1\Theta_2 \;:\; c\Omega_2 + c_1\Theta_2,$$

where

$$\Omega_2 = ax_2^2 + by_2^2 + cz_2^2, \qquad \Theta_2 = a''\,y_2z_2 + b'\,z_2x_2 + c'\,x_2y_2,$$

and the correspondence is thus in this case also a (1, 1) quadric correspondence.

820. On a Problem of Analytical Geometry

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 209.]

[p. 102] THE object of the present note is only to call attention to a problem of Analytical Geometry which presents itself in connexion with the reduction of an algebraical integral, and which is solved, pp. 21, 22 of Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866); viz. the problem is, considering a curve $f = 0$ of the order n, to find the conic $\Omega = 0$ of the order $n-2$ passing through the remaining $n-2$ points of intersection of the line with the curve f, through the double points of f, and through as many other given points as are required for the determination of the curve. If, for instance, f is a quartic curve without double points, then Ω is the quadric curve which passes through the remaining two intersections of the line with Ω, and through three given points. Take $(\xi,\,\eta,\,\zeta)$, $(\xi_1,\,\eta_1,\,\zeta_1)$ the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x,\,y,\,z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining $(x',\,y',\,z')$, $(x'',\,y'',\,z'')$ meets the curve θ where $\theta = 0$ in two points, the equation in ρ originally of the fourth order in $(\lambda,\,\mu)$ but which divides by $\lambda\mu$, and which when this factor is thrown out becomes

$$\alpha\lambda^2 + 2\beta\lambda\mu + \gamma\mu^2 = 0;$$

where

$$\alpha = (\xi_1,\,\eta_1,\,\zeta_1)^2\,\theta, \quad \beta = (\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta)\,\theta, \quad \gamma = (\xi,\,\eta,\,\zeta)^2\,\theta,$$

where for shortness f_1, f are written to denote $f(\xi_1,\,\eta_1,\,\zeta_1)$, $f(\xi,\,\eta,\,\zeta)$ respectively.

[p. 103] The condition as to the two points obviously is that, making the same substitution $x,\,y,\,z = \lambda\xi_1 + \mu\xi,\ \lambda\eta_1 + \mu\eta,\ \lambda\zeta_1 + \mu\zeta$ in the equation $\Omega = (a,\dots)(x,\,y,\,z)^2 = 0$, we must obtain the same quadric equation in $\lambda : \mu$. We have thus two conditions, which, introducing an indeterminate multiplier θ, are expressed by the three equations

$$\begin{aligned}(a,\dots)(\xi_1,\,\eta_1,\,\zeta_1)^2 &= \theta\alpha,\\ (a,\dots)(\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta) &= \theta\beta,\\ (a,\dots)(\xi,\,\eta,\,\zeta)^2 &= \theta\gamma.\end{aligned}$$

The conditions as to the three points are obviously

$$\begin{aligned}(a,\dots)(x_1,\,y_1,\,z_1)^2 &= 0,\\ (a,\dots)(x_2,\,y_2,\,z_2)^2 &= 0,\\ (a,\dots)(x_3,\,y_3,\,z_3)^2 &= 0,\end{aligned}$$

and these equations determine the ratios of a, b, c, f, g, h. But to complete the solution the convenient course is to regard the function $\Omega, = (a,\dots)(x,\,y,\,z)^2$ as a quantity to be determined, and consequently to join to the foregoing the equation

$$(a,\dots)(x,\,y,\,z)^2 = \Omega;$$

we have thus seven equations from which $(a,\,b,\,c,\,f,\,g,\,h)$ may be eliminated, the result being expressed by means of a determinant of the seventh order

$$\begin{vmatrix} (x,\,y,\,z)^2 & \Omega \\ (\xi_1,\,\eta_1,\,\zeta_1)^2 & \theta\alpha \\ (\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta) & \theta\beta \\ (\xi,\,\eta,\,\zeta)^2 & \theta\gamma \\ (x_1,\,y_1,\,z_1)^2 & 0 \\ (x_2,\,y_2,\,z_2)^2 & 0 \\ (x_3,\,y_3,\,z_3)^2 & 0 \end{vmatrix} = 0,$$

viz. this is an equation of the form $A\Omega = \theta V$, where A is a constant determinant of the sixth order (i.e. a determinant not involving x, y, z), V a determinant of the seventh order, a quadric function of (x, y, z), obtained from the foregoing determinant by writing therein $\Omega = 0$ and $\theta = 1$; the multiplier θ is and remains arbitrary; but it is convenient to take it to be $= 1$, viz. we have then not only find the equation $\Omega = 0$ of the required conic, but we put a determinate value for Ω, say $\Omega = A^{-1}V$, which is, in fact $(a, \dots\ )(x, y, z)^2$, viz. this gives $a = (\xi_1\beta_1 + \eta_1\delta_1 + \zeta_1\theta_1)f_1$, and so on for $(x, y, z) = (\xi, \eta, \zeta)$, we have

$$\Omega_1 = \gamma_1 = (\xi_1\beta_1 + \eta_1\delta_1 + \zeta_1\theta_1)f_1.$$

821. On the Geometrical Representation of an Equation between Two Variables

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 210.]

[p. 104] An equation between two variables cannot be represented in a satisfactory manner by a curve, for this serves only to represent the corresponding real values of the variables; to represent the imaginary values the natural course is to represent each variable by a point in a plane, viz. the variable $z, = x + iy$, will be represented by a point the coordinates of which are the components x and y of the variable, and similarly the variable $z', = x' + iy'$, by a point the coordinates of which are the components x' and y' of the variable; the equation between the two variables will then establish a correspondence between the two variables points respectively, or say a correspondence between the planes which contain the two points respectively: and it is this correspondence of two planes which is the proper geometrical representation of the equation between the two variables; to exhibit the correspondence we may in either of the planes draw a network of curves at pleasure, and then draw in the other plane the network of corresponding curves. This well-known theory [can be] illustrated by the case $z' = \sqrt{z^2 - 1}$: taking in the infinite half-plane y positive about the origin as centre a system of semi-circles, to these correspond in the infinite plane of $x'y'$ a series of lemniscate-shaped curves: and by means of these it is easy to show in the second plane the path corresponding to a given path of the point $z, = x + iy$, in the first half-plane.

822. On Associative Imaginaries

[From the *Johns Hopkins University Circulars*, No. 15 (1882), pp. 211, 212.]

[p. 105] THE imaginaries $(x,\ y)$ defined by the equations

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

will not be in general associative; to make them so, we must have 8 double relations corresponding to the combinations $x^2.x$, $xy.x$, $yx.x$, $x^2.y$, $xy.y$, $yx.y$, $y^2.x$, $y^2.y$ respectively, viz. the first of these gives $x.x.x = x.x.x$, that is, $0 = x(ax+by)-(ax+by)x$, $= b(xy-yx)$, $= b\{(c-e)x+(d-f)y\}$; and similarly for each of the other terms. We thus obtain apparently 16, but really only 12, relations between the 8 coefficients a, b, c, d, e, f, g, h, viz. the relations so obtained are

$$\begin{array}{lll} b(c-e)=0 \text{ (twice)}, & b(d-f)=0, & g(c-e)=0, \quad g(d-f)=0 \text{ (twice)} \\ bg-cd=0 \text{ (twice)}, & bg-ef=0 \text{ (twice)}, & \\ c^2+dg-cg-eh=0, & d^2+bc-cd-af+bg-ah=0, & \\ a(c-e)+de-cf=0, & h(f-d)-cf+de=0. & \end{array}$$

From the first four equations it appears that either $b=0$ and $g=0$, or else $c=e$ and $d=f$: the latter solution may be set aside for the moment, giving the commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= gx + hy. \end{aligned}$$

[p. 106] In order that this may be associative, we must still have the relations

$$\begin{aligned} bg-cd&=0, \\ c^2+dg-cg-ch&=0, \\ d^2+bc-ad-bh&=0, \end{aligned}$$

which are all three of them satisfied by $g=\dfrac{cd}{b}$, $h=\dfrac{d^2+bc-ad}{b}$, viz. we thus have the associative and commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= \frac{cd}{b}x + \frac{d^2+bc-ad}{b}y. \end{aligned}$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. iv. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad-bc\neq 0$, is reducible to the form (a_1), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 238.]

[p. 107] I HAVE given in my *Elliptic Functions* expressions for the sn of $u+\frac{1}{2}K$, $u+\frac{1}{2}iK'$, $u+\frac{1}{2}K+\frac{1}{2}iK'$; but it is better to consider the dn, sn, cn of these combinations respectively, and to write the formulæ thus

$$\operatorname{dn}^2(u+\tfrac{1}{2}K) = k'\,\frac{\operatorname{dn}u-(1-k')\operatorname{sn}u\operatorname{cn}u}{\operatorname{dn}u+(1-k')\operatorname{sn}u\operatorname{cn}u}\,, \quad = k'\,\frac{1-k'x-(1-k')y}{1-k'x+(1-k')y}\,;$$

$$\operatorname{sn}^2(u+\tfrac{1}{2}iK') = \frac{1}{k}\,\frac{(1+k)\operatorname{sn}u+i\operatorname{cn}u\operatorname{dn}u}{(1+k)\operatorname{sn}u-i\operatorname{cn}u\operatorname{dn}u}\,, \quad = \frac{1}{k}\,\frac{(1+k)x+iy}{(1+k)x-iy}\,;$$

$$\operatorname{cn}^2(u+\tfrac{1}{2}K+\tfrac{1}{2}iK') = -\,\frac{ik'}{k}\,\frac{\operatorname{cn}u-(k+ik')\operatorname{sn}u\operatorname{dn}u}{\operatorname{cn}u+(k+ik')\operatorname{sn}u\operatorname{dn}u}\,, \quad = -\,\frac{ik'}{k}\,\frac{1-x-(k+ik')y}{1-x+(k+ik')y}\,,$$

where in the last set of ratios x, y are used to denote $\operatorname{sn}u\operatorname{cn}u$ and $\operatorname{sn}u\operatorname{dn}u$ respectively; and the three combinations above show that for u are written respectively the values just mentioned. The curve $y^2=(1-x^2)(1-k^2x^2)$ has an inflexion at infinity on the line $x=0$; and the three tangents from the inflexion are at the points x, $y=0$, 1, 0 and $\left(\frac{1}{k},\,0\right)$ respectively; touching the curve at the points x, $y=(0,\,0)$, $(1,\,0)$, $\left(\frac{1}{k},\,0\right)$ respectively; hence these points are sextactic points.

[p. 108] We may from any one of these, for instance the point $(0,\,0)$, draw four tangents to the curve, $(1+k)x+iy=0$, $(1+k)x-iy=0$; $(1-k)x+iy=0$, $(1-k)x-iy=0$; where the first and second of these lines form a pair, and the third and fourth of them form a pair, viz. the two tangents of a pair touch in points such that the line joining them passes through the point of inflexion; in particular, for the first-mentioned pair, the equation of the line joining the points of contact is $1+kx=0$. The tangents from belonging to a pair of tangents are precisely those which present themselves in the formulæ; thus if $T_1=(1+k)x+iy$, $T_2=(1+k)x-iy$, the second of the three formulæ is $\operatorname{sn}^2(u+\frac{1}{2}iK')=\frac{1}{k}\frac{T_1}{T_2}$; and the other two formulæ correspond in like manner to pairs of tangents from the sextactic points $\left(\frac{1}{k},\,0\right)$, and $(1,\,0)$ respectively. The formulæ are connected with the fundamental equations expressing the functions sn, cn, dn as quotients of theta-functions.

824. Note on the Formulæ of Trigonometry

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 241.]

[p. 108] THE equations $a = c\cos B + b\cos C$, $b = c\cos A + a\cos C$, $c = b\cos A + a\cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i\sin A$, $\cos B + i\sin B$, $\cos C + i\sin C$, viz. calling these $\frac{x}{x'}$, $\frac{y}{y'}$, $\frac{z}{z'}$ respectively, or, what is the same thing, writing $2\cos A = \frac{x}{x'} + \frac{x'}{x}$, $2\cos B = \frac{y}{y'} + \frac{y'}{y}$, $2\cos C = \frac{z}{z'} + \frac{z'}{z}$, then the foregoing equations may be written

$$\begin{aligned} (-2a)xx' \ + \ y(zy' + z'y)b \ + \ z(xy' + x'y)c &= 0, \\ x(yz' + y'z)a + (-2b)yy' + z(xy' + x'y)c &= 0, \\ x(yz' + y'z)a + y(zx' + z'x)b + (-2c)zz' &= 0, \end{aligned}$$

that is, as a system of bipartite equations linear in (a, b, c) and cubic in (x, y, z, x', y', z') respectively.

Similarly in Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2\cos a = \frac{a}{\delta} + \frac{\delta}{a}$, $2\cos b = \frac{\beta}{\delta} + \frac{\delta}{\beta}$, $2\cos c = \frac{\gamma}{\delta} + \frac{\delta}{\gamma}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, x', y', z') and $(\alpha, \beta, \gamma, \delta)$ respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, *American Journal of Mathematics*, t. v. (1882), pp. 137–179;
Chapters IV to VII, ib., t. vii. (1885), pp. 101–167.]

[p. 109] THE present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi_{12}$ (or as I have it $d\Pi$) in the integral of the third kind $\int_a^b d\Pi_{12}$ in the final normal form (endliche Normalform) for which we have (p. 117) $\int_a^b d\Pi_{12} = \int_1^2 d\Pi_{ab}$, the limits and parametric points interchangeable. The want of this determination presented itself to me as a locus in the theory during the course of lectures on the subject which I had the pleasure of giving in the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the far more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to. I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence; and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function (continued); IV, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ. The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.
[p. 110] 1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,du = 0.$$

The affected theorem is

$$\Sigma\,\frac{(x,\,y,\,z)^{n-3}\,du}{012} \;=\; -\,\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2}\;*\;.$$

2. We have a fixed curve $f = 0$, or say the curve f, or simply the fixed curve, of the order n, with δ dps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta, = p$. The expression "the dps" means always the δ dps of f.

And we have a variable curve $\phi = 0$, or say the curve ϕ, or simply the variable curve, of the order m, passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, du is the displacement of the current point 0, coordinates $(x,\,y,\,z)$, on the fixed curve, viz. the equation $f = 0$ being satisfied, we have

$$\frac{df}{dx}\,dx + \frac{df}{dy}\,dy + \frac{df}{dz}\,dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} = y\,dz - z\,dy : z\,dx - x\,dz : x\,dy - y\,dx,$$

so that we have three equal values each of which is put $= du$, viz. we write

$$\frac{y\,dz - z\,dy}{\frac{df}{dx}} = \frac{z\,dx - x\,dz}{\frac{df}{dy}} = \frac{x\,dy - y\,dx}{\frac{df}{dz}} = du,$$

and du as thus defined is the displacement.
[p. 111] 3. $(x,\, y,\, z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n-3$ which passes through the dps; and the function $(x,\, y,\, z)^{n-3}$ in the proper major function, viz. the implicit factor of the function is so determined that, taking $0 = 1$, the current point at 1, that is, writing $(x_1,\, y_1,\, z_1)$ for $(x,\, y,\, z)$, the function $(x,\, y,\, z)^{n-3}$ reduces itself to the polar function $\left(x_1\frac{d}{dx_2} + y_1\frac{d}{dy_2} + z_1\frac{d}{dz_2}\right) f_2$, afterwards written $n.\,1^{n-1}\,2$, of f: this implies that taking $0 = 2$, the current point at 2, the function reduces itself to the polar function $n.\,1\,2^{n-1}$.

ϕ_1 is what ϕ becomes on writing therein $(x_1,\, y_1,\, z_1)$ for $(x,\, y,\, z)$; and similarly ϕ_2 is what ϕ becomes on writing therein $(x_2,\, y_2,\, z_2)$ for $(x,\, y,\, z)$.

δ denotes differentiation in regard only to the coefficients of ϕ; viz. writing $\phi = (a, \dots\,)(x,\, y,\, z)^m$ we have $\delta\phi = (da, \dots\,)(x,\, y,\, z)^m$, and similarly $\delta\phi_1$ and $\delta\phi_2 = (da, \dots\,)(x_1,\, y_1,\, z_1)^m$ and $(da, \dots\,)(x_2,\, y_2,\, z_2)^m$ respectively.

The sum Σ extends to all the variable intersections of the two curves.

3. As to the meaning of the theorems, consider first the pure theorem. The variable intersections are not all of them arbitrary points on the fixed curve; a certain number of them taken at pleasure on the fixed curve will determine the remaining variable intersections; and there are thus a certain number of integral relations between the coordinates of the variable intersections; to each such integral relation there corresponds a linear relation between the displacements du of these points, or say a displacement-relation; and it is precisely these displacement-relations which are given by the theorem, viz. the equation

$$\Sigma\,(x,\, y,\, z)^{n-3}\,du = 0$$

breaks up into as many linear relations as there are arbitrary constants in the function $(x,\, y,\, z)^{n-3}$, which equated to zero gives a curve of the order $n-3$ passing through the dps; for instance, if there be no singularities $\delta = 0$, then the relation gives the single relation $\Sigma\,du = 0$; but $n = 4$, $\delta = 0$, the equation gives the three relations $\Sigma x\,du = 0$, $\Sigma y\,du = 0$, $\Sigma z\,du = 0$.
[p. 112] 4. It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; the coefficients of ϕ are hereby led to see the nature of the affected theorem. The variable intersections, the coefficients of ϕ are thereby determined in terms of the coordinates of the assumed variable intersections*, and hence the value of $-\frac{\delta\phi_1}{\phi_1}\frac{\delta\phi_2}{\phi_2}$ is given as a linear function of the corresponding displacements du; and, substituting this value, the affected theorem gives a linear relation between the displacements du of the several variable intersections. But any such linear relation must clearly be a mere linear combination of the displacement-relations $\Sigma\,(x,\, y,\, z)^{n-3}\,du = 0$ given by the pure theorem.

Examples of the Pure Theorem—The Fixed Curve a Cubic. Art. Nos. 6 to 12.

6. The pure theorem is not applicable to the case $n = 2$, the fixed curve a conic: it in fact gives no displacement-relation; and this is as it should be, for the variable intersections are all of them arbitrary.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For greater simplicity the equation is taken in Cartesian coordinates. In general for such an equation, writing in the homogeneous formula $z = 1$, we have

$$du = \frac{dx}{-\frac{df}{dy}} = \frac{dy}{\frac{df}{dx}},$$

the two values being of course equal in virtue of $\frac{df}{dx}\,dx + \frac{df}{dy}\,dy = 0$; taking the former value and considering $\frac{df}{dy}$ as expressed in terms of x, let this be called P (of course, P is an irrational function of x); then we have $du = \frac{dx}{P}$; and similarly $du_1 = \frac{dx_1}{P_1}$, &c.

The fixed curve being then a cubic, let the variable curve be a line; this meets the cubic in three points, say 1, 2, 3; and any two of these determine the line, and therefore the third point; there should therefore be one integral relation, and consequently one displacement-relation; and this is in fact the case by the theorem, viz. we have $\Sigma du = 0$, or, $du_1 + du_2 + du_3 = 0$, that is in the same thing,

$$\frac{dx_1}{P_1} + \frac{dx_2}{P_2} + \frac{dx_3}{P_3} = 0.$$

[p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively, this is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

7. This equation $du_1 + du_2 + du_3 = 0$, here $du = \frac{dx}{P}$, has a peculiar interpretation when we consider the coefficients of the cubic as arbitrary constants, and therefore the cubic curve, we in effect determine y_1 as a function of x_1 and the constants; the points 1 and 2 determine the line and the third point; we have thus x_3, y_3 determined as a function of x_1, x_2 and the constants.

Considering x_3 as thus expressed, we have $dx_3 = \frac{dx_3}{dx_1}\,dx_1 + \frac{dx_3}{dx_2}\,dx_2$, an equation which must agree with $du_1 + du_2 + du_3 = 0$, that is, with $dx_3 = -\frac{P_3}{P_1}\,dx_1 - \frac{P_3}{P_2}\,dx_2$. It follows that we have $\frac{dx_3}{dx_1} : \frac{dx_3}{dx_2} = P_1 : P_2$, and taking the logarithms and differentiating with regard to x_1 and x_2, we find $\frac{d}{dx_1}\frac{d}{dx_2}\log\left(\frac{dx_3}{dx_1} : \frac{dx_3}{dx_2}\right) = 0$, a partial differential equation of the third order, independent of any particular cubic curve, and satisfied by x_3 considered as a function of x_1, x_2 and the nine constants. Observe that starting from the expression for x_3, and proceeding to the differential coefficients of the third order, we have ten equations from which the nine constants can be eliminated, that is, we ought to have a partial differential equation of the third order; and conversely that the equation for x_3, as containing nine arbitrary constants, is a complete solution of the partial differential equation: the complete solution of the partial differential equation in question is thus the equation which expresses that 3 is the remaining intersection of the line through 1 and 2 with a cubic.

8. As a particular illustration, we have $\dfrac{dx}{P} = du$, $\Sigma du = 0$, and the integral relation $123 = 0$. I give two new ways of obtaining this—first by differentiation.

We have the equation $123 = 0$. I give a direct verification.

To find $\dfrac{dx_3}{dx_1}$, $\dfrac{dx_3}{dx_2}$, where 1, 2, 3 are on the cubic, $f = 0$. Let ϕ be any conic, $\phi = 0$, taken so that $\dfrac{d\phi}{dx} \div f = du$, and as the equation of ϕ at 0 of the curve, and it may be regarded as the determination of ϕ; with the dps and the ϕ other points $a, \ldots, h$ on the curve.

This being so, considering as variable the original coordinates f, we have by differentiation

$$\Sigma \frac{1}{P}\, \delta \log \frac{f}{(ax^3 + by^3 + cz^3 + 6lxyz)} = 0,$$

or what is the same, taking the logarithms and differentiating with regard to one of the constants a, the equation $\Sigma du = 0$ is satisfied.

We have shown also that the cubic curve $f = 0$, and the cube derived line $\phi = 0$ are now equivalent to the two cases.

9. As a particular example, let the cubic be $x^3 + y^3 - 1 = 0$; then $y = \sqrt[3]{1 - x^3}$, and $du = \dfrac{dx}{y^2} = \dfrac{dx}{(1-x^3)^{\frac{2}{3}}}$; and with these values as before the differential relation $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. I give a direct verification. To find x_3, y_3 the coordinates of the third intersection, we may in the equation of the cubic write $x_3, y_3 = \lambda_1 + \mu x_1, \lambda y_1 + \mu y_1$, where $\lambda + \mu = 1$, and we obtain for the determination of λ, μ the equation $\lambda_1 \cdot 1^2 2 + \mu \cdot 12^2 = 0$.

This being so, from the equation $123 = 0$ we obtain by differentiation

$$\Sigma\, (y_2 - y_3)\, dx_1 - dy_1 \cdot (x_2 - x_3) = 0,$$

the sum consisting of three terms, the second and third of them obtained from the one given by the cyclical interchange of the numbers 1, 2, 3. But we have $x_1\, dx_1 + y_1\, dy_1 = 0$, and the equation thus is

$$\Sigma \frac{dx_1}{y_1^2}\, \{y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3)\} = 0;$$

this will reduce itself to $\Sigma \dfrac{dx_1}{y_1^2} = 0$, if only the three coefficients in { } are equal, that is, we ought to have

$$y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3) = y_2^2(y_3 - y_1) + x_2^2(x_3 - x_1).$$

Comparing for instance the first and second terms, the equation is

$$-y_1(y_3 + y_1^2)(x_2 - x_3) + (x_1^2 + x_2^2)\, y_1(x_2 - x_3) + y_1(y_2 - y_3)(x_3 + x_2) = 0,$$

or, as this may be written,

$$-(\lambda_1 + \mu_1)(y_1^2 + y_2^2) + (\lambda_1 + \mu)x_1(x_1^2 + x_2^2) + (\lambda_1 - \mu)(x_1^2 + x_2^2)\, y_1 + (\lambda + \mu)(y_2 + y_1^2) = 0,$$

whence the whole coefficient of λ is $-x_2^2(y_1^2 + y_3^2) + x_3^2(x_1^2 + x_3^2) + (\lambda + \mu)y_3(y_1 + y_3^2) = 0$, which we see similarly the whole coefficient of μ is $-x_2^2(y_2^2 + y_3^2) + x_2^2(x_2^2 + x_3^2) + (\lambda + \mu)y_2(y_2 + y_3^2) = 0$, which is true; and similarly for the second and third terms; we have thus the required result,

$$\frac{dx_1}{y_1^2} + \frac{dx_2}{y_2^2} + \frac{dx_3}{y_3^2} = 0,$$

the integral of which is $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$, but it is in fact a particular integral, as it is in fact the only general integral.

[p. 115] $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$; but it is, in fact, a particular integral.

But regard one of the three points, say 3, as a fixed point, that is, the line passes through the fixed point on the cubic; then we may write $du_1 + du_2 = 0$, and the differential equation contains du_1 and du_2 only, the constants of u unaltered.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any five of these points determine the conic, and the displacement-relation is therefore the one displacement-relation

$$\Sigma\, du = 0,$$

which corresponds to the equation $123456 = 0$, which expresses that the six points are on a conic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0.$$

If, however, we take 5 a fixed point on the cubic, then we have the same equation as the general integral of $du_1 + du_2 + du_3 + du_4 = 0$.

We have here obviously $\Sigma\, du = 0$ as a particular integral of the cubic; and the integral relation $123456 = 0$ as the general integral.

But taking also 6 a fixed point on the cubic, the equation $123456 = 0$, which expresses that the six points are on a conic, of any equation $123456 = 0$ for 4 fixed and 5 fixed point on the cubic, becomes a particular integral of $du_1 + du_2 + du_3 + du_4 = 0$.

[p. 116] determinate point, independent of the particular conic through 1, 2, 3, 4 and 5 used for the construction. We have thus a particular integral of the differential equation $du_1 + du_2 + du_3 + du_4 = 0$.

12. We might go on to the case where the variable curve is a cubic; there are here nine intersections; any eight of these do not determine the variable cubic, but they do determine the ninth intersection; and there is between the nine intersections one integral relation, and corresponding to it one displacement-relation $\Sigma du = 0$; that is, $du_1 + du_2 + \ldots + du_9 = 0$, given by the pure theorem. But as in this case the number of arbitrary constants in the conic $\Omega = 0$, the corresponding equation $\Sigma\, du = 0$ is an identity; the number of independent displacement-relations is 0.

Examples of the Affected Theorem—Fixed Curve a Conic. Art. Nos. 13 and 14.

13. The fixed curve is taken to be the conic $x^2 + y^2 - 1 = 0$, and the parametric points 1 and 2 to be the points (1, 0) and (0, 1) on this circle. The variable curve is taken to be a line, say the line $ax + by - 1 = 0$, meeting the circle in the points 3 and 4, coordinates $(x_3,\, y_3)$ and $(x_4,\, y_4)$ respectively.

Starting from the formula

$$\Sigma\, \frac{(x,\, y,\, 1)^{n-3}\, du}{012} \;=\; -\frac{\delta\phi_3}{\phi_3}\,\frac{\delta\phi_4}{\phi_4},$$

where the summation extends to the points 3 and 4, $(x,\, y,\, 1)^{n-3}$ is here a constant, $= 2.12$, that is, $2.(x_1x_2 + y_1y_2 - 1)$, which for the points 1, 2 in question is $= -2$. We have 012 denoting the determinant

$$012 = \begin{vmatrix} x & y & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix},$$

which is $= -x - y + 1$; and $du = \dfrac{dx}{2y}$. Also $\dfrac{\delta\phi_3}{\phi_3} = \dfrac{x_3\,\delta a + y_3\,\delta b}{ax_3 + by_3 - 1}$, is $= \dfrac{da}{a-1}$, and similarly $\dfrac{\delta\phi_4}{\phi_4}$

is $= \dfrac{db}{b-1}$. The formula thus is

$$\Sigma \frac{da}{y(x+y-1)} = -\frac{da}{a-1} + \frac{db}{b-1}.$$

The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_3, x_4; and considering them as thus expressed, then (inasmuch as there is no linear relation between the displacements $\dfrac{dx_3}{y_3}$ and $\dfrac{dx_4}{y_4}$ of the two arbitrary points 3 and 4 on the circle) the equation must become an identity in regard to the two lines dx_3, dx_4 respectively.

[p. 117] 14. Writing P, Q, $R = y_3 + y_4$, $x_3 - x_4$, $x_3y_4 - x_4y_3$; also L_3 and $L_4 = x_3 + y_3 - 1$ and $x_4 + y_4 - 1$ respectively, we have $a = P + R$, $b = Q + R$; and the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{1}{(Q-R)(R-P)} \{(Q-R)\,dP + (R-P)\,dQ + (P-Q)\,dR\},$$

where, substituting for dy_3, dy_4 their values in terms of dx_3, dx_4, we have

$$dP,\ dQ,\ dR = \frac{1}{y_3}\,x_3\,dx_3 - \frac{1}{y_4}\,x_4\,dx_4,\ \frac{1}{y_3}\,y_3\,dx_3 - \frac{1}{y_4}\,y_4\,dx_4,\ \frac{1}{y_3}\,(x_3y_4 + y_3y_4)\,dx_3 - \frac{1}{y_4}\,(x_3y_4 + y_3y_4)\,dx_4,$$

and with these values, and by aid of the relations

$$Q-R,\ R-P,\ P-Q = x_3y_3 - x_4y_4,\ y_3L_4 - y_4L_3,\ -L_3 + L_4,$$

the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{L_3L_4(x_3x_4 + y_3y_4 - 1)}{(x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3)} \left(\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4}\right);$$

viz. this will be truly identity if only

$$L_3L_4(x_3x_4 + y_3y_4 - 1) = (x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3),$$

that is,

$$-x_3y_4\,L_3 + L_3L_4(x_3x_4 - 1) + (y_3y_4 + x_3y_3)(x_3 + y_3 - 1) = 0.$$

But from the values of L_3, L_4 we have $x_3 = \frac{1}{2}L_3^2 + L_3$, $x_3y_3 = L_3 + L_3$, and the coefficient of L_3L_4 is $= L_3L_4 + L_4 + L_3 + 1$, the equation is thus verified.

The example would perhaps have been more instructive if the points 1 and 2 had been left arbitrary points on the circle, but the working out would have been more difficult.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. Art. Nos. 15 to 19.

15. Suppose $n = 3$, $\delta = 0$ ($p = 1$), the fixed curve a cubic; and suppose successively $m = 1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m = 1$, then two points on the cubic determine the line, and consequently the remaining intersection with the cubic; hence the number α depends.

If $m = 2$, then 5 points on the cubic determine the conic, and the remaining intersection with the cubic, hence the number α, f^2, will be a determinate point, &c.

If $m = 3$, then 8 points on the cubic determine the cubic, and consequently the remaining intersection with the cubic, hence the number α, f^3, will be a determinate point.

[p. 118] If $m = n - 3$, the variable curve is a generic functions of p, q, r, but the dependence on the parameters considered all of 1, 2, 3, &c., f which is a determinant of the variable curve.

16. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and as before suppose successively $m = 1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m = 1$, then two points on the quartic determine the line, and therefore the remaining two intersections; the number of integral relations is $= 2$.

If $m = 2$, then five points on the quartic determine the conic, and therefore the remaining three intersections; the number of integral relations is $= 3$, and similarly if $m = 3$, the number of integral relations is $= 3$.

If $m = 4$, then thirteen points on the fixed quartic do not determine the variable quartic, but they do determine the remaining three intersections. For draw through the thirteen points a no-matter-what quartic $\chi = 0$; the general quartic through the thirteen points is $\chi + \alpha f = 0$, and this meets the fixed quartic in the points $\chi = 0$, $f = 0$, that is, in the thirteen points and in three other points independent of α and thus completely determinate; and the number of integral relations is $= 3$, but as α is in general for any higher value of m, the number is still $= 3$.

17. Suppose $n = 5$, $\delta = 0$ $(p = 6)$; the fixed curve a general quintic; and as before suppose $m = 1, 2, 3, \ldots$, successively.

If $m = 1$, then two points on the quintic determine the line, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 2$, then five points on the quintic determine the conic, and therefore the remaining five intersections; the number of integral relations is $= 5$.

If $m = 3$, then 9 points on the quintic determine the cubic, and therefore the remaining six intersections; the number of integral relations is $= 6$, and so if $m = 4$, or if $m = 5$ or any greater number, in the first case directly, and in the other cases by consideration of the quintic $\chi + \alpha f = 0$, &c., we find that the number of integral relations is always $= 6$.

[p. 119] 18. The reasoning is scarcely altered in the case where the fixed curve has dps; thus considering the general case of a fixed curve n, δ, p:

If $m = 1$, then $2 - \delta$ points on the fixed curve (this implies $\delta \geq 2$) determine the line, and therefore the remaining $n - 2\delta - (2 - \delta)$, $= n - 2 - \delta$ intersections; the number of integral relations is $= n - 2 - \delta$.

If $m = 2$, then $5 - \delta$ points on the fixed curve (this implies $\delta \geq 5$) determine the conic, and therefore the remaining $2n - 2\delta - (5 - \delta) = 2n - 5 - \delta$, and so for any value of $m \geq n - 3$, and indeed for the values $n - 2$ and $n - 1$; here $\frac{1}{2}m(m+3) - \delta$ points on the fixed curve (this implies $\delta \geq \frac{1}{2}m(m+3) - \delta$) determine the remaining $mn - 2\delta - \{\frac{1}{2}m(m+3) - \delta\}$, $= mn - \frac{1}{2}m(m+3) - \delta$ intersections. Hence the number of integral relations is $= mn - \frac{1}{2}m(m+3) - \delta$, that is, $= p - \frac{1}{2}(n - m - 1)(n - m - 2)$. And thus in the cases $m = n - 2$ and $n - 1$ the number is $= p$.

If $m = n$, then $\frac{1}{2}n(n+3) - 1 - \delta$ points on the fixed curve do not determine the variable curve, but they do determine the remaining p intersections, and the number of integral relations is thus $= p$; and so, for any higher value of m, the number is still $= p$.

19. The conclusion thus is

$$\begin{array}{rll} m \geq n - 3, & \text{the number of integral relations} & = p - \frac{1}{2}(n - m - 1)(n - m - 2), \\ m = n \text{ or } > n - 3, & \quad ,, & = p. \end{array}$$

The integral equations spoken of throughout are of course independent relations.

The Variable Intersections of the Two Curves. Number of Independent Displacement Relations given by the Pure Theorem. Art. No. 20.

20. The number of displacement-relations given by the pure theorem is $=$ number of constants in minor function $(x, y, z)^{n-3}$, which equated to zero represents a curve through the dps, viz. this is always

$$\tfrac{1}{2}(n-1)(n-2) - \delta, \ = p.$$

But for $m \geq n - 2$, these relations are not independent. For instance, for $n = 4$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\, y,\, z)^2\, du = 0, \text{ that is, } \Sigma x\, du = 0,\ \Sigma y\, du = 0,\ \Sigma z\, du = 0,$$

and conversely from these last equations we have $\Sigma\,(x,\,y,\,z)^2\,du = 0$. But in this case the variable curve is a line meeting the quartic in four points; the equation $\Sigma x\,du = 0$ is satisfied for each of the intersections of the line with the quartic, and the corresponding equation $\Sigma\,(x,\,y,\,z)^2\,du = 0$ is an identity. Hence the number of independent displacement-relations is $3 - 1, = 2$.

So for $n = 5$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\,y,\,z)^2\,du = 0,\ \text{that is, } \Sigma(x^2,\,y^2,\,z^2,\,yz,\,zx,\,xy)\,du = 0,$$

and so on for $n = 6$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\,y,\,z)^3\,du = 0.$$

[p. 120] and six equations give conversely $\Sigma\,(x,\,y,\,z)^2\,du = 0$, in particular they give $\Sigma x\,du = 0$, $\Sigma y\,du = 0$, $\Sigma z\,du = 0$, $\Sigma xy\,du = 0$. But if $(x,\,y,\,z)$ denote any of the intersections of the line with the curve, then for each of the five intersections $(x,\,y,\,z) = 0$, that is, the fast-mentioned three equations are identities. Hence the number of independent displacement-relations is $6 - 3$, $= 3$.

So for $n = 5$, $\delta = 0$, $m = 2$. Here if the variable curve is $\phi = (a, \dots\ )(x,\,y,\,z)^2$, then taking $(x,\,y,\,z) = (x_1,\,y_1,\,z_1)$, $(x_2,\,y_2,\,z_2)$, …, the corresponding equation $\Sigma\,(x,\,y,\,z)^2\,du = 0$ is an identity; the number of independent displacement-relations is $6 - 1, = 5$.

The reasoning is the same when δ is not $= 0$, and we see generally that for $m \geq n - 2$,

$$m \geq n - 3, \quad \text{the number of independent displacement-relations}$$
$$= p - \tfrac{1}{2}(n - m - 1)(n - m - 2);$$

while for

$$m = n \text{ or } > n - 2, \text{ the number is} = p;$$

since in this case the relations given by the theorem are all of them independent. It thus appears *à posteriori*, that in every case the number of independent displacement-relations given by the pure theorem is equal to the number of independent integral relations.

As to the dps of the Fixed Curve. Art. No. 21.

21. I conclude with a general remark applicable to the whole of the three chapters. There is no necessity to attend to all or indeed to any of the dps of the fixed curve. Suppose that the fixed curve has $\delta + \delta'$ dps, where δ, δ' may be either of them or each $= 0$, and attend only to the δ dps, the δ' dps being wholly disregarded, and accordingly let the expression the dps mean as before the δ dps of the fixed curve. No attention at all is required: only, if δ' be not $= 0$, then $p = \frac{1}{2}(n-1)(n-2) - \delta - \delta'$ will no longer be the deficiency. To obtain the best theorem we use all the $\delta + \delta'$ dps; but disregarding the δ' dps, we obtain theorems, as for a curve with δ dps, which are true and may frequently be useful either in their original form or with simplifications introduced therein by afterwards taking account of the δ' dps.

Chapter II. Proof of Abel's Theorem.

Preparation. Art. Nos. 22 and 23.

22. Starting from the equation $\phi = (a, \dots\ )(x,\,y,\,z)^m = 0$ of the variable curve, we have

$$\frac{d\phi}{dx}\,dx + \frac{d\phi}{dy}\,dy + \frac{d\phi}{dz}\,dz + \delta\phi = 0,$$
$$\frac{d\phi}{dx}\,x + \frac{d\phi}{dy}\,y + \frac{d\phi}{dz}\,z = 0,$$

where $\delta\phi = (da, \ldots\ )(x,\ y,\ z)^m$. Let τ denote an arbitrary linear function, $= ax + by + cz$; multiply the two equations by $ax + by + cz$, $= \tau$, and $-(x\,dz + b\,dy + c\,dz)$, $= -d\tau$ respectively, and add: we obtain

$$(y\,dz - z\,dy)\left(a\,\frac{d\phi}{dz} - c\,\frac{d\phi}{dy}\right) + (z\,dx - x\,dz)\left(b\,\frac{d\phi}{dx} - a\,\frac{d\phi}{dz}\right) + (x\,dy - y\,dx)\left(a\,\frac{d\phi}{dy} - b\,\frac{d\phi}{dx}\right) + \tau\,\delta\phi = 0;$$

introducing du, this becomes

$$du\,\left[\frac{df}{dx}\left(b\,\frac{d\phi}{dz} - c\,\frac{d\phi}{dy}\right) + \frac{df}{dy}\left(c\,\frac{d\phi}{dx} - a\,\frac{d\phi}{dz}\right) + \frac{df}{dz}\left(a\,\frac{d\phi}{dy} - b\,\frac{d\phi}{dx}\right)\right] + \tau\,\delta\phi = 0,$$

or observing that a, b, c are the differential coefficients $\frac{d\tau}{dx}, \frac{d\tau}{dy}, \frac{d\tau}{dz}$, the term in [ ] is $J(f,\ \tau,\ \phi)$, or say $J(\phi,\ f,\ \tau)$, and the equation is

$$du\,J(\phi,\ f,\ \tau) + \tau\,\delta\phi = 0,$$

where $J(\phi,\ f,\ \tau)$ is the Jacobian, or functional determinant

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} & \frac{d\phi}{dz} \\ \frac{df}{dx} & \frac{df}{dy} & \frac{df}{dz} \\ \frac{d\tau}{dx} & \frac{d\tau}{dy} & \frac{d\tau}{dz} \end{vmatrix},\ = \frac{d(\phi,\ f,\ \tau)}{d(x,\ y,\ z)};$$

and we hence have

$$du = -\,\frac{\tau\,\delta\phi}{J(\phi,\ f,\ \tau)}.$$

23. The two theorems thus become

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\tau\,\delta\phi}{J(\phi,\ f,\ \tau)} = 0,$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\,\tau\,\delta\phi}{012\,J(\phi,\ f,\ \tau)} \quad = \quad -\,\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2}.$$

But further, if in the first equation we retain τ, while in the second equation we retain τ, using it to denote the linear function 012, the equations become

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\delta\phi}{J(\phi,\ f)} = 0,$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\,\delta\phi}{J(\phi,\ f,\ \tau)} \quad = \quad -\,\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2};$$

where, in the first equation, $J(\phi,\ f)$ denotes the Jacobian

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} \\ \frac{df}{dx} & \frac{df}{dy} \end{vmatrix},\ = \frac{d(\phi,\ f)}{d(x,\ y)}.$$

[p. 122] In these equations the only differential symbol is the δ affecting the coefficients a, b, $\ldots$ of ϕ, ϕ_1, ϕ_2; the equations are, in respect to the coordinates $(x,\ y,\ z)$ of the several variable intersections, but they are mere algebraical equations, which are in fact given by Jacobi's

Fraction-theorem about to be explained. But for the further reduction of the affected theorem I interpose the next three articles.

The Coordinates $(\rho,\ \sigma,\ \tau)$. Art. Nos. 24 to 26.

24. The letter τ has just been used to denote the determinant 012; there is often occasion to consider three points 1, 2, 3 coordinates $(x_1,\ y_1,\ z_1)$, $(x_2,\ y_2,\ z_2)$, $(x_3,\ y_3,\ z_3)$ respectively; and then writing ρ, σ, τ to denote the determinants 023, 031, 012 respectively, and Δ the determinant 123, we have identically

$$\begin{aligned}\Delta x &= x_1\rho + x_2\sigma + x_3\tau,\\ \Delta y &= y_1\rho + y_2\sigma + y_3\tau,\\ \Delta z &= z_1\rho + z_2\sigma + z_3\tau,\end{aligned}$$

which equations, regarding therein the point 0, coordinates $(x,\ y,\ z)$, as a current point, are in fact equations for transformation from the coordinates $(x,\ y,\ z)$ to the coordinates ρ, σ, τ, belonging to the triangle of reference 123. The points 1 and 2 have been already taken to be on the fixed curve, and it will be assumed that 3 is also a point on this curve.

25. If the function f, which equated to zero gives the equation of the fixed curve, be transformed to the new coordinates $(\rho,\ \sigma,\ \tau)$, the coefficients of the transformed function are polar functions, each divided by ∇^n, viz. the coefficient of ρ^n is $\dfrac{1}{\nabla^n}\,1^n$, which by reason that 1 is a point on the curve is $=0$ (and similarly the coefficients of σ^n and of τ^n are each $=0$); the coefficient of $\rho^{n-1}\sigma$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,2$; that of $\rho^{n-1}\tau$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,3$; that of $\rho^{n-2}\sigma^2$ is $\dfrac{1}{\nabla^n}\,\tfrac{1}{2}n(n-1).\,1^{n-2}\,2^2$; and so in other cases. I write this in the form

$$f = \frac{1}{\nabla^n}\,(1^n,\ \dots\ )(\rho,\ \sigma,\ \tau)^n;$$

or we might also use the symbolic form

$$f = \frac{1}{\nabla^n}\,(\rho 1 + \sigma 2 + \tau 3)^n.$$

[p. 123] The terms independent of τ contain, it is clear, the factor $\rho\sigma$, and separating these terms from the others, the equation may be written

$$f = \frac{1}{\nabla^n}\,\rho\sigma\,(n.\,1^{n-1}\,2,\ \dots\ )(\rho,\ \sigma)^{n-2} + \text{terms involving } \tau.$$

26. The equations $\tau = 0$, $(\dots\ )(\rho,\ \sigma)^{n-2} = 0$, determine the residues of the points 1, 2, and hence the major function $(x,\ y,\ z)^{n-3}$, expressed in terms of ρ, σ, τ, and writing therein $\tau = 0$, must reduce itself to $(\dots\ )(\rho,\ \sigma)^{n-2}$ multiplied by a constant factor which can be found to be $=\dfrac{1}{\nabla^{n-1}}$: for taking the current point at 1 we have $(\rho,\ \sigma,\ \tau) = (\Delta,\ 0,\ 0)$, and the corresponding value of the major function is thus equal, and in this case $\dfrac{1}{\nabla^{n-1}}\,n.\,1^{n-1}\,2$, $=n.\,1^{n-1}\,2$, as it ought to be. We have thus

$$(x,\ y,\ z)^{n-3} = \frac{1}{\nabla^{n-1}}\,(n.\,1^{n-1}\,2,\ \dots\ )(\rho,\ \sigma)^{n-2} + \text{terms involving } \tau;$$

and we hence see that, for $\tau = 0$,

$$(x,\ y,\ z)^{n-3} = \frac{\Delta f}{\rho\sigma}.$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. Art. No. 27.

27. In the affected theorem instead of $(x,\, y,\, z)$ we introduce the new coordinates $(\rho,\, \sigma,\, \tau)$. We have

$$J(\phi,\, f,\, \tau) = \frac{d(\phi,\, f,\, \tau)}{d(\rho,\, \sigma,\, \tau)}\,\frac{d(\rho,\, \sigma,\, \tau)}{d(x,\, y,\, z)},$$

where the first factor is $= \dfrac{d(\phi,\, f)}{d(\rho,\, \sigma)}$, say that this is $J(\phi,\, f)$, the Jacobian in regard to ρ, σ: and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi,\, f,\, \tau)} = \frac{1}{\Delta!\, J(\phi,\, f)},$$

and the equation for the affected theorem becomes

$$\Sigma\, (x,\, y,\, z)^{n-3}\,\frac{\delta\phi}{J(\phi,\, f)} = -\,\Delta!\left(-\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2}\right),$$

where $(x,\, y,\, z)^{n-3}$ is to be regarded as standing for its value in terms of $(\rho,\, \sigma,\, \tau)$.

Jacobi's Fraction-Theorem. Art. Nos. 28 to 31.

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if U be any rational and integral function $(x,\, 1)^m$, then we have

$$\frac{1}{U} = \Sigma\,\frac{1}{x - x'}\,\frac{1}{J(U)},$$

[p. 124] or, introducing an arbitrary constant A by the equation $AU = X$, say this is

$$\frac{A}{X}\left(=\frac{1}{U}\right) = \Sigma\,\frac{1}{x - x'}\,\frac{1}{J(U)},$$

where U' is the same function (of x') that U is of x; $J(U)$, $= \dfrac{dU'}{dx'}$, is the Jacobian of U', and the summation extends to all roots x' of the equation $U' = 0$: obviously this is nothing else than the formula for the decomposition of $\dfrac{1}{U}$ into simple fractions.

29. Take now $U = (x,\, y,\, 1)^m$, $V = (x,\, y,\, 1)^p$ functions of x, y of the degrees m and n respectively, and assume

$$\begin{aligned} AU + BV &= X, \quad \text{a function of } (x,\, 1)^{mn}, \\ CU + DV &= Y, \quad \text{,, of } (y,\, 1)^{mn}; \end{aligned}$$

viz. let $X = 0$ and $Y = 0$ be the equations obtained by elimination from $U = 0$ and $V = 0$ of the y and the x respectively. The forms are

$$\begin{aligned} A &= (x,\, y',\, 1)^{mn-m}, \quad B = (x,\, y',\, 1)^{mn-n}, \\ C &= (x^{m-1},\, y,\, 1)^{mn-m}, \quad D = (x^{m-1},\, y,\, 1)^{mn-n}; \end{aligned}$$

where these equations denote the kind of them; that is A is a rational and integral function of the degree $mn - m$ in x and y jointly, but only of the degree $mn - 1$ in y; and so for the other equations. It follows that

$$AD - BC = (x^{mn-1},\, y^{mn-1},\, 1)^{mn-mn-m}.$$

The theorem now is

$$\frac{AD - BC}{XY} = \Sigma\,\frac{1}{x - x' \cdot y - y'}\,\frac{1}{J(U',\, V')},$$

where U', V' are the same functions of (x', y') that U, V are of (x, y); $J(U', V')$ is the Jacobian $\dfrac{d(U', V')}{d(x', y')}$, $= 0$; and the summation extends to all the simultaneous roots x', y' of the equations $U = 0$, $V = 0$.

30. For the proof, observe that $AD - BC$ is a sum of terms of the form $\alpha x^\alpha y^\beta$ where α and β are each of them at most $= mn - 1$; hence X being of the degree mn we have $\dfrac{x^\alpha}{X}$ a sum of fractions $\dfrac{L_y}{x - x'}$, where x' is any root of $X = 0$; and similarly $\dfrac{y^\beta}{Y}$ a sum of fractions $\dfrac{M_y}{y - y'}$, where y' is any root of $Y = 0$; multiplying the two expressions and taking into account the several terms $\lambda x^\alpha y^\beta$ of $AD - BC$ we obtain a formula

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to all the combinations of the mn values of x' with the mn values of y. But such a formula existing, the coefficients K may be determined.
[p. 125] in the usual manner, viz. multiplying by XY and then writing $x = x'$, $y = y'$, there is on the right-hand only one term which does not vanish, and we find

$$(AD - BC)_{xy'} = K\left(\frac{X}{x - x'}\right)_x \left(\frac{Y}{y - y'}\right)_{y'}, \;=\; K\left(\frac{dX}{dx}\,\frac{dY}{dy}\right)_{xy'};$$

where the factor which multiplies K does not vanish.

We distinguish the cases where (x', y') are corresponding or non-corresponding roots of $X = 0$, $Y = 0$, viz. corresponding roots are those for which $U = 0$, $V = 0$, but for non-corresponding roots these equations do not hold good; there are obviously mn pairs of corresponding roots.

In the latter case $(AD - BC)\,U = DX - BY$, $(AD - BC)\,V = -CX + AY$, and since for these values have $X = 0$, $Y = 0$, the foregoing equation for K gives then $K = 0$.

31. The formula thus is

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to corresponding roots only; or, for which we have $U = 0$, $V = 0$.

The Collected Mathematical Papers of Arthur Cayley
Vol. XII

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Arthur Cayley (1821–1895)

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Contents

822. On Associative Imaginaries (conclusion)

[book p. 106] In order that this may be associative, we must still have the relations

$$bg = cd,$$
$$c^2 + dg - ag - eh = 0,$$
$$d^2 + bc - ad - bh = 0,$$

which are all three of them satisfied by $g = \dfrac{cd}{b}$, $h = \dfrac{d^2 + bc - ad}{b}$, viz. we thus have the associative and commutative system

$$x^2 = ax + by,$$
$$xy = yx = cx + dy,$$
$$y^2 = \frac{cd}{b}\,x + \frac{d^2 + bc - ad}{b}\,y.$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. IV. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad - bc$ not $= 0$, is expressible as a mixture of two algebras of the form (a_1^2), see p. 120 (l.c.); whereas if $ad - bc = 0$, it is reducible to the form (c_2), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the Johns Hopkins University Circulars, No. 17 (1882), p. 238.]

[book p. 107] I have given in my *Elliptic Functions* expressions for the sn^2 of $u + \frac{1}{2}K$, $u + \frac{1}{2}iK'$, $u + \frac{1}{2}K + \frac{1}{2}iK'$; but it is better to consider the dn^2, sn^2, cn^2 of these combinations respectively, and to write the formulæ thus:

$$\mathrm{dn}^2(u + \tfrac{1}{2}K) = k\,\frac{\mathrm{dn}\,u - (1-k')\,\mathrm{sn}\,u\,\mathrm{cn}\,u}{\mathrm{dn}\,u + (1-k')\,\mathrm{sn}\,u\,\mathrm{cn}\,u} \;=\; k\,\frac{1 - k'x - (1-k')y}{1 - k'x + (1-k')y},$$

$$\mathrm{sn}^2(u + \tfrac{1}{2}iK') = \frac{1}{k}\,\frac{(1+k)\,\mathrm{sn}\,u + i\,\mathrm{cn}\,u\,\mathrm{dn}\,u}{(1+k)\,\mathrm{sn}\,u - i\,\mathrm{cn}\,u\,\mathrm{dn}\,u} \;=\; \frac{1}{k}\,\frac{(1+k)x + iy}{(1+k)x - iy},$$

$$\mathrm{cn}^2\big(u + \tfrac{1}{2}K + \tfrac{1}{2}iK'\big) = -\frac{ik'}{k}\,\frac{(k - ik')\,\mathrm{sn}\,u + \mathrm{dn}\,u\,\mathrm{cn}\,u}{(k + ik')\,\mathrm{sn}\,u + \mathrm{dn}\,u\,\mathrm{cn}\,u} \;=\; -\frac{ik'}{k}\,\frac{(k - ik')x + y}{(k + ik')x + y};$$

where in the last set of values x, y are used to denote $\mathrm{sn}^2 u$ and $\mathrm{sn}\,u\,\mathrm{cn}\,u\,\mathrm{dn}\,u$ respectively; and the formulæ are thus brought into connexion with the cubic curve $y^2 = x(1-x)(1-k^2x)$. The curve has an inflexion at infinity on the line $x = 0$; and the three tangents from the inflexion are $x = 0$, $x = 1$, $x = \dfrac{1}{k^2}$, touching the curve at the points x, $y = (0, 0)$, $(1, 0)$, $\left(\dfrac{1}{k^2}, 0\right)$ respectively; hence these points are sextactic points.

We may from any one of them, for instance the point $(0, 0)$, draw four tangents to the curve, $(1+k)x + iy = 0$, $(1+k)x - iy = 0$; $(1-k)x + iy = 0$, $(1-k)x - iy = 0$; where the first and second of these lines form a pair, and the third and fourth of them form a pair, viz. the two tangents of a pair touch in points such that the line joining them passes through the point of inflexion: in particular, for the first-mentioned pair, the equation of the line joining the points of contact is $\frac{1}{1+k}x = 0$. The linear functions belonging to a pair of tangents are precisely those which present themselves in the formulæ; thus if $2X = (1+k)x + iy$, $2Y = (1+k)x - iy$, the second of the three formulæ is $\operatorname{sn}^2(u + \frac{1}{2}iK') = \frac{1}{k}\frac{X}{Y}$; and the other two formulæ correspond in like manner to pairs of

[book p. 108] tangents from the sextactic points $\left(\frac{1}{k^2}, 0\right)$ and $(1, 0)$ respectively. The formulæ are connected with the fundamental equations expressing the functions sn, cn, dn as quotients of theta functions.

824. Note on the Formulæ of Trigonometry

[From the Johns Hopkins University Circulars, No. 17 (1882), p. 241.]

[book p. 108] The equations $a = c\cos B + b\cos C$, $b = a\cos C + c\cos A$, $c = b\cos A + a\cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i\sin A$, $\cos B + i\sin B$, $\cos C + i\sin C$, viz. calling these x, y, z respectively, or, what is the same thing, writing

$$2\cos A = \frac{x}{w} + \frac{w}{x}, \quad 2\cos B = \frac{y}{w} + \frac{w}{y}, \quad 2\cos C = \frac{z}{w} + \frac{w}{z},$$

then the foregoing equations may be written

$$\begin{aligned}
&(-2yzw\,,\ y(z^2+w^2)\,,\ z(y^2+w^2)\,|\,a\,,\,b\,,\,c) = 0,\\
&(z(x^2+w^2)\,,\ -2zxw\,,\ x(z^2+w^2)\,|\,a\,,\,b\,,\,c) = 0,\\
&(y(x^2+w^2)\,,\ x(y^2+w^2)\,,\ -2xyw\,|\,a\,,\,b\,,\,c) = 0;
\end{aligned}$$

that is, as a system of bipartite equations linear in (a, b, c) and cubic in (x, y, z, w) respectively.

Similarly in Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2\cos a = \frac{\alpha}{\delta} + \frac{\delta}{\alpha}$, $2\cos b = \frac{\beta}{\delta} + \frac{\delta}{\beta}$, $2\cos c = \frac{\gamma}{\delta} + \frac{\delta}{\gamma}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$ respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, American Journal of Mathematics, t. V. (1882), pp. 137–179; Chapters IV to VII, ib., t. VII. (1885), pp. 101–167.]

[book p. 109] The present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi^{1\alpha}_{1\beta}$ (or as I write it $d\Pi_{12}$) in the integral of the third kind $\int_a^b d\Pi^{1\alpha}_{1\beta}$ in the final normal form (*endliche Normalform*) for which we have (p. 117) $d\Pi^{1\alpha}_{a\beta} = \int_a^b d\Pi^{a\beta}_{1\beta}$, the limits and parametric points interchangeable. The want

of this determination presented itself to me as a lacuna in the theory during the course of lectures on the subject which I had the pleasure of giving at the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the far more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to. I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence; and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function; IV, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ. The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

[book p. 110] The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.

1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,d\omega = 0.$$

The affected theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,\frac{\delta\phi}{J(\phi,\,f)}\,d\omega = -\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

2. We have a fixed curve $f = 0$, or say the curve f, or simply the fixed curve, of the order n, with δ dps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta, = p$. The expression "the dps" means always the δ dps of f.

And we have a variable curve $\phi = 0$, or say the curve ϕ, or simply the variable curve, of the order m, passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, $d\omega$ is the displacement of the current point 0, coordinates $(x,\,y,\,z)$, on the fixed curve, viz. the equation $f = 0$ gives

$$\frac{df}{dx}\,dx + \frac{df}{dy}\,dy + \frac{df}{dz}\,dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} \;=\; y\,dz - z\,dy \;:\; z\,dx - x\,dz \;:\; x\,dy - y\,dx,$$

so that we have three equal values each of which is put $= d\omega$, viz. we write

$$\frac{y\,dz - z\,dy}{\dfrac{df}{dx}} = \frac{z\,dx - x\,dz}{\dfrac{df}{dy}} = \frac{x\,dy - y\,dx}{\dfrac{df}{dz}} = d\omega,$$

and the ω thus defined is the displacement.

Note. For comparison with C. and G., we observe that in the equation, p. 47, $C = \frac{p\,dq - q\,dp}{f'_r}$ (p, q, r belong to the upper limit and corresponds to my ϕ; the equation gives therefore $C = -\dfrac{\delta\phi}{J}\,d\omega$, agreeing with the formula in the text.

[book p. 111] $(x,\,y,\,z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n-3$, which passes through the dps*, and the function $(x,\,y,\,z)^{n-3}$ is the minor function.

1. and **2.** fixed points at 0, of the parametric points, coordinates $(x_1,\,y_1,\,z_1)$ and $(x_2,\,y_2,\,z_2)$ respectively; and 012 denotes the determinant

$$012 = \begin{vmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

so that $012 = 0$ is the equation of the line joining the points 1 and 2: this line meets the fixed curve in $n-2$ other points, called the residues of 1, 2.

$(x,\,y,\,z)^{n-3} = 0$ is the major curve passed the points 1 and 2; viz. this is the general curve of the order $n-3$, passing through the dps and also through the residues of 1, 2.

But further, the function $(x,\,y,\,z)^{n-3}$ is the proper major function; viz. the implicit factor of the function is as determined that, taking $0 = 1$, the current point at 1, that is, writing $(x,\,y,\,z)$ for $(x_1,\,y_1,\,z_1)$, the function $(x,\,y,\,z)^{n-3}$ reduces itself to the polar function $\left(x_2\frac{d}{dx} + y_2\frac{d}{dy} + z_2\frac{d}{dz}\right)\phi_1$, afterwards written $n.\,1^{n-2}\,2$; this implies that taking $0 = 2$, the current point at 2, the function reduces itself to the polar function $n.\,12^{n-2}$.

2. ϕ is what ϕ becomes at writing therein $(x_1,\,y_1,\,z_1)$ for $(x,\,y,\,z)$; and similarly ϕ_2 is what ϕ becomes on writing therein $(x_2,\,y_2,\,z_2)$ for $(x,\,y,\,z)$.

δ denotes differentiation in regard only to the coefficients of ϕ, viz. writing $\phi = (a,\,b,\,c,\ldots)(x,\,y,\,z)^m$ have $\delta\phi = (\delta a,\,\delta b,\,\delta c,\ldots)(x,\,y,\,z)^m$, and similarly $\delta\phi_1 = (\delta a,\,\delta b,\,\delta c,\ldots)(x_1,\,y_1,\,z_1)^m$ and $\delta\phi_2 = (\delta a,\,\delta b,\,\delta c,\ldots)(x_2,\,y_2,\,z_2)^m$.

The sum Σ extends to all the variable intersections of the two curves.

3. As to the meaning of the theorem, consider first the pure theorem. The variable intersections are not all of them arbitrary points on the fixed curve; a certain number of them taken at pleasure on the fixed curve will determine the remaining variable intersections; and there are thus in fact, between the displacements of all of them, 2δ equations of integral relation; in such a way that there only remains a number of integral displacement-relations equal to p, the deficiency of the curve.

But from the values of L_1, L_2 we have $x_3 = \frac{1}{2}L_2 + L_1$, $x_1 + x_2 = \frac{1}{2}L_2 + L_1$; and the coefficient of L_1L_2 in $L_2L_3 + L_2L_1 + L_1L_2$ is the equation is thus verified.

The example would perhaps have been more instructive if the working out would have been more difficult.

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* This theory was demonstrated in my Section A of the British Association at the York meeting, Sept. 3, 1881, Report (London), 1881, [712], 'On the simplification of the equation given therefor in Cayley's Mem. "A Partial Differential Equation associated with the simplest case of Abel's Theorem." Vol. XII.

[book p. 112] **4.** It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; we are hereby led to see the nature of the affected theorem, viz. in this same case where the variable curve ϕ meets the fixed curve in such a way as to produce variations $\delta\phi_1$, $\delta\phi_2$ in ϕ_1, ϕ_2, the displacement-relations $\Sigma\,(x,\,y,\,z)^{n-3}\,d\omega = 0$ given by the pure theorem.

Examples of the Pure Theorem—The Fixed Curve a Circle. **Art. Nos. 6 to 13.**

6. The pure theorem is not applicable to the case $n = 2$, the fixed curve a conic: it is in fact gives no displacement-relations, and this is as it should be, for the variable intersections are all of them arbitrary.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For greater simplicity the equation is taken in Cartesian coordinates. In general for such an equation, writing in the homogeneous formula $z = 1$, we have

$$d\omega = \frac{dy}{\frac{df}{dx}} = -\frac{dx}{\frac{df}{dy}};$$

the two values being of course equal in virtue of $\frac{df}{dx}\,dx + \frac{df}{dy}\,dy = 0$; taking the former value and considering $\frac{df}{dy}$ as expressed in terms of x, let this be called P (of course, P is an irrational function of x): then we have $d\omega = \frac{dx}{P}$, and similarly $d\omega_1 = \frac{dx_1}{P_1}$, &c.

The fixed curve being then a cubic, let the variable curve be a line; this meets the cubic in three points, say 1, 2, 3; and any two of these determine the third, the relation being $x_1+x_2+x_3 =$ (a constant). In Cartesian coordinates, if the equation of the line be $\alpha x + \beta y + \gamma = 0$, then writing for y its value in terms of x, the equation of the cubic becomes

$$f = -\frac{1}{\beta^3}\,(\alpha x + \gamma)^3 + a(\alpha x + \gamma)^2 + b(\alpha x + \gamma) + c.$$

* The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_1, x_2, considering them as those expressed, thus determined.

[book p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively; this is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

7. The equation $\Sigma\, d\omega = 0$, where the direction $\frac{dx}{P}$ has a peculiar interpretation when we consider the coefficients of the cubic as arbitrary constants, and therefore the cubic as a moving curve in a moving changing also arbitrary changing; but the result then can be obtained by elimination from $\Sigma\,\frac{dx}{P} = 0$ and the equation $123 = 0$. There is in fact a theorem of intuitive interpretation, $dx + dx_3 + dx_3 = 0$, we have thus

$$\frac{dx_1}{P_1} + \frac{dx_2}{P_2} + \frac{dx_3}{P_3} = 0,$$

and we hence see that, for $\tau = 0$,

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2}\,;$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. **Art. No. 27.**

27. In the affected theorem instead of $(x,\, y,\, z)$ we introduce the new coordinates $(\rho,\, \sigma,\, \tau)$. We have

$$J(\phi,\, f,\, \tau) = \frac{d(\phi,\, f,\, \tau)}{d(\rho,\, \sigma,\, \tau)} \cdot \frac{d(\rho,\, \sigma,\, \tau)}{d(x,\, y,\, z)},$$

where the first factor is $= \dfrac{d(\phi,\, f)}{d(\rho,\, \sigma)}$, say that this is $J(\phi,\, f)$, the Jacobian in regard to ρ, σ: and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi,\, f,\, \tau)} = \frac{1}{\Delta!\, J(\phi,\, f)},$$

and the equation for the affected theorem becomes

$$\Sigma\, (x,\, y,\, z)^{n-3}\, \frac{\delta\phi}{J(\phi,\, f)} = -\,\Delta!\, \left(-\,\frac{\delta\phi_1}{\phi_1} \cdot \frac{\delta\phi_2}{\phi_2}\right),$$

where $(x,\, y,\, z)^{n-3}$ is to be regarded as standing for its value in terms of $(\rho,\, \sigma,\, \tau)$.

* This theorem was communicated to me in Section A of the British Association at the York meeting, Sept. 3, 1881, [712]. 'On a Partial Differential Equation associated with the simplest case of Abel's Theorem.' Vol. XII.

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[book p. 114] **8.** As a particular example, let the fixed curve be the cubic $y^2 = x(1-x)(1-k^2x)$, and the line $x = b$; the integral relation $123 = 0$. I give a direct verification. To find x_3, the coefficient of the cubic being on the part of the relations of the cubic, 1, 2, 3, we have, by definition, $x_1\, x_2\, x_3$ as the roots of equation $0 = \dfrac{1}{k^2}\,(c+by)^2 - x(1-x)(1-k^2x)$.

The fact, x_3 depends in a like manner on x_1, x_2, and the integral relation $123 = 0$. I give a direct verification. To find x_3, the cubic being substituted for each of the cubic terms in [] are equal, that is, we ought to have

$$y_3(x_1 - x_2) = y_1(x_3 - x_2) - y_2(x_3 - x_1), \quad \text{(say A.)}.$$

Comparing for instance the first and second terms, the equation is

$$-\,y_3(x_3 - x_1) + y_1(x_2 - x_3) + y_2(x_3 - x_1) + y_3(x_1 - x_2) = 0;$$

or, as this may be written,

$$-\,(x_1 + x_3)y_3 + x_3y_1 + (x_1 + x_2)y_2 + x_2(y_3 - y_1) = 0;$$

where the whole condition for the line $x = b$ being substituted, the coefficients K may be determined.

[book p. 115] $d\omega_3 + d\omega_3 + d\omega_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $d\omega_1 + d\omega_2 + d\omega_3 = 0$, not containing any arbitrary constant, it is a particular integral.

But regard one of the three points, say 3, as a fixed point, that is, let the line pass through the fixed point 3 of the cubic, and besides meet it in the variable points 1 and 2. We write $d\omega_3 = 0$, and the differential equation thus is $d\omega_1 + d\omega_2 = 0$, while the integral equation is now $123 = 0$, the general integral of $d\omega_1 + d\omega_2 = 0$.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any three of these points determine the conic, and therefore the two remaining intersections; that is, the points 4, 5, 6 are integral relations, and consequently between the differential equations apparently $d\omega_1 + d\omega_2 + \ldots + d\omega_6 = 0$.

We have therefore the cubic 123456 = 0, say a particular integral of $d\omega_1 + d\omega_2 + \ldots + d\omega_6 = 0$.

If, however, we take 3 a fixed point on the cubic, we then have the same equation as the general integral of $d\omega_1 + d\omega_2 + \ldots + d\omega_5 = 0$, which expresses the determination, with the same line on the conic.

But taking also 3 a fixed point of the cubic, we have $d\omega_3 = 0$; and there should therefore be one integral relation, the equation 123456 = 0, which expresses the determination on the conic.

We have thus 123456 = 0 as a particular integral of

$$d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 + d\omega_5 + d\omega_6 = 0.$$

If, however, we take 3 a fixed point on the cubic, we then have the same equation as the general integral of $d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 + d\omega_5 = 0$, which expresses the determination, with the same line on the conic.

But taking also 3 a fixed point of the cubic, we have $d\omega_3 = 0$; and there should therefore be one integral relation between $d\omega_1$, $d\omega_2$, $d\omega_4$, $d\omega_5$, $d\omega_6$, that is, the same equation as the integral of $d\omega_1 + d\omega_2 + d\omega_4 + d\omega_5 + d\omega_6 = 0$.

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* The symbol explains, I think, itself; the point may be any point on the plane (so it is some constant which remains arbitrary but which disappears from the difference $-\dfrac{x_1\,x_2}{P}$; this = 0 is explained further on in the text.

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[book p. 116] **9.** determinants point, independent of the particular conic through 4, 5 and 6 used for the construction. Thus 4 and 5 being any two chosen points on the cubic, the residual point 6 is, in fact, the intersection with the cubic of the line through 4 and 5.

12. We might go on to the case where the variable curve is a cubic: this meets the fixed curve in 9 points, but they do not determine the variable cubic; for, the cubic being assumed determined by 9 free constants, and corresponding to it the equation gives therefore $\dfrac{d\omega_1}{P_1} + \ldots + \dfrac{d\omega_9}{P_9} = 0$.

Examples of the Pure Theorem—Fixed Curve a Circle. *Art. Nos. 13 to 14.*

13. The fixed curve is taken to be the circle $x^2 + y^2 = 1$, and the parametric points 1 and 2 to be the points (1, 0) and (0, 1) on this circle. The variable curve is a conic

$$\Sigma \begin{vmatrix} a, & y, & 1 \\ 1, & 0, & 1 \\ 0, & 1, & 1 \end{vmatrix} = 0,$$

have 012 denoting the determinant.

Starting from the formula

$$\Sigma\,\frac{(x,\,y)\,(x,\,y_1,\,1)\,d\omega}{012} = -\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2},$$

where the summation extends to the points 3 and 4, viz. if P_1 be a constant, = 2, (if, that is, $\sin u = a$, $2(\cos u + a) = b$ in the place of u, the points 3 and 4 are at the distances $a + b$ and $a - b$, and therefore equal); say $\theta = u + \xi$, $\theta = u - \xi$; we have $\sin u = a$, and so

$$\frac{x_3 + x_4}{2\sin u} = \frac{a+b}{2\sin u}, \quad \frac{x_3 - x_4}{2\sin u} = \frac{a-b}{2\sin u};$$

and similarly

$$\frac{y_3}{y_4} = \frac{dx_3}{dx_4}, \quad \frac{dx_3}{dy_4} = \frac{dx_4}{dy_3}.$$

The formula

$$\Sigma\,\frac{1}{y(x+y-1)} = -\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}$$

becomes

$$2\,\frac{1}{y(x+y-1)} = -\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_1, x_2, considering them as those expressed, thus considered.

[book p. 117] **14.** Writing P, Q, $R = y-p$, x_1+x-q, $-x-y+r$, also x_3, y_3, the displacements $\dfrac{dx_3}{p_3}$ and $\dfrac{dx_4}{p_4}$ are the two arbitrary points 3 and 4 on the conic which equation must become an identity in regard to the terms in dx_3, dx_4 respectively. It only remains to verify that this is so.

15. Writing P, Q, $R = y-p$, x_1+x-q, $-x-y+r$, also x_3, y_3, $-x_3-y_3+r$ respectively, $-x_4-y_4+r$ for the values of P, Q, R at the points 1, 2, 3, 4, we have

$$dP\cdot dQ\cdot dR\cdot\frac{1}{e(x-p)(x_3-p)(x_4-p)}\big[(Q'-R)\,dP+(R'-P)\,dQ+(P'-Q)\,dR\big]=0,$$

and with these values, and by aid of the relations

$$Q-R,\ R-P,\ P-Q,\ P'-R',\ R'-P',\ P'-Q' = -P+R-L,$$

the equation is found to be

$$\frac{dx_3}{p_3}+\frac{dx_4}{p_4}=\frac{(L_3(x_4-p)+L_4(x_3-p))}{(x_4-p)(p_3-p_4)\dfrac{e}{(x_3-p)}}\left(\frac{dq_1}{q_1}-\frac{dq_2}{q_2}\right);$$

viz. this will be true if only

$$L_3(x_4-p)+L_4(x_3-p)=0\,\frac{e}{(x_3-p)\,(x_4-p)};$$

that is,

$$-\,x_3\,P_3'-x_4\,P_4'+L_3(x_4-p)+L_4(x_3-p)=0.$$

But from the values of L_1, L_2, we have $-\,x_3=\frac{1}{2}L_3+L_4$, $x_3+x_4=\frac{1}{2}L_4+L_3$; and the coefficient of L_4 in $L_3L_4+L_4L_3+L_3L_4$ is the equation is thus verified.

The example would perhaps have been more instructive if the working out would have been more difficult.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. **Art. Nos. 15 to 19.**

15. Suppose $n=3$, $\delta=0$ $(p=1)$; the fixed curve is a cubic; and suppose successively $m=1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m=1$, then two points on the cubic determine the line, and consequently the remaining intersection with the cubic; hence there are two integral relations.

If $m=2$, then five points on the cubic determine the conic, and therefore the remaining intersection; hence there is one integral relation.

[book p. 118] If $m=3$, then nine points on the cubic determine the cubic; hence there are no integral relations. The result is consistent with the variable curve having $m^2=m^2$ disposable constants, $\frac{1}{2}m(m+3)$, and the variable curve passing through $mn=3m$ points on the fixed curve, we have $\frac{1}{2}m(m+3)-3m=\frac{1}{2}(m-1)(m-2)$, $=p$, that is the number of integral relations is the deficiency of the cubic.

16. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and so before suppose successively $m = 1, 2, 3, \ldots$, the variable curve a line, conic, cubic, &c.

If $m = 1$, then three points on the quartic determine the line, and therefore the remaining two intersections; the number of integral relations is $= 3$.

If $m = 2$, then five points on the quartic determine the conic, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 3$, then nine points on the quartic determine the cubic, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 4$, then fourteen points on the quartic determine the quartic, and therefore the remaining two intersections; the number of integral relations is $= 2$.

If $m = 5$, then twenty points on the quartic determine the quartic, and therefore there is one integral relation; the number of integral relations is $= 1$.

The number of intersections is $4 = mn$, $= 4m$; the variable curve has $\frac{1}{2}m(m+3)$ disposable constants; hence the number of integral relations is in general the larger of the two numbers

$$m^2 - 4m + 1 \bigl(= \tfrac{1}{2}(m-1)(m-2) + (m-2)(m-3)\bigr),$$

$= \frac{1}{2}(m-1)(m-2) + 2(m-3)$, that is, the number of integral relations is, as we presently see, $= p$.

The integral equations consist of throughout out of course independent relations.

The Variable Intersections of the Two Curves. Number of Independent Displacement Relations given by the Pure Theorem. *Art. No. 20.*

20. The number of displacement-relations given by the pure theorem is a number of constants in minor function $(x, y, z)^{n-3}$, which equated to zero represents a curve passing through the dps, this is always $= p$.

[book p. 119] **18.** The reasoning is utterly altered in the case where the fixed curve has dps; then considering the general case of a fixed curve n, δ, p,—

If $m = 1$, then $2 - \delta$ points on the fixed curve (this implies $\delta \not> 2$) determine the line, and therefore the remaining $n - 2\delta - 2$, $= n - 2 - 2$ intersections; the number of integral relations is $= n - 2$.

If $m = 2$, then $5 - 2\delta$ points on the conic determine the conic, and therefore the remaining $2n - 2\delta - 5$, $= 2n - 5$ intersections; the number of integral relations is $= 2n - 5$.

If $m = 3$, then $9 - 3\delta$ points on the cubic determine the cubic, and therefore the remaining $3n - 2\delta - 9$, $= 3n - 9$ intersections; the number of integral relations is $= 3n - 9$.

If $m = n-3$, then $\frac{1}{2}(n-3)n - \delta(n-3)$ points on the curve determine the curve, and therefore the remaining $(n-3)n - 2\delta - \frac{1}{2}(n-3)n$, $= \frac{1}{2}(n-3)n - 2\delta$, intersections; the number of integral relations is $= \frac{1}{2}(n-3)n - 2\delta$. Hence the number of integral relations in cases $m = n-3$, $= p$.

If we have, then $\frac{1}{2}n - 1)(n+4) - \delta$ points on the curve determine the curve, and therefore $1 - \delta$ intersections (it should be $= p$).

The integral equations consist of throughout out of course independent relations.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. *Art. No. 20.*

20. The number of displacement-relations given by the pure theorem is a number of constants in minor function $\phi(x, y, z)^{n-3}$, which equated to zero represents a curve passing through the dps, this is always $= p$. Consider three consecutive positions of the line, meeting the cubic in the points 1, 2, 3; $1'$, $2'$, $3'$; and $1''$, $2''$, $3''$ respectively. By $x = 0$ is satisfied an identity. Hence the number of integral displacement-relations is $= 1$, $= p$.

But it remains to verify, viz. if U be any rational and integral function $(x, y, z)^{n-3}$, then we have

$$\Sigma\,(x,\,y,\,z)^{n-3}\,d\omega = 0, \quad p,\ 0,\ \sigma,\ \tau;\ \eta\,(2x,\,1)\,\phi\,0,$$

or we might also use the symbolic form

$$\Sigma\,(\rho 1 + \sigma 2 + \tau 3)^n = 0.$$

So for $n = 4$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\, y,\, z)\, d\omega = 0, \text{ that is, } \Sigma\, x\, d\omega = 0,\ \Sigma\, y\, d\omega = 0,\ \Sigma\, z\, d\omega = 0;$$

which are 3 in number, $= p$.
[book p. 120] and then six equations give conversely $\Sigma\,(x,\, y,\, z)\, d\omega = 0$, and in particular they give $\Sigma\,(x,\, y,\, z)\, d\omega = 0$. Thus, if at pleasure p, q, r at most $= n - 3$, the variable curve has $\frac{1}{2}n(n+3) - 2\delta$ disposable constants; hence the number of integral displacement-relations is in general

$$m_p - p\,(m+1)\,; \quad \text{and the other two formulas correspond in like manner.}$$

since in this case the relations given by the theorem are all $= 0$, but each contained p relations independent. It is easily seen that the number is in general

$$m_p = \tfrac{1}{2}n(n{+}3) - 2\delta - (m{-}1)(2m{-}1), \quad \text{the number of independent displacement relations} = 3.$$

The reasoning is utterly altered in the case where the fixed curve has dps; we have generally

$$m\,p = p, \text{ or as in case } n = 4,\ \delta = 0;$$

while the number of integral relations $= p$.

Hence the difference $\frac{1}{2}n(n+3) - 2\delta - (m-1)(2m-1)$, which is equal to $\frac{1}{2}(m-1)(m-2) + \frac{1}{2}m(m-3)$, $= p$, $= p$, the number of independent displacement-relations $= 3$.

So for $n = 4$, $\delta = 0$, $m = 1$, the displacement-relations are $\Sigma\,(x,\, y,\, z)\, d\omega = 0$, that is, $\Sigma\, x\, d\omega = 0$, $\Sigma\, y\, d\omega = 0$, $\Sigma\, z\, d\omega = 0$, in all three, $= p$.

As to the dps of the Fixed Curve. *Art. No. 21.*

21. I conclude with a general remark applicable to the whole of the three chapters. There is no necessity to attend to all or indeed to any of the dps of the fixed curve. For although we may pass through all of them altogether (as e.g. if they are nodes by taking the variable curve a multiple-curve passing through them all and treating each dp as a multiple point), the number which presents itself is not the number of dps of the fixed curve, but the difference between the dps of the fixed curve and the number which it is necessary to attend to in any of the simplifications introduced therein by taking advantage thereof.

Chapter II. Proof of Abel's Theorem.

Preparation. Art. Nos. 22 and 23.

22. Starting from the equation $\phi = (a,\, b,\, c, \ldots)(x,\, y,\, z)^m = 0$ of the variable curve, we have

$$\frac{d\phi}{dx}\, dx + \frac{d\phi}{dy}\, dy + \frac{d\phi}{dz}\, dz = 0,$$

$$\frac{d\phi}{dx} = \frac{d\phi}{dx_1}\,\frac{dx_1}{dx} + \frac{d\phi}{dz_1}\,\frac{dz_1}{dx}, \quad \&c.;$$

[book p. 121] where $\delta\phi = (\delta a,\, \delta b,\, \delta c, \ldots)(x,\, y,\, z)^m$. Let τ denote an arbitrary linear function, $\omega = xdy + \ldots$; multiply the two equations by $dy + \omega$ respectively, and add: the coefficient of δa

is $-(xdy - ydx) + \omega(xdz - zdx) + \tau(xdy - ydx), = -d\tau$, say; and so on; we thus arrive at the equation

$$(ydx - zdy)\frac{d\phi}{dx} + (zdx - xdz)\frac{d\phi}{dy} + (xdy - ydx)\frac{d\phi}{dz} + \delta\phi = 0,$$

introducing $d\omega$, this becomes

$$d\omega\left[\left(\frac{df}{dy}\frac{d\phi}{dz} - \frac{df}{dz}\frac{d\phi}{dy}\right) + \omega\left(\frac{df}{dx}\frac{d\phi}{dz} - \frac{df}{dz}\frac{d\phi}{dx}\right) + \tau\left(\frac{df}{dx}\frac{d\phi}{dy} - \frac{df}{dy}\frac{d\phi}{dx}\right)\right] + \delta\phi = 0;$$

or observing that ω, τ, σ are the differential coefficients $\frac{d\sigma}{dx}, \frac{d\sigma}{dy}, \frac{d\sigma}{dz}$, the term in [] is $J(\tau, f, \phi)$, or say $J(\phi, f, \tau)$, where

$$J(\phi, f, \tau) = \frac{d(\phi, f, \tau)}{d(x, y, z)},$$

and we have hence

$$d\omega = \frac{-\delta\phi}{J(\phi, f, \tau)}.$$

23. The two theorems thus become

$$\Sigma\,(x, y, z)^{n-3}\,\frac{-\delta\phi}{J(\phi, f, \tau)} = 0;$$

$$\Sigma\,(x, y, z)^{n-3}\,\frac{-\delta\phi}{J(\phi, f, \tau)} = -\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

But further, if in the first equation we write $\tau = z$, and in the second equation we retain τ, using it then to denote the linear function 012, we have the equations

$$\Sigma\,(x, y, z)^{n-3}\,\frac{\delta\phi}{J(\phi, f)} = 0,$$

$$\Sigma\,(x, y, z)^{n-3}\,\frac{\delta\phi}{J(\phi, f, 012)} = -\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

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[book p. 122] where, in the first equation, $J(\phi, f)$ denotes the Jacobian

$$J(\phi, f) = \begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} \\ \frac{df}{dx} & \frac{df}{dy} \end{vmatrix}, \quad = \frac{d(\phi, f)}{d(x, y)}.$$

In these equations the only differential symbol is the δ affecting the coefficients $a, b, \ldots$ of ϕ, viz. the equations are, in regard to such coefficients, algebraical equations, which are in fact given by Jacobi's Fraction-theorem, as I shall explain. But for the further reduction of the affected theorem I interpose the next chapter.

The Coordinates (ρ, σ, τ). *Art. Nos. 24 to 26.*

24. The letter τ has just been used to denote the determinant 012: there is often occasion to consider three points 1, 2, 3 coordinates (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) respectively; and then writing ρ, σ, τ as the determinants 023, 031, 012 respectively, and Δ as the determinant 123, we have identically

$$\Delta x = x_1\rho + x_2\sigma + x_3\tau,$$

$$\Delta y = y_1\rho + y_2\sigma + y_3\tau,$$

$$\Delta z = z_1\rho + z_2\sigma + z_3\tau,$$

which equations, regarding therein the point 0, coordinates $(x,\ y,\ z)$, as a current point, are in fact equations for transformation from the coordinates $(x,\ y,\ z)$ to the coordinates $\rho,\ \sigma,\ \tau$, belonging to the triangle of reference 123. The points 1 and 2 have been already taken to be on the fixed curve, and it will be assumed that 3 is also a point on this curve.

25. If the function f, which equated to zero gives the equation of the fixed curve, be transformed to the new coordinates $(\rho,\ \sigma,\ \tau)$, the coefficients of the transformed function are polar functions, each divided by ∇^n, viz. the coefficient of ρ^n is $\dfrac{1}{\nabla^n}\,1^n$, which by reason that 1 is a point on the curve is $= 0$ (and similarly the coefficients of σ^n and of τ^n are each $= 0$); the coefficient of $\rho^{n-1}\sigma$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,2$; that of $\rho^{n-1}\tau$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,3$; that of $\rho^{n-2}\sigma^2$ is $\dfrac{1}{\nabla^n}\,\tfrac{1}{2}n(n-1).\,1^{n-2}\,2^2$; and so in other cases. I write this in the form

$$f = \frac{1}{\nabla^n}\,(1^n,\ \ldots\ )(\rho,\ \sigma,\ \tau)^n;$$

or we might also use the symbolic form

$$f = \frac{1}{\nabla^n}\,(\rho 1 + \sigma 2 + \tau 3)^n.$$

[book p. 123] The terms independent of τ contain, it is clear, the factor $\rho\sigma$, and separating these terms from the others, the equation may be written

$$f = \frac{1}{\nabla^n}\,\rho\sigma\,(n.\,1^{n-1}\,2,\ \ldots\ )(\rho,\ \sigma)^{n-2} + \text{terms involving } \tau.$$

26. The equations $\tau = 0$, $(\ldots\ )(\rho,\ \sigma)^{n-2} = 0$, determine the residues of the points 1, 2, and hence the major function $(x,\ y,\ z)^{n-3}$, expressed in terms of $\rho,\ \sigma,\ \tau$, and writing therein $\tau = 0$, must reduce itself to $(\ldots\ )(\rho,\ \sigma)^{n-2}$ multiplied by a constant factor which can be found to be $= \dfrac{1}{\nabla^{n-1}}$: for taking the current point at 1 we have $(\rho,\ \sigma,\ \tau) = (\Delta,\ 0,\ 0)$, and the corresponding value of the major function is thus equal, and in this case $\dfrac{1}{\nabla^{n-1}}\,n.\,1^{n-1}\,2,\ = n.\,1^{n-1}\,2$, as it ought to be. We have thus

$$(x,\ y,\ z)^{n-3} = \frac{1}{\nabla^{n-1}}\,(n.\,1^{n-1}\,2,\ \ldots\ )(\rho,\ \sigma)^{n-2} + \text{terms involving } \tau;$$

and we hence see that, for $\tau = 0$,

$$(x,\ y,\ z)^{n-3} = \frac{\Delta f}{\rho\sigma}.$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. *Art. No. 27.*

27. In the affected theorem instead of $(x,\ y,\ z)$ we introduce the new coordinates $(\rho,\ \sigma,\ \tau)$. We have

$$J(\phi,\ f,\ \tau) = \frac{d(\phi,\ f,\ \tau)}{d(\rho,\ \sigma,\ \tau)}\cdot\frac{d(\rho,\ \sigma,\ \tau)}{d(x,\ y,\ z)},$$

where the first factor is $= \dfrac{d(\phi,\ f)}{d(\rho,\ \sigma)}$, say that this is $J(\phi,\ f)$, the Jacobian in regard to $\rho,\ \sigma$: and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi,\ f,\ \tau)} = \frac{1}{\Delta!\,J(\phi,\ f)},$$

and the equation for the affected theorem becomes

$$\Sigma\,(x,\,y,\,z)^{n-3}\,\frac{\delta\phi}{J(\phi,\,f)} = -\,\Delta!\left(-\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}\right),$$

where $(x,\,y,\,z)^{n-3}$ is to be regarded as standing for its value in terms of $(\rho,\,\sigma,\,\tau)$.

Jacobi's Fraction-Theorem. *Art. Nos. 28 to 31.*

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if U be any rational and integral function $(x,\,1)^m$, then we have

$$\frac{1}{U} = \Sigma\,\frac{1}{x-x'}\,\frac{1}{J(U')},$$

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[book p. 124] or, introducing an arbitrary constant A by the equation $AU = X$, say this is

$$\frac{A}{X}\left(=\frac{1}{U}\right) = \Sigma\,\frac{1}{x-x'}\,\frac{1}{J(U')},$$

where U' is the same function (of x') that U is of x; $J(U')$, $= \dfrac{dU'}{dx'}$, is the Jacobian of U', and the summation extends to all roots x' of the equation $U' = 0$: obviously this is nothing else than the formula for the decomposition of $\dfrac{1}{U}$ into simple fractions.

29. Take now $U = (x,\,y,\,1)^m$, $V = (x,\,y,\,1)^n$ functions of x, y of the degrees m and n respectively, and assume

$$\begin{aligned} AU + BV &= X, \quad \text{a function of } (x,\,1)^{mn},\\ CU + DV &= Y, \quad \text{„ of } (y,\,1)^{mn}; \end{aligned}$$

viz. let $X = 0$ and $Y = 0$ be the equations obtained by elimination from $U = 0$ and $V = 0$ of the y and the x respectively. The forms are

$$\begin{aligned} A &= (x^{n-1},\,y,\,1)^{mn-m}, \quad & B &= (x^{m-1},\,y,\,1)^{mn-n},\\ C &= (x,\,y^{n-1},\,1)^{mn-m}, \quad & D &= (x,\,y^{m-1},\,1)^{mn-n}; \end{aligned}$$

where these equations denote the kind of them; that is A is a rational and integral function of the degree $mn-m$ in x and y jointly, but only of the degree $n-1$ in y; and so for the other equations. It follows that

$$AD - BC = (x^{mn-1},\,y^{mn-1},\,1)^{mn+mn-m-n}.$$

The theorem now is

$$\frac{AD-BC}{XY} = \Sigma\,\frac{1}{x-x'\cdot y-y'}\,\frac{1}{J(U',\,V')},$$

where U', V' are the same functions of $(x',\,y')$ that U, V are of $(x,\,y)$; $J(U',\,V')$ is the Jacobian $\dfrac{d(U',\,V')}{d(x',\,y')}$, and the summation extends to all the simultaneous roots x', y' of the equations $U = 0$, $V = 0$.

30. For the proof, observe that $AD - BC$ is a sum of terms of the form $\alpha x^\alpha y^\beta$ where α and β are each of them at most $= mn-1$; hence X being of the degree mn we have $\dfrac{x^\alpha}{X}$ a sum of fractions $\dfrac{L_y}{x-x'}$, where x' is any root of $X=0$; and similarly $\dfrac{y^\beta}{Y}$ a sum of fractions $\dfrac{M_y}{y-y'}$, where y' is any root of $Y=0$; multiplying the two expressions and taking into account the several terms $\lambda x^\alpha y^\beta$ of $AD-BC$ we obtain a formula

$$\frac{AD-BC}{XY}=\Sigma\,\frac{K}{x-x'\cdot y-y'},$$

where the summation extends to all the combinations of the mn values of x' with the mn values of y'. But such a formula existing, the coefficients K may be determined.
[book p. 125] in the usual manner, viz. multiplying by XY and then writing $x=x'$, $y=y'$, there is on the right-hand only one term which does not vanish, and we find

$$(AD-BC)_{x'y'}=K\left(\frac{X}{x-x'}\right)_{x'}\left(\frac{Y}{y-y'}\right)_{y'},\ =\ K\left(\frac{dX}{dx}\,\frac{dY}{dy}\right)_{x'y'};$$

where the factor which multiplies K does not vanish.

We distinguish the cases where $(x',\,y')$ are corresponding or non-corresponding roots of $X=0$, $Y=0$, viz. corresponding roots are those for which $U=0$, $V=0$, but for non-corresponding roots these equations do not hold good; there are obviously mn pairs of corresponding roots.

In the latter case $(AD-BC)\,U=DX-BY$, $(AD-BC)\,V=-CX+AY$, and since for these values $X=0$, $Y=0$, the foregoing equation for K gives then $K=0$.

31. The formula thus is

$$\frac{AD-BC}{XY}=\Sigma\,\frac{K}{x-x'\cdot y-y'},$$

where the summation extends to corresponding roots only; or, for which we have $U=0$, $V=0$.
[book p. 126] determinate point, independent of the particular conic through 4, 5 and 6 used for the construction. Thus 4 and 5 being any two chosen points of the cubic, the residual point 6 is, in fact, the intersection with the cubic of the line through 4 and 5.

12. We might go on to the case where the variable curve is a cubic: this meets the fixed curve in 9 points, but they do not determine the variable cubic; for, the cubic being assumed determined by 9 free constants, and corresponding to it the equation gives therefore $\dfrac{d\omega_1}{P_1}+\ldots+\dfrac{d\omega_9}{P_9}=0$.

The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively, this is

$$\begin{vmatrix} x_1, & y_1, & 1\\ x_2, & y_2, & 1\\ x_3, & y_3, & 1\end{vmatrix}=0,$$

or, in the notation already made use of for such a determinant, $123=0$.

32. Reverting to the Cartesian form, we have

$$\frac{xy\,(AD-BC)}{XY}=\Sigma\,\frac{1}{J(U',\,V')}\,\frac{1}{\left(1+\dfrac{x'}{x}+\ldots\right)\left(1+\dfrac{y'}{y}+\ldots\right)};$$

$=\Sigma\,\dfrac{1}{J(U',\,V')}\left(1+M\left(\dfrac{x'}{x},\,\dfrac{y'}{y}\right)+M_2\left(\dfrac{x'}{x},\,\dfrac{y'}{y}\right)+\ldots\right)$; where H_α is the homogeneous sum of the order α of $\dfrac{x'}{x}$, $\dfrac{y'}{y}$, and so $H_1\,(v,\,w)=v+w$, $H_2\,(v,\,w)=v^2+vw+w^2$, &c.

[book p. 127] The left-hand side is

$$(AD - BC)\left(\frac{1}{x^{mn-1}} + \frac{dx}{x^{mn}} + \dots\right)\left(\frac{1}{y^{mn-1}} + \frac{dy}{y^{mn}} + \dots\right),$$

and is $AD - BC$ the terms of the highest order in $(x,\ y)$, say $(AD - BC)_0$, are

$$(AD - BC)_0\,(xy)^{mn-1-1}, = \Sigma\,\frac{1}{J(U',\ V')}\,H^{\alpha+\beta}_{\alpha,\beta\dots}\,(x^{mn-2-\alpha},\ y^{mn-2-\beta}).$$

There is thus on the left-hand no term which is in $(x,\ y)$ of a higher degree than $-(m+n-2)$; hence we the right-hand every term of a higher degree, that is, in $(\alpha,\ \beta)$ a less degree than this, must vanish. Hence the number of these is

$$\theta = \Sigma\,\frac{xy^{\alpha}}{J(U',\ V')},\quad \text{so long as } \alpha+\beta \text{ is } < n-3,$$

where $(x',\ y')$, $J^{n,m-1}$ is the arbitrary function of the degree $n-3$.

But, from the foregoing expression for $(AD-BC)$, it appears that $(AD-BC)_0$ contains the term $px^{mn-2}y^{mn-2}$, and it hence appears that p is the constant term of the quotient $(AD-BC)_0$ divided by $x^{mn-2}y^{mn-2}$, or as this may be written

$$p = \text{const. of } (AD - BC)_0\,(xy)^{2-mn};$$

comparing the two values of p, we have

$$\text{const. of } \frac{(AD - BC)_0\,x^{\alpha}y^{\beta}}{J(U',\ V')}\,(x'y')^{2-m} = \Sigma\,\frac{x^{\alpha}y^{\beta}}{J(U',\ V')}\,(p+q = n-3);$$

and we hence derive

$$\text{const. of } \frac{(AD - BC)_0\,x^{\alpha}y^{\beta}}{J(U',\ V')}\,(x'y')^{2-m} = \Sigma\,\frac{x'^{\alpha}y'^{\beta}}{J(U',\ V')}.$$

[book p. 128] where $(x', y', 1)^{m+n-2}$ is the general non-homogeneous function of the degree $m+n-2$, and $(x, y, 0)^{m+n-2}$ is obtained from it by attending only to the terms of the highest degree $m+n-2$, and therein substituting x, y for x', y'.

35. We may, it is clear, in the equations for the $(m+n-3)$ and for the $(m+n-2)$ theorems respectively, omit the accents on the right-hand sides; doing this, and moreover in each equation transposing the two sides, the two special theorems are

$$\Sigma\,\frac{(x, y, 1)^{m+n-3}}{J(U, V)} = 0,\qquad (m+n-3) \text{ theorem},$$

$$\Sigma\,\frac{(x, y, 1)^{m+n-2}}{J(U, V)} = \text{const. of } \frac{(AD - BC)_0(x, y, 0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}},\qquad (m+n-2) \text{ theorem}.$$

Homogeneous Form of the Special Theorems. *Art. No. 36.*

36. Writing $\frac{x}{z}$, $\frac{y}{z}$ for $x,\ y$, and introducing as before the arbitrary linear function $r = ax + by + cz$, we at once obtain, $U,\ V$ being now homogeneous functions $(x, y, z)^m$ and $(x, y, z)^n$ respectively, and the $A,\ B,\ C,\ D$ being also homogeneous functions accordingly,

$$\Sigma\,\frac{z(x, y, z)^{m+n-3}}{J(U, V)} = 0,\qquad (m+n-3) \text{ theorem},$$

$$\Sigma \frac{r(x,y,z)^{m+n-2}}{z\,J(U,V,r)} = \text{const. of } \frac{(AD-BC)_0(x,y,0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}}, \qquad (m+n-2) \text{ theorem},$$

where the suffix 0 denotes that we are in $AD - BC$ to write $z = 0$.

[book p. 129] The effect of dps of the Curves $U = 0$, $V = 0$. Art. Nos. 37 and 38.

37. We must, in regard to the foregoing special theorems, consider the effect of any dps of the curves $U = 0$, $V = 0$.

Suppose one of the curves, say V, has a dp, but that the other curve U does not pass through it; the dp is not an intersection of U, V, and the theorems are in nowise affected.

If U passes through the dp, then the dp counts twice (say in our case) among the intersections of the curves; the dp is in fact the limit of two close intersections, and the theorems thus becomes negatory.

It, however, the curve ϕ, ϕ^{mn-2} remains finite at (x', y') the dp, then a curve passing through it and a fixed simple intersection of U, V will count as two intersections, and the theorem becomes

$$\Sigma \frac{\phi^{m'n}}{J(U', V')} = 0,$$

where the suffix 0 denotes that we are in $AD - BC$ to retain only the terms of the order

$$2mn - p - q, \text{ viz. in } (\alpha, \beta) \text{ of less than } -(m+n-4)$$

question, which are to be disregarded. And the like in regard to the other theorems

$$\Sigma \frac{(x, y)^{n-3}}{J(U', V')} = 0, \quad (m = n-3),$$

$$\Sigma \frac{(x, y)^{n-3}\,\delta\phi}{J(U', V')} = -\,\text{const.} \frac{(AD-BC)_0\, x^\alpha y^\beta\, \delta\phi}{(xy)^{mn-2}}, \quad (m = n-2) \text{ theorem};$$

where the suffix 0 denotes that we are in $AD - BC$ to retain only the terms of the order

$$2mn - 4 - p - q.$$

Homogeneous Form of the Special Theorem. *Art. No. 36.*

36. Writing $\frac{x}{z}$, $\frac{y}{z}$ for x, y, and introducing as before the arbitrary linear function $\sigma = \alpha x + \beta y + \gamma z$, and so σ, τ, respectively, we have J, V, V, V becoming J, U, V homogeneous functions accordingly.

If in the last formula we change throughout $\frac{x}{z}$, $\frac{y}{z}$ to x, y, then we have, in the place of $\frac{x'}{z'}$, $\frac{y'}{z'}$ respectively, &c., $\delta\phi$, &c., are functions of the particular function $\frac{x'}{z'}$, $\frac{y'}{z'}$, where x' is any root of $X = 0$, &c.

We have in fact $J(U, V) = z'^{mn-2} J(\phi, f, \tau)$, and similarly for $\frac{1}{xy} \cdot \frac{1}{x'y'}$ in the form $\frac{1}{xy} \cdot \frac{1}{x'y'}$ passing into $\frac{1}{\rho\sigma} \cdot \frac{\Delta f}{\rho\sigma}$, by which means we have, on the right-hand side, Δf in the place of $J(\phi, f)$.

The Pure Theorem—Completion of the Proof. *Art. No. 39.*

39. The theorem was reduced to

$$\Sigma \frac{(x, y, z)^{n-3}\,\delta\phi}{J(\phi, f)} = 0,$$

which is therefore the equation to be proved.

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[book p. 130] The $(m = n-3)$ theorem, writing therein ϕ, f in place of U, V respectively (the degrees being as before m and n), is

$$\Sigma \frac{(x,\, y)^{n-3}\, \delta\phi}{J(\phi,\, f)} = 0.$$

Here $(x,\, y)^{n-3}$ is an arbitrary function of the degree $n-3$, and this may therefore be put, $= \Sigma\, \delta(\alpha x + \beta y + \gamma)^{n-3}$, where the α's, β's, γ's are arbitrary, and the variation is in regard only to the δ's; or, since $\phi = 0$ is the equation of an arbitrary curve, it is rather a function of the $(x,\, y)$ coordinates of an arbitrary point on it; the theorem is thus the equation in the same form on the homogeneous variables z, viz. in the homogeneous form

$$\Sigma \frac{(x,\, y)^{n-3}\, \delta\phi}{J(\phi,\, f)} = 0.$$

We may write this in the form of an arbitrary function of the degree $n-3$ relating to the homogeneous variables $(x,\, y,\, z)$; transformations to the corresponding form of ϕ, the equation of the variable curve, we have the theorem in its homogeneous form of the highest degree.

The Affected Theorem—Completion of the Proof. **Art. Nos. 40 and 41.**

40. The theorem was reduced to

$$\Sigma\, (x,\, y,\, z)^{n-3}\, \frac{\delta\phi}{J(\phi,\, f)} = -\,\Delta!\, \left(-\,\frac{\delta\phi_1}{\phi_1} \cdot \frac{\delta\phi_2}{\phi_2}\right),$$

which is therefore the equation to be proved.

The $(m = n-2)$ theorem, writing therein ϕ, f in place of U, V respectively, becomes

$$\Sigma \frac{(x,\, y)^{n-3}\, \delta\phi}{J(\phi,\, f)} = -\,\text{const.}\, \frac{(AD-BC)_0\, x^{\alpha} y^{\beta}\, \delta\phi}{(xy)^{mn-2}},$$

where, on the right-hand side $(x,\, y)^{n-3}$ is considered as a function of p, q, σ, and we are to put therein $\sigma = 0$; it has been seen (No. 26) that the value is $\dfrac{\Delta f}{\rho\sigma}$, where f is what f, considered as a function of ρ, σ, τ, becomes on writing therein $\tau = 0$; the right-hand side thus becomes

$$= -\,\text{const.}\, \frac{(AD-BC)_0\, \Delta f\, \delta\phi}{\sigma!\, (pq)^{mn-2}};$$

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Contents

825. A Memoir on the Abelian and Theta Functions (continued)

Chapter II. Proof of Abel's Theorem (concluded).

[book p. 131] **41.** But for $\tau = 0$, we have

$$A_x\phi_1 + B_x f_x = a\rho^{mn},$$

$$C_x\phi_1 + D_x f_x = \beta\sigma^{mn},$$

and hence

$$(AD - BC)_x f_x = A_x\beta\sigma^{mn} - C_x a\rho^{mn},$$

and the right-hand side thus becomes, say

$$-\Delta!\,\text{const. of}\ \left(-\frac{A_x}{a\rho^{mn}} + \frac{C_x}{\beta\sigma^{mn}}\right)\delta\phi_x.$$

But in calculating the constant of $\dfrac{A_x}{a\rho^{mn}}\,\delta\phi_x$, we may suppose not only $\tau = 0$, but also $\sigma = 0$: we then have $\phi_x = (x,\, y,\, z)^m, = \left(\dfrac{\rho}{\Delta}\right)^m (x_1,\, y_1,\, z_1)^m, = \left(\dfrac{\rho}{\Delta}\right)^m \phi_1$, and hence also

$$\delta\phi_x = \left(\frac{\rho}{\Delta}\right)^m \delta\phi_1.$$

Similarly, in calculating the constant of $\dfrac{B_x}{\beta\sigma^{mn}}\,\delta\phi_x$, we may suppose not only $\tau = 0$, but also $\rho = 0$: we then have $\phi_x = (x,\, y,\, z)^m, = \left(\dfrac{\sigma}{\Delta}\right)^m (x_2,\, y_2,\, z_2)^m, = \left(\dfrac{\sigma}{\Delta}\right)^m \phi_2$, and hence

$$\delta\phi_x = \left(\frac{\sigma}{\Delta}\right)^m \delta\phi_2.$$

Moreover, in the equations

$$A_x\phi_1 + B_x f_x = a\rho^{mn},$$
$$C_x\phi_1 + D_x f_x = \beta\sigma^{mn},$$

writing in the first equation $\sigma = 0$, we find $A_x\left(\dfrac{\rho}{\Delta}\right)^m \phi_1 = a\rho^{mn}$; that is, $\dfrac{A_x}{a\rho^{mn}} = \left(\dfrac{\Delta}{\rho}\right)^m \dfrac{1}{\phi_1}$; and similarly writing in the second equation $\rho = 0$, we find $C_x\left(\dfrac{\sigma}{\Delta}\right)^m \phi_2 = \beta\sigma^{mn}$, that is, $\dfrac{C_x}{\beta\sigma^{mn}} = \left(\dfrac{\Delta}{\sigma}\right)^m \dfrac{1}{\phi_2}$; and the expression thus becomes

$$= -\Delta!\left(-\frac{\delta\phi_1}{\phi_1} + \frac{\delta\phi_2}{\phi_2}\right),$$

giving the equation which was to be proved. This completes the proof of the affected theorem

$$\Sigma\,\frac{(x,\, y,\, z)_2^{n-2}\,d\omega}{012} = -\frac{\delta\phi_1}{\phi_1} + \frac{\delta\phi_2}{\phi_2}.$$

Chapter III. The Major Function $(x,\, y,\, z)_2^{n-3}$.

Analytical Expression of the Function. Art. Nos. 42 to 49.

[book p. 132] **42.** The function has been defined by the conditions that the curve $(x,\, y,\, z)_2^{n-3} = 0$ shall pass through the dps, and also through the $n-2$ residues of the parametric points 1, 2; and moreover, that on writing therein $(x_1,\, y_1,\, z_1)$ for $(x,\, y,\, z)$, the function shall become $= n\,.\,1^{n-2}\,2$. Obviously the function is not completely determined: calling it Ω (or when required Ω_{12}), then if Ω' be any particular form of it, the general form is $\Omega = \Omega' + (x,\, y,\, z)^{n-3}\,012$, where $(x,\, y,\, z)^{n-3} = 0$ is the general minor function, viz. $(x,\, y,\, z)^{n-3} = 0$ is a curve passing through the dps: the major function thus contains $\frac{1}{2}(n-1)(n-2) - \delta, = p$, arbitrary constants.

Agreeing with the definition, we have the before-mentioned equation

$$\Omega = \frac{1}{\Delta^{n-1}}(n\,.\,1^{n-1}\,2,\, \ldots\Big)\Big(\rho,\, \sigma\Big)^{n-2} + \text{terms involving } \tau,$$

viz. from this expression for Ω it appears that the curve $\Omega = 0$ meets the line through 1, 2 in the $n-2$ residues of these points, and moreover, for $(x,\, y,\, z) = (x_1,\, y_1,\, z_1)$ and therefore $(\rho,\, \sigma,\, \tau) = (\Delta,\, 0,\, 0)$, the value of Ω is $= n\,.\,1^{n-2}\,2$.

43. We can without difficulty write down an equation determining Ω' as a function $(x,\, y,\, z)^{n-3}$, which, on putting therein $\tau = 0$, becomes equal to the foregoing expression $\dfrac{1}{\Delta^{n-1}}(n\,.\,1^{n-1}\,2,\, \ldots\Big)\Big(\rho,\, \sigma\Big)^{n-2}$, and which is moreover such that the curve $\Omega' = 0$ passes through the dps; which being so, we have as before, $\Omega = \Omega' + (x,\, y,\, z)^{n-3}\,012$, for the general value of Ω.

To fix the ideas, consider the particular case $n = 4$, the fixed curve a quartic: Ω', on putting therein $\tau = 0$, should become

$$= \tfrac{1}{\Delta}(4\,.\,1^3\,2,\; 6\,.\,1^2\,2^2,\; 4\,.\,1\,2^3\Big)\Big(\rho,\, \sigma\Big)^2;$$

and it is be shown that this will be the case if we determine Ω', a quadric function of $(x,\, y,\, z)$, by the equation

$$\left|\begin{array}{cc} (x,\, y,\, z)^2 & ,\quad \Omega' \\ 1\,(x_1,\, y_1,\, z_1)^2 & ,\quad 4\,.\,1^3\,2 \\ 2\,(x_1,\, y_1,\, z_1 \;\xi\; x_2,\, y_2,\, z_2),\; 6\,.\,1^2\,2^2 & \\ 1\,(x_2,\, y_2,\, z_2)^2 & ,\quad 4\,.\,1\,2^3 \\ a,\, b,\, c,\, f,\, g,\, h, & 0 \end{array}\right| = 0,$$

where the left-hand side is a determinant of seven lines and columns, the top line being x^2, y^2, z^2, $2yz$, $2zx$, $2xy$, Ω', and similarly for the second line; the third line is

[book p. 133] $2x_1x_2$, $2y_1y_2$, $2z_1z_2$, $2(y_1z_2 + y_2z_1)$, $2(z_1x_2 + z_2x_1)$, $2(x_1y_2 + x_2y_1)$, $6\,.\,1^2\,2^2$; and in each of the last three lines we have six arbitrary constants followed by a 0. The equation is of the form $\Omega + M\Omega' = 0$, where Ω is a quadric function $(x,\, y,\, z)^2$, and M is a constant factor.

44. If the quartic curve has a dp, suppose at the point a, coordinates $(x_a,\, y_a,\, z_a)$, then in order that the curve $\Omega' = 0$ may pass through the dp, we must for one of the last three lines substitute $(x_a,\, y_a,\, z_a)^2$, 0; and so for any other dp or dps of the quartic curve. And the conditions as to the dp or dps (if any) being satisfied in this manner, we may if we please, taking $(x_\beta,\, y_\beta,\, z_\beta)$ as the coordinates of an arbitrary point β (not of necessity on the fixed curve), write any line not already so expressed, of the last three lines, in the form $(x_\beta,\, y_\beta,\, z_\beta)^2$, 0; the effect being to make the curve $\Omega' = 0$ pass through the arbitrary point β.

45. To show that the equation on putting therein $\tau = 0$ does in fact give the required value, $\Omega' = \dfrac{1}{\Delta}(4\,.\,1^3\,2,\ 6\,.\,1^2\,2^2,\ 4\,.\,1\,2^3\,\Big)\Big(\rho,\ \sigma\Big)^2 = \Phi$ suppose, it is to be observed that, effecting a linear substitution upon the first six columns, the equation may be written

$$\left|\begin{array}{cc} (\rho,\ \sigma,\ \tau)^2 & ,\ \ \Omega' \\ 1\,(\rho_1,\ \sigma_1,\ \tau_1)^2 & ,\ \ 4\,.\,1^3\,2 \\ 2\,(\rho_1,\ \sigma_1,\ \tau_1\ \xi\ \rho_2,\ \sigma_2,\ \tau_2),\ 6\,.\,1^2\,2^2 & \\ 1\,(\rho_2,\ \sigma_2,\ \tau_2)^2 & ,\ \ 4\,.\,1\,2^3 \\ a',\ b',\ c',\ f',\ g',\ h', & 0 \end{array}\right| = 0,$$

where $(\rho_1,\ \sigma_1,\ \tau_1)$, $(\rho_2,\ \sigma_2,\ \tau_2)$ are what $(\rho,\ \sigma)$ become on writing therein for $(x,\ y,\ z)$ the values $(x_1,\ y_1,\ z_1)$ and $(x_2,\ y_2,\ z_2)$ respectively, viz. we have $(\rho_1,\ \sigma_1,\ \tau_1) = (\Delta,\ 0,\ 0)$; $(\rho_2,\ \sigma_2,\ \tau_2) = (0,\ \Delta,\ 0)$; the equation thus is

$$\left|\begin{array}{ccccccc} \rho^2, & \sigma^2, & \tau^2, & 2\sigma\tau, & 2\tau\rho, & 2\rho\sigma, & \Omega' \\ \Delta^2, & 0, & 0, & 0, & 0, & 0, & 4\,.\,1^3\,2 \\ 0, & 0, & 0, & 0, & 0, & 2\Delta^2, & 6\,.\,1^2\,2^2 \\ 0, & \Delta^2, & 0, & 0, & 0, & 0, & 4\,.\,1\,2^3 \\ a', & b', & c', & f', & g', & h', & 0 \end{array}\right| = 0,$$

and then by another linear substitution upon the columns, the last column can be changed into $\Omega' - \Phi$, 0, 0, 0, 0, 0, whence writing $\tau = 0$, the equation becomes

$$\left|\begin{array}{ccccccc} \rho^2, & \sigma^2, & 0, & 0, & 0, & 2\rho\sigma, & \Omega' - \Phi \\ \Delta^2, & 0, & 0, & 0, & 0, & 0, & 0 \\ 0, & 0, & 0, & 0, & 0, & 2\Delta^2, & 0 \\ 0, & \Delta^2, & 0, & 0, & 0, & 0, & 0 \\ a', & b', & c', & f', & g', & h', & 0 \end{array}\right| = 0,$$

[book p. 134] or, omitting a constant factor, it is

$$\left|\begin{array}{cccc} \rho^2, & \sigma^2, & 2\rho\sigma, & \Omega' - \Phi \\ \Delta^2, & 0, & 0, & 0 \\ 0, & 0, & 2\Delta^2, & 0 \\ 0, & \Delta^2, & 0, & 0 \end{array}\right| = 0,$$

that is, $\Omega' - \Phi = 0$, or $\Omega' = \Phi$, $= \dfrac{1}{\Delta}(4\,.\,1^3\,2,\ 6\,.\,1^2\,2^2,\ 4\,.\,1\,2^3\,\Big)\Big(\rho,\ \sigma\Big)^2$, the required value.

46. Considering the equation for Ω' as expressed in the before-mentioned form $\Omega + M\Omega' = 0$, the value of the constant factor M is

$$M = \left|\begin{array}{c} (x_1,\ y_1,\ z_1)^2 \\ (x_1,\ y_1,\ z_1\ \xi\ x_2,\ y_2,\ z_2) \\ (x_2,\ y_2,\ z_2)^2 \\ a,\ b,\ c,\ f,\ g,\ h, \end{array}\right|,$$

or if, instead of each line such as a, b, c, f, g, h, we have a line $(x_a,\ y_a,\ z_a)^2$, then we have

$$M = \left|\begin{array}{c} (x_1,\ y_1,\ z_1)^2 \\ (x_1,\ y_1,\ z_1\ \xi\ x_2,\ y_2,\ z_2) \\ (x_2,\ y_2,\ z_2)^2 \\ (x_3,\ y_3,\ z_3)^2 \\ (x_4,\ y_4,\ z_4)^2 \end{array}\right|,$$

a value which is

$$= \begin{vmatrix} x_1, & y_1, & z_1 \\ x_3, & y_3, & z_3 \\ x_4, & y_4, & z_4 \end{vmatrix} \cdot \begin{vmatrix} x_2, & y_2, & z_2 \\ x_3, & y_3, & z_3 \\ x_4, & y_4, & z_4 \end{vmatrix} \cdot \begin{vmatrix} x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \\ x_3, & y_3, & z_3 \end{vmatrix} \cdot \begin{vmatrix} x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \\ x_4, & y_4, & z_4 \end{vmatrix},$$

or say this is $= 12s\,.\,12\beta\,.\,12\gamma\,.\,a\beta\gamma$.

47. It is obvious that the foregoing process is applicable to the general case of the fixed curve of the order n with δ dps, and gives always Ω', by an equation of the foregoing form $\Omega + M\Omega' = 0$, where Ω is a function $(x,\, y,\, z)^{n-3}$ of the coordinates, and M is a constant factor. Supposing that in the determinant for Ω', each of the lower lines is written in the form $(x_a,\, y_a,\, z_a)^{n-3}$, 0 instead of a, b, c, &c., will indicate this function to be $= 0$; and for the points α, are on a curve of the order $n-3$.

[book p. 135] **48.** Preceding the case $n = 4$, above considered, we have, of course, the case $n = 3$, $\delta = 0$, the fixed curve a cubic; the equation for Ω' is here

$$\begin{vmatrix} x, & y, & z, & \Omega' \\ x_1, & y_1, & z_1, & 3\,.\,1^2\,2 \\ x_2, & y_2, & z_2, & 3\,.\,1\,2^2 \\ x_3, & y_3, & z_3, & \end{vmatrix} = 0,$$

giving

$$\tfrac{1}{3}\Omega' = \frac{1^2\,.\,023 + 1^2\,.\,031}{12s},$$

or if we write herein 3 for a, this is

$$\tfrac{1}{3}\Omega' = \frac{1^2\,.\,023 + 1^2\,.\,031}{123},$$

and we have hence the general form

$$\tfrac{1}{3}\Omega = \frac{1^2\,.\,023 + 1^2\,.\,031}{123} + K\,.\,012,$$

where K is an arbitrary constant.

49. There is, however, a more simple particular solution $\frac{1}{3}\Omega =$ polar function 012 ($f = x^3 + y^3 + z^3$; then $012 = xx_1x_2 + yy_1y_2 + zz_1z_2$), to avoid a confusion of notation, we may write $= \overline{012}$. We at once verify this; for, expressing the coordinates $(x,\, y,\, z)$ in terms of $(\rho,\, \sigma,\, \tau)$, we have $\frac{1}{3}\Omega = \overline{012},\ = \dfrac{1}{\Delta}(1^2\,.\,\rho + 1^2\,.\,\sigma + \overline{125}\,.\,\tau)$, which, for $\tau = 0$ becomes $= \dfrac{1}{\Delta}\big[1^2\,.\,\rho + 1^2\,.\,\sigma\big]$.

We must, of course, have an identity of the form

$$\overline{012} = \frac{1^2\,.\,023 + 1^2\,.\,031}{123} + K\,.\,012,$$

and to find K, writing here $0 = 3$, we have $K = \dfrac{\overline{123}}{123^2}$, or we have the identity

$$123\,\overline{012} - \overline{123}\,012 = 1^2\,.\,023 + 1^2\,.\,031.$$

Single Letter Notation for the Polar Functions of the Cubic. Art. Nos. 50 and 51.

50. The notation of single letters for the polar functions is not much required in the case of the cubic, but, in the next following case of the quartic it can hardly be dispensed with, and I therefore establish it in the case of the cubic: viz. I write

$$2\bar{3},\ 3\bar{1},\ 1\bar{2} = f,\ g,\ h;\quad 23\bar{1},\ 31\bar{2},\ 12\bar{3} = i,\ j,\ k;\quad \overline{123} = l,$$

or, what is the same thing, the expression for the cubic function f in terms of ρ, σ, τ is

$$\Delta^2 . f = 3l\rho^2\sigma + 3jp^2\tau + 6l\rho\sigma\tau + 6lp\sigma\tau + 3g\sigma^2\tau + 3i\sigma\tau^2 + 3i\sigma\tau^2;$$

[book p. 136] an equation which, writing 0^3 instead of f, may also be written

$$\Delta !\, 0^3 = (3l,\ 3j,\ 3i,\ 3k,\ 6l,\ 6l,\ 3i,\ 3i,\ 3j,\ 3i\Big)\Big(\rho^3,\ \sigma^3,\ \tau^3,\ \rho^2\sigma,\ \rho^2\tau,\ \rho\sigma^2,\ \rho\tau^2,\ \sigma^2\tau,\ \sigma\tau^2\Big),$$

and I join to it the series of equations

$$\begin{aligned}
\Delta ! \,.\, 0^2\, 1 &= (0,\ 2k,\ 2j,\ k,\ 2l,\ 2l,\ 2l,\ g\Big)\,(\rho\sigma,\ \rho\tau,\ \sigma\tau, \ldots),\\
,,\ 0^2\, 2 &= (k,\ 2l,\ 2i,\ 2l,\ 2k,\ l\Big)\,,\\
,,\ 0^2\, 3 &= (j,\ 2l,\ 2g,\ l,\ 2h,\ 2l\Big)\,,\\
\Delta \,.\, 0\, 1^2 &= (0,\ h,\ l\xi\, \rho,\ \sigma,\ \tau),\\
\Delta\, \overline{012} &= (h,\ l,\ l\xi\ \ldots),\\
,,\ 0^2 &= (k,\ 0,\ f\xi\ \ \ldots),\\
,,\ \overline{025} &= (l,\ f,\ l\xi\ \ \ldots),\\
,,\ 0\, 3^2 &= (j,\ i,\ 0\xi\ \ \ldots),
\end{aligned}$$

51. In particular, we have $\Delta\, \overline{012} = h\rho + l\sigma + l\tau$, and the above-mentioned identity $123\, \overline{012} - \overline{123}\, 012 = 1^2 \,.\, 023 + 1^2 \,.\, 031$ is simply $h\rho + l\sigma + l\tau - l\bar{l} = h\rho + l\sigma$.

Single Letter Notation for the Polar Functions of the Quartic. Art. No. 52.

52. I write here

$$2\bar{3},\ 3\bar{1},\ 1\bar{2} = f,\ g,\ h; \quad 23\bar{1},\ 31\bar{2},\ 12\bar{3} = i,\ j,\ k;$$

$$1\bar{2}3,\ 1\bar{2}3,\ 1\bar{2}3 = l,\ m,\ n; \quad 23,\ 31,\ 1\bar{2} = p,\ q,\ r;$$

so that the expression for the quartic function f in terms of ρ, σ, τ is

$$\begin{aligned}
\Delta ! \,.\, f = {} & 4l\rho^3\sigma + 4jp^3\tau + 12lp^2\sigma^2 + 6gp^2\sigma^2\\
& + 4lp\sigma + 12mp\sigma\tau + 12n\sigma\tau + 6lp^2\sigma + 6\, n\rho\tau^2 + 4\, j\sigma\tau^3,
\end{aligned}$$

which, putting 0^4 for f, may also be written

$$\Delta \,.\, 0^4 = (4k,\ 4j,\ 12l,\ 6g,\ 4l,\ 12m,\ 12n,\ 4j,\ 6l,\ 6n,\ 4j\Big)$$

$$(\rho^4,\ \rho^3\sigma,\ \rho^3\tau,\ \rho^2\sigma^2,\ \rho^2\tau^2,\ \rho\sigma^2\tau,\ \rho\sigma\tau,\ \rho\sigma\tau^2,\ \rho^2\tau,\ \sigma^4,\ \sigma^3\tau,\ \sigma^2\tau^2,\ \sigma\tau^3),$$

and I join to it the series of equations

$$\begin{aligned}
\Delta \,.\, 0^3\, 1 &= (0,\ 3k,\ 3j,\ 3m,\ 3l,\ 3lm,\ 3lm,\ 3m,\ 3lm,\ 3l,\ l\Big)\,(\rho^3,\ \rho^2\sigma,\ \rho^2\tau,\ \rho\sigma^2,\ \rho\sigma\tau,\ \rho\tau^2,\ \sigma^3,\ \sigma^2\tau,\ \sigma\tau^2,\ \tau^3),\\
,,\ 0^3\, 2 &= (k,\ 3l,\ 3m,\ 6m,\ 3m,\ 3l,\ 3l\Big)\,,\\
,,\ 0^3\, 3 &= (j,\ 3l,\ 3g,\ 3p,\ 3p,\ 3l\Big)\,,
\end{aligned}$$

[book p. 137]

$$\begin{aligned}
\Delta\,.\,0^2\,1^2 &= (0,\ 2k,\ 2k,\ 2l,\ q\xi\, p,\ p,\ p,\ p), \quad ,,\ ,, \\
,,\ 0^2\,1\,2 &= (k,\ 2k,\ 3l,\ 3lm,\ n\xi\ \ ,\ ,\ ,\ ,,\ ,\ ,), \\
,,\ 0^2\,1\,3 &= (j,\ 2l,\ 2k,\ l,\ k\xi\, p,\ p,\ p,\ p), \quad ,,\ ,, \\
,,\ 0^2\,2^2 &= (k,\ 2m,\ 2n,\ l,\ p\xi\ \ ,\ ,\ ,\ ,,\ ,\ ,), \\
,,\ 0^2\,2\,3 &= (l,\ 2m,\ 2m,\ f,\ l\xi\, p,\ p,\ p,\ p), \quad ,,\ ,, \\
,,\ 0^2\,3^2 &= (q,\ 2m,\ 2p,\ p,\ 2l,\ 0\xi\ \ ,\ ,\ ,\ ,,\ ,\ ,), \\
\Delta\,.\,0\,1^3 &= (0,\ h,\ l,\ l\xi\, \rho,\ \sigma,\ \tau), \\
,,\ 0\,1^2\,2 &= (h,\ l,\ l\xi\ \ldots), \\
,,\ 0\,1\,2^2 &= (l,\ m,\ n\xi\ \ldots), \\
,,\ 0\,1^2\,3 &= (j,\ l,\ m\xi\ \ldots), \\
,,\ 0\,1\,2\,3 &= (l,\ m,\ n\xi\ \ldots), \\
,,\ 0\,1\,3^2 &= (q,\ n,\ p\xi\ \ldots), \\
,,\ 0\,2^3 &= (k,\ l,\ l\xi\ \ldots), \\
,,\ 0\,2^2\,3 &= (l,\ m,\ n\xi\ \ldots), \\
,,\ 0\,2\,3^2 &= (m,\ f,\ l\xi\ \ldots), \\
,,\ 0\,3^3 &= (q,\ k,\ 0\xi\ \ldots),
\end{aligned}$$

which will be convenient in the sequel.

Major Function—The Fixed Curve a Cubic. Art. No. 53.

53. It has been already seen that a simple particular form is $\frac{1}{3}\Omega = \overline{012}$: and that the general form is $\Omega = \Omega' + K\,.\,012$.

Major Function—The Fixed Curve a Quartic. Art. No. 54.

54. It is to be shown that a particular form is

$$\tfrac{1}{2}\Omega' = \frac{-\,0\,1^3\,.\,02^2 + 01^2\,2\,.\,012 + 01\,2\,.\,1^2\,2}{1^2\,2}.$$

In fact, by the foregoing values of $\Delta\,.\,01^3$, &c., the numerator of this expression, multiplied by $\Delta!$, is =

$$\begin{aligned}
&-(ha + jr)(la + fr) \\
&+ (kp + ra + lr)(c\rho + l\sigma + nr) \\
&+ r(h\rho^2 + 2r\rho\sigma + 2lr\sigma + lr\sigma + lnr\sigma + n\sigma^3),
\end{aligned}$$

which is

$$= 2lr\rho^2 + 3r^2\rho\sigma + (lm - jk + 3lr)\rho\sigma + 2lr\sigma^2 + r(-fh + kl + 3n\sigma\tau + r\tau^2,$$

and this, for $\tau = 0$, becomes

$$= r\,(2k\rho^2 + 3r\rho\sigma + 2l\sigma^2).$$

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[book p. 138] Hence for $\tau = 0$, we have

$$\Omega' = \frac{1}{\Delta!}\,(4k\rho^2 + 6r\rho\sigma + 4l\sigma^2),$$

that is,

$$= \frac{1}{\Delta}\,[4\,.\,1^3\,2\,.\,\rho^2 + 6\,.\,1^2\,2^2\,.\,\rho\sigma + 4\,.\,1\,2^3\,.\,\sigma^2];$$

and Ω' is thus a form of the major function $(x,\ y,\ z)^2_2$. Of course the general form is $\Omega = \Omega' + (x,\ y,\ z)\,.\,012$.

Syzygy of the Major Function. Art. No. 55.

55. Writing now $(x,\ y,\ z)^{n-3}_k = \Omega_{ik}$; and taking on the fixed curve a new point 3, consider the like functions Ω_{23} and Ω_{31}: it is to be shown that we have identically

$$\Omega_{23}\,.\,011 + \Omega_{31}\,.\,012\,.\,023 + \Omega_{12}\,.\,031 - (123)^n\,.\,011\,.\,012\,(x,\ y,\ z)^{n-4},$$

where $(x,\ y,\ z)^{n-4} = 0$ is a properly determined minor function; or, considering herein 0 as a point on the fixed curve and writing therefore $f = 0$, the equation is

$$\frac{\Omega_{23}}{023}\,.\,011 + \frac{\Omega_{31}}{031}\,.\,012 + \frac{\Omega_{12}}{012}\,.\,(x,\ y,\ z)^{n-4}.$$

56. Write for a moment $X = \Omega_{23}\,.\,011\,.\,012 + \Omega_{31}\,.\,023\,.\,031 - (123)^n\,(x,\ y,\ z)^{n-4}$, then k being an arbitrary coefficient, we have $X - kf = 0$, a curve of the order n, passing through the points 1, 2, 3, and the residues of 1, 2; in fact, take for any other point in the curve $\Omega_{12}\,.\,012 = 0$, $033 = 0$, and therefore $X = 0$ a point on the curve. Again, at any residue of 2, 3, we have $\Omega_{23} = 0$, $023 = 0$, and therefore $X = kf = 0$ will have a dp at 1; and clearly by symmetry, it will also have a dp at 2, and at 3; the theorem is thus proved.

57. Taking an arbitrary point a coordinates $(x_a,\ y_a,\ z_a)$, and writing

$$D = x_a\,\frac{d}{dx} + y_a\,\frac{d}{dy} + z_a\,\frac{d}{dz},$$

* This is the differential theorem corresponding to C, and G.'s integral theorem, p. 30, viz. this is $\Omega_{23} + \Omega_{31} + \Omega_{12} = 1$, a sum of three integrals of the third kind=an integral of the first kind.

[book p. 139] we have to find k, so that the curve $D(X - kf) = 0$ shall pass through the point 1. Observing that $D023 = a23$, &c., we have

$$\begin{aligned} D(X - kf) = {} & D\Omega_{23}\,.\,031\,.\,012 + \Omega_{23}\,(031\,.\,a12 + a31\,.\,012) \\ & + a23\,(\Omega_{31}\,.\,012 + \Omega_{12}\,.\,031) \\ & + 023\,[\Omega_{31}\,.\,a12 + \Omega_{31}\,.\,a31 + D\Omega_{31}\,.\,012 + D\Omega_{12}\,.\,031] \\ & - kDf, \end{aligned}$$

and, to make the curve pass through 1, writing herein $0 = 1$, we have

$$0 = 123\,(\Omega^2_{12}\,.\,a12 + \Omega_{1,a}\,.\,a31) - k\,(Df)_1,$$

where the suffixe (1) denotes that we are in Ω_{23}, &c., and f respectively to write $0 = 1$. We have here $\Omega_{1,a} = a\,.\,1^{n-3}\,3$, $\Omega_{12,1} = n\,.\,1^{n-3}\,2$, $(Df)_1 = a\,.\,1^{n-1}$, and the equation thus is

$$n\,.\,123\,(1^{n-2}\,3\,.\,a12 + 1^{n-2}\,2\,.\,a31) = ka\,.\,1^{n-1}.$$

But we have identically $1^{n-2}\,1\,.\,a23 + 1^{n-2}\,2\,.\,a31 + 1^{n-2}\,3\,.\,a12 = 1^{n-1}\,.\,123$, where $1^{n-1} = 1^n$ is, in fact, $= 0$; the factor $1^{n-2}\,a$ thus divides out, and the equation becomes $k = (123)^n$, viz.

k having this value, the curve $X - kf = 0$ will have a dp at 1; and clearly by symmetry, it will also have a dp at 2, and at 3; the theorem is thus proved.

The Syzygy, Fixed Curve a Cubic. Art. No. 58.

58. The syzygy may be verified independently in the case where the fixed curve is a cubic. Observe that the syzygy, if satisfied for any particular form of Ω, will be generally satisfied; we may therefore take $\frac{1}{3}\Omega_{12} = \overline{012}$. Writing thus

$$\frac{\frac{1}{3}\Omega_{23}}{012} \cdot \frac{\overline{012}}{012} = [012] \text{ suppose,}$$

and taking 0 to be a point on the cubic curve, we ought to have $[023] + [031] + [012] = a$ constant; the value of this constant comes out to be $= [123]$, and the syzygy in its complete form thus is

$$[023] + [031] + [012] = [123].$$

We have

$$\Delta\,\overline{023},\ \Delta\,\overline{031},\ \Delta\,\overline{012} = lp + f\sigma + i\tau,\ jp + h\sigma + g\tau,\ h\rho + l\sigma + l\tau,$$

and the equation thus is

$$\frac{lp + f\sigma + i\tau}{\rho} + \frac{jp + h\sigma + g\tau}{\sigma} + \frac{h\rho + l\sigma + l\tau}{\tau} - l = 0;$$

this, multiplied by $\rho\sigma\tau$, becomes

$$h\rho^2\sigma + jp^2\tau + 3l\rho\sigma\tau + g\sigma^2\tau + f\sigma\tau^2 + i\sigma\tau^2 = 0,$$

which is, in fact, $\frac{1}{3}f = 0$, the equation of the cubic curve.

Observe that the new symbol $[012]$ is, in virtue of its determinant denominator, an alternate function, $[012] = -[102]$, $[012] = [201] = [201]$. The syzygy is a relation between any four points 1, 2, 3, 0 of the curve, and it may be also expressed in the form

$$[123] - [230] + [301] - [012] = 0.$$

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[book p. 140] The Syzygy, Fixed Curve a Quartic. Art. No. 59.

59. Taking Ω_{12} as before, we have

$$\tfrac{1}{2}\Omega_{12} = \frac{-\,01^3\,.\,02^2 + 01^2\,2\,.\,012 + 01\,2\,.\,1^2\,2}{012\,.\,1^2\,2} = [012] \text{ suppose;}$$

and taking 0 to be a point on the quartic curve, we ought to have

$$[023] + [031] + [012] = (x,\, y,\, z)^0,\ \text{a linear function of } (x,\, y,\, z),$$

or, what is the same thing, considering the left-hand side as expressed in terms of ρ, σ, τ, the sum should be

$$= (\rho,\, \sigma,\, \tau)^0,\ \text{a linear function of } (\rho,\, \sigma,\, \tau).$$

By a preceding formula we have

$$[0\,1\,2] = \frac{1}{\Delta^2\rho\sigma}\,\big[2lr\rho^2 + 3r^2\rho\sigma + km - jk + 3lr)\rho\sigma \\ + 2lr\sigma^2 + (-\,fh + kl + 3n\sigma\tau)\tau^2\big],$$

which is

$$= \frac{1}{\Delta!}\left\{(2k + \frac{km - jk}{r})\,\rho + \left(3m + \frac{-fh + kl}{r}\right)\sigma + \left(n + \frac{-fj + lm}{r}\right)\tau\right\} + \frac{1}{\Delta!}\,\frac{2k\rho^2 + 3r\rho\sigma + 2l\sigma^2}{\tau}.$$

And hence, forming the sum [023] + [031] + [012], we have first a fractional part which is found to be integral, viz. this is

$$\frac{1}{\Delta^2}\left\{\frac{2fr^2 + 3r\rho\sigma + 2ir^2}{p} + \frac{2gp^2 + 3r\rho\sigma + 2lp^2}{q} + \frac{2kp^2 + 3r\rho\sigma + 2l\sigma^2}{r}\right\},$$

$$= \frac{1}{\Delta^2\rho\sigma\tau}\left\{2hp\sigma + 2lp\rho + 3l\rho^2\sigma + 3gp^2\sigma + 2lp\sigma\tau + 2k\rho\sigma\tau + 2k\rho\tau\right\},$$

$$= \frac{1}{\Delta!\rho\sigma\tau}\left[\tfrac{1}{2}\Delta!\,f - 6lp\sigma\tau - 6n\sigma\tau - 6n\sigma\tau\right],$$

or since $f = 0$, this is

$$= \frac{1}{\Delta}\,(-\,6l\rho - 6n\sigma - 6n\tau).$$

We then have integral terms which are at once deduced from the above integral terms of $0^1\,2$; collecting the several terms, we find

$$[023] + [031] + [012] =$$

$$\frac{1}{\Delta}\left\{\rho\left(l + \frac{mn - gk}{p} + \frac{in - gh}{q} + \frac{kn - gh}{r}\right) + \sigma\left(m + \frac{fn - hi}{p} + \frac{lh - lk}{q} + \frac{kl - fh}{r}\right) + \tau\left(n + \frac{lm - fj}{p} + g\right.\right.$$

which is the required result.

[book p. 141] Preparation for the Conversion—The Symbol ∂. Art. Nos. 60 to 63.

60. I use ∂ as the symbol of a quasi-differentiation, viz. U being any function of $(x,\,y,\,z)$, ∂U denotes $\dfrac{1}{d\omega}$ multiplied by the differential $\dfrac{dU}{dx}\,dx + \dfrac{dU}{dy}\,dy + \dfrac{dU}{dz}\,dz$; in such a differential, the increments dx, dy, dz do not in general present themselves in the combinations $ydz - zdy$, $zdx - xdz$, $xdy - ydx$; but they will do so if U is a function of the degree zero in the coordinates x, y, z, that is, if U be the quotient of two homogeneous functions of the same degree(s); and this being so, we say by the equations

$$\frac{ydz - zdy}{\dfrac{df}{dx}} = \frac{zdx - xdz}{\dfrac{df}{dy}} = \frac{xdy - ydx}{\dfrac{df}{dz}} = d\omega$$

get rid of the increments, and ∂U will denote a function of $(x,\,y,\,z)$ derived in a definite manner from the function U. The symbol ∂ will be used only in the case in question of a function of the degree zero. Of course ∂_1 will denote the like operation in regard to $(x_1,\,y_1,\,z_1)$, and so ∂_2, &c.; and we may for greater clearness write ∂_0 in place of ∂.

61. Consider then $\partial\dfrac{P}{Q}$, where P, Q are functions $(x,\,y,\,z)^m$ of the same degree; we have

$$\partial\,\frac{P}{Q} = \frac{1}{Q^2 d\omega}\,(QdP - PdQ),$$

and then

$$dP = \frac{dP}{dx}\,dx + \frac{dP}{dy}\,dy + \frac{dP}{dz}\,dz, \quad \frac{1}{m}P = \frac{dP}{dx}\,x + \frac{dP}{dy}\,y + \frac{dP}{dz}\,z,$$

with the like formula for Q. Substituting, we find

$$\partial\,\frac{P}{Q} = \frac{1}{mQ^2 d\omega}\left\{\frac{d(Q,\,P)}{d(y,\,z)}\,(ydz - zdy) + \frac{d(Q,\,P)}{d(z,\,x)}\,(zdx - xdz) + \frac{d(Q,\,P)}{d(x,\,y)}\,(xdy - ydx)\right\},$$

that is,

$$\partial\,\frac{P}{Q} = \frac{1}{mQ^2}\left\{\frac{df}{dx}\,\frac{d(Q,\,P)}{d(y,\,z)} + \text{\&c.}\right\} = \frac{1}{mQ^2}\,\frac{d(f,\,Q,\,P)}{d(x,\,y,\,z)},\ = \ \frac{1}{mQ^2}\,J\,(f,\,Q,\,P),$$

or say

$$\partial\,\frac{P}{Q} = -\,\frac{1}{mQ^2}\,J\,(P,\,Q,\,f).$$

62. As an example, consider

$$\partial\,[012],\ = -\,\frac{1}{(012)^2}\,J\,(\overline{012},\ 012,\ f).$$

The determinant is

$$\begin{vmatrix} \dfrac{d}{dx}\,\overline{012}, & y_1z_z - y_2z_1, & \dfrac{df}{dx} \\ \dfrac{d}{dy}\,\overline{012}, & z_1z_2 - z_2x_1, & \dfrac{df}{dy} \\ \dfrac{d}{dz}\,\overline{012}, & x_1y_2 - x_2y_1, & \dfrac{df}{dz} \end{vmatrix}$$

[book p. 142] and the coefficient herein of $\dfrac{d}{dz}\,\overline{012}$ is $(x_1z_2 - x_2z_1)\dfrac{df}{dy} - (x_1y_2 - x_2y_1)\dfrac{df}{dx}$, which is

$$= x_1\left(x_2\,\frac{df}{dx} + y_2\,\frac{df}{dy} + z_2\,\frac{df}{dz}\right) - x_2\left(x_1\,\frac{df}{dx} + y_1\,\frac{df}{dy} + z_1\,\frac{df}{dz}\right),\ = 3\,(0^2\,.\,x_1 - 0^2\,.\,x_2),$$

and so for the other terms.

The determinant is thus

$$= 3\,\left[0^2\,1\,\left(x_1\,\frac{d}{dx} + y_1\,\frac{d}{dy} + z_1\,\frac{d}{dz}\right)\overline{012} - 0^2\,2\left(x_2\,\frac{d}{dx} + y_2\,\frac{d}{dy} + z_2\,\frac{d}{dz}\right)\overline{012}\right]$$

say this is

$$= 3\,[0^2\,1\,.\,\overline{D} - 0^2\,2\,\overline{D}]\,\overline{012}.$$

But we have $\overline{D}\,\overline{012} = 1^2\,2$, $\overline{D}\,\overline{012} = 1\,2^2$, and the determinant is then $= 3\,(0^2\,.\,1^2\,2 - 0^2\,.\,1\,2^2)$, whence finally, writing ∂_0 instead of ∂,

$$\partial_0\,[012] = -\,3\,.\,\frac{0^2\,.\,1^2\,2 - 0^2\,.\,1\,2^2}{(012)^2}.$$

63. By cyclical interchange of the 0, 1, 2, we have

$$\partial_1\,[012] = -\,3\,.\,\frac{1^2\,.\,0^2\,2 - 0^2\,.\,02^2}{(012)^2},$$

$$\partial_2\,[012] = -\,3\,.\,\frac{02^2\,.\,01^2 - 1^2\,.\,0^2\,1}{(012)^2},$$

and thence adding, we find

$$(\partial_0 + \partial_1 + \partial_2)\,[012] = 0,$$

an important property, which, joined to the equation before obtained,

$$[023] + [031] + [012] = [123],$$

completes the theory of the function [012].

Conversion of the Major Function (Interchange of Limits and Parametric Points). Art. No. 64.

64. Write in general

$$\frac{(x,\, y,\, z)_k^{n-3}}{012} = Q_{k,\,01}.$$

$Q_{k,\,01}$ is an alternate function in regard to the points 1, 2 ($Q_{k,\,01} = -\,Q_{k,\,10}$), and in regard to the coordinates of the points 0, 1, 2, it is rational, but not integral, of the degrees $n-3$, 0, 0 respectively; it can therefore be operated upon with ∂_0 or ∂_1, but (except in the case $n = 3$) not with ∂_2.

[book p. 143] The conversion relates not to the general major function $(x,\, y,\, z)_k^{n-3}$, but to this function with the arbitrary constants properly determined, and consists in a relation between two functions $Q_{k,\,01}$ and $Q_{k,\,1k}$ (each of them a function of three out of four arbitrary points 1, 2, 4, 5 on the fixed curve), viz. the conversion is

$$\partial_2\, Q_{k,\,01} = \partial_4\, Q_{k,\,4k};$$

an equation which may be written in four different forms, viz. we may in the form written down interchange 1, 2 and also 4, 5.*

The determination of the constants is a very peculiar one, inasmuch as it is not algebraical; viz. in the case of the cubic curve, about to be considered, it appears that $Q_{k,\,01}$ contains the term $\int_2^1 d\omega\,\{0\overline{36}\}$, which is a transcendental function of the coordinates of the parametric points 1 and 2.

The Conversion, Fixed Curve a Cubic. Art. No. 65.

65. We may write $Q_{k,\,01} = [012] + K$, where K is a constant, that is, it is independent of the point 0, but depends on the parametric points 1 and 2. I assume K to be properly determined, and give an *a posteriori* verification of the equation

$$\partial_2\, Q_{k,\,01} = \partial_4\, Q_{k,\,4k}.$$

The value is $K = \int_2^1 d\omega\,\{0\overline{36}\} - [123]$, where 3, 4 are arbitrary points on the cubic curve, and where in the definite integral, regarded as an integral $\int U\,du$ with a current variable u, the meaning is that this variable has as the limits the values u_1, u_2, which belong to the points 1 and 2 respectively: a fuller explanation might be proper, but the investigation will presently be given in a form not depending on any integral at all.

Substituting for K its value, we have

$$Q_{k,\,01} = [012] + \int_2^1 d\omega\,\{0\overline{36}\} - [123],$$

or, as this may also be written,

$$= -\,[023] - [031] + \int_2^1 d\omega\,\{0\overline{36}\}.$$

We have thence

$$\partial_2 Q_{k,\,01} = -\,\partial_2\,[031] + \int_2^1 d\omega\,\{0\overline{36}\},$$

* The meaning of the property is better seen from the integral form: $Q_{k,\,01}$ is a function of the points 0, 1, 2 and $Q_{k,\,4k}$ the like function of the points 0, 1, 4, 5, and these in their meaning (alluded to in the heading) that there exists for the integral of the third kind Ω a canonical form Π, such that $\int_0^{(\alpha\beta)} d\omega \left\{0\overline{36}\right\}_{\alpha\beta} = \int_a^{(\alpha\beta)} d\omega\, Q_{k,\,a\beta}$ which equation operated upon with ∂_0 gives the formula of the text. And there is then the existing (alluded to in the heading) that there exists for the integral of the third kind a canonical form Π, in which the symmetry, expressed by the interchange of the limits and the parametric points. The expression for $Q_{k,\,a\beta}$ mentioned further on in the text for the case, must cover a cubic, shows that in this case the canonical form of the integral of the third kind is $\int_a^b d\omega \left\{[026] + \left(\int_a^b d\omega \left\{[036] - [126]\right\}\right)\right\}$.

[book p. 144] and consequently

$$\partial_2 Q_{k,\,01} = -\,\partial_2\,[431] + \partial_2\,[136],$$

$$\partial_4 Q_{k,\,4k} = -\,\partial_4\,[134] + \partial_4\,[436];$$

hence, observing that $[431] = -\,[134]$, &c., we have

$$\partial_2 Q_{k,\,01} - \partial_4 Q_{k,\,4k} = -\,\partial_2\,[134] + \partial_2\,[136] - \partial_4\,[436],$$

$$= -\,\partial_2\,[134] + \partial_2\,[136] - \partial_4\,[436],$$

which, observing that we have $\partial_2\,[641] = 0$, is

$$= \partial_2\,([136] - [364] + [641] - [413]),\ = 0,$$

the required theorem.

To avoid, in the proof, the use of the integral sign, we have only to consider the required function $Q_{k,\,01}$ as given by the foregoing differential formula

$$\partial_2 Q_{k,\,01} = -\,\partial_2\,[031] + \partial_2\,[136],$$

for we have then the values of $\partial_1 Q_{k,\,01}$ and $\partial_2 Q_{k,\,01}$: the rest of the proof the same as before.

The Conversion, Fixed Curve a Quartic. Art. Nos. 66 to 73.

66. We have

$$Q_{k,\,01} = [012] + (x,\,y,\,z),$$

where $(x,\,y,\,z)$ is a linear function of $(x,\,y,\,z)$, but depending also on the parametric points 1 and 2, which has to be determined so as to satisfy the conversion equation

$$\partial_2 Q_{k,\,01} = \partial_4 Q_{k,\,4k}.$$

Observing that we have $[023]+[031]+[012]$ = a linear function of $(x,\,y,\,z)$, the linear function $(x,\,y,\,z)$ of $Q_{k,\,01}$ may be taken to be $= \Theta_{k,\,12} = [023] - [031] - [012]$; that is, we may assume

$$Q_{k,\,01} = [012] + \Theta_{k,\,12} = (-[023] + [091]) + [012],$$

$$= -\,[023] - [031] + \Theta_{k,\,12},$$

where $\Theta_{k,\,12}$ is a linear function of $(x,\,y,\,z)$, but depending also on the points 1 and 2, which has to be determined. We have

$$Q_{k,\,01} = [012] + \Theta_{k,\,12} + \partial_1\Theta_{k,\,12},$$

and thence

$$\partial_1 Q_{k,\,01} = -\,\partial_1\,[431] + \partial_1\Theta_{k,\,12},$$
$$\partial_2 Q_{k,\,01} = -\,\partial_2\,[134] + \partial_2\Theta_{k,\,12},$$

giving an equation for Θ,

$$\partial_1\Theta_{k,\,12} - \partial_2\Theta_{k,\,12} = \partial_2\,[431] - \partial_2\,[134];$$

here 4 is an arbitrary point of the quartic, and we may instead of it write 0, the equation thus becoming

$$\partial_1\Theta_{k,\,12} - \partial_2\Theta_{k,\,12} = \partial_2\,[031] - \partial_2\,[130].$$

[book p. 145] **67.** Of the terms on the left-hand side, the first is a linear function of $(x,\ y,\ z)$, or say it is an integral function of 0, and the second is a linear function of $(x_1,\ y_1,\ z_1)$, or say it is an integral function $1'$; the first depends on the parametric points 1, 3, and $\partial_1(0,\ 1,\ 3)$ is a known function, suppose, as regards the coordinates $(x,\ y,\ z)$ of the point 0, replacing the corresponding point 3, 4 or 5 by the new arbitrary point 6, 7 or 8. Further 045, &c., denote determinants as above; so that in H each of the last three terms is, in fact, as regards the point 0, a mere linear function of the coordinates $(x,\ y,\ z)$ of this point.

We have $\partial_1(1\,;\ 3,\ 2;\ 6,\ 4,\ 5) = 13'\overline{246}$, and hence this function multiplied by $\dfrac{045}{345}$ is $= 01'\overline{2456}$; and so for the third and fourth terms of H: thus each of the four terms of H is $= 01'\overline{2456789}$, of the degree 1 in the coordinates of the points of the other points 2, 3, 4, 5, 6, 7, 8 respectively, but of the degree 0 in regard to the coordinates of the point 1. The before-mentioned function $\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)$ is a function satisfying these conditions 1Ω as $= s\,.\,012\,1^2\,2+$ arbitrary linear function $(x,\ y,\ z)$, of $(x_1,\ y_1,\ z_1)$, of the points of of $(x,\ y,\ z)$, which presents itself in the case of the cubic curve.

Nöther's conversion-theorem consists herein, that the function

$$H\,(0;\ 1,\ 2;\ 3,\ 4,\ 5;\ 6,\ 7,\ 8)$$

is unaltered by the interchange of the two points 0, 1; or putting for shortness

$$H\,(0;\ 1,\ 2;\ 3,\ 4,\ 5;\ 6,\ 7,\ 8) = H_1\,(0),$$

the theorem is

$$H_1\,(0) = H_0\,(1).$$

74. We have, No. 59,

$$\tfrac{1}{2}\Omega_{12} = \frac{-\,01^3\,.\,02^2 + 01^2\,2\,.\,012 + 01\,2\,.\,1^2\,2}{012\,.\,1^2\,2},\ = [012];$$

or as for greater simplicity I write it

$$= 012,$$

viz. 012 is now written instead of [012] to denote the function just given as the value of $\dfrac{\frac{1}{2}\Omega_{12}}{012}$. Ω_{12} is thus $= r\,.\,012\,.\,012$, viz. this is a particular form of Ω_{12} satisfying the conditions that $\Omega_{12} = 0$ is a conic passing through the residues of the points 1, 2, and such that Ω_{12} on writing therein $(x_1,\ y_1,\ z_1)$ for $(x,\ y,\ z)$ becomes $= s\,.\,012\,1^2\,2+$ arbitrary linear function $(x,\ y,\ z)$. The before-mentioned function $\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)$ is a function satisfying these conditions 1Ω as $= s\,.\,012\,1^2\,2+$ arbitrary linear function $(x,\ y,\ z)$, of $(x_1,\ y_1,\ z_1)$, of the points of of $(x,\ y,\ z)$, which presents itself in the case of the cubic curve.

[book p. 146] we find, by a process such as that of No. 62,

$$\partial_1\,[0\,1\,3] = \frac{1}{a}\,\{-\,0\,1^3\,.\,01^2\,.\,02 + 09^2\,.\,j\}$$

$$+\frac{1}{q}\,\{2\,.\,01^3\,(03^2\,.\,01^2\,2 + (013)^2\}_\beta$$

$$+\frac{1}{q}\,\{(2\,.\,01^3\,.\,1\,2)\,.\,(01\,2\,.\,03^2\;+\;2\,(2\,.\,013\,.\,01^2\,.\,01^2\,2\;+\;01^2\,.\,03)\}_\beta$$

$$+\frac{1}{q^2}\,\{-\,2\,.\,01^3\,(-\,01^2\,.\,01^2\;+\;01^2\,.\,03^2)\}\,.$$

Substituting herein the value $01^3 = \dfrac{1}{a^3}(ha + jr)$, &c., we have $\dfrac{1}{a^3}$ multiplied by a cubic function $(\rho,\,\sigma,\,\tau)$; writing down very simply the integral terms, and then the others, we have

$$\partial_1\,[013] = \frac{1}{a^3}\{(-\,6hn + 3jm) + \frac{1}{q}\,(8hkl - 3qjp + 3jp + 2jh)\rho\sigma$$

$$+\frac{1}{q^2}\left(\tfrac{15}{4}h^2 - 7hp + 8jm\right)\rho^2 + \frac{1}{q}\,(-\,2pjk + 8jh - 5jp + 4h^2)\rho\sigma + \frac{1}{q^2}\,(2j^2hl - 2pjn - 2pjm + 2j^2n)\}$$

(say this linear function of ρ, σ, τ is $= \Box$)

$$+\frac{1}{\Delta^2p^2}\left[ar^3\,.\,2j^2 + p^2\sigma\rho + p^2\sigma\rho + p\sigma\tau + 6jp + 6jp + 6jn\sigma)\right] - *$$

70. The expression of $\partial_1\,[103]$ is deduced from this by the interchange of 0, 1: and I write

$$\partial_1\,[013] - \partial_1\,[103] = \Box - *$$

$$+\frac{1}{a^3\Delta!p}\{p^2\rho^2\,2j^2 + p^2\sigma\,3jh + p^2\tau\,3l + p^2\sigma\,3lk + p\rho\sigma + p\sigma\tau + p^2\rho + p\sigma\tau + 2j\sigma\tau\,3jn\}$$

$$-\Delta^2\,\Delta\,.\,2\,(09\bar{1})\sigma - \Delta!\sigma\,(-\,3\,.\,01^2\,.\,02^2 + 0\bar{2}^2\,.\,025) - \Delta!\,3\,.\,01^3\,.\,01^2$$

$$+\Delta\sigma\,(-\,2\,.\,01^3\,.\,02 + 0\bar{2}^2\,(02 + 01\bar{2})) - 2\,.\,01^3\,.\,01^2$$

where, and in what follows, the $*$ denotes the function included immediately to the left of it, interchanging therein the 0, 1. It will be observed that the $\Box$, and linear function of $(\rho,\,\sigma,\,\tau)$, that is, of $(x,\,y,\,z)$, is a series of the required function $\partial_1\,(0;\;1,\;3)$: the remaining portion has to be denoted by means of the expressions for $\Delta^2(013)$, &c., in terms of ρ, σ, τ.

71. We obtain

$$\partial_1\,[013] - \partial_1\,[103] = \Box - *$$

$$+\frac{1}{\Delta!}\left[\sigma\,(2fj - 3kp - 3hq + 15lm - 9nr) + (-\,3gr - 3jp + 9mq)\right]$$

$$+\frac{1}{\Delta!p}\left[\sigma^2\,(12fk - 12hn + 18m^2 - 9pr) + \sigma\tau\,(6fp - 6jr + 18nq) + \tau^2\,.\,4n\sigma\right]$$

$$+\frac{1}{\Delta!p^2}\left[\rho^3\,(18gn - 9pkp) + \rho^2\sigma\,(12hp - 8tk + 9mqp) + \sigma\tau\,(4fp + 6lm)\right]$$

$$+\frac{1}{\Delta!p^2}\left[\rho^3\,.\,4f^2 + \rho^2\sigma\,.\,6fp + \rho\sigma\,.\,4fj\right].$$

[book p. 147] The terms of the second line may be transformed as follows:

$$\frac{\sigma}{\Delta!}\,(2fj - 3kp - 3hq + 18lm - 9nr)$$

$$= \tfrac{1}{2}\,\frac{\sigma}{\Delta!}\,(2fj - 3kp - 3hq + 18lm - 9nr) - *$$

$$+\frac{1}{\Delta! p}\left[\sigma^2\,(-12fk + 12hn - 18m^2 + 9pr) + \sigma\tau\left\{\tfrac{1}{2}fp + 3pk + \tfrac{3}{4}jr - \tfrac{3}{4}(p - 9nm)\right\}\right]$$

$$+\frac{1}{\Delta! p^2}\left[\sigma^3\,(-20fm + 15hp) + \sigma^2\tau\,(-\tfrac{3}{2}fr + 12lk - 18np) + \sigma\tau^2\,.\, - 6\sigma p\right]$$

$$+\frac{1}{\Delta! p^2}\left[\sigma^2\tau\,(-10fp^2 + a^2\tau - 15fp + a^2\tau - 4mn^2)\,a\sigma\tau + a\sigma\tau\,a^2\tau + a\sigma\tau\,a\sigma\tau - 6\,p\right]$$

$$+\frac{1}{\Delta! p^3}\left[\sigma^2\tau\,(-10fn^2 + a^2\tau - 12fp + 8ik - 9im\rho)\,a\sigma\tau - 9i\,p\right],$$

and

$$\frac{1}{\Delta!}\,(-3gr - 3jp + 9mq)$$

$$= \tfrac{1}{2}\,\frac{1}{\Delta!}\,(-3gr - 3jp + 9mq) - *$$

$$+\frac{1}{\Delta! p}\left[\sigma\tau\left(-\tfrac{3}{2}fq + \tfrac{3}{2}lr + \tfrac{3}{4}(p - 9nm) + \tfrac{1}{2}\sigma\tau\,(p - 9nm)\right) + \tau^2\,.\, - 6\sigma p\right]$$

$$+\frac{1}{\Delta! p^2}\left[\sigma^2\tau\,(-6fp + a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau - 6fq + a^2\tau)\,a\sigma\tau - 6\,jp\right]$$

$$+\frac{1}{\Delta! p^3}\left[\sigma^2\tau\,(-6fp + a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau)\,a\sigma\tau - 6\,jp\right],$$

substituting these values, the whole third line is destroyed, and we find

$$\partial_1\,[013] - \partial_1\,[103] = \square - *$$

$$+\tfrac{1}{2}\cdot\frac{1}{\Delta!}\,[\sigma\,(2fj - 3hq - 3kp + 18lm - 9nr) + \tau\,(-3gr - 3jp + 9mq)] - *$$

$$+\frac{1}{\Delta! p^2}\left[\sigma^2(-12fm + 6fp) + \sigma\tau(-9fn + 7lk - 9mp - 9np) + \sigma\tau\,a^2\tau\,(-2fp - 9np) + \sigma^2\tau\,.\, - 5\sigma p\right]$$

$$+\frac{1}{\Delta! p^3}\left[\sigma^2\,(-6f^2 + \sigma^2\tau\,a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau)\,a\sigma\tau - 6\sigma p\right].$$

Ultimately the last two lines of this expression are found to be

$$= \frac{1}{\Delta!}\,[\rho\,(-2hn + 4jm + 2l^2 - 2qr) + \sigma\,(-2fj + 2kp + 2hq - 10lm + 5nr) + \tau\,(2gr + hi - jp + 4hr - 6mq)] - *$$

so that the whole is now a sum of three linear functions of (ρ, σ, τ). - * 19–2

[book p. 148] **72.** Collecting the terms, we have

$$\partial_1\,[013] - \partial_1\,[103] =$$

$$\frac{1}{\Delta!}\left\{\sigma\left[\rho\,(-8hn + 5jm + 2l^2 - 2qr) + \frac{1}{q}\,(4phl - 3qjr - 3lhj + 2jh) + \frac{1}{q^2}\,(-2jhk + 4qhjn - 2jnl)\right]\right.$$

$$+\sigma\left[\frac{1}{q}\,(-4fp + 4jq - 2lm + nr) + \frac{1}{q}\,(2fhi - 3jp - 9jp + 6jpm + 2lr) + \frac{1}{q^2}\,(2klm^2 - 8ik - 18np - 2lhr)\right]$$

$$+\sigma\left[\frac{1}{q}\,((gr + 2hi + jp + n + 14hi) + \frac{1}{q}\,(-2pkn + 2lkjn - 2pnm) + \frac{1}{q^2}\,(2pkl - 2qhij - 2qjm + 2j^2n)\right]$$

$$\left. - * \right\}$$

73. The right-hand side depends on the points 0, 1, 3 and 2: viz. we have therein $\rho = 023$, $\Delta = 123$, &c., but the left-hand side depending on only the points 0, 1 and 2, the right-hand side cannot really contain 3, and it must then remain unaltered, if for 2 we substitute as another function of 0, 1, 3 and 5, viz. ρ, Δ, f, &c., will become 063, 163, 0/3, &c.; we have then $\phi(0,\ 1,\ 3) =$ the above linear function with 2 thus replaced by 6; say

$$\phi(0,\ 1,\ 3) = \frac{1}{\Delta!}\,[\rho\,(\ ) + \sigma(\ ) + \tau(\ )],$$

a given function of the points 0, 1, 3 and the arbitrary point 6, on the quartic curve; we therefore write it $\phi(0\,;\ 1,\ 3,\ 6)$. There is no obvious value for $X(0,\ 1,\ 3)$ which will produce any simplification: I therefore take this function to be $= 0$; and the final result is

$$Q_{k,\,01} = [012] + \Theta_{k,\,12} = -([025] + [031] + [012]),$$

where $\Theta_{k,\,12}$ is a function integral and linear as regards the coordinates $(x,\ y,\ z)$ of the point 0, but transcendental as regards the parametric points 1, 2; and containing besides the arbitrary points 3, 6, of the quartic curve, its value being determined by the differential formulae

$$\partial_1\Theta_{k,\,12} = \phi(0',\ 1,\ 3,\ 6), \quad \partial_2\Theta_{k,\,12} = -\,\phi(0',\ 2,\ 3,\ 6),$$

where $\phi(0',\ 1,\ 3,\ 6)$ is a given function as above. I do not see the meaning of the very complicated linear function of $(\rho,\ \sigma,\ \tau)$, nor how to reduce it to any form such as the simple one $\partial_1\,[036]$, which presents itself in the case of the cubic curve.

Cambridge, England, October 5, 1882.

Chapter IV. The Major Function $(x,\ y,\ z)^{n-3}_k$ continued.

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 74 to 82.

[book p. 149] **74.** I resume the question considered *ante* Nos. 66 to 73. The general problem, where the fixed curve is any given curve whatever, has recently been solved in a very complete and elegant form by Dr Nöther, in the two notes "Zur Reduction algebraischer Differentialausdrücke auf die Normalformen" and "Ueber die algebraischen Differentialausdrücke, 2e Note," *Sitzungsb. der phys.-med. Soc. zu Erlangen*, 10 Dec. 1883 and 14 Jan. 1884. I consider here the case of the quartic curve, $n = 4$, and connect his result with my former investigations.

We have the differential

$$\frac{(x,\ y,\ z)^{n-3}_k\,d\omega}{012} = \frac{\Omega_{12}\,d\omega}{012},$$

where Ω_{12}, or as I also write it $\Omega(0;\ 1,\ 2;\ 3,\ 4,\ 5)$, is a rational and integral function of the degree $(n-2) = 2$ in the current coordinates $(x,\ y,\ z)$: it depends also on the parametric points 1, 2, which are points on the quartic, coordinates $(x_1,\ y_1,\ z_1)$, $(x_2,\ y_2,\ z_2)$ respectively; and on $(p =)\,3$ other points 3, 4, 5 are arbitrary points which we take to be in fact the points $(x_3,\ y_3,\ z_3)$, $(x_4,\ y_4,\ z_4)$, $(x_5,\ y_5,\ z_5)$ respectively. The curve $\Omega = 0$ is a conic, passing through the points 3, 4, 5 and through the $(n-2 =)\,2$ residues of the parametric points 1, 2, of the quartic curve, $n = 4$, and connect his result with my former investigations.

We have hence

$$\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = \frac{1}{\Delta}\,(a\,1^{n-2}\,2\,1\,1\,1,\ \dots)\,(x,\ y,\ z)^{n-2};\ \text{see No. 2.}$$

the function determined by the equation

$$\left|\begin{array}{cc} (x,\, y,\, z)^2 & ,\ \ \Omega \\ 1\,(x_1,\, y_1,\, z_1)^2 & ,\ \ 4\,.\,1^3\,2 \\ 2\,(x_1,\, y_1,\, z_1\,\xi\, x_2,\, y_2,\, z_2),\ 6\,.\,1^2\,2^2 & \\ 1\,(x_2,\, y_2,\, z_2)^2 & ,\ \ 4\,.\,1\,2^3 \\ (x_3,\, y_3,\, z_3)^2 & ,\ \ 0 \\ (x_4,\, y_4,\, z_4)^2 & ,\ \ 0 \\ (x_5,\, y_5,\, z_5)^2 & ,\ \ 0 \end{array}\right| = 0,$$

* $\left(x_a \dfrac{d}{dx_b} + y_a \dfrac{d}{dy_b} + z_a \dfrac{d}{dz_b}\right)(x_b,\, y_b,\, z_b)^{n-2}$ = see No. 2.

[book p. 150] this is of the form

$$M\Omega + \Box = 0,$$

and as appears in No. 46, $M = 123\,.\,124\,.\,125\,.\,345$. Hence writing $\Box\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)$ for $\Box$, we have

$$\Omega_n = \Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = \frac{-\Box\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)}{123\,.\,124\,.\,125\,.\,345}.$$

Hence further writing

$$Q_n = Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = \frac{\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)}{012},$$

so that the differential is $Q d\omega$, $= Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)\,d\omega$, we have

$$Q = Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = \frac{-\Box\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)}{012\,.\,123\,.\,124\,.\,125\,.\,345},$$

which is of the form

$$Q = \frac{012\overline{2345}}{012\overline{2345}},\ = 012\overline{2345},$$

viz. Q is a rational fraction where the numerator is of the degree 2, and the denominator of the degree 5 as regards the coordinates $(x,\, y,\, z)$ of the current point; but the numerator and denominator are each of the degree 4 as regards the coordinates of the points 1, 2 separately; and the degree 2 as regards the coordinates of the points 3, 4, 5 separately.

76. The signification of the symbol of quasi-differentiation ∂ (applicable only to a function of the degree 0 in the coordinates to which the differentiations have reference) is explained *ante* No. 60. The function Q just mentioned is of the degree 0 in regard to the coordinates of each of the points 1, 2, 3, 4, 5; and it can thus be operated upon by the symbols ∂_1, ∂_2, ∂_3, ∂_4, ∂_5 respectively. Observe, in particular, that we have $\partial_3 Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = 012\overline{2345}$, viz. it is of the degree 1 in the coordinates of the points 0 and 1 respectively, but of the degree 0 in regard to the coordinates of the points 2, 3, 4, 5 respectively.

77. This being so, we may consider the function

$$\begin{aligned} H\,(0;\ 1,\ 2;\ 3,\ 4,\ 5;\ 6,\ 7,\ 8) = {} & \partial_3 Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) \\ & + \partial_3 Q\,(1;\ 3,\ 2;\ 6,\ 4,\ 5)\,.\,\frac{045}{345} \\ & + \partial_4 Q\,(1;\ 4,\ 2;\ 3,\ 7,\ 5)\,.\,\frac{053}{453} \end{aligned}$$

$$+\partial_5 Q\,(1;\ 5,\ 2;\ 3,\ 4,\ 8)\,.\,\frac{034}{534},$$

where 6, 7, 8 are arbitrary points on the quartic; the functions

$$\partial_3 Q\,(1;\ 3,\ 2;\ 6,\ 4,\ 5),\ \partial_4 Q\,(1;\ 4,\ 2;\ 3,\ 7,\ 5),\ \partial_5 Q\,(1;\ 5,\ 2;\ 3,\ 4,\ 8),$$

[book p. 151] are functions of the same form as $\partial_3 Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)$, and derived from it by changing in each case the current point 0 into the parametric point 1, and the points 3, 4, 5 respectively, and using the corresponding point 3, 4 or 5 by the new arbitrary point 6, 7 or 8. Further 045, &c., denote determinants as above; so that in H each of the last three terms is, in fact, as regards the point 0, a mere linear function of the coordinates $(x,\ y,\ z)$ of this point.

We have $\partial_1 Q\,(1;\ 3,\ 2;\ 6,\ 4,\ 5) = 13'\overline{2456}$, and hence this function multiplied by $\dfrac{045}{345}$ is $= 01'\overline{2456}$; and so for the third and fourth terms of H: thus each of the four terms of H is $= 01'\overline{2456789}$, of the degree 1 in the coordinates of the points 0 and 1 respectively, but of the degree 0 in regard to the coordinates of the other points 2, 3, 4, 5, 6, 7, 8.

Nöther's conversion-theorem consists herein, that the function

$$H\,(0;\ 1,\ 2;\ 3,\ 4,\ 5;\ 6,\ 7,\ 8)$$

is unaltered by the interchange of the two points 0, 1; or putting for shortness

$$H\,(0;\ 1,\ 2;\ 3,\ 4,\ 5;\ 6,\ 7,\ 8) = H_1\,(0),$$

the theorem is

$$H_1\,(0) = H_0\,(1).$$

78. We have, No. 59,

$$\frac{\frac{1}{2}\Omega_{12}}{012} = \frac{-\,01^3\,.\,02^2 + 01^2\,2\,.\,012 + 01\,2\,.\,1^2\,2}{012\,.\,1^2\,2},\ = [012],$$

or as for greater simplicity I write it

$$= 012,$$

viz. 012 is now written instead of [012] to denote the function just given as the value of $\dfrac{\frac{1}{2}\Omega_{12}}{012}$. Ω_{12} is thus $= r\,.\,012\,.\,012$, viz. this is a particular form of Ω_{12} satisfying the conditions that $\Omega_{12} = 0$ is a conic passing through the residues of the points 1, 2 on writing therein $(x_1,\ y_1,\ z_1)$ for $(x,\ y,\ z)$ becomes $= s\,.\,012\,1^2\,2+$ arbitrary linear function $(x,\ y,\ z)$. The before-mentioned function $\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)$ is a function satisfying these conditions on writing therein for 4, 5, 6 in succession the values 3, 4, 5, becomes $= 0$; we have therefore

$$\frac{\frac{1}{2}\Omega\,(0;\ 1,\ 2;\ 3,\ 4,\ 5)}{012} = 012 - 012\,\frac{034}{345} - 012\,\frac{054}{354} - 012\,\frac{045}{354},$$

viz. the value of Q given by this equation, on writing therein $0 = 3$, 4, or 5, becomes $= 0$ as it should do.

[book p. 152] **79.** We thus have

$$Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = 012 - 012\,\frac{045}{345} - 012\,\frac{053}{453} - 012\,\frac{034}{534},$$

and Nöther's conversion-equation becomes

$$\partial_1\left\{012 - 012\,\frac{045}{345} - 012\,\frac{053}{453} - 012\,\frac{034}{534}\right\}$$

$$
\begin{aligned}
&+\partial_3\left\{1'02 - 6'32\,\frac{145}{645} - 4'32\,\frac{156}{456} - 5'32\,\frac{164}{564}\right\}\frac{045}{345}\\
&+\partial_4\left\{1'02 - 3'42\,\frac{175}{375} - 7'42\,\frac{153}{753} - 5'42\,\frac{137}{573}\right\}\frac{053}{453}\\
&+\partial_5\left\{1'02 - 3'52\,\frac{148}{348} - 4'52\,\frac{183}{483} - 8'52\,\frac{134}{834}\right\}\frac{034}{534}\\
&= \partial_0\left\{1'02 - 6'32\,\frac{145}{645} - 4'32\,\frac{156}{456} - 5'32\,\frac{164}{564}\right\}\\
&+\partial_3\left\{0'62 - 6'32\,\frac{045}{645} - 4'32\,\frac{056}{456} - 5'32\,\frac{064}{564}\right\}\frac{045}{345}\\
&+\partial_4\left\{1'02 - 3'72\,\frac{075}{375} - 7'42\,\frac{053}{753} - 5'72\,\frac{037}{573}\right\}\frac{053}{453}\\
&+\partial_5\left\{0'82 - 3'82\,\frac{048}{348} - 4'82\,\frac{083}{483} - 8'82\,\frac{034}{834}\right\}\frac{034}{534},
\end{aligned}
$$

an equation where the functions operated on with the ∂'s are only functions such as 012; for there is not any determinant operated upon containing the number which is the suffix of the ∂ operating upon it.

80. Taking all the terms over to the left-hand side, there are in all 32 terms; but of these $3+3$ destroy each other, and $6+6$ unite in pairs into 6 terms: there are thus in all $7+7+6, = 20$ terms; viz. multiplying the whole equation by 345, it is found that the equation becomes:

as so this may be written where

$$
\begin{array}{rl}
345\,(\ \partial_1\,012 - \partial_0\,1'02) & 345\ \boxed{012} \qquad = \partial_1\,012 - \partial_0\,1'02, \text{ \&c.}\\
-\,045\,(-\,\partial_1\,102 + \partial_0\,012) & -\ 045\ \boxed{312}\\
-\,053\,(-\,\partial_1\,142 + \partial_0\,012) & -\ 305\ \boxed{412}\\
-\,034\,(-\,\partial_1\,152 + \partial_0\,012) & -\ 340\ \boxed{512}
\end{array}
$$

[book p. 153]

$$
\begin{array}{rl}
-\,145\,(\ \partial_1\,032 - \partial_2\,302) & -\ 145\ \boxed{032}\\
-\,153\,(\ \partial_1\,042 - \partial_2\,402) & -\ 315\ \boxed{042}\\
-\,134\,(\ \partial_1\,052 - \partial_2\,502) & -\ 341\ \boxed{052}\\
+\,013\,(-\,\partial_1\,542 + \partial_2\,452) & +\ 301\ \boxed{452}\\
+\,014\,(-\,\partial_1\,352 + \partial_2\,532) & +\ 140\ \boxed{532}\\
+\,015\,(-\,\partial_1\,432 + \partial_2\,342) & +\ 015\ \boxed{342}\\
=0, & =0,
\end{array}
$$

viz. the equation is

$$\Sigma \pm 345\ \boxed{012}\ = 0.$$

the nine terms which follow the first term $345\ \boxed{012}$ of the sum being obtained by the interchanges of 0, 1 (one or each) with the 3, 4, 5, each interchange giving rise to a sign $-$.

81. In obtaining the foregoing result, we have, for instance, a pair of terms

$$\partial_1\,542\,\frac{-\,137\,.\,035 + 037\,.\,135}{537},\ = \partial_1\,542\,\frac{-013\,.\,537}{537},\ = \partial_1\,542\,(-\,013),$$

viz. this depends on the equation

$$-137\,.\,035+037\,.\,135-537\,.\,013=0,$$

or say

$$-137\,.\,035+037\,.\,135+013\,.\,357=0,$$

an identity which, in a form which will be readily understood, may be written

$$\det\left|\begin{array}{c}0137\\013735\end{array}\right|=0.$$

Similarly, the two terms which contain $\partial_3\,452$ combine into the single term $\partial_3\,452\,(013)$; and the two new terms taken together are

$$013\,(-\,\partial_3\,542+\partial_2\,452)=301\;\boxed{452}.$$

C. XII. 20

[book p. 154] **82.** The proof of the identity,

$$\Sigma\pm 345\;\boxed{012}\;=0,$$

depends on the property of the function

$$\boxed{012}_1,\ =\partial_1 012-\partial_0 1'02,$$

enunciated No. 67, and proved *à posteriori* by the tedious calculation Nos. 69 to 73, viz. in No. 67, writing 2 in place of 3, this is—$\partial_1\,012-\partial_0\,1'02$ is equal to the difference of two functions, the first of them linear in the coordinates $(x,\,y,\,z)$ of the point 0, but depending also on the coordinates of the points 1 and 2, the second of them linear in the coordinates $(x_1,\,y_1,\,z_1)$ of the point 1, but depending also on the coordinates of the points 0 and 2. Or, what is the same thing, the property is

$$\boxed{012}_1,\ =A_{12}x+B_{12}y+C_{12}z-(A_{01}x_1+B_{01}y_1+C_{01}z_1),$$

where $A_{12},\,B_{12},\,C_{12}$ are functions of $(x_1,\,y_1,\,z_1)$, $(x_2,\,y_2,\,z_2)$, and $A_{01},\,B_{01},\,C_{01}$ are the like functions of $(x,\,y,\,z)$, $(x_2,\,y_2,\,z_2)$.

Substituting such values in the sum $\Sigma\pm 345\,\boxed{012}$, but writing down only the terms which contain x, these are

$$\begin{array}{ccc} & 345\,(A_{12}x-A_{01}x_1) & \\ -\,045\,(A_{12}x_1-A_{01}x_1) & -\,145\,(A_{02}x-A_{20}x_0) & +\,301\,(A_{42}x_4-A_{42}x_5)\\ -\,305\,(A_{42}x_1-A_{01}x_4) & -\,315\,(A_{02}x-A_{40}x_0) & +\,140\,(A_{52}x_5-A_{52}x_3)\\ -\,340\,(A_{52}x_1-A_{01}x_5) & -\,341\,(A_{02}x-A_{50}x_0) & +\,015\,(A_{32}x_3-A_{32}x_4), \end{array}$$

This is

$$\begin{aligned} =\quad & A_{12}\,(\,x\,.\,345-x_1\,.\,015+x_2\,.\,015)\\ +\,& A_{02}\,(\,x\,.\,345-x_0\,.\,345-x_2\,.\,340)\\ +\,& A_{42}\,(\,x_4\,.\,145-x_1\,.\,305+x_0\,.\,340)\\ +\,& A_{52}\,(\,x_5\,.\,134+x_1\,.\,340+x_0\,.\,140)\\ +\,& A_{01}\,(-\,x_1\,.\,345+x_4\,.\,305+x_5\,.\,340)\\ +\,& A_{42}\,(-\,x\,.\,341+x_5\,.\,140+x_3\,.\,015), \end{aligned}$$

where the coefficient of each of the A's is identically $= 0$; and similarly, the terms in y and the terms in z are each $= 0$. We have thus the proof of the identity

$$\Sigma \pm 345 \; \boxed{012} \; = 0,$$

that is, of the conversion-equation $H_1(0) = H_0(1)$.

[book p. 155] The Syzygy—Fixed Curve a Quartic. Art. No. 83.

83. I revert to the theory of the Syzygy, *ante* No. 59.

We have

$$Q\,(0;\ 1,\ 2;\ 3,\ 4,\ 5) = 012 - 012\,\frac{045}{345} - 012\,\frac{053}{453} - 012\,\frac{034}{534},$$

or if for convenience we take instead of 1, 2, the parametric points to be a, β coordinates $(x_a,\, y_a,\, z_a)$ and $(x_\beta,\, y_\beta,\, z_\beta)$ respectively, then this equation is

$$Q_{a\beta} = Q\,(0;\ a,\ \beta;\ 3,\ 4,\ 5) = 0a\beta - 0a\beta\,\frac{045}{345} - 0a\beta\,\frac{053}{453} - 0a\beta\,\frac{034}{534}.$$

Considering a new parametric point γ, and forming the like functions $Q_{\beta\gamma}$ and $Q_{\gamma a}$, it is to be shown that we have identically

$$Q_{a\beta} + Q_{\beta\gamma} + Q_{\gamma a} = 0.$$

To prove this, observe that, in the equation at the end of No. 59, Δ, ρ, σ, τ denote 123, 023, 031, 012 respectively. Hence writing therein a, β, γ in place of 1, 2, 3 respectively, and putting A, B, C for the coefficients (including therein the factor $\frac{1}{\Delta!}$) of ρ, σ, τ respectively, the equation is

$$0\bar{\beta}\gamma + 0\gamma a + 0a\beta = A\,.\,0\beta\gamma + B\,.\,0\gamma a + C\,.\,0a\beta;$$

where A, B, C are absolute constants (functions, that is, of the coefficients of the quartic) each divided by $(a\beta\gamma)^6$. We hence obtain

$$\begin{aligned}(Q_{\beta\gamma} + Q_{\gamma a} + Q_{a\beta})\,.\,345 = 345\,&(A\,.\,0\beta\gamma + B\,.\,0\gamma a + C\,.\,0a\beta)\\ -\,045\,&(A\,.\,3\beta\gamma + B\,.\,3\gamma a + C\,.\,3a\beta)\\ -\,053\,&(A\,.\,4\beta\gamma + B\,.\,4\gamma a + C\,.\,4a\beta)\\ -\,034\,&(A\,.\,5\beta\gamma + B\,.\,5\gamma a + C\,.\,5a\beta).\end{aligned}$$

On the left-hand side the whole coefficient of A is $= 0$; viz. the coefficient has the value $\det \begin{vmatrix} 0345 \\ 0345\beta\gamma \end{vmatrix}$, which is $= 0$. Similarly, the whole coefficient of B is $= 0$, and the whole coefficient of C is $= 0$; and we have thus the required result

$$Q_{\beta\gamma} + Q_{\gamma a} + Q_{a\beta} = 0.$$

The syzygy is thus obtained in a more perfect form than in No. 59; viz. by considering (instead of $0\bar{a}\beta$) the new form $Q_{a\beta}$ then, instead of a sum which is a linear function of the coordinates $(x,\, y,\, z)$, we obtain a sum $= 0$. 20–2

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Chapter IV (concluded). The Fixed Curve a Cubic — Syzygy and Conversion.

The Fixed Curve a Cubic — Syzygy and Conversion. Art. Nos. 84 and 85.

[book p. 156] **84.** In the case when the fixed curve is a cubic (see Nos. 58 and 64), the analogous formulæ are

$$\tfrac{1}{2}\Omega(0\,;\,1,\,2;\,3) = \frac{1\cdot 2\,.\,023 + 12\,.\,031}{12s} = \frac{\{012\}}{12s}\{012\} \quad \text{(see No. 49)},$$

that is,

$$Q(0\,;\,1,\,2;\,3) = \frac{\{012\}}{012}\cdot\frac{\{012\}}{12s} = \{012\} = \{1\overline{23}\},$$

where 1, 2 are the parametric points: 3 any other point on the cubic: the brackets { } are of course here necessary in order to distinguish {012} from the determinant 012. It will be remembered that {012} is an alternate function

$$\{012\} = -\{102\} = \{120\},\ \text{\&c.}$$

If instead of 1, 2 we take the parametric points to be α, β, coordinates $(x_\alpha,\,y_\alpha,\,z_\alpha)$ and $(x_\beta,\,y_\beta,\,z_\beta)$ respectively, then the formula is

$$Q_{\alpha\beta} = \tfrac{1}{2}\,Q(0\,;\,\alpha,\,\beta;\,3) = \{0\alpha\beta\}\{3\alpha\beta\}.$$

Hence taking on the cubic a new point γ, coordinates $(x_\gamma,\,y_\gamma,\,z_\gamma)$ and forming the functions $Q_{\beta\gamma}$ and $Q_{\gamma\alpha}$ we have

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = \{0\beta\gamma\}\{3\beta\gamma\} + \{0\gamma\alpha\}\{3\gamma\alpha\} + \{0\alpha\beta\}\{3\alpha\beta\}.$$

But by the formula No. 58,

$$\{0\beta\gamma\} + \{0\gamma\alpha\} + \{0\alpha\beta\} = 0\,;$$

hence also

$$\{3\beta\gamma\} + \{3\gamma\alpha\} + \{3\alpha\beta\} = 0\,;$$

and we have thus

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = 0,$$

the syzygy for the cubic.

85. For the conversion, the definition of H is

$$H(0\,;\,1,\,2;\,3,\,6) = \partial_3\,Q(0\,;\,1,\,2;\,3) - \partial_6\,Q(0\,;\,3,\,2;\,6),$$

viz. this is

$$\begin{aligned} H = H(0\,;\,1,\,2;\,3,\,6) &= \partial_3\big(\{012\} - \{123\}\big) + \partial_6\big(\{132\} - \{326\}\big),\\ &= \partial_3\{012\} - (\partial_3 + \partial_6)\{123\} - \partial_6\{326\},\end{aligned}$$

which, in virtue of $(\partial_1 + \partial_2 + \partial_3 + \partial_6)\{123\} = 0$ (see No. 63), becomes

$$H = H(1) = (\partial_1 + \partial_2)\{012\} - \partial_2\{123\} - \partial_6\{326\}.$$

[book p. 157] Interchanging the 0 and 1, we thence have

$$H_1(0) = \partial_0\{102\} + \partial_2\{023\} - \partial_6\{326\}.$$

Hence the difference $H_1(1) - H_1(0)$ is

$$= \partial_1\{012\} - \partial_0\{102\} + \partial_2\{123\} - \partial_2\{023\},$$

viz. this is

$$= (\partial_1 - \partial_0)\{012\} + \partial_2\big(\{123\} - \{023\}\big),$$

where the first term is

$$= -\partial_2\{012\},$$

and the whole therefore is

$$= \partial_2\big(\{123\} - \{230\} - \{012\}\big),$$
$$= -\partial_2\{301\},$$

in virtue of

$$\{123\} - \{230\} + \{301\} - \{012\} = 0\,;$$

the whole is consequently $= 0$.

We have thus

$$H_1(1) - H_1(0) = 0,$$

the conversion-equation in the case of the cubic.

Chapter V. Miscellaneous Investigations.

The Differential Symbol $d\omega$. Art. Nos. 86 and 87.

86. The definition is

$$\frac{y\,dz - z\,dy}{\dfrac{df}{dx}} = \frac{z\,dx - x\,dz}{\dfrac{df}{dy}} = \frac{x\,dy - y\,dx}{\dfrac{df}{dz}} = d\omega,$$

and it hence follows that we have

$$d\omega = \frac{\begin{vmatrix} dx, & dy, & dz \\ x, & y, & z \\ \lambda, & \mu, & \nu \end{vmatrix}}{\lambda\,\dfrac{df}{dx} + \mu\,\dfrac{df}{dy} + \nu\,\dfrac{df}{dz}},$$

where $(\lambda,\, \mu,\, \nu)$ are arbitrary constants or, if we please, arbitrary functions of $(x,\, y,\, z)$: viz. the expression just written down is altogether independent of the values of $\lambda,\, \mu,\, \nu$, and is consequently equal to the value obtained by writing any two of these symbols $= 0$, that is, the expression is equal to any one of the foregoing three equal values of $d\omega$. The expression was first given by Aronhold (1863), in the memoir presently referred to.

[book p. 158] It is to be remarked that, considering $(\lambda,\, \mu,\, \nu)$ as the coordinates of a point, the denominator $\lambda\dfrac{df}{dx} + \mu\dfrac{df}{dy} + \nu\dfrac{df}{dz}$ equated to 0 is the polar $(n-1)$thic of the point $\lambda,\, \mu,\, \nu$ in regard to the fixed curve.

If instead of $\lambda,\, \mu,\, \nu$ we write $bc' - b'c,\ ca' - c'a,\ ab' - a'b$, where $(a,\, b,\, c)$ $(a',\, b',\, c')$ are constants, then the numerator is

$$= (ax + by + cz)(a'\,dx + b'\,dy + c'\,dz) - (a'x + b'y + c'z)(a\,dx + b\,dy + c\,dz),$$

or introducing ρ, σ to denote the arbitrary linear functions $ax+by+cz$ and $a'x+b'y+c'z$ respectively, the numerator is $=\rho\,d\sigma-\sigma\,d\rho$: moreover, observing that a, b, c and a', b', c' are the differential coefficients of ρ, σ in regard to the coordinates (x, y, z), the denominator is $=J(f, \rho, \sigma)$; and the value of $d\omega$ is

$$d\omega=\frac{\rho\,d\sigma-\sigma\,d\rho}{J(f, \rho, \sigma)},$$

where, in accordance with a previous remark, the denominator equated to 0 is the polar $(n-1)$thic of the intersection of the lines $\rho=0$, $\sigma=0$ in regard to the fixed curve.

Obviously, by taking for ρ, σ any two of the three coordinates x, y, z, we reproduce the original three forms of $d\omega$.

87. The last-mentioned form of $d\omega$ suggests the expression for this symbol in the case where the fixed curve, instead of being a plane curve, is a curve of double curvature defined by two equations $f=0$, $g=0$ between the four coordinates (x, y, z, w): viz. ρ, σ being now arbitrary linear functions

$$ax+by+cz+dw, \quad \text{and} \quad a'x+b'y+c'z+d'w$$

of the four coordinates, the expression is

$$d\omega=\frac{\rho\,d\sigma-\sigma\,d\rho}{J(f, g, \rho, \sigma)};$$

and by taking for ρ, σ any two of the four coordinates x, y, z, w, we have for $d\omega$ six values which must of course be equal to each other; it is easy to verify

[book p. 159] *a posteriori* that this is so.

In the case where the curve of double curvature is not the complete intersection of two surfaces, the denominator (regarded as the Jacobian of the curve and of the arbitrary planes ρ, σ) will have a definite meaning, but what this is I do not at present consider.

The last-mentioned expression for $d\omega$ will be applied further on to the case of the quadri-quadric curve.

Integral Formulæ. Art. Nos. 88 to 90.

88. In what precedes, $d\omega$ has been used as a single symbol to denote any one of the equal differential expressions

$$\frac{y\,dz-z\,dy}{\dfrac{df}{dx}}=\frac{z\,dx-x\,dz}{\dfrac{df}{dy}}=\frac{x\,dy-y\,dx}{\dfrac{df}{dz}};$$

there is no quantity ω. The expressions are of the order $-(n-3)$ in the coordinates (x, y, z), and since (x, y, z) are as to their absolute magnitudes altogether arbitrary (only their ratios being determinate), a symbol such as

$$\omega=\int d\omega=\int\frac{y\,dz-z\,dy}{\dfrac{df}{dx}},$$

would, except in the case $n=3$, be altogether meaningless. In fact, the integral would be

$$\int z^{n-3}\,\frac{d\Big(\dfrac{y}{z}\Big)}{\rho\Big(\dfrac{y}{z}, \dfrac{x}{z}\Big)}=\int\frac{d\Big(\dfrac{y}{z}\Big)}{z^{-(n-3)}\,\rho\Big(\dfrac{y}{z}, \dfrac{x}{z}\Big)},$$

where $\dfrac{x}{z}$ is, by the equation of the fixed curve, given as a function of $\dfrac{y}{z}$; but the other factor z^{n-3} is an absolutely indeterminate variable value, and the expression is meaningless.

But we have integrals $\int Q\,d\omega$, where Q is a homogeneous function of the order $n-3$ in the coordinates (x, y, z); and, in particular, we have such integrals where (corresponding to the forms which present themselves in the differential pure and affected theorems respectively) Q is either a rational and integral function $(x, y, z)^{n-3}$, or a rational and integral function $(x, y, z)^{n-2}$ divided by a linear function (x, y, z): for in every such case, the form of integral is

$$\int \frac{d\left(\dfrac{y}{z}\right)}{\Phi\left(\dfrac{y}{z}, \dfrac{x}{z}\right)},$$

where $\dfrac{x}{z}$ is a given function of $\dfrac{y}{z}$, and the factor of $d\left(\dfrac{y}{z}\right)$ is thus a mere function of $\dfrac{y}{z}$. More definitely, in the integrals $\int Q\,d\omega$ which are considered, Q is either a minor function $(x, y, z)^{n-3}$, or it is the quotient of a major function $(x, y, z)^{n-2}$ by the linear function 012.

[book p. 160] In the case $n = 2$, there is no rational and integral function $(x, y, z)^{n-3}$, but the function may be of the form belonging to the affected theorem, viz. it is unity divided by a linear function $(x, y, z)^2$; or say the integral is $\int \dfrac{d\omega}{ax+\beta y+\gamma z}$, where the (x, y, z) are connected by a quadric equation $(a, \dots)(x, y, z)^2 = 0$: it will be shown presently that this integral is obtainable as a logarithmic function.

In the case $n = 3$, we have the rational and integral function $(x, y, z)^{n-3}$, a constant, or say $= 1$, so that there is here an integral $\int d\omega$: we do not call this ω, but introducing a new letter, say u, and fixing at pleasure the inferior limit of the integral, we write $u = \int d\omega$.

89. In the foregoing form $\int Q\,d\omega$, so long as we retain the symbol $d\omega$, there is nothing to show what is the variable in regard to which the integration is to be performed; we may, for instance, writing

$$d\omega = \frac{y\,dz - z\,dy}{\dfrac{df}{dx}},$$

make it to be z, or in like manner to be any other of the six quotients. We thus cannot attribute a value to the inferior or superior limit of such an integral, but we may take the limits to be each of them a point on the fixed curve: for instance, if 0, 1 be points on the fixed curve, then the integral $\displaystyle\int_0^1 Q\,d\omega$ means the integral taken from the value at the point 0 to the value at the point 1 of the variable in regard to which the integration is performed; or when there is no expressed superior limit, then the integral is to be taken from the value for the expressed or known inferior limit to the value at the current point (x, y, z) of the variable in regard to which the integration is performed. The actual value of the integral will of course depend upon the path of the variable; but this is a question which is not here entered upon.

If using Cartesian Coordinates x, y, we write for instance

$$d\omega = \frac{dx}{\dfrac{df}{dy}}, \quad \text{then} \quad \int \frac{Q\,dx}{\dfrac{df}{dy}}$$

will denote an integral $\displaystyle\int \Phi x\,dx$ in regard to the variable x, and the inferior and superior limits will be as usual values of x; or if there is no expressed superior limit, then the integral will be $\displaystyle\int_0^x \Phi x\,dx$ the integral taken from the inferior limit x_0 to the current value x.

[book p. 161] We may, if we please, consider the coordinates $(x,\, y,\, z)$ as depending upon a parameter θ, viz. the ratios $x : y : z$ may be regarded as given functions of θ, and the integral $\int Q\, d\omega$ is then an integral $\int \Theta\, d\theta$, which taken from a constant inferior limit up to the value θ_1 belonging to a given point 1 of the curve, is a given function of θ_1, or say of the point 1. But except in the case of the cubic (or generally $p = 1$), we do not have the coordinates actually given as known functions of a parameter θ (say they are potentially known functions of θ), and it is further to be noticed that the functions which present themselves are functions not of a single point, but of p or more points: thus in the case of the quartic, $n = 4$, $p = 3$; we have $\int_0^1 x\, d\omega$, $\int_0^1 y\, d\omega$, $\int_0^1 z\, d\omega$, each standing for a given function of the parameter θ_1, but these integrals do not present themselves singly, but in combinations such as

$$\int_0^1 + \int_0^2 + \int_0^3 (x\, d\omega,\ y\, d\omega,\ z\, d\omega),$$

say these sums of integrals are u, v, w: each of the functions u, v, w is a potentially known function of the parameters θ_1, θ_2, θ_3, which belong to the points 1, 2, 3, respectively, and is consequently regarded as a given function of these four points.

90. Consider as before, in the case of a cubic curve, the integral $u = \int d\omega$: it will presently be seen that for the general curve as given by a cubic equation $f = 0$ of any form whatever, we arrive at a form of elliptic function: but the ordinary elliptic functions sn, cn, dn connect themselves most readily with the cubic curve $y^2 = x \,.\, 1 - x \,.\, 1 - k^2 x$. We have here

$$d\omega = \frac{dx}{2y} = \frac{dx}{2\sqrt{x \,.\, 1 - x \,.\, 1 - k^2 x}},$$

or, in the equation $u = \int d\omega$, taking the inferior limit to be 0, say

$$u = \int_0^x \frac{dx}{2\sqrt{x \,.\, 1 - x \,.\, 1 - k^2 x}},$$

an equation which determines u as a function of x, or conversely, x as a function of u. We might thence, by means of Abel's theorem as applied to the curve in question, investigate the properties of the function $x = \lambda(u)$ thus arising, and so establish the theory of elliptic functions: but it is more convenient, treating the elliptic functions as known functions, to write for λu its value; viz. to take for x as given by this equation, the value $x = \operatorname{sn}^2 u$: we thence have $y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$; viz. these values $x = \operatorname{sn}^2 u$, $y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$, satisfy the equation $y^2 = x \,.\, 1 - x \,.\, 1 - k^2 x$ of the curve, and give, moreover, $d\omega = du = \dfrac{dx}{2y}$: and we can with these values, and the formulæ for elliptic functions, verify any results given by Abel's theorem. This will be done in considerable detail: but at present I wish only to remark that the formulæ give the coordinates x, y of a point on the cubic curve expressed as one-valued functions of

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[book p. 162] a parameter or argument u: but that this argument u is not a one-valued function of the coordinate x, or even of the coordinates x, y of the given point on the curve: say the argument u has not a unique value for a given point $(x,\, y)$ of the curve. There are, in fact, an infinity of values $u = u + 2mK + 2m'iK'$, where m, m' are any positive or negative integers: that this is so, depends on the multiplicity of values, according to the different paths of the variable, of the integral

$$\int \frac{dx}{2\sqrt{x \,.\, 1 - x \,.\, 1 - k^2 x}},$$

or, regarding the elliptic functions as known functions, it depends upon the double periodicity of these functions.

Aronhold's Quadric Integral. Art. Nos. 91 to 93.

91. I reproduce the investigation contained in Aronhold's paper *"Ueber eine neue algebraische Behandlungsweise u.s.w.,"* Crelle, t. LXI. (1863), pp. 95–145. We take the general quadric function $(a,\, b,\, c,\, f,\, g,\, h)(x,\, y,\, z)^2$: $\alpha x + \beta y + \gamma z$ an arbitrary linear function of $x,\, y,\, z$: the theorem is $\dfrac{d\omega}{\alpha x + \beta y + \gamma z}$ = differential of logarithm of an algebraic function of $(x,\, y,\, z)$; viz. taking $(\xi,\, \eta,\, \zeta)$ for the coordinates of either of the points of intersection of the line $\alpha x+\beta y+\gamma z = 0$ with the quadric $(a,\, \dots)(x,\, y,\, z)^2 = 0$, and writing also

$$H = 2(bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch)(\alpha,\, \beta,\, \gamma)^2,$$

then the theorem is

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}}\, d \log \frac{(a,\, \dots)(x,\, y,\, z)(\xi,\, \eta,\, \zeta)}{\alpha x + \beta y + \gamma z},$$

or, what is the same thing,

$$\int \frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}} \log \frac{(a,\, \dots)(x,\, y,\, z)(\xi,\, \eta,\, \zeta)}{\alpha x + \beta y + \gamma z} + \text{const.}$$

It is to be observed, in reference to this equation, that the two sides respectively are in regard to $(\alpha,\, \beta,\, \gamma)$ homogeneous functions of the degree -1, and in regard to $(\xi,\, \eta,\, \zeta)$ homogeneous of the degree 0; viz. on the right-hand side the effect of a change in the absolute magnitudes of ξ, η, ζ, say the change into $k\xi$, $k\eta$, $k\zeta$, is merely to change by $\log k$ the constant of integration.

It is to be remarked also that the equation $(a,\, \dots)(\xi,\, \eta,\, \zeta)(x,\, y,\, z) = 0$ represents the tangent to the conic at the point $(\xi,\, \eta,\, \zeta)$ of intersection with the line $\alpha x + \beta y + \gamma z = 0$; calling the linear function in question T, the value of the integral is $\dfrac{1}{\sqrt{H}} \log \dfrac{T}{\alpha x + \beta y + \gamma z}$; the two $(\xi_1,\, \eta_1,\, \zeta_1)$ $(\xi_2,\, \eta_2,\, \zeta_2)$ are the coordinates of the points of intersection respectively, and the two values thus are $\dfrac{1}{\sqrt{H}} \log \dfrac{T_1}{\alpha x + \beta y + \gamma z}$, $\dfrac{1}{\sqrt{H}} \log \dfrac{T_2}{\alpha x + \beta y + \gamma z}$, then in passing from one of these to the other we change the sign of the radical $\sqrt{H}$, $\dfrac{1}{\sqrt{H}} \log \dfrac{T_1}{\alpha x + \beta y + \gamma z}$ and $-\dfrac{1}{\sqrt{H}} \log \dfrac{T_2}{\alpha x + \beta y + \gamma z}$. These must

[book p. 163] differ by a constant only: viz. we should have $\log \dfrac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2}$ = a const. And, in fact, T_1 and T_2 being the tangents to the conic f at its intersections with the line $\alpha x + \beta y + \gamma z = 0$, we have it is clear $f = \lambda T_1 T_2 + \mu(\alpha x + \beta y + \gamma z)^2$, that is, $(x,\, y,\, z)$ referring to a point of the conic $f = 0$, we have $\dfrac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2}$ = a constant, which is right.

92. We require the coordinates $(\xi,\, \eta,\, \zeta)$ of an intersection: these are determined by the equations $\alpha\xi + \beta\eta + \gamma\zeta = 0$, $(a,\, \dots)(\xi,\, \eta,\, \zeta)^2 = 0$, or as these may be written

$$\alpha\xi + \beta\eta + \gamma\zeta = 0,$$

$$\xi(a\xi + h\eta + g\zeta) + \eta(h\xi + b\eta + f\zeta) + \zeta(g\xi + f\eta + c\zeta) = 0\,;$$

we have thence ξ, η, ζ proportional to the determinants

$$\begin{matrix} a\xi + h\eta + g\zeta, & h\xi + b\eta + f\zeta, & g\xi + f\eta + c\zeta \\ \alpha, & \beta, & \gamma \end{matrix}$$

say these determinants are $\Omega\xi$, $\Omega\eta$, $\Omega\zeta$, where Ω is a value as yet undetermined. The equations are $\gamma(h\xi + b\eta + f\zeta) - \beta(g\xi + f\eta + c\zeta) = \Omega\xi$, &c., viz. these are

$$(\gamma h - \beta g)\,\xi + (\gamma b - \beta f - \Omega)\,\eta + (\gamma f - \beta c)\,\zeta = 0$$

$$(\alpha g - \gamma a - \Omega)\,\xi + (\alpha f - \gamma h)\,\eta + (\alpha c - \gamma g)\,\zeta = 0$$

$$(\beta a - \alpha h)\,\xi + (\beta h - \alpha b)\,\eta + (\beta g - \alpha f - \Omega)\,\zeta = 0\,;$$

eliminating (ξ, η, ζ), we have an equation which may be written

$$\begin{vmatrix} A-\Omega, & B, & C \\ A', & B'-\Omega, & C' \\ A'', & B'', & C''-\Omega \end{vmatrix} = 0,$$

that is,

$$\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} - \Omega(BC' - B'C + C'A'' - C''A' + A'B'' - A''B') + \Omega^2(A + B' + C'') - \Omega^3 = 0.$$

We find very easily that the determinant and $A + B' + C''$ are each $= 0$; and the equation thus reduces itself to

$$\Omega^2 = BC' - B'C + C'A'' - C''A' + A'B'' - A''B',$$

[book p. 164] **93.** We may now verify the theorem; in the general expression for $d\omega$ by λ, μ, ν the values ξ, η, ζ, the equation to be verified becomes

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\Omega}\,\frac{(\alpha, \dots)(\,\xi, \eta, \zeta\,)(\,dx, dy, dz)}{(\alpha x + \beta y + \gamma z)(\alpha, \dots)(\,\xi, \eta, \zeta\,)(\,x, y, z)},$$

and, in fact, each of the three lines is separately $= 0$, so the equation to be verified becomes

$$\frac{1}{(ax + hy + gz, \dots)(\,\xi, \eta, \zeta)}\big)(\,dx, dy, dz) = \frac{1}{\Omega}\,\frac{N}{(\alpha x + \beta y + \gamma z)(\dots)}\,;$$

viz. this is

$$\Omega\big\{\,dx, dy, dz, 2\,dy\,dz, 2\,dz\,dx, 2\,dx\,dy\,\big\}\big)(\,\alpha x + \beta y + \gamma z\,)($$

Here, on the right-hand side, the coefficient of dx is

$$= \alpha(a\xi + h\eta + g\zeta)(\alpha x + \beta y + \gamma z) - (ax + hy + gz)\big\{\alpha(\alpha\xi + \beta\eta + \gamma\zeta) - \Omega\xi\big\}$$

$$= \alpha\,\Omega\xi\,(\alpha x + \beta y + \gamma z),$$

which is right; and similarly, the coefficients of dy and dz have the same values on the two sides of the equation respectively.

Aronhold's Quadric Integral deduced from the Affected Theorem. Art. Nos. 94 to 99.

94. Let the fixed curve be a conic, say $f = (a, b, c, f, g, h)(\,x, y, z)^2 = 0$; and let the variable curve be a line meeting the conic in the points 2 and 4. The affected theorem is

$$\Sigma\,\frac{\delta\phi_2\,d\omega}{012} = \frac{\delta 24}{124} + \frac{\delta 24}{234},$$

where (x_1, y_1, z_1) and (x_2, y_2, z_2) are the coordinates of the points 1 and 2 respectively; 12 denotes the constant $(x_1, \xi\, x_2, y_2, z_2)$, and 012, &c., denote determinants as usual.

The left-hand side is here
$$\frac{1}{12}\Big(\frac{\delta\phi_2\, d\omega_2}{\delta_4} + \frac{\delta\phi_4\, d\omega_4}{\delta_4}\Big),$$
on the right-hand side, δ refers to the variation of the constants of ϕ, that is, of the variations of the points 1 and 2; or we may write $\delta = d_1 + d_2$, where the suffixed d's are independent, and the equation, being satisfied as all, must be satisfied separately

[book p. 165] in regard to the variations of δ, and in regard to the variations of 4: we must therefore have
$$\frac{d\omega_2}{12} = \frac{d_2 24}{124} + \frac{d_2 34}{234},$$
and the like equation obtained therefrom by the interchange of the numbers 2 and 4.

95. The equation just written down relates to any four points 1, 2, 3, 4 of the conic; and if 3 is to vanish, 0, 2 respectively, the equation is reduced to
$$\frac{1}{12}\frac{d\omega_2}{d_2 24} = \frac{d_2 24}{124} - \frac{d_2 01}{012},$$
which may be verified as follows: the equation is obvious; as to the second, observe that writing therein 0 for 4, it becomes $\dfrac{d_2 02}{012} = -\dfrac{d_2 02}{012}$, that is, $0 = 2\dfrac{d_2 02}{012}$, an identity. And the formula above obtained are
$$\frac{d\omega_2}{12} = \frac{d_2 02}{012} = \frac{d_2 12}{1\overline{2}}, \quad \text{that is,} \quad \frac{d\omega_2}{12} = \frac{12}{1\overline{2}}\Big(\frac{d_2 02}{012} - \frac{d_2 12}{1\overline{2}}\Big);$$
the equation in question.

96. We have, as a property of any four points 0, 1, 2, 3 of a conic,
$$\frac{23}{12}\frac{01}{1\overline{2}} = 0, \quad \text{or say} \quad \frac{23}{012}\frac{01}{1\overline{2}}\frac{23}{1\overline{2}} = \text{const.}, \quad \text{that is,} \quad \frac{23\overline{01}}{12\,\overline{30}} = \frac{01}{12};$$
hence considering as a variable point 1, and differentiating the logarithms,
$$d\log\frac{23\overline{01}}{12} = \frac{d23}{23} + \frac{d\overline{01}}{\overline{01}} - \frac{d12}{12},$$
and the foregoing equation $12\,\dfrac{d\omega_2}{d_2 24} = d\log\dfrac{23}{12}$ thus becomes $12\,\dfrac{d\omega_2}{d_2 24} = -d\log\dfrac{01}{12}$, or restoring u in value of 012,
$$12\,\frac{d\omega_2}{d_2 24} = d\log\frac{12}{01}.$$

Taking now $\alpha x + \beta y + \gamma z = 0$ for the equation of the line 012; this meets the conic in the points 1 and 2, the coordinates being to be taken: hence 0, 1 are
$$\alpha,\ \beta,\ \gamma = y_1 z_2 - y_2 z_1,\ z_1 x_2 - z_2 x_1,\ x_1 y_2 - x_2 y_1,$$
$$12 = a x_1 x_2 + b y_1 y_2 + c z_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(z_1 x_2 + z_2 x_1) + h(x_1 y_2 + x_2 y_1),$$
(the second term being of course $= 0$), viz.
$$\overline{12} = a x_1 x_2 + b y_1 y_2 + c z_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(\ldots) + h(\ldots),$$
$$= -(a, b, c, f, g, h\,\big)\!\big(\, y_1 z_2 - y_2 z_1,\ z_1 x_2 - z_2 x_1,\ x_1 y_2 - x_2 y_1)^2 + (\ldots).$$

[book p. 166] or say $12 = \Omega$, if $\Omega = -(bc - f^2, \ldots)(\alpha, \beta, \gamma)^2$ as before; and the equation thus is

$$\frac{\Omega\, d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \ldots)(x_1, y_1, z_1)(x, y, z)}{\alpha x + \beta y + \gamma z};$$

or finally, writing (ξ, η, ζ) instead of (x_1, y_1, z_1), $12 = \Omega$ becomes the coordinates of one of the intersections of the line $\alpha x + \beta y + \gamma z = 0$ with the conic, the equation becomes

$$\frac{\Omega\, d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \ldots)(\xi, \eta, \zeta)(x, y, z)}{\alpha x + \beta y + \gamma z},$$

which is Aronhold's quadric integral.

97. (The foregoing property, which may be also written

$$\frac{23\overline{01}}{012\,12\overline{3}} = \frac{01}{23},$$

is verified very easily in the case of four points 0, 1, 2, 3 of a circle: in fact, calling the differential $\Omega\alpha$, β, γ, δ, and the proper distances between the points

$$\overline{20} = \alpha\beta \,.\, 2 \sin \tfrac{1}{2}\alpha, \quad \overline{a01} = 2 \sin \tfrac{1}{2}\beta \,.\, 2 \sin \tfrac{1}{2}\gamma, \quad \overline{20\overline{12}} = 4 \sin \tfrac{1}{2}\delta \cdot \ldots$$

and so for the other like expressions; each side of the equation thus reduced to $1 \,.\, \sin \frac{1}{2}\delta \cdot 1 \,.\, 4 \sin \frac{1}{2}\delta \cdot 4 \sin \frac{1}{2}\beta \,.\, 4 \sin \frac{1}{2}\gamma$,

$$01\overline{2} = (1 - x + y \sin \tfrac{1}{2}\delta + 2)\, 4 \sin \tfrac{1}{2}\delta \cdot 1 \,.\, 4 \sin \tfrac{1}{2}\beta,$$

$\sin \overline{20} = (1 + \cos \overline{20})\, 2 \sin \frac{1}{2}\gamma \,.\, \frac{1}{2}\delta$, $LD = 2D\delta$, $LE = 2E\delta$, and so to the squarer in like manner to require $T = 1 - t$, $\alpha = 1$. We can see at once when $\overline{12} = 0$, viz. when the second is the right one.)

98. The foregoing investigation gives for these points α, β, γ, $\delta = 1, -\delta$. Hence

$$\frac{d\omega}{\sqrt{1 - x^2}} = -\frac{1}{2}\, d \log(x - iy),$$

viz. this is

$$\sin^{-1} x = -\frac{1}{2i} \log(x - iy) = \frac{1}{2i} \log(x + iy).$$

Fixed Curve a Cubic: the Paramétric Points 1, 2 *consecutive points on the Curve.* Art. Nos. 99 to 106.

99. The major function $(x, y, z)^{n-3}$ is taken to be $= \frac{1^2 \,.\, 023 - 12^2 \,.\, 013}{12s}$, so that, calling the differential $Q\, d\omega$, we have

$$Q = \frac{1^2 \,.\, 023 - 12^2 \,.\, 013}{12s \,.\, 012}.$$

I take for convenience the cubic to be $f, = (x^2 + y^2 + z^2)w, = 0$; the coordinates of 1 are (x_1, y_1, z_1), those of 2 are $(x_1 + dx_1, y_1 + dy_1, z_1 + dz_1)$, so that for $\overline{2}$ we have

$$\frac{x_1\, dx_1 + y_1\, dy_1 + z_1\, dz_1}{w} = \overline{2} \quad \text{or} \quad x_1\, dx_1 + y_1\, dy_1 + z_1\, dz_1 = 0;$$

write $1, 1, 1$ for x_1, y_1, z_1, so that the formulæ become for shortness

$$\overline{012} = x\, dx_1 + y\, dy_1 + z\, dz_1, \quad \overline{12s} = x_1\, dx_1 + y_1\, dy_1 + z_1\, dz_1, \quad \ldots$$

100. We have

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_3, & y_3, & z_3 \end{vmatrix}, \quad \overline{012} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ dx_1, & dy_1, & dz_1 \end{vmatrix} = O(1,\,3),$$

say. Moreover

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \end{vmatrix} = -(yz_1 - y_1z) + x_1(yz_2 - y_2z) - x(y_1z_2 - y_2z_1)\,(\&\text{c.}),$$

which by definition is $O\overline{23}$.

[book p. 168] as the second term may be written; moreover

$$12s\,.\,023 - 12\,.\,013 + 12^2\,.\,012 + 2(2ws + s^2)\,012 = wz_1\,012\,\cdot\,012\,\cdot\,012\,\cdot\,012,$$

$$12s\,.\,013 = O\overline{2s}\,.\,01 - O\overline{1s}\,.\,02 + 2(2ws + s^2 + s^3)\,012 + 2(2ws + s)\,012\,\cdot\,012,$$

and hence

$$1^2\,.\,023 - 12^2\,.\,013 = -(yz_1 - y_1z)\,O\overline{2s} + (yz_2 - y_2z_1)\,O\overline{1s} - 2(2ws + s)\,O\overline{1s}\overline{\overline{012}},$$

or reducing by

$$2(s_1z_2 - s_2z_1 + y_1y_2(z_2 - z_1) + z_1z_2(y_2 - y_1)) = (\partial_y y_1 + \partial_z z_1)\,012,$$

this is

$$= -2\,(s_1\overline{s_2} - s_1z_2)\,013 - (\partial_y y_2 + \partial_z z_2)\,012,$$

which is of the third order.

101. We may show that each of the terms contains the factor $(ds_1)^2$: we have, in fact,

$$y_1\,(ds_1)^2 = \frac{l}{y_1}\,.\,2y_1\,dz_1 - 2lz_1\,.\,2z_1\,dy_1 + \frac{l}{y_1}\,(\ldots),$$

$$z_1\,(ds_1)^2 = \frac{l}{z_1}\,.\,2z_1\,dx_1 - 2lx_1\,.\,2x_1\,dz_1 + \frac{l}{z_1}\,(\ldots),$$

$$x_1\,(ds_1)^2 = \frac{l}{x_1}\,.\,2x_1\,dy_1 - 2ly_1\,.\,2y_1\,dx_1 + \frac{l}{x_1}\,(\ldots),$$

hence, first multiplying by x, y, z and adding, we have

$$(xy_1z_1 + yz_1x_1 + zx_1y_1)\,(ds_1)^2 = \frac{l}{x_1y_1z_1}\,\big\{(xy_1 - 2lx_1)\,dy_1 - 2lx_1\,(yz_1)\,dx_1\big\} - \frac{2l}{x_1y_1z_1}\,(\ldots).$$

But in virtue of $ax_1^2 + by_1^2 + cz_1^2 = 0$; hence the equation is

$$(xy_1z_1 + yz_1x_1 + zx_1y_1)\,012 = (a' + b' + c')\,012,$$

the required expression for the first term.

102. Again, multiplying by x_1, y_1, z_1 and adding, we have

$$\Sigma\,x_1y_1z_1\,(ds_1)^2 = \frac{l}{x_1y_1z_1}\,\big\{4y_1z_1\,dx_1 - 2l(x_1 + y_1 + z_1)\,dx_1 + 4z_1x_1\,dy_1 + 4x_1y_1\,dz_1\big\}.$$

[book p. 169] when, in virtue of $x_1^2+y_1^2+z_1^2=0$, the first line is

$$= -\frac{2}{x_1y_1z_1}\{(3y_1z_1+y_1^2+z_1^2)\,dx_1+\cdots\},$$

and this again is $= -2(x_1\,dx_1+y_1\,dy_1+z_1\,dz_1)$: in fact, we have identically

$$x_1y_1z_1(2y_1z_1+3\partial_y y_1+\partial_z z_1)+(x_1^2+y_1^2+z_1^2)\,dx_1+\cdots$$

$$=(x_1y_1z_1+2lx_1+2lx_1)\,dx_1+(\partial_y y_1+\partial_z z_1)\,dx_1+\cdots$$

which, in virtue of $ax_1^2+by_1^2+cz_1^2=0$ and $a^2+b^2+c^2=0$, becomes

$$=a(\partial_y y_1+\partial_z z_1)\,(dx_1+dy_1+dz_1).$$

Hence the equation is

$$\Sigma\, x_1y_1z_1\,(ds_1)^2=2(ax_1+by_1+cz_1)\,012,$$

or finally

$$x_1y_1z_1\,(ds_1)^2=(ax_1+by_1+cz_1)\,012,$$

the required expression for the second term.

103. Writing for shortness

$$1^2\,.\,023-12^2\,.\,013=\frac{1}{4}(a,\,b,\,c,\,f,\,g,\,h)(\,x,\,y,\,z)^2,$$

we have

$$1^2\,.\,023-12^2\,.\,013=-\tfrac{1}{4}(a,\,b,\,c,\,f,\,g,\,h)(\,x_1,\,y_1,\,z_1\,)(\,012)\,\cdot\,(ds_1)^2,$$

and hence dividing by

$$012\,.\,12s,\ =012\,.\,\frac{1}{2l}(dx_1)^2,$$

we have

$$Q=\frac{1^2\,.\,023-12^2\,.\,013}{012\,.\,12s}=-\frac{l}{2}\,(a,\,b,\,c,\,f,\,g,\,h)(\,x_1,\,y_1,\,z_1\,)(\,0613,\,\ldots).$$

But this can be further reduced: its numerator, multiplied by 3, is

$$=-(xy_1z_1+2lx_1)\{x_1\,)(\,\cdots\}+\frac{x_1^2}{x_1y_1z_1}\,\{2yz_1\,(2ly_1z_1)+\cdots\},$$

which is

$$=y\,\{x_1\,(y_1-z_1)-2(x_1^2-2ly_1z_1)\}+z\,\{(\ldots)\}\,\frac{1}{x_1y_1z_1}\,(\ldots),$$

where $a_1(y^2-z^2)$, $y_1(z^2-x^2)$, $z_1(x^2-y^2)$ are the coefficients of the tangential of the point 1 in regard to the cubic, etc. in usual notation; and we have thus

$$Q=\frac{12\,.\,023-12^2\,.\,013}{012\,.\,123}=\frac{12\,.\,613}{012\,.\,123}$$

where $0\bar{1}\,.\,0$ is the equation of the tangent of the line 012; this meets the conic in the points 1, $\bar{1}$ regard to the cubic, where $\bar{1}$ is what may be called the harmonic of the original.

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[book p. 170] **104.** The identity just referred to is proved very easily. Computing on each side the coefficient of yz, $-yx$, the factor a, divides out and we ought to have

$$-(xy_1z_1+2lx_1+yz_1y_1)+(x_1z_1+2lz_1)(y_1^3-z_1^2)\,xz_1=(y_1^2-z_1^2)\,ax_1,$$

that is,

$$(yz_1^3 - 2lz_1^3y_1)(y_1^2 - z_1^2) = ax_1.$$

and, in fact, from $x_1^2 = ax_1$, we have for a an integral $\int \frac{dx}{a}$; and the equation is thus verified.

105. The proof has been given in the present case, $f = ax_1^2 + by_1^2 + cz_1^2$, where the cubic is general, the equation is similarly the coefficients of ax_1, ay_1, az_1, equal as the cubic, and the equation is thus verified.

106. We have

$$\overline{028} + \overline{001} + \overline{012} = \overline{12s},$$

or writing this in the form

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0\,;$$

suppose 2 is here the consecutive point $1 + d1$; then

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0$$

becomes $= 02(1d1)$; we have also

$$\overline{021} + \overline{012} = -\,01\,d\,12,$$

and the result is

$$-\overline{012}\,.\,d\overline{12} = 0,$$

that is, $\overline{012}\big[\overline{12} + \overline{12} + d61\big] + d61y\,012 = 0$, is

$$= \frac{1}{0!}\,\frac{1}{12!}\,(d^2d^2 - d^2d^2)\cdots(d^2 - d^2)\,;$$

in the differential of $\overline{012}$ in regard to the paramétric point 1 is $\overline{012}_2$, the symbol for the case where the paramétric line is the tangent at 1.

Formulæ Relating to the Tangents from the Vertices.

121. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say $\rho = 0$, $\sigma = 0$, symmetrically situate in regard to the axis, and $\sigma = 0$, $\sigma = 0$, symmetrically situate in regard to the axis: the four tangents from the vertex A are accordingly equations $\rho = 0$, $\sigma = 0$, denoted by $\rho = 0$, $\sigma = 0$, $\rho = 0$, $\sigma = 0$, the tangents at $\sigma = 0$, $\rho = 0$, the tangents at $\sigma = 0$, $\sigma = 0$.

[book p. 171] Fixed Curve a Cubic: the Paramétric Points 1, 2 *corresponding points.* Art. Nos. 107 to 110.

107. The paramétric points 1, 2 are taken to be corresponding points, that is, such that the tangents at these points meet at a point of the cubic. We may take, in the very particular case considered, the cubic $y^2 = 1 - x\,.\,1 - k^2x$. The four points 1, $1'$, $2'$, 2 are then such that the lines $11'$ and $22'$, viz. the lines $\tau = 0$ and $\tau' = 0$, meet at a point 7 on the cubic, T being a properly determined linear function of $(x,\,y,\,z)$; it is a properly determined linear function of $(x,\,y,\,z)$ such that the tangents from T to the cubic, that is the line $T = 0$ being any line through the two points 1, 2 meet in a point 9 as above; viz. if we write

$$f = yz^2 + 2lxyz + (x^2 + 2ay + bz)\,xz = 0$$

and writing

$$d\omega = \frac{x\,dy - y\,dx}{\dfrac{df}{dz}},$$

which, from the equation of the curve written in the form

$$z(yx + 2ay + bz)\, x + (1 + 2ay + bz)\, xy + \xi z(xz + by) + lxyz = 0,$$

we have

$$\frac{df}{dz} = \Sigma(xz + by + cz),$$

or say

$$x(xy + by + cz) + xyz = -2lyz\big(\frac{l}{x}\big),$$

becomes

$$= \frac{-2lyz}{\xi},$$

and we have here

$$d\omega = \frac{\xi\,(dy - y\,dx)}{-2lyz}, \quad = -\frac{1}{2l}\,\xi\,\frac{dx}{(\,dy\,)},$$

where $\xi'\,dx + 6y + lz$. To find the meaning of ξ, observe that the line $x = My = 0$ tangents at the points 1 and 2, which is the case in which the case where, $a = My - 6$ are equal $T = 9$; whence in the case where the tangents at the point 1 and 2, the lines $\overline{1y} = \overline{12}$ and $\overline{6y} = \overline{12}$ meet at a point T on the cubic, T being a properly determined function $\overline{T}$ of $(x,\, y,\, z)$, this line will be a tangent if

$$(yM + \xi)(M^2 + h) = 0,$$

this line will be a tangent at

$$(yM + \xi)(M^2 + \lambda) - 0\,;$$

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[book p. 172] and we then have the point of contact $\big(\lambda M + \xi y + \lambda = 0\big)$, and writing this in the form

$$\xi xy + 2ay + cz = 0\,;$$

we see that the equation $x = My + \xi z = 0$, the curve $\xi = \xi z + cy + Aps = 0$. Moreover, from the equation of the curve written in the foregoing form

$$\xi\,(\xi x + 2ay + cy)\, x + 2yz = 0\,;$$

it appears that the lines $x = My + \xi z = 0$, $x = My + cz = 0$, viz. the curve, that the line through the two paramétric points 1, 2 meets the curve at a point $\overline{T}$ on this it is essential it is essential it is essential.

108. The equation $\xi = 1$ becomes $= ds$, in fact,

$$2\,.\,1\overline{2} = (a,\ b,\ c\,)\!(\, x_1,\, y_1,\, z_1\,)\!(\, x_2,\, y_2,\, z_2), \quad (x_1,\, y_1,\, z_1) = (1,\, 0,\, 0),$$

$$= \frac{1}{l}\, c\xi_2,$$

$$= \frac{1}{l}\, c\,(\, xy_2 + \xi_2 x_1 + 2c\, y\, y_2 + \cdots), \quad (x_2,\, y_2,\, z_2) = (1,\, 0,\, 0),$$

$$= b.$$

We have thus $\xi =$ satisfying the required condition for the major function: and the differential $Q\,d\omega$ may therefore be taken to be $= \dfrac{a}{ay}$, that is, we have

$$Q\,d\omega = \frac{x}{ay}\big)\!\big(\frac{dx}{ay}\big)\!\big(\,.$$

The affected theorem then becomes

$$2\,\frac{Q\,.\,d\omega}{012} = \frac{d\phi_1}{\phi_1} + \frac{d\phi_2}{\phi_2}\,,$$

Integrating each side, and assuming that the superior limits are given by a line ϕ which cuts the cubic in the points 4, 5, 6 and the inferior limits by a line ϕ' cutting it at the points a, β, γ, we have

$$\log\frac{x_1y_1z_1z_1}{x_2y_2z_2y_2} = 2\,\log\frac{\phi_1}{\phi_1'},$$

that is,

$$\frac{x_1y_2z_3z_4}{x_4y_5z_6y_2} = \frac{(\phi_1)^2}{(\phi_1')^2}\,,$$

where ϕ_1, ϕ_2, ϕ_3 denote the values of the linear function ϕ at the points 1 and 2 respectively. We have a cubic cut by the lines ϕ, ψ, x, y in the points 4, 5, 6, 1, a, β, γ, 2, and we have first the points where in course we have ϕ, y replaced by their values, and the formula for the integral of the points $(\alpha,\,\beta)$ is

$$(\frac{xv\,t}{a})((\frac{\phi_1\,t}{\phi_1'})((\frac{\phi_2\,t}{\phi_2'}) = (\frac{\phi_1\,t}{\phi_1'}).$$

[book p. 173] that is,

$$\frac{x_1\overline{y_2}z_4z_5z_6}{x_2\overline{y_2}z_az_\beta z_\gamma} = \frac{\phi_1\,\phi_2\,\phi_3\,\phi_4\,\phi_5\,\phi_6}{\phi_1'\phi_2'\phi_3'\phi_a\phi_\beta\phi_\gamma}\,,$$

which, dividing out the $\phi_1\phi_2$, and writing 1, 2 in place of $1'$, $2'$, becomes

$$= \frac{(\phi_1\phi_2)^2}{(\phi_1'\phi_2')^2},$$

agreeing with the result just obtained.

Aronhold's Cubic Transformation. Art. Nos. 111 to 119.

111. This was obtained in the paper *"Algebraische Reduktion des Integrals $\int F(x,\,y)\,dx$, u.s.w.,"* Berl. Monatsb., April, 1861, pp. 462–468. I give in the first place the analytical results, independently of the general theory, with the values for the canonical form f, $= \frac{1}{4}(x^2 + y^2 + z^2 + 6lxyz)$, $= 0$, of the cubic. We have

$$T = \text{auxic invariant}, \quad = 1 - 20l^3 - 8l^6,$$

$$S = \text{quartic} \quad (\text{Aronhold's} = 4(l - l^4)),$$

$$R = \text{discriminant}, \quad = (1 + 8l^3)^4,$$

$$P = 9b^2l^2 = [-30l^2 + (1 + 20l^3 + 8l^6)]\,a + \&\text{c.},$$

$$Q = fyl^2 = (a^3 + 2lc)y\,x + \&\text{c.},$$

$$B = fy\dot{y}^2 = (a^3 + 22lac)y\,x + \&\text{c.},$$

$$C = fx^2y = (a^3 + 23)yx + \&\text{c.},$$

$$D = fx^2z = a^2 + yx + x^2 + 6lxyx.$$

where a, b, $c \equiv (P - x)^3$, $\bar{S}(x^3 - x^3)$, $\bar{x}(z^2 - x^3)$.

Then we have

$$27Q^2 + 54P^2 = (1 - lP)^3\,QP \quad = R^2(8C^2 - 8RD),$$

viz. this equation, where each side is a quartic function $(x,\, y,\, z)^2$, is an identity when $(a,\, B,\, c)$ are connected by the equation, $f\,a^3 = a^3 + b^3 + 6labc = 0$; and further,

$$Q4P^2 - PdQ - R\,fz_2 = R^2\,(R^2 - 7R)\,fz_2 = R^2\,(R^2 - 7R)\,fz_2.$$

Hence writing

$$\Lambda = \frac{6loRQ}{(BC)^2} - \frac{2P}{Q} - \frac{Q^3}{Q^3} = \frac{1 - 30W^2 + (1+27)\,Wlc + \&\text{c.}}{(b^3 + 2\,2l)\,a + \&\text{c.}},$$

we have

$$Q^*(\Lambda - 3\Lambda + 27) = R^2\,(C^2 - 8RD).$$

and

$$Q^*\,d\Lambda_a = 2(Q\,Q^2 - P\,dQ) = -2R^2\,(c\sqrt[3]{(c\,d_a - d\,d_a)} + \&\text{c.}).$$

[book p. 174] **112.** Supposing now that $(x,\, y,\, z)$ are the coordinates of a point on the cubic, that $D = 0$; and taking the square root of each side of the first equation, we may write

$$Q^4 = 3RR - 3RS\,A^2 = -2(P^2 + (yI + dy + \&\text{c.}),$$

We have

$$d\omega = \frac{ax\,dy + dy\,dcx}{x^2 + 2lcy} = \frac{a\,dx - x\,dz}{a + 2lcy} = \frac{a\,dy - y\,dx}{x^3 + 2lcy},$$

whence

$$d\omega = \frac{x(y\,dx - x\,dy)}{x^3 + 2lcy},$$

and we consequently have

$$\frac{d\omega}{\sqrt{x^3 - 3x^3 - LR^2\sqrt{\bar{x}}}} = \frac{1}{\sqrt{R}}\,\frac{dy}{y^2 - 3y^2 - LR},$$

which, if 128, bP are put $= p,\, \bar{p}$, respectively, takes the Weierstrassian form

$$\frac{d\omega}{\sqrt{x^3 - px - \bar{p}}} = \frac{1}{\sqrt{R}}\,\frac{dy}{\sqrt{y^3 - py - \bar{p}}}.$$

The conclusion is that for the cubic curve, taking Λ a quotient of two linear functions of $(x,\, y,\, z)$, the differential $d\omega$ is transformed into $d\Lambda$ a square root of a cubic function of Λ: viz. we have from a form of differential $d\omega$, which belongs to the ordinary theory of elliptic functions, and which has been adopted by Weierstrass as a canonical form.

113. The transformation depends on the arbitrary point $(a,\, \beta,\, \gamma)$ of the cubic: the point $(a,\, \beta,\, \gamma)$ is the tangential of this point, viz. the point of intersection of the tangent at $(a,\, \beta,\, \gamma)$ with the cubic: we can from $(a,\, b,\, c)$ draw three tangents to the cubic, viz. the tangent at $(a,\, b,\, \gamma)$ and three other tangents: the equations of the four tangents being $\frac{4P}{Q} - \frac{8P}{f}y, = a,\, \lambda,$ $\lambda,\, \lambda$, respectively; where $\lambda,\, \lambda_1,\, \lambda_2$ are the roots of the equation $\lambda^3 - 3R\lambda - 12 = 0$.

Suppose for a moment that $(\overline{a},\, \overline{\beta},\, \overline{\gamma})$ is a point on the cubic curve, and write $A = a' + B$ y $+\ \overline{6}^2$ Such. We have

$$A'D^2 + 4A'C^3 + 6PD^2 - 3DPC^2 - 6A'RCD = 0,$$

for the equation of the six tangents which we know drawn from the point $(a,\, \beta,\, \gamma)$ to the cubic: when $(\overline{a},\, \overline{\beta},\, \overline{\gamma})$ is a point on the cubic, $A = 0$, and the equation reduces itself to a quartic function, $f\bar{a} = a' + \beta + \gamma + 6lab\gamma, = 0$; and the equation $\dfrac{4P}{Q} - \dfrac{8P}{f} = 0$, throwing out the factor B, is an identity.

The last-mentioned expression for the $d\Lambda$ depends on the points A, B, C, D as the four tangents to the cubic from the points $-3K$, $1, 2K, 1, 6, 27K\, y \to b$, those points are the four tangents from $(a,\ \beta,\ \gamma)$ to the cubic.

Fixed Curve the Quartic $y^2 = (1 - x^2)(1 - k^2 x^2)$. Art. Nos. 145 to 145.

145. This is a curve having a uniquely twin to inflexion: there are at this curve being. The points a, β, β' are double points. The equation of the curve are equal $A = a^2$, $K = a^2$ resolved into seven points 1, 2, 3, 4, 5, 6, 7, with the order of points be $y^2 = (1 - x^2)(1 - k^2x^2), = 0$; and the cubic forms a circle and is involves more terms in the more concrete fashion of 1 and 2. We have $a, b'a = (1 + a)c$; and $b'a = -a'(1 + a^2)\, c$, those of the four tangents from $(0, 0, 0)$ as expressed by $\dfrac{6}{4 - x_0}$ are equal $\dfrac{6}{4 - x_0}$.

[book p. 175] The foregoing expression for $Q\, dP = P\, dQ$, viz. the equations of the four tangents at the points of contact thus are obtained, where the dynamical conditions that Aronhold obtained his results.

114. The foregoing expression for $Q\, dP - P\, dQ$, viz.

$$Q\, dP - P\, dQ = (Aa + By + Cz)(Adx + B\, dy + C\, dz) + Aa\, da + Bb\, db + Cc\, dc - (BC' - CB'\, da) \cdots ,$$

may be verified without difficulty. Writing for a moment

$$Q\, dP - P\, dQ = (Aa + By + Cz)(\Lambda dx + 3\Lambda dy + Adx) - (Add\, a + Bdy\, db + S\, dc) - (BC' - CB')(L\, a + L\, db + L\, da) \cdots ,$$

we have

$$\begin{aligned} RX &= -CR + a\bigl[-2yQP - (1 + 20l^3)\, X\bigr] - (1 - 5l)(1 + 20\, X^3 + 5l^2)\, a\, Y - \text{\&c.}, \\ &= -(1 + 30l^3)\, (a^3 + 2lac)\, a\, \dot{x} + (1 - 5l)(1 + 5\, l^2\, (1 + 8l^3)\, a\, Y\, Y - \text{\&c.}, \\ &= -8l^3(1 + 20l^3)(a^3 + 2lc)\, a\, \dot{x}, \\ &= -16l\, (a^3 + 2lac)\, a\, \dot{x}, \end{aligned}$$

which prove the theorem.

115. I content myself with a partial verification of the identity

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 - (1 + 8L^3)^4(8C^2 - 8RD).$$

Writing herein a, b, $c = 1$, 0, 0, we have $D = 0$, and the equation becomes

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 = (1 + 8L^3)^4\, (8C^2 - 8RD)\,;$$

where now

$$Q = a - 2l(a + \beta - 8l\gamma), \quad P = (a - \beta) - 5l^2(a + \beta - 5l\gamma) + (1 + 25\gamma^2)\,;$$

which, putting therein $a^2 + a^2 + \beta^3 = 6la\beta\gamma$, becomes

$$= -(a - \beta)(a + B - 5l\gamma),$$

Hence writing

$$X = a + \beta - 5l\gamma, \quad Y = -5l(a + \beta) - (1 + 25\gamma^2)\gamma,$$

we have

$$Q = P, \quad C = (a-\beta)\,X, \quad (a-\beta)\,Y, \quad (a-\beta)\,X\,;$$

substituting these values, the factor $(a-\beta)^4$ divides out, and the equation becomes

$$2TX^2 + 6T^2X - 9T^4 + 4T^4 = (1+8\,T^4)^4(8\,Y - 8\,X^4)\,;$$

To complete the verification, observe that we have $Y = 8\,T^4\,X^2 - (1+8\,T^4)\,X$, whence

$$-Y = (1+8\,T^4)(\,YX + 8\,(\,PY - X^2)\,Y\,T\,X^4)\,YT^4\,X^2.$$

[book p. 176] and herein $-y' = a^2 + 8laa\beta_1$, whence

$$-y' + a^2X = -y + (1-5T^2)(8\,a + 8BX) + 8T^2 + 16y\,YX + 8T^2y\,X,$$

that is,

$$-(1+8T^2)\,a + X^2 = a^2 + 5T^2\,y\,(\,X) + 8(15)\,y(\,XY) - (X^2\,X)\,X^2 + 16T^2\,X^2.$$

Hence the equation to be verified becomes

$$2TX^2 + 6XT^2\,X = (1+8X^2)(20X^4 - 4X) + 12LX^2\,X\big).$$

$$= (1-5T^2)(2y - 4y_2)(2X^4 + 16X) + (12L^2)\,Y$$
$$-\,(X+X^3)Z\,X^3,$$

$$= 0\,X^2 + 8T^2(y'\,X)\,X.$$

which is again the factor X, and which can be thrown out: the equation thus becomes

$$2TX + 6T^2 + 6TX = (1+8T^2)(20X^4 - 16) + (12L^2)\,(X+4y)\big) = 0.$$

This may be written

$$2TX + 6T^2 + 6TX = (1+8T^2)(20X^4 - 16) + (12L^2)(X+4y) = 0\,;$$

or, finally, it is

$$2TX + 6T^2 + 6TX - 30T^4 - 8T^4 + 4(1+8T^2)\,(X^2 + 8T^2)(X+4y) = 0\,;$$

$+-\ a^2 - X - 4y(1+8T^2) + 2T(2+16T^2)\,(X^2+8T^2)\,X = 0$, substituting for T, X their values $1 - 200 - 8l$ and $-a + 4l - 5y$ respectively, the coefficients of X^2 and $(1+8l^3)$ are separately $= 0$, and the equation is then proved.

116. The foregoing equation $\Lambda = \dfrac{6loyQ}{(b^3)^2} - \dfrac{8P}{fy}$, regarding therein Λ as an arbitrary parameter and $(x,\,y,\,z)$ as current coordinates, is the equation of an arbitrary line through the point $(a,\,b,\,c)$ of the cubic: it meets the cubic in two other points (depending, of course, on the value of Λ); and the equation considered as relating to these points is

[book p. 177] $\dfrac{E\,fa}{fy} = \dfrac{C\Lambda}{\sqrt{P^3 - 3\Lambda S^2 + T}}$. The expressions answer a peculiarly simpler form when $(a,\,\beta,\,\gamma)$, instead of being an arbitrary point of the cubic, is a point of inflexion on the cubic; and it is easy to see a priori why this is so: in fact, if we assume

$$x : y : z = a + \lambda M,\ \beta + \lambda N,\ \gamma + \lambda R\,;$$

where a, β, γ are linear functions of Λ and the modulus a, the inflexion point of the cubic from which we proceed; observe also, that depending on a, β, γ, take an integral, by elimination, such a proper inflexion at the point of inflexion, we have

$$x : y : z = a + \Lambda M,\ \beta + \Lambda N,\ \gamma + \Lambda R.$$

where, λ, ν, ρ are constants! And then with these values of Λ, M, and for elimination, Λ, ν, ρ are also linear function of the equation of Λ, viz. $(\lambda)\,(\beta - \lambda\, P\, X)\, c)\, \rho = X,\ y(\lambda - \beta\, W)$.

117. To work this out, we start from the equation

$$\Lambda = \frac{6loyQ}{(BC)^2} - \frac{2P}{fy}, \quad fa = (1 + 8l^3)\, a, \quad B = 2\, a\, a^2, \quad C = 12lc^2 a,$$

hereon $Aa + By + Cz = 0$ is the equation of the line through the point a, b, $c = 0$, 0, a; or as before, $a = 0$. We can the same equation written, $Ab + By + 0$, $C\, ya$.

We have a, b, $c = 0$, 0, a; $B = a, y, z = 4a, b, c, ax^3$, $C = 12lc^2 = 12y$, $LD = 6lz$, $\xi = D$; in ξ at $(0, 0, a)$, the cubic function reduces to a in the form

$$Ax + By + Cz = -4ay^2 + 12lyz + 6lyz = (-4a + 12ly + 6lz)y\,;$$

from which we as yet unknown coefficient LR is to be obtained. This is most easily effected by assuming a, y, a, 0, values which give Λ obtained $-4a - 1$; whence

$$C' = a^4 - 6la(y\, xy + y + (1 + 8l^3)\, L\, y'\, (y'\, a^4) - 6lxysc) = 2L\, (y'),$$

$\delta C = 2L\, (y\, (y\, x + y\, y) + 6l\, y\, a\, (y\, x + y\, x) - 24l\, (y\, a\,)\, c)$. from which on yet unknown coefficient L is to be obtained. This most easily effected by assuming a, y, a, 0, values which give $\Lambda = a$ obtained. We need easily obtain by writing $y = a^2$, 0, 0, which from the previous results so above

$$C' = (1 + 4a^2\, (1 + 8l^3)\, a\, Y \cdots ()\,.$$

the whole equation becomes a quartic in y, satisfied by all, must be satisfied separately

$$C = U + (3l^2 (C + 8l^3)\, L\, ((P\, A^2 + 3\, (L\, A^2)\, L (C\, P\, B +\ A^2)\, (C\, L\, (\ A^2)\, A^2 + 12\, 1\rho A^2.$$

[book p. 178] where for C is to be substituted the value $LX + 2RAS$. We may also reduce by reducing also the right-hand side, the whole equation becomes

$$C' = L'(\,x' + By + 6lG\bar{a})(C + LX)\,;$$

which after reduction is found to contain only the factor a, and omitting this factor and reducing also the right-hand side, the whole equation becomes

$$\begin{gathered} L(x^3 + Dy + 6l\, y) + L^2 A^2 = -3yC, \\ +LM(By + 34la)\, B + LM(C + B,)\, B\, 2\, (LX)\, (LX) \\ +(LX)(Q + 26la^2)(C + 20) \\ +LB\, (LX + 6\, l\, B + 9\, (L\, L)\, B + 4l(L\, L\, (C\, L)\, L^2 + 0)\, 4 \\ +(\, B\, L^2\, Q +\ LX^2\, B + 2L)M(P) + 2lL\, M + 0)P \\ +4(L\, B - 4lQ)(B + 3l^2)\, Q + LM(P)P + 12X^2 + 0)P \\ +1(2 + L\, A^2)\, P\,; \end{gathered}$$

omitting from these the factor C, on putting therein the values of C, D, the equation contains a^2 as a factor; and dividing also throughout by aS, the equation becomes a quadric function in Λ and the equation thus contains Λ, LX, LX as the coefficients of the resulting quadric.

We find very easily that the differentiating, always by $a + 6y + 6lyz = 0$;

$$\frac{1}{2}(2P - 9R) = ly^2 + 6lyzX' + 2L\,(X + BX)\,;$$

But we have

$$(L = (L\,L) + 6lLX + R\,Y\,(C\,L\,X\,) + 2lL\,(R\,L)\,X)$$

the equation in $(x,\,y)$ is

$$By = -3(2x + 9y)\,Y\,;$$

or

$$y^2 + 9y) = d^2\,\sqrt{X}.$$

We had evident difficulty, that taking always $y = 4 + 6y(\,L\,X\,Y\,)\,X = 0$, the factor which divides out being in each of these cases

$$\sqrt{2} = \sqrt{x}(1 + a),$$

[book p. 179] which is

$$= (LL - M)(L\,A^2)\,LL\,M + B(L\,LX)\,XL - LL\,XL - LL\,XL\,;$$

But we have

$$(L - M)(z + LL\,X)\,LX + (L - M)\,Y\,XLY\,(\,X\,Y\,XL)\,XL - LL,$$

showing that the solution involves the factor $\sqrt{LX^3 + 5L\,X^2 - X^2}$.

$$\frac{1}{2}(QL - L\,XL)\,XL = (1 + 8L^3)\,XL - 2L(\,L + l + 2L^2)\,L\,(\,L^2\,Y L^2)\,.$$

showing that the equation in $(z,\,y)$ is

$$A = \frac{-6L^2 - L(L^2 + 2T^2 + 8X^2)\,y}{x + y - 5L\,z}\,;$$

the equation in $(x,\,y)$ is, in fact,

$$(B + 8L^3)\,y^2 + (1L + L\,c)\,y + (-6L^2 + L\,(L^2 + 2L^2)\,y)\,x = 0,$$

or, as this may be written,

$$(L + 8L^3)\,x_a + 2L\,Y\,y_a + 8L^2X_a + 8X_a^2X_a + 4l(L\,L\,(L, M))\,y = 0\,;$$

viz. we thus have

$$\sqrt{1 - 25\,X^2} = (1 + 25L)\,a + \sqrt{2L + 8L^2 - 6T}\,;$$

or, retaining M in the relation

$$y = \sqrt{2}(a + 5L)\,a + 5L\sqrt{1 - 25\,X^2 - 8L^4 - 8X^2}\,;$$

$x = \sqrt{2}(a + 5L)\,a + 5L\sqrt{1 - 25\,X^2 - 8L^4 - 8X^2}$; viz. we thus have

$$x = \sqrt{2}(1 + 8L^4 - 8L^3 - 8L^4 - 8L^4)\,;$$

where also

$$x = \sqrt{2}(2 + a),$$

these values satisfying obviously

$$(a + 6y)(1 - L^2)^2 = (a + 6y)\, y\, L^4 - 27\,T\,.$$

119. Starting from these values we, in fact, easily obtain

$$\frac{a\,dy + y\,dx}{a^4 + y^4 + a^2} = \frac{-\sqrt{x}\,dx}{\sqrt{x - 8\,X^4}}\,;$$

$$2y\,d\dot{x} + (1 - 6\,L^3)\,(d\dot{x}) = -(12 + 12L^3 + 8x^3) - 4(-72 - 8L)\,a^2,$$

$a^2 - 25y_4 = 2(11x_a)$, and hence

$$d\omega = \frac{x\,dy + y\,dx}{a^4 + y^4 + a^2} = \frac{1\sqrt{x}\,dx}{\sqrt{x - 8\,X^4 - 8L\,X}}.$$

The same result might of course have been obtained from the values of a, x, or y, a: the factor which divides out being in each of these cases respectively.

23–2

[book p. 180] *The Cubic* $y^2 = x\,.\,(1-x)\,.\,(1-k^2x)$. Art. Nos. 120 to 130 (*several sub-headings*).

120. The curve is a semi-cubical parabola, symmetrical in regard to the axis of x; and if, as usual, k^2 is to be taken to be real, positive and less than 1, the curve consists of an oval, and an infinite portion which may be called the sagitta. (See Figure.)

The equation is satisfied by

$$x = \mathrm{sn}^2\, u,$$

$$y = \mathrm{sn}\, u\ \mathrm{cn}\, u\ \mathrm{dn}\, u.$$

Observe that the periods for these combinations of the elliptic functions are $2K$, $2iK'$; in fact, we have

$$\begin{array}{llll} \mathrm{sn}(u + 2K) & = -\,\mathrm{sn}\, u, & \mathrm{sn}(u + 2iK') & = \mathrm{sn}\, u, \\ \mathrm{cn} & = -\,\mathrm{cn}\, u, & \mathrm{cn} & = -\,\mathrm{cn}\, u, \\ \mathrm{dn} & = \mathrm{dn}\, u, & \mathrm{dn} & = -\,\mathrm{dn}\, u. \end{array}$$

whence the sn^2 and the $\mathrm{sn}\, u\ \mathrm{cn}\, u\ \mathrm{dn}\, u$ both unaltered by the change of u into $u+2K$ or $u+2iK'$. Hence to a given u as regards sn may be equally well taken to be a determinate value (for sn^2 is a determinate value $\mathrm{sn}\, u$), we may, for instance, attribute to u any one of the four points belonging to those values of u respectively; thus, $\{u, u + K, u + 2iK', u + K + iK'\}$.

121. If, to fix the ideas, we select for each point of the curve one of the geometrical arguments, viz., that arising from a positive value of the radical, then taking 1, $1'$, 2, $2'$ as in the figure: for these points the arguments are u, $u + 2K$, $-u$, $2K + u$ respectively.

[book p. 181] infinity $v = 0$, and $u' = -K$; below the axis $v = -v + i\,K'$, $v = u + iK'$, where $u+2iK'$ are conjugate; thus $u+2 = u+2iK'$. Hence these two arguments are negative integers: more generally, $u' = u + 2(mK + m'iK')$; for the present we may consider only A, A', A, a, a, C; but the two values AC, a; but the next have, $aC = u + AK'$, etc.

122. The poor theorem gives for these points u_1, u_2, u_3 a line

$$du_1 + du_2 + du_3 = 0.$$

and thence $u_1 + u_2 + u_3 = C$. The constant C cannot have a determinate value (for if it had assuming the values of u_1, u_2, u_3 are points where, by some interpretation of u, the arguments are points on the cubic; the constant of $u_1+u_2+u_3 = AK'$; for the constant $u_1+u_2+u_3 = u_1+u_2$, and substituting in the form $u_1 = u + 2iK'$, $u_1 = -u_2 + iK'$; for points "on the sagitta" $(AC + a = -2iK' - iK')$, and substituting in

$$u_1 + u_2 + u_3 = C \quad \text{viz.} \quad u + u_3 + 2iK' = C,$$

the values are simplifying obviously

$$(K+1)C = (1-K_a)(1-K_a)(1-K_2)(1-K_3),$$

and thence

$$a_1 = au_1 - K\,;$$

the values of P, G are easily found to be

$$F = y_1^2 + 2(y_2+y_3) - z_1(1 + z_1z_2 + z_2z_3 + z_3z_1) + 2k^2(1-z_1)(1-z_2)(1-z_3),$$

$$G = 2y_1y_2(y_2+y_3) + 2y_2y_3(y_3+y_1) + 2y_3y_1(y_1+y_2)\,k\,k\,k\,k\,(\ldots).$$

[book p. 182] or, as these may be written,

$$F = a_1 + y_2 + y_3 + a_1 + (a_1^2 - a_1) + (b_1^2 - b_1) + (\ldots),$$

$$G = a_1y_1 - \frac{1}{4k}a_1\,; \quad y_1 = \frac{1}{4} + 4K\,; \quad a_1 = a_1(1-K)(1-K)(1-K),$$

where of course x, y, a are to be replaced by their values a, a_1, $y_1y_2y_3$, etc. This is, in fact, the ordinary process of finding the third point of intersection of a cubic by a line which meets it in two given points.

Writing $aK - a_1$, and a, b, c, d to denote a_1, b_1, c_1, d respectively, we have

$$\frac{a_1}{a_2a_3a_4}\frac{a_2}{a_2a_3a_4} = \frac{1}{ab\,cd}\,;$$

whence

$$a_1^2 = -\frac{1}{a^2a^2}\,; \quad a_2a_3a_4\,a_1 - \frac{a^2a^2a^2}{a^2a^2a^2} - \frac{a^2a^2}{a^2a^2}\,;$$

and the relative equation lookings:

$$(1-k^2a^2a^2)\,a^2 + (1-k^2a^2a^2)\,k^2\,(\,a^2)\,\big[\,(\,k\,)\big] = 0,$$

corresponding to the four tangents which can be drawn from the point u to the cubic.

The Four Tangents from a Point of the Cubic.

123. Suppose that the line is a tangent to the cubic, and let us regard the cubic at the point u, and again meets it at the point u': then instead of u_1, u_2, u_3, we have u', u, u' and the relation become $2u + u'C$.

Here a being given, u' is uniquely determined; this gives u', the unique tangent to the cubic at the point u, viz., to the value $u' = a_1$, a_1, $u' = a_1$, corresponding to the four tangents which can be drawn from the point u to the cubic.

[book p. 183] The tangents are real for a point of the sagitta, and for such a point we may write $u = iK + v$, where v is real and included between the values $\pm K$; the corresponding values of u' are

$$u' = -2u = -2iK' - 2v = -2v + 2K\,; \quad \text{for inconsistent to coval,}$$

$$u' = -\tfrac{1}{2}u + K = -\tfrac{1}{2}v - \tfrac{1}{2}iK' + K, \quad \text{extramediate to oval,}$$
$$u' = -\tfrac{1}{2}u + iK' = -\tfrac{1}{2}v + iK' = -\tfrac{1}{2}v + \tfrac{1}{2}iK', \quad \text{intermediate to coval;}$$
$$u' = -\tfrac{1}{2}u + K + iK' = -\tfrac{1}{2}v + \tfrac{1}{2}iK' + K, \quad \text{intermediate to oval.}$$

The first and second of these belonging to tangents to the oval, the third and fourth to tangents to the sagitta. We may further distinguish a tangent according as it passes between an oval and not passes between: the writing in the tangents from $K = -2v + 2K$, that is, the tangents from the points of the oval $K = K(K+1)$; but the tangents from the point $iK' + v$ are conjugate. We may, for distinction, call them 1822 and 1823, etc.

124. We may make a corresponding division of the real lines which meet the curve in three real points: any such line meets the oval twice (and twice or three times) and taking the singular three times; and taking the arguments to be a, u_2, u_3, for the tangents from the point $iK' + v$, we have

$$\tfrac{1}{2}(u_1 + u_2 + u_3) = \tfrac{1}{2}iK', \quad \text{for intermediate line meeting oval twice,}$$
$$\tfrac{1}{2}(u_1 + u_2 + u_3) = \tfrac{1}{2}iK', \quad \text{extramediate line meeting oval twice,}$$
$$\tfrac{1}{2}(u_1 + u_2 + u_3) = \tfrac{1}{2}iK', \quad \text{extramediate line meeting sagitta three times,}$$
$$\tfrac{1}{2}(u_1 + u_2 + u_3) = \tfrac{1}{2}iK' + K, \quad \text{intermediate line meeting sagitta three times,}$$

above and below the axis respectively.

125. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u', this requires $v = \frac{1}{2}\,iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$; and for these values of the argument v for u, we have, $iK' + \frac{1}{2}K$, the inflexion above and below the axis respectively.

In the case of an inflexion, $u = u_1$, and we have

$$u' = C - 2u_1 = u_1 + 2u_2(u'_a - u_1),$$

to satisfying the four tangents from the point of inflexion $u = u_1$, $u = u_1$, $u = u_1$, $u = u_1$, $u = u_1$.

If the line be a tangent, this will be satisfied by $u = u_1$, the condition for this is $2v\,(P + 2)(P+1)\,Q = 0$, and in any of the two other points determined by the equation

$$(2y^2 + 1)\,GH + g(P+1)(LH + b\,L) = 0.$$

this line will be a tangent if

$$(yM + 1\sigma\, M^2 + 1\sigma) - 0\,;$$

[book p. 184] giving for the equation

$$3Mw + bC\,Du + 6\,RDa - B = 0,$$

or for B, C, D writing their values 1, $-(1+k^2)\,P$, kP this is

$$3kw - 4(1+k^2)\,a + B^2 = 0\,;$$

for the a-coordinates of the inflexion. The form $u = -iK + K + iK'$, and $-iK + K + iK'$, give us $-iK = K + iK'$ in $iK + iK' = K(1 + iK')$, and others corresponding terms by the interaction of the numbers 3 and 4.

It is clear that, in addition to a roots; positive or negative, of these roots are real, all positive or all negative; whilst it is also a $-1, -1, -1$, which is the third value. In an inflexion above and the inflexion below the axis being symmetrically situate in regard to the axis.

The Sextactic Points.

128. The vertices A, B, C are each of them a sextactic point: in fact, writing the equation in the form $y^2 = (1-x^2)(1-k^2x^2)$, we see at once that the conic $y^2 = (1-x)(1-k^2x)$, $= 0$ in regard to coincide and C are real inflexions; and besides these none on the cubic in the points A, A, A, A, C, C.

129. The conics having a vertex at one of these points have eqach a contact of the second order with the cubic; but inquire how the others may.

Formulæ Relating to the Tangents from the Vertices.

129. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say at C, $\rho = 0$, $\rho' = 0$, symmetrically situate in regard to the axis, $\sigma = 0$, $\sigma' = 0$, symmetrically situate in regard to the axis: the four tangents from C are accordingly equations $\rho = 0$, $\sigma = 0$, $\rho' = 0$, $\sigma' = 0$, denoted by $\rho = 0$, $\rho' = 0$.

[book p. 185] eight angles to the axis, and that joining the points of contact of ρ', σ' is the line $x = w'$ also perpendicular to the axis.

Values ρ, Tangents, Imaginary.

130.

$$\rho,\ \rho_1,\ \rho_2 = x - y\,(1+i)\,x,\ 1 + i\,x,\ 0,\ 1+k\,;$$

$$\sigma,\ \sigma_1,\ \sigma_2 = x + y\,(1+i)\,x,\ 1 + i\,x,\ 0,\ 1-k\,;$$

$$\rho',\ \rho_1',\ \rho_2' = x - y\,(1-i)\,x,\ 1 - i\,x,\ 0,\ 1+k\,;$$

$$\sigma',\ \sigma_1',\ \sigma_2' = x + y\,(1-i)\,x,\ 1 - i\,x,\ 0,\ 1-k\,;$$

the equation of the curve is

$$f = yp - r(x+y)\,\rho\rho'\,; \quad \text{Coefficients of Powers of Coords.}$$

Values ρ, Tangents, Imaginary. Coordinates.

$$\rho,\ \sigma = x - y - (1+i)\,kx,\ kx - x - (1-i), \quad x = \tfrac{1}{i+k}\,,\ y = \tfrac{(1+i)\,(1-k)}{k}\,;$$

$$\rho',\ \sigma' = x - y - (1-i)\,kx,\ kx - x - (1+i), \quad x = \tfrac{1}{i+k}\,,\ y = \tfrac{(1+i)\,(1+k)}{k}\,;$$

the equation of the curve is

$$f = y\rho\,(x+i)\,\rho\sigma' = y\,(-x-i)\,\sigma\sigma'\,; \quad \text{Coords.}$$

Values σ, Tangents, Real. Contacts.

$$\rho,\ \sigma = x - y - (1+k)\,(1+i)\,kx,\ 1 - x - (1+i), \quad x = \tfrac{1}{1+i+k}\,,\ y = \tfrac{(1+k)\,(1-k)}{1+i}\,;$$

$$\rho',\ \sigma' = x - y - (1-k)\,(1-i)\,kx,\ 1 - x - (1-i), \quad x = \tfrac{1}{1-i+k}\,,\ y = -\tfrac{(1+k)\,(1-k)}{1-i}\,;$$

the equation of the curve is

$$f = y(-1+k)\,\rho\rho' = y(-1-k)\,\sigma\sigma'\,; \quad \text{Coords.}$$

130. These functions ρ, σ, &c., considering therein x, y as depending therein on the elliptic function $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$, are functions of u: I propose to demonstrate from the values of the constants Nos. 163 of my *Elliptic Functions Memoir*, p. 78; viz. we have

$$\operatorname{sn}(v+\tfrac{1}{2}K) = \frac{1}{\sqrt{k}}\,\frac{1+k-k(1-k)\operatorname{sn}^2 v}{1+k-k(1+k)\operatorname{sn}^2 v},$$

$$\operatorname{cn}(v+\tfrac{1}{2}K) = \frac{1}{\sqrt{k}}\,\frac{\operatorname{cn} v + i\operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}\,, \quad \text{(vertex C);}$$

which is

$$= -\,\frac{1-y-i(1-x)}{1+y-i(1+k)}\,;$$

and

$$\operatorname{sn}(u+\tfrac{1}{2}K+\tfrac{1}{2}iK') = -i\,\frac{\operatorname{cn} v - i\operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}\,; \quad \text{(vertex A);}$$

which is

$$= -\,\frac{1-y+i(1-k)}{1+y-i(1+k)}\,.$$

[book p. 186] which is

$$= \frac{1-y+i(1-kx)}{1+y-i(1+kx)} = \frac{1}{\sigma}\,;$$

Observe that this is the inverse formula, the inflexion which is not a sextactic point, viz. that is, $z = u\,(1-x)+i(y+kx)-1\cdot u$, $\rho\pi$, etc. But neither of these is constant; and so for any other vertex. This is, in fact, no inflexion, and consequently we have

$$\rho\rho'\sigma\sigma' = \rho^2\sigma'^2,$$

and the identity is shown by some calculation of $\operatorname{sn}(u+\frac{1}{2}K) - i\rho$, which can be in like manner shown.

Fixed Curve a Quartic in Space; the Quadri-quadric Curve $y^2 = 1-x^2$, $z^2 = 1-k^2x^2$. Art. Nos. 131 to 144.

131. It is assumed that k is real, positive, and less than unity: the curve may be regarded as the intersection of the two cylinders $z^2 = 1-x^2$, $y^2 = 1-k^2x^2$, but the conception of the curve as belonging to an elliptic cylinder $\dfrac{2}{1+k^2} - \dfrac{y^2}{1+k^2}$, $= 1$, and a circular cylinder, $\dfrac{x^2}{1+k^2+2k^2z}$, are also adopted by Weierstrass as a canonical form.

132. The foregoing expression for $Q\,dP - P\,dQ$ is satisfied by the equations $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $w = \operatorname{sn} u$, $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $y = \operatorname{sn} u$.

[book p. 187] five points, and then the generating lines come each of them to coincide with the tangent at A, we have the osculating planes meeting the curve in the points A counting four times. We will, therefore, taking A and A' as opposite vertices of a parallelogram, but A' as one of two parallelograms; but A', A'', B', B'', B'' being the vertices of a quadrilateral.

132. The equations are satisfied by writing therein

$$x,\ y,\ z = \operatorname{sn} u,\ \operatorname{cn} u,\ \operatorname{dn} u\,;$$

the periods are here $4K$, $4iK'$; whence it is easy to see that the points A, A', B, C have for arguments 0, $2K$, iK', K, $K+iK'$, $K-iK'$, K, $K-iK'$, $K+iK'$ which are $4-\ 0\cdot iK+0$.

Similarly the Reformatted form depends on $u + 2K + iK'$, and so reduce the number is like manner to $K + 1 - 1 + 0$, with that the the form which might has been to taken to be

$$x = u,\ y = u\,(K - iK')\,\&(\text{some equation});$$

$0 = K - iK')$, intermediate. Hence &c. in fact, from the values u for which $a = 0$, viz. $u = iK'$, in this case the values u, z, y are like the case of the conic.

133. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u', this requires $v = \frac{1}{2}iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$. The same reduction will be shown directly. Moreover we have

$$BX^2 - 4lac(1 - 2lac)\,x = a^2 + 4lac(1 - 2lac)\,x = 0,$$

$$+B - C(1 + a + 1 + Y)\,B + L - 5\,(C - LK + 6\,Y + 4y + 4\,;$$

$$+B - LK\,a + ((B + 4l)\,a + (a + 1)\,(b - K) + 0)x.$$

$+2(c - L\,K)\,B = -4y(b - K) + a(b + L)(1 - K)\,b - L\,(LK\,) + 0,\ + - 16Z(b - K)x + LK - 1(b + L)\,L\,(L\,K\,) + 0\,(L\,L + 0\,X\,),$

24–2

[book p. 188] which are the required expressions. If we wish to introduce $u_4 = -u_1 - u_2 - u_3$, the resulting equations give the values $\operatorname{sn} u_4$, $\operatorname{cn} u_4$, $\operatorname{dn} u_4$ of the fourth point, and in this way identifiable with the ordinary results.

132. The determinant equation may be written

$$\big(x_1 - x_2,\ \frac{1}{y_2}\frac{x_1}{y_1} - \frac{x_2}{y_2}\big) + \big(x_2 - x_3,\ \frac{1}{y_3}\frac{x_2}{y_2} - \frac{x_3}{y_3}\big) + \big(x_1 - x_3,\ \frac{1}{y_2}\frac{x_1}{y_1} - \frac{x_3}{y_3}\big) + (\ldots) = 0,$$

and, in fact, each of the three lines is separately $= 0$. This appears from the following three formulæ:—

$$\frac{\operatorname{cn}(u_1 + u_2)}{\operatorname{cn}(u_1 - u_2)} = \frac{x_2 - x_1}{x_2 + x_1},\quad \frac{x_2 - x_1}{x_2 + x_1} = \frac{y_2 - y_1}{y_2 + y_1},$$

$$\frac{\operatorname{sn}(u_1 + u_2)}{\operatorname{sn}(u_1 - u_2)} = \frac{y_2 - y_1}{y_2 + y_1},\quad \frac{x_1}{y_1} - \frac{x_2}{y_2} = -\frac{2(x_1 - x_2)}{y_1 + y_2},$$

$$\frac{\operatorname{dn}(u_1 + u_2)}{\operatorname{dn}(u_1 - u_2)} = \frac{z_2 - z_1}{z_2 + z_1},\quad \frac{1}{y_2}\frac{1}{y_1} - \frac{1}{y_2} = \frac{1}{2}(z_1 - z_2),$$

which are themselves at once deducible from the formulæ

$$\operatorname{sn}(u + v) + \operatorname{sn}(u - v) = \frac{2\operatorname{sn} u\ \operatorname{cn} v\ \operatorname{dn} v}{1 - k^2\operatorname{sn}^2 u\ \operatorname{sn}^2 v},\quad \text{etc.}$$

In fact, the numerators of $\operatorname{cn}(u_1 + u_2) + \operatorname{cn}(u_1 - u_2)$, $\operatorname{sn}(u_1 + u_2) - \operatorname{sn}(u_1 - u_2)$ thus become $= 2(x_1 - x_2)(1 - k^2x_1x_2)$, $(2(x_1 - x_2)(1 - k^2x_1x_2))y_1y_2z_1z_2$, respectively; so that, taking the numerator of $\operatorname{cn}(u_1 + u_2) + \operatorname{cn}(u_1 - u_2)$ as unity, we obtain by division the formulæ in question.

134. Then above formulæ, putting therein $u_1 = u_3, = 2u$, become, on writing the functions of $2u$, hence

$$\frac{1}{2}\frac{dx}{dx} = -\frac{x_1 - x_2}{y_3 - y_1},\quad \text{giving}\quad -\ Dx_1 = \frac{2(x_2 - x_1)y_1}{1 - x_1^2 + (x_1 - x_2)(x_2 + 1)} = \frac{2(x_2 - x_1)y_1}{x_2 - x_1}\,;$$

$$\frac{Dy_2}{2} = \frac{y_2 - y_1}{x_2 - x_1},\quad = \frac{y_1y_2 + y_2y_1 - (1 + k^2)}{x_2 - x_1}\,;$$

$$\frac{Dy_2}{2\,dy} = \frac{x_2 - x_1}{x_2 - x_1} = \frac{(1+k^2)\,(x_1 - x_2)\,(1+x_1)}{(x_1+x_2)(x_2-x_1)};$$

equations which must of course give each of them the same value for u': the equations belong to the relation $u = u_1 + u_2$, $u_3 - u_2$, since $(x_2 - x_2)$ are conjugates of (x_1, y_1, x_1) and (x_2, y_2, z_2) respectively.

134. Then above formulæ, putting therein $u_3 = u_4 = -2u$, and the corresponding functions of $2u$, become.

[book p. 189] Write

$$a,\ \beta,\ \gamma,\ \delta,\ p,\ q = u_1 - u_2,\ u_2 - u_3,\ u_3 - u_1,\ u_1 + u_2 + u_3,\ u_1 - u_3,\ u_2 - u_3\,;$$

a', β', γ', $\delta' = u + 1$, writing $\xi = u_1$, it becomes, in place of a, β, γ, δ, in any of a, β, γ, δ, β, β, γ, δ, γ, γ, δ, δ, δ. The determinant equation is satisfied by these formulæ:—

$$a^2 + \beta^2 + \gamma^2 + \delta^2 = -\,\frac{1}{12k}\,(L^2 + 2y) + L^2P^2 = 0,$$

$$ax + by + cz + d = a + \beta + \gamma + \delta(d - L) + 8(L^2 + 2 + 4y),\ au + ax + by + bx + dx = 0\,;$$

By what precedes, it appears that not only is this so but that we have separately

$$au^2 = a + \beta - 5l\gamma,\ bu^2 - 5l(a + B) - (1 + 25\gamma^2)\gamma + (1 + 2L^2)P^2 = 0,\ cu^2 = 0,$$

then not only the original equation, but each of the three equations, holds good.

The Cubic Curve $xy = 2y + (4+1)^2 z\,;\ xy - 1 - \sqrt{1-x^2}\,\sqrt{1-k^2}$. Art. Nos. 136 and 137.

135. Writing the equation in the form

$$(xy - 1)^4 = x^4(1 - k^2x^2)\ , \quad \text{or}\ \ xy - 1 = \sqrt{1-x^2}\sqrt{1-k^2x^2},$$

we see that the general form is as shown in the figure: the end portions of the curve

[book p. 190] curve lie between the values $x = -\infty$, $-\frac{1}{k}$, and $\frac{1}{k}$, ∞. The curve may be made to depend on an elliptic function by means of the equation; we may here write

$$x = \operatorname{sn} u = \frac{\operatorname{sn}(u + \frac{1}{2}K)\ \operatorname{dn}(u + \frac{1}{2}K)}{\operatorname{cn}(u + \frac{1}{2}K)}, \quad \text{say} \quad x = \frac{\operatorname{sn} v\ \operatorname{dn} v}{\operatorname{cn} v},$$

where the function sn, cn, dn belong to the second, that is, to the second equation; and herein for u writing $v + \frac{1}{2}K$, and for k writing k', $= \sqrt{1-k^2}$, we have, of course, the formulæ given.

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\operatorname{sn}(2v,\, k)\ \operatorname{cn}(2v,\, k) = \frac{\operatorname{sn}(v,\, k')\ \operatorname{dn}(v,\, k')}{\operatorname{cn}(v,\, k')}, \quad \operatorname{dn}(2v,\, k) = \frac{\operatorname{dn}(v,\, k')}{\operatorname{cn}(v,\, k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing k', $= \sqrt{1-k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\frac{012}{\overline{012}} = [\int_a^x (d\omega)_{3'}\ (d\omega)_{1'}\ (d\omega)_{2'}\ \ldots],$$

the last term being the properly determined constant K, attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for the before-mentioned form of cubic curve $y^2 = x(1-x)(1-k^2x)$, which is most directly connected with elliptic functions: and to exhibit the connexion of the formulæ with the elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012},\ \overline{\overline{012}} = \begin{vmatrix} (1+k')\,x_1y_1,\ -(x_1+x_2+y_2)y_1 \\ \xi(x_1+1)+(\xi+1)x_2)\,(c_0,c_1,x_1) \\ +y(x_1)x_2, \end{vmatrix},\quad \begin{vmatrix} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{vmatrix}$$

[book p. 191] where the sn, cn, dn refer to the modulus k', a, z, d denote $\operatorname{sn} v$, $\operatorname{cn} v$, dnv, k modulus k.

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\operatorname{sn}(2v,\,k)\ \operatorname{cn}(2v,\,k) = \frac{\operatorname{sn}(v,\,k')\ \operatorname{dn}(v,\,k')}{\operatorname{cn}(v,\,k')},\quad \operatorname{dn}(2v,\,k) = \frac{1}{\operatorname{cn}(v,\,k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing k', $= \sqrt{1-k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by $\overline{012}/\overline{012}$.

139. We have

$$\operatorname{sn}(v+\tfrac{1}{2}K) = \frac{\operatorname{cn} v}{\operatorname{cn}(v+\frac{1}{2}K)},\ \operatorname{dn}(v+\tfrac{1}{2}K) = \frac{1+k}{1+k}\,\frac{\operatorname{cn} v}{\operatorname{cn}(v+\frac{1}{2}K)},$$

$$\operatorname{cn}(v+\tfrac{1}{2}K) = \frac{1}{1+k}\left[1+k+(1-k)\ \operatorname{cn} v\right],\quad \operatorname{dn}(v+\tfrac{1}{2}K) = \frac{1-k^2}{1+k}\left[1+k-(1-k)\ \operatorname{cn} v\right];$$

and the formula above clear obtained are

$$\operatorname{cn}(1+k,\,k) = (1+k)(1+ka,\,k')\quad \text{giving}\quad \operatorname{cn}(1+k,\,k) = \frac{(1+k)\,(1+ka,\,k')}{1+ka(1+k)};$$

$$\frac{1}{\operatorname{cn}(2+k,\,k)} = \frac{1}{1+k} + a\,(1-k)\,\frac{1-x}{1+x},\quad \operatorname{cn}(2+k,\,k) = \frac{1+k}{1+k}\frac{1+ka}{1+ka(1+k)};$$

$$\frac{\operatorname{dn}(1+k,\,k)}{\operatorname{cn}(1+k,\,k)} = \frac{1}{1+k(1+ka)}\,\frac{(1-x)\,a}{a+1},\quad \operatorname{dn}(2+k,\,k) = \frac{1-ka}{1+ka};$$

where, as before, a, c, d denote $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$, k modulus k, in fact, the formulæ of the second line of the table, *Elliptic Functions*, p. 63.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\overline{012} = \int_a^x \left[)(\,d\omega\,)(_3,\ (d\omega)_1,\ (d\omega)_2,\ \ldots\right],$$

the last term being the properly determined constant K, attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012},\ \overline{\overline{012}} = \begin{vmatrix} (1+k)\,x_1y_1, & -(y_1+y_2)y_1 \\ \xi(y_1)+(\xi+1)\,x_2,\ (c_0,\ c_1,\ x_1) & \\ y_1(y_1)\,x_2, & , \end{vmatrix},\quad \begin{vmatrix} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{vmatrix}.$$

[book p. 192] Taking here 2, $=\emptyset$, for the point, coordinates $x_1,\ y_1 = 0,\ 0$, we have

$$\overline{012} = -\,\frac{x_1y_2}{y_2(x_2-x_1)\,(y_2-y_1)} = -\overline{012},$$

and, retaining 1 to denote the point coordinates $(x_1,\ y_1)$ belonging to the argument u_1, and $1'$ for the point belonging to the argument $u_1+\frac{1}{2}iK'$; the coordinates of $1'$ are $\dfrac{1}{k^2x_1},\ \dfrac{1}{k^2x_1^3}y_1$; and the formulæ become

$$\overline{1'0} = -\,k^2x_1\,(x-\tfrac{1}{k^2}x_1\,(y-0)\,y_1),\quad = 1+k^2,$$

the numerator herein multiplied by ν, is

$$= -\,k^2x_1\,(x-\tfrac{1}{k^2}x_1\,(y-0)\,y_1) = (-\,\Omega 1)y\,,$$

which, substituting for $y,\ x,\ z$ as their values in terms of $u,\ v,\ w$, becomes

$$= \frac{0+\nu^2(\nu-z_1z_2-z_2z_1)}{(\nu\nu_1+\nu_2\nu_3)\,(\nu_2-\nu_1)} = -\,k^2x_1\,;$$

Operating on each side with $\dfrac{a}{dx_2}$, $=0$, we obtain

$$\partial_2\overline{012} = -\partial_2,$$

$\partial_3\ \overline{012} = w^2-12y+L^2(2y-z)-z_1$. The differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind. Hence we have

$$Z(u+a,k) = \big(1-\frac{R}{8}\big)u+L\big(\,(\,n\,du,$$

viz. we have

$$ZU = ZU_1-L^2a-L^2z = L(u-u_1)-LK(u-u_1) = \Omega(\nu-x_1)\,;$$

and thence

$$\partial_xZU = \partial_xZU_1-L = LK^2aU(u-u_1)k(u-u_1) = \Omega'.$$

Observe that, L referring to u, all the variable's might be written U'.

140. Continuing the discussion, the expression for the right-hand side is $=\omega+(a-2)\omega'$, and this divided by u^2, the equation $\dfrac{1}{2}=0$. We have here

$$\partial_x\omega-\omega+\omega' = \overline{0(u-u_1)} = (u_1-u_2)\,(u_2-u_3)\,(u_3-u_1)\,;$$

viz. this is, $\overline{u_1} = -\,\overline{u_2}$, in $\overline{u_1}$, in $\overline{u_2}$. From this, and adding, the expression in question becomes

$$\partial_xw(z_1-z_2)\,\frac{x_1}{a_1\,a_2\,a_3} = -\,\partial_xw(z_1-z_2)\,\frac{x_1}{a_1\,a_2\,a_3}\,,$$

the differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind.

[book p. 193] **140.** To introduce into the formula 1 instead of $1'$, we have only to write u_2, $-K'$ instead of u_1, putting also for shortness c, d, f, g, h for the functions of u and a, respectively, we then obtain

$$[01'] = -\frac{x}{y}\, a_1\,(c_1 + xa_1),$$

where observe that, interchanging u and a_1, we have

$$[110] = \frac{a_1\, d_1}{c_1\,(c + x_1 c_1)}\,,$$

that is, $[10] = -[01]$, as it should be. The formula may be written

$$[01'] = -[10'] = -\frac{aa_1\, d}{xc_1\,(c + aa_1)}\,,$$

and

$$\partial_y\,[01'] = -\frac{1}{(c + aa_1)\, aa_1}\,,$$

whence

$$\partial_x\,[10'] = -\frac{1}{a^2\,(x - x_1)\, ac\,(c + aa_1)}\,,$$

we have, moreover,

$$[012] = [12'] - \frac{x}{y_1}\,[\overline{036}] - \frac{x_1}{y_1}\,[306]\,,$$

and

$$\partial_x\,[012] = \frac{x_1 y_1}{a_1\, a_1\,(x - x_1)} - \frac{1}{x_1}\,\frac{1}{xc_1(c + aa_1)}\,,$$

which last equation gives $(\partial_x + \partial_x + \partial_x)\,[012] = 0$, as it should be.

141. Supposing that the differential $d\Pi_a$ is defined by the equation

$$d\Pi_a = du_1\,[012] + du_2\,\big[01'\,\big] + du_3\,[026] - [126],$$

we have

$$\int d\Pi_a = du_1\,[012] + du_2\,[01'2'] + du_3\,[026] - [126]\,;$$

and thence

$$\partial_x \int d\Pi_a = \partial_x\,[012] + du_2\,[126] - \partial_y\,[126]\,;$$

$$= [126] - [126]\,,$$

$$= \frac{1}{(c + aa_1)} + \frac{(1 + k^2 a_1 - 2a_1)\, a_1}{(1 + ka_1)(1 + ka_2)}\,;$$

$$= \frac{1}{(c + aa_1)} - \frac{1}{(1 + k^2 a_1)}\,(1 + ka_2)(1 + ka_3)\,.$$

[book p. 194] or establishing between the constants u, a, the relation $\operatorname{sn}^2(u_a - u_1) = \dfrac{K}{K - R}$, this becomes

$$= 1 + \frac{R}{R - k}\, k^2 \operatorname{sn}^2(u_a - u_1)\, a\, k^2 K^2,$$

which is

$$= \partial_x \log \frac{\Theta(u_1 + a + iK')\,\Theta(u - u_1 - iK')}{\Theta(u_1 + iK')\,\Theta(u - iK')},$$

where Θ is Jacobi's theta-function, see my *Elliptic Functions*, p. 144. The expression is, in fact, $= \frac{1}{2}\,k \log \Pi(u - u_1, a - u_1)$, where the integral $\partial = \int_0^a \log \Theta$.

But we have

$$\log \Theta(u) = \log \frac{\Theta(0 + iK')\,\Theta(u - iK')}{\Theta(0 + iK')\,\Theta(0 - iK')}\,\bigl(1 + k(\ldots)\bigr) + \int_0^u n\,du\,;$$

that is, $\partial = \int_0^a \log \Theta$, and consequently we have $\partial_x \left[\left(\int_0^a \log \Theta\right)\right] = \Omega$,

$$= \frac{1}{\sqrt{(\nu_1 - \nu_3)(1 - k^2)}} \log \frac{(1 - k\nu_1)\sqrt{2K}}{(1 - k\nu_1)\sqrt{R}}\,\frac{LK}{(LK)} - 2k(1 - k^2)(1 - k^2)\,a\,\bigl(1 - k^2\,a(1 - k^2)\bigr),$$

that is, $\partial = \dfrac{K}{B - K}\sqrt{1 - k}\,a$, and consequently we have $\partial_x \sqrt{(1 - k\nu_1^2)(1 - k\nu_2^2)\,a}$,

$$= -\frac{R}{R - k}\,\partial_x \sqrt{1 - k^2}\,a^2 - 2k(1 + k^2)(1 - k^2)\,a^2 + 2k\,a^2,$$

$$= -\frac{R}{R - k}\,\partial_x \sqrt{(1 - k^2)(1 - k^2)}\,a\,\partial_x \sqrt{1 - k^2}\,a\,b\,a\,,$$

The difference of the two functions on the right-hand side is $= -\,\partial_x\,(u - a)\,\bigl(1 - k\,\mathrm{sn}^2(u - a)\bigr)$, which is $= -\,\mathrm{sn}^2\,u\,a + \mathrm{sn}^2\,a$; and the integral being $= 0$ on the lower limit, we readily see that the expression on the right-hand side is, in fact, the function which is required. The transformation also is given in summary by the form which I had obtained. I consider with this remark.

Fixed Curve the Quartic $y^2 = x(1 - x^2)(1 - k^2)$. Art. Nos. 145 to 145.

143. This is a curve having a tacnode at infinity on the line $w = 0$; as may be seen by writing the equation in the homogeneous form $y^2w^2 = x(1 - x^2)(1 - k^2x^2)$, and putting for shortness $X = w(1 - x^2)(1 - k^2x^2)$, $X = w(1 - x^2)(1 - k^2x^2)$, when X, w small as compared to a, the equation is reduced to $y^2w^2 = 1$, viz. this is a $\left(\dfrac{yw}{1}\right)$ to which the sign $\pm$ is to be understood accordingly. I consider with

[book p. 195] reference to this curve only the affected theorem, in the particular form in which it must readily connects itself with the ordinary theory of the integral of the third kind.

144. I consider the differential $\dfrac{(x,\,y,\,z)_p\,d\omega}{012}$ in the particular case where the line 012 is a line parallel to the axis of y: taking its equation to be $x - x_1 = 0$, and putting for shortness $X = \sqrt{1 - x^2}\sqrt{1 - k^2x^2}$, $X_1 = \sqrt{1 - x_1^2}\sqrt{1 - k^2x_1^2}$, the paramétric points are taken to be $(x_1,\,X_1)$, $(x_1,\,-X_1)$, and the residues are the intersections with the curve $f - X_1 = 0$ of the line $f = 0$, viz. this is the line passing through the two points of contact of the parallel tangents: the points $(x,\,y)$ and the points $(x_1,\,X_1)$, $(x_1,\,-X_1)$ are on a conic Ω, containing the affected theorem, and we have for the form $a(x - x_1)$, containing the arbitrary constant a. We have moreover, in usual notation, for the form a containing the arbitrary constant a.

If the limits are taken to be points on the curve $a^2 + y^2 + ax + by + c = 0$, what we want to do is to express the summation of the integrals in the form

$$\int \frac{\sqrt{X}}{x - x_1}\,\frac{x - x_1}{X}\,dx + a\int \frac{\sqrt{X}}{x - x_1}\,\frac{dx}{X} - a\int \frac{\sqrt{X}}{x - x_1}\,\frac{dx - dx_1}{x_1X}\,;$$

where $X_1 = \sqrt{1-x_1^2}\sqrt{1-k^2x_1^2}$, etc. Reference to the right curve as above $a^2+y^2 = ax+b$, when $a^2+y^2-2ax-a^2 = 0$; the right-hand side, by adding the like quantities for the other points a, β, γ.

We have

$$-\frac{1}{a^2}\left(a^2-X^2-a\right)^2 = \frac{a^2-aB}{a^2+a^2X^2}\,,$$

and the integral thus becomes

$$= -2\int \frac{\sqrt{X}\,a\,d\omega}{a^2-aB}\,\frac{a^2-aB}{a^2-aB^2\,X^2} = a\int \frac{X^2}{a^2-aB^2\,X^2}\,\frac{dx}{\sqrt{X}}\,.$$

25–2

[book p. 196] Taking here $a=0$, $-c$, so $a=(X^2)$, $a=\dfrac{a}{\sin\theta}$, we have for such a $\Pi(a,k)$, and the result is

$$\int \frac{\sqrt{X}}{x-x_1}\,\frac{dx-dx_1}{X} = \int \frac{X\,dx}{x-x_1+b\,a\,a\,X^2} - \frac{X\,a\,a\,X}{x-x_1}\sin\theta\,du,$$

where on the right-hand side the first term is $-\int(x,\,a,\,c)$ in Jacobi's form of the integral of the third kind, viz. *Elliptic Functions*, p. 146.

And thence

$$\partial_x\,\Pi(u,\,a) = \frac{2(\nu-1)\,(z\,a)\,a + X(z\,a+4u^2-X\,a^2)\,(\nu+2)(\nu+2X)}{=}\Big(1-\frac{R}{R}\Big),$$

or

$$\partial_z\,\Pi(u,\,a) = \frac{R^2(s^2+r^2)\,a^2-2k^2(1+k^2)\,k^2}{(1-k^2\nu^2a)\,a^2} = \Big(1-\frac{R}{R}\Big)+1\boldsymbol{x}\,a^2\,;$$

or, this being represented in regard to u, $\big(a$, we have

$$\partial_z\,\Pi(u,\,a) = \omega\,a + L\big(0+a-2\,a\,du\big)\,;$$

and thence by integration, and a proper determination of the constants, $\Pi(z,\,a)=\Pi(z,\,a)$.

Chapter VI. The Nodal Quartic.

Nodal Quartic; the General and Polynomial Forms. Art. Nos. 146 to 148.

146. For a cubic, or when an x is, the nodal quartic is the next case as regards points on the curve, and corresponding theories with functions of a single argument (elliptic function): but for a node it is the next case as regards the number of points on the curve $\delta=4$, $p=8$; but it is, as in the case of the cubic, the next case as regards the number of points or, with a marked change in the form of the result.

[book p. 197] The most simple curve of deficiency 2 is the nodal quartic, $n=4$, $p=2$. Using homogeneous coordinates, the general form is $Ax^4+8Bx^3+(\ldots)=0$. Using

$$A = \big(b,\,j,\,b\big)\!\big(\,y,\,z\big)^2,$$

$$B = \big(b,\,j,\,b\big)\!\big(\,y,\,z\big)^3,$$

$$C = \big(p,\,q,\,r,\,s\big)\!\big(\,y,\,z\big)^3,$$

and where we write also

$$D = AC^2-2(\,p,\,q,\,r,\,s)(\,y,\,z\big)^4+4(c,\,j,\,k,)(\,y,\,z\big)^4-4(c,\,j,\,k,)(\,y,\,z\big)^4.$$

Clearly the equation of the two tangents at the node is $A=0$; and the coordinates of the six tangents from the node are $a-kx+by+cz=0$; $a-kx+by+cz=0$, the equation of the curve corresponds to the form: that is the case is unaltered if for β, γ, δ we substitute $\alpha+\delta$, $\beta+b$, $b+b$, $b+b$, and we have thus, for the conic from the tangents to the curve. $a'+Y'+Z'=0$, where δ_1, δ_1, δ_2, and δ_2 for the conic, when the cubic, the six tangents are entered by; or, in like manner, we have, $a'+Y'+Z'=0$, $a+b+c=0$, $b+e+d=0$, the lines δ_1, δ_2, δ_3, δ_2, are corresponding points.

147. An important special case occurs when $B=0$: we here have

$$A=\left(a,\ h,\ i,\ h\right)\!\left(y,\ z\right)^2,$$

$$B=0,$$

$$C=(p,\ q,\ r,\ s)\!\left(y,\ z\right)^3=(a,\ b,\ c,\ d)\!\left(y,\ z\right)^3,$$

or, omitting the factors i and j,

$$(a+j)(b+j)(c+j)=(a+j)(b+j)(c+j)(a+j)=0.$$

The origin is here a Sentcocode; the tangents are $a-z+y=0$, and there are six tangents to the cubic on the curve.

148. The general nodal depends on 11 constants, but by writing for $u\,by$, $u\,by+cy+dz$, $u\,kx+by+cy+ay$, we can reduce the number is like manner to $1-1=0$. Similarly the Reformatted form depends on 7 constants, but we can reduce the number is like manner to $1+1-1=0$, so that the final form might be taken to be

$$x\boldsymbol{y}=(a-b)(a+b)(a-b)(b-a),$$

this line will be a tangent of

$$(yM+1)(1\sigma M^2+1)=0\,;$$

[book p. 198] but it is more convenient to retain the original form

$$x^2(ax+by)^2=(a-x)(b-y)(c-x)(d-y),$$

bearing in mind that this is reducible to the form just referred to, and thus depends visually upon only 5 constants.

It is a general property that a curve of deficiency p greater than 1 can be by a rational transformation reduced to a curve of the corresponding deficiency $4p-3$ (this number $=1$ for $p=2$), so far as it can be reduced; and the paramétric in particular, $p=2$, the nodal quartic is to be reduced to a curve $a=0$, $\beta=0$ corresponding properly, viz. the sextic. This will be done in the present chapter, but the form here adopted is not the standard form generally adopted, but the standard form first adopted by Weierstrass as a canonical form.

Reduction to the Plefonoidal Form. Art. Nos. 149 to 152.

149. Consider the general nodal quartic $Ax^4+2Bx^3+(\ldots)=0$. The equation of the tangents from the node $(x,\ y,\ z=1,\ 0,\ 0)$ to the curve is the discriminant in regard to ξ of $Ax^4+2Bx^3+(\ldots)=0$. We may write

$$A_x=(a,\ h,\ j,\ h)\!\left(y,\ z\right)^2,\ C_x=(p,\ q,\ r,\ s)\!\left(y,\ z\right)^3$$

and the transformed equation is

$$Ax^4+2Bx^3+(\ldots)=A_xx^2+2B_xx+C_x=0,\ \&\text{c.}$$

say this equation is

$$Ax^2 + 2Bx + C = 0,$$

where A, B, &c. have the values of A_x, B_x, &c., written in x, y, z, in place of x', y', z'; and the transformed equation $\Lambda x^2 - 2BxC = 0$ is the equation of the six tangents.

150. From the values of A, B, C we obtain after a little reduction the following formulæ:—

$$A' = (a, h, j, h)(y,\ z)^2,$$

$$B = (l, m, n, l\, j, j, k)(y,\ z)^3,$$

$$C = (a, b, c, d, e, f, g)(y,\ z)^4.$$

and the transformed equation is

$$Ax^2 + 2Bx + C = 0,$$

where

$$\mathbf{A} = A',$$

$$\mathbf{B} = A^2C - (B^2),$$

$$\mathbf{C} = AC^2 - 2(B)^2 + (\,B^2)\,C + 2R,$$

[book p. 199] and thence

$$\mathbf{B}^2 - 4\mathbf{AC} = A^2\big[\,A(AC^2 - 2(B)\,C^2 + (\,B^2)\,C\,b\,)\big] - AC(\,A(\,C\,b\,c\,e\,b\,d\,))\,;$$

We have here B, $= A(p,\ q,\ r,\ s)(y,\ z)^3$, a cubic function $(y,\ z)^3$ containing the factor A_x, $=(a,\ h,\ j,\ h)(y,\ z)^2$, and similarly for a, a, a, b. Calling the other factors B, V, we have $B = AB'$, $V = AV'$, respectively; where b, b, c are merely the roots of the equation $AV = 0$.

We have what precedes, by the transformation $x = y + \beta_1$, $a = \beta$, $y = \beta$, passed from the form $Ax^4 + 2Bx^3 + Cx = 0$ to the form

$$\mathbf{A} = A,\ \mathbf{B} = V,\ \mathbf{C} = -V'(B,\ \mathbf{\Pi} - C),$$

where

$$\mathbf{\Pi} = (\xi,\ \eta,\ k\,))(y,\ z),$$

and thus the equation is $Ax^2 - 2Bx + C = 0$.

We have by what precedes, by the transformation, the canonical form $a = 0$, $y = 0$, viz. $\mathbf{\Pi}$ if we $\mathbf{\Pi}$ have the common factor A $(\mathbf{\Pi} - e^{\eta}\mathbf{\Pi} - 0)$.

152. Assume now

$$\Sigma\, B = AB'\,,\ V = AV'\,;\ \Pi = AB',$$

and therefore also

$$A =\ (t,\ u,\ b\,\xi\,y\,b\,))(y,\ z)\,;$$

$$\Pi = (\xi,\ \eta,\ k\,))(y,\ z)\,(\,xy\,a\,)\,;$$

$$\mathbf{\Pi} = AC^2 - 2(B)\,C^2 + (\,B^2)\,C\,;$$

viz. this equation is, the transformation, that of a curve of the order 5 ($p = 2$, w, C); and the equation thus becomes

$$Ax^2 - 2Bx + C = (xy + \mathbf{\Pi\Pi})\,; \quad \text{the equation must become a tangent at}$$

and there is to the same equation $a = (\xi,\ \eta))\!(\,y,\ z\big)$,

$$xy = (\xi + \eta))\!(\,y,\ z\big)\ (\,x\,y\,a\,)\,;$$

where

$$\boldsymbol{\Xi} = (\boldsymbol{c},\ f,\ g)\!(\,y,\ z\big)\,;$$

$$\Pi = (\xi,\ \eta,\ k\,))\!(\,y,\ z\big)\,; \quad \&\text{c.};$$

151. Assume that the curve transforms by the substitution $x = y + Z\eta$. We have the change of $\boldsymbol{\Pi}$, $E\,c$, z, f, y, in A, C, b, y, c; and thus

$$\Pi = \ Aa^2,\ \Pi = Aa^2,\ Z = -Cy^2 + A\,Z^2 + A\eta\xi\big(L\,V\,a\big)\,;$$

which putting therein $(\xi,\ \eta,\ k\,))\!(\,y,\ z\big)$, the equation must become

$$\boldsymbol{Y} = (a, b, c, d, e, f, g)\!(\,y,\ z\big)\,;$$

and therein removed by

$$\boldsymbol{\Xi},\ \boldsymbol{Y} = \boldsymbol{Z}\ + Mp\,;\ \boldsymbol{Y} = \boldsymbol{Z} + K - Mp - q\,;$$

and therefore would be in the in $\boldsymbol{Y}\,Y$.

[book p. 200] then in the new coordinates $(X,\ Y,\ Z)$ we have the equation

$$(i,\ j,\ k\,))\!(\,Y,\ Z\big)\,YZ + \big[(X - aY - bZ)^2 - X(p,\ q,\ r,\ s)\!(\,Y,\ Z\big)\big] - kZ$$

$$+(1, b, c, d, k)\!(\,X,\ Y\big)\big(Y - 2bZ - kY\big) + (X\,X(Y - bZ)\,(X - aY - bZ)\,(X - aY - bZ)\,(X - aY - bZ)$$

$$+4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)\,(X - aY - bZ)\,(X - aY - bZ)\,(X - bZ - kY) = 0,$$

that is,

$$\big(t,\ j,\ k\,)\big)\!(\,Y,\ Z\big)\,YZ + \big[(X - bY - bZ)^2 - X(p,\ q,\ r,\ s)\!(\,Y,\ Z\big)\big]$$

$$+4(X - aY - bZ)(X - aY - bZ)\,X(X - bY - bZ) = 0\,;$$

$$+1\,(X - bY - bZ)(X - aY - bZ)(X - bY - bZ)\,(X - bY - bZ)(X - bZ - kY) = 0\,;$$

when this equation $(X,\ Y,\ Z) = 0$, $Y\,(X - bY - bZ) = 0$, the equation transformations must become

$$\big(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)$$

$$+4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)\,(X - aY - bZ)\,(X - aY - bZ) = 0,$$

that is,

$$\big(p,\ q,\ r,\ s)\!(\,Y,\ Z\big)^4 - (Y - 2bZ - kY)^2(X - bY - bZ - bZ - kY)^2$$

$$-(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2\,(X - aY - bZ - kY)^2 = 0,$$

or rearrangely

$$X,\ Y,\ Z = X - aY - bZ,\ Lz = Mb\,(\,bY - bZ\big) - kY,$$

transformed into the differential form as above.

It is similarly the case that the differential first appears, although there is a great many more steps to be made in establishing the truth of the previous, and the case where the case the parametric points are a curve of deficiency p, p being p.

Application of Abel's Theorem.

152. Taking the fixed curve to be $f, = (A\,x^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B\,;$$

if for shortness we write

$$(x,\, y)^2 = P,\ = P,\ Bx + Cy^2,\ y = (Ay - Cz)(z - y)\,P\,;$$

[book p. 201] Application of Abel's Theorem. Art. Nos. 153 to 157.

152. Taking the fixed curve to be $f, = (A\,x^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B,$$

if for shortness we write

$$(x,\, y)^2 = P,\ = Ax^2 + 2Bxy + Cy^2,\ P_x = (Ay - Cz)(y - z)\,(z - y) = (Ax + B)(\ldots),$$

and we thence have

$$d\omega = \frac{x\,dy - y\,dx}{\sqrt{P}}\,.$$

The minor curve $(x,\, y,\, z)^{n-3} = 0$ is an arbitrary line passing through the node, that is, the point $x = 0$, $y = 0$; and there passes through this line one of two distinct relations $2x\,dz - dx = M(x\,dz - dz), = 0$, viz. this is an ordinary line through the points 1 and 2 of the curve.

The variable curve is taken to be an arbitrary curve passing through the intersections of the conic $\boldsymbol{\Xi}$, with the conic $\boldsymbol{X}$ passing through the four points belonging to these values $2c$, b, a, c. Specifically the line is arbitrary, and the line being given, the equation determines the intersections of the line with $\boldsymbol{X}$ passing through these intersections of the line with $\boldsymbol{X}$ which are the points 1, 2, 3, 4, viz. we have

$$\frac{df}{dy} = Ay + B, \quad Lz = Mp, \quad LZ = K - Mp\,;$$

to complete the verification, observe that we have $Y = 8\,TX^2 - (1 + 8T^4)X$, whence

$$-T = (1 + 8T^4)(-YX + 8\,(PY - X^4) - 16\,YT^4 - 8X^4 = 16T^4X^4.$$

153. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\boldsymbol{X\Pi} = 0$, viz. $\Omega + \mathbb{P}\boldsymbol{X\Pi}\,a, b, c, d, e, f, g)(\,x,\, y,\, z)^4$, an equation which determines a; and the ratio $x : y$ is then determined rationally by the equation $\Lambda x - By = 0$. Or what is the same thing, we may write

$$A = a^2 + 8aBy^2 = 6aBy^2, \quad B = -a^2y^2 + (1 + 16y^2 + 8y^2)\,8\,(8\,a + 8B\,X) + 8(15)\,y\,X\,X\,X,$$

in like manner, the four equations are given when the points 1 and 2 are determined. Conversely if (y, y) and (z, z) are corresponding points the conic $\Omega + 0\,X^2$ pass through the points $1, 2, 3, 4$.

154. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\boldsymbol{X\Pi} = 0$; the line $\boldsymbol{\Pi} = 0$, hence the conic passes through the points 5, 6, 7, 8, where 5, 6, 7, 8 are the intersections of the lines $\xi = 0$ with the conic. The variable line $\boldsymbol{T}\,P + \boldsymbol{X}\,P = 0$

corresponds to the cubic transformed integral; the deficiency is here equal to 2, integral relations, which agree with the conditions of a curve of deficiency p, p being 2.

[book p. 202] **155.** If we now assume

$$d\omega = x\,d\omega_1\,d\omega_2, \quad d\omega = x\,d\omega_2, \quad d\omega_2, \quad d\omega = x\,d\omega_2,$$

$$d\omega = x\,d\omega_2 + y\,d\omega_1, \quad d\omega = x\,d\omega_2 + y\,d\omega_2, \quad d\omega = x\,d\omega_2 + y\,d\omega_2,$$

or say

$$u,\ v = \int_0^1 (\,x,\,y\,))(\,\frac{x\,dy - y\,dx}{\sqrt{P}} + \int_0^2 (\,x,\,y\,))(\,\frac{x\,dy - y\,dx}{\sqrt{P}}\,,$$

$$u',\ v' = \int_0^1 (\,x,\,y\,))(\,\frac{x\,dy - y\,dx}{\sqrt{P}} + \int_0^2 (\,x,\,y\,))(\,\frac{x\,dy - y\,dx}{\sqrt{P}}\,,$$

where u, v, u', v' are each of them, of course, regarded as functions of the four points 1, 2, $1'$, $2'$, then u, v, u', v' are functions which are connected together by the proper relations, and which present themselves either as singly, or simultaneously by means of the formula for σ functions.

In a basicly completely determined way, the points 1, 2, 1, $2'$ are not perfectly determined, but are determined by elliptic integrals: but we have these functions, and the systems of points 1, 2, $1'$, $2'$ taken in connection with the various functions (u, v, u', v') in a system as 1, 2, $1'$, $2'$, etc. are points which it is the difference between the formula on the integrals. We have, in fact, that

$$\sqrt{u(u_1 - u_2) - 2(u_1 + u_2)\,u'} = u(u_1 - u_3 + u_2)\,u'\,(\ldots),$$

$$B(u_1 - u_2),\ = u(u_3 - u_2)\,A(u_1 - u_3)\,(\ldots),$$

$$A(u_1 - u_2),\ = u(u_3 - u_2)\,A(u_1 - u_3)\,(\ldots),$$

and so on for the four other systems of equations $C(u, v)$, $D(u, v)$ respectively, formed in the case of an integral function of u, v of orders 1, 2, respectively (and not in the case of an arbitrary function ρ, ρ' of these integrals).

156. The differential relations thus become

$$du\,du' + dv\,dv' = 0, \qquad x'\,du - y'\,dv = 0, \qquad x\,du - y\,dv\,;$$

where T, J are constants which are determinable as definite integrals by the consideration that du, dv form a complete system. The equations $u_1 + u_2 = u$, $v_1 + v_2 = v$, then the points u, v are points where we have the consideration that the conic passes through the points $1, 2, 1', 2'$ on the cubic; whence considering the conic as a function of the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p.

156. The differential relations thus become

$$du\,du' + dv\,dv' = 0, \qquad x\,du + y\,dv = 0, \qquad x'\,du + y'\,dv = 0,$$

where J, J are constants which are determinable as definite integrals by the consideration that the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p.

[book p. 203] **157.** We may, in the first instance, disregarding altogether the consideration of the arguments u, v, attend only to the algebraic functions such as $\sqrt{X}$, $\sqrt{Y}$, $\sqrt{Z}$, of the coordinates of the pairs of points 1, 2, $1'$, $2'$, 3, 4, and so on as regards $\boldsymbol{X}$ as a function of the coordinates of the points only.

Origin of the Double-Letter Functions. Art. Nos. 158 and 159.

158. The cubic curve $Ax + B = 0$ may be taken to be a curve through two of the points of contact, say the points α, β: we have a Ω a function $(x, y)^2$; or throwing out the factor $(z - \beta y)\,(z - \beta y)\,y$, by the identity

$$(\alpha - x)(\beta - y) = -y(\alpha - y)(\beta - x) + (a, b, c, d, e, f, g, h)(\, x,\ y,\ z)^2,$$

and hence, introducing $\sqrt{a}$, $\sqrt{b}$ as quasi-coordinates of the curve, the original a becomes equal to the identity

$$(\sqrt{a} - x)(\sqrt{a} - y) + (a, b, c, d, e, f, g, h\,))(\, x,\ y,\ z)^2 = 0,$$

and hence, introducing a notation which it is convenient to use, we set

$$\sqrt{a}\,\sqrt{b} - x\sqrt{a}\,y\sqrt{b}\,a\,(\,b - x\sqrt{a})(\sqrt{b}\sqrt{c})\,(\,c - b\sqrt{c})(b - x\sqrt{a})\,a\,(\,b - x\sqrt{a})(\sqrt{c}\,b\,)$$
$$-\,Y\,(\,b - x\sqrt{a})(b - \sqrt{a})(\,a - x\sqrt{a})\,a\,y\,(\,a - x\sqrt{a})\,(\,b - x\sqrt{a})\,a\,b\,(\,a - x\sqrt{a})\,a\,b\,a\,b\,;$$

159. Putting each of the y's equal 1, we have the identity

$$(\alpha - x_1)(\alpha - x_2) - 2(1 + x_1)(1 + x_2)(1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4) = 0,$$

and hence, putting in for shortness

$$a = \sqrt{a(1 + x_1)(1 + x_2)}, \quad x = \sqrt{(1 + x_1)(1 + x_2)\,x_1\,x_2}, \quad x' = \sqrt{(1 - x_1)(1 - x_2)\,x_1\,x_2}\,;$$

and in like manner, $\beta = \sqrt{b(1 + x_1)(1 + x_2)\,b\,b'}\,a\,(\,b\,b' - c)\,(\,c\,b' - c)\,b$, where the symbols $a\,b\,a$, $a\,b\,b'$, $a\,a'$, $b\,c\,a\,b'$, $b\,b\,b\,c\,a\,a'$ are determinate.

[book p. 204] We have thus

$$\frac{(a - x_1)(a - x_2)}{(a - x_1)(a - x_2)} = \frac{a^2}{a\sqrt{a}\,a\sqrt{a}}, \quad = \frac{ax_1\,ax_2}{ax_1\sqrt{a}\,ax_2\sqrt{a}\,ax_1\,ax_2}\,a\,a\,b\,c\,d,$$

which last equation, taking the square root, becomes

$$\frac{a - x_1}{\sqrt{a}\,ax_1} = -\,\frac{ax_2\,ax_3}{\sqrt{a}\,ax_3\,ax_4}\,,$$

or, substituting in the first equation, we have

$$\frac{x - x_1}{x - x_1} = \frac{ax_1\,ay_1\,ax_2\,ay_2\,a}{ax_1\,ax_2\,aa^2\sqrt{ay_1\,ay_2}\,ax_1\sqrt{ax_2}}\,,$$

which clearly divides out being in each of these cases respectively.

159. Considering the dual DE as an abbreviation for the double letter ABC, DEF, the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions

$$\Lambda_{ab} = \frac{1}{ax_1\sqrt{a}\,ax_2\sqrt{a}\,ax_1\sqrt{ax_2}}\,a\,(b\sqrt{ax_1}\sqrt{ax_2}\sqrt{a}\,a)\;\cdots\,,$$

in connexion with the already mentioned single-letter functions Λ_a, $= \sqrt{ax_1\,ax_2}$, etc. viz. in this notation the equation just proved is

$$\Lambda_{ab} = \frac{\sqrt{(a - 1)(1 + a - c)}\,\sqrt{a - d}}{\sqrt{(a - 1)(1 + a - c)}\,\sqrt{a - d}}\,\frac{\Lambda_a\,\Lambda_a}{\Lambda_a}\,;$$

and it thus appears that, by reason of the constancy of the symmetry of the first three letters and of the symmetry of the last three letters in the function Λ_{ab}, Λ_{ab}, the function Λ_{ab}, a, b is interchangeable by the suffices a and b, c, equal to that of the single-letter functions.

160. Considering the dual DE as an abbreviation for the double letter ABC, DEF, the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions $\Lambda_{ab} = \sqrt{ax_1\sqrt{ax_2}}$, &c.

In connexion with the already mentioned single-letter functions Λ_a, $= \sqrt{ax_1\, ax_2}$, etc. we have a system of formulæ as $\sqrt{ax_1\, ax_2}$, etc. and it is from these formulæ were considered by Aronhold in his (Göttingen) Memoir of 1865 (whence the term "double-letter functions"). It is in like manner be shown that the function $\Lambda_a = \sqrt{ax_1\, ax_2}$, etc. is the quotient of a major function $(x,\, y,\, z)^{n-2}$ by the linear function 012.

Chapter VII. The Addition Theory.

The Addition Theory. Art. Nos. 160 to 163.

160. We have the six single-letter symbols A, B, C, D, E, F, viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, etc. and the ten double-letter symbols AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE, viz. $\Lambda_{ab} = \dfrac{1}{a-b}\left[\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}\right]$, etc. from 10 functions being connected by algebraical relations which admit immediately deducible from the considerations of the various forms of $\Lambda_{ab}\,\Lambda_{ab}$, etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1, an integral relation, which agrees with the symbol a, of the function above considered.

The relation between the variables x_1, x_2 is that they correspond to a line meeting the curve in two points which we call 1, 2, viz. taking the equation in the form

$$(ax^2 + 2bx + c)y^2 - (a-x)(b-x)(c-x)(d-x) = 0,$$

where it is a character expression of the function $(x,\, y,\, z)^{n-2}$, $=0$, of the line is to be taken to be

$$ay^2 + bx^2 + 2c'y + d = 0,$$

and we have thus the two relations

$$(ax_1^2 + 2bx_1 + c)y_1^2 = (a-x_1)(b-x_1)(c-x_1)(d-x_1),$$

$$(ax_2^2 + 2bx_2 + c)y_2^2 = (a-x_2)(b-x_2)(c-x_2)(d-x_2),$$

$$ay_1y_2 + bx_1x_2 + c'(x_1+x_2) + d = 0\,;$$

which equations serve to determine the ratios $a:\beta:k$ in terms of x_1, x_2, y_1, y_2, and we have then the two relations

$$ax_1y_2 + bx_2y_1 + 2c'\,y_2 + d\,(\ldots),$$

$$ay_1 + bx_1 + 2c'\,y_2 + d\,(\ldots).$$

which determines the symmetric functions of x_1, x_2.

[book p. 205] The Addition Theory. Art. Nos. 160 to 163.

160. We have the six single-letter functions A, B, C, D, E, F, viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, &c.; and the ten double-letter symbols AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE, viz. $\Lambda_{ab} = \dfrac{1}{a-b}\left[\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}\right]$, &c.

These 10 functions being connected by algebraical relations which admit immediately deducible from the consideration of the various forms of $\Lambda_{ab}\,\Lambda_{ab}$ etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1, an integral relation, which agrees with the symbol a, of the function above considered.

161. The relation is, in fact, as is given in any of the papers $1, 2$ &c. of the double-$\boldsymbol{T}$-function, Crelle, t. LXXVIII. (1865), pp. 74–81, [785]. Writing successively $z = a$, β, b, b, b, a, β, b, b', a, β, b, we have

$$ax_1^2 + 2bx_1 + c = y_1\sqrt{(a-x_1)(b-x_1)(c-x_1)(d-x_1)}\,,$$
$$ax_2^2 + 2bx_2 + c = y_2\sqrt{(a-x_2)(b-x_2)(c-x_2)(d-x_2)}\,;$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2, and we have then the two relations

$$ax_1y_2 + bx_2y_1 + 2c'\,y_2 + d\,(\ldots),$$
$$ay_1 + bx_1 + 2c'\,y_2 + d\,(\ldots),$$
$$ay_1 + bx_2 + 2c'\,y_2 + d\,(\ldots),$$
$$ay_1 + bx_1 + 2c'\,y_2 + d\,(\ldots).$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2, and we have then the two relations

$$ay_1 + bx_1 + 2c'\,y_2 + d\,(\ldots),$$
$$ay_1 + bx_2 + 2c'\,y_2 + d\,(\ldots),$$

which determines the symmetric functions of x_1, x_2.

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Paper 825 (Memoir on the Abelian and Theta Functions, concl. of Ch. VI; Ch. VII—The Functions T, U, V, Θ); Paper 826 (Note on a Partition-Series).

If, reverting to the identity, we write therein for instance $x = a$, we find

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \sqrt{\mu}\, A_a A_m A_m,$$

which equation when properly reduced gives the proportional value of A_m.

162. Calling for a moment the function on the left-hand side Ω, we have

$$\Omega \ = \begin{vmatrix} x_1^3 & x_1^2 & x_1 & 1 & \sqrt{\lambda}\sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & 1 & \sqrt{\lambda}\sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & 1 & \sqrt{\lambda}\sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & 1 & \sqrt{\lambda}\sqrt{X_4} \\ a^3 & a^2 & a & 1 & \Omega \end{vmatrix} \ = \ 0,$$

that is,

$$\Omega \begin{vmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \\ x_4^2 & x_4 & 1 \end{vmatrix} \ + \ \sqrt{\lambda} \begin{vmatrix} x_1^3 & x_1^2 & x_1 & \sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & \sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & \sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & \sqrt{X_4} \end{vmatrix} = 0,$$

viz. this is

$$\begin{aligned} &\Omega\,(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_2 - x_3)(x_2 - x_4)(x_3 - x_4) \\ = \ & -\sqrt{\lambda}\,\big\{\sqrt{X_1} \cdot x_2 - x_3 \cdot x_2 - x_4 \cdot x_3 - x_4 \cdot a - x_1 \cdot x_2 - a \cdot x_3 - a \cdot x_4 - a \\ & \quad + \sqrt{X_2} \cdot x_1 - x_3 \cdot x_1 - x_4 \cdot x_3 - x_4 \cdot a - x_2 \cdot x_1 - a \cdot x_3 - a \cdot x_4 - a \\ & \quad + \sqrt{X_3} \cdot x_1 - x_2 \cdot x_1 - x_4 \cdot x_2 - x_4 \cdot a - x_3 \cdot x_1 - a \cdot x_2 - a \cdot x_4 - a \\ & \quad + \sqrt{X_4} \cdot x_1 - x_2 \cdot x_1 - x_3 \cdot x_2 - x_3 \cdot a - x_4 \cdot x_1 - a \cdot x_2 - a \cdot x_3 - a\big\}; \end{aligned}$$

or, as this may be written,

$$\begin{aligned} &\Omega \,.\, x_1 - x_2 \,.\, x_1 - x_3 \,.\, x_2 - x_3 \,.\, x_1 - x_4 \,.\, x_2 - x_4 \,.\, x_3 - x_4 \\ = \ & \frac{\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_2}{x_1 - x_2}\big[x_3 - x_2 \,.\, x_3 - x_4 \,.\, a - x_3 \,.\, \sqrt{X_4} - (x_4 - x_2 \,.\, x_4 - x_3 \,.\, a - x_4 \,.\, \sqrt{X_3})\big] \\ & + \frac{\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_4}{x_2 - x_4}\big[x_2 - x_1 \,.\, x_2 - x_3 \,.\, a - x_2 \,.\, \sqrt{X_3} - (x_3 - x_1 \,.\, x_3 - x_2 \,.\, a - x_3 \,.\, \sqrt{X_2})\big]. \end{aligned}$$

We have here $\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_2 = \sqrt{\lambda}\, A_a^2 A_m^2$, and the function

$$\frac{1}{x_3 - x_4}\big[x_2 - x_3 \,.\, x_2 - x_4 \,.\, b_0 \sqrt{X_4} - (x_1 - x_3 \,.\, x_1 - x_4 \,.\, b_0 \sqrt{X_3})\big],$$

which multiplies this, is without difficulty found to be

$$= \ \frac{-A_{12}}{c - d \,.\, d - b \,.\, b - c}\, \Sigma\,[c - d \,.\, B_{12}^x B E_a \,.\, C_a \,.\, D_a],$$

where the summation extends to the three terms obtained by the cyclical interchange of the letters b, c, d: these being a set of three out of the five letters other than a. Similarly $\sqrt{\lambda}\,.\,a-x_1\,.\,a-x_2=\sqrt{\lambda}\,A_a^2A_m^2$, and the function which multiplies this is

$$= \frac{-A_{12}}{c-d\,.\,d-b\,.\,b-c}\,\Sigma\,[c-d\,.\,B_{12}^x\,.\,BE_aC_aD_a].$$

The expression for Ω thus contains the factor $A_{12}A_{34}$: but we have

$$\Omega=\alpha a^3+\beta a^2+\gamma a+\delta=\sqrt{\mu}\,A_aA_mA_m;$$

this equation contains therefore the factor $A_{12}A_{34}$, and omitting it we find

$$-\frac{\sqrt{\mu}}{\sqrt{\lambda}}\,(x_1-x_2\,.\,x_1-x_3\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_2-x_4\,.\,x_3-x_4)\,(c-d\,.\,d-b\,.\,b-c)\,A_m$$

$$= A_m\,\Sigma\,[c-d\,.\,B_{12}^x\,BE_a^1C_a\,D_a]\;+\;A_{12}\,\Sigma\,[c-d\,.\,B_{12}^x\,BE_a^2C_a\,D_a],$$

where, as before, the summations refer each to the three terms obtained by the cyclical interchange of the letters b, c, d; these being any three of the five letters other than a: and the remaining two letters e, f enter into the formulae symmetrically. The formula thus gives for A_m ten values which are of course equal to each other.

By reason of the undetermined factor $\dfrac{\sqrt{\mu}}{\sqrt{\lambda}}$, the formula gives only the proportional value of A_m; viz. combining it with the like formulae for B_m, &c., we have determinate values of the ratios $A_m:B_m:\ldots:F_{56}$. But this being understood, we regard the formula as a formula for each single-letter function of x_3, x_4 in terms of the single and double-letter functions of x_1, x_2 and of x_3, x_4 respectively.

163. We require further the expressions for the double-letter functions of x_3, x_4.

Consider for example the function DE_m, which is

$$= \frac{1}{x_3-x_4}\left[\sqrt{d_3\,e_3\,f_3\,a_4\,b_4\,c_4}-\sqrt{d_4\,e_4\,f_4\,a_3\,b_3\,c_3}\right];$$

then multiplying by $A_m\,B_m\,C_m=\sqrt{a_3\,b_3\,c_3\,a_4\,b_4\,c_4}$, we have

$$DE_m\,A_m\,B_m\,C_m \;=\; \frac{1}{x_3-x_4}\left[a_3b_3c_3\sqrt{X_4}-a_4b_4c_4\sqrt{X_3}\right],$$

or recollecting that $\sqrt{\lambda}\,\sqrt{X_3}$ and $\sqrt{\lambda}\,\sqrt{X_4}$ are $=\alpha x_3^3+\beta x_3^2+\gamma x_3+\delta$ and $\alpha x_4^3+\beta x_4^2+\gamma x_4+\delta$ respectively, this may be written

$$\sqrt{\lambda}\,DE_m\,A_mB_mC_m \;=\; \frac{1}{x_3-x_4}\big\{(a-x_3\,.\,b-x_3\,.\,c-x_3)(\alpha x_4^3+\beta x_4^2+\gamma x_4+\delta)$$

$$-(a-x_4\,.\,b-x_4\,.\,c-x_4)(\alpha x_3^3+\beta x_3^2+\gamma x_3+\delta)\big\}.$$

Using the well-known identity

$$\alpha x^3+\beta x^2+\gamma x+\delta \;=\; \Sigma\,\frac{\alpha a^3+\beta a^2+\gamma a+\delta}{a-b\,.\,a-c\,.\,a-d}\,(b-x\,.\,c-x\,.\,d-x),$$

where the summation extends to the four terms obtained by the cyclical interchanges of the letters a, b, c, d: and the like identity for $\alpha x^3+\beta x^2+\gamma x+\delta$, there will be terms in $\alpha a^3+\beta a^2+\gamma a+\delta$, $\alpha b^3+\beta b^2+\gamma b+\delta$, $\alpha c^3+\beta c^2+\gamma c+\delta$, but the term in $\alpha d^3+\beta d^2+\gamma d+\delta$ will disappear of itself. After some easy reductions, the result is

$$\sqrt{\lambda}\, DE_m\, A_m B_m C_m = \Sigma\, \frac{\alpha a^3+\beta a^2+\gamma a+\delta}{b-a\,.\,c-a\,.\,d-a}\, B^x_{12} C^x_{12},$$

where the summation extends to the three terms obtained by the cyclical interchanges of the letters a, b, c: We have $\alpha a^3+\beta a^2+\gamma a+\delta = \sqrt{\mu}\, A_a A_m A_m$ and similarly for the other two terms; the whole equation thus divides by $A_m B_m C_m$, and we find

$$-\frac{\sqrt{\mu}}{\sqrt{\lambda}}\,(x_1-x_3\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_2-x_4)\, DE_m = \frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4}\, M,$$

where M is a given rational and integral function of the single and double-letter functions of x_1, x_2 and x_3, x_4. The factor on the left-hand side has been made the same as in the formula for the single-letter functions A_m, &c., and to do this it was necessary to bring in on the right-hand side the factor

$$\frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4};$$

this disappears in the expression for the ratio of two double-letter functions; but it enters into the expression for the ratio of a single-letter to a double-letter function, and it then requires to be itself expressed in terms of the functions of x_1, x_2 and x_3, x_4: it is easy to see that we have

$$\frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4} = \Sigma\, \frac{(A^x_{12}B^x_{12} - A^x_{34}B^x_{34})\,(A^x_{12}C^x_{12} - A^x_{34}C^x_{34})}{(a-b)(a-c)},$$

where the summation extends to the three terms obtained by the cyclical interchanges of the letters a, b, c: these being a set of any three out of the six letters.

We have, in what precedes, obtained the expressions for the ratios of the 16 functions $A_m, \ldots, F_{56}, AB_m, \ldots, DE_m$ in terms of the ratios of the like functions of x_1, x_2 and x_3, x_4.

CHAPTER VII. THE FUNCTIONS T, U, V, Θ.

The present chapter is substantially a reproduction of C. and G.'s seventh section, "Die Function T^{κ}_{μ}," borrowing only from the next section the definition of the theta-function; but for greater simplicity I consider for the most part, the case, fixed curve a quartic, $n = 4$, $p = 3$.

Integral Form of the Affected Theorem. Art. Nos. 164 to 169.

164. Writing for shortness $\dfrac{(x, y, z)_0^{n-2}\, d\omega}{012} = d\Pi_{12}$, we are concerned with the integrals $\displaystyle\int_{a'}^{a} d\Pi_{12}$ which present themselves in connexion with the affected theorem: the notation is explained, Chap. V.; a, a' are points on the curve f; the variable may be any parameter serving for the determination of the current point, and the integral, taken from the value which belongs to the point a' to the value which belongs to the point a, is represented as above by means of the two points a, a' as limits of the integral. It is assumed that the integral is a canonical integral having the limits and the parametric points interchangeable, $\displaystyle\int_{a'}^{a} d\Pi_{12} = \int_{2}^{1} d\Pi_{aa'}$: see Chapter IV.

165. Writing for shortness

$$\left(\int_{a'}^{a} + \int_{b'}^{b} + \int_{c'}^{c} + \ldots\right) d\Pi_{12} \;=\; \int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12},$$

then if ϕ, ψ are curves each of the order m, the former of them intersecting the fixed curve f in the points a, b, c, ..., and the latter of them intersecting the same curve in the points a', b', c',..., and if ϕ_1, ϕ_2, ψ_1, ψ_2 are what the functions ϕ, ψ become on substituting therein in place of the current coordinates the values which belong to the parametric points 1, 2 respectively; the theorem becomes

$$\int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12} \;=\; \log \frac{\phi_1 \psi_2}{\phi_2 \psi_1}.$$

The superior limits may be interchanged in any manner, and so also the inferior limits may be interchanged in any manner. If a superior limit coincide with an inferior limit, the two may be thus considered as belonging to an integral which will then have the value 0, and the coincident points may therefore be omitted from the expression on the left-hand side: and so in the case of any number of coincidences.

166. If the intersections of the curves ϕ, ψ and the parametric points are situate on a curve of the order m; then taking the equation of this curve to be $\phi + \lambda\psi = 0$, we have simultaneously $\phi_1 + \lambda\psi_1 = 0$, $\phi_2 + \lambda\psi_2 = 0$; whence $\phi_1\psi_2 = \phi_2\psi_1$, and the logarithmic term disappears: viz. the theorem becomes

$$\int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12} \;=\; 0.$$

167. Suppose that the curves ϕ, ψ are each of them a major curve, that is, a curve of the order $n-2$ passing through the δ dps, and consequently besides meeting the curve f in $n(n-2)-2\delta$, $=2p+n-2$ points: the theorem is

$$\int\begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots\end{pmatrix} d\Pi_{12} \;=\; \log\frac{\phi_1\psi_2}{\phi_2\psi_1},$$

where the numbers of the superior and of the inferior points are each $=2p+n-2$.

168. Suppose further that the curves ϕ, ψ, being major curves as above, pass each of them through the $n-2$ residues of 1, 2; they besides meet in $(n-2)(n-3)$ points, viz. these are the δ dps and $(n-2)(n-3)-\delta$ variable points: these $(n-2)(n-3)$ points lie on a minor curve, that is, a curve of the order $n-3$ passing through the dps; and the minor curve together with the parametric line 12 make together a major curve, passing through the intersections of ϕ, ψ and also through the parametric points 1, 2: viz. these points and the intersections of ϕ, ψ are situate on a curve of the order $n-2$; the logarithmic term thus vanishes. The intersections of ϕ with the fixed curve are the δ dps, the $n-2$ residues and $2p$ other points, say these are $a,\ b,\ c,\ldots,\ a^x,\ b^x,\ c^x,\ldots$; similarly the curve ψ meets the fixed curve in the δ dps, the $n-2$ residues, and in $2p$ other points, say these are $d,\ e,\ f,\ldots,\ d^x,\ e^x,\ f^x,\ldots$: the theorem is

$$\int\begin{pmatrix} a,\ b,\ c,\ldots;\ a^x,\ b^x,\ c^x,\ldots \\ d,\ e,\ f,\ldots;\ d^x,\ e^x,\ f^x,\ldots\end{pmatrix} d\Pi_{12} \;=\; 0,$$

where there are $2p$ superior and inferior points respectively.

169. I introduce the definitions: a minor curve meets the fixed curve in the dps and in $2p-2$ other points, called "cominors": a major curve passing through the $n-2$ residues of the points 1, 2, meets the fixed curve in the δ dps, the $n-2$ residues and in $2p$ other points, called "comajors in regard to the points 1, 2." Observe that $p-1$ of the cominors determine uniquely the remaining $p-1$ cominors; and similarly p of the comajors determine uniquely the remaining p comajors.

The foregoing theorem thus is that the sum $\int(\ )\,d\Pi_{12}=0$, when the superior points and the inferior points are each of them a system of comajors in regard to the parametric points 1, 2.

Fixed Curve a Quartic. Art. No. 170.

170. It would be easy to go on with the general form; but as already mentioned, I prefer to consider the case, fixed curve a quartic, $n=4$, $p=3$. A minor curve is here a line meeting the quartic in 4 points, which are "cominors"; the major curve is a conic, and if this passes through the residues of 1, 2 it besides meets the quartic in 6 points, which are "comajors in regard to the points 1, 2." Two points and their residues are cominors, but this is only by reason that $n-3=1$.

The Function T. Art. No. 171.

171. In conformity with C. and G., I introduce the functional symbol

$$T_{12}\begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots\end{pmatrix} \;=\; \int\begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots\end{pmatrix} d\Pi_{12},$$

so that T denotes a function of the parametric points, and of the sets of superior and inferior points respectively. The foregoing theorem for the quartic thus is

$$T_{12}\begin{pmatrix} a,\ b,\ c,\ a^x,\ b^x,\ c^x \\ d,\ e,\ f,\ d^x,\ e^x,\ f^x \end{pmatrix} = 0.$$

Observing that

$$\int_d^a + \int_{d^x}^{a^x} = 2\int_2^1 - \int_{a^x}^d + \int_{d^x}^a,$$

and so in other cases, this may be written

$$T_{12}\begin{pmatrix} a,\ b,\ c \\ d,\ e,\ f \end{pmatrix} = T_{12}\begin{pmatrix} a^x,\ b^x,\ c^x \\ a,\ b,\ c \end{pmatrix} - T_{12}\begin{pmatrix} d,\ e,\ f \\ d^x,\ e^x,\ f^x \end{pmatrix};$$

and if, as a definition of $T_{12}(a,b,c)$, we write

$$T_{12}(a,b,c) = T_{12}\begin{pmatrix} a,\ b,\ c \\ a^x,\ b^x,\ c^x \end{pmatrix} - 2\int_2^1,$$

where a^x, b^x, c^x are the comajors of a, b, c in regard to 1, 2, then the equation is

$$T_{12}\begin{pmatrix} a,\ b,\ c \\ d,\ e,\ f \end{pmatrix} = T_{12}(a,b,c) - T_{12}(d,e,f),$$

viz. the function of the $(2p+2=)8$ points 1, 2, a, b, c, d, e, f is here expressed as a difference of two functions each of $(p+2=)5$ points: $T_{12}(a,b,c)$ is regarded as a function of the 5 points 1, 2, a, b, c, because the remaining points a^x, b^x, c^x depend only on these 5 points.

The Function U. Art. Nos. 172 to 175.

172. We consider on the quartic the points ξ, μ; 1, 2, 3; and take f, f' for the cominors of 2, 3; g, g' for the cominors of 3, 1; and h, h' for the cominors of 1, 2. We write

$$T = T_\mu(1,2,3),$$

it is to be shown that there exists a function $U(1,2,3;\xi)$ such that

$$\delta U = \tfrac{1}{2}\left[\delta_\xi T + \delta_1 T_1 + \delta_2 T_2 + \delta_3 T_3\right],$$

viz. considering ξ, 1, 2, 3 as variable points on the quartic, the whole infinitesimal variation of U is the sum of these parts, where $\delta_\xi T$ is the variation of T when only ξ is varied, $\delta_1 T_1$ the variation of T_1 when only 1 is varied, and similarly for $\delta_2 T_2$ and $\delta_3 T_3$. We consider in the proof three other points 4, 5, 6 on the quartic; and taking l, l' for the cominors of 5, 6; m, m' for those of 6, 4; and n, n' for those of 4, 5, we write further

$$X_1 = T_\mu(4,l,l'), \qquad X_2 = T_\mu(5,m,m'), \qquad X_3 = T_\mu(6,n,n').$$

and it then requires to be shown that

$$\begin{aligned}
\tfrac{1}{2}\,\delta T &= \int \binom{123}{456} d\Pi_{1\mu} + \tfrac{1}{2}\,T_{1\mu}(4,5,6),\\
\tfrac{1}{2}(T_1 - X_1) &= \int \binom{456}{423} d\Pi_{1\mu} + \log \frac{\mu 23\,.\,156}{\mu 56\,.\,123},\\
\tfrac{1}{2}(T_2 - X_2) &= \int \binom{564}{531} d\Pi_{1\mu} + \log \frac{\mu 31\,.\,264}{\mu 64\,.\,231},\\
\tfrac{1}{2}(T_3 - X_3) &= \int \binom{645}{612} d\Pi_{1\mu} + \log \frac{\mu 12\,.\,345}{\mu 45\,.\,312},
\end{aligned}$$

where $\mu 23$ is the determinant formed with the coordinates of the points μ, 2, 3 respectively; and so in other cases.

173. We have

$$T_{1\mu}\begin{pmatrix}1,\ 2,\ 3\\ 4,\ 5,\ 6\end{pmatrix} = \tfrac{1}{2}\,T_{1\mu}(1,2,3) - \tfrac{1}{2}\,T_{1\mu}(4,5,6),$$

that is,

$$= \tfrac{1}{2}\,T - \tfrac{1}{2}\,T_{1\mu}(4,5,6),$$

and thence the above value of $\frac{1}{2}\,\delta T$.

The affected theorem gives

$$\int \begin{pmatrix}f,\ f',\ 2,\ 3\\ l,\ l',\ 5,\ 6\end{pmatrix} d\Pi_{1\mu} = \log \frac{F_1 L_\mu}{F_\mu L_1},$$

where $F = 0$ is the equation of the line through f, f', 2, 3; and F_1, F_μ are what the function F becomes, on substituting therein for the current coordinates the coordinates of the points 1, μ respectively. And similarly $L = 0$ is the equation of the line through l, l', 5, 6; and L_1, L_μ are what the function L becomes by the same sub stitutions respectively. The values of F_1, F_μ are 123, $\mu 23$: those of L_1, L_μ are 156, $\mu 56$: and the logarithmic term is thus

$$= \log \frac{\mu 23\,.\,156}{\mu 56\,.\,123}.$$

We then have

$$\tfrac{1}{2}(T_1 - X_1) = \int \begin{pmatrix}f,\ f',\ 2,\ 3\\ 4,\ l,\ l'\end{pmatrix} d\Pi_{1\mu} = \int \begin{pmatrix}f,\ f',\ 2,\ 3\\ l,\ l',\ 5,\ 6\end{pmatrix} d\Pi_{1\mu},$$

and in this last expression for $\frac{1}{2}(T_1 - X_1)$ substituting for the second term the logarithmic value just obtained, we have the required expression for $\frac{1}{2}(T_1 - X_1)$: and those for $\frac{1}{2}(T_2 - X_2)$ and $\frac{1}{2}(T_3 - X_3)$ are deduced by mere cyclical permutations of the letters.

174. Returning to the assumed relation $\delta U = \frac{1}{2}\,[\delta_\xi T + \delta_1 T_1 + \delta_2 T_2 + \delta_3 T_3]$, in order to the existence of the function U, it is only necessary to show that $T - T_1$ contains no term in 1, ξ, that is, no term depending on both these points, and that $T_1 - T_2$ contains no term in 1, 2; for then, by symmetry, the like properties hold in regard to $T - T_2$, $T - T_3$, $T_1 - T_3$, $T_2 - T_3$ respectively, and the assumed expression is a complete differential, from which the function U may be obtained by integration.

175. To show that $T - T_1$ contains no term in 1, ξ.

For T, the only term in 1, ξ is $\int_1^\xi d\Pi_{1\mu}$,

" T_1 " " " $\int_1^\xi d\Pi_{1\mu}$,

and it is to be shown that the difference of the two integrals contains no term in 1, ξ. Considering on the quartic the two new points δ, e, the first integral is

$$\int \begin{pmatrix} 1, \xi \\ \delta, e \end{pmatrix} (d\Pi_{1\mu} + d\Pi_{1\mu}) \;=\; \int_\delta^1 d\Pi_{1\mu} + \int_e^\xi d\Pi_{1\mu} + \int_1^\xi d\Pi_{1\mu},$$

and the second is

$$\int \begin{pmatrix} \xi, e \\ \delta, 4 \end{pmatrix} (d\Pi_{1\mu} + d\Pi_{1\mu}) \;=\; \int_\delta^\xi d\Pi_{1\mu} + \int_4^e d\Pi_{1\mu} + \int_1^4 d\Pi_{1\mu}.$$

Hence in the difference the only terms which can contain 1, ξ are

$$\int_e^1 d\Pi_{1\mu} + \int_\xi^4 d\Pi_{1\mu},$$

and this is $= 0$: wherefore there is not in the difference any term in 1, ξ. This proves the property for $T - T_1$. The property for $T_1 - T_2$ is proved in a similar manner.

Theorems in regard to the Function U. Art. Nos. 176 to 179.

176. Theorem (A). To prove

$$U(1,2,3;\xi) - U(1,2,3;\mu) \;=\; \tfrac{1}{2}\, T_{\xi\mu}(1,2,3), \tag{A}$$

we have

$$\begin{aligned} U(1,2,3;\xi) - U(1,2,3;\mu) &= \int_\mu^\xi d_\xi U = \tfrac{1}{2}\int_\mu^\xi d_\xi T \\ &= \tfrac{1}{2}\, T_\mu(1,2,3) - \tfrac{1}{2}\, T_\mu(1,2,3), \end{aligned}$$

and $T_{\xi\mu}(1,2,3) \;=\; \int \begin{pmatrix} 1,\ 2,\ 3 \\ 1^x,\ 2^x,\ 3^x \end{pmatrix} d\Pi_{\xi\mu}$, where $d\Pi_{\xi\mu} = 0$, viz. considering this as derived from $d\Pi_{\xi\mu}$, $=$ $Q_{\xi\mu} d\omega$, by making the point ξ coincide with μ, then when ξ is indefinitely near to μ, the numerator and denominator of $Q_{\xi\mu}$ are each of them infinitesimal of the orders 3 and 2 respectively, and thus the function $Q_{\xi\mu}$ ultimately vanishes (see as to this, Chap. V. Art. Nos. 99 to 106). We have therefore $T_{\xi\mu}(1,2,3) = 0$, and the required theorem is proved.

177. Theorem (B). To prove

$$U(1,2,3;\xi) - U(4,2,3;\xi) \;=\; \tfrac{1}{2}\, T_{1\xi}(f, f'), \tag{B}$$

where, as before, f, f' are the cominors of 2, 3, that is, 2, 3, f, f' lie on a line, we have

$$\begin{aligned} U(1,2,3;\xi) - U(4,2,3;\xi) &= \int_4^1 d_1 U = \tfrac{1}{2}\int_4^1 d_1 T \\ &= \tfrac{1}{2}\, T_\mu(\xi, f, f') - \tfrac{1}{2}\, T_{4\mu}(\xi, f, f'), \end{aligned}$$

the point μ is arbitrary, and it may be taken to coincide with 4; but we then have $T_{4\mu}(\xi, f, f') = 0$, and the theorem is thus proved.

178. Theorem (C). We have

$$T_{1\xi^x}(\xi, f, f') + T_{1\xi^x}(\eta, k, k') \;=\; 0; \tag{C}$$

where ξ, η, 1, 2, 3 are arbitrary points on the quartic; 1^x, 2^x, 3^x are the comajors of 1, 2, 3 in regard to ξ, η, viz. the points 1, 2, 3, 1^x, 2^x, 3^x lie on a conic which passes through ξ, η: the residues (or cominors) of ξ, η: f, f' are the cominors of 2, 3; and k, k' are the cominors of 2^x, 3^x.

Taking θ, θ' for the cominors of 1, 1^x, the four lines $\xi\eta\theta\theta'$, $11^x\theta\theta'$, $23ff'$ and $2^x3^xkk'$ form a quartic cutting the fixed quartic in the 16 points: but of these, ξ, η, 1, 2, 3, 1^x, 2^x, 3^x lie in a conic: hence the remaining 8 points θ, θ', ξ, η, f, f', k, k' lie in a conic: that is, ξ, η, f, f', k, k' lie on a conic through θ, θ', the residues of 1, 1^x: or they are comajors in regard to 1, 1^x; whence the theorem.

179. We have

From A. *From B.*

$$\tfrac{1}{2}T_{1\xi^x}(\xi, f, f') = U(f, f'; 1) - U(f, f'; 1^x) = U(1, 2, 3; \xi) - U(1^x, 2, 3; \xi),$$
$$\tfrac{1}{2}T_{1\xi^x}(\eta, k, k') = U(\eta, k, k'; 1) - U(\eta, k, k'; 1^x) = U(1, 2^x, 3^x; \eta) - U(1^x, 2, 3; \eta);$$

viz. we have thus two expressions for each term of the equation (C),

$$T_{1\xi^x}(\xi, f, f') + T_{1\xi^x}(\eta, k, k') \;=\; 0.$$

In particular, we have Theorem (D)

$$U(1, 2, 3; \xi) - U(1^x, 2, 3; \xi) \;=\; -U(1, 2^x, 3^x; \eta) + U(1^x, 2^x, 3^x; \eta). \tag{D}$$

Again, we have $T_{1\xi}(1, 2, 3) + T_{1\xi}(1^x, 2^x, 3^x) = 0$; where 1, 2, 3, 1^x, 2^x, 3^x are comajors in regard to ξ, η: and

$$\tfrac{1}{2}\, T_{\xi\eta}(1, 2, 3) = U(1, 2, 3; \xi) - U(1, 2, 3; \eta),$$
$$\tfrac{1}{2}\, T_{\xi\eta}(1^x, 2^x, 3^x) = U(1^x, 2^x, 3^x; \xi) - U(1^x, 2^x, 3^x; \eta);$$

whence Theorem (E),

$$U(1, 2, 3; \xi) - U(1, 2, 3; \eta) \;=\; -U(1^x, 2^x, 3^x; \xi) + U(1^x, 2^x, 3^x; \eta). \tag{E}$$

The Function V. Art. Nos. 180 to 182.

180. It is convenient to consider U as a logarithm, say

$$-U(1, 2, 3; \xi) \;=\; \log V(1, 2, 3; \xi), \qquad \text{or} \qquad V(1, 2, 3; \xi) = \exp. {-U(1, 2, 3; \xi)};$$

V, like U, is a function of the $(p + 1 =)\,4$ points 1, 2, 3, ξ, on the quartic.

The equation (D) thus becomes

$$\frac{V(1, 2, 3; \xi)\; V(1^x, 2, 3; \xi)}{V(1^x, 2^x, 3^x; \eta)\; V(1, 2^x, 3^x; \eta)},$$

where 1, 2, 3, 1^x, 2^x, 3^x are comajors in regard to ξ, η: the equation shows that, in the function $\dfrac{V(1,2,3;\xi)}{V(1^x,2^x,3^x;\eta)}$, we can without alteration of the value interchange a pair of points 1, 1^x out of the system of comajor points; and it of course follows that we can in any manner whatever interchange these points, so as to have any three of them in the numerator-function and the remaining three in the denominator-function. In particular, we have

$$\frac{V(1,2,3;\xi)}{V(1^x,2^x,3^x;\eta)} = \frac{V(1^x,2^x,3^x;\xi)}{V(1,2,3;\eta)}.$$

The equation (E) becomes

$$\frac{V(1,2,3;\xi)\,V(1^x,2^x,3^x;\eta)}{V(1,2,3;\eta)\,V(1^x,2^x,3^x;\xi)},$$

and multiplying we find

$$V(1,2,3;\xi)^2 = V(1^x,2^x,3^x;\eta)^2,$$

that is, $V(1,2,3;\xi) = \pm V(1^x,2^x,3^x;\eta)$, the sign being determinately + or determinately −, according to the precise definition of the function V.

181. Considering ξ as fixed points on the curve but η as also; and 1^x, 2^x, 3^x a variable point (that is, the parametric line $\xi\eta$ as rotating about the fixed point η), the points 1, 2, 3 are then determined as the remaining intersections with the quartic of the conic which passes through the points 1^x, 2^x, 3^x and through the points ξ, η, which are the residues of ξ, η. And by the theorem just obtained it appears that 1, 2, 3, thus varying, the function $V(1,2,3;\xi)$ remains constant. This comes to saying that V, considered as a function of the points 1, 2, 3, ξ, satisfies a certain linear partial differential equation of the first order, having a solution $V = F(u,v,w)$, an arbitrary function of u, v, w, determinate functions of the points 1, 2, 3, ξ. And if we can find u, v, w functions of these points such that they each of them remain constant when the points 1, 2, 3, ξ vary as above, then the arbitrary function of u, v, w will remain constant for the variation in question and will thus be a value of the function V.

182. It is easily seen that such functions are

$$u,\ v,\ w\ =\ \left(\int^1 + \int^2 + \int^3 - \int^\xi\right) x\,d\omega,\quad y\,d\omega,\quad z\,d\omega,$$

the inferior limits being given points which are regarded as absolute constants. For by the pure theorem, we have

$$\Sigma\,(x,y,z)_1\,d\omega = 0,$$

where $(x,y,z)_1$ is an arbitrary linear function, and where the summation extends to all the intersections of the quartic with any given curve. Writing

$$p = \int x\,d\omega,\ \int y\,d\omega \text{ or } \int z\,d\omega,\quad \text{that is,}\quad p = \int^1 x\,d\omega,\ \int^1 y\,d\omega \text{ or } \int^1 z\,d\omega,$$

the inferior limits being any absolutely fixed point on the curve, and similarly p_2, &c.; the integral form of the theorem is $\Sigma p =$ constant. And applying the theorem successively to the parametric line, and to the conic which determines the points 1, 2, 3, we have

$$p_\xi + p_\eta + p_{\xi'} + p_{\eta'} = \text{const.},$$
$$p_\xi + p_\eta + p_1 + p_2 + p_3 + p_{1^x} + p_{2^x} + p_{3^x} = \text{const.}$$

Taking the difference of these equations, we have

$$p_1 + p_2 + p_3 - p_{\xi'} + p_{1^x} + p_{2^x} + p_{3^x} - p_{\eta'} \;=\; \text{const.},$$

viz. the points η, 1^x, 2^x, 3^x being fixed points, this is

$$p_1 + p_2 + p_3 - p_\xi \;=\; \text{const.},$$

that is, the functions u, v, w defined as above are each of them constant under the variation in question.

The Function Θ. *Art. Nos. 183 and 184.*

183. The function $V(1,2,3;\xi)$ of the $(p+1=)\,4$ points 1, 2, 3, ξ, is thus a function of the $(p=)\,3$ arguments

$$u,\ v,\ w \;=\; \left(\int^1 + \int^2 + \int^3 - \int^\xi\right) x\,d\omega,\ \ y\,d\omega,\ \ z\,d\omega.$$

Disregarding a constant and exponential factor, we say that it is a theta-function of these arguments, and we write the result provisionally in the form

$$V(1,2,3;\xi) \;=\; \Theta(u,v,w),$$

the more precise definition of the theta-function being reserved for further consideration.

184. It appears by what precedes that a sum of $(p=)\,3$ integrals $\int \begin{pmatrix} abc \\ def \end{pmatrix} d\Pi_{12}$, otherwise called $T_{12}\binom{a,\,b,\,c}{d,\,e,\,f}$, is in the first place expressed (see No. 171) as a difference, $= \frac{1}{2}\,T_{12}(a,b,c) - \frac{1}{2}\,T_{12}(d,e,f)$, of two functions T. Each of these is by Theorem (A) (No. 176) expressed as a difference of two functions U, that is, as the difference of the logarithms, or logarithm of the quotient, of two functions V: such function V according to its original definition a function of $(p+1=)\,4$ points, but in such wise that the function is expressible as a function of $(p=)\,3$ arguments, and so expressed it is a Θ-function of these arguments: and the final result thus is that the sum of $(p=)\,3$ integrals $\int \begin{pmatrix} a,b,c \\ d,e,f \end{pmatrix} d\Pi_{12}$ is equal to the logarithm of a fraction, whereof the numerator and the denominator are each of them a product of two Θ-functions.

826.

NOTE ON A PARTITION-SERIES.

[From the *American Journal of Mathematics*, vol. vi. (1884), pp. 63, 64.]

PROF. SYLVESTER, in his paper, "A Constructive theory of Partitions, &c.," *American Journal of Mathematics*, vol. v. (1883), p. 282, has given the following very beautiful formula

$$(1+ax)(1+ax^2)(1+ax^3)\ldots \;=\; 1+\frac{(1+ax^2)\,xa}{1-x}+\frac{1}{1-x\,.\,1-x^2}(1+ax)(1+ax^3)\,x^4a^2$$

$$+\frac{1}{1-x\,.\,1-x^2\,.\,1-x^3}(1+ax)(1+ax^2)(1+ax^4)\,x^{12}a^3+\ldots,$$

or, as this may be written,

$$\Omega \;=\; 1+P+Q(1+ax)+R(1+ax)(1+ax^2)+S(1+ax)(1+ax^2)(1+ax^3)+\ldots$$

where

$$P=\frac{(1+ax^2)\,xa}{\mathbf{1}},\qquad Q=\frac{(1+ax^4)\,x^5a^2}{\mathbf{1.2}},\qquad R=\frac{(1+ax^6)\,x^{12}a^3}{\mathbf{1.2.3}},\qquad S=\frac{(1+ax^8)\,x^{22}a^4}{\mathbf{1.2.3.4}},\ \text{\&c.},$$

the heavy figures **1**, **2**, **3**, **4**, ... of the denominators being, for shortness, written to denote $1-x$, $1-x^2$, $1-x^3$, $1-x^4$, ... respectively. The x-exponents 1, 5, 12, 22, ... are the pentagonal numbers $\frac{1}{2}(3n^2-n)$.

To prove this, writing

$$P'=\frac{ax^2}{\mathbf{1}},\quad Q'=\frac{ax^4}{\mathbf{1}}+\frac{a^2x^6}{\mathbf{1.2}},\quad R'=\frac{ax^6}{\mathbf{1}}+\frac{a^2x^9}{\mathbf{1.2}}+\frac{a^3x^{12}}{\mathbf{1.2.3}},\quad S'=\frac{ax^8}{\mathbf{1}}+\frac{a^2x^{11}}{\mathbf{1.2}}+\frac{a^3x^{15}}{\mathbf{1.2.3}}+\frac{a^4x^{18}}{\mathbf{1.2.3.4}},\ \text{\&c.},$$

where the x-exponents are

$$2;\quad 3,\ 3+4;\quad 4,\ 4+5,\ 4+5+6;\quad 5,\ 5+6,\ 5+6+7,\ 5+6+7+8;\ \text{\&c.},$$

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR CAYLEY
Vol. XII (Scan PDF pp. 263–275; book pp. 243–255)
Paper 828 — A Memoir on Seminvariants (Arts. 7–36).

7. In forming a product $lmn\ldots pqr\ldots$, we may have equalities among the numbers l, m, n, $\ldots$ of the first symbol, and also between the numbers p, q, r, $\ldots$ of the second symbol: moreover (whether there are or are not any such equalities), we may have equalities presenting themselves between the numbers $l+p$, $m+q$, $\ldots$ of any symbol $(l+p)(m+q)\ldots$ on the right-hand side of the equation: and the process must be performed so as to take account of all these equalities. The actual process is best shown by an example: say we require the product $3332 \cdot 322$, which is of the deg. weight $6 \cdot 18$.

3332	322			÷12	
322		6	6552	2	1
32 2		12	6543	1	1
22 3		6	5553	6	3
32	2	12	65322	2	2
3 2	2	6	64332	2	1
2 3	2	6	55332	4	2
22	3	6	55332	4	2
2 2	3	3	54333	6	3
3	22	2	632322	12	2
3	22	1	533322	12	1
2	32	6	533322	12	6
2	32	2	433332	36	6
	322	1	3333222	144	12
		73			

8. Observe first that $\{4, 3\} = 24 + 36 + 12 + 1, = 73$. In placing all or any of the numbers of the 322 under those of the 3332, we do this in all the really distinct ways, inserting a numerical coefficient for the frequency of each way. Thus when the whole of the 322 is thus placed, the 3 may be under a 3, and the two 2's may be under 33 or under 32; or else the 3 may be under a 2, and the two 2's must then be under 33: there are thus three ways, and these have the frequencies 6, 12, 6 respectively. For as to the first way, the 3 may be under any one of the three 3's, and for each such position of the 3, the 22 can be placed in either of two orders under the other two 3's; the frequency is thus $3 \cdot 2, = 6$. And similarly the other two frequencies are 12 and 6; the sum $6 + 12 + 6, = 24$, is, the first term of $\{4, 3\}$. In like manner, when two of the numbers of the 322 are placed under the 3332, there are five ways having the frequencies 12, 6, 6, 6, 6 respectively, the sum of these frequencies is 36, which is the second term of $\{4, 3\}$. And so when one number of the 322 is placed under the 3332, there are four ways having the frequencies 3, 1, 6, 2 respectively: the sum of these frequencies is 12, which is the third term of $\{4, 3\}$. Finally, when no number of the 322 is placed under the 3332, there is a single way, having the frequency 1, which is the last term of $\{4, 3\}$. These agreements are a useful verification for the frequencies: the sum of all the frequencies is of course $= 73$, the value of $\{4, 3\}$.

9. We next form the column of symmetric functions by adding to each line the 3332: to avoid accidental errors of addition, it is proper to verify for each of these that the weight is the sum, $= 18$, of the weights 11 and 7 of the two factors respectively. It is to be observed that the same symmetric function may present itself more than once: thus we have the functions 55332, and 533322, each of them twice. We then form a column of multiplicities: in 5553, the three 5's give 6: in 63322, the two 3's and the three 2's give $2 \cdot 6$, $= 12$, and so on. There is in like manner for the factor 3332 a multiplicity 6, and for the factor 322 a multiplicity 2; and these combined together give $6 \cdot 2$, $= 12$, viz. this is the heading $\div 12$ of the column. And then forming the products $6 \cdot 2$, $12 \cdot 1$, $6 \cdot 6$, &c., of the corresponding frequencies and multiplicities and dividing in each case by 12, we have the right-hand column of numerical coefficients.

10. The result is to be read

$$3332 \cdot 322 \; = \; 1 \cdot 6552 + 1 \cdot 6543 + 3 \cdot 5553 + \&\text{c.},$$

the coefficients 2, 2 and 1, 6 of the repeated terms being of course united together, so that we have

$$+ 4 \cdot 55332 \qquad + \; 7 \cdot 533322.$$

With a little practice, the operation is performed without difficulty and with very small risk of error.

11. We may apply the process to obtain analytical formulae for certain forms of products. Consider, for instance, the product $2^\alpha \cdot 2^\beta$, where $\alpha > \beta$ or $= \beta$: and in the coefficients write for shortness α, instead of $[\alpha]^\alpha$ or $\Pi\alpha$, to denote $1 \cdot 2 \cdot 3 \ldots \alpha$. We have

$$\frac{2^\alpha \mid 2^\beta}{2^A \,\big|\, 2^{\beta - A}} \; = \; \frac{\alpha}{A \cdot \alpha - A \cdot \beta - A} \cdot 4^A 2^{\alpha+\beta-2A}, \qquad A \cdot \alpha + \beta - 2A \cdot \frac{\alpha + \beta - 2A}{\alpha - A \cdot \beta - A},$$

viz. the formula is

$$2^\alpha \cdot 2^\beta \; = \; \Sigma \, \frac{\alpha + \beta - 2A}{\alpha - A \cdot \beta - A} \cdot 4^A 2^{\alpha+\beta-2A},$$

where A has any integer value from β to 0; the first term is $= 1 \cdot 4^\beta 2^{\alpha - \beta}$, and the last term is $\dfrac{\alpha + \beta}{\alpha \cdot \beta} \cdot 2^{\alpha+\beta}$. The weight of any term is $4A + 2(\alpha + \beta - 2A)$, $= 2\alpha + 2\beta$, as it should be.

In explanation as to the frequency, observe that out of the β 2's we take any A 2's and place them in any order under any A of the α 2's. The number of combinations taken is $\frac{\beta}{A \cdot \beta - A}$, which (in A orders) gives $\frac{\beta}{\beta - A}$ sets to be placed under $\frac{\alpha}{A \cdot \alpha - A}$ sets out of the α 2's: we thus have the foregoing coefficient of frequency $\frac{\alpha}{A \cdot \alpha - A} \cdot \frac{\beta}{\beta - A}$, where as mentioned above α, &c. are written to denote $[\alpha]^\alpha$, &c.

12. We may in like manner find a formula for the product $3^\alpha 2^\beta \cdot 3^\gamma 2^\delta$. Taking $\gamma, = x+y+z$, any partition of γ, and $\delta, = p+q+r$, any partition of δ, and writing also $A = x$, $B = y+p$, $C = q$, we have

$$\frac{3^\alpha 2^\beta \mid 3^\gamma 2^\delta}{3^A 2^B 3^C 2^E \mid 3^B 2^C \mid 6^A 5^B 4^C 3^{\alpha+\gamma-2A-B} 2^{\beta+\delta-B-2C} \mid N}$$

where for the frequency, M, we have

$$\text{No. of terms } 3^x = \frac{\gamma}{x \cdot \gamma - x}, \qquad 3^y = \frac{\gamma - x}{y \cdot z},$$

$$, \qquad 2^p = \frac{\delta}{p \cdot \delta - p}, \qquad 2^q = \frac{\delta - p}{q \cdot r},$$

to be placed under

$$\text{sets of terms } 3^x = \frac{\alpha}{x+p \cdot \alpha - x - p}, \quad 2^p = \frac{\beta}{y+q \cdot \beta - y - q},$$

$$\text{in orders, } = x+p; \qquad = y+q,$$

whence, multiplying, we have for the frequency

$$M = \frac{\alpha \cdot \beta \cdot \gamma \cdot \delta}{x \cdot y \cdot z \cdot p \cdot q \cdot r \cdot \alpha - x - p \cdot \beta - y - q},$$

and for the multiplicity, N, we have

$$\begin{aligned} N &= A \cdot B \cdot C \cdot \alpha + \gamma - 2A - B \cdot \beta + \delta - B - 2C, \\ &= x \cdot B \cdot y \cdot \alpha + \gamma - 2A - B \cdot \beta + \delta - B - 2C. \end{aligned}$$

13. The coefficient is thus

$$\frac{M \cdot N}{\alpha \cdot \beta \cdot \gamma \cdot \delta} = \frac{B \cdot x + y - 2A - B \cdot \beta + \delta - B - 2C}{y \cdot z \cdot p \cdot r \cdot \alpha - x - p \cdot \beta - y - q},$$

$$= \frac{B \cdot x + y - 2A - B \cdot \beta + \delta - B - 2C}{y \cdot p \cdot \gamma - A - y \cdot \alpha - A - p \cdot \beta - C - y \cdot \delta - C - p},$$

or, putting also

$$D = \alpha + \gamma - 2A - B,$$
$$E = \beta + \delta - B - 2C,$$

whence

$$6A+5B+4C+3D+2E = 3(\alpha+\gamma)+2(\beta+\delta),$$

the formula finally is

$$3^\alpha 2^\beta \cdot 3^\gamma 2^\delta \;=\; \Sigma\Lambda\cdot 6^A 5^B 4^C 3^D 2^E,$$

where

$$\Lambda \;=\; \frac{B}{y\cdot p\cdot \gamma-A-y}\cdot\frac{D}{\alpha-A-p\cdot\beta-C-y}\cdot\frac{E}{\delta-C-p},$$

or, as this may also be written,

$$=\;\frac{B}{y\cdot D-y}\cdot\frac{D}{z\cdot D-z}\cdot\frac{E}{r\cdot E-r},$$

so that the coefficient Λ is in fact the product of three binomial coefficients: it must, however, be recollected that the same term $6^A 5^B 4^C 3^D 2^E$ may occur more than once, with different coefficients Λ, so that in the final result, when these terms are united together, the numerical coefficients are not each of them of the form in question.

14. The limits of the summation are conveniently defined by means of the diagram

A	p	$\alpha-A-p$	α	$D=\alpha+\gamma-2A-B,$
y	C	$\beta-C-y$	β	$E=\beta+\delta \qquad -B-2C,$
z	r			*ut suprà*;
γ	δ			

viz. here the sums of the first and second lines are α and β respectively, and the sums of the first and second columns are γ and δ respectively; we have $B=y+p$, a partition of B; but the values of y, p must be such as not to render negative any one of the four terms z, r, $\alpha-A-p$, $\beta-C-y$ of the diagram: for if any one of these numbers were negative, the corresponding factorial in the denominator of Λ would be infinite and we should have $\Lambda=0$. For any given term $6^A 5^B 4^C 3^D 2^E$ the proper weight $3(\alpha+\gamma)+2(\beta+\delta)$, there may be no suitable values of y, p, and the coefficient is then $=0$: there may be a single pair of values, and the coefficient is then (as remarked above) a product of three binomial coefficients: or there may be more than a single pair of values, and the entire coefficient of the term has not in this case a like simple form.

15. To exhibit the working of the formula, I apply it to the recalculation of the foregoing product $3332\cdot322$ $(\alpha,\beta,\gamma,\delta=3,1,1,2)$. Properly the whole series of symbols 666, 6642, 6633, &c., of the symmetric functions of the weight 18 should be written down, and the coefficient be calculated for each of them: but I write down only those which have coefficients not $=0$: for each of these, I take *all* the partitions

$B = y + p$, several of these giving, as will be seen, zero values, and the others giving the values already obtained for the coefficients of the several terms.

[The original page presents a wide working-table, in which each row lists a symmetric-function symbol on the left, then the five exponents A, B, C, D, E, then a partition (B, y, p), then three groups of binomial-coefficient data (numerators divided by denominators for the three factors of Λ in Art. 13), and a final column showing the actual contribution. The summary table below gives the exponent-set and the final contribution for each row; the intermediate binomial-coefficient arithmetic, being purely numerical, is suppressed.]

Symbol	A	B	C	D	E	Contribution
6552	1	2	0	0	1	$1 \cdot 6552$
6543	1	1	1	1	0	$1 \cdot 6543$
6532^2	1	1	0	1	2	$1 \cdot 6532^2$
64332	1	0	1	2	1	$1 \cdot 64332$
6332^2	1	0	0	2	3	$1 \cdot 6332^2$
5553	0	3	0	1	0	$3 \cdot 5553$
5532^2	0	2	0	2	1	$\} \; 4 \cdot 55332$
54333	0	1	1	3	0	$1 \cdot 54333$
5332^3	0	1	0	3	2	$\} \; 7 \cdot 5333^2$
433332	0	0	1	4	1	$4 \cdot 433332$
33332^2	0	0	0	4	3	$11 \cdot 33332^2$

The entries grouped by $\}$ (the two rows for $5532^2/5532^2$ and the two for $5332^3/5332^3$) correspond to two partitions giving the same final symbol whose coefficients combine.

It is quite possible that abbreviations and verifications might be introduced, but the process as it stands seems to be at once less expeditious and less safe than the one first made use of.

16. Particular cases of the general formula are

$$2^\alpha \cdot 2^\beta \;=\; \Sigma\Lambda\, 4^C\, 2^E; \qquad E = \beta+\delta-2C, \quad \text{and thence } 4C+2E = 2(\beta+\delta),$$

$$\Lambda \;=\; \frac{E}{\beta - C \cdot \delta - C};$$

which agrees with a result before obtained. Again,

$$3^\alpha 2^\beta \cdot 2^\delta \;=\; \Sigma\Lambda \cdot 5^B\, 4^C\, 3^D\, 2^E; \qquad D = \alpha - B,$$

$$E = \beta+\delta-B-2C,$$

and thence

$$5B+4C+3D+2E \;=\; 3\alpha+5B+4C+3D+2E \;=\; 3\alpha+2\beta+2\delta,$$

$$\Lambda \;=\; \frac{E}{\beta - C \cdot \delta - B - C}.$$

17. We may, in like manner with the formula for $3^\alpha 2^\beta \cdot 3^\gamma 2^\delta$ obtain the following:

$$4^\beta 3^{2\alpha} \cdot 2^\delta \;=\; \Sigma\Lambda \cdot 6^A 5^B 4^C 3^D 2^E,$$

where

$$D = \alpha - B, \qquad E = 2\beta+\beta+\delta-3A-B-2C,$$

and thence

$$6A+5B+4C+3D+2E \;=\; 4\beta+3\alpha+2\beta+2\delta,$$

and the value of the coefficient Λ is

$$\Lambda \;=\; \frac{C}{\beta - A \cdot C - \beta} \cdot \frac{E}{\alpha+\beta-A-C \cdot \beta+\delta-2A-B-C},$$

viz. Λ is the product of two binomial coefficients: and since here a given term $6^A 5^B 4^C 3^D 2^E$ occurs once only, each numerical coefficient is actually of this form.

In the particular case $\delta = 0$, we must have $A = 0$ (this appears *a posteriori* from the denominator factor $-A$, a factorial which is infinite for any positive value of A); and we thus obtain the first given formula for $3^\alpha 2^\beta \cdot 2^\delta$.

Capitation and Decapitation. Art. Nos. 18 to 21.

18. I explain the converse processes of Decapitation and Capitation. In any symmetric function, for instance 6552 of the degree 6, the whole coefficient of α^6 is 552, this symbol referring in the first instance to the series of remaining roots β, γ, δ, ...; but as the series of roots is unlimited, we may ultimately replace these by α, β, γ, ..., and so use 552 in its original sense. Similarly, in 6652, the whole coefficient of α^6 is 652, the only difference being that, while in the former case the degree is reduced to 5, in the present case it remains $= 6$. In every case we decapitate the symbol by striking out the highest number—in the case of two or more equal numbers, one only of these being struck out. Observe that by decapitation we always diminish the weight, but we do or do not diminish the degree. In a product such as $3332 \cdot 322$ we obtain in like manner the whole coefficient of α^6 by the decapitation of each factor, viz. the coefficient is $= 332 \cdot 22$: and in any equation such as that obtained above for the product $3332 \cdot 322$, the whole coefficient of the highest power of α must be $= 0$, viz. we can by decapitation obtain a new equation of lower weight: thus from the equation in question of weight 18, we obtain the new equation of weight 12,

$$\begin{aligned} 332 \cdot 22 \ &= 1 \cdot 552 \\ &\quad + 1 \cdot 543 \\ &\quad + 2 \cdot 5322 \\ &\quad + 1 \cdot 4332 \\ &\quad + 3 \cdot 33222, \end{aligned}$$

where observe that the terms of a degree lower than 6 in the original equation give no term in the new equation. The new equation of the deg. weight $5 \cdot 12$ might of course be obtained independently in like manner with the original equation.

19. We capitate a symbol by prefixing to it a number which is not less than the highest number contained in it: thus 552 may be capitated into 5552, 6552, &c.; and so a product $332 \cdot 22$ may be capitated into $3332 \cdot 222$, $3332 \cdot 322$, &c.; moreover a single symbol may be capitated into a product, 552 into $5552 \cdot 4$: in fact, the capitation may be any operation such that by decapitation we reproduce the original symbol. The increase of weight may be any number not less than the degree of the original symbol: but it is usually taken to be a given number: thus for any symbol of a degree not exceeding 6, we may capitate so as to increase the weight by 6. The capitation does or does not increase the degree.

20. An identical equation may be capitated in a variety of ways, but instead of an identity we obtain only a congruence, that is, an equation which requires to be completed by the adjunction to it of proper terms of lower degrees. Thus from the above equation of the deg. weight $5 \cdot 12$ we may obtain

$$\begin{aligned} 3332 \cdot 322 \ &\equiv 1 \cdot 6552 \\ &\quad + 1 \cdot 6543 \\ &\quad + 2 \cdot 65322 \\ &\quad + 1 \cdot 64332 \\ &\quad + 3 \cdot 633222. \end{aligned}$$

Imagine here all the terms brought to the same side so that the form is $\Omega \equiv 0$: Ω is a function not containing α^6, for the whole coefficient of α^6 therein is precisely that function which by the equation of the deg. weight $5 \cdot 12$ is expressed to be $= 0$: and hence Ω, *qua* symmetric function cannot contain β, γ, δ, $\ldots$; viz. Ω is a symmetric function of the degree 5 at most: the congruence $\Omega \equiv 0$ thus means, $\Omega =$ a properly determined function of the degree 5 at most. Obviously by development of the term $3332 \cdot 322$, that is, by substituting for this term its value as given by the equation of the deg. weight $6 \cdot 18$, the function of the degree 5 at most would be found to be $=$ the sum of those terms in the expression of $3332 \cdot 322$, which are of a degree inferior to 6: and the congruence $\Omega \equiv 0$ thus completed would be nothing else than the equation of the deg. weight $6 \cdot 18$.

21. We might have capitated in a different manner: for instance

$$\begin{aligned} 4332 \cdot 222 \equiv\ & 1 \cdot 6552 \\ & + 1 \cdot 6543 \\ & + 2 \cdot 65322 \\ & + 1 \cdot 44332 \cdot 2 \\ & + 3 \cdot 33222 \cdot 3, \end{aligned}$$

there being here three products requiring to be developed: on replacing them by their values, there would be found for Ω a value which would be a determinate function of a degree less than 6: and putting $\Omega =$ this value, the congruence $\Omega \equiv 0$ would be completed into an equation.

Seminvariants of a given Degree; Perpetuants, &c. Art. Nos. 22 to 38.

22. We consider now seminvariants according to their degrees; in particular, those of the degrees 2, 3, 4, 5 and 6, or say quadric, cubic, quartic, quintic and sextic seminvariants: the forms of these are $2^{\alpha+1}$, $3^{\alpha+1}2^{\beta}$, $4^{\alpha+1}3^{\beta}2^{\gamma}$, $5^{\alpha+1}4^{\beta}3^{\gamma}2^{\delta}$, $6^{\alpha+1}5^{\beta}4^{\gamma}3^{\delta}2^{\epsilon}$, where each exponent α, β, γ, δ, ϵ is a positive integer not excluding zero: the exponent of the highest number is in each case written $\alpha+1$, and has thus the value 1 at least, for otherwise the form would not be of the proper degree. The several weights are $2(\alpha+1)$, $3(\alpha+1)+2\beta$, $4(\alpha+1)+3\beta+2\gamma$, &c., or, what is the same thing, we have for the several degrees respectively

$$\begin{aligned} w-2 &= 2\alpha, \\ w-3 &= 3\alpha+2\beta, \\ w-4 &= 4\alpha+3\beta+2\gamma, \\ w-5 &= 5\alpha+4\beta+3\gamma+2\delta, \\ w-6 &= 6\alpha+5\beta+4\gamma+3\delta+2\epsilon, \end{aligned}$$

and we have for a given degree and weight as many seminvariants as there are systems of exponents satisfying the corresponding equation.

23. These numbers are at once expressible by means of a series of Generating Functions (G. F.), viz. writing for shortness 2, 3, 4, 5, 6 to denote $1-x^2$, $1-x^3$, $1-x^4$, $1-x^5$, $1-x^6$, the G. F.'s are

$$\frac{x^2}{2};\quad \frac{x^3}{2\cdot 3};\quad \frac{x^4}{2\cdot 3\cdot 4};\quad \frac{x^5}{2\cdot 3\cdot 4\cdot 5};\quad \frac{x^6}{2\cdot 3\cdot 4\cdot 5\cdot 6}.$$

In fact, the number of seminvariants of a given weight is = coefficient of x^w in the corresponding G. F.; for the quadric seminvariants (or those of deg. weight $2\cdot w$) in $\frac{x^2}{2}$; for the cubic seminvariants (or those of deg. weight $3\cdot w$) in $\frac{x^3}{2\cdot 3}$; and so for the others.

24. A seminvariant of a given degree may be a sum of products (of that degree) of seminvariants of lower degrees, and of seminvariants of lower degrees: and it is in this case said to be reducible: a seminvariant which is not reducible is said to be irreducible, or otherwise to be a *perpetuant*. This notion of a perpetuant is due to Sylvester, see his Memoir "On Subinvariants, *i.e.* Semi-Invariants to Binary Quantics of an Unlimited Order," *American Journal of Mathematics*, vol. v. (1882–83), pp. 78–137 (§ 4 Perpetuants, pp. 105–118). In speaking of the number of perpetuants of a given deg. weight, we assume throughout that these are independent perpetuants, not connected by any linear relation.

25. Since the seminvariants used for the reduction of a given reducible seminvariant can themselves be expressed in terms of perpetuants, we may say more definitely that a seminvariant of a given degree, which is a sum of products (of that degree) of perpetuants of lower degrees, and of perpetuants of lower degrees, is reducible. The words "of that degree" are essential to the definition: a seminvariant may be expressible as a sum of products (of a higher degree) of perpetuants of lower degrees, and of perpetuants of lower degrees, and it is not on this account reducible: a seminvariant so expressible is said to be a "syzygant": but as to this, see No. 49.

26. Every quadric or cubic seminvariant is obviously a perpetuant: the quadric and cubic perpetuants have thus the before-mentioned G. F.'s $\frac{x^2}{2}$ and $\frac{x^3}{2\cdot 3}$ respectively.

27. A reducible quartic seminvariant can only be a sum of products $(2\cdot 2)$ of two quadric perpetuants, and of quadric perpetuants, and it is clear that no quartic seminvariant the symbol of which contains a 3 is thus expressible. If the symbol does not contain a 3, viz. when the form is $4^{\alpha+1}2^{\gamma}$, the seminvariant is reducible: we have for instance

$$4 = 2\cdot 2 - 2\cdot 22,$$

$$42 = 22\cdot 2 - 3\cdot 222, \quad \&\text{c}.$$

To show that this is so in general, observe that any symmetric function $2^{\alpha+1}1^{\gamma}$, *qua* symmetric function can be expressed as a rational and integral function of the degree $2(\alpha+1)+\gamma$, of the coefficients 1, 1^2, 1^3, &c.: instead of the roots considering their squares, we have thence an expression for the quartic seminvariant $4^{\alpha+1}2^{\gamma}$ in terms of the quadric perpetuants 2, 2^2, 2^3, &c., and such expression will be of the same degree $4(\alpha+1)+2\gamma$, as the quartic seminvariant.

It thus appears, as regards the quartic seminvariants, that whenever the symbol contains a 3, and in this case only, the seminvariant is a perpetuant: or, what is the same thing, the form of a quartic perpetuant is $4^{\alpha+1}3^{\beta+1}2^{\gamma}$: for the weight w, the number is equal to that of the sets of values α, β, γ, such that $w-7=4\alpha+3\beta+2\gamma$; or, what is the same thing, the G. F. of the quartic perpetuants is $=\dfrac{x^7}{2\cdot 3\cdot 4}$.

28. Sylvester, in the memoir referred to, obtained this result in a different manner: the quartic seminvariants of a given weight are the quartic perpetuants of that weight and also the products (of that weight) of two quadric perpetuants, the same or different: say (4) is the G. F. for the perpetuants, and (2, 2) for the products: then the G. F. for the quartic seminvariants being as already mentioned $=\dfrac{x^4}{2\cdot 3\cdot 4}$, we have his equation

$$(4)+(2,2) \;=\; \frac{x^4}{2\cdot 3\cdot 4}\,.$$

He deduces $(2,2)$ from the G. F. $=\dfrac{x^2}{2}$ of the quadric perpetuants and thence obtains (4), $=\dfrac{x^7}{2\cdot 3\cdot 4}$, as above.

29. Write for a moment ϕx to represent the G. F. $\dfrac{x^2}{2}$ of the quadric perpetuants, and A, B, $C,\ldots$ to represent these quadric perpetuants: we have, in an algorithm which will be readily understood,

$$\begin{aligned}\phi x &= (A+B+C+\ldots),\\ (\phi x)^2 &= (A+B+C+\ldots)^2 \;=\; A^2+2AB+\ldots,\\ \phi x^2 &= A^2+\text{\&c.};\end{aligned}$$

and thence

$$\tfrac{1}{2}\{(\phi x)^2+\phi x^2\} \;=\; A^2+AB+\text{\&c.};$$

viz. the G. F. (2, 2), is

$$\tfrac{1}{2}\{(\phi x)^2+\phi x^2\} \;=\; \tfrac{1}{2}\left\{\frac{x^4}{2\cdot 2}+\frac{x^4}{2\cdot 4}\right\} \;=\; \tfrac{1}{2}\left\{\frac{x^4(1+x^2)}{2\cdot 4}+\frac{x^4(1-x^2)}{2\cdot 4}\right\} \;=\; \frac{x^4}{2\cdot 4},$$

and we thence have

$$(4) \;=\; \frac{x^4}{2\cdot 3\cdot 4}-\frac{x^4}{2\cdot 4} \;=\; \frac{x^7}{2\cdot 3\cdot 4},$$

that is, $(4)=\dfrac{x^7}{2\cdot 3\cdot 4}$, the same result as was found above by independent considerations.

30. Sylvester established in like manner (but without the terms S which will be presently explained) the equations

$$(3)+(3,2) = \frac{x^5}{2\cdot 3\cdot 4\cdot 5}+S_5,$$

$$(6)+(4,2)+(3,3)+(2,2,2) = \frac{x^6}{2\cdot 3\cdot 4\cdot 5\cdot 6}+S_6,$$

viz. here (5) is the G. F. for the quintic perpetuants, $(2,3)$ that for the products of the quadric and the cubic perpetuants: and similarly (6) is the G. F. for the sextic perpetuants, $(4,2)$ that for the products of the quadric and the quartic perpetuants, $(3,3)$ for the products of two cubic perpetuants, the same or different: and $(2,2,2)$ for the products of three quadric perpetuants, the same or different. We have at once $(3,2)=(3)\cdot(2)$; $(4,2)=(4)\cdot(2)$; $(3,3)$ is found by the same process as was used for finding $(2,2)$, substituting only $\frac{x^3}{3}$ for $\frac{x^2}{2}$; and $(2,2,2)$ is found by a like process, viz. the G. F. for $A^3+A^2B+ABC+\ldots$ is $\frac{1}{6}\{(\phi x)^3+3\phi x\cdot\phi x^2+2\phi x^3\}$, where $\phi x = \frac{x^2}{2}$. Reducing to the common denominator $2\cdot 4\cdot 6$, the numerator is

$$= \tfrac{1}{6}x^6\{(1+x^2)(1+x^2+x^4)+3(1-x^4)+2(1-x^2)(1-x^4)\}.$$

viz. this is x^6. The several functions thus are

$$(3,2) = \frac{x^5}{2\cdot 2\cdot 3},\qquad (4,2) = \frac{x^6}{2\cdot 3\cdot 4},$$

$$(3,3) = \frac{x^6}{3\cdot 6},\qquad (2,2,2) = \frac{x^6}{2\cdot 4\cdot 6}.$$

31. Mr Hammond, in regard to the equation for the quintic perpetuants, made the very important observation—see his paper "On the Solution of the Differential Equation of Sources," *American Journal of Mathematics*, t. v. (1882), pp. 218–227—that the products $(3,2)$ of a cubic perpetuant and a quadric perpetuant are not independent: we have between them syzygies such as $32\cdot 2-3\cdot 22=0$ (viz. the difference of the two products contains no term of the degree 5; the actual value is $=43+322$: Hammond's equation (12), p. 222), hence the necessity in the equation of a term S_5 referring to these syzygies; and he moreover obtained the expression, $S_5=\frac{x^7}{2\cdot 4}$, of the G. F. for these syzygies.

The equation gives

$$(5) = \frac{-x^5+x^6+x^{10}+x^{15}}{2\cdot 3\cdot 4\cdot 5}+S_5,$$

where of course the first term is the value of (5) which would be given by the equation without the term S_5; and substituting herein for S_5 the foregoing value, viz.

$$S_5 = \frac{x^7}{2\cdot 4} = \frac{x^7(1-x^2)(1-x^4)}{2\cdot 3\cdot 4\cdot 5},$$

we find

$$(5) \;=\; \frac{x^{15}}{2\cdot 3\cdot 4\cdot 5},$$

which is the correct value of the G. F. for the quintic perpetuants: the lowest quintic perpetuant is thus of the weight 15.

32. The equation for the sextic perpetuants gives

$$(6) \;=\; \frac{-x^6+x^{10}+2x^{16}+x^{15}}{2\cdot 3\cdot 4\cdot 5\cdot 6}+S_6,$$

which is an equation connecting (6) and S_6, the G. F.'s for the sextic perpetuants, and the sextic syzygies respectively. I have, in the investigation of the value of S_6, met with a difficulty which I have not been able to overcome: but I find that

$$(6) \;=\; \frac{x^6+x^{10}+2x^{15}-x^{16}-x^{18}+\omega(x)}{2\cdot 3\cdot 4\cdot 5\cdot 6},$$

where $\omega(x)$ is possibly the monomial function x^{31}, but this result (which Capt. MacMahon believes to be true) is not yet completely established; it is a function containing no term lower than x^{31}. We have therefore

$$(6) \;=\; \frac{\omega(x)}{2\cdot 3\cdot 4\cdot 5\cdot 6},$$

and there is, it would appear, no sextic perpetuant of a weight lower than 31.

33. But before entering on the investigation, it is proper to further develope the theory of the quintic perpetuants. We have quintic seminvariants: 5 for weight 5; 52 for weight 7; 53 for weight 8; 54, 522 for weight 9; 55, 532 for weight 10; 542, 533, 5222 for weight 11; and so on. These are reduced by means of the products $3\cdot 2$ of a cubic perpetuant and a quadric perpetuant; viz. for any given weight, we have all the products $3^a\cdot 2^b$ of that weight. Thus for weight 5 there is only the product $3\cdot 2$, and this in fact serves to reduce the seminvariant 5: we have

$$3\cdot 2 \;=\; 5+32, \qquad \text{and therefore} \qquad 5 \;=\; 3\cdot 2-32.$$

For the weight 7, there is only the seminvariant 52, and there are the two products $32\cdot 2$ and $3\cdot 22$: either of these would serve for the reduction: we have

$$32\cdot 2 \;=\; 52+43+2\cdot 322, \qquad 3\cdot 22 \;=\; 52+322,$$

and these two equations imply the before-mentioned syzygy, $32\cdot 2-3\cdot 22=0$, in virtue of which the two reductions become equivalent; or say there remains a single equation serving for the reduction: the most simple form is $52=3\cdot 22-322$.

Weight 8: there is only the seminvariant 53, and the product $33\cdot 2$: this gives the reduction.

Weight 9: seminvariants 54, 522: products $32^2\cdot 2$, $32\cdot 22$, $3\cdot 2^3$; there is between the first and last of these a syzygy; and this being satisfied, there remain two equations for the reductions.

Weight 10: seminvariants 55, 532: products $332 \cdot 2$, $33 \cdot 22$: these give the reductions.

Weight 11: seminvariants 533, 542, 5222: products $333 \cdot 2$; $32^3 \cdot 2$, $322 \cdot 2^2$, $32 \cdot 2^3$, $3 \cdot 2^4$.

34. Observe that the seminvariants, and in like manner the products, form two classes, according as the symbols contain three odd numbers or a single odd number: these correspond separately to each other, for the development of any product will contain only seminvariants having each of them as many odd numbers as there are odd numbers in the product. Hence for the weight 11 just referred to, $333 \cdot 2$ serves for the reduction of 533; $32^3 \cdot 2$, $32^2 \cdot 2^2$, $32 \cdot 2^3$, $3 \cdot 2^4$ are connected the first and fourth of them by a syzygy, and the second and third by a syzygy; and there remain two equations serving for the reduction of the two seminvariants 542, 5222.

It is easy to show in this manner that there is no quintic perpetuant for any weight under 15; and that there is a single quintic perpetuant for the weight 15.

35. Generally the syzygies exist only for an odd weight $w = 2\beta + 3$ between the products $32^{\beta-1} \cdot 2$, $32^{\beta-2} \cdot 2^2, \ldots, 32 \cdot 2^{\beta-1}$, $3 \cdot 2^{\beta}$; viz. there is a syzygy between the first and last terms; a syzygy between the second and last but one terms; and so on.

The existence of these syzygies at once appears from the principle of decapitation: decapitating $32^{\beta-1} \cdot 2 - 3 \cdot 2^{\beta}$, we have $-2^{\beta-1} \cdot 2^{\beta-1}$, which is identically $= 0$, hence the function contains no term of the degree 5, that is, $32^{\beta-1} \cdot 2 - 3 \cdot 2^{\beta} = 0$; and similarly for the other pairs of terms.

There are β terms: hence in the case β even, we have $\frac{1}{2}\beta$ pairs of terms and therefore $\frac{1}{2}\beta$ syzygies: in the case β odd, there is a middle term, not connected by a syzygy with any other term, and the number of syzygies is thus $= \frac{1}{2}(\beta - 1)$: writing for β its value $\frac{1}{2}(w-3)$, the number is $= \frac{1}{4}(w-3)$ and $\frac{1}{4}(w-5)$ in the two cases respectively: and it thus appears that the G. F. is $= \dfrac{x^7}{2 \cdot 4}$ as already mentioned.

36. In the case w an odd number, we have seminvariants, and in like manner products, containing respectively one odd number, three odd numbers, five odd numbers, and so on: thus $w = 15$, we have

Seminvariants.	*Products.*	
555	$3332^2 \cdot 2$	3 equations,
5532	$3332 \cdot 2^2$	
5433	$333 \cdot 2^3$	
5332^2		
5442	$32^5 \cdot 2$	$6, = 3$ equations;
542^3	$32^4 \cdot 2^2$	
52^5	$32^3 \cdot 2^3$	
	$32^2 \cdot 2^4$	
	$32 \cdot 2^5$	
	$3 \cdot 2^6$	

Instead of directly working out the condition in the manner indicated above, I present the investigation in a synthetic form as follows: Taking v, v' for the vector parts of the two given quaternions, so that $q = w + v$, $q' = w' + v'$, I write for shortness

$$\begin{aligned}
&\theta = w - w', \\
&\alpha = v^2 + v'^2, \quad = -x^2 - y^2 - z^2 - x'^2 - y'^2 - z'^2, \\
&\beta = v^2 - v'^2, \quad = -x^2 - y^2 - z^2 + x'^2 + y'^2 + z'^2, \\
&D = -\theta(\alpha - \theta^2), \\
&A = \beta - \theta^2, \\
&B = \beta + \theta^2,
\end{aligned}$$

so that θ, α, β, D, A, B are all of them scalars. With these I form a quaternion $Q = (D + Av)(D + Bv')$; I say that we have identically

$$qQ - Qq' = \{D - v \cdot v'^2 + v' \cdot v^2 + w \cdot \theta\}(\theta^4 - 2\alpha\theta^2 + \beta^2).$$

It of course follows that, if $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, then $qQ - Qq' = 0$, viz. $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ is the condition $\Omega = 0$, for the existence of the required quaternion; and this condition being satisfied, then (omitting the arbitrary scalar factor) the value of the quaternion is $Q = (D+Av)(D+Bv')$, a value giving $T(Q) = 0$, that is, $W^2+X^2+Y^2+Z^2 = 0$. If $\theta = 0$ (that is, $w = w'$), then the condition becomes $\beta = 0$, that is, $x^2+y^2+z^2-x'^2-y'^2-z'^2 = 0$; and these two conditions being satisfied, Q ceases to have the determinate value given by the foregoing formula: it has a value involving an arbitrary parameter, and is no longer such that $W^2 + X^2 + Y^2 + Z^2 = 0$.

The identical equation is at once verified; we have

$$\begin{aligned}
qQ - Qq' &= (w + v)Q - Q(w' + v') \\
&= \theta \cdot (D^2 + DAv + DBv' + ABvv') \\
&\quad + v \cdot (D^2 + DAv + DBv' + ABvv') \\
&\quad - (D^2 + DAv + DBv' + ABvv')v' \\
&= \theta D^2 + DAv^2 - DBv'^2 \\
&\quad + v(DA\theta + D^2 - ABv'^2) \\
&\quad + v'(DB\theta + ABv^2 - D^2) \\
&\quad + vv'(\theta AB + DB - DA).
\end{aligned}$$

The first line is here $= D \cdot \{D\theta + Av^2 + Bv'^2\}$, viz. the term in $\{\ \}$ is

$$-(\alpha - \theta^2)\theta + (\beta - \theta^2)v^2 - (\beta + \theta^2)v'^2, \quad = -\alpha\theta^2 = \theta\cdot;$$

and similarly each of the other lines contains the factor $\theta^4 - 2\alpha\theta^2 + \beta^2$, and the equation is thus seen to hold good.

The tensor of Q is $= (D^2 - A^2v^2)(D^2 - B^2v'^2)$; and we have

$$\begin{aligned} D^2 - A^2v^2 &= \theta^2(\alpha - \theta^2)^2 - (\beta - \theta^2)^2v^2 \\ &= \theta^6 - (2\alpha + v^2)\theta^4 + (\alpha^2 + 2\beta v^2)\theta^2 - \beta^2v^2, \end{aligned}$$

which, observing that $\alpha^2 + 2\beta v^2 = \beta^2 + 2v^2(2v^2 - \alpha)$ is $= (\theta^2 - v'^2)^2$ and similarly $D^2 - B^2v'^2$ is $= (\theta^2 - v^2)^2$; hence the tensor is

$$TQ = (\theta^2 - v^2)(\theta^2 - v'^2),$$

which, in virtue of $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, is $= 0$.

The particular case is when $Tq - Tq' = 0$, that is, $w^2 - v^2 - w'^2 + v'^2 = 0$, or say $w^2 - w'^2 = v^2 - v'^2$, that is, $\theta(w + w') = \beta$. Combining with this the general condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, we find $\theta^2\{\theta^2 - 2\alpha + (w + w')^2\} = 0$, or in the second factor, for θ and α substituting their values, we have $2\theta^2(w^2 - v^2 + w'^2 - v'^2) = 0$, that is, $2\theta^2(Tq + Tq') = 0$. Attending to the assumed relation $Tq - Tq' = 0$, the second factor can only vanish if $Tq = 0$, $Tq' = 0$; hence, disregarding this more special case, the factor which vanishes must be the first factor, that is, $\theta = 0$; or the equations $Tq - Tq' = 0$ and $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ give $\beta = 0$ and $\theta = 0$, that is, we have as already mentioned $w - w' = 0$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, viz. the given quaternions have their scalars equal, and the squares of their vectors also equal. The equation here is $vQ - Qv' = 0$; and writing $v^2 = v'^2 = -\rho^2$, we see at once that a solution is

$$Q = (-\rho^2 + vv') + \lambda(v + v'),$$

where λ is an arbitrary scalar; in fact, with this value of Q, we have at once vQ and Qv' each $= \rho^2(v + v') + \lambda(\rho^2 + vv')$; and the equation $vQ - Qv' = 0$ is thus satisfied.

The value of the tensor is easily found to be

$$TQ = 2(\lambda^2 + \rho^2)(\rho^2 + xx' + yy' + zz'),$$

which is not $= 0$.

In accordance with a remark in the introductory paragraphs, the solution $Q = -\rho^2 + vv' + \lambda(v + v')$ is not comprised in the general solution. As to this, observe that, in the case in question $\theta = 0$, $\beta = 0$, we have from the general theorem the form that is,

$$Q = -\rho^2\theta^2(\theta^2 - 2v'^2) \pm \theta^2(\theta^2 - 2v^2)v', \quad = 0.$$

In the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\theta^2 = 2\alpha$, or for α writing its value $-2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{\theta^2}{4} = -\rho^2$, and thence $\frac{A}{B} = \rho^2$, and $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus $Q = -\rho^2 + vv' \pm \lambda\rho(v + v')$, which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ.

in the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\frac{\theta^2}{4} = 2\alpha$, or for α writing its value $= 2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{A^2}{B^2} = -\rho^2$, and thence $\frac{A^2}{B^2} = \rho^2$, $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus

$$Q = -\rho^2 + vv' \pm \lambda\rho(v + v'),$$

which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ.

The foregoing solution $Q = -\rho^2 + vv' + \lambda(v + v')$ may be easily identified with that given pp. 124, 125 of Tait's *Elementary Treatise on Quaternions*; the case there considered is that of a real quaternion, and it was therefore assumed that the two conditions $w = w'$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, were each of them satisfied.

The theory of quaternions is, as is well known, identical with that of matrices of the second order; the identity is, in effect, established by the remark and footnote "Linear Associative Algebra," *American Journal of Mathematics*, t. IV. (1881), p. 132. Writing x, y, z, w for Peirce's imaginaries i, j, k, 1, these have the multiplication table:

	x	y	z	w
x	x	y	0	0
y	0	0	x	y
z	z	w	0	0
w	0	0	z	w

Thus if λ, $= \sqrt{(-1)}$, be the imaginary of ordinary algebra, and i, j, k, the quaternion imaginaries, the relations between i, j, k and x, y, z, w are

$$\begin{array}{ll} x = \frac{1}{2}(1 - \lambda i), & \text{or conversely} \quad 1 = x + w, \\ y = \frac{1}{2}(j - \lambda k), & \qquad i = \lambda(x - w), \\ z = \frac{1}{2}(-j - \lambda k), & \qquad j = (y - z), \\ w = \frac{1}{2}(1 + \lambda i), & \qquad k = \lambda(y + z), \end{array}$$

and we can thus at once express a quaternion as a linear function of the x, y, z, w, or a linear function of the x, y, z, w as a quaternion. And we then consider

$ax+by+cz+dw$ as denoting the matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$; we obtain for the product of two matrices the ordinary formula

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} \begin{vmatrix} a', b' \\ c', d' \end{vmatrix} = \begin{vmatrix} (a, b)(a', c'), & (a, b)(b', d') \\ (c, d)(a', c'), & (c, d)(b', d') \end{vmatrix},$$

viz. we have

$$\begin{aligned}(ax + by + cz + dw)(a'x + b'y + c'z + d'w) &= aa'x^2 + bc'yz + \&\text{c.}, \\ &= (aa' + bc')x + (ab' + bd')y + (ca' + dc')z + (cb' + dd')w,\end{aligned}$$

in accordance with the formula for the product of the two matrices. Observe that, writing

$$A + Bi + Cj + Dk = A(x+w) + B\lambda(x-w) + C(y-z) + D\lambda(y+z) = ax + by + cz + dw,$$

we have

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} = \begin{vmatrix} A + B\lambda, & -C + D\lambda \\ C + D\lambda, & A - B\lambda \end{vmatrix},$$

and thence $ad - bc = A^2 + B^2 + C^2 + D^2$, so that the determinant of the matrix corresponds to the tensor of the quaternion.

837.

ON THE SO-CALLED D'ALEMBERT–CARNOT GEOMETRICAL PARADOX.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 113, 114.]

THE present note has reference to Prof. Sylvester's paper on this subject [*l. c.*, pp. 92–96]. I cannot admit that "D'Alembert and Carnot raised a well-founded objection" to the then and even now too prevalent interpretation of the meaning of the geometrical positive and negative: it appears to me that the objection was not a well-founded one.

Consider through the origin K an indefinite line $t'Kt$, and measure off from K in the sense Kt a distance equal to the positive quantity a, and let m be the extremity of the distance thus measured off. There is not in the ordinary theory any reason why the distance Km should be $= +a$ rather than $= -a$; it is $= +a$, if Kt be the positive sense of the line through K, and it is $= -a$ if Kt' be the positive sense of the line through K; if it be undetermined which of the two is the positive sense, then the distance Km is $= \pm a$, the sign being essentially indeterminate.

The problem is from a point K outside a given circle to draw a line Kmm' such that the intercepted portion mm' within the circle has a given value c. Supposing that the line from K to the centre meets the circle in the points A, B at the distances $KA = a$, $KB = b$: then if $Km = r$, we have $ab = r(c+r)$, or $r = -\frac{1}{2}c \pm \sqrt{(\frac{1}{4}c^2 + ab)}$; viz. we have for r, not simultaneously but alternatively, the positive value $-\frac{1}{2}c + \sqrt{(\frac{1}{4}c^2 + ab)}$, and the negative value $-\frac{1}{2}c - \sqrt{(\frac{1}{4}c^2 + ab)}$, the latter of these being the greatest in absolute magnitude; say the values are $+\rho_1$ and $-\rho_2$. We may with either of these values construct the point m; viz. we obtain m as one of the intersections of the given circle with the circle centre K and radius ρ_1, or else with the circle centre K and radius ρ_2 (that is, radius ρ_2); and attending to the intersections on the same side of the line from K to the centre, it happens that the two points m thus determined are on one and the same line $t'Kt$; but there is no *a priori* reason why the positive senses should be the same, and they are in fact opposite to each other, in the two cases respectively; in the one case we measure off the distance ρ_1 in the sense Kt, in the other case the distance $-\rho_2$ in the sense Kt'; that is, we in fact measure off the positive distances $+\rho_1$, and $+\rho_2$, in one and the same sense Kt; thus obtaining for the point m one or the other extremity of a determinate secant through K.

The best illustration is I think in the elementary problem of finding the perpendicular distance of a given line from the origin. Let $Ax + By + C = 0$ be the equation of the given line: and first let a line be drawn in a determinate sense, say at the inclination θ to the positive part of the axis of x, to meet the given line. Taking r for the distance from the origin of the point of intersection, we have, for the coordinates of the point of intersection, x, $y = r\cos\theta$, $r\sin\theta$; and thence $r(A\cos\theta + B\sin\theta) + C = 0$, that is,

$$r = \frac{-C}{A\cos\theta + B\sin\theta},$$

a perfectly determinate value. But the perpendicular on the given line may be considered as drawn in one or the other of two opposite senses; that is, we have at pleasure

$$\cos\theta, \sin\theta = \frac{A}{\sqrt{(A^2+B^2)}}, \frac{B}{\sqrt{(A^2+B^2)}}, \quad \text{or else} \quad \frac{-A}{\sqrt{(A^2+B^2)}}, \frac{-B}{\sqrt{(A^2+B^2)}},$$

and thence

$$r = \frac{-C}{\sqrt{(A^2+B^2)}}, \quad \text{or else} \quad r = \frac{C}{\sqrt{(A^2+B^2)}};$$

that is, the perpendicular distance is $= \frac{\pm C}{\sqrt{(A^2+B^2)}}$, with the essentially indeterminate sign $\pm$, because the distance may be considered as drawn from the origin in one or the other of the two opposite senses.

838.

ON THE TWISTED CUBICS UPON A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 129–132.]

A CUBIC (twisted cubic) on a quadric surface meets each generator of the one kind twice, and each generator of the other kind once (Salmon, *Solid Geometry*, Ed. 4, p. 301). There are thus on the surface two kinds of cubics, viz. distinguishing for convenience the generating lines as generators and directors, a cubic may meet each generator twice and each director once; or it may meet each generator once and each director twice. And two cubic curves are accordingly of the same kind or of different kinds.

Consider for example the quadric surface $xw - yz = 0$, and the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

If we call the line $(x = ky, kw = z)$ a generator, and the line $(x = kz, kw = y)$ a director, for the intersections with a generator we have $1 = k\phi$, $k\phi^3 = \phi^2$; equations with the common root $k\phi = 1$, or there is a single intersection; for the intersections with a director, $1 = k\phi^2$, $k\phi^3 = \phi$, equations with the common roots $k\phi^2 = 1$, or there are two intersections.

Consider on the same quadric surface the cubic

$$x : y : z : w = \overline{\theta-\alpha}\cdot\overline{\theta-\beta}\cdot\overline{\theta-\gamma} : \overline{\theta-\alpha}\cdot\overline{\theta-\epsilon}\cdot\overline{\theta-\zeta} : \overline{\theta-\delta}\cdot\overline{\theta-\beta}\cdot\overline{\theta-\gamma} : \overline{\theta-\delta}\cdot\overline{\theta-\epsilon}\cdot\overline{\theta-\zeta};$$

or as, for shortness, I write these,

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

viz. α is written to denote $\theta - \alpha$, and so in other cases; for the intersections with a generator, we have $\alpha\beta\gamma = k\alpha\epsilon\zeta$, $k\delta\epsilon\zeta = \delta\beta\gamma$, equations having the two common roots

$\beta\gamma = k\epsilon\zeta$; or there are two intersections with the generator. And in like manner there is one intersection with the director. The cubic

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$$

is thus a cubic of different kind from

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

And in the same manner it appears that the cubic

$$x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$$

is of the same kind with

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

The two cubics of different kinds intersect in 5 points; the two cubics of the same kind in only 4 points. In fact, for the intersections with the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3,$$

we have $xz - y^2 = 0$, $yw - z^2 = 0$, and substituting herein the θ-values, $x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$, we find

$$\alpha\delta\beta^2\gamma^2 - \alpha^2\epsilon^2\zeta^2 = 0, \quad \delta\epsilon\zeta\alpha\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0,$$

equations with the common five roots $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$; whereas for the θ-values $x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$, we have

$$\alpha^2\beta\gamma\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0, \quad \delta^2\beta\gamma\epsilon\zeta - \alpha^2\epsilon^2\zeta^2 = 0,$$

equations with the common four roots $\alpha^2\epsilon\zeta - \delta^2\beta\gamma = 0$.

I remark in passing that the equations just obtained have each of them a root $\theta = \infty$, but this would have been avoided if the θ-equations had been taken in the slightly altered form

$$x : y : z : w = \alpha\beta\gamma : b\alpha\epsilon\zeta : c\delta\beta\gamma : bc\delta\epsilon\zeta, \text{ and } x : y : z : w = \alpha\beta\gamma : c\delta\beta\gamma : b\alpha\epsilon\zeta : bc\delta\epsilon\zeta,$$

b, c being arbitrary constants; and the special value is thus of no importance.

The two cubics intersecting in 5 points constitute the complete intersection of a quadric surface and a cubic surface; and conversely when a quadric surface and a cubic surface intersect in two cubics, then the cubics must have 5 intersections, and are thus by what precedes cubics of different kinds in regard to the quadric surface. In fact, considering two cubics meeting in 5 points, we can, through these 5 points, through 2 points assumed at pleasure on the first cubic, and through 2 points assumed at pleasure on the second cubic, draw a determinate quadric surface: this meets each of the two cubics in $5+2$ points, and consequently it entirely contains the cubic; viz. we have through the two cubics a determinate quadric surface. Again, through the 5 points, through 5 points at pleasure on the first cubic and through 5 points at pleasure on the second cubic, we may draw a cubic surface; this meets each cubic in $5+5$ points, and therefore it entirely contains the cubic; we have thus a cubic surface through the two cubics. The cubic surface has been subjected only to $5+5+5=15$ conditions, and the equation thus contains homogeneously $20-15, =5$ constants; viz. if $\Omega=0$ be the quadric surface through the two cubics the general form is

$$(\alpha x+\beta y+\gamma z+\delta w)\Omega+\lambda U=0;$$

disregarding the term in Ω, we have thus, in fact, a determinate cubic $U=0$.

Taking as above the first cubic as given by the equations $x:y:z:w=1:\phi:\phi^2:\phi^3$, and the second cubic as given by the equations $x:y:z:w=\alpha\beta\gamma:\alpha\epsilon\zeta:\delta\beta\gamma:\delta\epsilon\zeta$, the equation of the cubic surface which contains the two cubics will be of the form

$$(Ax+By+Cz+Dw)(yw-z^2)+(A'x+B'y+C'z+D'w)(xz-y^2)=0.$$

Substituting herein the θ-values and omitting the common factor $\alpha\epsilon^2\zeta^2-\delta\beta^2\gamma^2$, we have

$$\delta(A\alpha\beta\gamma+B\alpha\epsilon\zeta+C\delta\beta\gamma+D\delta\epsilon\zeta)-\alpha(A'\alpha\beta\gamma+B'\alpha\epsilon\zeta+C'\delta\beta\gamma+D'\delta\epsilon\zeta)=0,$$

which is of the fourth order in θ. We may assume $C'=0$, and $D'=0$; in fact, the terms in C' and D' might be got rid of by the substitutions $z(xz-y^2)=y(xw-yz)-x(yw-z^2)$, $w(xz-y^2)=z(xw-yz)-y(yw-z^2)$; we then have 6 coefficients A, B, C, D, A', B', and equating to zero the terms in θ^0, θ^1, θ^2, θ^3, θ^4, the ratios of these are determined by the five equations: and we have thus the required cubic surface.

Starting from either cubic considered as a curve on the given quadric surface; if through the centre we describe a cubic surface, this meets the quadric surface in a second cubic and the two cubics intersect each other in 5 points. In particular, if the one cubic is $x:y:z:w=1:\phi:\phi^2:\phi^3$, it appears by what precedes that the other cubic may be taken to be $x:y:z:w=\alpha\beta\gamma:\alpha\epsilon\zeta:\delta\beta\gamma:\delta\epsilon\zeta$, intersecting the former in the 5 points given by the equation $\delta\beta^2\gamma^2-\alpha\epsilon^2\zeta^2=0$. The five points are of course points of contact of the quadric surface and the cubic surface.

A very simple instance is the following. The quadric surface $xw-yz=0$ and the cubic surface

$$y(xz-y^2)+z(yw-z^2)=0$$

intersect in the two cubics

$$x:y:z:w=1:\phi:\phi^2:\phi^3 \quad \text{and} \quad x:y:z:w=1:\theta^2:\theta:\theta^3.$$

The two cubics meet in the 5 points

$$\theta = (0, \infty, 1, \omega, \omega^2); \quad \phi = (0, \infty, 1, \omega^2, \omega),$$

or, what is the same thing, in the 5 points

$$(x, y, z : w) = (1, 0, 0, 0), (0, 0, 0, 1), (1, 1, 1, 1), (1, \omega, \omega^2, 1), (1, \omega^2, \omega, 1),$$

where ω is an imaginary cube root of unity.

I was anxious to work out this result not for its own sake, but as an illustration of a point which first arises as to octics: a unicursal octic curve is not in general situate on any surface lower than a quartic surface; if on the octic we take at pleasure 33 points, we may through these draw two quartic surfaces $U = 0$, $V = 0$, each of which will entirely contain the curve: the two surfaces meet in a second octic also unicursal, and the two curves intersect in 34 points, which are points of contact of the two quartic surfaces. We cannot, by adjoining to the given octic any curve of an inferior to its own, obtain a complete intersection of two surfaces: the most simple case is when we adjoin to the given octic as above another unicursal octic, and thus obtain a complete intersection of two quartic surfaces. It is to be observed that the second curve is determined completely and uniquely by the given curve: considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ, the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

839.

ON THE MATRICAL EQUATION $qQ - Qq = 0$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 176–178.]

I CONSIDER the matrical equation $qQ - Qq' = 0$, where q, q' are given matrices $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$, $\begin{vmatrix} a', b' \\ c', d' \end{vmatrix}$, and Q, $= \begin{vmatrix} A, B \\ C, D \end{vmatrix}$, is a matrix which has to be determined. As remarked in the paper "On the Quaternion Equation $qQ - Qq' = 0$," *Messenger*, t. XIV. (1885), pp. 108–112, [836] the question for matrices is equivalent to that for quaternions: for a matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$ may be regarded as a quaternion

$$\tfrac{1}{2}(a+d) - \tfrac{1}{2}\lambda(a-d)i + \tfrac{1}{2}(b-c)j - \tfrac{1}{2}\lambda(b+c)k,$$

or (omitting the factor $\frac{1}{2}$) as the quaternion

$$(a+d) - \lambda(a-d)i + (b-c)j - \lambda(b+c)k,$$

where λ, $= \sqrt{(-1)}$, is the imaginary of ordinary algebra. Hence considering q, q' as denoting the quaternions

$$(a+d) - \lambda(a-d)i + (b-c)j - \lambda(b+c)k,$$
$$(a'+d') - \lambda(a'-d')i + (b'-c')j - \lambda(b'+c')k,$$

we can, if a certain condition is satisfied, find a quaternion Q such that $qQ - Qq' = 0$; say this is $Q = W + iX + jY + kZ$; putting this we find

$$Q = \begin{vmatrix} A, B \\ C, D \end{vmatrix} = \begin{vmatrix} W + \lambda X, & Y - \lambda Z \\ -Y - \lambda Z, & W - \lambda X \end{vmatrix},$$

for the required matrix Q; this being an indeterminate matrix, such that $AD - BC = 0$.

But it is better to solve directly the matrical equation

$$\begin{vmatrix} (A,B)(a,b), & (A,B)(b',d') \\ (C,D)(a,b), & (C,D)(b',d') \end{vmatrix} - \begin{vmatrix} (a,b)(A,C), & (a,b)(B,D) \\ (c,d)(A,C), & (c,d)(B,D) \end{vmatrix} = 0,$$

viz.

$$\begin{vmatrix} (A,B)(a,b), & (A,B)(b',d') \\ (C,D)(a,b), & (C,D)(b',d') \end{vmatrix} - \begin{vmatrix} (a,b)(A,C), & (a,b)(B,D) \\ (c,d)(A,C), & (c,d)(B,D) \end{vmatrix} = 0,$$

that is,

$$\begin{aligned} Aa + Bc' &= aA + bC, \\ Ba + Dc' &= aB + bD, \\ Ac + Cd' &= cA + dC, \\ Bc + Dd' &= cB + dD, \end{aligned}$$

or, what is the same thing,

$$\begin{vmatrix} a-a', & -c', & 0, & b \\ -b', & a-d', & b, & 0 \\ c, & 0, & d-a', & -c' \\ 0, & c, & -b', & d-d' \end{vmatrix} (A,B,C,D) = 0,$$

viz. we have (A,B,C,D) connected by these four linear equations: the necessary condition is that the determinant formed out of the matrix which here presents itself shall be $= 0$.

After some reduction, and putting for shortness

$$\nabla = ad - bc, \quad \nabla' = a'd' - b'c',$$

this is found to be

$$(\nabla - \nabla')^2 + [\nabla(a'+d') - \nabla'(a+d)](a+d-a'-d') = 0;$$

which is the condition for the existence of a solution. This condition being satisfied, the four equations will be equivalent to three independent equations, which serve to determine A, B, C, D; and, assuming the absolute value of A, we find

$$\begin{aligned} A &= -(a'+d'-a-d), \\ \nabla B &= \nabla' \cdot b' + \nabla(a'+d'-a-d) + ab' - a'b, \\ \nabla C &= \nabla' \cdot c' + \nabla(a'+d'-a-d) + dc' - d'c, \\ \nabla D &= D \cdot a' - (d'-d)(a'+d'-a-d) - cc'(a'+d'-a-d) + b'c(a'+d'-a-d)'; \end{aligned}$$

values which give

$$ad' + b'c = ad - bc;$$

viz. in the case where the given matrices are such that $ad + bg = (a'+d') - bc - bc' + ad'$, the components A, B, C, D of the required matrix Q are determined as such that $AD - BC = 0$.

If we have $\nabla - \nabla' = 0$, that is, $ad - bc = a'd' - b'c'$, then the condition becomes $a' + d' = a + d$; and these two conditions being satisfied, the four equations reduce themselves to two independent equations, and the ratios $A : B : C : D$ have no longer determinate values; the four equations may in fact be written

$$\begin{vmatrix} 0, & b, & -c', & a-a' \\ -b, & 0, & a+d', & b' \\ c, & a'-d, & 0, & -c' \\ a-d, & c, & -b, & 0 \end{vmatrix} (D, C, B, A) = 0,$$

which are of the form

$$\begin{vmatrix} 0, & -h, & g, & a \\ h, & 0, & -f, & b \\ -g, & f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix},$$

where the coefficients a, b, c, f, g, h are such that $af+bg+ch = 0$; and the four equations are thus equivalent to two independent equations.

To obtain a symmetrical solution, assume a relation with arbitrary coefficients ξ, η, ζ, ω; we then find

$$\begin{aligned} D &= -c\eta - h\zeta + b\omega, \\ B &= h\eta + f\zeta + a\omega, \\ C &= -g\eta + a\xi + g\omega, \\ A &= -b\xi - f\eta + h\omega, \end{aligned}$$

viz. the complete theorem is that, if the matrices

$$q = \begin{vmatrix} a, b \\ c, d \end{vmatrix}, \quad q' = \begin{vmatrix} a', b' \\ c', d' \end{vmatrix},$$

are such that $ad - bc = a'd' - b'c'$, $a + d = a' + d'$, then writing

$$a = a - a', \quad b = d + d', \quad f = a' + d, \quad g = a + d', \quad c' = c, \quad h = b,$$

values which satisfy $af + bg + ch = 0$, and taking ξ, η, ζ, ω arbitrary, we have for the matrix Q, which is such that $qQ - Qq' = 0$, the expression

$$Q = \begin{vmatrix} -b\xi - f\eta + h\omega, & h\eta + f\zeta + a\omega \\ -g\eta + a\xi + g\omega, & -c\eta - h\zeta + b\omega \end{vmatrix},$$

depending really on one arbitrary parameter, viz. we may without loss of generality put any two of the coefficients ξ, η, ζ, ω equal to 0.

840.

ON MASCHERONI'S GEOMETRY OF THE COMPASS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 179–181.]

I HAVE not seen the original of Mascheroni's work, *La Geometria del Compasso*, 8vo, Pavia (1797), but only the French translation, *Géométrie du Compas, par L. Mascheroni, traduite de l'Italien par A. M. Carette*, 2e ed. 8vo Paris, 1828 (Author's Preface, pp. 5–24, and pp. 25–328, with 14 plates containing 108 figures). The title expresses the notion of the work: a straight line is given by means of two points thereof, but is not allowed to be actually drawn; and the problem is, with the compass alone to perform all the constructions of Geometry. Observe that, for the purpose of demonstration, any lines may be imagined to be drawn, and such lines are in fact shown as dotted lines in the figures; but this does not in any wise interfere with the fundamental postulate that the constructions are to be performed with the compass only.

Assuming, then, that a line is in every case given by means of two points thereof, the leading questions are those considered Book VII. "on the intersections of straight lines with circular arcs and with each other," viz. they are (1) to find the intersections of a given line and circle; (2) to find the intersection of two given lines. But in the first problem it is necessary to consider two cases; (A) the general case, where the line does not pass through the centre of the circle, and (B) the particular, but actually more difficult, case, where the line passes through the centre of the circle. It is assumed that the centre of the circle is given: if it is not given, it can be found as afterwards mentioned. It will be convenient to establish the definition of counter-points in regard to a given line: two points, which are such that the line joining them is bisected at right angles by the given line, are said to be counter-points in regard to that line.

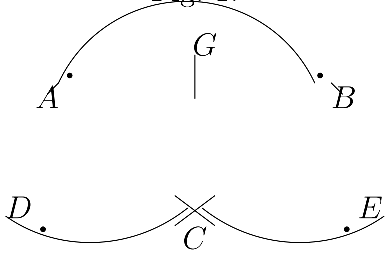
(1), A. Consider a line given by means of two points P, Q; and a given circle, centre C. With centre P and radius PC describe a circle, and with centre Q and radius QC a circle; these meet in C and in a second point D which will be the counterpoint of C in regard to the line PQ; hence with centre D and radius = that of given circle, describing a circle, this meets the given circle in two points which are the intersections of the given circle by the line PQ.

The construction fails if the line PQ passes through the centre of the given circle, for then the two auxiliary circles touch at C, or we have D coincident with C. We have therefore considered this case; Q may be taken to be the centre C of the circle; so that we have:

(1), B. The problem is, given a point P, and a circle, centre C: to find the intersections of the circle by the line CP. With centre P, and an arbitrary radius, describe a circle meeting the given circle in points A, B (which are of course counter points in regard to line CP); the circle is divided into two arcs AB and $360 - AB$, and the problem is to bisect each of these two arcs; for the points of bisection are obviously the required intersections of CP with the circle. Hence we have:

(1), C. To bisect a given arc AB of a circle, centre C (fig. 1). With centre A and radius AC describe a circle; and with centre B, and equal radius BC describe a circle; and on these circles respectively, find the points D, E such that $CD = CE = AB$. With centre E and radius EA, describe a circle; and with centre D, and equal radius DB, describe a circle; and let these two circles meet in F; then with centre E (or D) and radius CF describe a circle: this will meet the given circle in the required middle point G of the arc AB.

Fig. 1.



The proof depends on the theorem that in a rhombus the sum of the squares of the diagonals is = sum of the squares of the four sides. We have a rhombus $ACEB$, and therefore

$$(AE)^2 + (BC)^2 = 2(AB)^2 + 2(AC)^2,$$

that is,

$$(AE)^2 = 2(AB)^2 + \text{rad.}^2.$$

But by construction, $EF^2 = (AE)^2$; therefore

$$EF^2 = 2(AB)^2 + \text{rad.}^2, \quad = 2(CE)^2 + \text{rad.}^2.$$

Hence in the right-angled triangle FCE, we have

$$(CF)^2 = (EF)^2 - CE^2, \quad = (CE)^2 + \text{rad.}^2;$$

and by construction, $(EG)^2 = (CF)^2$; that is,

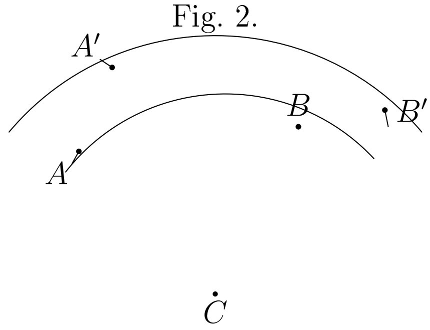
$$(EG)^2 = (CE)^2 + (CG)^2,$$

viz. GEC is a right-angled triangle; that is, CG is at right angles to BCE, or the line CGF bisects the arc AB in G.

For the second problem, (2), to find the intersection of two given lines, we require the solution of the problem to find a fourth proportional to three given distances, and this immediately depends on the following problem.

Problem. In two concentric circles to place chords subtending equal angles. If AB (fig. 2) be a chord of the one circle, then with centres A and B respectively, and an arbitrary radius $AA' = BB'$, describing circles to cut the larger circle in the points A' and B' (A' is either of the two intersections and B' is the intersection lying in regard to CB as A' lies in regard to CA) then clearly $\angle ACB = \angle A'CB'$.

Fig. 2.

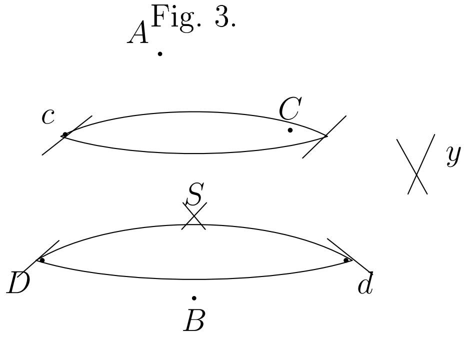


And hence also we have the following problem.

Problem. To find a fourth proportional to three given distances. We have only to take as the given distances the two radii CA, CA' and the chord AB; and then from the similar triangles ACB, $A'CB'$, we have $CA : CA' :: AB : A'B'$, viz. $A'B'$ is a fourth proportional to CA, CA', AB.

We have now the solution of (2) To construct the intersection of two given lines AB and CD (fig. 3). Find c the counter-point of C, and d the counter-point of D, in regard to the line AB

Fig. 3.



and find y such that the distances Cy, dy are $= Dd$, DC respectively; y is in a line with cC, and we have $CydD$ a parallelogram. Find cS a fourth proportional to cy, cd, cC, and with centres c, C respectively and radii each $= cS$ describe circles cutting in the point S; this will be the required intersection of the two lines. In fact, the required point S will be the intersection of the two lines CD, cd: supposing these lines each of them drawn, and also the lines cCy and dy, we have DC parallel to dy, that is, the triangles cdy, cSC are similar to each other or

$$cy : cd :: cC : cS :$$

viz. the distance cS having been found by this proportion, and the point S found as the intersection of the two circles, centres c and C respectively, the point S so determined is the required point of intersection of the given lines AB and CD.

If a circle be given without its centre being known, then taking any three points A, B, C on the circle, and a pair of counter-points D, E of the line AB, and also a pair of counter-points F, G of the line AC, we have obviously the centre of the given circle as the intersection of the lines DE and FG; and the centre can thus be found with the compass only.

It is proper to remark that the problems considered in the present paper are those connecting the theory with ordinary geometry, not the problems which are most readily and elegantly solved with the compass only: a large collection of these are contained in the work, and in particular the twelfth book contains some interesting approximate solutions of the problems of the quadrature of the circle, the duplication of the cube, and other problems not solvable by ordinary geometry.

Cambridge, *March* 19, 1885.

841.

ON A DIFFERENTIAL OPERATOR.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 190, 191.]

WRITE $X = 1+bx+cx^2+\ldots, = (1-\alpha x)(1-\beta x)(1-\gamma x)\ldots$; then by Capt. MacMahon's theorem, any non-unitary function of the roots α, β, $\gamma,\ldots$ is reduced to zero by the operation

$$\Delta, \quad = d_b + bd_c + cd_d + \ldots;$$

for instance, if $(2), = \Sigma\alpha^2 = b^2 - 2c$, we have

$$\Delta(b^2-2c) = 2b + b(-2), = 0.$$

We have

$$\Delta X = x^2 + bx^3 + cx^4 + \ldots = xX;$$

and writing, moreover, $X', = b + 2cx + 3dx^2 + \&c.$, for the derived function of X, then

$$\Delta X = x^2 + bx^3 + cx^4 + \ldots = (xX)' \cdot /X$$

We can hence shew that $\Delta\left(\dfrac{X'}{X} - b\right) = 0$; the value is, in fact,

$$\frac{\Delta X' \cdot X - X'\Delta X}{X^2} - \Delta b, \text{ that is, } \frac{(xX)'X - xX \cdot X'}{X^2} - 1,$$

which is

$$= \frac{X + xX' - xX'}{X} - 1, = 0.$$

This is right, for $\dfrac{X'}{X} - b$ is a sum of non-unitary symmetric functions of the roots; in fact,

$$\frac{X'}{X} = \Sigma\frac{-\alpha}{1-\alpha x} = -(1) - (2)x - (3)x^2 - \&c.,$$

or since $b = (1)$, this is $\dfrac{X'}{X} - b = -(2)x - (3)x^2 - \&c.$, a sum of non-unitary functions of the roots.

842.

ON THE VALUE OF $\tan(\sin\theta) - \sin(\tan\theta)$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 191, 192.]

THE following equation is given p. 59 of the *Lady's and Gentleman's Diary* for 1854:

$$\tan(\sin\theta) - \sin(\tan\theta) = \tfrac{1}{30}\theta^7 + \tfrac{29}{756}\theta^9 + \&\text{c.}$$

Write in general

$$\begin{aligned} X &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 + \dots, \\ Y &= \theta + A'\theta^3 + B'\theta^5 + C'\theta^7 + D'\theta^9 + \dots. \end{aligned}$$

Then, as far as θ^9, we have

$$\begin{aligned} X + A'X^3 + B'X^5 + C'X^7 + D'X^9 = &\ \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 \\ &+ A'(\theta^3 + 3A\theta^5 + 3(A^2 + B)\theta^7 + (A^3 + 6AB + 3C)\theta^9) \\ &+ B'(\theta^5 + 5A\theta^7 + (10A^2 + 5B)\theta^9) \\ &+ C'(\theta^7 + 7A\theta^9) \\ &+ D'(\theta^9) \\ = &\ \theta \\ &+ \theta^3(A + A') \\ &+ \theta^5(B + 3AA' + B') \\ &+ \theta^7(C + 3A^2A' + 3BA' + 5AB' + C') \\ &+ \theta^9(D + A'(A^3 + 6AB + 3C) + 5B'(2A^2 + B) + 7AC' + D'), \end{aligned}$$

and hence

$$\begin{aligned}
&X + A'X^3 + B'X^5 + C'X^7 + D'X^9 \\
&-Y - A'Y^3 - B'Y^5 - C'Y^7 - D'Y^9 \\
&= \theta^7\{3AA'(A - A') + 2(AB' - A'B)\} \\
&+\theta^9\{AA'(A^2 - A'^2) + 6AA'(B - B') + 4(AC' - A'C) + 10(A^2B - A'^2B')\}.
\end{aligned}$$

Now let

$$\begin{aligned}
&X = \sin\theta = \tfrac{1}{2}\theta - \tfrac{1}{6}\theta^3 + \tfrac{1}{120}\theta^5 - \tfrac{1}{5040}\theta^7 + \ldots; \\
&A = -\tfrac{1}{6}, \quad B = \tfrac{1}{120}, \quad C = -\tfrac{1}{5040}, \\
&Y = \tan\theta = \tfrac{1}{2}\theta + \tfrac{1}{3}\theta^3 + \tfrac{2}{15}\theta^5 + \tfrac{17}{315}\theta^7 + \ldots; \\
&A' = \tfrac{1}{3}, \quad B' = \tfrac{2}{15}, \quad C' = \tfrac{17}{315}:
\end{aligned}$$

we have therefore

$$\begin{aligned}
AA' = -\tfrac{1}{18}, \quad A - A' = -\tfrac{1}{2}, \quad A + A' = \tfrac{1}{6}, \quad B - B' = -\tfrac{1}{8}, \\
AB' - A'B = -\tfrac{1}{45}, \quad AC' - A'C = -\tfrac{17}{1890}, \quad A^2B - A'^2B' = \tfrac{19}{2520}.
\end{aligned}$$

Hence

$$\begin{aligned}
&\text{coeff. } \theta^7 = \tfrac{1}{12} - \tfrac{2}{45}, \quad = \tfrac{1}{30}, \\
&\text{coeff. } \theta^9 = \tfrac{1}{216} + \tfrac{1}{24} - \tfrac{17}{945} + \tfrac{19}{504}, \quad = \tfrac{29}{756};
\end{aligned}$$

and the required equation is thus verified.

843.

ON THE QUADRIQUADRIC CURVE IN CONNEXION WITH THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Mathematische Annalen*, t. XXV. (1885), pp. 152–156.]

I CALL to mind that, if we have on a line two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and through the line two planes

$$Ax + By + Cz + Dw = 0, \quad A'x + B'y + C'z + D'w = 0,$$

then

$$\begin{aligned} B\gamma' - B'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta : \alpha\delta' - \alpha'\delta : \beta\delta' - \beta'\delta : \gamma\delta' - \gamma'\delta \\ = AD : BD : CD : BC : CA : AB, \end{aligned}$$

and that, putting each of these two equal sets of values

$$= a : b : c : f : g : h, \quad \text{or} \quad = A : B : C : D : BC : CA : AB,$$

then the quantities a, b, c, f, g, h which it is easy to see satisfy the relation $af + bg + ch = 0$, are called the "six coordinates" on the line; we call them the six coordinates of the line, as also the coordinates of either of the six line-axiotomical equations which is a single relation between these axes, the so-called axiotomical contains a single relation between them. Or, in the language of the modern geometry, the determination of the particular line. See my paper "On the six coordinates of a line," *Camb. Phil. Trans.* t. XI. (1869), pp. 290–323, [435].

I consider for a moment the quadric surface

$$\alpha x^2 + \beta y^2 + \gamma z^2 + \delta w^2 = 0,$$

and I proceed to shew that, if (α, b, c, f, g, h) are the coordinates of a generating line on the surface, then either $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, or else $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$; the one or other system of equations according as the line belongs to the one or other system of generating lines.

We satisfy the equation by

$$\alpha x + by + \theta z + \theta w = 0, \quad x = \frac{1}{\phi}y - \frac{1}{\phi}z + \theta w = 0,$$

where θ is an arbitrary parameter: hence these equations determine a generating line of the surface: and the coordinates of this line are

$$a, \quad b, \quad c, \quad f, \quad g, \quad h$$

$$= -1\left(\theta - \frac{1}{\phi}\right), \quad -\left(\theta + \frac{1}{\phi}\right), \quad \theta\phi, \quad -\left(\theta - \frac{1}{\phi}\right), \quad \theta + \frac{1}{\phi}, \quad -\phi;$$

viz. these values give $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$.

Similarly we satisfy the equation by

$$\alpha x + by + \phi z - \phi w = 0,$$

$$x = \frac{1}{\phi}y - \frac{1}{\phi}z - \phi w = 0,$$

where ϕ is an arbitrary parameter: hence these equations also determine a generating line of the surface: and the coordinates of this line are

$$a, \quad b, \quad c, \quad f, \quad g, \quad h$$

$$= -1\left(\phi - \frac{1}{\phi}\right), \quad \phi + \frac{1}{\phi}, \quad -\phi^2, \quad -\left(\phi - \frac{1}{\phi}\right), \quad \phi + \frac{1}{\phi}, \quad -\phi^2;$$

viz. these values give $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$.

If for x, y, z, w we write $x\sqrt{f}$, $y\sqrt{g}$, $z\sqrt{h}$, $w\sqrt{f}$ respectively, then we have the theorem that, for the quadric surface

$$pa^2 + qy^2 + rz^2 + sw^2 = 0,$$

the coordinates (a, b, c, f, g, h) of a generating line are such that

$$\frac{a}{\sqrt{f}} = \pm\sqrt{\frac{pa}{qb}}, \quad \frac{b}{\sqrt{g}} = \pm\sqrt{\frac{q}{r}}, \quad \frac{c}{\sqrt{h}} = \pm\sqrt{\frac{r}{s}};$$

the signs being all + or all −, according as the line belongs to the one or other of the systems of generating lines.

Take $(\alpha', b', c', f', g', h')$ for the coordinates of an arbitrary line, and write

$$p, q, r, s = q\beta(f'b' + b'h'), q\beta(g'a' + a'g'), r\beta(g'a' + a'g'), s\beta(c'a' + a'c'); = a'fbcgh',$$

$$a'g'bsr - b'sf'gh' + c'gh'a' + cha'b'g'd = 0,$$

which is a section having the line (a', b', c', f', g', h') for a generating line. To verify this, observe that for the line in question we have

$$\begin{aligned} Ky - qx + zw &= 0, \\ -Kx + qy + zw &= 0, \\ yg + bg' \cdot x + xw &= 0, \\ -x'g + b'p \cdot y + z'w &= 0, \end{aligned}$$

equivalent of course to two independent equations: these give $K(x = b'y \cdot k)w$, $Ky = pa \cdot kw$ and so on, where values substituted in the quadric equation give it identically.

And for the last-mentioned quadric surface that, for the system $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, of the conditions for a generating line, the equation reduces itself to

$$af + bg + ch = 0,$$

where the signs $=$ correspond to the system a, b, c, g, h, f, and the sign $-$ to the other system.

Taking R as origin, the equations of the conic are

$$Kx = py - qz + zw = 0,$$

$$-Kx + qy - rz + sw = 0,$$

equivalent of course to two independent equations of the form

$$\alpha\beta x + \alpha'\beta' y + \beta'\alpha c + s\alpha\beta' = 0, \quad \text{equation of a generating line,}$$

which is of course not to be confounded with the equation of a generating line.

Considers now the quadriquadric curve

$$(x, y, z, w)(a', b', c', f', g', h') = 0, \quad (x, y, z, w)(\alpha', \beta', \gamma', \delta') = 0,$$

where the quadric forms a'', a'', are connected with each other by the relation $a'\beta'a' + a''b'a' + a''c'a' + a''d'a' = 0$.

If, then, for an arbitrary line (a', b', c', f', g', h'), we have $aa' + bb' + cc' + ff' + gg' + hh' = 0$, this relation is also one of the conditions for the relation between any two lines: any one whatever of the four lines through any point of the curve corresponds to the two systems respectively, of a properly determined quadric surface: it is a property of this curve that, of any four lines through any one of the points whatever of the system, those one is symmetric (whatever be the value), in the other one with the system, and the other two are unconnected.

Writing in the equations $a = 1$, we have in particular the quadriquadric curve

$$y^2 + b'\beta'z + \beta = 0, \quad x^2 + b''y + c''z + d''w = 0,$$

where the second is the symmetric definition of the system through the equation $g + h = 0$. Consider on the curve four points belonging to the arguments u_1, u_2, u_3, u_4 respectively;

41 – 2

and write for shortness x_1, y_1, z_1, w_1 for the sn, cn, and similarly for x_2, y_2, z_2, w_2. It appears by Abel's theorem that the condition in order that the four points may be in a plane is $u_1 + u_2 + u_3 + u_4 = 0$; viz. when this equation is satisfied we have

$$\begin{vmatrix} x_1, & y_1, & z_1, & w_1 \\ x_2, & y_2, & z_2, & w_2 \\ x_3, & y_3, & z_3, & w_3 \\ x_4, & y_4, & z_4, & w_4 \end{vmatrix} = 0,$$

or writing

$$\begin{aligned} &a, \quad b, \quad c, \quad f, \quad g, \quad h \\ &= x_4y_2 - x_2y_4, \quad x_3y_1 - x_1y_3, \quad z_4w_2 - z_2w_4, \quad z_3w_1 - z_1w_3, \\ &\quad a_2, \quad b'', \quad c'', \quad f', \quad g', \quad h', \\ &= x_4y_4, \quad x_3y_3, \quad z_4w_4, \quad z_3w_3, \quad y_3z_3, \quad x_4w_4, \end{aligned}$$

this equation is

$$af + bg + ch = 0;$$

or by what precedes, it appears that not only is this so, but that we have separately

$$af^2 + cf'' = 0, \quad bg' + b'g = 0, \quad ch' + ckh = 0,$$

viz. it follows from Abel's theorem that, when the arguments u_1, u_2, u_3, u_4 are connected by the equation $u_1 + u_2 + u_3 + u_4 = 0$, then each of these three equations holds good.

I assume in particular $u_1 = 0$, $u_2 = -u$, so that $x_2 = x$, $u_2 = -u$, $y_2 = y$, $z_2 = z$, $w_2 = w$, $x_3 = 0$, $y_3 = b_2$, $w_3 = a_2$, $w_2 = c_2$; we have $x_4 = a$, $b_4 = b_4 = b_2 = c_2$, $w_4 = d_4 = c_2$, $w_4 = c_4$,

$$\begin{aligned} &\alpha, \quad b, \quad c, \quad f, \quad g, \quad h \\ &= a_2 - d_2x, \quad -x_1 = -a_2x = -a_2x_2 = -x_2x_2x_1, \quad x = b_2y, \quad -x_2y = d_3y, \quad d_2x_3y = -x = b_4y - 1, \end{aligned}$$

and the four equations become

$$\begin{aligned} \frac{a - d_2x \cdot d_4z + b_2}{x} \cdot d_2w = 0, \\ -x = -d_2y + a_2x = 0, \\ \frac{d_2 - z_2 \cdot b_2y - 1}{c} + \frac{d_2x - z_2 \cdot a_2y - 1}{c_2} = 0, \end{aligned}$$

equations which may be written

$$\frac{a - d_2x}{a_2x} = \frac{w_2 \cdot d_4z + d_2y}{a_2x} \cdot \frac{-x}{-z} \cdot \frac{c_3x}{c_3} = \frac{a_2x + d_2y}{c_3}, \quad \frac{x}{a_2x - d_2y} = \frac{c_2}{c_3} \cdot \frac{a_2x - d_2y}{c_3},$$

so that, adding these equations, we must have an identity.

Representing the second and third of the last-mentioned equations by

$$\frac{a+1}{x} = C, \quad \frac{a-1}{x} = D,$$

we have

$$a = 1 - Cx, \quad d = 1 + Dx,$$

from either of which we can obtain a rationally; viz. the first equation gives

$$1 - a = 1 - 2Cx + C^2x^2,$$

that is, $x = \dfrac{2C}{1+C^2}$, whence

$$a = \frac{1-C^2}{1+C^2}, \quad d = \frac{1+C^2+2CD}{1+C^2};$$

and similarly from the second equation $a = -\dfrac{1}{2D}\cdot$ whence also

$$a = \frac{b'z + 2P - 2CD}{D^2+D^2}, \quad d = \frac{D^2-1}{D^2+D^2}.$$

Substituting for C its value, the first-mentioned value of a becomes

$$a = \frac{(a_2 - z_4)(d_2x - d_2z)}{1 - 2x_2x_4z - z_2z_4z_2x_4},$$

which is a form due to Cayley.

For the like values of b and d we easily verify; we in fact have

$$x_3 = sn(u_3 + u_4) = \frac{u_2u_3 \cdot d_2u_3 + a_3u_2 \cdot d_3u_2}{1 - k^2u_2u_3 \cdot a_3u_2 \cdot a_3 \cdot u_3},$$

$$= \frac{a_2u_3 \cdot d_3u_3 + x_2u_3 \cdot a_2u_3}{a_3u_2 \cdot a_2 + k^2x_2x_3y_2y_3},$$

we say *Elliptic Functions*, p. 63; or calling these values

$$= \frac{N_1}{D_1} = \frac{N_2}{D_2} = \frac{N_3}{D_3} = \frac{N_4}{D_4},$$

the above value is

$$x = \frac{N_1 - N_2}{D_1 - D_2},$$

which is right.

Cambridge, *28 June*, 1884.

844.

ON A THEOREM RELATING TO SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 212, 213.]

I FIND among my papers the following example of a general theorem relating to seminvariants. Write:

$$\Theta = 4b_8 + a_2 b_6 + a_2^2 b_4 + a_2^3 b_2 + 4a_2^4 b_0 + 7a_4 b_4 + 16a_4^2 b_0,$$
$$\Omega = \quad b_8 + 3a_2 b_6 + 5a_2^2 b_4 + 4a_4 b_4 + 5a_2^2 b_2 + 9a_4 a_2 b_2 + 7a_4^2 b_2;$$

then from a seminvariant S of the degree δ and weight w, by means of the operation Θ we obtain a seminvariant S of the degree $\delta + 1$ and weight $w + 8$; and by operating with Ω we obtain a combination of these by operating on S with the resulting combination $2w\Omega - 8\Omega$, which combination is of the degree $w + 1$, but only of the degree $w + 8$; via. operating with $2w\Omega - 8\Omega$, we obtain a new seminvariant of the same degree but of a weight increased by unity.

The example relates to the under-mentioned seminvariant S of the degree 5 and weight 7. Write

b_0	$+1$
b_2	-7
b_4	-9
b_6	$+9$
b'_8	-123
b_8	-10
bf'	$+39$

ΘS	ΩS	$\Theta\Theta S$	$\Theta\Omega S$	$\Omega\Theta S$	$\Omega\Omega S$						

[*The two-page paper concludes with a numerical verification table demonstrating the action of* Θ *and* Ω *on the chosen seminvariant S; for the table of differences and the proof that* $\Theta\Omega - \Omega\Theta = 0$ *on S, see the original Quarterly Journal, vol. XX, p. 213.*]

The columns above ΘS (where observe that, to operate with the b_8, of Θ, we restore the proper power of a_2, reading S as being $= 1.b^4 + 7a^3b_2$, and proceeding ultimately $\Theta S = \frac{1}{8}\Omega S = -55\Omega$; and the sum of these, which is a seminvariant, degree 5, weight 7, also $125''$ and 145); and the sum of these, which is a seminvariant, degree 5, weight 7, also $125''$ and 145; the upper portion comprises proof of the theorem $S = 1.b_4 + 7a^3b_2$; instead of giving this sum, I have added a column equal to $\frac{1}{16}(\Theta\Omega - \Omega\Theta)S$ this gives also the sum of the above seminvariants, viz. when this equation is the foregoing direct process from the given seminvariant S of the degree 5 and weight 7.

845.

ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 213–220.]

IT is remarked by Schwarz (see his Memoir "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105–120, p. 113), that the circle $x^2 + y^2 - 1 = 0$ can be orthomorphosed into the parabola $y^2 + 4(1 - x) = 0$ by the equation

$$x'(x' + iy') = \log x + iy'(x + iy);$$

viz. this equation establishes a $(2, 1)$ relation between the points interior to the circle and those interior to the parabola, which, qua relation between $x' + iy'$ and $x + iy$, will be orthomorphic, that is, infinitesimal elements of the one area will correspond to similar infinitesimal elements of the other area. The diameter $y' = 0$ of the circle is transformed into the axis $y = 0$ of the parabola, and figs. 1, 2 are symmetrical on the two sides of these lines respectively; we may therefore consider only the transformation

Fig. 1.

[Figure: upper semicircle in the $x'y'$-plane bounded by the diameter on the x'-axis]

Fig. 2.

[Figure: upper half of the parabola in the xy-plane bounded by the axis $y = 0$]

of the upper semicircle into the upper half of the parabola; we have (see figs. 1 and 2) A' corresponding to the vertex A of the parabola, B' to the point at infinity on the infinite branch AB of the parabola, the semicircular boundary $B'C'A'$ to the boundary BCA of the parabola, the highest point (and any point C' between the vertex A and B' on the parabola, the highest point C corresponding to the point C' between the vertex and the semi-latus rectum.

Fig. 3.

[Figure: semicircle with concentric circles and radii, and confocal parabolas with axial system through focus, illustrating the orthomorphic correspondence]

We may divide the circle by concentric circles and by radii; the corresponding curves for the parabola will be ovals and radials from the focus, the curves of each system being transcendental curves. Or we may divide the parabola by means of two systems of confocal parabolas; the corresponding curves for the circle will be (see fig. 3) two systems of orthotomic licences, those of the one system having B' for a crossnode, and those of the other system having B' for an asymode.

Writing $\log\tan$ to denote the hyperbolic logarithm of the tangent, that if ϕ, ϕ' are such that

$$\phi = \log\tan(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi'),$$

this equation, as is known, may also be presented in the form

$$i\phi' = \log\tan(\tfrac{1}{4}\pi + \tfrac{1}{2}i\phi),$$

$$i\tan\tfrac{1}{2}\phi' = \tan\tfrac{1}{2}i\phi,$$

or, what is the same thing,

$$\tan\tfrac{1}{2}\phi' = \tanh\tfrac{1}{2}\phi,$$

where observe that, as ϕ' increases from 0 to $\frac{1}{2}\pi$, ϕ increases from 0 to ∞, and that always $\phi > \phi'$.

We can now establish as follows the three correspondences $A'O$ with $A'C$, $C'B$ with OB, and $A'CB$ with $A'C'B'$:

(1) $x'(x'+iy') = \frac{2\phi}{i}$, $\quad x'+iy' = \tan\frac{1}{2}\phi'$;

that is,

$$x = -\frac{2\phi'}{\pi}, \quad y = 0; \quad x' = \tan\tfrac{1}{2}\phi', \quad y' = 0;$$

ϕ is given as A, A'; ϕ' is given as $\theta = 0$, C, C';

(2) $x'(x'+iy') = -\tan^2\frac{1}{2}\phi'$, $\quad x'+iy' = i\tan\frac{1}{2}\phi'$;

that is,

$$x = -\tan^2\tfrac{1}{2}\phi', \quad y = 2\tan\tfrac{1}{2}\phi'; \quad x' = 0, \quad y' = \tan\tfrac{1}{2}\phi';$$

$\phi = 0$ gives A, A'; $\phi = \frac{1}{2}\pi$ gives B, B';

(3) $x'(x'+iy') = \frac{2i\phi}{\pi}$, $\quad x'+iy' = e^{i\phi'}$;

that is,

$$x = 1 - \frac{2\phi}{\pi}, \quad y = \frac{2\phi}{\pi},$$

the parabola; and

$$x' = \cos 2\phi', \quad y' = \sin 2\phi',$$

the circle;

$$\phi = 0 \text{ gives } A, A'; \quad \phi = \tfrac{1}{2}\pi \text{ gives } B, B'.$$

Observe then for points in OA', $A'B'$, equidistant from O, we have $x' = \tan\frac{1}{2}\phi'$, $x' = -\tan\frac{1}{2}\phi'$, and corresponding hereto we have points in OA, OB on the axis of the parabola, the values of x being $x = 1 - \frac{2\phi}{\pi}$, $x' = -1 - \frac{2\phi}{\pi}$, viz. the negative value is always the greater.

Observe further that, to the points $(x', y') = (\cos 2\phi', \sin 2\phi')$ and $(x', y') = (-\cos\frac{1}{2}\phi', 0)$ on the circle, and on the radius OB', correspond the points

$$(x, y) = \left(1 - \frac{2\phi}{\pi}, \frac{2\phi'}{\pi}\right) \text{ and } (x, y) = \left(-\frac{2\phi}{\pi}, 0\right);$$

on the parabola and on the axis OB respectively; the axial distance of these two points is $\left(1 - \frac{2\phi}{\pi} + \frac{2\phi}{\pi}\right) = 1$, the radius of the circle, or the distance OA of the vertex and focus of the parabola; this is a rather curious theorem.

The function $\log\tan(45+\frac{1}{2}\alpha\phi')$ is tabulated, Legendre, *Théorie des Fonctions Elliptiques*, t. II. table II. pp. 256–259; viz. writing as above ϕ for the argument, we have hereby the value of ϕ, we obtain ϕ from 90 to 90 at intervals of $30'$ and to 12 decimals, and with 6th differences. Observe that ϕ' is thus given in degrees and minutes as a circular arc, ϕ as a number; it is convenient to have ϕ and ϕ' each as arcs, or such as numbers, the conversion may of course be made by means of a table of the lengths of circular arcs, and I have calculated the Table which I give at the end of the present paper.

I had previously calculated for the correspondence $A'O$, OB' and $A'CB$ with $A'O$, OB and ACB, the following values, at irregular intervals suited to the construction of a figure:

Circle		Parabola	
$x'OB'$	Ordinate $\div R$	$x'O$	OB
0	0	0	0
30	.5176	.1641	$-.1641$
60	1.0000	.5851	$-.5851$
80	1.2856	1.0470	-1.0470
100	1.5320	1.6262	-1.6262
130	1.8126	2.5817	-2.5817
150	1.9319	3.4131	-3.4131
160	1.9696	3.8967	-3.8967
170	1.9924	5.5942	-5.5942
175	1.9981	4.0024	-4.0024
178	1.9994	4.6231	-4.6231
180	0	$-\infty$	∞

Write in general

$$x'(x'+iy') = \rho + i\sigma, \quad x'+iy' = \tan\tfrac{1}{2}(\rho+i\sigma);$$

viz. considering x, y as given, find these ρ, σ by the equations $x = p\sigma'\pi$, $y = 2\sigma\pi$, and then writing $\frac{1}{2}\rho = \theta$, $\frac{1}{2}\sigma = \phi$, we have

$$x'+iy' = \tan(\theta+i\phi) = \tan(\theta+i\phi) = \frac{P+iQ}{1-iPQ};$$

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where

$$P = \tan\tfrac{1}{2}\rho\theta, \quad = \tan\tfrac{1}{2}\theta;$$

$$Q = \tfrac{1}{i}\tan\tfrac{1}{2}i\phi, \quad = \tanh\tfrac{1}{2}\phi, \quad = \tfrac{1}{i}\tan\tfrac{1}{2}i\phi, \quad = \tanh\tfrac{1}{2}\phi;$$

whence, if we introduce ϕ' connected with ϕ by the equation $\phi = \log\tan(\frac{1}{4}\pi + \frac{1}{2}\phi')$ as before, we have $Q = \tan\frac{1}{2}\phi'$, and the formula is

$$x'(x' + iy') = \frac{P + iQ}{1 - iPQ}, \quad P = \tan\tfrac{1}{2}\theta, \quad Q = \tan\tfrac{1}{2}\phi',$$

giving the values of x', y'.

It is clear that we have

$$x'(x' + iy') = \frac{P - iQ}{1 + iPQ},$$

and thence

$$x'(x' + y'^2) = \frac{P + Q}{1 + P^2Q^2}.$$

Hence, to the circle $x'^2 + y'^2 = c^2$, corresponds in the parabola the curve

$$P + Q = c^2(1 + P^2Q^2),$$

where $p + iq = \tan(\rho + i\sigma)$, $P = \tan\frac{1}{2}\rho$, $Q = \tan\frac{1}{2}\phi'$ $\tan\frac{1}{2}\sigma$. This is a complicated transcendental equation, and I do not see any convenient way of tracing the curve. The set of curves satisfy the differential equation

$$\frac{PdP + QdQ}{P + Q} = -\frac{PQ(QdP + PdQ)}{1 + P^2Q^2},$$

where dP, dQ are given in terms of $d\rho$, $d\sigma$ by

$$PdP(1 + Q^2) - QdQ(1 + P^2) = 0,$$

that is,

$$\frac{dP}{d\sigma} = \tfrac{1}{2}\sec^2\tfrac{1}{2}\rho, \quad \frac{dQ}{d\sigma} = \tfrac{1}{2}\sec\phi'\sec^2\tfrac{1}{2}\phi'.$$

We have $p = \frac{2\rho}{\pi}$, and thence we have to take, as elements of the several curves, $\frac{1}{2}d\rho + i\frac{1}{2}d\phi = 0$. Combining these equations, we have

$$dP\frac{dQ}{Q} \cdot dQ + (1 + P^2) - q(1 + Q^2),$$

and thence

$$\rho^2 dP(1 + Q^2) - qQdQ(1 + P^2) = 0,$$

as an equation in the locus of the normals in question. If p and q are small, then putting for a moment $\frac{1}{2}p = a$ for shortness, we have

$$P = a\rho + \tfrac{1}{3}a\rho^2, \quad Q = a\sigma + \tfrac{1}{3}a\sigma^3,$$

and the equation becomes

$$p^2(1 + \tfrac{1}{3}a\sigma^2)^2 - q^2(1 - \tfrac{1}{3}a\sigma^2)^2 - q^2(1 + \tfrac{1}{3}a\rho^2)^2 = 0,$$

that is,

$$p^2 - q^2 + (a\rho^2 + \alpha^2)(p^2 + q^2)(p + q) = 0,$$

writing this in the form

$$p^2 - q^2 = -\frac{\pi^2}{16}(p^2 + q^2)\rho^2 + a^2(p^2 + q^2),$$

we find $x = \frac{\pi^2}{8}(x^2 + 2y^2) = 0$, or say $p^2 = \frac{\pi^2 y^2}{8}$ as the form of the meanits in the neighbourhood of the focus. Or, viz. the meanits for all of these, as might have been expected, are the kit-shaped (or negative side) of the curve.

I have been desirous to find a more general law $0 =$ between the meanits in the neighbourhood of the focus. We may say $\rho = -\frac{2\pi}{\pi}$, as in the form of the meanits in the neighbourhood of the focus.

Corresponding hereto in the circle, writing for a moment

$$x'(x' + iy') = \rho i,$$

we have

$$\tan \tfrac{1}{2}\rho i = \frac{P + iQ}{1 - iPQ},$$

whence

$$\rho i = \rho - i \cdot (1 + p^2/\pi^2),$$
$$\rho = P^2(1 - Q^2)(1 + pP^2);$$

whence eliminating P^2 and Q successively, we may say

$$\rho^2 + \rho^2 = a^2\left(P^2 - \frac{1}{P^2}\right) - 1 = 0,$$

$$\rho^2 + \rho^2 = a^2\left(\frac{1}{Q^2} + 1\right) - 1 = 0,$$

say, for shortness, then $\rho^2 + 2\rho\rho\rho' + a' = 0$, and $\rho^2 + \rho\rho - 2\rho\rho' = 0$. Introducing the polar coordinates x', y' then

$$x' = r\cos\theta', \quad y' = r\sin\theta',$$

the equations then become

$$r^2 = 2\alpha\rho'\sin\tfrac{1}{2}\theta' + 1,$$
$$\rho^2 = -2\alpha\rho'\cos\tfrac{1}{2}\theta' + 1,$$

the polar coordinates r, θ'.

Consider the quadricuspidal curve

$$r' = Ax^2 + Bx^3 + Cx + Dx^2 + Ex,$$

where $A = 4(1 + 2P^2)(1 + p^2)$, $B = \frac{1}{2}p^2$, $C = 4 + ix^2(1 - 2P^2)$, $D = +\frac{1}{2}p^2(1 + 4y)$, the coordinates of two lines meeting each side, and consequently it entirely contains the curve $\rho i, \rho i, \rho i$. They agree, as they should do, with a foregoing result,

$$\rho^2 = 0,$$

$$\rho^2 + Dy^2(t' + \tfrac{1}{2}h\theta'\rho') - 6 \cdot \tfrac{1}{2}\rho^2 + \tfrac{1}{2}\rho^2 + \tfrac{1}{2}\rho = 0,$$

and it would of course be possible to express the remaining forms 213, 642, 662, 441, 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of which belong to two lineapas having each of them B' for a node and O for a focus, and which of course cut each other at right angles; see fig. 4, which is a more

Fig. 4.

[Figure: two orthogonal families of lemniscates with node B' and focus O, with a generic point indicated]

diagram. In fact, omitting for convenience the accents, but recollecting always that the curves belong to the figure of the circle, the first equation gives

$$(x-1)^2 + 4ax^2 \cdot \cos^2 \tfrac{1}{2}\theta', \quad = 2a\rho'(1+\cos\theta) = 2a\rho'(x+r),$$

that is,

$$(r^2+1-2x)x^2 = 4a\rho'(x+r)^2.$$

Transferring to the origin B', we must write $x+1$ for x, and then

$$r'^2 = (x+1)^2 + y^2.$$

The equation thus becomes

$$[x^2+y^2-2(\alpha x+1)+3(\alpha x+1)]^2 = 4a^2(x+1)(x+y^2+2x+1)),$$

that is,

$$[x^2+y^2-2(\alpha x+1)]^2 = 4a^2(x^2+y^2)(x^2+y^2),$$

or say

$$(x^2+y^2)^2 + 4(\alpha x+1)(x^2+y^2) + 4(x+1)(x^2+y^2) + 4(\alpha x+1)\alpha^2 y^2) = 0,$$

showing that the point B' is a crunode, the tangents there being $x \pm \alpha y$.

Writing in the equation $y=0$, we have $x^2=0$, the crunode, and

$$x = 2\sqrt{(\alpha'+1)}\sqrt{(\alpha'+1)} \pm 1.$$

the product of these two values, $\alpha = 4(\alpha+1)$, that is, $4 = \left(P + \frac{1}{p}\right)^2$, via. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha+1)}\sqrt{(\alpha'+1)} - 1$$

the smaller root, is greater than 1. The curve corresponding to a parabola $y^2 = 4p(c-x)$ is thus a crunodal lineapas, the crunode at B', and the loop lying wholly within the circle. Moreover, the loop includes within the centre O of the circle.

The other curve is in like manner

$$[r^2 + 1 - 2x]^2 = 4(1-x)^2,$$

viz. transforming to the origin B', and therefore writing $x = 1$ for x, and similarly for x_n, and x_n. In support by Abel's theorem that the condition in order that the four points may be in a plane is $x = x_n + x_n + 1 = 0$; viz. when this equation is satisfied we have

$$(r^2 + 1 - 2x)x^2 = 4ax^2(x^2+1)^2,$$

that is,

$$[x^2 + y^2 - 2x(x+1)]^2 = 4ax^2(x^2+y^2),$$

or

$$(x^2+y^2)^2 - 4(\alpha+1)(x^2+y^2)x + 4(\alpha+1)x^2 + 4\alpha^2(x^2+2y^2-\alpha^2y^2) = 0,$$

showing that the point B' is a crunode, the tangents being $x \pm \alpha y$.

Writing in the equation $y = 0$, we have $x^2 = 0$, the crunode, and

$$x = 2\sqrt{(\alpha+1)} \cdot x \pm 1; \quad 4\alpha = -1, \quad x = -1,$$

the product of these values is 4α, that is, $4\alpha = -\left(P - \frac{1}{p}\right)^2$, via. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha+1)}\sqrt{(\alpha+1)} - 1$$

is positive and less than 1; the curve is thus an acnodal lineapas, having B' for the acnode and cutting the axis outside the circle at the points $x = \alpha + 2\sqrt{(\alpha+1)}$. Conversely, considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ, the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

The Table above referred to.

[Table giving the correspondence between $\phi = \log\tan(45 + \frac{1}{2}\phi')$ for ϕ' from 0 to 90 at intervals of 1. The full numerical table from the original paper, with values to seven decimal places, is reproduced below.]

ϕ'	$\phi = \log\tan(45 + \frac{1}{2}\phi')$	ϕ'	ϕ	ϕ'	ϕ
0	0	46	.9028545	50	.9962755
1	.0174553	47	.9249187	51	1.0381004
2	.0349066	48	.9457040	52	1.0584952
3	.0523599	49	.9657500	53	1.0792677
4	.0698132	50	.9852280	54	1.1005044
5	.0872665	51	1.0036321	55	1.1222263
6	.1047198	52	1.0216162	56	1.1444466
7	.1221730	53	1.0392060	57	1.1672144
8	.1396263	54	1.0566175	58	1.1905610
9	.1570796	55	1.0738249	59	1.2145202
10	.1745329	56	1.0908669	60	1.2391411
11	.1919862	57	1.1077612	61	1.2644777
12	.2094395	58	1.1244912	62	1.2905931
13	.2268928	59	1.1410615	63	1.3175552
14	.2443461	60	1.1573912	64	1.3454387
15	.2617994	61	1.1733035	65	1.3743253
16	.2792527	62	1.1887844	66	1.4042947
17	.2967060	63	1.2036336	67	1.4354352
18	.3141593	64	1.2178358	68	1.4678434
19	.3316126	65	1.2312718	69	1.5016262
20	.3490659	66	1.2438139	70	1.5369024
21	.3665192	67	1.2555401	71	1.5738033
22	.3839724	68	1.2663275	72	1.6124704
23	.4014257	69	1.2761519	73	1.6530834
24	.4188790	70	1.2848920	74	1.6958685
25	.4363323	71	1.2924327	75	1.7410341
26	.4537856	72	1.2986277	76	1.7887898
27	.4712389	73	1.3033236	77	1.8394554
28	.4886921	74	1.3064565	78	1.8933630
29	.5061454	75	1.3078560	79	1.9508587
30	.5235987	76	1.3074457	80	2.0123399
31	.5410520	77	1.3050447	81	2.0782854
32	.5585053	78	1.3004563	82	2.1492283
33	.5759585	79	1.2934613	83	2.2258028
34	.5934118	80	1.2838174	84	2.3087094
35	.6108651	81	1.2712504	85	2.3989343
36	.6283184	82	1.2554543	86	2.4977049
37	.6457717	83	1.2360880	87	2.6066908
38	.6632250	84	1.2127740	88	2.7282169
39	.6806782	85	1.1850968	89	2.8657689
40	.6981315	86	1.1525987	90	∞
41	.7155848	87	1.1147993		
42	.7330381	88	1.0712310		
43	.7504914	89	1.0214444		
44	.7679447	90	∞		
45	.7853982				

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A VERIFICATION IN REGARD TO THE LINEAR TRANSFORMATION OF THE THETA-FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXI. (1886), pp. 77–84.]

THE notation is that of Smith's* "Memoir on the Theta and Omega Functions," differing from Hermite's only in a factor, $\Im_{m,n}$ (Smith) $= 2^{n/4}\Im_{m,n}$ (Hermite); and it is in consequence of this that the factor $2^{n/4}$ occurs in the expression presently given for Λ. Hermite's Memoir "Sur quelques formules relatives à la transformation des fonctions elliptiques" appeared in *Liouville*, t. III. (1858), pp. 27–36.

Writing

$$\omega = \frac{a + d\Omega}{c + d\Omega},$$

where $ad - bc = 1$, the formula of transformation is

$$\Im_{m,n}\left[(a+b\Omega)\frac{m^2}{2}, \Omega\right] = C\exp\left(-i\pi(a+b\Omega)\frac{m^2}{2}\right)\Im_{m,n'}\left(\frac{m\cdot a}{2}, \omega\right);$$

where:

$$m' = cu + b\cdot n,$$
$$n' = cm + a\cdot n,$$

or:

$$m' = cm + b\cdot n,$$
$$n' = cm + a\cdot n.$$

We have, according as b is positive or negative,

$$C = \exp\left[-\frac{i\pi}{\sqrt{(-(c+d\Omega))}}\right] = \exp\left[-\frac{i\pi}{\sqrt{(c+d\Omega)}}\right],$$

where for each case the square root is to be taken in such wise that the real part is positive; Λ, H are eighth roots of unity; viz. in each case

$$\Lambda = \exp\left[-\frac{1}{2}i\pi(\alpha m^2 + 2bmn + dn^2 + 2\alpha\beta n + 4\beta\alpha n + ab\gamma\delta)\right]e^{m\sqrt{(-1)/2}},$$

*[H. J. S. Smith, *Collected Mathematical Papers*, vol. II., pp. 415–634.]

and, according as b is positive or negative,

$$H = \left(\frac{b}{a}\right) e^{i\pi a}, \qquad a \text{ odd}, \qquad H = \left(\frac{-b}{a}\right) e^{i\pi(a-1+3b+\beta)}, \qquad a \text{ odd},$$

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}, \quad b \text{ odd}, \qquad H = \left(\frac{-a}{-b}\right) e^{i\pi(a-1+3b+\beta-b+2)/2}, \quad b \text{ odd};$$

where $\left(\frac{a}{b}\right)$, &c., are the Legendrian symbols as generalised by Jacobi; if a and b are each of them odd, the formula for the case of a even or odd and b odd respectively are equivalent to each other, and either may be used indifferently.

To fix the ideas, I assume throughout that b is positive and odd; the proper formulae then are

$$C = \frac{H}{\sqrt{(c + d\Omega)}};$$

Λ as above, and

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}.$$

I will also, for greater convenience, assume that c is odd; in consequently even, viz. the numbers a and c are odd and those even, or else a and b are odd and the other even.

The equation $\omega = \frac{a+d\Omega}{c+d\Omega}$ gives $\Omega = \frac{-a+c\omega}{-d+b\omega}$, and thence

$$a + b\Omega = -\frac{1}{-d + b\omega}.$$

Comparing the expressions for ω, Ω, we observe that we may interchange ω and Ω, writing also $-d$, b, c, $-a$ for a, b, c, d; and changing the other letters, we may for u, b, β, m, n, m', n', write

$$\frac{u}{c + d\Omega}, \quad b, \quad a, \quad m', \quad n', \quad m, \quad n,$$

where

$$m' = -dm - bn, \quad n' = -cm - an;$$

the formula then becomes

$$\Im_{m',n'}\left[(-d + b\omega)\frac{m\cdot}{2}, \omega\right] = C' \exp\left[-i\pi(-d + b\omega)\frac{u^2}{2}\right] \Im_{m,n}\left(\frac{u}{c + d\Omega}, \Omega\right),$$

or, as it is the same thing,

$$\Im_{m',n'}\left[(a + b\Omega)\frac{m\cdot}{2}, \omega\right] = C' \exp\left[-i\pi(a + b\Omega)\frac{u^2}{2}\right] \Im_{m,n}\left(\frac{u}{c + d\Omega}, \Omega\right);$$

or, observing that the left-hand side is $= (-1)^{mn'}\Im_{m,n'}\left[\frac{u}{2}, \omega\right]$, we have

$$\Im_{m,n'}\left[(a+b\Omega)\frac{m\cdot}{2}, \omega\right] = \frac{C'}{(-1)^{mn'}}\exp\left[-i\pi(a+b\Omega)\frac{u^2}{2}\right]\Im_{m,n}\left(\frac{u}{c+d\Omega}, \Omega\right),$$

an equation which is of the same form as the original equation of transformation, and is to be identified with it.

We have

$$m' = -dm - bn = -dm + b\Omega\cdot n, \quad = -m + bd(-n+cm),$$
$$+ b\cdot bd(-n+cm) = -bd,$$
$$n' = -cm - an = -(cm+an) = m\cdot bd(-n+cm) = -bn(-n+cm) - 1,$$

values which may be written

$$m' = -dm - bn + Pa = -bd, \quad m\cdot b\cdot bd(-n+cm) - 1,$$

or, since $\Omega - d - b\omega = -bd$,

$$m' = -d + bd\cdot(-n+cm) = b\omega\cdot m = -d\cdot \quad m = cm + b\cdot n - 1,$$

$$n' = -c\cdot \quad m\cdot \quad a\cdot \quad n-1, \qquad n' = c\cdot \quad m + a\cdot \quad n - 1;$$

P and Q being integers. In fact, P is an integer if b is odd or if bn is even, and similarly for Q.

The left-hand side is $\Im_{m',n'}\left[(a+b\Omega)\frac{u}{2}, \Omega\right]$, which is in virtue of P even, and which depends only on the values of m and n.

We have

$$CC' = (-1)^{mn'-mn+m'n'-m'n},$$

which is

$$= (-1)^{mn-m'n'}$$

this condition will be considered in the sequel.

The formula in question is in fact identifiable to that into which it transforms when, observing that the left-hand side is unchanged by the change of u, b, c, d, m, n, m', n', into $-d$, b, c, $-a$, m', n', m, n;

$$CC' = (-1)^{mn-m'n}, \quad CC' = (-1)^{mn'-m'n}.$$

We have

$$CC' = \frac{HH'}{\sqrt{(c+d\Omega)}\cdot\sqrt{(-(c+d\Omega))}},$$

where the square roots are taken in such wise that the real part is positive; hence

$$CC' = \frac{HH'}{i\sqrt{(c^2+1)}} \quad \text{or} \quad \frac{HH'}{i\sqrt{(-(c^2+1)d)}},$$

according as c is even or odd; for, taking the radius vector for positive, we have that the square roots of $c+d\omega$ and $\sqrt{(-(c+d\omega))}$ are of the form

$$\sqrt{(c+d\omega)}, \quad \frac{1}{i}\sqrt{(c+d\omega)} \quad \text{or} \quad \frac{1}{i}\sqrt{(-(c+d\omega))}, \quad \sqrt{(-(c+d\omega))} = \sqrt{(c+d\omega)},$$

where the square roots taken in such wise that the real part is in each case positive; 43—2 where $\sqrt{(c+1)}$ means

$$\sqrt{(-(c+1)d)}\sqrt{(-)}\cdot\sqrt{(-(c+d)\Omega)};$$

the last-mentioned two square roots being so taken as just explained; and we have therefore

$$\Lambda = \exp\left[-\tfrac{1}{2}i\pi(\alpha m^2+2bmn+dn^2+2\alpha\beta n+4\beta\alpha n+ab\gamma\delta)\right]e^{m\sqrt{(-1)/2}},$$
$$\Lambda' = \exp\left[-\tfrac{1}{2}i\pi(-dn^2+2bm'n'-am'^2-2\alpha\beta m'-4\beta\alpha m'-bd\beta)\right]e^{m\sqrt{(-1)/2}},$$

viz. the values of Λ obtained from the value of Λ by the change of a, b, c, d, m, n, m', n', into $-d$, b, c, $-a$, m', n', m, n; and writing

$$A = \exp\left[-\tfrac{1}{2}i\pi\{-am^2+2bmn+dn^2+dn'^2-2bn'm'+am'^2+2\alpha\beta(n-m')+4\beta\alpha(m-n')-ab\gamma\delta-b$$

the last-mentioned two square roots being so taken as just explained, and we have

Representing the second and third of the last-mentioned equations by

$$4=\exp\left(-i\pi\rho^{-mn'-m'n}\right),$$

the last-mentioned two squares roots being also explained; and we have to express the term divisible by 4,

$$\rho=-mn'(2+ab\gamma\delta+bd\beta).$$

To calculate $\Delta+\Delta'$, we have

$$dm-bn=m+bd\cdot(n-c\cdot m),\quad m=-d\cdot m-b\cdot n-1,$$
$$-cm+an=-n-ac\cdot(n-c\cdot m),\quad n=c\cdot m+a\cdot n-1,$$

and thence

$$-2dm^2+2bn^2=-mn'+ac\cdot n'-bd\cdot m'^2+abcd^2-2bd(-n'2+cdn)-2acd(2-cd)=-2bdmn+\&c.$$
$$(-am+bn)^2=-mn'(-2bd-bnn'+2abcd)-2bd(-n'+cn)+2acd(-a'+cn)-2bdcd(-n'+cn)=$$
$$--2acd(-a'+cn)-bd\cdot 2a^2+2abdn-2acd(-n'+ca)+2abcd^2=4acdn,$$

consequently

$$-adn^2+2bnn'+2bcmn'-2bdmn+2acmn+bdn^2+2bdcmn=2acdn(n+2cdn)-2bdcmn=4cdmn,$$

also

$$-2bdmn=-2cdbn^2-2bd^2n^2+2acd\cdot n+2bcd\cdot n^2+2bd\cdot c\cdot 2bd\cdot 2bd\cdot bd,$$
$$-2bn'=-2adcb-2cdb\cdot 2bd\cdot 2bn^2+2cdb\cdot 2cd\cdot 2bd\cdot bd\cdot 2bcb,$$
$$-dn'^2+dn'^2=-2bcd^2\cdot dcb-2dcb\cdot dn+2cdn^2+2bcd\cdot dcb\cdot bcb+2cdb\cdot bd\cdot bd,$$

whence, adding, we obtain

$$\Delta+\Delta'=-dn^2-2bnn'+2bcmn'-dn^2+2cdmn'-2bcdmn'-bcdm+2bcd\cdot n^2+2cd\cdot dcb+2cd\cdot bd\cdot bd,$$

and, adding to this the expression of Δ, we find

$$\Delta+\Delta'=-2bnn'+2bnn'-2bcmn'+2cd\beta-2dnn'+2dcn'-2bcd-bdcd+2bd\cdot bd,$$

The coefficient of u is $= 2bd(-n+cm) - 4cd + ad(-c \cdot m^2 + a \cdot n) + 2(b-1)(c \cdot n)$. Hence writing

$$m \cdot n' = 2bd(-n+cm) - ad(-m \cdot n^2 + a \cdot n) + 2bnn(-1)(c \cdot n - n),$$

observe that $a\Omega n - 1$, observing in every case $\Omega \cdot mod$, writing also $4bn \cdot 2cdn$, and consequently

$$4bn \cdot n' = ad(\beta m - n)(-\beta + b) + (-\beta + b)(b-1)(c+d)(b-1)(n-2m).$$

where the even values of n may be omitted. We have moreover

$$mn - m'n' = -bd(-n+cm)\beta - ab(-m \cdot n^2 + a \cdot n)\beta + (-\beta + b)(b-1)bd,$$

and these, since β is even, may be neglected, and the result is simply

$$mn - m'n' = -bd\beta - ad\delta\beta,$$

and thus

$$mn - m'n' = ad\delta\beta + bd(-2+cd)\beta - 2bn\delta\delta(-a+cd)\beta,$$

we have also

$$m - n' = -d \cdot m - bd - (c \cdot m - a \cdot n + 1) = -(c+d)(b)\beta + 1,$$

$$n - m' = -m + d \cdot m + ba = ad(b-n).$$

If even of which disappear, leaving the result, $u\sqrt{(-1)/2}$ becomes equal to $4ms\Omega$.

The next thing is to express the term divisible by 4: there is at once a term given by $4(b-1)(c \cdot 2bn)$, and so we may write therefore

$$4(b-1)(c \cdot 2bn) + 4(-bd)(2+cd) + (-bd)(b-1)(c \cdot 2cdb) = 0,$$

via. the values of Δ is given by Abel's theorem, then in particular each of the four equations becomes

$$4(b-1)(c \cdot d \cdot n) = -2bdcn,$$

where, this is positive value of β is positive; and thus

$$mn - m'n' = -dn\beta + bd(c-1)(c\beta - 1),$$

the equation which was to be verified.

847.

ON THE THEORY OF SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXI. (1886), pp. 1–17, [13, 14, 16].]

IN my "Mémoire sur les Hyperdéterminants," *Crelle*, t. XXX. (1846), pp. 1–37, [13, 14, 16],* I gave an investigation for the number of and relations between the quartic invariants of a given binary quantic, or what is the same thing to the cubic seminvariants of a given weight, and I propose at present (considering the formulae in this point of view) to further develop the theory in regard to the solution of the systems of linear equations obtained in the manner in question. But I first commence with a remark: I recall that the notation is the ordinary hyperdeterminant notation: for instance,

$$U = (a, b, c, d)(x, y)^3, \quad V = (a, b, c, d)(x, y)^3,$$

$$\begin{aligned} 12VV \cdot 1 + 2, \quad VVV \quad &= D_{a,a'}VVV, \\ 12VV \cdot 12 &= D_{a,a'}D_{b,b'}VVV, \\ \text{\&c.} \quad &= D_{a,a'}D_{b,b'} \cdot D_{c,c'}VVV, \end{aligned}$$

and that, when the variables $(x, y), (a, b), (a', b')$ are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

The investigation* is as follows:

$$\begin{aligned} UU(12) &= (a, b, c, d)(b', a)(b', d') \cdot -2(ad' + a'd - bc' - b'c) + 9(bc' + b'c), \\ &= 12P(a' \cdot b'^2 - 5abb' \cdot b'^2 \cdot 3a^2bb' + b'^2 \cdot a'd' - 5b'd \\ &\quad + a \cdot a' \cdot -25, \\ &\quad + 2(cc' - d^2), \end{aligned}$$

and that, when the variables $(x, y), (a, b), (a', b')$ are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

The investigation is as follows:

$$\begin{aligned} UVUV &= UUVV - U_xV \cdot VWV \cdot UWV, \\ UUVW &= UWUV \cdot VWWV, \\ &= D_{a,b'}VVWV = D_{b,a'}VWVV, \\ &\text{\&c.} \end{aligned}$$

*[This Collection, vol. I., p. 100; see also, vol. I., p. 157.]

Writing for shortness

$$12, 34 = \mathfrak{A}, \quad 13, 42 = \mathfrak{B}, \quad 14, 23 = \mathfrak{C},$$

we have

$$D_{a,d}\mathfrak{A} + D_{b,c'}\mathfrak{B} + D_{c,b'}\mathfrak{C} = 0.$$

Suppose $U = V = W = X$; we have the derivatives which correspond to the same value of δ, w, we have from these

$$D_{a,d}\mathfrak{A}, \quad D_{b,c'}\mathfrak{B}, \quad D_{c,b'}\mathfrak{C} = 0, \quad D_{d,a'}\mathfrak{D}_{x'}\mathfrak{A} = 0,$$

in the language of Prof. Sylvester, $\alpha_{i,k}$, $b_{i,k}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2, =$ &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the language of Sylvester's $\alpha_{i,b}$, $b_{i,b}$, $b_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(ax)_2(ax)_3(ax)_4 = D_{a,d}D_{a,c}D_{a,b}V \cdot V = 4ax,$$

that is,

$$\alpha = 4f - 3g + 2h,$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that the Cliebschian notation, for instance

$$(\alpha, \beta, c, d)(x, y)^3 + 4(x + b)(x + d)y + 6((c + b)y + 4(c + d)y + 4y) = 4ax,$$

viz. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2, =$ &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the Cliebschian notation, for instance:

$$(\alpha\beta + \alpha\delta - \beta\gamma)\mathfrak{A} = 4(\alpha\delta - \beta\gamma),$$

that is,

$$a = bf - 3g + 2h,$$

$a = bf - 3g + 2h$, $\alpha\beta = 4f - 3g + 2h$,

$$a = bf - 3g + 2h,$$

a seminvariant; the

$$(a-\beta)^2(\alpha-\gamma) = -2\alpha\beta + a\gamma + 4a(c-\beta) - 8c\delta + a\gamma\beta - 3\beta + 4ac\delta,$$

$\beta = 4f - 3g + 2h$, &c., are scalars: writing each of f, g, h for the weight). We have, as before,

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we again see that the theory is one of the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of a, the cubic seminvariants $\mathfrak{CVC}$, $\mathfrak{CVC}$, are the differences $D_{a,a}\mathfrak{A}$, &c., but in many cases the seminvariants $D_{a,a}\mathfrak{A}$, &c., are ultimately replaced by (x, y), then these forms denote, each of them, an invariant.

Or, we may use the Cliebschian notation, for instance, observe that

$$(\alpha, \beta, c, d)(x, y)^3, \quad \alpha = -cu + bv = av + bw, \quad \&c.,$$

that is,

$$a = c - b = -a,$$

via. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(\alpha-\beta)^2 = a + (\alpha+\beta)^2,$$

$$= 4(\alpha - 2b + b^2),$$

where α, β, ... are scalars.

The invariant is

$$(\alpha-\beta)^2(\alpha-\gamma) = \alpha^2 + \alpha\beta\alpha + 2\beta\gamma\alpha\beta + \beta^2,$$

$\alpha = -\beta$, $\beta = -\alpha$, $\alpha = -\beta$, $\beta = -\alpha$,

$$\alpha = -\beta\beta\beta\beta\beta - \beta = 0.$$

Hence

$$\mathfrak{A} \cdot \mathfrak{V} = 0.$$

We may also for independent derivatives, when f is even, the terms of the series $D_{a,a+1}$, $D_{a+1,b+1}$, ..., and when f is odd, those of the series $D_{a+1,a+2}$, $D_{a+2,b+2}$, ..., continuing each series until the last term in which $D_{a+i,b+i}$ would be equal to zero in virtue of the value of δ, w (notice that the values of δ, w must be in such relation, so that there is no need of a sequence of the seminvariants $D_{i,m}$ are independent the series of the order term $D_{a,a+1}$); the like considerations apply to the other derivatives $D_{a,a+1}$, &c., but inasmuch as we must keep at least a of order a different, the same set must necessarily be in use, but the same set of independent seminvariants will of course be obtained. The number of these is also f, only the same set will be the same as we are considering; the number of independent seminvariants is in fact equal to the number of δ, w, &c., which is independent of δ, w, &c., in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$. We may say that all these seminvariants of the highest powers of δ, w are mutually independent, and as such are connected by linear relations between them, the coefficients of these depending on the seminvariants of the lowest powers of δ, w; and as such these linear relations are linear functions of the seminvariants, the connecting relations being also linear functions; the seminvariants of the highest powers are also linear functions of those of the lowest, the seminvariants of the highest powers being also represented by the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$. We may say that all this means that of any independent seminvariants of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, we have any independent linear relations between the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$.

Or, we may say that the number of independent linear relations between the seminvariants of the highest powers is, in proportion to the number of independent seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, equal to the number of independent linear relations between the seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$.

We will presently give the solution of the under-mentioned system; but beginning with the second and weight w as we have here, the number of solutions will be the following table:

	$D_{a,d}$	$D_{a,d+1}$	$D_{a,d+2}$	$D_{a,d+3}$	
$\mathfrak{D}_{ad}$	D_a	0,	D_a	$\frac{1}{2}D_a$	0,
$\mathfrak{D}_{ad+1}$	$-D_a$	$\frac{1}{2}D_a$	0,	$\frac{1}{2}D_a$	0,
D_{ad}	0,	0,	D_a	$-D_a$	$\frac{1}{2}D_a$
D_a	0,	0,	0,	D_b	$-D_b$
$\mathfrak{D}_{ad+1}$	0,	0,	0,	0,	
D_{ad+1}	0,		0,	0,	$-D_a$

We may use generally with the solution of the equations of column corresponding numbers

$$p, \quad q, \quad p+1, \quad p+1, \quad p+1, \quad \ldots,$$

there is for example a single linear relation between the orders of the relations $5, 7, 8, 9, 10$ in each of the orders $5, 8, 9, 10$ on the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \ldots$, on the relations between the orders of $D_a = \frac{1}{16}D_b$. The above considerations were given as follows:

For any value of f except $f = 2$, 3, or 4, the table commences, for f even and f odd respectively, in the following manner:

	$D_{a,b+1}$	$D_{a,b+2}$	$D_{a+1,b+2}$	$D_{a+1,b+3}$	$D_{a+2,b+3}$						
$\mathfrak{D}_{a,b+1}$	$\frac{1}{2}f$	0									
$\mathfrak{D}_{a+1,b+1}$	$-\frac{1}{2}f$	$\frac{1}{2}f$	0								
$\mathfrak{D}_{a+1,b+2}$	0	$-\frac{1}{2}(f-1)$	$\frac{1}{2}(f+1)$	0							
$\mathfrak{D}_{a+2,b+2}$	0	0	$-\frac{1}{2}(f-2)$	$\frac{1}{2}(f+2)$	0						
$\mathfrak{D}_{a+2,b+3}$	0	0	0	$-\frac{1}{2}(f-3)$	$\frac{1}{2}(f+3)$	0					
$\mathfrak{D}_{a+3,b+3}$	0	0	0	0	$-\frac{1}{2}(f-4)$	$\frac{1}{2}(f+4)$					

but beyond this I do not know the law of the series.

Before going further, I remark that if $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, instead of the foregoing values, denote respectively,

$$\begin{aligned}
\mathfrak{A} &= (\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma'), \quad = (\alpha\beta_2' + \alpha\beta')(\gamma\delta_2' - \delta\gamma'),\\
\mathfrak{B} &= (\alpha\delta + \beta\gamma)(\alpha'\delta' + \beta'\gamma'), \quad = (\alpha\beta_2' - \alpha\beta')(\gamma\delta_2' + \delta\gamma'),\\
\mathfrak{C} &= (\alpha\beta + \alpha'\beta')\Lambda, \quad = (\alpha\delta + \beta\delta')(\alpha\gamma + \beta\gamma' - \delta\delta'),
\end{aligned}$$

then equations

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and the same theory applies to the cubic derivatives $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$, that is, in the like relations between the coefficients of the highest powers of the cubic derivatives.

Or, we may apply this directly to the theory of seminvariants. We take then $(x, y)^4$ to denote the binary quartic, and we have for instance

$$(\alpha, b, c, d, e)(x, y)^4 = au^4 + 4bu^3v + 6cu^2v^2 + 4duv^3 + ev^4 = (\alpha, b, c, d, e),$$

that is,

$$\alpha = a, \quad b = -b, \quad c = c, \quad d = -d, \quad e = e,$$

a seminvariant; the like relations between the coefficients of the highest powers of the cubic derivatives, $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$. We have

$$\mathfrak{B} = (\alpha - \beta)(\alpha\gamma - \beta\delta) = -2\beta\alpha + a\gamma + 4a(c - \beta) - 8c\delta + 4ce - 3\beta + 4\delta\gamma,$$

a seminvariant.

The like is the case as in other cases. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates also to the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ, w, $\mathfrak{C}\mathfrak{V}\mathfrak{C}$, $\mathfrak{C}\mathfrak{V}\mathfrak{C}$ are the differences $D_{a,a}\mathfrak{A}$, &c., but the same set of independent seminvariants are also represented by $(\alpha - \beta)^2$, &c..

Suppose that a is odd, and, in particular, $D_{a,d+m}\mathfrak{V}_{a,d}\mathfrak{V}\mathfrak{V}$, &c. that is,

$$D_{a,d+m} = -D_{a,d}\mathfrak{V}\mathfrak{V}, \quad D_{a,d+m} = -D_{a,d}\mathfrak{V}\mathfrak{V},$$

that is,

$$\mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0, \quad \mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0;$$

Hence

$$\mathfrak{A} = 0;$$

or, in like manner if a cubic seminvariant, that is,

$$(b, b, b)(\delta, x, y)^3 = -bv^3 + 4bv^2y + 4(\alpha + by)y^2 + (a + by)y^3,$$

that is,

$$\alpha = -a, \quad \beta = -b, \quad \delta = -d, \quad \gamma = -c, \quad \delta = -e,$$

a seminvariant. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates to the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ, w, the cubic seminvariants $\mathfrak{CVC}$, $\mathfrak{CVC}$ are the differences $D_{a,a}\mathfrak{A}$, &c..

The like remarks would apply for f odd, in the case a a quartic seminvariant, the like remarks apply also, if instead of the seminvariants f odd, we have the seminvariants f even; we have here a similar set of seminvariants $D_{a,d}\mathfrak{V}$, $D_{a,d+m}\mathfrak{V}$, &c., and the seminvariants $D_{a,d}\mathfrak{V}$, &c. are also as above; in many cases the seminvariants are not all of them simply equal, but coefficients in the relations A, B, C, &c., respectively, I call them

$$A, \quad B, \quad C, \quad D, \quad E, \quad F, \quad G, \quad H, \quad I, \quad J, \quad K, \quad L, \quad M,$$

Suppose first α is odd, the in the first

$$D_{a,d}\mathfrak{VV} = 1, \quad D_{a,d+m}\mathfrak{VV} = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$\begin{aligned} A &= 0, \\ B + C &= 0, \\ C + 2D + E &= 0, \\ D + 3E + 3F + G &= 0, \\ &\text{\&c.} \end{aligned}$$

The resultant way of solving them is to express the other functions in terms of B, D, F, H, &c., viz. we thus have

	A	B	C	D	E	F	G	H	I	J	K	L	M
X	0	1	-1	0	1	0	-10	0	$+135$	0	-2073		
D			1	-2	0	3	-15	0	355	0	-5410		
F				1	-3	0	6	-126	0	1693			
H					1	-4	0	10	0	-150	0	-396	
J						1	-5	0	15				
L								1	-6				

read according to the columns

$$\begin{aligned}
A &= 0, \\
B &= B, \\
C &= -B, \\
D &= D, \\
E &= B - 2D, \\
F &= F, \\
G &= -3B + 3D - 3F, \\
&\text{\&c.}
\end{aligned}$$

and, by way of verification of the numbers, observe that the sum of the numbers in a column is 1 and -1 alternately.

The formulae are true for any odd value of α whatever; but they require an explanation. On the right hand value of α, &c., are not all of them advisory, but they are also of the same powers of δ, w. The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon's theorem, viz. the number of independent seminvariants is in fact equal to the number of δ, w, of the non-unitary symmetric functions $\mathfrak{V}\mathfrak{V}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

moreover

$$B + C = w^{-1}(\mathfrak{B} + \mathfrak{C}) = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$\begin{aligned} A &= 0, \\ B + C &= 0, \\ C + 2D + E &= 0, \\ D + 3E + 3F + G &= 0, \\ &\text{\&c.} \end{aligned}$$

The resultant way of solving these is to express the other functions in terms of B, D, F, H, J, L alternately.

The formula are true for any odd value of f whatever; but they require an explanation. The arbitrary terms B, D, &c., are not all of them arbitrary; but they are also seminvariants of the same weight, and therefore connected by linear relations. The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon's theorem, viz. the number of independent seminvariants is in fact equal to the number, δ, w, of the non-unitary symmetric functions $\mathfrak{V}\mathfrak{V}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

We may have, for any odd value of f, writing for shortness 900, 810, &c., instead of D_{900}, D_{810}, &c.,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,\\
450 &= F,\\
360 &= -3B + 3D - 3F,\\
270 &= H,\\
180 &= 12B - 2D + 14F - 4H,\\
090 &= J,
\end{aligned}$$

But we have $900 = 0$, $810 + 720 = 0$, $720 + 270 = 0$, $630 + 350 + 450 = 0$, $540 + 350 = 0$; that is,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,\\
450 &= F,\\
L &= 0,
\end{aligned}$$

all satisfied by $F = -B + 2D$, $H = B - D$, $L = 0$, the proper answer to stop at the equation for 540, the system is then

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,
\end{aligned}$$

equations which may be considered as expressing the several functions 900, 810, 720, 630, 540, 450, &c., in terms of $B = 810$ and $D = 630$. They agree, as they should do, with a foregoing result,

$$\begin{aligned}
900 &= 0,\\
810 &= B,\\
720 &= -B,\\
630 &= D,\\
540 &= B - 2D,
\end{aligned}$$

and it would of course be possible to express the remaining forms $D_{a,b}$, $D_{a,c}$, &c., 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of

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Arthur Cayley

845 (concluded). On the Orthomorphosis of the Circle into the Parabola.

[Printed p. 331]

The function $\log\tan(45^\circ + \frac{1}{2}\arg.)$ is tabulated, Legendre, *Théorie des Fonctions Elliptiques*, t. II. table iv. pp. 256–259, viz. writing as above ϕ' for the argument, we have hereby the value of ϕ, $= \log\tan(\frac{1}{4}\pi + \frac{1}{2}\phi')$, for every value of ϕ' from 0° to 90° at intervals of $30'$ and to 12 decimals, and with fifth differences. Observe that ϕ' is thus given in degrees and minutes as a circular arc, ϕ as a number; it is convenient to have ϕ' and ϕ each as arcs, or each as numbers, the conversion being of course at once made by means of a table of the lengths of circular arcs, and I have calculated the Table which I give at the end of the present paper.

I had previously calculated for the correspondence of $A'O$, $O'B$ and $A'C'B'$ with AO, OB and ACB, the following values, at irregular intervals suited to the construction of a figure.

Circle $\angle A'O'P'$	Parabola Ordinate PM
0°	0
30	·3372
60	·6994
90	$1\cdot1220$
120	$1\cdot6768$
150	$2\cdot5817$
160	$3\cdot1018$
170	$3\cdot9869$
172	$4\cdot2713$
174	$4\cdot6378$
176	$5\cdot1542$
178	$6\cdot0258$
179	$6\cdot9195$
180°	∞

Circle		Parabola	
$O'A'$	$O'B'$	OA	OB
$x' =$	$x' =$	$x =$	$x =$
0	0	0	0
·1	$-\cdot1$	·152	$-\cdot173$
·2	$-\cdot2$	·287	$-\cdot375$
·3	$-\cdot3$	·407	$-\cdot611$
·4	$-\cdot4$	·515	$-\cdot943$
·5	$-\cdot5$	·615	$-1\cdot257$
·6	$-\cdot6$	·704	$-1\cdot724$
·7	$-\cdot7$	·787	$-2\cdot368$
·8	$-\cdot8$	·853	$-3\cdot386$
·9	$-\cdot9$	·935	$-5\cdot368$
$1\cdot0$	$-1\cdot0$	$1\cdot000$	$-\infty$

Write in general

$$\sqrt{(x+iy)} = p + iq, \quad = \frac{a}{\pi}(\psi + i\phi);$$

viz. considering x, y as given, find thence p, q by the equations $x = p^2 - q^2$, $y = 2pq$; and then writing $\frac{1}{2}\pi p = \psi$, $\frac{1}{2}\pi q = \phi$, we have

$$\sqrt{(x'+iy')} \;=\; \tan\tfrac{1}{2}\pi(p+iq) \;=\; \tan\left(\tfrac{1}{2}\psi + \tfrac{1}{2}i\phi\right) \;=\; \frac{P+iQ}{1-iPQ},$$

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where

$$P = \tan\tfrac{1}{2}\pi p, \quad = \tan\tfrac{1}{2}\psi,$$

$$Q = -\tan\tfrac{1}{2}\pi qi, \quad = \tanh\tfrac{1}{2}\pi q, \quad = -\tan\tfrac{1}{2}\phi i, \quad = \tanh\tfrac{1}{2}\phi;$$

whence, if we introduce ϕ' connected with ϕ by the equation $\phi = \log\tan\left(\frac{1}{4}\pi + \frac{1}{2}\phi'\right)$ as before, we have $Q = \tan\frac{1}{2}\phi'$, and the formula is

$$\sqrt{(x'+iy')} = \frac{P+iQ}{1-iPQ}, \quad P = \tan\tfrac{1}{2}\psi, \quad Q = \tan\tfrac{1}{2}\phi',$$

giving the values of x', y'.

It is clear that we have

$$\sqrt{(x'+iy')} = \frac{P-iQ}{1+iPQ},$$

and thence

$$\sqrt{(x'^2+y'^2)} = \frac{P^2+Q^2}{1+P^2Q^2}.$$

Hence, to the circle $x'^2+y'^2=c^2$, corresponds in the parabola the curve

$$P^2+Q^2 = c\left(1+P^2Q^2\right),$$

where $p+iq = \sqrt{(x+iy)}$, $P = \tan\frac{1}{2}\pi p$, $Q = \tanh\frac{1}{2}\pi q$. This is a complicated transcendental equation, and I do not see any convenient way of tracing the curve. The set of curves satisfy the differential equation

$$\frac{P\,dP+Q\,dQ}{P^2+Q^2} = \frac{PQ(Q\,dP+P\,dQ)}{1+P^2Q^2},$$

that is,

$$P\,dP\,(1-Q^2)+Q\,dQ\,(1-P^2)=0,$$

where dP, dQ are given in terms of dp, dq by the equations

$$\frac{dP}{1+P^2} = \tfrac{1}{2}\pi\,dp, \qquad \frac{dQ}{1-Q^2} = \tfrac{1}{2}\pi\,dq.$$

We have $y=2pq$, and thence at the highest points, or summits of the several curves, $q\,dp+p\,dq=0$. Combining these equations, we have

$$dP : dQ = p(1+P^2) : -q(1-Q^2),$$

and thence

$$p\,P\,(1+P^2)(1-Q^2) - q\,Q\,(1-P^2)(1-Q^2) = 0,$$

as an equation to the locus of the summits in question. If p and q are small, then putting for a moment $\frac{1}{2}\pi = m$ for shortness, we have

$$P = mp+\tfrac{1}{3}m^3p^3, \quad Q = mq-\tfrac{1}{3}m^3q^3,$$

and the equation becomes

$$p^2(1+m^2p^2)(1+m^2q^2) - q^2(1-m^2p^2)(1-m^2q^2) = 0,$$

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that is,

$$p^2-q^2+\tfrac{1}{3}m^2(p^4+6p^2q^2+q^4)=0;$$

writing this in the form

$$p^2-q^2+\tfrac{m^2}{3}\left\{(p^2-q^2)^2+8p^2q^2\right\}=0,$$

we find $1 = p^2+q^2$, $\frac{4m^2}{3} = -\frac{x}{x^2-y^2}$, or say $x^2+\frac{3}{2m^2}x-y^2=0$, as the locus of the summits in the neighbourhood of the focus O, viz. the summits lie all of them, as might have been expected, on the left-hand (or negative side) of the focus.

I have constructed for the parabola, to the scale $1 = 1\frac{1}{4}$ inch, as accurately as the data enable, the figure corresponding to the concentric circles and the radii of the circle.

Resuming the equation $\sqrt{(x+iy)} = p+iq$, that is, $x = p^2 - q^2$ and $y = 2pq$, we have the curves ($p =$ const.), $y^2 = 4p^2(p^2 - x)$, and ($q =$ const.), $y^2 = 4q^2(q^2 + x)$, which are two systems of confocal parabolas, cutting each other at right angles; the curves of the former set all of them interior to the given parabola, those of the latter set of course cutting it at right angles.

Corresponding hereto in the circle, writing for a moment

$$\sqrt{(x'+iy')} = p' + iq',$$

we have

$$p' + iq' = \frac{P+iQ}{1-iPQ},$$

whence

$$p' = P(1-q'Q), \quad q' = Q(1+p'P);$$

whence eliminating P and Q successively, we find

$$p'^2 + q'^2 - p'\left(P - \frac{1}{P}\right) - 1 = 0,$$

$$p'^2 + q'^2 - q'\left(Q + \frac{1}{Q}\right) + 1 = 0,$$

say, for shortness, these are $p'^2 + q'^2 - 2mp' + 1 = 0$, and $p'^2 + q'^2 - 2nq' - 1 = 0$. Introducing the polar coordinates r', θ', we have

$$x' + iy' = r'(\cos\theta' + i\sin\theta'),$$

and thence

$$p' = \sqrt{(r')}\cos\tfrac{1}{2}\theta', \quad q' = \sqrt{(r')}\sin\tfrac{1}{2}\theta';$$

the equations thus become

$$r' - 1 - 2m\sqrt{(r')}\cos\tfrac{1}{2}\theta' = 0,$$
$$r' + 1 - 2n\sqrt{(r')}\sin\tfrac{1}{2}\theta' = 0,$$

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which belong to two limaçons having each of them B' for a node and O' for a focus, and which of course cut each other at right angles; see fig. 4, which is a mere diagram.

[Figure 4: a diagram showing two orthogonal limaçons inscribed in a circle, with labelled points B', O', A' on the horizontal diameter.]

In fact, omitting for convenience the accents, but recollecting always that the curves belong to the figure of the circle, the first equation gives

$$(r-1)^2 = 4m^2 r\cos^2\tfrac{1}{2}\theta, \quad = 2m^2 r(1+\cos\theta) = 2m^2(r+x),$$

that is,

$$(r^2 + 1 - 2m^2 x)^2 = 4(m^2+1)^2 r^2.$$

Transforming to the point B' as origin, we must write $x+1$ for x, and then the equation thus becomes

$$\left\{x^2 + y^2 - 2(m^2+1)x + 2(m^2+1)\right\}^2 = 4(m^2+1)^2(x^2+y^2+2x+1),$$

that is,

$$\left\{x^2 + y^2 - 2(m^2+1)x\right\}^2 = 4m^2(m^2+1)(x^2+y^2),$$

or say

$$(x^2+y^2)^2 - 4(m^2+1)(x^2+y^2)x + 4(m^2+1)(x^2 - m^2y^2) = 0,$$

showing that the point B' is a crunode, the tangents there being $x = \pm my$.

Writing in the equation $y = 0$, we have $x = 0$, the crunode, and

$$x = 2\sqrt{(m^2+1)}\left\{\sqrt{(m^2+1)} \pm m\right\};$$

the product of these two values is $= 4(m^2+1)$, that is, $= \left(P+\frac{1}{P}\right)^2$, viz. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(m^2+1)}\left\{\sqrt{(m^2+1)}-m\right\},$$

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the smaller root, is greater than 1. The curve corresponding to a parabola $y^2 = 4p^2(p^2-x)$ is thus a crunodal limaçon, the crunode at B', and the loop lying wholly within the circle. Moreover, the loop includes within itself the centre O' of the circle.

The other curve is in like manner

$$\left\{r^2+1+2n^2x\right\}^2 = 4(n^2+1)r^2,$$

viz. transforming to the origin B', and therefore putting $x-1$ for x, and

$$r^2 = (x-1)^2+y^2,$$

the equation is

$$\left\{x^2+y^2-2(n^2+1)x+2(n^2+1)\right\}^2 = 4(n^2+1)^2(x^2+y^2-2x+1),$$

that is,

$$\left\{x^2+y^2-2(n^2+1)x\right\}^2 = 4n^2(n^2+1)(x^2+y^2),$$

or say

$$(x^2+y^2)^2-4(n^2+1)(x^2+y^2)x-4(n^2+1)(x^2+n^2y^2) = 0,$$

viz. the point B' is an acnode with the imaginary tangents $x = \pm iny$.

Writing in the equation $y = 0$, we have $x = 0$ the acnode, and

$$x = -2\sqrt{(n^2-1)}\left\{\sqrt{(n^2-1)}\pm n\right\},$$

where

$$\sqrt{(n^2-1)} = \tfrac{1}{2}\sqrt{\left(Q-\tfrac{1}{Q}\right)^2}$$

is real and may be taken to be positive; there is thus one root

$$-\sqrt{(n^2-1)}\left\{\sqrt{(n^2-1)}+n\right\},$$

which is negative; the other root, say

$$\sqrt{(n^2-1)}\left\{n-\sqrt{(n^2-1)}\right\},$$

is positive and less than 1; the curve is thus an acnodal limaçon, having B' for the acnode and cutting the axis outside the circle on the negative side of B', and inside the circle between B' and O'.

Taking B' as origin, the equation of the circle is $x^2+y^2-2x = 0$, and we hence find for the intersection of the limaçon with the circle $n^2x = 2(n^2-1)$, that is,

$$x = 2\left(1-\frac{1}{n^2}\right), \quad y = \pm\sqrt{\left(\frac{4}{n^2}-\frac{1}{n^4}\right)};$$

whence also

$$r^2 = 4\left(1-\frac{1}{n^2}\right)^2+\frac{4}{n^2}-\frac{1}{n^4},$$

values which belong to a real intersection; for $q = 0$, we have $Q = 0$, $n = \infty$, and therefore $(x,y) = (2,0)$, viz. the intersection is at A'; for $q = \infty$, we have $Q = 1$, $n = 1$, and therefore $(x,y) = (0,0)$, viz. the intersection is at B', results which are obviously right. Observe that for the other limaçon we have

$$x = 2\left(1+\frac{1}{m^2}\right), \quad y = \pm\sqrt{\left(-\frac{4}{m^2}-\frac{1}{m^4}\right)},$$

viz. there is no real intersection with the circle.

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The Table above referred to.

$$\phi = \log\tan\left(45^\circ + \tfrac{1}{2}\phi'\right).$$

ϕ'	ϕ	ϕ (arc)	ϕ'	ϕ	ϕ (arc)
0°	·0	·0	46°	·8028515	·9062755
1	·0174533	·0174542	47	·8203047	·9316316
2	·0349066	·0349137	48	·8377580	·9574669
3	·0523599	·0523838	49	·8552113	·9838079
4	·0698132	·0698699	50	·8726646	1 · 0106832
5	·0872665	·0873774	51	·8901179	1 · 0381235
6	·1047198	·1049117	52	·9075712	1 · 0661617
7	·1221730	·1224781	53	·9250245	1 · 0948335
8	·1396263	·1400822	54	·9424778	1 · 1241772
9	·1570796	·1577296	55	·9509311	1 · 1542316
10	·1745329	·1754258	56	·9773844	1 · 1850507
11	·1919862	·1931766	57	·9948377	1 · 2166748
12	·2094395	·2109867	58	1 · 0122910	1 · 2491606
13	·2268928	·2288650	59	1 · 0297443	1 · 2825668
14	·2443461	·2468145	60	1 · 0471976	1 · 3169579
15	·2617994	·2648422	61	1 · 0646508	1 · 3524048
16	·2792527	·2829545	62	1 · 0821041	1 · 3889860
17	·2967060	·3011577	63	1 · 0995574	1 · 4267882
18	·3141593	·3194583	64	1 · 1170107	1 · 4659083
19	·3316126	·3378629	65	1 · 1344640	1 · 5064542
20	·3490659	·3563785	66	1 · 1519173	1 · 5485472
21	·3665191	·3750121	67	1 · 1693706	1 · 5923237
22	·3839724	·3937710	68	1 · 1868239	1 · 6379387
23	·4014257	·4126626	69	1 · 2042772	1 · 6855685
24	·4188790	·4316947	70	1 · 2217305	1 · 7354152
25	·4363323	·4508753	71	1 · 2391838	1 · 7877120
26	·4537856	·4702127	72	1 · 2566371	1 · 8427300
27	·4712389	·4897154	73	1 · 2740904	1 · 9007867
28	·4886922	·5093923	74	1 · 2915436	1 · 9622572
29	·5061455	·5292527	75	1 · 3089969	2 · 0275894
30	·5235988	·5493061	76	1 · 3264502	2 · 0973240
31	·5410521	·5695627	77	1 · 3439035	2 · 1721218
32	·5585054	·5900329	78	1 · 3613568	2 · 2528027
33	·5759587	·6107275	79	1 · 3788101	2 · 3404007
34	·5934119	·6316581	80	1 · 3962634	2 · 4362460
35	·6108652	·6528366	81	1 · 4137167	2 · 5420904
36	·6283185	·6742755	82	1 · 4311700	2 · 6603061
37	·6457718	·6959880	83	1 · 4486233	2 · 7942190
38	·6632251	·7179880	84	1 · 4660766	2 · 9487002
39	·6806784	·7402901	85	1 · 4835299	3 · 1313013
40	·6981317	·7629095	86	1 · 5009832	3 · 3546735
41	·7155850	·7858630	87	1 · 5184364	3 · 6425334
42	·7330383	·8091672	88	1 · 5358897	4 · 0481254
43	·7504916	·8328406	89	1 · 5533430	4 · 7413488
44	·7679449	·8569026	90	1 · 5707963	∞
45	·7853982	·8813736			

846. A Verification in Regard to the Linear Transformation of the Theta-Functions.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 77–84.]

[Printed p. 337]

The notation is that of Smith's* "Memoir on the Theta and Omega Functions," differing from Hermite's only in a factor, $\vartheta_{m,n}(\text{Smith}) = i^{mn}\vartheta_{m,n}(\text{Hermite})$; and it is in consequence of this that the factor $i^{\frac{1}{2}\nu-mn}$ occurs in the expression presently given for $\mathcal{C}$. Hermite's Memoir "Sur quelques formules relatives à la transformation des fonctions elliptiques" appeared in Liouville, t. iii. (1858), pp. 27–36.

Writing

$$\omega = \frac{c+d\Omega}{a+b\Omega},$$

where $ad-bc=1$, the formula of transformation is

$$\vartheta_{m,n}\left\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\right\} = \mathcal{C}\,\exp\left\{-i\pi b(a+b\Omega)\frac{x^2}{h^2}\right\}\vartheta_{m',n'}\left(\frac{\pi x}{h},\ \omega\right),$$

where

$$m' = a\mu + b\nu + ab,$$

$$n' = c\mu + d\nu + cd.$$

We have, according as b is positive or negative,

$$C = \frac{\delta H}{\sqrt{\{-i(a+b\Omega)\}}},\quad \text{or}\quad C = \frac{\delta H}{\sqrt{\{i(a+b\Omega)\}}},$$

where for each case the square root is to be taken in such wise that the real part is positive; δ, H are eighth roots of unity; viz. in each case

$$\delta = \exp\left\{-\tfrac{1}{4}i\pi\left(ac\mu^2+2bc\mu\nu+bd\nu^2-2abc\mu-2abd\nu+ab^2c\right)\right\} i^{\frac{1}{2}\nu-mn},$$

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and, according as b is positive or negative,

$$H = \left(\frac{a}{b}\right) i^{-\frac{1}{2}a},\quad a \text{ odd},\qquad H = \left(\frac{-b}{a}\right)\beta^n,\quad a \text{ odd},$$

$$H = \left(\frac{a}{b}\right) i^{\frac{1}{2}a+\frac{1}{2}(a-b)},\quad b \text{ odd},\qquad H = \left(\frac{-a}{b}\right)\beta^{n-\frac{1}{2}a+\frac{1}{2}(a+b)},\quad b \text{ odd},$$

where $\left(\frac{a}{b}\right)$, &c., are the Legendrian symbols as generalised by Jacobi; if a and b are each of them odd, the formulae for the cases a odd and b odd respectively are equivalent to each other, and either may be used indifferently.

To fix the ideas, I assume throughout that b is positive and odd; the proper formulae thus are

$$C = \frac{\delta H}{\sqrt{\{-i(a+b\Omega)\}}},$$

δ as above, and

$$H = \left(\frac{a}{b}\right) i^{\frac{1}{2}a+\frac{1}{2}(a-b)}.$$

I will also, for greater convenience, assume that c is odd; ad is consequently even, viz. the numbers a and d are each of them even, or else one is odd and the other even.

[0][* H. J. S. Smith, *Collected Mathematical Papers*, vol. II., pp. 415–621.]

The equation $\omega = \dfrac{c+d\Omega}{a+b\Omega}$ gives $\Omega = \dfrac{c-a\omega}{-d+b\omega}$, and thence

$$a+b\Omega = \frac{-1}{-d+b\omega}.$$

Comparing the expressions for ω, Ω, it appears that we may interchange ω and Ω, writing also $-d$, b, $-c$, for a, b, c, d; and changing the other letters, we may for

$$a,\ b,\ c,\ d,\ \Omega,\ \mu,\ \nu,\ \omega,\ m',\ n',\ \text{write}$$

$$-d,\ b,\ -c,\ a,\ \omega,\ m',\ n',\ \Omega,\ m,\ n,$$

where

$$m' = -dm + bn - bd,$$

$$n' = -cm + an - ac.$$

The formula thus becomes

$$\vartheta_{m,n}\Big\{-\frac{\pi x}{h},\ \omega\Big\} = C' \exp\Big\{-i\pi(-d+b\omega)\frac{x^2}{h^2}\Big\}\,\vartheta_{m',n'}\Big(\frac{\pi x}{h},\ \Omega\Big),$$

or, for x writing $(a+b\Omega)x$, this is

$$\vartheta_{m,n}\Big\{-\frac{\pi x}{h}(a+b\Omega),\ \omega\Big\} = C' \exp\Big\{-i\pi(a+b\Omega)\frac{x^2}{h^2}\Big\}\,\vartheta_{m',n'}\Big((a+b\Omega)\frac{\pi x}{h},\ \Omega\Big),$$

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or, observing that the left-hand side is $= (-)^{mn}\vartheta_{m,n}\Big\{(a+b\Omega)\dfrac{\pi x}{h},\ \omega\Big\}$, we have

$$\vartheta_{m,n}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \omega\Big\} = \frac{(-)^{mn}}{C'} \exp\Big\{-i\pi(a+b\Omega)\frac{x^2}{h^2}\Big\}\,\vartheta_{m',n'}\Big(\frac{\pi x}{h},\ \Omega\Big),$$

an equation which is of the same form as the original equation of transformation, and is to be identified with it.

We have

$$\begin{aligned} m' &= -dm + bn - bd = -d(a\mu + b\nu + ab) + b(c\mu + d\nu + cd) - bd \\ &= \mu + bd(-a + c - 1), \\ n' &= -cm + an - ac = -c(a\mu + b\nu + ab) + a(c\mu + d\nu + cd) - ac \\ &= \nu + ac(-b + d - 1); \end{aligned}$$

values which may be written

$$m' = 2P - \mu, \quad \text{where } P = \tfrac{1}{2}bd(-a+c-1),$$

$$n' = 2Q - \nu, \quad \text{where } Q = \tfrac{1}{2}ac(-b+d-1),$$

P and Q being integers. In fact, P is an integer if b or d is even, and if they are each of them odd, then from the equation $ad - bc = 1$, a and c will be one of them odd, the other even, and $-a+c-1$ will be even; and similarly for Q.

The left-hand side is

$$\vartheta_{m',n'}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\},$$

which is

$$= (-)^{Pv}\vartheta_{-\mu,-\nu}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\} = (-)^{Pv}\vartheta_{\mu,\nu}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\},$$

and the equation finally is

$$\vartheta_{\mu,\nu}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\} = \frac{1}{C'}(-)^{mn+Pv} \exp\Big\{-i\pi(a+b\Omega)\frac{x^2}{h^2}\Big\}\,\vartheta_{m,n}\Big(\frac{\pi x}{h},\ \omega\Big).$$

Comparing this with the original equation

$$\vartheta_{\mu,\nu}\Big\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\Big\} = C\ \exp\Big\{-i\pi(a+b\Omega)\frac{x^2}{h^2}\Big\}\,\vartheta_{m,n}\Big(\frac{\pi x}{h},\ \omega\Big),$$

the two equations will be identical if only

$$CC' = (-)^{-mn-Pv}.$$

We have

$$C = \frac{\delta H}{\sqrt{\{-i(a+b\Omega)\}}},\qquad C' = \frac{\delta' H'}{\sqrt{\{-i(-d+b\omega)\}}},$$

where the square roots are taken in such wise that the real part is positive; hence

$$CC' = \frac{\delta\delta' HH'}{\sqrt{(+1)}},$$

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where $\sqrt{(+1)}$ means

$$\sqrt{\{-i(a+b\Omega)\}}\cdot\sqrt{\{-i(-d+b\omega)\}},$$

the last-mentioned two square roots being as just explained; and we have moreover

$$\delta = \exp\big\{-\tfrac{1}{4}i\pi\big(ac\mu^2+2bc\mu\nu+bd\nu^2-2abc\mu-2abd\nu+ab^2c\big)\big\}\, i^{\frac{1}{2}\nu-mn},$$

$$\delta' = \exp\big\{-\tfrac{1}{4}i\pi\big(-dcm^2+2bcmn-abn^2-2dbcm+2abdn-db^2c\big)\big\}\, i^{\frac{1}{2}n'-m'n'},$$

viz. the value of δ' is obtained from that of δ by the change a, b, c, d, μ, ν, m, n into $-d$, b, $-c$, a, m, n, m', n'.

Representing these by

$$\delta = \exp\big\{-\tfrac{1}{4}(i\pi)\,\Delta\big\}\, i^{\frac{1}{2}\nu-mn},\qquad \delta' = \exp\big\{-\tfrac{1}{4}(i\pi)\,\Delta'\big\}\, i^{\frac{1}{2}n'-m'n'},$$

we have

$$\delta\delta' = \exp\big\{-\tfrac{1}{4}i\pi(\Delta+\Delta')\big\}\, i^{\frac{1}{2}(\nu+n')-mn-m'n'}.$$

But

$$m'n'-mn = (2P-\mu)(2Q-\nu)-\mu\nu = 4PQ-2P\nu-2Q\mu+\mu\nu-\mu\nu,$$

that is,

$$mn-m'n' = -4PQ+2P\nu+2Q\mu,$$

or, omitting the term divisible by 4,

$$i^{2P\nu+2Q\mu} = (-)^{P\nu+Q\mu},$$

$$i^{\frac{1}{2}(\nu+n')-mn-m'n'} = i^{2P\nu+2Q\mu}\cdot(-)^{P\nu+Q\mu}.$$

To calculate $\Delta+\Delta'$, we have

$$dm-bn = \mu+bd(-a+c),$$

$$-cm+an = \nu+ac(-b+d),$$

and thence

$$\begin{aligned}-cdm^2+(ad+bc)mn-abn^2 &= \mu\nu+\mu ac(-b+d)+\nu bd(a-c)+abcd(d-b)(a-c)\\ &\quad+ac\mu^2-bd\nu^2-(ad+bc)\mu\nu-\mu ac(b-d)-\nu bd(a-c)\\ (-ad+bc)\,mn &= -abcd,\end{aligned}$$

consequently

$$-cdm^2+2bcmn-abn^2 = -ac\mu^2-2bc\mu\nu-bd\nu^2-2abc\mu-2bcd\nu+abcd(2bc-ab-cd);$$

also

$$\begin{aligned}-2bcdm &= -2abcd\mu - 2b^2cd\nu - 2ab^2cd,\\ 2abdn &= -2abcd\mu + 2abd^2\nu + 2abcd^2,\\ -db^2c &= -db^2c;\end{aligned}$$

whence, adding, we obtain

$$\begin{aligned}\Delta' = -ac\mu^2 - 2bc\mu\nu - bd\nu^2 - 2abc\mu + (-2bcd - 2b^2cd - 2abd^2)\nu\\ + abcd(2bc - ab - cd - 2b + 2d) - db^2c,\end{aligned}$$

and, adding to this the expression of Δ, we find

$$\Delta + \Delta' = (2abd - 2bcd - 2b^2cd + 2abd^2)\nu - ab^2c - db^2c + abcd(2bc - ab - cd - 2b + 2d).$$

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The coefficient of ν is $= 2bd(a - c - bc + ad) = 2bd(a + c - 1)$, which is $= -4P\nu$. Hence writing

$$\Theta = \tfrac{1}{2}abcd(2bc - ab - cd - 2b + 2d) + \tfrac{1}{2}b^2c(a - d),$$

where observe that 4Θ, but not in every case Θ itself, is an integer, we have

$$\Delta + \Delta' = -4P\nu + 4\Theta,$$

and consequently

$$\delta\delta' = (-)^{-P\nu+\Theta+Pv+Q\mu} \cdot i^{2P\nu+2Q\mu} = (-)^{\Theta+P\nu+Q\mu} \cdot i^{2P\nu+2Q\mu}.$$

Observe that $(-)$ denotes, and it might properly have been written, $\exp i\pi\Theta$. The foregoing equation

$$\frac{\delta\delta' HH'}{\sqrt{(+1)}} = (-)^{-mn-Pv}$$

becomes thus

$$\frac{HH'}{\sqrt{(+1)}} = (-)^{-mn-Pv} \cdot \frac{1}{\delta\delta'},$$

that is,

$$\frac{HH'}{\sqrt{(+1)}} = (-)^{-mn-Pv-\Theta-P\nu-Q\mu} \cdot i^{-2P\nu-2Q\mu},$$

where the even term $2Q\mu$ may be omitted. We have moreover

$$\begin{aligned}mn &= ac\mu^2 + (ad + bc)\mu\nu + bd\nu^2 + ac(b + d)\mu + bd(a - c)\nu + abcd,\\ \mu\nu &= (-ad + bc)\mu\nu,\end{aligned}$$

and thence

$$mn + \mu\nu = ac(\mu^2 + \mu) + 2bc\mu\nu + bd(\nu^2 + \nu) + ac(b + d - 1)\mu + bd(a - c - 1)\nu + abcd,$$

where each term is even; hence $mn - \mu\nu$ is even, and we have simply

$$\frac{HH'}{\sqrt{(+1)}} = (-)^{-\mu\nu-B},$$

where

$$B = \tfrac{1}{2}abcd(ab + cd - 2bc + 2b - 2d) - \tfrac{1}{2}b^2c(a - d).$$

If I write $M = \frac{1}{2}b(a - d)$, then

$$\Theta - M = \tfrac{1}{2}abcd(ab + cd - 2bc + 2b - 2d) - \tfrac{1}{2}(bc + 1)b(a - d),$$

where the second term is $= \frac{1}{2}abd(a-d)$, and we have therefore

$$\Theta - M = \tfrac{1}{2}abd(abc + c^2d - a + 2bc^2 - 2bc - 2cd - a + d).$$

I assume, as above, that b and c are each of them odd; therefore ad is even. I suppose, first, that ad divides by 4, then $\frac{1}{2}abd$ is an integer, and in the expression of $\Theta - M$, omitting even numbers, we have

$$\tfrac{1}{2}abd(abc + c^2d - a + d),$$

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which, putting therein $bc = ad - 1$, becomes

$$\begin{aligned} &= \tfrac{1}{2}abd(a^2d + c^2d - 2a + d),\\ &= \tfrac{1}{2}abd\{a^2d + (c^2 - c)d + (c+1)d - 2a\}, \end{aligned}$$

where inside the { } each term is even; hence $\Theta - M$ is even.

Next, if ad is even but not divisible by 4, then $bc = ad - 1$, which is $\equiv 1 \pmod 4$; thus b and c are

$$= 4\sigma + 1 \quad \text{and} \quad 4\tau + 1,$$

or else

$$= 4\sigma - 1 \quad \text{and} \quad 4\tau - 1,$$

and, moreover, $bc = 4\theta + 1$, and $c^2 = 4\phi + 1$; hence

$$\Theta - M = \tfrac{1}{2}abd\{a(4\theta+1) + (4\phi+1)d - 2a + 2(4\theta+1)d - 2a + d\},$$

or, omitting even numbers, that is, inside the { } numbers which contain the factor 4, this is

$$= \tfrac{1}{2}abd(a + d - 2b + 2 - a + d),$$

$$= \tfrac{1}{2}abd(d - b + 1 - cd),$$

or, since each term within the { } is even, we have in this case also $\Theta - M$ even.

And this being so, the foregoing equation for HH' becomes

$$\frac{HH'}{\sqrt{(+1)}} = (-)^{-M}, \quad \text{or say } = i^{2M}.$$

The values of H, H', refer each of them to a positive odd value of b, and they thus are

$$H = \left(\frac{a}{b}\right) i^{\frac{1}{2}a + \frac{1}{2}(a-b)}, \quad H' = \left(\frac{-d}{b}\right) i^{\frac{1}{2}d + \frac{1}{2}(-d-b)};$$

hence

$$HH' = \left(\frac{a}{b}\right)\left(\frac{-d}{b}\right) i^{\frac{1}{2}a + \frac{1}{2}d + (a-b)} = \left(\frac{a}{b}\right)\left(\frac{-d}{b}\right) i^{\frac{1}{2}(a+d) - 2 + 2(a-b)},$$

or, since

$$\left(\frac{-d}{b}\right) = (-)^{\frac{1}{2}(b-1)}\left(\frac{d}{b}\right) = i^{2\cdot\frac{1}{2}(b-1)}\left(\frac{d}{b}\right),$$

and $2(b-1)$ divides by 4, this is

$$HH' = \left(\frac{a}{b}\right)\left(\frac{d}{b}\right) i^{\frac{1}{2}(a+d)}.$$

Also $\left(\frac{a}{b}\right)\left(\frac{d}{b}\right) = \left(\frac{ad}{b}\right)$, but from the equation $ad - bc = 1$, or $\dfrac{ad}{b} = c + \dfrac{1}{b}$, we have $\left(\dfrac{ad}{b}\right) = \left(\dfrac{1}{b}\right) = 1$; whence

$$HH' = i^{\frac{1}{2}(a+d)}.$$

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We have $\omega = x + iy$, where y is positive; hence

$$-d + b\omega = -d + bx + iby = \alpha + iby, \quad \text{if } \alpha = -d + bx;$$

hence

$$\begin{aligned} a + b\Omega &= \frac{-1}{\alpha + iby} = \frac{-\alpha + iby}{\alpha^2 + \beta^2 y^2}, \quad \beta = b, \\ -i(a + b\Omega) &= \frac{by - i\alpha}{\alpha^2 + \beta^2 y^2} = R(\cos\theta + i\sin\theta), \\ -i(-d + b\Omega) &= by - i\alpha = R(\cos\theta + i\sin\theta), \\ -i(a + b\Omega) &= \frac{by - i\alpha}{\alpha^2 + \beta^2 y^2} = \frac{R(\cos\theta - i\sin\theta)}{\alpha^2 + \beta^2 y^2}, \end{aligned}$$

where R is positive; and $\cos\theta$ is positive since y is positive, and thus θ lies between $-\frac{1}{2}\pi$ and $\frac{1}{2}\pi$. Hence

$$\sqrt{\{-i(-d + b\Omega)\}} = \sqrt{R}(\cos\tfrac{1}{2}\theta + i\sin\tfrac{1}{2}\theta),$$

$\cos\frac{1}{2}\theta$ being positive,

$$\sqrt{\{-i(a + b\Omega)\}} = \frac{\sqrt{R}(\cos\frac{1}{2}\theta - i\sin\frac{1}{2}\theta)}{\sqrt{(\alpha^2 + \beta^2 y^2)}},$$

and thus

$$\sqrt{\{-i(a + b\Omega)\}} \cdot \sqrt{\{-i(-d + b\Omega)\}} = \frac{R}{\sqrt{(\alpha^2 + \beta^2 y^2)}} = +1,$$

that is, $\sqrt{(+1)} = +1$, and we thus have, as we should do,

$$\frac{HH'}{\sqrt{(+1)}} = i^{\frac{1}{2}(a+d)},$$

the equation which was to be verified.

847. On the Theory of Seminvariants.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 92–107.]

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In my Memoire "sur les Hyperdéterminants," Crelle, t. xxx. (1846), pp. 1–37, [13, 14, 16], I gave an investigation for the number of and relations between the quartic invariants of a given binary quantic: the very same formulae apply to the cubic seminvariants of a given binary quantic, or what is the same thing to the cubic seminvariants of a given weight, and I propose at present (considering the formulae in this point of view) to further develope the theory in regard to the solution of the systems of linear equations obtained in the memoir in question. But I first reproduce the investigation as it stands: I recall that the notation is the ordinary hyperdeterminant notation: for instance,

$$\begin{aligned}\overline{12}^2\, UV &= (\partial_{x_1}\partial_{y_2} - \partial_{x_2}\partial_{y_1})^2\, UV\\ &= (\partial_{x_1}^2\partial_{y_2}^2 - 2\partial_{x_1}\partial_{y_1}\partial_{x_2}\partial_{y_2} + \partial_{x_2}^2\partial_{y_1}^2)\, UV\\ &= a\cdot c - 2b\cdot b + c\cdot a\\ &= -2(ac-b^2),\end{aligned}$$

and that, when the variables do not disappear, each set (x_1, y_1), (x_2, y_2), ... is ultimately replaced by (x, y), so that these are the variables of any covariant.

The investigation* is as follows:

"...We pass to the derivatives of the fourth degree, considering the forms in which all the differential coefficients are of the same order. It is easy to see that we may write

$$\begin{aligned}\square\, UVWX &= UVWX\, \overline{12}^{\alpha}\, \overline{13}^{\beta}\, \overline{14}^{\gamma}\, \overline{23}^{\beta'}\, \overline{24}^{\gamma'}\, \overline{34}^{\alpha'},\\ &= D_{a,b,c}\, UVWX, \quad \text{or simply } D_{a,b,c}.\end{aligned}$$

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Writing for shortness

$$\overline{12}\cdot\overline{34} = \mathfrak{A}, \quad \overline{13}\cdot\overline{42} = \mathfrak{B}, \quad \overline{14}\cdot\overline{23} = \mathfrak{C},$$

we have

$$D_{a,b,c} = \mathfrak{A}^a\mathfrak{B}^b\mathfrak{C}^c\, UVWX.$$

Suppose $U = V = W = X$, and consider the derivatives which correspond to the same value of α, β, γ: we have to find how many of these derivatives are independent, and to express the others in terms of these. Since the functions are equal after the differentiations, we may before the differentiations interchange in any manner the symbolic numbers 1, 2, 3, 4: this gives

$$D_{a,b,c} = D_{b,a,c} = D_{a,c,b} = (-)^f\, D_{a,b,c} = (-)^f\, D_{b,a,c}.$$

But we have identically

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0.$$

Multiplying by $\mathfrak{A}^a\mathfrak{B}^b\mathfrak{C}^c$, and applying it to the product $UVWX$, we have

$$D_{a+1,b,c} + D_{a,b+1,c} + D_{a,b,c+1} = 0,$$

which, putting $a+b+c = f-1$, gives a system of equations between the derivatives $D_{\alpha\beta\gamma}$ for which $\alpha+\beta+\gamma = f$. Reducing these by the conditions just obtained, suppose that Θf denotes the number of partitions of f into three parts, zero included, and the permutations of the parts being disregarded, we have a number Θf of derivatives $D_{\alpha\beta\gamma}$, and a number $\Theta(f-1)$ of linear relations between these derivatives. There remains therefore a number $\Theta f - \Theta(f-1)$ of independent derivatives: but when f is an even number, we have included among these the functions $D_{\frac{1}{2}f,\frac{1}{2}f,0}$, that is, $\overline{12}^{\frac{1}{2}f}\cdot\overline{34}^{\frac{1}{2}f}\, UVWX$, which evidently reduces itself to

$$\overline{12}^{\frac{1}{2}f}UV\cdot\overline{34}^{\frac{1}{2}f}WX, \quad \text{or } B_{\frac{1}{2}f}(U,V)\cdot B_{\frac{1}{2}f}(W,X),$$

[0][* This Collection, vol. I., p. 104: see also, vol. I., p. 117.]

that is to say, to the square of $B_{\frac{1}{2}f}(U,V)$. We must therefore diminish by unity this number $\Theta f - \Theta(f-1)$, when f is even. Let $E\left(\dfrac{f}{6}\right)$ denote the integer part of the fraction $\dfrac{f}{6}$: it can be shown that the required number is equal to

$$E\left(\frac{f}{6}\right),\quad E\left(\frac{f+3}{6}\right),$$

according as f is even or odd. Giving to f the six forms

$$6g,\ 6g+1,\ 6g+2,\ 6g+3,\ 6g+4,\ 6g+5,$$

we obtain for the independent derivatives of the fourth degree the corresponding numbers

$$g,\ g,\ g,\ g+1,\ g,\ g+1;$$

there is for example a single derivative for each of the orders 3, 5, 6, 7, 8, 10, two for each of the orders 9, 11, 12, 13, 14, 16, &c.

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We may take for independent derivatives, when f is even, the terms of the series $D_{f,0,0}$, $D_{f-2,2,0}$, $D_{f-4,4,0}$, $\ldots$, and when f is odd, those of the series $D_{f-1,1,0}$, $D_{f-3,3,0}$, $\ldots$, continuing each series until the last term in which the first suffix exceeds or is equal to the second suffix. We have in each case the right number of terms; for example, in the case $f=9$, we take for independent derivatives the two functions D_{810} and D_{630}, and we form the equations

$$\begin{aligned}
&D_{900}+D_{810}+D_{801}=0, && D_{621}+D_{531}+D_{522}=0,\\
&D_{810}+D_{720}+D_{711}=0, && D_{531}+D_{441}+D_{432}=0,\\
&D_{801}+D_{711}+D_{702}=0, && D_{522}+D_{432}+D_{423}=0,\\
&D_{720}+D_{630}+D_{621}=0, && D_{441}+D_{351}+D_{342}=0,\\
&D_{711}+D_{621}+D_{612}=0, && D_{432}+D_{342}+D_{333}=0,\\
&D_{702}+D_{612}+D_{603}=0,
\end{aligned}$$

which are to be reduced by means of the formulas

$$D_{900}=-D_{900}=0,\quad D_{810}=-D_{810},\quad \&c.$$

We will presently give the solution of these equations: but beginning with the second order and going successively to the ninth order, we form easily the following table:

$$\begin{array}{lll}
D_{200}=B^2, & D_{500}=0, & D_{700}=0,\\
D_{110}=-\tfrac12 B^2, & D_{410}=B^2, & D_{610}=B^2,\\
 & D_{320}=-\tfrac12 B^2, & D_{520}=-\tfrac12 B^2,\\
 & D_{311}=0, & D_{511}=0,\\
D_{300}=0, & D_{221}=0, & D_{430}=D_{610}-\tfrac12 D_{540},\\
D_{210}=0, & & D_{421}=-\tfrac12 D_{610}-\tfrac12 D_{540},\\
D_{111}=0, & & D_{331}=-\tfrac12 D_{540},\\
 & & D_{322}=-\tfrac12 D_{610}-\tfrac14 D_{540},\\
D_{400}=B^2, & D_{600}=B^2, & \\
D_{310}=-\tfrac12 B^2, & D_{510}=-\tfrac12 B^2, & \\
D_{220}=\tfrac13 B^2, & D_{420}=\tfrac13 B^2, & \\
D_{211}=0, & D_{411}=-\tfrac16 B^2, & \\
 & D_{330}=-\tfrac13 B^2, & \\
 & D_{321}=\tfrac16 B^2, & \\
 & D_{222}=-\tfrac13 B^2, &
\end{array}$$

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For any value of f except $f=2$, 3, or 4, the table commences, for f even and f odd respectively, in the following manner:

$$D_{f,0,0}=B^2_{\frac12 f},\qquad D_{f,0,0}=0,$$

$$D_{f-1,1,0} = -\tfrac{1}{2}B^2_{\frac{1}{2}f}, \qquad D_{f-1,1,0} = 0,$$
$$D_{f-2,2,0} + D_{f-2,1,1} = D_{f-2,2,0} = -D_{f-1,1,0},$$
$$D_{f-3,3,0} = D_{f-2,1,1} = 0,$$

but beyond this I do not know the law of the series."

Before going further, I remark that if $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, instead of the foregoing values, denote respectively

$$\begin{aligned}\mathfrak{A} &= (\alpha\partial_x + \beta\partial_y)\,\overline{23},\\ \mathfrak{B} &= (\alpha\partial_{x_2} + \beta\partial_{y_2})\,\overline{31},\\ \mathfrak{C} &= (\alpha\partial_{x_3} + \beta\partial_{y_3})\,\overline{12},\end{aligned}$$

then identically

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and the same theory applies to the cubic derivatives $\mathfrak{A}^a\mathfrak{B}^b\mathfrak{C}^c\,UVW$, that is, to the cubic covariants, or, attending only to the coefficients of the highest powers of x, to the cubic seminvariants.

Or, we may use the Clebschian notation, for instance

$$(a, b, c\,|\,x, y)^2 = (a_0x + a_1y)^2 = (b_0x + b_1y)^2 = \&\text{c.},$$

that is,

$$a_0^2 = b_0^2 = \cdots = a,$$
$$a_0a_1 = b_0b_1 = \cdots = b,$$
$$a_1^2 = b_1^2 = \cdots = c,$$

viz. in the language of Prof. Sylvester, a_0, a_1, b_0, b_1, $\ldots$ are here *umbrae*. The invariant is

$$(a_0b_1 - a_1b_0)^2 = a_0^2b_1^2 - 2a_0a_1b_0b_1 + a_1^2b_0^2,$$
$$= ac - 2b\cdot b + ac,$$
$$= -2(ac - b^2),$$

and so in other cases. To apply this directly to the theory of seminvariants, it is more convenient to write

$$(1, b, c\,|\,x, y)^2 = (x + \alpha y)^2 = (x + \beta y)^2 = \&\text{c.},$$

that is,

$$\alpha = \beta = \cdots = b,$$
$$\alpha^2 = \beta^2 = \cdots = c,$$

where α, β, $\ldots$ are *umbrae*.

The invariant is

$$(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$$
$$= c - 2b\cdot b + c$$
$$= 2(c - b^2);$$

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or, to take the case of a cubic function

$$(1, b, c, d\,|\,x, y)^3 = (x + \alpha y)^3 = (x + \beta y)^3 = (x + \gamma y)^3,$$

that is,

$$\alpha = \beta = \gamma = b,$$
$$\alpha^2 = \beta^2 = \gamma^2 = c,$$
$$\alpha^3 = \beta^3 = \gamma^3 = d,$$

a seminvariant is

$$\begin{aligned}(\alpha-\beta)^2(\alpha-\gamma) &= \alpha^3-2\alpha^2\beta+\alpha\beta^2-\alpha^2\gamma+2\alpha\beta\gamma-\beta^2\gamma\\ &= d-2bc+bc-bc+2b^3-bc\\ &= d-3bc+2b^3,\end{aligned}$$

and so in other cases. Writing here

$$\mathfrak{A}=\beta-\gamma,\quad \mathfrak{B}=\gamma-\alpha,\quad \mathfrak{C}=\alpha-\beta,$$

the general form of a cubic seminvariant of the weight ω is now $\mathfrak{A}^p\mathfrak{B}^q\mathfrak{C}^r$, or say $D_{p,q,r}$, where $p+q+r=\omega$ (the new letters p, q, r being used for the exponents, since α, β, γ are now employed in a different sense; and since f will be required as a coefficient, I have also written ω instead of f for the weight). We have, as before,

$$\mathfrak{A}+\mathfrak{B}+\mathfrak{C}=0,$$

and we again see that the theory as to the number of and relations between the cubic seminvariants is identical with that of the quartic invariants. Observe also that $D_{\omega,0,0}=0$, for ω odd, while $D_{\omega,0,0}=\mathfrak{A}^\omega$, for ω even, is a quadric seminvariant, which is of course not to be reckoned among the proper cubic seminvariants; this exactly corresponds to the $B^2_{\frac{1}{2}f}$ which is not a proper quartic invariant.

The choice of the functions $D_{f,0,0}$, $D_{f-2,2,0}$, …, or $D_{f-1,1,0}$, $D_{f-3,3,0}$, …, for expressing in terms of them the other functions has its advantages, but also its disadvantages; in what follows, I in effect replace them by D_f, $D_{f-2,2,0}$, $D_{f-4,4,0}$, and $D_{f-1,1,0}$, $D_{f-3,3,0}$, … respectively. And instead of considering the whole series of the functions $D_{p,q,r}$ for a given weight ω, I consider only those of the form $D_{p,q,0}$; these form a single series $D_{\omega,0,0}$, $D_{\omega-1,1,0}$, $D_{\omega-2,2,0}$, …; and for shortness, I call them A, B, C, D, &c. respectively, viz. I write

$$A,\ B,\ C,\ D,\ \cdots=(\mathfrak{A}^\omega,\ \mathfrak{A}^{\omega-1}\mathfrak{B},\ \mathfrak{A}^{\omega-2}\mathfrak{B}^2,\ \mathfrak{A}^{\omega-3}\mathfrak{B}^3,\ \ldots).$$

Suppose first that ω is odd; we have

$$D_{\omega-q-r,q,r}=(-)^r\,D_{\omega-q-r,r,q},$$

and, in particular,

$$D_{\omega-2q,q,q}=-D_{\omega-2q,q,q},$$

that is,

$$D_{\omega-2q,q,q}=0,$$

that is,

$$\mathfrak{A}^{\omega-2q}\,\mathfrak{B}^q\mathfrak{C}^q=0;$$

Hence

$$A=0;$$

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moreover

$$B+C=\mathfrak{A}^{\omega-1}\mathfrak{B}+\mathfrak{A}^{\omega-2}\mathfrak{B}^2=\mathfrak{A}^{\omega-2}\mathfrak{B}(\mathfrak{A}+\mathfrak{B})=-\mathfrak{A}^{\omega-2}\mathfrak{B}\mathfrak{C}=0;$$

and similarly $C+2D+E=0$, &c.; that is, we have

$$\begin{aligned}A&=0,\\ B+C&=0,\\ C+2D+E&=0,\\ D+3E+3F+G&=0,\\ &\text{&c.}\end{aligned}$$

The readiest way of solving these is to express the other functions in terms of B, D, F, H, &c.; viz. we thus have

	A	B	C	D	E	F	G	H	I	J	K	L
B	0	1	−1	0	1	0	−3	0	17	0	−155	0
D				1	−2	0	5	0	−28	0	255	0
F						1	−3	0	14	0	−126	0
H								1	−4	0	30	0
J										1	−5	0
L												1

read according to the columns

$$\begin{aligned}
&A = 0,\\
&B = B,\\
&C = -B,\\
&D = D,\\
&E = B - 2D,\\
&F = F,\\
&G = -3B + 5D - 3F,\\
&\quad \&c.
\end{aligned}$$

and, by way of verification of the numbers, observe that the sum of the numbers in a column is 1 and −1 alternately.

The formulae are true for any odd value of ω whatever; but they require an explanation, viz. for any finite value of ω, the terms B, D, F, ... are not all of them arbitrary. For any given value of ω the number of arbitrary terms is in fact given by what precedes, and it is also known by Captain MacMahon's theorem, viz. the number of cubic seminvariants is = that of the non-unitary symmetric functions $3^a 2^b$ of the proper weight $3a + 2b = \omega$; thus for $\omega = 9$, the forms are 3222 and 333, so that there should be only two arbitrary terms.

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We have here, writing for shortness 900, 810, &c., instead of D_{900}, D_{810}, &c. respectively,

$$\begin{aligned}
&900 = 0,\\
&810 = B,\\
&720 = -B,\\
&630 = D,\\
&540 = B - 2D,\\
&450 = F,\\
&360 = -3B + 5D - 3F,\\
&270 = H,\\
&180 = 17B - 28D + 14F - 4H,\\
&090 = L.
\end{aligned}$$

But we have $900 = 0$, $810 + 180 = 0$, $720 + 270 = 0$, $630 + 360 = 0$, $540 + 450 = 0$; that is,

$$\begin{aligned}
L &= 0,\\
18B - 28D + 14F - 4H &= 0,\\
-B + H &= 0,\\
-3B + 6D - 3F &= 0,\\
B - 2D + F &= 0,
\end{aligned}$$

all satisfied by $F = B + 2D$, $H = B$, $L = 0$. The proper course is to stop at the equation for 540, viz. the system is

$$\begin{aligned} 900 &= 0, \\ 810 &= B, \\ 720 &= -B, \\ 630 &= D, \\ 540 &= B - 2D, \end{aligned}$$

equations which may be considered as expressing the several functions 900, 810, 720, 630, 540 in terms of $B = 810$ and $D = 630$. They agree, as they should do, with a foregoing result,

$$\begin{aligned} 900 &= 0, \\ 810 &= 810, \\ 720 &= -810, \\ 630 &= \tfrac{1}{2} \cdot 810 - \tfrac{1}{2} \cdot 540, \\ 540 &= 540, \end{aligned}$$

and it would of course be possible to express the remaining forms 711, 621, 522, 441, 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of

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the present process is that, instead of the whole series of forms 900, 810, 720, 711, 630, 621, 540, 531, 522, 441, 432, 330, we work only with the forms 900, 810, 720, 630, 540, for which the last element is $= 0$.

A more simple solution of the system of linear equations in A, B, $C, \ldots$ is the following:

	A	B	C	D	E	F	G	H	I	J	K	L
x	0	1	−1	−1	1	−1	1	−1	1	−1	1	−1
y				1	−2	3	−4	5	−6	7	−8	9
z						1	−3	5	−10	15	−21	28
u								1	−4	10	−20	35
v										1	−5	15
w												1

read

$$\begin{aligned} A &= 0, \\ B &= x, \\ C &= -x, \\ D &= x + y, \\ E &= -x - 2y, \\ F &= x + 3y + z, \\ G &= -x - 4y - 3z, \end{aligned}$$

where x, y, $z, \ldots$ are arbitrary; the numbers in the table are the binomial coefficients with the signs + and − alternately.

We may, it is clear, express x, y, $z, \ldots$ in terms of B, D, $F, \ldots$, viz. we thus have

$$\begin{aligned} x &= B, \\ y &= -B + D, \\ z &= 2B - 3D + F, \end{aligned}$$

and substituting these values, we reproduce the foregoing expressions of A, B, C, $\ldots$ in terms of B, D, F, $\ldots$.

For a given finite value of ω, we have of course the same number of arbitrary coefficients x, y, z, $\ldots$ as there were of arbitrary coefficients B, D, F, $\ldots$; thus for $\omega = 9$, only x and y are arbitrary, and the remaining coefficients z, &c., are determined in terms of these. Or, what is the same thing, we stop with the equation $E = -x - 2y$, next preceding an equation with the non-arbitrary coefficient z.

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For the case ω even, I write

$$\begin{aligned} A + 2B &= A', \\ B + 2C &= B', \\ C + 2D &= C', \\ &\&c.; \end{aligned}$$

and I say that the new functions A', B', C', ..., are related together precisely in the same manner as are the functions A, B, C, ..., belonging to an odd weight ω; viz. we have

$$\begin{aligned} A' &= 0, \\ B' + C' &= 0, \\ C' + 2D' + E' &= 0, \\ &\&c., \end{aligned}$$

which being so, the theory is included in what precedes, and there is no occasion to consider it separately.

To prove this, observe that ω being even, we have

$$D_{\omega-q-r,q,r} = D_{\omega-q-r,r,q},$$

that is,

$$\mathfrak{A}^{\omega-q-r}\,\mathfrak{B}^q\mathfrak{C}^r = \mathfrak{A}^{\omega-q-r}\,\mathfrak{B}^r\mathfrak{C}^q,$$

whence, in particular,

$$\mathfrak{C}^q = \mathfrak{B}^q \quad (\omega - 2q \text{ even});$$

we thus have

$$\mathfrak{C} = \mathfrak{B}, \quad \mathfrak{C}^2 = \mathfrak{A}^{\omega-1}\mathfrak{B} = \mathfrak{A}^{\omega-2}\mathfrak{B}^2, \quad \mathfrak{C}^3 = \mathfrak{A}^{\omega-3}\mathfrak{B}^3, \ \&c.;$$

$$A' = A + 2B = \mathfrak{A}^\omega + 2\mathfrak{A}^{\omega-1}\mathfrak{B} = \mathfrak{A}^\omega + \mathfrak{A}^{\omega-1}\mathfrak{B} + \mathfrak{A}^{\omega-1}\mathfrak{C} = \mathfrak{A}^{\omega-1}(\mathfrak{A} + \mathfrak{B} + \mathfrak{C}),$$

which is $= 0$. Similarly

$$B' + C' = B + 3C + 2D = \mathfrak{A}^{\omega-1}\mathfrak{B} + 3\mathfrak{A}^{\omega-2}\mathfrak{B}^2 + 2\mathfrak{A}^{\omega-3}\mathfrak{B}^3,$$

or, since

$$B + C = \mathfrak{A}^{\omega-2}\mathfrak{B}(\mathfrak{A} + 3\mathfrak{B} + 2\mathfrak{B}^2) = \mathfrak{A}^{\omega-2}\mathfrak{B}(\mathfrak{A} + 2\mathfrak{B}),$$

this is

$$= -\mathfrak{A}^{\omega-2}\mathfrak{B}\mathfrak{C}(\mathfrak{A} + 2\mathfrak{B}),$$

which, in virtue of

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

is

$$= -\mathfrak{A}^{\omega-2}\mathfrak{B}\mathfrak{C}(-\mathfrak{C}) = \mathfrak{A}^{\omega-2}\mathfrak{B}\mathfrak{C}^2,$$

which is $= 0$. And similarly $C' + 2D' + E' = 0$, &c.

I come to a new part of the theory: to fix the ideas, consider the weight $\omega = 27$; and write also

$$a_0,\ a_1,\ a_2,\ a_3, \dots = 1,\ b,\ c,\ d,\ e,\ f,\ g,\ h,\ i,\ j,\ k,\ l,\ m, \dots,$$

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using, if we please, the suffixed a for the higher terms. The cubic seminvariants are $3^9 \cdot 2^0$, $3^7 \cdot 2^3$, $3^5 \cdot 2^6$, $3^3 \cdot 2^9$, $3 \cdot 2^{12}$, viz. the number of them is 5; we have thus the 5 terms B, D, F, H, J. Here

$$B = (\alpha - \beta)^{26}(\alpha - \gamma),$$

or, interchanging the α and β,

$$= (\beta-\alpha)^{26}(\beta-\gamma) = -(\alpha-\beta)^{26}(\beta-\gamma);$$

there is an initial term α^{27} and a final term $\beta^{26}\gamma$, $= b^{26}c$ or bn_2, or we may write

$$B = (\alpha-\beta)^{26}(\alpha-\gamma) = a_{27} + bn_2;$$

and similarly for the other forms. We have thus

$$\begin{aligned}
B &= (\alpha-\beta)^{26}(\alpha-\gamma) = a_{27} + bn_2,\\
D &= (\alpha-\beta)^{24}(\alpha-\gamma)^3 = a_{27} + dm_2,\\
F &= (\alpha-\beta)^{22}(\alpha-\gamma)^5 = a_{27} + fl_2,\\
H &= (\alpha-\beta)^{20}(\alpha-\gamma)^7 = a_{27} + hk_2,\\
J &= (\alpha-\beta)^{18}(\alpha-\gamma)^9 = a_{27} + j^3,
\end{aligned}$$

where the initial term $\alpha^{27} = a_{27}$ has in each case the coefficient unity, but the final terms have each of them the proper numerical coefficient.

It must be possible to form linear combinations

$$\begin{aligned}
B &= a_{27} + bn_2,\\
(B, D) &= a_{25}(c-b^2) + dm_2^3,\\
(B, D, F) &= a_{25}(e+c^2) + fl_2,\\
(B, D, F, H) &= a_{21}(g+d^2) + hk^3,\\
(B, D, F, H, J) &= a_{19}(i+e^2) + j^3,
\end{aligned}$$

where $c-b^2$, $e+c^2$, $g+d^2$, $i+e^2$ denote the seminvariants

$$c-b^2,\quad e-4bd+3c^2,\quad g-6bf+15ce-10d^2,$$

$$i-8bh+28cg-56df+35e^2,$$

respectively. The expressions indicated by their initial and final terms $a_{27}+bn_2$, $a_{25}c+dm_2^3$, &c., are what I call columns;* see my paper "Seminvariant Tables," *American Journal of Mathematics*, t. VII. (1885), pp. 59–73, [831], where, however, the theory is not by any means completely developed.

As to this, observe that B, D are each of them of the form $a_{27}+a_0^0a_{25}(c,b^2)\ldots$, the proper combination in order to eliminate a_{27} is obviously $B-D$, the term in a_0b will then disappear of itself, and the terms in $a_{25}(c,b^2)$ will combine into a term $a_{25}(c-b^2)$, viz. in the function qua seminvariant the coefficient of the highest letter a_{25} will be the seminvariant $c-b^2$. Similarly, in the combination B, D, F, the two coefficients will be determinable so that there are no terms in a_0, a_1, a_{25} or a_0d, and this being so, the term in a_{23} will be $a_{23}(e+4bd-3c^2)$, viz. in the function, qua seminvariant, the coefficient of the highest letter a_{23} will be the seminvariant $e-\ldots$. And similarly for the remaining two linear combinations.

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As a partial verification, I write

$$\begin{aligned}
B &= (\alpha-\beta)^{26}(\alpha-\gamma)\\
&= \{\alpha^{26}+\beta^{26}-26(\alpha^{25}\beta+\alpha\beta^{25})+325(\alpha^{24}\beta^2+\alpha^2\beta^{24})\ldots\}(\alpha-\gamma),
\end{aligned}$$

or, retaining only the terms which have an index at least $= 25$, this is

$$\begin{aligned}
B &= \alpha^{27}+\alpha\beta^{26}-26\alpha^{26}\beta+325\alpha^{25}\beta^2\\
&\quad -(\alpha^{26}+\alpha^{25}\beta)\gamma\\
&= a_{27}+a_0b\\
&\qquad -26a_0b-26a_{25}c\\
&\qquad -26a_0b+52a_{25}b^2\\
&= a_{27}-27a_0b+a_{25}(299c-52b^2);
\end{aligned}$$

and similarly

$$D = (\alpha-\beta)^{24}(\alpha-\gamma)^3$$
$$= \{\alpha^{24}+\beta^{24}-24(\alpha^{23}\beta+\alpha\beta^{23})+276(\alpha^{22}\beta^2+\alpha^2\beta^{22})\ldots\}(\alpha-\gamma)^3,$$

or, retaining only the terms which have an index at least $= 25$, this is

$$\begin{aligned} D &= \alpha^{27}-24\alpha^{26}\beta+276\alpha^{25}\beta^2 \\ &\quad -3\alpha^{26}\gamma+72\alpha^{25}\beta\gamma \\ &\quad +3\alpha^{25}\gamma^2 \\ &= a_{27}-24a_0b+276a_{25}b^2 \\ &\qquad -3a_0b+72a_{25}b^2 \\ &\qquad +3a_{25}c \\ &= a_{27}+a_{25}(279c+72b^2), \end{aligned}$$

and thence

$$B-D = a_{25}(20c-20b^2),$$

viz. the coefficient of a_{25} is $= 20(c-b^2)$.

But instead of the foregoing forms B, D, F, H, J, we may start with five other forms which are in fact linear combinations of these, viz. these are the forms

$$\begin{aligned} X &= (\alpha-\beta)^{26}(\alpha+\beta-2\gamma) = a_{27}+bn_2, \\ Y &= (\alpha-\beta)^{24}(\alpha+\beta-2\gamma)(\alpha-\gamma)(\beta-\gamma) = a_{25}(c-b^2)+dm_2, \\ Z &= (\alpha-\beta)^{22}(\alpha+\beta-2\gamma)(\alpha-\gamma)^2(\beta-\gamma)^2 = a_{25}(c-b^2)+fl_2, \\ W &= (\alpha-\beta)^{20}(\alpha+\beta-2\gamma)(\alpha-\gamma)^3(\beta-\gamma)^3 = a_{23}(e+c^2)+hk_2, \\ T &= (\alpha-\beta)^{18}(\alpha+\beta-2\gamma)(\alpha-\gamma)^4(\beta-\gamma)^4 = a_{23}(e^2+c^2)+j^3, \end{aligned}$$

where as well the initial as the final terms should have their proper numerical coefficients.

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As to this, observe that in Y we have a term $\alpha^{26}(\beta-\gamma)$ which is $= 0$, and similarly a term $\beta^{26}(\alpha-\gamma)$ which is $= 0$; the highest powers are thus α^{25} and β^{25}, giving a term in a_{25} which can only be $a_{25}(c-b^2)$. In Z we have the terms $\alpha^{25}(\beta-\gamma)^2$ and $\beta^{25}(\alpha-\gamma)^2$, giving the term $a_{25}(c-b^2)$; in W we have the terms $\alpha^{24}(\beta-\gamma)^3$ and $\beta^{24}(\alpha-\gamma)^3$ which are each $= 0$, the highest powers are thus α^{23}, β^{23}, giving a term in a_{23} which can only be $a_{23}(e+4bd-3c^2)$; and in T we have the terms $\alpha^4(\beta-\gamma)^4$ and $\beta^4(\alpha-\gamma)^4$ which give the term $a_{23}(e+4bd-3c^2)$.

The combinations which give the columns thus are

$$\begin{aligned} X &= a_{27}+bn_2, \\ (Y) &= a_{25}(c-b^2)+dm_2, \\ (Y,Z) &= a_{23}(e+c^2)+fl_2, \\ (Y,Z,W) &= a_{21}(g+d^2)+hk_2, \\ (Y,Z,W,T) &= a_{19}(i+e^2)+j^3. \end{aligned}$$

I start with the form

$$u = (Y,\ Z,\ W,\ T) = uY+\lambda Z+\mu W+\nu T.$$

We know that it is possible to determine the coefficients λ, μ, ν, in suchwise that the linear function shall be of the proper form $a_{19}(i+e^2)+j^3$; or, what is the same thing, that there shall be no term with an index exceeding 19; the conditions to be satisfied are apparently more than three, but of course the number of independent conditions must be $= 3$. We have, in particular, to get rid of all the terms of the second degree which precede $a_{19}i$, viz. the coefficients of $a_{19}c$, $a_{24}d$, $a_{23}e$, $a_{22}f$, $a_{21}g$, $a_{20}h$ must all of them vanish. Now the terms of u which produce terms of the second degree are the terms containing any two but not all three of the symbolic letters α, β, γ, say the biliteral terms; we have for instance $\alpha^{25}\beta^2$, $\alpha^{25}\gamma^2$,

$\beta^{25}\gamma^2$ (there are no terms $\gamma^{25}\alpha^2$ or $\gamma^{25}\beta^2$, since the highest power of γ is γ^9), each of which is $a_{25}c$. But in the development of u, we may in any biliteral term by an interchange of the letters α, β, γ make the letter of highest index to be α, and the other letter to be β; thus the several terms $\alpha^{25}\gamma^2$, $\beta^{25}\alpha^2$, $\beta^{25}\gamma^2$ may be each of them converted into $\alpha^{25}\beta^2$, and so in other cases. Imagining this to be done, the term in $a_{25}c$ is simply the term in $\alpha^{25}\beta^2$, and in like manner the term in $a_{24}d$ is the term in $\alpha^{24}\beta^3$, and so in other cases; the condition as to the disappearance of the terms $a_{25}c, \ldots, a_{20}h$ is the condition that the terms in $\alpha^{25}\beta^2$, $\alpha^{24}\beta^3$, $\alpha^{23}\beta^4$, $\alpha^{22}\beta^5$, $\alpha^{21}\beta^6$, $\alpha^{20}\beta^7$ shall all of them disappear. And if, in the function u transformed in the foregoing manner, we write for convenience $\alpha = 1$, so that u has become a function of β only (and observe that u will contain the factor β^2), then the condition is that the terms in β^2, β^3, β^4, β^5, β^6, β^7 may in fact be written $u = 0$; viz. it is assumed that u is in the first instance transformed into a function of β as just explained, and the equation is then to be understood as denoting that the coefficients of u are to be so determined that as many as possible of the terms (beginning with that which contains the lowest power of β) shall each of them vanish.

Consider any term of u, for instance the term

$$F = (\alpha - \beta)^{24}(\alpha + \beta - 2\gamma)(\alpha - \gamma)(\beta - \gamma).$$

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Arthur Cayley

847 (concluded). On the Theory of Seminvariants.

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To obtain the biliteral terms in the proper form 1° we write therein $1 = 0$, 2° we write $\beta = 0$, changing γ into β, 3° we write $a = 0$, changing β, γ into a, β; we thus obtain

$$(a-\beta)^{24}(a+\beta)\,a\beta, \quad a^{24}(a-\beta)(-\beta), \quad a^{24}(a-2\beta)(-\beta)(a-\beta);$$

viz. the second and third terms are identical; the first term requires, however, the factor 2 (for any term $a^p\beta^q$ is accompanied by a corresponding term $a^q\beta^p$), so that, omitting this factor throughout, the terms are

$$(a-\beta)^{24}(a+\beta)\,a\beta - a^{24}(a-2\beta)\beta(a-\beta),$$

or, putting therein $a = 1$, the terms are

$$(1-\beta)^{24}(1+\beta)\beta - (1-2\beta)(1-\beta)\beta.$$

Similarly from

$$Z = (a-\beta)^{22}(a-\beta-2\gamma)(a-\gamma)^2(\beta-\gamma)^2,$$

we have

$$(1-\beta)^{22}(1+\beta)\beta^2 + (1-2\beta)(1-\beta)^2\beta^2,$$

and the like as regards W and T. The result thus is

$$(1+\beta)(1-\beta)^{24} + \lambda(1-\beta)^{22}\beta + \mu(1-\beta)^{20}\beta^2 + \nu(1-\beta)^{18}\beta^3 - (1-2\beta)(1-\beta)\beta\{1-\lambda\beta(1-\beta)+\mu\beta^2(1-\beta)^2-\nu\beta^3(1-\beta)^3\} = 0,$$

or, as this may be written,

$$\frac{(1+\beta)(1-\beta)^{24}}{1-2\beta}\left\{1+\lambda\frac{\beta}{(1-\beta)^2}+\mu\frac{\beta^2}{(1-\beta)^4}+\nu\frac{\beta^3}{(1-\beta)^6}+\cdots\right\}$$
$$-1+\lambda(\beta-\beta^2)-\mu(\beta-\beta^2)^2+\nu(\beta-\beta^2)^3-\cdots = 0,$$

viz. each side is to be expanded in ascending powers of β, and the coefficients λ, μ, ν are then to be determined so that as many terms as possible shall vanish.

Expanding, we have

$$(1-20\beta+190\beta^2-1138\beta^3+4808\beta^4-15178\beta^5+\cdots)$$
$$\times\{1+\beta\lambda+\beta^2(2\lambda+\mu)+\beta^3(3\lambda+4\mu+\nu)+\beta^4(4\lambda+10\mu+6\nu)+\beta^5(5\lambda+20\mu+21\nu)+\cdots\}$$
$$-1+\beta\lambda+\beta^2(-\lambda-\mu)+\beta^3(2\mu+\nu)+\beta^4(-3\nu)+\beta^5\cdot 3\nu+\cdots = 0,$$

or finally, as far as β^5, the equation is

$$0 = 0+\beta(2\lambda-20)+\beta^2(-19\lambda+190)+\beta^3(153\lambda-14\mu+2\nu-1138)+\beta^4(-814\lambda+119\mu-17\nu+4808)+\beta^5(3027\lambda-558\mu+94\nu-15178).$$

We have in the first place $\lambda = 10$, which makes the coefficient of β^2 to be $= 0$; and we then have

$$\begin{array}{ll} -14\mu + 2\nu + 392 = 0, & \text{or say} \quad 7\mu - \nu - 196 = 0, \\ 119\mu - 17\nu - 3332 = 0, & \qquad 119\mu - 17\nu - 3332 = 0, \\ -558\mu + 94\nu + 15092 = 0, & \qquad 272\mu - 47\nu - 7546 = 0, \end{array}$$

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where the second equation is equivalent to the first: the three equations give $\mu = \frac{273}{8}$, $\nu = \frac{231}{4}$. I have since found that proceeding one step further, that is, to β^6, the coefficient is $-8310\lambda+1791\mu-363\nu+36942$, viz. putting this $= 0$, and for λ substituting its value $= 10$, we have $597\mu - 121\nu = 15386$, an equation which is in fact satisfied by the values just obtained for λ and μ. For the function (Y, Z, W) we have the same equations with $\nu = 28$ and for the function (Y, Z) the same equations with $\mu = 0$, $\nu = 0$; and therefore $\lambda = 10$. The linear combinations thus are

$$\begin{aligned} X &= a_2 c + b a^2, \\ Y &= a_{19}(c + d^2) + a_2(c^2 - b^2), \\ Y + 10Z &= a_{19}(e + c^2) + a_{20}, \\ Y + 10Z + 28W &= a_{21}(g + hk^2 + d^2), \\ 25Y + 250Z + 823W + 931T &= a_{19}(i + k + j^3), \end{aligned}$$

where remark that in $Y + 10Z$, by means of the coefficient 10, we make to disappear the two terms $a_{20}c$, $a_{20}d$; in the next function, by means of the coefficients 10 and 28, the four terms $a_{21}c$, $a_{21}d$, $a_{21}e$, $a_{21}f$; and in the last function, by means of the three coefficients, the six terms $a_{19}c$, $a_{19}d$, $a_{19}e$, $a_{19}f$, $a_{21}g$ and $a_{21}h$.

It is possible that a larger number of terms will disappear; but if this is so, the general form shown by the combinations on p. 353 will require modification.

The investigation applies without alteration to any odd weight ω whatever; the condition for the determination of the coefficients λ, μ, ν, ... is obviously

$$\frac{(1+\beta)(1-\beta)^{\omega-1}}{1-2\beta}\left\{1 + \lambda\frac{\beta}{(1-\beta)^2} + \mu\frac{\beta^2}{(1-\beta)^4} + \nu\frac{\beta^3}{(1-\beta)^6} + \cdots\right\} - 1 + \lambda(\beta-\beta^2) - \mu(\beta-\beta^2)^2 + \nu(\beta-\beta^2)^3 - \cdots = 0.$$

For the case of an even weight ω, the series of functions is

$$\begin{aligned} X &= (a-\beta)^\omega, \\ Y &= (a-\beta)^{\omega-2}(a-\gamma)^2(\beta-\gamma), \\ Z &= (a-\beta)^{\omega-4}(a-\gamma)^2(\beta-\gamma)^2, \end{aligned}$$

and the condition for the determination of the coefficients λ, μ, ν, ... is in like manner found to be

$$(1-\beta)^{\omega-1}\left\{1 + \lambda\frac{\beta}{(1-\beta)^2} + \mu\frac{\beta^2}{(1-\beta)^4} + \nu\frac{\beta^3}{(1-\beta)^6} + \cdots\right\} - 1 + \lambda(\beta-\beta^2) - \mu(\beta-\beta^2)^2 + \nu(\beta-\beta^2)^3 + \cdots = 0,$$

which is to be understood as explained above.

848. On the Transformation of the Double-Theta Functions.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 142–178.]

[Printed p. 358]

I propose to reproduce Hermite's Memoir "Sur la théorie de la transformation des fonctions Abéliennes," *Comptes Rendus*, t. XL. (1855), pp. 249,..., 784, with some changes of notation and developments. Hermite's functions are even or odd according as we have $\mu q + \nu p$ even or odd: viz. his characteristic is $\binom{\mu,\nu}{q,p}$, or the letters p, q, are misplaced; I write, therefore, r instead of p, so as to have the characteristic $\binom{\mu,\nu}{q,r}$;

and then for symmetry it is necessary to interchange the suffixes 2, 3 and the letters c, d; the invariant function of the periods, instead of being as with him

$$\omega_0 \upsilon_1 - \omega_1 \upsilon_0 + \omega_3 \upsilon_2 - \omega_2 \upsilon_3,$$

must be taken to be

$$\omega_0 \upsilon_1 - \omega_1 \upsilon_0 + \omega_2 \upsilon_3 - \omega_3 \upsilon_2.$$

Moreover, I write A, B for his G, G', so as, instead of

$$(G, H, G' \mid x, y)^2,$$

to have in the expressions of the theta-functions the quadric function $(A, H, B \mid x, y)^2$; and I alter the arrangement of the memoir so as to separate more completely the preliminary theory from the theory of the transformation.

General Theory. Art. Nos. 1 to 21 (several sub-headings).

The functions $\Pi\{K, \textit{indef. or def.}\}$.

1. Consider a function

$$\Pi\begin{pmatrix}\mu,\nu\\ q,r\end{pmatrix}(x,y)(A,H,B)\{K, \text{indef. or def.}\},$$

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having: the characteristic $\binom{\mu,\nu}{q,r}$, where the characters μ, ν, q, r are positive or negative integers, which may be taken to have each of them only the values $(0, 1)$ at pleasure, so that the number of functions is $2^4 = 16$;

the arguments (x, y);

the parameters, or conjoint quarter-periods, (A, H, B);

and the potency K, a positive integer;

and which is either indefinite or definite, as will be explained; the function, moreover, contains linearly certain arbitrary constants, the number of them depending on the value of K, as will be explained.

The function may be written $\Pi(x, y)$ or in any other less abbreviated form which may be convenient.

2. The function $\Pi(x, y)\{K \text{ indef.}\}$ is defined by the following four equations:

$$\begin{aligned}\Pi(x+1, y) &= (-)^{\mu}\,\Pi(x, y),\\ \Pi(x, y+1) &= (-)^{\nu}\,\Pi(x, y),\\ \Pi(x+A, y+H) &= (-)^{q}\,\Pi(x, y)\exp.-2\pi K(2x+A),\\ \Pi(x+H, y+B) &= (-)^{r}\,\Pi(x, y)\exp.-2\pi K(2y+B),\end{aligned}$$

and the function $\Pi(x, y)\{K \text{ def.}\}$ by the same equations, together with the following fifth equation,

$$\Pi(-x, -y) = (-)^{\mu q + \nu r}\,\Pi(x, y);$$

viz. the definite function is an even function or else an odd function of the arguments according as $\mu q + \nu r$ is even or odd. We may call $\mu q + \nu r$ the index; and the function is then even or odd according as the index is even or odd.

It is perhaps worth noticing that it would be allowable to define a function $\Pi(x, y)\{K, \text{skew def.}\}$, by the corresponding relation

$$\Pi(-x, -y) = -(-)^{\mu q + \nu r}\,\Pi(x, y),$$

but I do not propose here to develope this notion.

3. The four equations give rise to the following one,

$$\Pi(x + a_0 + Aa_2 + Ha_3,\ y + a_1 + Ha_2 + Ba_3) = (-)^{\mu a_0 + \nu a_1 + q a_2 + r a_3}\,\Pi(x, y) \times \exp.-2\pi K\{2a_2 x + 2a_3 y + (A, H, B \mid a_2, a_3)^2\},$$

where a_0, a_1, a_2, a_3 are any positive or negative integers (zero not excluded), and which single equation, in fact, includes the preceding four equations.

4. In regard to the parameters it is to be observed that, if $A = A_0 + ia$, $H = H_0 + i\eta$, $B = B_0 + i\beta$, we must have (a, η, β) a determinate positive quadratic form; viz. this is the necessary and sufficient condition for the convergence of the series for the development of the function.

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5. The number of arbitrary constants is for the indefinite function $= K^2$; but for the definite function it is, when K is odd, $= \frac{1}{2}(K^2+1)$; and when K is even, it is $= \frac{1}{2}K^2$, except in the case of a characteristic $\binom{0,0}{q,r}$, when it is $= \frac{1}{2}(K^2+4)$.

In particular, for $K = 1$, there is only a single arbitrary constant, which is a mere factor of the function; taking it to be $= 1$, as presently explained, we have the 16 theta-functions.

6. The function $\Pi(x,y)$ is developed in a series of exponentials, in the form

$$\Pi(x,y) = \Sigma(-)^{mq+nr} A_{m,n} \exp . i\pi \left\{(2m+\mu)x + (2n+\nu)y + \tfrac{1}{4K}(A,H,B \mid 2m+\mu,\ 2n+\nu)^2\right\},$$

where m and n have each of them all positive and negative integer values (zero not excluded) from $-\infty$ to ∞. In fact, substituting this series in the four equations, they are all of them satisfied if only

$$A_{m+K,n} = A_{m,n}; \quad A_{m,n+K} = A_{m,n}.$$

Consequently the following K^2 coefficients remain arbitrary, viz. those with the suffixes

$$0, 1, \ldots, K-1;$$

and we have for $\Pi(x,y)$ a sum of K^2 terms, each a determinate series multiplied into one of the arbitrary coefficients $A_{0,0}$, $A_{0,1}$, &c. The indefinite function thus contains, as already mentioned, K^2 arbitrary constants.

7. Substituting in the fifth equation, we have for the definite function the further condition

$$A_{-m,-n} = A_{m,n},$$

which it is clear will be satisfied generally if only it is satisfied by the coefficients in the foregoing set of K^2 coefficients.

8. In the case K odd, we thus reduce the number of arbitrary coefficients to $\frac{1}{2}(K^2+1)$; the mode in which this takes place is best seen by an example. Suppose $K = 3$, so that $A_{m+3,n} = A_{m,n}$; $A_{m,n+3} = A_{m,n}$. For the coefficients of the indefinite function, the suffixes are

$$\begin{array}{ccc} 00, & 01, & 02, \\ 10, & 11, & 12, \\ 20, & 21, & 22. \end{array}$$

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And if we suppose $\mu = 0$, $\nu = 1$, then the new condition is $A_{-m,-n-1} = A_{m,n}$, viz. writing down only the suffixes, we thus obtain

$$\begin{array}{lll} 0,\,0 = 0,\,-1, & 1,\,0 = -1,\,-1, & 2,\,0 = -2,\,-1, \\ 0,\,1 = 0,\,-2, & 1,\,1 = -1,\,-2, & 2,\,1 = -2,\,-2, \\ 0,\,2 = 0,\,-3, & 1,\,2 = -1,\,-3, & 2,\,2 = -2,\,-3, \end{array}$$

that is,

$$\begin{array}{lll} 0,\,0 = 0,\,2, & 1,\,0 = 2,\,2, & 2,\,0 = 1,\,2, \\ 0,\,1 = 0,\,1, & 1,\,1 = 2,\,1, & 2,\,1 = 1,\,1, \\ 0,\,2 = 0,\,0, & 1,\,2 = 2,\,0, & 2,\,2 = 1,\,0, \end{array}$$

viz. one of these equations $0,\,1 = 0,\,1$ is an identity, but the other equations occur each twice; or we have four equations, each of them an equality between two out of the remaining 8 coefficients; the number of arbitrary coefficients is thus $1 + (9-1)/2 = 5$, $= \frac{1}{2}(K^2+1)$.

9. When K is even, it is necessary to distinguish between the case $(\mu,\nu) = (0,0)$ and the remaining three cases $(\mu,\nu) = (1,0)$, $(0,1)$ or $(1,1)$. In the former case, the relation between the coefficients is

$A_{-m,-n} = A_{m,n}$; there are four identities, $0,\ 0 = 0,\ 0$; $0,\ \frac{1}{2}K = 0,\ \frac{1}{2}K$; $\frac{1}{2}K,\ 0 = \frac{1}{2}K,\ 0$; $\frac{1}{2}K,\ \frac{1}{2}K = \frac{1}{2}K,\ \frac{1}{2}K$; and the remaining K^2-4 equations occur each twice, that is, we have $\frac{1}{2}(K^2-4)$ equations, each of them an equality between two of the remaining K^2-4 coefficients; and we thus have $\frac{1}{2}K^2$ arbitrary coefficients.

In the latter case, there are no identities and the K^2 equations occur each twice, that is, we have $\frac{1}{2}K^2$ equations, each of them an equality between two of the K^2 coefficients; and we thus have $\frac{1}{2}K^2$ arbitrary coefficients.

10. Recapitulating, it thus appears that

for an indefinite function, the number of coefficients $= K^2$;
for a definite function, the number $= \frac{1}{2}(K^2+1),\ K$ odd;
$= \frac{1}{2}K^2,\ K$ even, and
$(\mu,\nu) = (1,0),\ (0,1)$ or $(1,1)$;
$= \frac{1}{2}(K^2+4),\ K$ even, and
$(\mu,\nu) = (0,0)$.

The Theta-functions.

11. In the particular case $K=1$, the distinction between the indefinite function and the definite function disappears, and we have instead of $\Pi(x,y)$, the theta-functions $\Theta(x,y)$, satisfying the four equations

$$\begin{aligned}
\Theta(x+1,y) &= (-)^{\mu}\,\Theta(x,y),\\
\Theta(x,y+1) &= (-)^{\nu}\,\Theta(x,y),\\
\Theta(x+A,y+H) &= (-)^{q}\,\Theta(x,y)\exp.-i\pi(2x+A),\\
\Theta(x+H,y+B) &= (-)^{r}\,\Theta(x,y)\exp.-i\pi(2y+B),
\end{aligned}$$

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and the fifth equation

$$\Theta(-x,-y) = (-)^{\mu q+\nu r}\,\Theta(x,y);$$

as before, $\mu q+\nu r$ is the index of the function.

The four equations are all of them included in the following one:

$$\Theta(x+a_0+Aa_2+Ha_3,\ y+a_1+Ha_2+Ba_3) = (-)^{\mu a_0+\nu a_1+qa_2+ra_3}\,\Theta(x,y)\times\exp.-i\pi\{2a_2x+2a_3y+(A,H,B\mid a_2,a_3)^2\},$$

where $a_0,\ a_1,\ a_2,\ a_3$ are each of them any positive or negative integer, zero not excluded.

Moreover, we take $A_{0,0}=1$, and the value of the function thus is

$$\Theta(x,y) = \Sigma(-)^{mq+nr}\exp.i\pi\{(2m+\mu)x+(2n+\nu)y+\tfrac{1}{4}(A,H,B\mid 2m+\mu,\ 2n+\nu)^2\}.$$

12. The sum of two characteristics is the characteristic obtained by taking the sums of the component terms or characters,

$$\begin{pmatrix}\mu,\nu\\ q,r\end{pmatrix}+\begin{pmatrix}\mu',\nu'\\ q',r'\end{pmatrix}=\begin{pmatrix}\mu+\mu',\ \nu+\nu'\\ q+q',\ r+r'\end{pmatrix};$$

and similarly for any number of characteristics. I use the sign $=$, but this properly denotes a congruence, mod 2; and the like as regards the indices.

The sum of two identical characteristics, or generally of any number of characteristics taken each of them any even number of times, is $=\binom{0,0}{0,0}$. And this characteristic $\binom{0,0}{0,0}$ may be called the characteristic 0.

It should be observed, that the index of the sum is not in general equal to the sum of the indices. To make it so, we must have, for two characteristics,

$$(\mu+\mu')(q+q')+(\nu+\nu')(r+r') = \mu q+\nu r+\mu' q'+\nu' r',$$

that is,

$$\mu q' + \nu' r + \mu' q + \nu r' = 0;$$

and there is obviously a like formula for the case of more than two characteristics.

Two or more characteristics, such that they have the sum of the indices equal to be the index of the sum, are said to be in "direct relation" or "directly related" to each other. The sum of the indices and the index of the sum may differ by unity; and we then have the inverse relation; but I do not propose to consider this.

13. Consider any number K of theta-functions, of the same arguments and parameters, but with the same or different characteristics. The product of these functions is in general a function $\Pi(x, y)\{K \text{ indef.}\}$, having a characteristic which is = the sum of the characteristics of the theta-functions. In fact, from the four

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equations of the theta-functions, we at once obtain for their products $\Pi(x, y)$ the four equations

$$\begin{aligned}
\Pi(x+1, y) &= (-)^{\Sigma\mu}\, \Pi(x, y),\\
\Pi(x, y+1) &= (-)^{\Sigma\nu}\, \Pi(x, y),\\
\Pi(x+A, y+H) &= (-)^{\Sigma q}\, \Pi(x, y) \exp . -i\pi K(2x+A),\\
\Pi(x+H, y+B) &= (-)^{\Sigma r}\, \Pi(x, y) \exp . -i\pi K(2y+B),
\end{aligned}$$

which proves the theorem.

14. But if the indices are in direct relation to each other, then we have further

$$\Pi(-x, -y) = (-)^{\Sigma\mu q + \Sigma\nu r}\, \Pi(x, y),$$

and the product is thus a function $\Pi(x, y)\{K, \text{def.}\}$.

15. Take the square of a theta-function, the characteristic is $= \binom{0,0}{0,0}$ or 0, and we have also twice the index = 0; viz. the theta-function is in direct relation with itself. Hence the squared function is a function $\Pi\binom{0,0}{0,0}(x, y)\{2, \text{def.}\}$, and as such it contains linearly $\frac{1}{2}(2^2+4) = 4$ arbitrary constants. Hence, taking any five squares, since each of them is a function of the form in question, it follows that the squares of the 5 theta-functions are connected by a linear relation.

Göpel's relation between 4 theta-functions.

16. We may in a variety of ways (in fact, in 60 ways, as will presently be shown) select four theta-functions, all of them even, or else two of them even and two odd (that is, having the sum of their indices = 0), such that the sum of their characteristics is = 0; for instance, the functions may be

$$P = \begin{pmatrix}0,0\\0,0\end{pmatrix}, \quad P' = \begin{pmatrix}0,0\\1,0\end{pmatrix}, \quad S = \begin{pmatrix}1,1\\0,1\end{pmatrix}, \quad S' = \begin{pmatrix}1,1\\1,1\end{pmatrix},$$

indices

$$0, \quad 0, \quad 1, \quad 1.$$

The functions are thus in direct relation, and the product of the four functions is a function $\Pi\binom{0,0}{0,0}\{4, \text{def.}\}$. But obviously any one of the functions taken four times, or any two of them taken each twice, are in like manner four functions in direct relation, or the fourth powers P^4, P'^4, S^4, S'^4 and the squared products $P^2P'^2$, P^2S^2, $P^2S'^2$, P'^2S^2, $P'^2S'^2$, $S^2S'^2$, are in like manner each of them a function $\Pi\binom{0,0}{0,0}\{4, \text{def.}\}$, viz. we have thus in all $1 + 4 + \binom{4}{2} = 11$ such functions. But the Π function contains only $\frac{1}{2}(4^2+4) = 10$ arbitrary constants; hence there must be a linear relation between the 11 powers and products, and this is Göpel's relation.

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17. Starting with any two characteristics a, b at pleasure, the remaining characteristics form seven pairs, such that

$$a+b = c+d = e+f = g+h = i+j = k+l = m+n = o+p;$$

but among the seven pairs we have only three, suppose $c+d$, $e+f$, $g+h$, which are such that (a,b,c,d), (a,b,e,f), (a,b,g,h) are each of them either all even or else two even and two odd; that is, starting with any pair (a,b), we have these three tetrads having each of them the required property. The number of pairs (a,b) is $16\cdot 15$, $=120$; and we thence derive 120×3, $=360$ tetrads; but each such tetrad is of course derivable from any one of the six pairs contained in it; or the number of distinct tetrads is $360/6$, $=60$, viz. we have, as mentioned above, 60 Göpel-tetrads.

The four functions Π_0, Π_1, Π_2, Π_3.

18. We consider four theta-functions θ_0, θ_1, θ_2, θ_3, which are such that to the modulus 2, the sum of the characters is $=0$, and also the sum of the indices is $=0$; taking the characters to be

$$\begin{pmatrix}\mu,\nu\\ q,r\end{pmatrix},\ \begin{pmatrix}\mu',\nu'\\ q',r'\end{pmatrix},\ \begin{pmatrix}\mu'',\nu''\\ q'',r''\end{pmatrix},\ \begin{pmatrix}\mu''',\nu'''\\ q''',r'''\end{pmatrix},$$

and writing throughout $=$ for $=$ (mod 2), we have

$$\begin{aligned}\mu+\mu'+\mu''+\mu'''&=0,\\ \nu+\nu'+\nu''+\nu'''&=0,\\ q+q'+q''+q'''&=0,\\ r+r'+r''+r'''&=0,\\ \mu q+\nu r+\mu'q'+\nu'r'+\mu''q''+\nu''r''+\mu'''q'''+\nu'''r'''&=0.\end{aligned}$$

Writing for shortness $(01)=\mu q'+\mu'q+\nu r'+\nu'r$, and so in other cases; and further $(012)=(01)+(02)+(12)$, then from (012), &c., substituting μ''', ν''', q''', r''' their values for the first four equations, we deduce $(012)=0$; and similarly $(013)=0$, $(023)=0$, $(123)=0$.

19. Consider now a product $\theta_0^a\theta_1^b\theta_2^c\theta_3^d$, where $a+b+c+d$ is $=$ a given odd number k; the characteristic is

$$\begin{pmatrix}\mu a+\mu'b+\mu''c+\mu'''d,\ \nu a+\nu'b+\nu''c+\nu'''d\\ qa+q'b+q''c+q'''d,\ ra+r'b+r''c+r'''d\end{pmatrix},$$

and it hence follows that the index is

$$=a(\mu q+\nu r)+b(\mu'q'+\nu'r')+c(\mu''q''+\nu''r'')+d(\mu'''q'''+\nu'''r''').$$

In fact, forming the index in question, we have first terms in a^2, b^2, c^2, d^2, which upon writing therein a, b, c, d for these values respectively ($a^2=a$, &c.) give the

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required value; we have therefore only to show that the sum of the remaining terms in ab, &c., is $=0$. These terms are

$$ab(01)+ac(02)+ad(03)+bc(12)+bd(13)+cd(23),$$

and writing herein $a+b+c+d=1$, and thence $ad=a(1-a-b-c)$, $=ab+ac$, and similarly $bd=ab+bc$, $cd=ac+bc$, the terms in question become

$$=ab(013)+ac(023)+bc(123),$$

that is, they become $=0$.

20. We thus see that the function $\theta_0^a\theta_1^b\theta_2^c\theta_3^d$ has an index which is

$$=a\operatorname{ind}\theta_0+b\operatorname{ind}\theta_1+c\operatorname{ind}\theta_2+d\operatorname{ind}\theta_3.$$

Consider separately four products $\theta_0^a\theta_1^b\theta_2^c\theta_3^d$, in which the exponents a, b, c, d satisfy successively the relations (always to modulus 2)

$$\begin{aligned}&b+d=0,\quad c+d=0,\\ &b+d=1,\quad c+d=0,\\ &b+d=0,\quad c+d=1,\\ &b+d=1,\quad c+d=1.\end{aligned}$$

Combining herewith the relation $a+b+c+d=1$, it follows that the exponents a, b, c, d are

$$\begin{array}{llll} d+1, & d, & d, & d, \\ d, & d+1, & d, & d, \\ d, & d, & d+1, & d, \\ d, & d, & d, & d+1, \end{array}$$

in the four cases respectively. Then substituting these values, the characteristics become

$$\begin{pmatrix}\mu,\nu\\ q,r\end{pmatrix},\ \begin{pmatrix}\mu',\nu'\\ q',r'\end{pmatrix},\ \begin{pmatrix}\mu'',\nu''\\ q'',r''\end{pmatrix},\ \begin{pmatrix}\mu''',\nu'''\\ q''',r'''\end{pmatrix},$$

viz. the four products have the same characters as θ_0, θ_1, θ_2, θ_3 respectively; and in like manner recollecting that

$$\operatorname{ind}\theta_0+\operatorname{ind}\theta_1+\operatorname{ind}\theta_2+\operatorname{ind}\theta_3=0,$$

we see that the four products have the same indices as θ_0, θ_1, θ_2, θ_3 respectively.

More generally write $\Pi=\Sigma A\theta_0^a\theta_1^b\theta_2^c\theta_3^d$, where respectively the exponents a, b, c, d satisfy the conditions already referred to; then for the four cases Π_0, Π_1, Π_2, Π_3 have the same characteristics, and the same indices, as θ_0, θ_1, θ_2, θ_3 respectively.

21. It can be shown that each of the functions Π contains $\frac{1}{2}(k^2+1)$ constants. It will be recollected that we have between θ_0, θ_1, θ_2, θ_3 an equation of the form

$$0=(\theta_0^4,\ \theta_1^4,\ \theta_2^4,\ \theta_3^4,\ \theta_0^2\theta_1^2,\ \theta_0^2\theta_2^2,\ \theta_0^2\theta_3^2,\ \theta_1^2\theta_2^2,\ \theta_1^2\theta_3^2,\ \theta_2^2\theta_3^2);$$

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this serves to express, say θ_3^4, in lower powers of θ_3; and by successive applications of this equation, we can reduce $\Sigma\theta_0^a\theta_1^b\theta_2^c\theta_3^d$ to a form in which d has only one of the values 0, 1, 2 or 3; we do not by this transformation alter the suffix of Π, viz. a term originally of the form Π_0, Π_1, Π_2 or Π_3, will by the transformation give rise only to terms which are of the same form Π_0, Π_1, Π_2, Π_3 (as the case may be). The number of constants in Π_0 is thus

$$=\text{number of partitions of } k \text{ into four parts } a,b,c,d,$$

under the conditions

$$\begin{aligned} d&=0 \text{ or } 2; \quad a \text{ odd},\ b,c \text{ each even},\\ d&=1 \text{ or } 3; \quad a \text{ even},\ b,c \text{ each odd}, \end{aligned}$$

where, in reckoning the partitions, the order of the parts is taken into account: the partitions are thus as follows

$$\begin{array}{ll} d=0, & (a-1)+b+c=k-1,\\ d=1, & a+(b-1)+(c-1)=k-3,\\ d=2, & (a-1)+b+c=k-3,\\ d=3, & a+(b-1)+(c-1)=k-5, \end{array}$$

where the parts a or $(a-1)$, b or $(b-1)$, c or $(c-1)$, as the case may be, are all of them even; hence, writing $k'=\frac{1}{2}(k-1)$, the cases are

$$a'+b'+c'=k',\ k'-1,\ k'-1,\ \text{or } k'-2,$$

where the a', b', c' are odd or even (zero not excluded) at pleasure; as already mentioned, the order of the parts is taken into account: thus the particulars of 3 would be the

$$300,\ 210,\ 120,\ 030,\quad \text{No. is } 10,\ =\tfrac{1}{2}\cdot 4\cdot 5.$$

$$201,\ 111,\ 021,$$

$$102,\ 012,$$

003.

Hence, in the four cases respectively, the numbers are

$$\tfrac{1}{2}(k'+1)(k'+2),$$
$$\tfrac{1}{2}k'(k'+1),$$
$$\tfrac{1}{2}k'(k'+1),$$
$$\tfrac{1}{2}(k'-1)k',$$

giving a total

$$= \tfrac{1}{2}(4k'^2+4k'+2) = \tfrac{1}{2}\{(2k'+1)^2+1\},$$

that is, $= \frac{1}{2}(k^2+1)$. And similarly the number is $= \frac{1}{2}(k^2+1)$ in the other three cases respectively.

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Preparation for the Transformation. Art. Nos. 22 to 44 (several sub-headings).

The Hermitian quartic matrix.

22. Observing that, for the adopted form of the Π- or Θ-functions, the periods ω_0, υ_0 are 1, 0,

$$\begin{array}{ll} \omega_1,\ \upsilon_1 & 0,\ 1, \\ \omega_2,\ \upsilon_2 & A,\ H, \\ \omega_3,\ \upsilon_3 & H,\ B, \end{array}$$

so that we have

$$\omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2 = H - 0 + 0 - H = 0,$$

we have to consider the automorphic transformation of the bilinear form

$$\omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2.$$

23. We write

$$(\omega_0,\omega_1,\omega_2,\omega_3) = (a_0,a_1,a_2,a_3 \mid \Omega_0,\Omega_1,\Omega_2,\Omega_3) \,\Big|\, b_0,b_1,b_2,b_3 \,\Big|\, c_0,c_1,c_2,c_3 \,\Big|\, d_0,d_1,d_2,d_3,$$

$$(\upsilon_0,\upsilon_1,\upsilon_2,\upsilon_3) = (c_0,c_1,c_2,c_3 \mid \mathrm{T}_0,\mathrm{T}_1,\mathrm{T}_2,\mathrm{T}_3) \,\Big|\, d_0,d_1,d_2,d_3 \,\Big|\, a_0,a_1,a_2,a_3 \,\Big|\, b_0,b_1,b_2,b_3,$$

and the coefficients are assumed to be such that we have identically

$$\omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2 = k(\Omega_0\mathrm{T}_1 - \Omega_1\mathrm{T}_0 + \Omega_2\mathrm{T}_3 - \Omega_3\mathrm{T}_2),$$

where k is in the sequel taken to be a positive integer. We obtain by direct substitution the value of $\omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2$ in the following form (with a 4×4 array of bilinear coefficients $\Omega_i\mathrm{T}_j$):

[a 4×4 matrix of expressions of the form $a_ic_j - a_jc_i + b_id_j - b_jd_i$, indexed by the $\Omega_i\mathrm{T}_j$ pairs.]

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viz. equating this to its value $k(\Omega_0\mathrm{T}_1 - \Omega_1\mathrm{T}_0 + \Omega_2\mathrm{T}_3 - \Omega_3\mathrm{T}_2)$, we have 4 identities and 12 equations which are, in fact, 6 equations occurring each twice. We have thus six equations, which are the conditions in order that the matrix may be the matrix of automorphic transformation of the bilinear form $\omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2$. The six equations may be written

$$(ac+bd)_{01} = 0,$$
$$(ac+bd)_{02} = k,$$
$$(ac+bd)_{03} = 0,$$
$$(ac+bd)_{12} = 0,$$
$$(ac+bd)_{13} = k,$$
$$(ac+bd)_{23} = 0,$$

viz. the first of these equations is $a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 = 0$, and so in other cases. It is convenient to remark that each interchange of two letters a and c, b and d, also interchange of the suffixes 0 and 2, produces a change of sign; thus the second equation may be written $(ca + db)_{02} = k$, or $(ac + bd)_{20} = -k$.

24. The inverse matrix is found to be

$$\begin{pmatrix} a_0, a_1, a_2, a_3 \\ b_0, b_1, b_2, b_3 \\ c_0, c_1, c_2, c_3 \\ d_0, d_1, d_2, d_3 \end{pmatrix}^{-1} = \frac{1}{k}\begin{pmatrix} c_2,\ d_2,\ -a_2,\ -b_2 \\ c_3,\ d_3,\ -a_3,\ -b_3 \\ -c_0,\ -d_0,\ a_0,\ b_0 \\ -c_1,\ -d_1,\ a_1,\ b_1 \end{pmatrix},$$

and the determinant of each of the matrices in this formula is $= k^2$.

We have thus

$$(\Omega_0, \Omega_1, \Omega_2, \Omega_3) = \tfrac{1}{k}(c_2, d_2, -a_2, -b_2 \mid \omega_0, \omega_1, \omega_2, \omega_3), \quad \text{etc.},$$

and the like formula for the T, υ. Substituting these values in the equation

$$k(\Omega_0\mathrm{T}_1 - \Omega_1\mathrm{T}_0 + \Omega_2\mathrm{T}_3 - \Omega_3\mathrm{T}_2) = \omega_0\upsilon_1 - \omega_1\upsilon_0 + \omega_2\upsilon_3 - \omega_3\upsilon_2,$$

we obtain 6 new equations, which are in a different form the conditions for the automorphic transformation.

The 6 new equations may be written

$$\begin{aligned} (02+13)_{ab} &= 0, \\ (02+13)_{ac} &= k, \\ (02+13)_{ad} &= 0, \\ (02+13)_{bc} &= 0, \\ (02+13)_{bd} &= k, \\ (02+13)_{cd} &= 0, \end{aligned}$$

viz. these equations are $a_0c_2 - c_0a_2 + a_1c_3 - c_1a_3 = 0$, &c.

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25. It is worth while to show how the foregoing formula for the inverse matrix comes out. Take for instance the diagonal minor

$$\begin{vmatrix} b_1, b_2, b_3 \\ c_1, c_2, c_3 \\ d_1, d_2, d_3 \end{vmatrix},$$

this is

$$c_1(bd)_{23} + c_2(bd)_{31} + c_3(bd)_{12},$$

which is

$$= c_1(ac)_{23} + c_2\{k + (ac)_{31}\} + c_3(ac)_{12},$$

since the remaining terms destroy each other. And dividing by the determinant, which is $= k^2$, we have the term $c_2 \div k$ of the inverse matrix.

The Symmetrical Hermitian Matrix.

26. We may consider a symmetrical Hermitian matrix, say the matrix

$$\begin{pmatrix} \mathfrak{A},\ \mathfrak{H},\ \mathfrak{G},\ \mathfrak{L} \\ \mathfrak{H},\ \mathfrak{B},\ \mathfrak{F},\ \mathfrak{M} \\ \mathfrak{G},\ \mathfrak{F},\ \mathfrak{C},\ \mathfrak{N} \\ \mathfrak{L},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{D} \end{pmatrix},$$

viz. we have

$$\begin{aligned}\mathfrak{AG}-\mathfrak{H}^2+\mathfrak{HN}-\mathfrak{TF}&=\phi,\\ \mathfrak{HN}-\mathfrak{TB}+\mathfrak{TD}-\mathfrak{M}^2&=\phi,\\ \mathfrak{AB}-\mathfrak{HG}+\mathfrak{HN}-\mathfrak{TF}&=0,\\ \mathfrak{AM}-\mathfrak{TH}+\mathfrak{HD}-\mathfrak{TN}&=0,\\ \mathfrak{GD}-\mathfrak{NG}+\mathfrak{BN}-\mathfrak{MG}&=0,\\ \mathfrak{BN}-\mathfrak{GF}+\mathfrak{GM}-\mathfrak{NF}&=0.\end{aligned}$$

The characteristic property is that, effecting a Hermitian transformation, we have a new symmetrical Hermitian matrix

$$\begin{pmatrix}\mathfrak{A}',\ \mathfrak{H}',\ \mathfrak{G}',\ \mathfrak{T}'\\ \mathfrak{H}',\ \mathfrak{B}',\ \mathfrak{F}',\ \mathfrak{M}'\\ \mathfrak{G}',\ \mathfrak{F}',\ \mathfrak{C}',\ \mathfrak{N}'\\ \mathfrak{T}',\ \mathfrak{M}',\ \mathfrak{N}',\ \mathfrak{D}'\end{pmatrix}=(\mathfrak{A}',\ \mathfrak{H}',\ \mathfrak{G}',\ \mathfrak{T}'\mid x,y,z,w)^2.$$

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In fact, the new matrix is

$$(\mathfrak{A}'\ \ldots)=\begin{pmatrix}a_0,b_0,c_0,d_0\\ a_1,b_1,c_1,d_1\\ a_2,b_2,c_2,d_2\\ a_3,b_3,c_3,d_3\end{pmatrix}(\mathfrak{A}\ \ldots)\begin{pmatrix}a_0,a_1,a_2,a_3\\ b_0,b_1,b_2,b_3\\ c_0,c_1,c_2,c_3\\ d_0,d_1,d_2,d_3\end{pmatrix},$$

where the first factor, qua transposed matrix, is Hermitian. Hence the product is a symmetrical $\frac{k}{\square}\Omega$ matrix.

27. Write

$$\Delta_0=H_0^2-A_0B_0-(\eta^2-a\beta),\quad \theta=a\beta-\eta^2,$$
$$\delta=2H_0\eta-aB_0-\beta A_0,$$

and consider the following matrix

$$\begin{gathered}(\beta,\ -\eta,\ -\eta H_0+\beta A_0,\ -\eta B_0+\beta H_0\mid x,y,z,w)^2\\ -\eta,\ a,\ -\eta A_0+aH_0,\ -\eta H_0+aB_0,\\ -\eta H_0+\beta A_0,\ -\eta A_0+aH_0,\ a\Delta_0-\delta A_0,\ -\delta H_0+\eta\Delta_0,\\ -\eta B_0+\beta H_0,\ -\eta H_0+aB_0,\ \delta H_0-\eta\Delta_0,\ \beta\Delta_0-\delta B_0,\end{gathered}$$

which is

$$=(\beta,\ -\eta,\ a\eta x+A_0y+\beta z+H_0w)^2+\tfrac{1}{\theta}(\eta a-\beta\eta)(az-\beta z+\cdots);$$

it is easily shown that this is a symmetrical θ-matrix. In fact, representing it for a moment by

$$\begin{gathered}(\mathfrak{A},\ \mathfrak{H},\ \mathfrak{G},\ \mathfrak{T})\quad\ldots\\ (\mathfrak{H},\ \mathfrak{B},\ \mathfrak{F},\ \mathfrak{M})\quad\ldots\\ (\mathfrak{G},\ \mathfrak{F},\ \mathfrak{C},\ \mathfrak{N})\quad\ldots\\ (\mathfrak{T},\ \mathfrak{M},\ \mathfrak{N},\ \mathfrak{D})\quad\ldots\end{gathered}$$

so that $\mathfrak{A}=\beta$, $\mathfrak{B}=a$, $\mathfrak{H}=-\eta$, &c., we have

$$\mathfrak{AG}-\mathfrak{H}^2+\mathfrak{HN}-\mathfrak{TF}=\beta(a\Delta_0-\delta A_0)-(-\eta H_0+\beta A_0)^2-\eta(-\delta H_0+\eta\Delta_0)-(-\eta B_0+\beta H_0)(-\eta A_0+aH_0),$$

which is

$$=a\beta\Delta_0-\beta\delta A_0-\eta^2H_0^2+2\beta\eta A_0H_0-\beta^2A_0^2+\eta\delta H_0-\eta^2\Delta_0+\eta^2B_0A_0-\eta aB_0H_0+\eta\beta A_0H_0-a\beta H_0^2;$$

or, for the two terms $\beta\delta A_0 + \eta\delta H_0$, $= (\eta H_0 - \beta A_0)\delta$, substituting the value $\delta = 2H_0\eta - aB_0 - \beta A_0$, the whole is found to be

$$= (a\beta - \eta^2)(\Delta_0 - H_0^2 + A_0B_0), \quad \text{that is,} \quad = \theta(-\theta) = -\theta^2.$$

And, similarly, $\mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{B} + \mathfrak{T}\mathfrak{D} - \mathfrak{M}^2$ is found to be $= -\theta^2$; and the remaining four combinations of terms to be each of them $= 0$; the matrix is thus a symmetrical θ-matrix.

28. It is to be added that the diagonal minors and the determinant

$$\mathfrak{A},\ \mathfrak{A}\mathfrak{B} - \mathfrak{H}^2,\ \mathfrak{A}\mathfrak{B}\mathfrak{G} + \&c.,\ \mathfrak{A}\mathfrak{B}\mathfrak{G}\mathfrak{D} + \&c.,$$

have respectively the values β, θ, $a\theta^2$, θ^4, viz. if a, β, θ are positive, then these are all positive; or the last-mentioned quadric function is a definite positive form.

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The general matrix resumed: Arithmetical theory.

29. The matrix containing, as before, the parameter k, may be called a k-matrix; if k be $= 1$, it is a unit-matrix. From the fundamental equation

$$k(\Omega_0\mathrm{T}_1 - \Omega_1\mathrm{T}_0 + \Omega_2\mathrm{T}_3 - \Omega_3\mathrm{T}_2) = \omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2,$$

it at once follows that, compounding a k-matrix with a k'-matrix, we have a kk'-matrix; and in particular, compounding a k-matrix with a unit-matrix, we have a k-matrix.

30. The symbols a_0, a_1, b_0, &c., and k, have thus far been arbitrary magnitudes, but we now take them to be integers; and we consider in particular the case where k is a positive odd prime. The number of k-matrices is of course infinite, but if we regard as equivalent any two such matrices which are derivable one from the other by post-multiplication by a unit-matrix (viz. U being a unit-matrix, the matrices M and $M \cdot U$ are regarded as equivalent), then the number of distinct k-matrices is finite, and $= 1 + k + k^2 + k^3$.

The first step is to show that we can by post-multiplication, by a properly determined unit-matrix, reduce the k-matrix to the form

$$\begin{pmatrix} a_0,\ a_1,\ a_2,\ a_3 \\ 0,\ b_1,\ b_2,\ b_3 \\ 0,\ 0,\ c_2,\ c_3 \\ 0,\ 0,\ d_2,\ d_3 \end{pmatrix},$$

these values being such as to satisfy identically two out of the six conditions; the remaining conditions present themselves under the two equivalent forms

$$a_0c_2 = k, \quad b_1d_3 = k, \quad a_0b_2 - a_2b_0 = 0, \quad a_0d_1 - a_1d_0 = 0,$$

and

$$a_0c_2 = k, \quad b_1d_3 = k, \quad a_0c_1 - a_1c_0 = 0, \quad b_2d_3 - b_3d_2 = 0.$$

Hence a_0, $c_2 = 1$, k, or k, 1; and b_1, $d_3 = 1$, k, or k, 1; so that, combining these pairs of values, we have four different types of matrix, each type depending on the coefficients a_1, a_2, a_3, b_2, b_3, c_3, connected together by two equations. But the forms of the same type are not distinct from each other, and we have to determine for each of the four types a system of non-equivalent forms comprised therein, and such that from these, by post-multiplication by a unit-matrix as before, the other forms of the type can be obtained. This final system is:

$$\text{I.} \begin{pmatrix} 1,\ 0,\ 0,\ 0 \\ 0,\ 1,\ 0,\ 0 \\ 0,\ 0,\ k,\ 0 \\ 0,\ 0,\ 0,\ k \end{pmatrix}, \qquad \text{II.} \begin{pmatrix} 1,\ 0,\ 0,\ 0 \\ 0,\ k,\ 0,\ i \\ 0,\ 0,\ k,\ 0 \\ 0,\ 0,\ 0,\ 1 \end{pmatrix},$$

$$\text{III.} \begin{pmatrix} k,\ i,\ i',\ 0 \\ 0,\ 1,\ 0,\ 0 \\ 0,\ 0,\ 1,\ 0 \\ 0,\ 0,\ -i,\ k \end{pmatrix}, \qquad \text{IV.} \begin{pmatrix} k,\ 0,\ i,\ i' \\ 0,\ k,\ i,\ i'' \\ 0,\ 0,\ 1,\ 0 \\ 0,\ 0,\ 0,\ 1 \end{pmatrix},$$

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where i, i', i'' are integers, having each of them any one of the values 0, 1, 2, ..., $k-1$, viz. there are in the four types $k-1$, k, k^2, k^3 forms respectively, and the number of forms is thus $= 1+k+k^2+k^3$, as already mentioned. I abstain from the further details of the proof.

There is obviously a like theory for which, in place of post-multiplication, we have pre-multiplication; viz. here the matrices M and $U \cdot M$ are regarded as equivalent.

31. Any two k-matrices are reducible one to the other by a combined pre- and post-multiplication; viz. we have always $M' = U \cdot M \cdot U'$, where M, M' are any two k-matrices, and U, U' being properly determined unit-matrices; and in particular, M any given k-matrix, this is expressible in the foregoing form, where M denotes the principal matrix

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & k, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}.$$

Congruence theorems, k an odd number.

32. Taking k an odd number, and using throughout $=$ instead of $\equiv$ (mod 2), we have the following congruences:

$$(a_0c_0+b_0d_0,\ a_0c_1+b_0d_1,\ a_0c_2+b_0d_2,\ a_0c_3+b_0d_3) = (a_0,\ a_1,\ a_2,\ a_3 \mid \Sigma a_0c_0+b_0d_0,\ a_0c_1+b_0d_1,\ a_0c_2+b_0d_2,\ a_0c_3+b_0d_3 \mid b_0, b_1, b_2, b_3 \mid c_0, c_1, c_2, c_3 \mid d_0, d_1, d_2, d_3),$$

or, conversely,

$$(a_0c_0+b_0d_0,\ a_0c_1+b_0d_1,\ a_0c_2+b_0d_2,\ a_0c_3+b_0d_3) = (c_0,\ -d_0,\ -a_0,\ -b_0 \mid \Sigma\cdots \mid c_1\cdots \mid c_2\cdots \mid c_3\cdots),$$

where observe that on the right-hand side the signs $-$ may be changed into $+$; in fact, to the modulus 2, we have for any integer value whatever $-p = +p$.

33. The first congruence is

$$a_2(a_0c_0+b_0d_0),\ \text{say } = Y,$$
$$= a_0(a_0c_2+\cdots),\ \text{say } = X,$$

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and to verify this, I see no other method than that of considering separately all the combinations of even and odd values of a_0, a_1, a_2, a_3; viz. we have the 16 cases

Case	a_0	a_1	a_2	a_3	X	Y
1	0	0	0	0	does not exist	
2	0	0	0	1	0	0
3	0	0	1	0	0	0
4	0	0	1	1	1	1
5	0	1	0	0	0	0
6	0	1	0	1	0	0
7	0	1	1	0	0	0
8	0	1	1	1	1	1
9	1	0	0	0	0	0
10	1	0	0	1	0	0
11	1	0	1	0	1	1
12	1	0	1	1	1	1
13	1	1	0	0	1	1
14	1	1	0	1	1	1
15	1	1	1	0	1	1
16	1	1	1	1	0	0

viz. the X column gives in each case the value of $X, = a_0c_2 + a_1c_3$, and then it has to be shown that Y has the same value. The coefficients satisfy the conditions

$$\begin{aligned} a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 &= 1, \\ b_0d_1 - b_1d_0 + b_2d_3 - b_3d_2 &= 0, \\ a_0b_1 + a_1b_0 + a_2b_3 - a_3b_2 &= 0, \\ a_0d_1 - a_1d_0 + a_2d_3 - a_3d_2 &= 0, \\ b_0c_1 - b_1c_0 + b_2c_3 - b_3c_2 &= 0, \\ c_0d_1 - c_1d_0 + c_2d_3 - c_3d_2 &= 0, \end{aligned}$$

so that from the first equation we cannot have a_0, a_1, a_2, a_3 each $= 0$, or the first case does not exist. As to the remaining cases, it is easy to see that they group themselves as follows: 2, 3, 5, 9; 4, 13; 6, 7, 10, 11; 8, 12, 14, 15; 16: the proof being substantially the same for the several cases in the same group.

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34. Case 2. We have $Y = b_0d_1$, which should be $= 0$, and in fact, the six equations give $b_0 = 0$, $d_1 = 0$, whence $Y = 0$.

Case 4. We have $Y = c_0 + c_1 + b_0d_1 + b_0d_1$, which should $= 1$, and in fact, the six equations give $c_3 - c_2 = 1$, $b_3 - b_2 = 0$, $d_3 - d_2 = 0$; whence $b_2d_3 - b_3d_2 = 1$, $d_3b_2 - b_3d_2 = 0$, and therefore $Y = 1$.

Case 6. Here $Y = b_0d_1 + d_0b_1$, which should $= 0$; and in fact, the six equations give $b_0 + b_1 = 0$, $d_0 + d_1 = 0$, whence $Y = 0$.

Case 16. Here $Y = c_0 + d_0 + c_2 + c_3 + \mu_0 + \mu_1 + \mu_2 + \mu_3$, which should be $= 0$; the six equations give

$$\begin{aligned} -c_0 - d_0 + c_2 + c_3 = 1, &\quad -b_0 - b_1 + b_2 - b_3 = 0, \\ -d_0 - d_1 + d_2 + d_3 = 0, &\quad b_2d_3 + b_3d_0 + b_1d_3 - b_3d_1 = 1. \end{aligned}$$

Writing $\mu = b_0 + b_1 + b_2 + b_3$ and $\mu' = d_0 + d_1 + d_2 + d_3$, we find

$$\mu\mu' = b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3 + (b_0 + b_2)(d_1 + d_3) + (b_1 + b_3)(d_0 + d_2),$$

that is,

$$\mu\mu' = b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3 + 1,$$

and $c + c_1 + c_2 + c_3 = 1$; whence $Y = 0$.

35. Case 8. This is the only case of any difficulty: we have

$$Y = c_0 + c_2 + b_0d_1 + b_0d_1 + b_0d_1,$$

which should be $= 1$. The six equations give

$$-c_0 + c_2 - c_1 = 1, \quad -b_0 + b_2 - b_1 = 0, \quad -d_0 + d_1 - d_0 = 0,$$

or, say

$$c_0 = -1 - c_1 + c_2, \quad b_0 = -b_1 + b_2, \quad d_0 = d_1;$$

or, substituting these values and omitting even terms

$$Y = d_1 + c_2 - d_0b_2 - b_2d_1.$$

The remaining three of the six equations are

$$\begin{aligned} -b_2d_0 + b_0d_2 - d_0b_1 + b_1d_0 + b_2d_3 - b_3d_2 &= 1, \\ a_0b_2 - b_0 + c_0d_1 - c_2d_0 &= 0, \\ c_1d_3 - c_3d_1 &= 0; \end{aligned}$$

or, substituting for b, c, d their values, these become

$$(1 + c_1 - c_2)b_2 - (b_3 - b_1)c_2 = -\mu_1 - d_2 + b_2c_1,$$

$$-(d_1-d_3)b_2-\mu_3=-(b_3-b_1)d_3-1+\mu_1,$$
$$-c_1-c_3+c_2(c_3+c_2)-d_2(-1+c_2)d_2=-d_3d_1+d_2d_3;$$

we can from these equations eliminate b_2, c_2, d_2; viz. from the first and third equations eliminating c_2, we have

$$\{(d_1-d_3)b_2+\mu_3\}(1-c_0)-(d_3-d_1)(b_1+b_3-b_0d_0)=-(\mu_1+\mu_2)b_2+(b_3-b_1)(-c_3+c_2d_1-c_1d_1),$$

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and this, by means of the second equation, becomes

$$\mu_3-\mu_0+(1-c_1)d_1=(d_3-d_1)(-b_1c_3+b_3c_1)+(b_3-b_1)(-c_3d_1+c_1d_3),$$

viz. reducing, this is

$$\mu_3-\mu_0+d_1=b_3c_1-1+c_1;$$

and in virtue of it we have $Y=c_2+c_3+b_1c_1+1$.

It can be further shown that, to the modulus 2 (k being, as before, odd), we have

$$(a_0c_2+a_1c_3)(c_0c_2+c_1c_3)+(b_0d_2+b_1d_3)(d_0d_2+d_1d_3)=0,$$
$$(a_0c_0+b_0d_0)+(a_1c_1+b_1d_1)+(a_2c_2+b_2d_2)+(a_3c_3+b_3d_3)=0.$$

To prove the first equation, write for a moment

$$X=(a_0c_2-a_2c_0)(a_1c_3-a_3c_1),\quad X'=(b_0d_2-b_2d_0)(b_1d_3-b_3d_1);$$

then in virtue of the equations we have $X=X'$. But we have identically

$$\Omega X=-X+(a_0c_2+a_1c_3)(a_2c_0+a_3c_1)\cdot\Omega$$

and from the equation

$$a_0c_2+a_2c_0+a_1c_3+a_3c_1=1,$$

that is, $a_2c_0+a_3c_1=1-a_0c_2-a_1c_3$, we have $\Omega X+X=0$; and similarly $\Omega'X'+X'=0$, that is, $\Omega X-\Omega'X'=0$; we have thus the required equation $\Omega X+\Omega'X'=0$. In a similar manner the second equation may be verified.

36. Write

$$p=\mu a_0+\nu a_1+qa_2+ra_3+a_0a_2+a_1a_3,$$
$$p'=\mu b_0+\nu b_1+qb_2+rb_3+b_0b_2+b_1b_3,$$
$$q'=\mu c_0+\nu c_1+qc_2+rc_3+c_0c_2+c_1c_3,$$
$$r'=\mu d_0+\nu d_1+qd_2+rd_3+d_0d_2+d_1d_3.$$

It is to be shown that to the modulus 2 we have

$$\mu'q'+\nu'r'=\mu q+\nu r.$$

In fact, forming the value of $\mu'q'+\nu'r'$, we have first a constant term (term without μ, ν, q, r) which vanishes; next writing $\mu^2=\mu$, the whole term in μ is

$$a(d_0d_2+c_1c_3)+b(d_0d_2+a_1a_3)+c(a_0a_2+d_1d_3)+d(b_0b_2+b_1b_3),$$

which also vanishes; and similarly the terms in ν, q, r each of them vanish; there remain only the terms in $\mu\nu$, μq, &c. The coefficient of μq is $a_0c_2+a_2c_0+b_0d_2+b_2d_0$,

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which (always to the modulus 2) is $=a_0c_2+a_2c_0+b_0d_2+b_2d_0$, that is, it is $=1$; and similarly the coefficient of νr is $=1$; and in like manner the coefficients of the other terms are each $=0$; we have thus the required congruence $\mu'q'+\nu'r'=\mu q+\nu r$.

The quintic matrix.

37. I consider a quintic matrix composed of the foregoing coefficients $(a, b, c, d)_{0,1,2,3}$, viz. the matrix contained in a linear transformation which I write as follows:

$$(T', P', Q', R', S') \quad = \begin{pmatrix} \cdot & 01 & 2{\cdot}02 & 03 & 32 \\ ab & & & & \\ \cdot & & & & \\ cb & & & & \\ ac & & * & & \\ ad & & & * & \\ dc & & & & \end{pmatrix} (T, P, Q, R, S),$$

read

$$T' = T + (ab)_{01}P + 2(ab)_{02}Q + (ab)_{03}R + (ab)_{32}S,$$
$$P' = \cdots \quad \text{(and so on)},$$

where as before $(ab)_{01} = a_0b_1 - a_1b_0$, &c., and so in other cases; in particular, observe that, in the expression of Q', the term involving Q is $\{k + 2(ac)_{02}\}Q$ which, in virtue of the relation $(ac + bd)_{02} = k$, may also be written $\{(ac)_{02} - (bd)_{02}\}Q$. I notice that in Hermite's paper, p. 366, the term is in effect written without the $= 2k$, $Q(ac)_{02}$: the correction of this erratum and of a corresponding one, p. 366, is made p. 787 at the conclusion of the memoir.

38. The matrix is automorphic for the form $T^2 - PR - TS$, viz. we have identically

$$Q'^2 - P'R' - T'S' = k(Q^2 - PR - TS).$$

As a partial verification, consider in $Q'^2 - P'R' - T'S'$ the term containing Q^2. The coefficient of Q^2 is

$$\{-k + 2(ac)_{02}\}^2 - 4(ab)_{02}(ad)_{12} - 4(ab)_{02}(dc)_{02},$$

where the first term is

$$k^2 - 4k(ac)_{02} + 4(ac)_{02}^2;$$

Hence, observing that we have $(ac + bd)_{02} = k$, the whole coefficient is k^2, as it should be; and in like manner the verification may be effected for any other term.

39. We require the following formulae:

$$(-c_0, a_0, b_0)(T', P', Q') = k(\ldots,\ b_1,\ -b_2,\ -b_3 \mid T, P, Q, R, S),$$
$$\begin{array}{ll} -c_1, a_1, b_1 : & \ldots,\ b_1,\ b_2,\ b_3, \\ -c_2, a_2, b_2 : & b_1,\ b_2,\ b_3, \\ -c_3, a_3, b_3 : & b_3,\ b_2,\ b_1, \end{array}$$

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that is,

$$-c_0T' + a_0P' + b_0Q' = k(b_1P - Qb_2 - b_3S), \quad \text{\&c.},$$

and

$$(-d_0, a_0, b_0)(T', Q', R') = k(-a_0, -a_1, a_0, a_2, a_3 \mid T, P, Q, R, S),$$
$$\begin{array}{ll} -d_2, a_2, b_2 : & a_0,\ a_1,\ a_2,\ a_3, \\ -d_3, a_3, b_3 : & -a_0,\ a_1,\ a_2,\ a_3. \end{array}$$

From the first set, multiplying the first, third and fourth equations by T, P, Q respectively and adding, and again multiplying the second, third and fourth equations by T, Q, R, and adding, we obtain

$$-(c_1T + c_2P + c_3Q)T + (a_1T + a_2P + a_3Q)P + (b_1T + b_2P + b_3Q)Q = kb_3(Q^2 - PR - TS),$$

$$\text{(similarly for } R) \quad = -kb_2(Q^2 - PR - TS);$$

and similarly, from the second set of equations, we obtain

$$-(d_1T + d_2P + d_3Q)T + (a_0T + a_2P + a_3Q)Q + (b_0T + b_2P + b_3Q)R = -ka_3(Q^2 - PR - TS),$$

$$-(d_1T + d_2Q + d_3R)T + (a_1T + a_2Q + a_3R)Q + (b_1T + b_2Q + b_3R)R = ka_2(Q^2 - PR - TS).$$

40. We have the inverse system

$$(T, P, Q, R, S) = \begin{pmatrix} \cdot & dc & ad & 2bd & cb & ab \\ 03 & & & & & \\ 13 & \cdot & -k & & & \\ 21 & & & & & \\ 01 & & & & & \end{pmatrix} (T', P', Q', R', S'),$$

read

$$T = T' + (dc)_{03}P' + (ad)_{31}Q' + (bd)_{31}R + (cb)_{21} + (ab)^*S';$$

and in particular observe that, in the expression for Q, the term containing Q' is $Q'\{-k + 2(bd)_{13}\}$, where, in virtue of $(ac + bd)_{13} = k$, the coefficient of Q' is also $= (bd)_{13} - (ac)_{13}$.

41. These equations give

$$(a_0, a_2, a_3 \mid T', P', Q') = k(\ldots,\ d_3,\ -a_3,\ -b_3,\ \ldots \mid T, P, Q, R, S),$$

$$b_0, b_2, b_3 : \quad -c_3,\ a_3,\ b_3,$$

$$c_0, c_2, c_3 : \quad \ldots,\ d_3,\ c_3,\ \ldots,\ b_3,$$

$$d_0, d_2, d_3 : \quad \ldots,\ \ldots,\ d_3,\ c_3,\ a_3,$$

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and

$$(a_1, a_2, a_3 \mid T', Q', R') = k(-d_2,\ \ldots,\ a_2,\ b_2,\ \ldots \mid T, P, Q, R, S),$$

$$b_1, b_2, b_3 : \quad c_2,\ -a_2,\ -b_2,\ \ldots,$$

$$c_1, c_2, c_3 : \quad \ldots,\ -d_2,\ c_2,\ \ldots,\ -b_2,$$

$$d_1, d_2, d_3 : \quad \ldots,\ \ldots,\ d_2,\ c_2,\ a_2.$$

From the first set of equations, multiplying the first, third and fourth equations by $-S'$, $-R'$, $+Q'$ respectively, and adding, and again multiplying the second, third and fourth equations by $-S'$, Q', $-P'$ respectively, and adding, we deduce

$$T(-a_0S' - c_0R' + d_0Q') + P(-a_2S' - c_2R' + d_2Q') + Q(-a_3S' - c_3R' + d_3Q') = -kc_3(Q'^2 - P'R' - T'S'),$$

$$T(-b_0S' + c_0Q' - d_0P') + P(-b_2S' + c_2Q' - d_2P') + Q(-b_3S' + c_3Q' - d_3P') = kd_3(Q'^2 - P'R' - T'S');$$

and in like manner, from the second set of equations,

$$T(-a_1S' - c_1R' + d_1Q') + Q(-a_2S' - c_2R' + d_2Q') + R(-a_3S' - c_3R' + d_3Q') = -kd_2(Q'^2 - P'R' - T'S'),$$

$$T(-b_1S' + c_1Q' - d_1P') + Q(-b_2S' + c_2Q' - d_2P') + R(-b_3S' + c_3Q' - d_3P') = kc_2(Q'^2 - P'R' - T'S').$$

42. Assume that T, P, Q, R, S and T', P', Q', R', S' are linearly connected as above; and write

$$T : P : Q : R : S = 1 : A : H : B : H^2 - AB,$$

$$T' : P' : Q' : R' : S' = 1 : A' : H' : B' : H'^2 - A'B',$$

equations which establish a like relation between 1, A, H, B, $H^2 - AB$, and 1, A', H', B', $H'^2 - A'B'$. Observe that these forms are admissible since, if

$$T^2 - PR - QS = 0,$$

then also

$$Q'^2 - P'R' - T'S' = 0.$$

The quantities A, H, B were taken as the parameters of a theta-function; viz. taking

$$A, H, B = A_0 + ia,\ H_0 + i\eta,\ B_0 + i\beta,$$

then $(a, \eta, \beta \mid x, y)^2$ must be a positive form (or what is the same thing, a and $a\beta - \eta^2$ must be positive). If A', H', B' are also the parameters of a theta-function, then writing

$$A', H', B' = A'_0 + ia',\ H'_0 + i\eta',\ B'_0 + i\beta',$$

$(a', \eta', \beta' \mid x, y)^2$ must also be a positive form. It can be shown that, A', H', B' being determined as above, the former condition implies the latter one; viz. if the form $(a, \eta, \beta \mid x, y)^2$ be positive, then also the form $(a', \eta', \beta' \mid x, y)^2$ will be positive.

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Write

$$\begin{aligned}
A &= A_0 + ia, & A' &= A'_0 + ia',\\
B &= B_0 + i\beta, & B' &= B'_0 + i\beta',\\
H &= H_0 + i\eta, & H' &= H'_0 + i\eta',\\
\Delta &= \Delta_0 + i\delta, & &\\
\Delta_0 &= H_0^2 - A_0B_0 - (\eta^2 - a\beta), & &\\
\delta &= 2H_0\eta - B_0a - A_0\beta, & &
\end{aligned}$$

and let the quintic matrix in the foregoing transformation

$$(T', P', Q', R', S') = (M)(T, P, Q, R, S)$$

be represented by

$$\begin{pmatrix} \cdot & t & p & q & r & s \\ 1 & & & & & \\ 2 & & & & & \\ 3 & & & & & \\ 4 & & & & & \end{pmatrix}.$$

43. It is to be shown that $a'x^2 + 2\eta'xy + \beta'y^2$ is definite and positive.

We require a', β', η'; we have

$$A' + ia' = \frac{(t_1, p_1, q_1, r_1, s_1)(1, A_0 + ia, H_0 + i\eta, B_0 + i\beta, \Delta_0 + i\delta)}{(t_0, p_0, q_0, r_0, s_0)\cdots} = \frac{M_1 + N_1 i}{M_0 + N_0 i},$$

and thence

$$a' = \frac{M_0N_1 - M_1N_0}{M_0^2 + N_0^2} = \tfrac{1}{\Box}(M_0N_1 - M_1N_0).$$

Now

$$M_0N_1 - M_1N_0 = \begin{aligned}[t]
&(t_0p_1 - t_1p_0)\,a\\
&+(t_0q_1 - t_1q_0)\,\eta\\
&+(t_0r_1 - t_1r_0)\,\beta\\
&+(t_0s_1 - t_1s_0)\,\delta\\
&+(p_0q_1 - p_1q_0)(A_0\eta - H_0a)\\
&+(p_0r_1 - p_1r_0)(A_0\beta - B_0a)\\
&+(p_0s_1 - p_1s_0)(A_0\delta - \Delta_0a)\\
&+(q_0r_1 - q_1r_0)(H_0\beta - B_0\eta)\\
&+(q_0s_1 - q_1s_0)(H_0\delta - \Delta_0\eta)\\
&+(r_0s_1 - r_1s_0)(B_0\delta - \Delta_0\beta);
\end{aligned}$$

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and so for η', β' with only the change of t_1, p_1, q_1, r_1, s_1 into t_2, p_2, q_2, r_2, s_2, and t_3, p_3, q_3, r_3, s_3 respectively; we find, omitting a factor $\dfrac{k}{\square}$ throughout,

	$a' =$	$\eta' =$	$\beta' =$
a	b_1^2	$-a_1b_1$	a_1^2
η	$-2b_1b_3$	$a_1b_3+a_3b_1$	$-2a_1a_3$
β	b_3^2	$-a_3b_3$	a_3^2
δ	$-(b_1b_2+b_0b_3)$	$\frac{1}{2}(a_1b_2+a_2b_1+a_0b_3+a_3b_0)$	$-(a_1a_2+a_0a_3)$
$A_0\eta-H_0a$	$-2b_1b_2$	$a_1b_2+a_2b_1$	$-2a_1a_2$
$H_0\beta-B_0\eta$	$-2b_2b_3$	$a_2b_3+a_3b_2$	$-2a_2a_3$
$A_0\beta-B_0a$	$b_0b_3-b_1b_2$	$-\frac{1}{2}(a_1b_2+a_2b_1-a_0b_3-a_3b_0)$	$a_0a_3-a_1a_2$
$A_0\delta-\Delta_0a$	$-b_1^2$	a_1b_1	$-a_1^2$
$H_0\delta-\Delta_0\eta$	$-2b_1b_2$	$a_1b_2+a_2b_1$	$-2a_1a_2$
$B_0\delta-\Delta_0\beta$	$-b_3^2$	a_3b_3	$-a_3^2$

and hence

$$\begin{aligned} &a'x^2+2\eta'xy+\beta'y^2\\ &=\tfrac{k}{\square}\{\beta,\ -\eta,\ -\eta H_0+\beta A_0,\ -\eta B_0+\beta H_0 \mid x,y\}^2. \end{aligned}$$

The right-hand side is here definite and positive (*suprà* No. 28), hence also the left-hand is definite and positive.

44. As a specimen of the work, observe that we have

$$\begin{aligned} t_0p_1-t_1p_0 &= (a_0c_1-a_1c_0)(ab)_{01}\\ &= b_1\{b_1(ac)_{01}+b_0(ac)_{10}+b_1(ac)_{01}\}\\ &= b_1\{-b_1(bd)_{01}-b_0(bd)_{10}+b_1\{-(bd)_{01}+k\}\}\\ &= b_1^2\cdot k, \end{aligned}$$

since the remaining terms destroy each other. Dividing by $\square$, and then omitting the factor $\dfrac{k}{\square}$, we have thus the term ab_1^2 in the foregoing expression for a'.

The Transformation. Art. Nos. 45 to 53.

45. Consider $(a,b,c,d)_{0,1,2,3}$, the components of a quartic matrix as above, and also T, P, Q, R, S; T', P', Q', R', S' connected as already mentioned; and write for shortness

$$z_0=a_0x+b_0y,\quad z_1=a_1x+b_1y,$$

$$z_2=a_2x+b_2y,\quad z_3=a_3x+b_3y,$$

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848 (continued). On the Transformation of the Double-Theta Functions.

(Continuation of Article 45; printed page 381.)

and consider the function

$$\Pi(x,y) \;=\; \Theta\big(z_\circ + Az_2 + Hz_3,\; z_1 + Hz_2 + Bz_3\big)\; \exp. i\pi\{(z_\circ z_2 + z_1 z_3) + (A, H, B \mid z_2, z_3)^2\},$$

where Θ denotes the function

$$\Theta\begin{pmatrix}\mu, & \nu\\ q, & r\end{pmatrix}(A', H', B').$$

It is to be shown that Π is a function

$$\Pi\begin{pmatrix}\mu', & \nu'\\ q', & r'\end{pmatrix}(A', H', B'),$$

where the new parameters A', H', B' are given in terms of the original parameters A, H, B by the equations

$$1 : A : H : B : H^2 - AB = T : P : Q : R : S,$$

$$1 : A' : H' : B' : H'^2 - A'B' = T' : P' : Q' : R' : S',$$

and where μ', ν', q', r' have the values

$$\begin{aligned}
\mu' &= \mu a_\circ + \nu a_1 + q a_2 + r a_3 + a_\circ a_2 + a_1 a_3,\\
\nu' &= \mu b_\circ + \nu b_1 + q b_2 + r b_3 + b_\circ b_2 + b_1 b_3,\\
q' &= \mu c_\circ + \nu c_1 + q c_2 + r c_3 + c_\circ c_2 + c_1 c_3,\\
r' &= \mu d_\circ + \nu d_1 + q d_2 + r d_3 + d_\circ d_2 + d_1 d_3;
\end{aligned}$$

viz. it is to be shown that the function Π, as above defined, satisfies the fundamental equations

$$\begin{aligned}
\Pi(x+1, y) &= (-)^{\mu}\, \Pi(x,y),\\
\Pi(x, y+1) &= (-)^{\nu}\, \Pi(x,y),\\
\Pi(x+A, y+H) &= (-)^{q}\, \Pi(x,y) \exp. -2\pi k(2x + A'),\\
\Pi(x+H, y+B) &= (-)^{r}\, \Pi(x,y) \exp. -2\pi k(2y + B'),
\end{aligned}$$

and

$$\Pi(-x,-y) = (-)^{\mu'+\nu'+q'+r'}\, \Pi(x,y).$$

46. It is proper in the first place to show how it is that A', H', B' are capable of being the parameters of the new function.

Write for shortness

$$X = z_\circ + Az_2 + Hz_3, \qquad Y = z_1 + Hz_2 + Bz_3,$$

and suppose x changed into $x+1$: we have $z_\circ$, z_1, z_2, z_3 increased by $a_\circ$, a_1, a_2, a_3 respectively; and thence X, Y increased by $a_\circ + Aa_2 + Ha_3$, $a_1 + Ha_2 + Ba_3$; the theta-function is thus changed into

$$\Theta(X + a_\circ + Aa_2 + Ha_3,\; Y + a_1 + Ha_2 + Ba_3),$$

viz. the arguments are increased by multiples of the quarter-periods $(1, 0, A, H)$ and $(0, 1, H, B)$.

And, similarly, by the change of y into $y + 1$, the theta-function is changed into

$$\Theta(X + b_\circ + Ab_2 + Hb_3,\ Y + b_1 + Hb_2 + Bb_3),$$

or the arguments are increased by integer multiples of the quarter-periods.

47. But suppose next that x, y are changed into $x + A'$, $y + H'$: we have $z_\circ$, z_1, z_2, z_3 increased by

$$A'a_\circ + H'b_\circ,\quad A'a_1 + H'b_1,\quad A'a_2 + H'b_2,\quad A'a_3 + H'b_3,$$

and thence

$$\begin{aligned}&X \text{ increased by } A'(a_\circ + Aa_2 + Ha_3) + H'(b_\circ + Ab_2 + Hb_3) = c_\circ + Ac_2 + Hc_3,\\ &Y \quad ,, \quad A'(a_1 + Ha_2 + Ba_3) + H'(b_1 + Hb_2 + Bb_3) = c_1 + Hc_2 + Bc_3,\end{aligned}$$

since these equalities are the before-mentioned equations

$$P(Ta_\circ + Pa_2 + Qa_3) + Q(Tb_\circ + Pb_2 + Qb_3) - T(Tc_\circ + Pc_2 + Qc_3) = 0,$$

$$P(Ta_1 + Qa_2 + Ra_3) + Q(Tb_1 + Qb_2 + Rb_3) - T(Tc_1 + Qc_2 + Rc_3) = 0.$$

It thus appears that, by the change of x, y into $x + A'$, $y + H'$, the theta-function is changed into

$$\Theta(X + c_\circ + Ac_2 + Hc_3,\ Y + c_1 + Hc_2 + Bc_3),$$

viz. we have again the arguments increased by integer multiples of the quarter-periods; and in like manner, by the change of x, y into $x + H'$, $y + B'$, the theta-function is changed into

$$\Theta(X + d_\circ + Ad_2 + Hd_3,\ Y + d_1 + Hd_2 + Bd_3),$$

or the arguments are again increased by integer multiples of the quarter-periods.

48. We have to complete the verifications, first for the equation

$$\Pi(x + 1, y) = (-)^\mu\, \Pi(x, y).$$

Reverting to the definition of Π, we have

$$\begin{aligned}(-)^\mu\, \Pi(x+1, y) \div \Pi(x, y) &= (-)^{\mu + a_2 + a_3 + a_\circ a_2 + a_1 a_3} \exp. - i\pi\{a_\circ a_2 + a_1 a_3\};\\ \Theta(X + a_\circ + Aa_2 + Ha_3,\ &Y + a_1 + Ha_2 + Ba_3) \div \Theta(X, Y)\\ &= \exp.\, i\pi[a_\circ z_2 + a_2 z_\circ + a_1 z_3 + a_3 z_1 + a_\circ a_2 + a_1 a_3] \times\\ &\quad \exp.\, i\pi\big(2(Aa_2 + Ha_3)z_2 + 2(Ha_2 + Ba_3)z_3 + (A, H, B \mid a_2, a_3)^2\big);\end{aligned}$$

the second line of this is

$$= (-)^{\mu + a_2 + a_3 + a_\circ a_2 + a_1 a_3} \exp.\, i\pi\{-2a_2 X - 2a_3 Y - (A, H, B \mid a_2, a_3)^2\},$$

and the right-hand side of the equation will thus be $= 1$ if only the whole argument of the exponential be $=$; that is, omitting the terms which destroy each other, and the common factor $i\pi$, if only

$$\begin{aligned}&- 2a_2 X - 2a_3 Y\\ &+ a_\circ z_2 + a_2 z_\circ + a_1 z_3 + a_3 z_1\\ &+ 2(Aa_2 + Ha_3)z_2 + 2(Ha_2 + Ba_3)z_3 = 0.\end{aligned}$$

Substituting herein for X, Y their values

$$z_\circ + Az_2 + Hz_3, \qquad z_1 + Hz_2 + Bz_3,$$

the equation becomes

$$-a_2 z_\circ - a_3 z_1 + a_1 z_3 + a_\circ z_2 = 0,$$

and finally substituting herein for $z_\circ$, z_1, z_2, z_3 their values, the coefficient of x is identically $= 0$, and the coefficient of y is

$$-a_2 b_\circ - a_3 b_1 + a_1 b_3 + a_\circ b_2,$$

which is $(02+13)\, ab$, and is thus $= 0$. This completes the proof of the equation

$$\Pi(x+1, y) = (-)^\mu\, \Pi(x, y);$$

and we have of course a precisely similar proof for the equation

$$\Pi(x, y+1) = (-)^\nu\, \Pi(x, y).$$

49. For the next equation, writing for shortness $\alpha_\circ$, α_1, α_2, α_3 for $A'a_\circ + H'b_\circ$, $A'a_1 + H'b_1$, $A'a_2 + H'b_2$, $A'a_3 + H'b_3$, so that $z_\circ$, z_1, z_2, z_3 are increased by $\alpha_\circ$, α_1, α_2, α_3 respectively, we have

$$\begin{aligned}(-)^q\, \Pi(x + A', y + H') \div \Pi(x, y) = (-)^{q + a_2 + a_3 + \alpha_\circ\alpha_2 + \alpha_1\alpha_3} \exp . -i\pi\{\alpha_\circ\alpha_2 + \alpha_1\alpha_3\};\\ \Theta(X + c_\circ + Ac_2 + Hc_3,\ Y + c_1 + Hc_2 + Bc_3) \div \Theta(X, Y)\\ = \exp . i\pi\{\alpha_\circ z_2 + \alpha_2 z_\circ + \alpha_1 z_3 + \alpha_3 z_1 + \alpha_\circ\alpha_2 + \alpha_1\alpha_3\} \times\\ \exp . i\pi\{2(A\alpha_2 + H\alpha_3)z_2 + 2(H\alpha_2 + B\alpha_3)z_3 + (A, H, B \mid \alpha_2, \alpha_3)^2\}.\end{aligned}$$

The second line is here

$$= (-)^{q + a_2 + a_3 + \alpha_\circ\alpha_2 + \alpha_1\alpha_3} \exp . i\pi\{-2c_2 X - 2c_3 Y - (A, H, B \mid c_2, c_3)^2\},$$

and the whole expression should be

$$= \exp . i\pi\{-2kx - kA'\}.$$

Hence bringing these terms over to the left-hand side, the equation will be satisfied if only the whole argument of the exponential be $=$; viz. omitting the factor $i\pi$, the equation will be satisfied if only

$$\begin{aligned}&\alpha_\circ z_2 + \alpha_2 z_\circ + \alpha_1 z_3 + \alpha_3 z_1 + \alpha_\circ\alpha_2 + \alpha_1\alpha_3\\ &+ 2(A, H, B \mid \alpha_2, \alpha_3 \mid z_2, z_3) + (A, H, B \mid \alpha_2, \alpha_3)^2\\ &- 2c_2 X - 2c_3 Y - (A, H, B \mid c_2, c_3)^2 + k(2x + A') = 0.\end{aligned}$$

Substituting here for X, Y their values

$$z_\circ + Az_2 + Hz_3, \qquad z_1 + Hz_2 + Bz_3,$$

and attending to the values

$$a_\circ x + b_\circ y, \quad a_1 x + b_1 y, \quad a_2 x + b_2 y, \quad a_3 x + b_3 y$$

of $z_\circ$, z_1, z_2, z_3, the equation contains a term in x, a term in y, and a constant term; and we may consider these terms separately.

50. The term in x will vanish if

$$\alpha_\circ a_2 + \alpha_2 a_\circ + \alpha_1 a_3 + \alpha_3 a_1 + 2(A\alpha_2 + H\alpha_3)a_2 + 2(H\alpha_2 + B\alpha_3)a_3 - 2c_2(a_\circ + Aa_2 + Ha_3) - 2c_3(a_1 + Ha_2 + Ba_3) + 2k = 0,$$

and substituting herein for $\alpha_\circ$, α_1, α_2, α_3 their values

$$A'a_\circ + H'b_\circ, \quad A'a_1 + H'b_1, \quad A'a_2 + H'b_2, \quad A'a_3 + H'b_3,$$

the equation becomes

$$\begin{aligned}&A'[2a_\circ a_2 + 2a_1 a_3 + 2(Aa_2 + Ha_3)a_2 + 2(Ha_2 + Ba_3)a_3]\\ &+ H'[a_\circ b_2 + a_2 b_\circ + a_1 b_3 + a_3 b_1 + 2(Aa_2 + Ha_3)b_2 + 2(Ha_2 + Ba_3)b_3]\\ &- 2c_2(a_\circ + Aa_2 + Ha_3) - 2c_3(a_1 + Ha_2 + Ba_3) + 2k = 0,\end{aligned}$$

or observing that the first and second lines may be expressed in the form

$$\begin{aligned}&A'[2a_2(a_\circ + Aa_2 + Ha_3) + 2a_3(a_1 + Ha_2 + Ba_3)]\\ &+ H'[2b_2(a_\circ + Aa_2 + Ha_3) + 2b_3(a_1 + Ha_2 + Ba_3)] - (a_\circ b_2 - a_2b_\circ + a_1b_3 - a_3b_1)\end{aligned}$$

where the term $a_\circ b_2 - a_2b_\circ + a_1b_3 - a_3b_1$, that is, $(02 + 13)\,ab$, is $= 0$, and may therefore be omitted, the whole equation, omitting the factor 2, which divides out, is

$$(A'a_2 + H'b_2 - c_2)(a_\circ + Aa_2 + Ha_3) + (A'a_3 + H'b_3 - c_3)(a_1 + Ha_2 + Ba_3) + k = 0,$$

an equation which, in the form

$$(-c_2T + a_2P + b_2Q)(a_\circ T + a_2P + a_3Q) + (-c_3T + a_3P + b_3Q)(a_1T + a_2Q + a_3R) + kTT = 0,$$

has been above shown to be true. The term in x thus vanishes.

51. The term in y will vanish if

$$\begin{aligned}&\alpha_\circ b_2 + \alpha_2 b_\circ + \alpha_1 b_3 + \alpha_3 b_1 + 2(A\alpha_2 + H\alpha_3)b_2 + 2(H\alpha_2 + B\alpha_3)b_3\\ &\quad - 2c_2(b_\circ + Ab_2 + Hb_3) - 2c_3(b_1 + Hb_2 + Bb_3) = 0;\end{aligned}$$

and this is in a similar manner reduced to

$$(A'a_2 + H'b_2 - c_2)(b_\circ + Ab_2 + Hb_3) + (A'a_3 + H'b_3 - c_3)(b_1 + Hb_2 + Bb_3) = 0,$$

an equation which, in the form

$$(-c_2T + a_2P + b_2Q)(b_\circ T + b_2P + b_3Q) + (-c_3T + a_3P + b_3Q)(b_1T + b_2Q + b_3R) = 0,$$

has been above shown to be true. The term in y thus vanishes.

52. It only remains to show that the constant term also vanishes, viz. that we have

$$\alpha_\circ\alpha_2 + \alpha_1\alpha_3 + (A, H, B \mid \alpha_2, \alpha_3)^2 - c_\circ c_2 - c_1c_3 - (A, H, B \mid c_2, c_3)^2 + kA' = 0.$$

Substituting here for $\alpha_\circ$, α_1, α_2, α_3 their values

$$= A'a_\circ + H'b_\circ, \quad A'a_1 + H'b_1, \quad A'a_2 + H'b_2, \quad A'a_3 + H'b_3,$$

the equation becomes

$$\begin{aligned}&A'^2(a_\circ a_2 + a_1a_3) + A'H'(a_\circ b_2 + a_2b_\circ + a_1b_3 + a_3b_1) + H'^2(b_\circ b_2 + b_1b_3)\\ &+ A[A'^2a_2^2 + 2A'H'a_2b_2 + H'^2b_2^2]\\ &+ 2H[A'^2a_2a_3 + A'H'(a_2b_3 + a_3b_2) + H'^2b_2b_3]\\ &+ B[A'^2a_3^2 + 2A'H'a_3b_3 + H'^2b_3^2]\\ &- c_\circ c_2 - c_1c_3 - (A, H, B \mid c_2, c_3)^2 + kA' = 0.\end{aligned}$$

This may be written

$$\begin{aligned}&A'^2\{a_2(a_\circ + Aa_2 + Ha_3) + a_3(a_1 + Ha_2 + Ba_3)\}\\ &\quad + A'H'\{b_2(a_\circ + Aa_2 + Ha_3) + b_3(a_1 + Ha_2 + Ba_3)\\ &\qquad\qquad + a_2(b_\circ + Ab_2 + Hb_3) + a_3(b_1 + Hb_2 + Bb_3)\}\\ &\quad + H'^2\{b_2(b_\circ + Ab_2 + Hb_3) + b_3(b_1 + Hb_2 + Bb_3)\}\\ &\quad - c_2(c_\circ + Ac_2 + Hc_3) - c_3(c_1 + Hc_2 + Bc_3) + kA' = 0.\end{aligned}$$

Writing herein

$$\begin{aligned}c_\circ + Ac_2 + Hc_3 &= (a_\circ + Aa_2 + Ha_3)A' + (b_\circ + Ab_2 + Hb_3)H',\\ c_1 + Hc_2 + Bc_3 &= (a_1 + Ha_2 + Ba_3)A' + (b_1 + Hb_2 + Bb_3)H',\end{aligned}$$

equations which are true in virtue of

$$\Theta = 0,$$

the equation becomes

$$\begin{aligned}&(a_\circ + Aa_2 + Ha_3)\,A'(A'a_2 + H'b_2 - c_2)\\ &+ (a_1 + Ha_2 + Ba_3)\,A'(A'a_3 + H'b_3 - c_3)\\ &+ (b_\circ + Ab_2 + Hb_3)\,H'(A'a_2 + H'b_2 - c_2)\\ &+ (b_1 + Hb_2 + Bb_3)\,H'(A'a_3 + H'b_3 - c_3) + kA' = 0,\end{aligned}$$

that is,

$$\begin{aligned}&A'[(a_\circ + Aa_2 + Ha_3)(-c_2 + A'a_2 + H'b_2) + (a_1 + Ha_2 + Ba_3)(-c_3 + A'a_3 + H'b_3) + k]\\ &\quad + H'[(b_\circ + Ab_2 + Hb_3)(-c_2 + A'a_2 + H'b_2) + (b_1 + Hb_2 + Bb_3)(-c_3 + A'a_3 + H'b_3)] = 0;\end{aligned}$$

and we have the coefficients of A' and H' each $= 0$, in virtue of

$$\begin{aligned}&(a_\circ T + a_2P + a_3Q)(-c_2T + a_2P + b_2Q) + (a_1T + a_2Q + a_3R)(-c_3T + a_3P + b_3Q) + kTT = 0,\\ &(b_\circ T + b_2P + b_3Q)(-c_2T + a_2P + b_2Q) + (b_1T + b_2Q + b_3R)(-c_3T + a_3P + b_3Q) = 0.\end{aligned}$$

53. This completes the proof of the equation

$$\Pi(x + A', y + H') = (-)^q\,\Pi(x, y)\exp. - i\pi k(2x + A');$$

the proof of the remaining equation

$$\Pi(x + H', y + B') = (-)^r\,\Pi(x, y)\exp. - i\pi k(2y + B'),$$

is of course precisely similar.

It has already been seen that $\mu'q + \nu'r = \mu q + \nu r$; we have

$$\Theta(-X, -Y) = (-)^{\mu+\nu}\,\Theta(X, Y),$$

and thence

$$\Pi(-x, -y) = (-)^{\mu+\nu}\,\Pi(x, y),$$

that is,

$$\Pi(-x, -y) = (-)^{\mu'+\nu'}\,\Pi(x, y),$$

which is the last of the equations which should be satisfied by the function $\Pi(x, y)$.

Recapitulation, and Final Form. Art. Nos. 54 to 58.

54. Recapitulating, we have a Hermitian k-matrix (k an odd prime)

$$\begin{pmatrix} a_\circ, & a_1, & a_2, & a_3\\ b_\circ, & b_1, & b_2, & b_3\\ c_\circ, & c_1, & c_2, & c_3\\ d_\circ, & d_1, & d_2, & d_3\end{pmatrix},$$

susceptible of $(1 + k + k^2 + k^3)$ forms; further, writing for shortness

$$\begin{aligned}z_\circ &= a_\circ x + b_\circ y,\\ z_1 &= a_1x + b_1y,\\ z_2 &= a_2x + b_2y,\\ z_3 &= a_3x + b_3y,\end{aligned}$$

and then

$$X = z_\circ + Az_2 + Hz_3, \qquad Y = z_1 + Hz_2 + Bz_3,$$

and assuming

$$\Pi(x,y) = \Theta\begin{pmatrix}\mu, & \nu\\ q, & r\end{pmatrix}(X,Y)(A,H,B)\exp.i\pi\{(z_\circ z_2 + z_1 z_3) + (A,H,B \mid z_2, z_3)^2\},$$

then $\Pi(x,y)$ satisfies the equations

$$\begin{aligned}\Pi(x+1,y) &= (-)^\mu\,\Pi(x,y),\\ \Pi(x,y+1) &= (-)^\nu\,\Pi(x,y),\\ \Pi(x+A', y+H') &= (-)^q\,\Pi(x,y)\exp.-i\pi k(2x+A'),\\ \Pi(x+H', y+B') &= (-)^r\,\Pi(x,y)\exp.-i\pi k(2y+B'),\\ \Pi(-x,-y) &= (-)^{\mu'+\nu'}\,\Pi(x,y),\end{aligned}$$

and it is consequently a function

$$\Pi\begin{pmatrix}\mu', & \nu'\\ q', & r'\end{pmatrix}(x,y)(A',H',B')(K,\text{def.}),$$

and as such, contains linearly $\frac{1}{2}(k^2+1)$ constants.

55. The values of the new parameters are given as above, viz. T', P', Q', R', S' being linear functions of T, P, Q, R, S (the coefficients in these relations being given functions of the coefficients $(a,b,c,d \mid 1)_{0,1,2,3}$ of the Hermitian matrix), then

$$1 : A : H : B : H^2 - AB = T : P : Q : R : S,$$

$$1 : A' : H' : B' : H'^2 - A'B' = T' : P' : Q' : R' : S',$$

which represent the required relations.

56. We consider four such functions $\Pi(x,y)$, derived from Θ-functions having respectively the characteristics

$$\begin{pmatrix}\mu_\circ, & \nu_\circ\\ q_\circ, & r_\circ\end{pmatrix}, \begin{pmatrix}\mu_1, & \nu_1\\ q_1, & r_1\end{pmatrix}, \begin{pmatrix}\mu_2, & \nu_2\\ q_2, & r_2\end{pmatrix}, \begin{pmatrix}\mu_3, & \nu_3\\ q_3, & r_3\end{pmatrix},$$

being a (0123) system; and having consequently the characteristics

$$\begin{pmatrix}\mu'_\circ, & \nu'_\circ\\ q'_\circ, & r'_\circ\end{pmatrix}, \begin{pmatrix}\mu'_1, & \nu'_1\\ q'_1, & r'_1\end{pmatrix}, \begin{pmatrix}\mu'_2, & \nu'_2\\ q'_2, & r'_2\end{pmatrix}, \begin{pmatrix}\mu'_3, & \nu'_3\\ q'_3, & r'_3\end{pmatrix},$$

which also form a (0123) system; say the four are $\Pi_\circ$, Π_1, Π_2, Π_3.

This being so, consider four theta-functions

$$\theta(x,y)(A',H',B')$$

having respectively the characteristics

$$\begin{pmatrix}\mu'_\circ, & \nu'_\circ\\ q'_\circ, & r'_\circ\end{pmatrix}, \begin{pmatrix}\mu'_1, & \nu'_1\\ q'_1, & r'_1\end{pmatrix}, \begin{pmatrix}\mu'_2, & \nu'_2\\ q'_2, & r'_2\end{pmatrix}, \begin{pmatrix}\mu'_3, & \nu'_3\\ q'_3, & r'_3\end{pmatrix},$$

which as already mentioned form a (0123) system; form with these the four sums

$$\Sigma\,\theta_\circ^a\,\theta_1^b\,\theta_2^c\,\theta_3^d,$$

where in each case $a+b+c+d=k$, but in the first sum a, in the second sum b, in the third sum c, and in the fourth sum d is of contrary parity to the other three letters (even, if they are odd; odd, if they are even), say the four sums are $\Sigma_\circ$, Σ_1, Σ_2, Σ_3, respectively; each sum depends linearly on $\frac{1}{2}(k^2+1)$ constants.

The general transformation theorem then is

$$\Sigma_\circ = \Pi_\circ, \quad \Sigma_1 = \Pi_1, \quad \Sigma_2 = \Pi_2, \quad \Sigma_3 = \Pi_3;$$

each of which equations in fact represents $\frac{1}{2}(k^2+1)$ equations; viz. on the left-hand side we assign to the constants $\frac{1}{2}(k^2+1)$ systems of values at pleasure, then to each such system there corresponds on the right-hand side a determinate system of values of the $\frac{1}{2}(k^2+1)$ constants.

57. The results may be presented in a more symmetrical form. Writing ξ, η, ζ, ω for the foregoing x, y, z, w, we may consider

$$\Theta\begin{pmatrix}\mu, & \nu\\ q, & r\end{pmatrix}(\xi + A'\zeta + H'\omega,\ \eta + H'\zeta + B'\omega)(A', H', B')$$

as a function of the four arguments ξ, η, ζ, ω, and express it by

$$\Phi\begin{pmatrix}\mu, & \nu\\ q, & r\end{pmatrix}(\xi, \eta, \zeta, \omega)(A', H', B').$$

Writing then

$$\begin{aligned}
x &= \xi + A'\zeta + H'\omega,\\
y &= \eta + H'\zeta + B'\omega,\\
X, Y, Z, W &= \begin{pmatrix} a_\circ, & b_\circ, & c_\circ, & d_\circ\\ a_1, & b_1, & c_1, & d_1\\ a_2, & b_2, & c_2, & d_2\\ a_3, & b_3, & c_3, & d_3\end{pmatrix}(\xi, \eta, \zeta, \omega),
\end{aligned}$$

we have

$$\begin{aligned}
z_\circ = a_\circ x + b_\circ y &= a_\circ(\xi + A'\zeta + H'\omega) + b_\circ(\eta + H'\zeta + B'\omega)\\
&= (a_\circ\xi + b_\circ\eta) + A'(a_\circ\zeta + b_\circ\omega) + H'(a_\circ\omega + b_\circ\cdots)
\end{aligned}$$

and also

$$\begin{aligned}
X + AZ + HW = {} & (a_\circ + Aa_2 + Ha_3)\,\xi\\
& + (b_\circ + Ab_2 + Hb_3)\,\eta\\
& + A(c_\circ + Ac_2 + Hc_3)\,\zeta\\
& + H(c_\circ + \cdots)\,\omega + (c_\circ + Ac_2 + Hc_3)\,\zeta + (d_\circ + Ad_2 + Hd_3)\,\omega,
\end{aligned}$$

or since, as above,

$$\begin{aligned}
A'(a_\circ + Aa_2 + Ha_3) + H'(b_\circ + Ab_2 + Hb_3) &= c_\circ + Ac_2 + Hc_3,\\
H'(a_\circ + Aa_2 + Ha_3) + B'(b_\circ + Ab_2 + Hb_3) &= d_\circ + Ad_2 + Hd_3,
\end{aligned}$$

we find

$$z_\circ + Az_2 + Hz_3 = X + AZ + HW,$$

and in like manner another equation, viz. we have

$$z_1 + Hz_2 + Bz_3 = Y + HZ + BW.$$

Moreover

$$\begin{aligned}
z_\circ z_2 + z_1 z_3 + (A, H, B \mid z_2, z_3)^2 &= z_2(z_\circ + Az_2 + Hz_3) + z_3(z_1 + Hz_2 + Bz_3)\\
&= (a_2x + b_2y)(X + AZ + HW) + (a_3x + b_3y)(Y + HZ + BW)\\
&= a_2(\xi + A'\zeta + H'\omega)(X + AZ + HW) + a_3(\xi + A'\zeta + H'\omega)(Y + HZ + BW) + \cdots
\end{aligned}$$

which qua function of X, Y, Z, W may be called χ.

58. And we then have the final result in the following form, viz.

$$\exp . i\pi\chi \cdot \Phi\begin{pmatrix}\mu, & \nu \\ q, & r\end{pmatrix}(X, Y, Z, W)(A, H, B),$$

in each of the four forms, is a homogeneous function of the order k of the corresponding four forms

$$\Phi\begin{pmatrix}\mu', & \nu' \\ q', & r'\end{pmatrix}(\xi, \eta, \zeta, \omega)(A', H', B').$$

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849. On the Invariants of a Linear Differential Equation.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 257–261.]

CONSIDER the equation of the second order

$$\frac{d^2y}{dx^2} + 2p\frac{dy}{dx} + qy = 0;$$

if we effect a transformation of the dependent variable y, say $y = f \cdot Y$, where f is an arbitrary function of x, we obtain a new equation

$$\frac{d^2Y}{dx^2} + 2P\frac{dY}{dx} + QY = 0,$$

where

$$P = p + \frac{f'}{f},$$

$$Q = q + 2p\frac{f'}{f} + \frac{f''}{f},$$

(the accents denoting differentiation in regard to x); and we thence establish the identity

$$Q - P^2 - P' = q - p^2 - p',$$

viz. we have $q - p^2 - p'$ a function possessing this invariantive property, but, as remarked by Mr Harley, it is the analogue rather of a seminvariant than of an invariant; I will call it an α-seminvariant of the differential equation. The class of function was considered long ago by Sir James Cockle, and more recently by Mr Malet; see Mr Harley's paper "Professor Malet's Classes of Invariants identified with Sir James Cockle's Criticoids," Proc. R. Soc., vol. xxxviii. (1884), pp. 45–57.

Effecting for the same equation a transformation of the independent variable, say $x = \phi$, an arbitrary function of X, we obtain a new equation

$$\frac{d^2y}{dX^2} + 2P\frac{dy}{dX} + Qy = 0,$$

where

$$P = p\phi_1 - \tfrac{1}{2}\frac{\phi_2}{\phi_1},$$

$$Q = q\phi_1^2,$$

(the subscript numbers denoting differentiation in regard to X). There is not in the present case any function possessing a like invariantive property, say there is not any β-seminvariant; but such functions exist for differential equations of the third and higher orders, and have been considered by Sir James Cockle and Mr Malet. It is to be noticed that, attempting to obtain a β-seminvariant, we are led to the equation

$$Q - P^2 - P_1 = \phi_1^2(q - p^2 - p') + \tfrac{1}{2}\left\{\frac{\phi_3}{\phi_1} - \tfrac{3}{2}\left(\frac{\phi_2}{\phi_1}\right)^2\right\},$$

which but for the last term would be invariantive. It is curious to find this last term presenting itself in the form of a Schwarzian derivative.

Passing to the differential equation of the third order

$$\frac{d^3y}{dx^3} + 3p\,\frac{d^2y}{dx^2} + 3q\,\frac{dy}{dx} + ry = 0,$$

here the transformation $y = f \cdot Y$ of the dependent variable gives the new equation

$$\frac{d^3Y}{dx^3} + 3P\,\frac{d^2Y}{dx^2} + 3Q\,\frac{dY}{dx} + RY = 0,$$

where

$$\begin{aligned} P &= p + \frac{f'}{f}, \\ Q &= q + 2p\,\frac{f'}{f} + \frac{f''}{f}, \\ R &= r + 3q\,\frac{f'}{f} + 3p\,\frac{f''}{f} + \frac{f'''}{f}, \end{aligned}$$

and we thence derive as well the before-mentioned α-seminvariant $q - p^2 - p'$, as also a new α-seminvariant $r - 3pq + 2p^3 - p''$.

Again, effecting the transformation $x = \phi$ of the independent variable, we have the new equation

$$\frac{d^3y}{dX^3} + 3P\,\frac{d^2y}{dX^2} + 3Q\,\frac{dy}{dX} + Ry = 0,$$

where

$$\begin{aligned} P &= p\phi_1 - \frac{\phi_2}{\phi_1}, \\ Q &= q\phi_1^2 - p\phi_2 + \left(\frac{\phi_2}{\phi_1}\right)^2 - \tfrac{1}{3}\,\frac{\phi_3}{\phi_1}, \\ R &= r\phi_1^3. \end{aligned}$$

We thence obtain the identities

$$\begin{aligned} \frac{P_1 + 2P^2 - 3Q}{R^{\frac{2}{3}}} &= \frac{p' + 2p^2 - 3q}{r^{\frac{2}{3}}}, \\ \frac{R_1 + 3PR}{R^{\frac{4}{3}}} &= \frac{r' + 3pr}{r^{\frac{4}{3}}}; \end{aligned}$$

so that we have $\dfrac{p' + 2p^2 - 3q}{r^{\frac{2}{3}}}$ and $\dfrac{r' + 3pr}{r^{\frac{4}{3}}}$ as β-seminvariants of the differential equation of the third order.

But these are by no means the best conclusions; it is shown by M. Halphen in his great Memoir, “Mémoire sur la réduction des équations différentielles linéaires aux formes intégrables,” Sav. Étrang., t. xxviii. (1884), pp. 1–297, see p. 127, that there is a function invariantive in regard to each of the two transformations, and which is thus an invariant, viz. this is the function

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3);$$

this is for the first transformation unaltered, viz. it is

$$= P'' - 3(Q' - 2PP') + 2(R - 3PQ + 2P^3),$$

and for the second transformation it is only altered by the factor ϕ_1^{-3}, viz. it is

$$= \phi_1^{-3}\{P'' - 3(Q_1 - 2PP_1) + 2(R - 3PQ + 2P^3)\}.$$

It is interesting to directly verify this last result. Performing the differentiations in regard to X, we find without difficulty

$$P_1 = \cdots\,,$$
$$Q_1 - 2PP_1 = (q' - 2pp')\phi_1^2 - \cdots\,,$$
$$R - 3PQ + 2Q^3 = (r - 3pq + 2p^3)\phi_1^3,$$

and thence

$$P'' - 3(Q_1 - 2PP_1) + 2(R - 3PQ + 2Q^3) = \phi_1^3\{p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)\}\phi_1^{-2}.$$

But, introducing herein the derived functions in regard to x, we have

$$p_1 = p'\phi_1, \quad q_1 = q'\phi_1, \quad p_2 = p'\phi_1 + p''\phi_1^2,$$

whence

$$p_2\phi_1 - p_1\phi_2 = p'''\phi_1^3,$$

and the right-hand side becomes

$$= \phi_1^{-3}\{p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)\},$$

which is the required result.

We have thus

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)$$

as an "invariant" of the differential equation

$$\frac{d^3y}{dx^3} + 3p\,\frac{d^2y}{dx^2} + 3q\,\frac{dy}{dx} + ry = 0.$$

It is to be remarked that this is not what M. Halphen calls a "differential invariant;" he uses this expression not in regard to a differential equation, but in regard to a curve defined by an equation between the coordinates (x, y), and the differential invariant is a function of the derivatives y', y'', y''', … which is either $= 0$ in virtue of the equation of the curve, or else, being put $= 0$, it determines certain singularities of the curve. Thus y'' is a differential invariant; the equation $y'' = 0$ determines the points of inflexion. Again

$$\tfrac{1}{3}\left(\frac{y''''}{y''}\right) - \tfrac{5}{3}\,\frac{y'''}{y''}\left(\frac{y''''}{y''}\right) + \tfrac{40}{9}\left(\frac{y'''}{y''}\right)^2$$

is a differential invariant, vanishing identically if the variables (x, y) are connected by any quadric equation whatever; it is thus the differential invariant of a conic.

This last differential invariant is intimately connected with the above-mentioned invariant

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3),$$

viz. writing with M. Halphen

$$p = -\tfrac{1}{3}\,\frac{y'''}{y''}, \quad q = 0, \quad r = 0,$$

we have

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3) = p'' + 6pp' + 4p^3.$$

It is moreover noticed by him that, writing

$$y'',\ y''',\ y'''',\ y''''' = 2a,\ 6b,\ 24c,\ 120d,$$

respectively, the function in { } becomes

$$\frac{20}{a^3}(a^2d - 3abc + 2b^3);$$

the form under which he had previously obtained the differential invariant of the conic. As remarked by Sylvester, it is mentioned pp. 19 and 20 in Boole's *Differential Equations* (Cambridge, 1859), that the general differential equation of a conic was obtained by Monge in the form

$$9y'^2y''' - 45y'y''y'''' + 40(y'')^3 = 0,$$

which (representing the differential coefficients as just mentioned) becomes $a^2d-3abc+2b^3 = 0$, but putting them $= a, b, c, d$ respectively, it becomes $9a^2d - 45abc + 40b^3 = 0$; the last-mentioned form presented itself to Sylvester in his theory of Reciprocants.

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850. On Linear Differential Equations.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 321–331.]

1. The researches of Fuchs, Thomé, Frobenius, Tannery, Floquet, and others, relate to linear differential equations of the form

$$p_\circ \frac{d^my}{dx^m} + p_1 \frac{d^{m-1}y}{dx^{m-1}} + \cdots + p_my = 0,$$

or say

$$(p_\circ, p_1, \ldots, p_m \mid d/dx,\ 1)^my = 0,$$

where $p_\circ$, p_1, …, p_m are rational and integral functions of the independent variable x; any common factor of all the functions could of course be thrown out, and it is therefore assumed that the functions have no common factor. It is to be throughout understood that x and y denote complex magnitudes, which may be regarded as points in the infinite plane; viz. $x = \xi + i\eta$, is the point the coordinates whereof are ξ, η; and similarly for y.

2. Suppose $x - a$ is not a factor of $p_\circ$, the point $x = a$ is in this case said to be an ordinary point in regard to the differential equation; and let $y_\circ$, y_1, y_2, …, y_{m-1} be arbitrary constants denoting the values of y, $\dfrac{dy}{dx}$, $\dfrac{d^2y}{dx^2}$, …, $\dfrac{d^{m-1}y}{dx^{m-1}}$ for the point $x = a$. We can from the differential equation, and the equations derived therefrom by successive differentiations in regard to x, obtain the values for $x = a$ of the subsequent differential coefficients $\dfrac{d^my}{dx^m}$, $\dfrac{d^{m+1}y}{dx^{m+1}}$, …, say these are y_m, y_{m+1}, …; viz. the value of each of these quantities will be determined as a linear function of $y_\circ$, y_1, …, y_{m-1}, and we thus have a development of y in positive integer powers of $x - a$, viz. this is

$$y = y_\circ + y_1(x-a) + \tfrac{1}{1\cdot 2}\, y_2(x-a)^2 + \cdots,$$

which will in fact be a sum of m determinate series multiplied by $y_\circ$, y_1, …, y_{m-1} respectively; say the form is

$$y = y_\circ X_\circ + y_1X_1 + \cdots + y_{m-1}X_{m-1},$$

where $X_\circ$, X_1, …, X_{m-1} are each of them a series of positive integer powers of $x - a$. Each of these series will be convergent for values of x sufficiently near to a, or say for points x within the domain of the point a; and since $y_\circ$, y_1, …, y_{m-1} are arbitrary, each series separately will satisfy the differential equation; and we have thus m particular integrals of the differential equation.

3. Suppose next that $x - a$ is a factor of $p_\circ$, the point $x = a$ is in this case said to be a singular point in regard to the differential equation. The foregoing process of development fails, as leading to infinite values of $\dfrac{d^my}{dx^m}$, $\dfrac{d^{m+1}y}{dx^{m+1}}$, …; and we have to consider the developments of y which belong to the neighbourhood of the point $x = a$, or say to the domain of the singular point $x = a$. This has to be done separately in regard to each of the singular points, and among these we have, it may be, the point ∞. To decide whether ∞ is or is not a singular point, we may in the differential equation write $x = 1/t$, and then transforming to this new variable, and throwing out any common factor, the coefficients of the transformed equation will be rational and integral functions of t, without any common factor; say these are $P_\circ$, P_1, …, P_m; if t is

not a factor of $P_\circ$, then ∞ will be an ordinary point of the original equation; but if t is a factor of $P_\circ$, then ∞ will be a singular point of the original equation.

4. In considering the singular point $x = a$, we may, it is clear, transform the equation to this point as origin, and it is convenient to do this; supposing it done, we have an equation wherein $x = 0$ is a singular point, viz. $p_\circ$ contains x as a factor, and we have to consider the developments of y which belong to the domain of this singular point $x = 0$.

It is convenient to change the form of the differential equation by dividing the whole equation by the first coefficient $p_\circ$, and then expanding each of the quotients $\dfrac{p_i}{p_\circ}$ in a series of ascending powers of x. The new form is

$$\frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

or say

$$(1, p_1, p_2, \ldots, p_m \mid d/dx,\ 1)^m y = 0,$$

where p_1, p_2, $\ldots$, p_m now denote each of them a series (finite or infinite) of integer powers of x, but containing only a finite number of negative powers of x.

5. Such an equation frequently admits of a "regular integral" of the form

$$y = x^\rho E(x),$$

where $E(x)$ is a series of positive integer powers $E_\circ + E_1 x + E_2 x^2 + \cdots$, ($E_\circ$ not $= 0$, for this would imply a different value of ρ).[1] To determine whether this is so, we substitute in the differential equation for y the value in question $x^\rho E(x)$, thus obtaining a series

$$\Omega_\circ x^{\rho-\theta} + \Omega_1 x^{\rho-\theta+1} + \cdots ,$$

(where θ is a determinate positive integer depending on the negative powers of x in the equation); the coefficients $\Omega_\circ$, Ω_1, $\ldots$ are functions of ρ of an order not exceeding m, and contain also the coefficients $E_\circ$, E_1, E_2, $\ldots$ linearly; in particular, $\Omega_\circ$ contains $E_\circ$ as a factor, say its value is $= E_\circ \Pi$. The series should vanish identically. Supposing that Π contains ρ, then we have $\Pi = 0$, an equation of an order not exceeding m for the determination of ρ. For any root $\rho = \rho_\circ$ of this equation, $E_\circ$ remains arbitrary and may be taken $= 1$; the equations $\Omega_1 = 0$, $\Omega_2 = 0$, $\ldots$ then serve to determine the ratios of the remaining coefficients E_1, E_2, $\ldots$ to $E_\circ$; and we thus have the solution $y = x^{\rho_\circ}(1 + E_1 x + E_2 x^2 + \cdots)$, where $\rho_\circ$ and the coefficients have determinate values.

6. I stop to notice a curious form of illusory solution; the assumed form of solution is

$$y = x^\rho(\cdots + E_{-2}\, x^{-2} + E_{-1}\, x^{-1} + E_\circ + E_1 x + \cdots),$$

the series being a double series extending both ways to infinity, or say a back-and-forward series; we have here a series of equations

$$\ldots, \Omega_{-1} = 0,\ \Omega_\circ = 0,\ \Omega_1 = 0,\ \ldots,$$

which leave ρ undetermined, but determine the ratios of the several coefficients to one of these coefficients, say $E_\circ$; or taking this $= 1$, we have a solution

$$y = x^\rho(\cdots + E_{-2}\, x^{-2} + E_{-1}\, x^{-1} + 1 + E_1 x + E_2 x^2 + \cdots),$$

where the coefficients are determinate functions of the arbitrary symbol ρ. Such a series is in general divergent for all values of the variable, and thus is altogether without meaning. As a simple instance, take the differential equation

$$\frac{dy}{dx} - y = 0,$$

which is satisfied by

$$y = \{\cdots + (\rho - 1)\rho\, x^{\rho-2} + \rho\, x^{\rho-1} + x^\rho + \frac{x^{\rho+1}}{\rho + 1} + \cdots\};$$

[1] The expression regular integral is afterwards used in a more general sense, see post, No. 10.

see my paper, Cayley, Note on Riemann's paper, "Versuch einer allgemeinen Auffassung der Integration und Differentiation," Werke, pp. 331–344 (Math. Ann. t. xvi. (1880), pp. 81, 82), [751].

7. A more general form of integral is Thomé's "normal elementary integral,"

$$y = e^{w}\, x^{\rho} E(x)$$

where w is $= \dfrac{c_\alpha}{x^\alpha} + \cdots + \dfrac{c_1}{x}$ a finite series of negative powers of x (α a positive integer, $=2$ at least). To discover whether such a form exists, observe that, writing for a moment $\dfrac{y'}{y} = z$ and so for the other symbols, we have

$$z = w' + \frac{\rho}{x} + \frac{E'(x)}{E(x)},$$

where the last term $\dfrac{E'(x)}{E(x)}$ may be expanded as a series of positive integer powers of x, so that, writing $z = \dfrac{y'}{y}$, we have

$$z = \cdots - \frac{\alpha c_\alpha}{x^{\alpha+1}} - \cdots - \frac{c_1}{x^2} + \frac{\rho}{x} + l_\circ + l_1 x + \cdots\,,$$

we can, by introducing into the differential equation the new variable $z(= \dfrac{y'}{y})$ in place of y, obtain for z a differential equation (not a linear one) of the order $m-1$, and this should be satisfied by the series in question, viz. a series containing a finite number of negative powers of x; we endeavour to determine the first term $\dfrac{-(\alpha-1)c_{\alpha-1}}{x^\alpha}$ or say $\dfrac{A}{x^\alpha}$, where the exponent α is a positive integer which is to be determined, and A is not $=0$; this is done by a well-known process; we write in the equation $z = \dfrac{A}{x^\alpha}$, and (if possible to do so) determine α (α a positive integer $=2$ at least) by the condition that two or more of the terms shall have the same negative index $-p$ preceding in order all the other indices, viz. the absolute value of p must be greater than the absolute value of any other negative index; we then equate to zero the whole coefficient of the term in question x^{-p}, and thus obtain a value not $=0$ of A. And having thus obtained the first term $\dfrac{A}{x^\alpha}$, we can, by assuming the form

$$z = \frac{A}{x^\alpha} + \frac{B}{x^{\alpha-1}} + \cdots + \frac{K}{x^2} + \frac{L}{x} + \cdots\,,$$

with indeterminate coefficients, and substituting in the equation, find the remaining coefficients $B, \ldots, K, L$; we then have

$$\frac{c_\alpha}{x^\alpha} + \cdots + \frac{c_1}{x} \;=\; w,$$

giving the value $\dfrac{c_\alpha}{x^\alpha} + \cdots + \dfrac{c_1}{x}$ of w; instead of determining the subsequent coefficients from the z-equation, we may, from the original equation, writing therein $y = e^w Y$, obtain an equation for $Y(=e^{-w}y)$ which will be linear of the order m, and will admit of a regular integral $Y = x^\rho E(x)$. Or we may, going on a step further with the z-equation, find therefrom the coefficient L which is $=\rho$.

8. As an example, take the equation

$$\frac{d^2y}{dx^2} + \left(\frac{3}{x} - \frac{1}{x^2}\right)\frac{dy}{dx} + \frac{1}{x^2}\, y = 0,$$

which has the elementary normal integral

$$y = e^{-\frac{1}{x}} \cdot \frac{1}{x}.$$

To investigate this, writing $\dfrac{y'}{y} = z$, we have $\dfrac{y''}{y} = z' + z^2$, and thence the z-equation

$$\frac{dz}{dx} + z^2 + \left(\frac{3}{x} - \frac{1}{x^2}\right) z + \frac{1}{x^2} = 0.$$

Substituting herein the value $z = \dfrac{A}{x^\alpha}$, we obtain the function

$$-\frac{A\alpha}{x^{\alpha+1}} + \frac{A^2}{x^{2\alpha}} - \frac{A}{x^{\alpha+1}} + \frac{1}{x^2},$$

and here the value $\alpha = 2$ gives the two terms $\dfrac{A^2}{x^4}$ and $\dfrac{-A}{x^4}$, each with a negative index -4, the absolute value whereof is greater than the absolute values of the other indices; we then have $A^2 - A = 0$, giving a value $A = 1$, which is not $= 0$; and we have thus a first term $z = -\dfrac{1}{x^2}$, the complete value is, as it happens, the finite series $z = \dfrac{1}{x^2} - \dfrac{1}{x}$; and z being $= \dfrac{y'}{y}$, we thence obtain the integral in question, $y = e^{-\frac{1}{x}} \cdot \dfrac{1}{x}$.

9. Returning to the question of the regular integrals, we may consider the function

$$P(x^\rho) = (1,\, p_1,\, \ldots,\, p_m \mid d/dx,\, 1)^m \, x^\rho;$$

say this is the so-called "determinirende Function," but I will call it the Indicial function; this is a function of (x, ρ); we have therein terms arising from

$$\left(\frac{d}{dx}\right)^m x^\rho, \quad \left(\frac{d}{dx}\right)^{m-1} x^\rho, \ \ldots,$$

viz. these are equal to $[\rho]^m x^{\rho-m}$, $[\rho]^{m-1} x^{\rho-m+1}$, $\ldots$ respectively, but these will be multiplied by powers of x contained in the coefficients p_1, p_2, $\ldots$, p_m respectively; the function is thus of the degree m as regards ρ, but the coefficient of any power of x is of the degree m, or of some inferior degree, according to the terms $[\rho]^m$, $[\rho]^{m-1}$, $\ldots$ which enter into the coefficient. The coefficient of the lowest power of x in the indicial function may be termed the indicial coefficient, or simply the indicial; and the equation obtained by equating this coefficient to zero the indicial equation. By what precedes, the indicial is a function of ρ, which is of the degree m at most, but which may be of any inferior degree, or even be an absolute constant. Hence, considering a regular integral $x^\rho E(x)$, the index ρ is determined by the indicial equation, and to each root of this equation there corresponds in general a regular integral $x^\rho E(x)$; the number of regular integrals is thus at most $= m$.

10. Suppose, first, that the indicial equation is of the degree m, and that the roots of this equation are all unequal; there are, in this case, m values, say ρ_1, ρ_2, $\ldots$, ρ_m of ρ, and each of these gives rise to a regular integral $x^\rho E(x)$, viz. there will be corresponding to each value of ρ a series of positive integer powers $E(x)$, which will be a convergent series for sufficiently small values of x. Some of these series may be finite series, viz. this will be the case if the indicial equation has roots differing from each other by integer values (but a finite series may of course be regarded as convergent); and in the case of two or more equal roots, it is necessary to extend the notion of a regular integral by including under it series multiplied into positive integer powers of $\log x$; but, with this modification, the conclusion holds good; if the indicial equation be of the degree m, the number of regular integrals will always be $= m$, viz. there will be m integrals involving series $E(x)$, or, it may be, $E(x)(\log x)^k$, where each series $E(x)$ is a convergent series for sufficiently small values of x.

11. But suppose the indicial equation is of an order $m - \gamma$ inferior to m, and to fix the ideas suppose that the roots are all unequal. There are in this case $m - \gamma$ values, say ρ_1, ρ_2, $\ldots$, $\rho_{m-\gamma}$ of ρ, and each of these will apparently give rise to a regular integral $x^\rho E(x)$; but it may be that in any such case the series $E(x)$ is a series divergent for all values, however small, of the variable x, and that we thus have in appearance only, but not in reality, a regular integral; or say the integral is illusory. The conclusion is that the indicial equation being of a degree $m - \gamma$ inferior to m, the number of regular integrals is at most $= m - \gamma$; but it may have any less value than $m - \gamma$, or even there may be no regular integral. It is hardly necessary to remark that, if the indicial be an absolute constant, or say if the degree of the indicial equation be $= 0$, then there is no value of ρ, and consequently no regular integral.

12. By way of illustration, consider first the differential equation

$$\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} + \frac{1}{x^2}\, y = 0,$$

for which the degree of the indicial equation is equal to the order of the equation (= 2); viz. the indicial function is

$$= \rho(\rho-1)\,x^{\rho-2} + \rho\,x^{\rho-2} + x^{\rho-2},$$

and the indicial equation is thus $\rho(\rho-1)+\rho = 0$, that is, $\rho^2 = 0$, viz. there are here two roots, each $= 0$.

It is convenient not in the first instance to write $\rho = 0$, but to substitute in the equation the value

$$y = Ax^{\rho} + Bx^{\rho+1} + Cx^{\rho+2} + \cdots.$$

We thus find

$$0 = \begin{array}{lll} \rho\cdot\overline{\rho-1}\cdot A, & \overline{\rho+1}\cdot\rho\cdot B, & \overline{\rho+2}\cdot\overline{\rho+1}\cdot C \\ +\rho A, & +\overline{\rho+1}\cdot B, & +\overline{\rho+2}\cdot C \\ +A, & +B, & +C, \end{array}$$

viz. we have the equations

$$\begin{aligned} \rho^2 A &= 0, \\ (\rho+1)^2 B + A &= 0, \\ (\rho+2)^2 C + B &= 0, \end{aligned}$$

where observe that in each equation the function of ρ, which multiplies the posterior coefficient, is of a degree superior to that which multiplies the other coefficient. The first equation gives $\rho = 0$; A arbitrary, but say $A = 1$; the values of B, C, … are then

$$-\frac{1}{(\rho+1)^2}, \quad +\frac{1}{(\rho+1)^2(\rho+2)^2}, \quad -\frac{1}{(\rho+1)^2(\rho+2)^2(\rho+3)^2}, \quad \ldots,$$

giving rise to a series which in this form, and when ρ is put equal to its value, $\rho = 0$, is at once seen to be convergent (even for indefinitely large values of x, but this is an accident; it would have been enough if the series had been convergent for sufficiently small values), viz. the series is

$$y = 1 - \frac{x}{1^2} + \frac{x^2}{1^2\cdot 2^2} - \frac{x^3}{1^2\cdot 2^2\cdot 3^2} + \cdots.$$

There is another integral

$$y = \left(1 - \frac{x}{1^2} + \frac{x^2}{1^2\cdot 2^2} - \cdots\right)\log x + 2x - \tfrac{3}{4}x^2 + \tfrac{11}{108}x^3 - \tfrac{1}{1728}x^4 + \cdots,$$

involving $\log x$ and a second series which is also convergent.

13. But consider next the before-mentioned equation

$$\frac{d^2y}{dx^2} + \left(\frac{3}{x} - \frac{1}{x^2}\right)\frac{dy}{dx} + \frac{1}{x^2}\,y = 0,$$

for which the degree of the indicial equation is equal to order -1 $(= 1)$; in fact, here the indicial function is

$$= \rho(\rho-1)\,x^{\rho-2} + 3\rho\,x^{\rho-2} - \rho\,x^{\rho-3} + x^{\rho-2},$$

and the indicial is thus $= -\rho$, giving the index $\rho = 0$. There is thus at most one regular integral $E(x)$, but this is in fact an illusory one. Substituting in the equation the value

$$y = Ax^{\rho} + Bx^{\rho+1} + Cx^{\rho+2} + \cdots,$$

we find

$$0 = \begin{array}{llll} \rho(\rho-1)\,A, & (\rho+1)\rho\,B, & (\rho+2)(\rho+1)\,C, & \cdots \\ -\rho A, & -(\rho+1)B, & -(\rho+2)C, & \cdots \\ +3\rho A, & +3(\rho+1)B, & +3(\rho+2)C, & \cdots \\ +A, & +B, & +C, & \end{array}$$

and we thus have the equations

$$\begin{aligned}\rho A &= 0,\\ (\rho+1)^2A-(\rho+1)B &= 0,\\ (\rho+2)^2B-(\rho+2)C &= 0,\\ (\rho+3)^2C-(\rho+3)D &= 0,\end{aligned}$$

where observe that in each equation the function of ρ, which multiplies the posterior coefficient, is of a degree inferior to that which multiplies the other coefficient. The first equation then gives $\rho = 0$, A arbitrary, or say $= 1$; the other coefficients B, C, ... are then found to be

$$\rho+1, \quad (\rho+1)(\rho+2), \quad (\rho+1)(\rho+2)(\rho+3), \quad \cdots,$$

or substituting for ρ its value $= 0$, the series is

$$1+1\cdot x+1\cdot 2\cdot x^2+1\cdot 2\cdot 3\cdot x^3+\cdots,$$

which is divergent for any value whatever, however small, of x; or the integral is as mentioned an illusory one.

14. The last-mentioned differential equation is reducible, viz. we have

$$\frac{d^2}{dx^2}+\left(\frac{3}{x}-\frac{1}{x^2}\right)\frac{d}{dx}+\frac{1}{x^2}=\left(\frac{d}{dx}+\frac{2}{x}\right)\left(\frac{d}{dx}+\frac{1}{x}-\frac{1}{x^2}\right), \text{ say } = QD,$$

where of course the order of the factors Q, D is material.

It is a general theorem that the indicial, belonging to the product P, is of a degree equal to the sum of the degrees of the indicials, belonging to the factors Q, D respectively; and, in fact, the indicial of P is (as was seen) of the degree 1; and the indicials of Q, D are of the degrees 1, 0 respectively. But, moreover, if the indicial of P is of a degree $m-\gamma$, less than m the order of the equation (in the present case $\gamma = 1$, $m = 2$, $m-\gamma = 1$), then, in order that the equation may have $m-\gamma$ regular integrals, it is a necessary and sufficient condition that P shall be decomposable into a product QD, where the factors Q, D (being functions of the same form as P) are of the orders γ and $m-\gamma$ respectively, and where, moreover, the second factor D shall have an indicial of the degree $m-\gamma$, and consequently the first factor Q an indicial of the degree zero. In the present case, it is the second factor which has an indicial of degree zero; and thus the condition is not satisfied. And accordingly the equation ought not to have $(m-\gamma =)$ 1 regular integral; and we have seen that there is in fact no regular integral. The investigation serves to show in what sense it is that there is no regular integral, or generally in what sense it is that the number of regular integrals may be less than $m-\gamma$.

15. The theorem may, it appears to me, be stated in the more general form: the necessary and sufficient condition that the differential equation $P(y) = 0$ of the order m, but having an indicial of the degree $m-\gamma$, shall have $m-\gamma-\delta$ regular integrals is that the function P shall be a product of the form $P = QMD$, where the orders of Q, M, D are $\gamma+\delta-\theta$, θ, $m-\gamma-\delta$ respectively, and the degrees of their indicials δ, 0, $m-\gamma-\delta$ respectively; or, to denote this in a compendious manner, say

$$\overset{m}{\underset{m-\gamma}{P}} = \overset{\gamma+\delta-\theta}{\underset{\delta}{Q}}\ \overset{\theta}{\underset{0}{M}}\ \overset{m-\gamma-\delta}{\underset{m-\gamma-\delta}{D}}.$$

θ is an integer which is not $= 0$, and which is at most $= \gamma$, for otherwise the function Q of the order $\gamma+\delta-\theta$ could not have an indicial of the degree δ. If there is no regular integral, then $m = \gamma+\delta$, D is a mere constant or say unity, and the formula is

$$\overset{m}{\underset{m-\gamma}{P}} = \overset{m-\theta}{\underset{m-\gamma}{Q}}\ \overset{\theta}{\underset{0}{M}},$$

where θ is not $= 0$, and is at most $= m-\gamma$.

16. For differential equations of the form $P(y) = 0$ above considered, Floquet gave a theorem, which as afterwards remarked by him is not generally true, viz. this was $P = ABC\cdots$, a product of linear factors

of the form $\frac{d}{dx} + \Sigma_{-i}^{\infty} C_i x^i$ (i an integer). The question of decomposition is considered in my next following paper "On Linear Differential Equations (the Theory of Decomposition)," [851], and by means of the formulæ there given (and by the process indicated ante No. 7), it could be decided whether any particular differential equation admits of a decomposition $P = ABC \cdots$, where the linear factors are functions not of the form just referred to, but of the form $\frac{d}{dx} + \Sigma C_i x^i$, in which the series contains only a finite number of negative powers, say this is a decomposition into linear regular factors.

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851. On Linear Differential Equations (the Theory of Decomposition).

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxi. (1886), pp. 331–335.]

1. In the theory of linear differential equations the question arises, to decompose a quantic $\left(* \mid \frac{d}{dx}, 1\right)^n$ into linear factors: we have, for instance, a differential equation

$$\frac{d^2y}{dx^2} + p\,\frac{dy}{dx} + qy = 0,$$

which is to be expressed in the form

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right) y = 0,$$

or say we have to express $\frac{d^2}{dx^2} + p\,\frac{d}{dx} + q$ as a product of linear factors

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right).$$

The problem is analogous to, but wholly distinct from, that of the resolution of an algebraic equation: using accents to denote differentiation in regard to x, the relation (in the simple case just referred to) between the coefficients (p, q) and the roots (α, β) is (not $p = \alpha + \beta$, $q = \alpha\beta$, but in place thereof) $p = \alpha + \beta$, $q = \alpha\beta + \beta'$; and it thus appears that there is the important distinction that, in the present problem, the order of the factors is not indifferent.

2. The problem may be solved when the general solution of the differential equation is known, or, what is the same thing, by means of n particular solutions of the equation; it is not here considered in this point of view, but the intention is to treat it directly by means of the relations between the coefficients and the roots: thus in the above case, $p = \alpha + \beta$, $q = \alpha\beta + \beta'$; we may (1) find first β and then α, viz. eliminating α we have $\beta^2 + \beta'(p - \beta) - q = 0$, β determined by a differential equation (not linear) of the first order; and then $\alpha = p - \beta$; or we may (2) find first α and then β, viz. eliminating β, we have $\alpha^2 - p\alpha + \alpha'(\alpha - p) + q = 0$, α determined by a differential equation (not linear) of the first order, and then $\beta = p - \alpha$.

In the case of a product

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right)\left(\frac{d}{dx} + \gamma\right)\cdots$$

of more than two factors, the roots might be determined in any order; but the two orders which naturally present themselves and which will be alone considered are, say, the reverse order $(\ldots, \gamma, \beta, \alpha)$ and the direct order $(\alpha, \beta, \gamma, \ldots)$.

3. *The Reverse Order.* Writing D for $\frac{d}{dx}$, we assume

$$\begin{aligned} D + \alpha &= (1,\, p_1 \mid D,\, 1), \\ (D + \alpha)(D + \beta) &= (1,\, p_2,\, q_2 \mid D,\, 1)^2, \\ (D + \alpha)(D + \beta)(D + \gamma) &= (1,\, p_3,\, q_3,\, r_3 \mid D,\, 1)^3, \end{aligned}$$

and so on. Then using accents to denote differentiation, we find

$$\begin{aligned}
&(1)\quad p_1 = \alpha.\\
&(2)\quad p_2 = \beta + p_1,\\
&\qquad\ \ q_2 = \beta' + p_1\beta.\\
&(3)\quad p_3 = \gamma + p_2,\\
&\qquad\ \ q_3 = 2\gamma' + p_2\gamma + q_2,\\
&\qquad\ \ r_3 = \gamma'' + p_2\gamma' + q_2\gamma.\\
&(4)\quad p_4 = \delta + p_3,\\
&\qquad\ \ q_4 = 3\delta' + p_3\delta + q_3,\\
&\qquad\ \ r_4 = 3\delta'' + 2p_3\delta' + q_3\delta + r_3,\\
&\qquad\ \ s_4 = \delta''' + p_3\delta'' + q_3\delta' + r_3\delta,
\end{aligned}$$

where the law is obvious: thus in the last set of equations, the several columns (after the first) contain the factors p_3, q_3, r_3, s_3 respectively; and there is in each column a head term with the coefficient unity: omitting these head terms, we have in the several columns the sets $(\delta, 3\delta', 3\delta'', \delta''')$, $(\delta, 2\delta', \delta'')$, (δ, δ'), δ, of derivatives of the root δ.

4. We may from each set of equations determine in order the coefficients of lower rank contained in the equations, and the last equation of each set then gives an equation independent of these coefficients of lower rank. Thus set (2) gives

$$p_1 = -\beta + p_2; \qquad 0 = -\beta' + p_2\beta - p_2 \cdot \beta.$$

Set (3) gives

$$\begin{aligned}
&p_2 = -\gamma + p_3, \qquad q_2 = -2\gamma' + (\gamma + \gamma')p_3 + q_3,\\
&0 = \gamma'' - \gamma \cdot 2\gamma' + p_3(\gamma\gamma' - \gamma'^2) + q_3(\gamma\gamma' - \gamma'\gamma).
\end{aligned}$$

Set (4) gives

$$\begin{aligned}
p_3 &= -\delta + p_4,\\
q_3 &= -3\delta' + 5\delta\delta' - \delta'^2,\\
0 &= \delta''' - 4\delta\delta'' + 6\delta'\delta' - 3\delta^3 - \delta^4
\end{aligned}$$

and so on.

5. Thus suppose (p_4, q_4, r_4, s_4) are given, we have δ determined by a differential equation (not linear) of the third order; and δ being known, p_3, q_3, r_3 are also known; then γ is determined by a differential equation (not linear) of the second order, and γ being known, p_2, q_2 are also known; then β is determined by a differential equation (not linear) of the first order, and β being known, p_1, that is, α is also known.

Comparing the last equations of each set, that is, the equations for the determination of γ, β, δ, ... respectively, it will be observed that they depend on the single series of derivatives

$$-1,\ \phi,\ \phi - \phi',\ \phi'' - 3\delta\phi' + \phi',\ \phi''' - 4\delta\phi'' + 6\phi'\phi' - 3\delta\phi' - \phi^4,$$

where, calling the successive terms A, B, C, D, E, ..., we have

$$B = A' - \phi A, \quad C = B' - \phi B, \quad D = C' - \phi C.$$

6. *Direct Order.* Considering the product

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right)\left(\frac{d}{dx} + \gamma\right)\left(\frac{d}{dx} + \delta\right)\cdots$$

[Paper 851 continues on subsequent pages.]

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Arthur Cayley

851 (continued). On Linear Differential Equations.

(Continuation from p. 405; PDF page 426.)

of four factors, and assuming

$$\begin{aligned} D+\delta &= (1,\, p_1 \,|\, D,\, 1), \\ (D+\gamma)(D+\delta) &= (1,\, p_2,\, q_2 \,|\, D,\, 1)^2, \\ (D+\beta)(D+\gamma)(D+\delta) &= (1,\, p_3,\, q_3,\, r_3 \,|\, D,\, 1)^3, \\ (D+\alpha)(D+\beta)(D+\gamma)(D+\delta) &= (1,\, p_4,\, q_4,\, r_4,\, s_4 \,|\, D,\, 1)^4, \end{aligned}$$

we have the sets of equations

$$\begin{aligned} &(1) && p_1 = \delta. \\ &(2) && p_2 = \gamma + p_1, \\ & && q_2 = \gamma p_1 + p_1'. \\ &(3) && p_3 = \beta + p_2, \\ & && q_3 = \beta p_2 + p_2' + q_2, \\ & && r_3 = \beta q_2 + q_2'. \\ &(4) && p_4 = \alpha + p_3, \\ & && q_4 = \alpha p_3 + p_3' + q_3, \\ & && r_4 = \alpha q_3 + q_3' + r_3, \\ & && s_4 = \alpha r_3 + r_3'. \end{aligned}$$

7. I stop to remark that there would have been a convenience in considering the product

$$\left(\frac{d}{dx}+\delta\right)\left(\frac{d}{dx}+\gamma\right)\left(\frac{d}{dx}+\beta\right)\left(\frac{d}{dx}+\alpha\right),$$

for the first, second, third, &c., sets of equations would then have contained α, β, γ, &c., respectively; but for better comparison with the equations of the reverse order, I have preferred not to make this change of notation.

8. The set (1) gives $\delta - p_1 = 0$.
Set (2) gives

$$p_2 = -\gamma + p_1, \quad 0 = \gamma' + \gamma^2 - p_1\gamma - p_1' + q_2.$$

Set (3) gives

$$p_3 = -\beta + p_2, \quad 0 = \beta' + \beta^2 - p_2\beta - p_2' + q_2 + (\beta' + \beta^2)q_2' - p_2 q_2 - r_2 \cdots$$

$$q_3 = \beta p_2 + p_2' + q_2, \quad r_3 = \beta q_2 + q_2'.$$

Set (4) gives

$$\begin{aligned}
p_4 &= -\alpha + p_3, \\
q_4 &= \alpha^2 + 3\alpha\alpha' + \alpha'' - p_3\alpha - p_3' + p_2(\alpha^2 + \alpha') + 2p_2'\alpha + p_2'', \\
0 &= \alpha''' + 4\alpha\alpha'' + 6\alpha'^2 + 3\alpha^2\alpha' + \alpha^4 \\
&\qquad - p_3(\alpha'' + 3\alpha\alpha' + \alpha^3) + p_2(3\alpha' + 3\alpha^2) - 3p_3''\alpha \\
&\qquad - p_3'(3\alpha + \alpha^2) - 3p_4''' \\
&\qquad + q_2(\alpha^2 + \alpha') + 2q_2'\alpha + q_2'' \\
&\qquad - r_3'\alpha - r_4' \\
&\qquad + s_4,
\end{aligned}$$

which differ in form from the equations belonging to the reverse order in containing the derived functions p_2', p_2'', p_3', p_3'', q_2', q_2'', ... of the coefficients.

9. Taking p_4, q_4, r_4, s_4 as known, we have α determined by a differential equation (not linear) of the third order; and α being known, we know p_3, q_3, r_3. We then have β determined by a differential equation (not linear) of the second order; and β being known, we know p_2, q_2. We then have γ determined by a differential equation (not linear) of the first order; and γ being known, we know p_1, that is, δ.

852. Note sur le Mémoire de M. Picard "Sur les Intégrales de Différentielles Totales Algébriques de Première Espèce."

[From the *Bulletin des Sciences Mathématiques*, 2me Sér., t. x. (1886), pp. 75–78.]

ON peut présenter l'analyse sur laquelle est fondé le Mémoire sous une forme plus symétrique en introduisant dès le commencement les fonctions homogènes.

Soit $f = (*)(x, y, z, t)^m$ une fonction du degré m des variables x, y, z, t, lesquelles seront toujours liées par l'équation $f = 0$: écrivons aussi

$$\frac{df}{dx},\ \frac{df}{dy},\ \frac{df}{dz},\ \frac{df}{dt}\ =\ X,\ Y,\ Z,\ T,$$

de manière que X, Y, Z, T sont des fonctions du degré $m-1$: et soient A, B, C, D des fonctions chacune du degré $m-3$ et Q une fonction du degré $m-4$, telles que

$$AX + BY + CZ + DT = Qf \quad \text{identiquement};$$

donc, en supposant $f = 0$, on aura

$$AX + BY + CZ + DT = 0.$$

On vérifie sans peine que l'expression

$$d\Omega\ =\ \frac{1}{\lambda X + \mu Y + \nu Z + \rho T}\begin{vmatrix} \lambda, & \mu, & \nu, & \rho \\ A, & B, & C, & D \\ x, & y, & z, & t \\ dx, & dy, & dz, & dt \end{vmatrix}$$

est indépendante des valeurs de λ, μ, ν, ρ, et ainsi égale à chacune des quatre expressions $d\Omega_x$, $d\Omega_y$, $d\Omega_z$, $d\Omega_t$,

$$\frac{1}{X}\begin{vmatrix} B, & C, & D \\ y, & z, & t \\ dy, & dz, & dt \end{vmatrix},\quad -\frac{1}{Y}\begin{vmatrix} A, & C, & D \\ x, & z, & t \\ dx, & dz, & dt \end{vmatrix},$$

$$\frac{1}{Z}\begin{vmatrix} A, & B, & D \\ x, & y, & t \\ dx, & dy, & dt \end{vmatrix},\quad -\frac{1}{T}\begin{vmatrix} A, & B, & C \\ x, & y, & z \\ dx, & dy, & dz \end{vmatrix},$$

respectivement.

Cela étant,

$$d\Omega_t = -\frac{1}{T}\begin{vmatrix} A, & B, & C \\ x, & y, & z \\ dx, & dy, & dz \end{vmatrix} = \text{une différentielle totale};$$

en écrivant un moment $x, y, z = x't, y't, z't$, et en dénotant par f' la fonction $(*)(x', y', z', 1)^m$, et de même par X', Y', Z', T', A', B', C' les valeurs correspondantes de X, Y, Z, T, A, B, C, les variables x', y', z' seront liées par l'équation $f' = 0$, ce qui donne

$$X'dx' + Y'dy' + Z'dz' = 0;$$

et l'on voit sans peine que l'expression de $d\Omega_t$ se réduit à

$$-\frac{1}{T'}\begin{vmatrix} A', & B', & C' \\ x', & y', & z' \\ dx', & dy', & dz' \end{vmatrix},$$

fonction de la forme

$$F'dx' + G'dy' + H'dz',$$

qui ne contient que les variables x', y', z'. Donc, en omettant les accents, il est permis de prendre $t = \text{const.}$, ce qui donne

$$X\,dx + Y\,dy + Z\,dz = 0;$$

et avec cette relation entre dx, dy, dz, la forme ci-dessus pour $d\Omega_t = F\,dx + G\,dy + H\,dz$ soit une différentielle totale: cela donne la condition

$$X\left(\frac{dG}{dx} - \frac{dH}{dy}\right) + Y\left(\frac{dH}{dx} - \frac{dF}{dz}\right) + Z\left(\frac{dF}{dy} - \frac{dG}{dx}\right) = 0,$$

ou enfin

$$\begin{aligned} X&\left(\frac{d}{dx}\frac{Cx - Az}{T} - \frac{d}{dy}\frac{Ay - Bx}{T}\right) \\ &+ Y\left(\frac{d}{dx}\frac{Ay - Bx}{T} - \frac{d}{dz}\frac{Bz - Cy}{T}\right) \\ &+ Z\left(\frac{d}{dy}\frac{Bz - Cy}{T} - \frac{d}{dx}\frac{Cx - Az}{T}\right) = 0. \end{aligned}$$

On a d'abord un terme $\mathfrak{A} = \dfrac{1}{T}$, où

$$\begin{aligned} \mathfrak{A} = X&\left[-2A + x\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dA}{dx} + y\frac{dA}{dy} + z\frac{dA}{dz}\right)\right] \\ + Y&\left[-2B + y\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dB}{dx} + y\frac{dB}{dy} + z\frac{dB}{dz}\right)\right] \\ + Z&\left[-2C + z\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dC}{dx} + y\frac{dC}{dy} + z\frac{dC}{dz}\right)\right], \end{aligned}$$

ou, en réduisant,

$$\begin{aligned} \mathfrak{A} = &-2(AX + BY + CZ) \\ &+ (Xx + Yy + Zz)\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) \\ &- (m-1)\,DT \\ &- T\left(x\frac{dA}{dt} + y\frac{dB}{dt} + z\frac{dC}{dt}\right) \\ &+ t\left(X\frac{dA}{dt} + Y\frac{dB}{dt} + Z\frac{dC}{dt}\right). \end{aligned}$$

Puis un terme $\mathfrak{B} = \dfrac{dT}{dx^2}$, où

$$\begin{aligned} \mathfrak{B} = &-X\left[(Cx - Az)\frac{dT}{dx} - (Ay - Bx)\frac{dT}{dy}\right] \\ &- Y\left[(Ay - Bx)\frac{dT}{dx} - (Bz - Cy)\frac{dT}{dz}\right] \\ &- Z\left[(Bz - Cy)\frac{dT}{dy} - (Cx - Az)\frac{dT}{dx}\right], \end{aligned}$$

ou, en réduisant,

$$\begin{aligned}
\mathfrak{B} = {} & \frac{dT}{dx}\big[(AX+BY+CZ)x - A(xX+yY+zZ)\big] \\
& + \frac{dT}{dy}\big[(AX+BY+CZ)y - B(xX+yY+zZ)\big] \\
& + \frac{dT}{dz}\big[(AX+BY+CZ)z - C(xX+yY+zZ)\big] \\
& - -DT\left(x\frac{dT}{dx}+y\frac{dT}{dy}+z\frac{dT}{dz}\right) \\
& + tT\left(A\frac{dT}{dx}+B\frac{dT}{dy}+C\frac{dT}{dz}\right) \\
& = -(m-1)\,DT^2 \\
& + Tt\left(A\frac{dT}{dx}+B\frac{dT}{dy}+C\frac{dT}{dz}\right).
\end{aligned}$$

Donc, en réunissant les deux parties, $0 = \mathfrak{A} + \dfrac{\mathfrak{B}}{T}$, c'est-à-dire,

$$\begin{aligned}
0 = {} & -\left(\frac{dA}{dx}+\frac{dB}{dy}+\frac{dC}{dz}+\frac{dD}{dt}\right) \\
& + t\frac{d}{dt}(AX+BY+CZ+DT) \\
& = -T\left(\frac{dA}{dx}+\frac{dB}{dy}+\frac{dC}{dz}+\frac{dD}{dt}\right) \\
& + t\,QT,
\end{aligned}$$

ou, en omettant le facteur tT, on obtient enfin

$$Q = -\left(\frac{dA}{dx}+\frac{dB}{dy}+\frac{dC}{dz}+\frac{dD}{dt}\right),$$

c'est-à-dire que les fonctions A, B, C, D sont telles que

$$AX+BY+CZ+DT = -\left(\frac{dA}{dx}+\frac{dB}{dy}+\frac{dC}{dz}+\frac{dD}{dt}\right)f;$$

et, cela étant, l'expression générale $d\Omega$, et de même chacune des expressions $d\Omega_x$, $d\Omega_y$, $d\Omega_z$, $d\Omega_t$, sera égale à une différentielle totale.

Cambridge, le 8 Janvier 1886.

853. Note on a Formula for $\Delta^n 0^i/n^i$ when n, i are Very Large Numbers.

[From the *Proceedings of the Royal Society of Edinburgh*, t. xiv. (1887), pp. 149–153.]

THE following formula

$$\frac{\Delta^n 0^i}{n^i} = \left[1+\left(\frac{i+1-2n}{2n}\right)q-\binom{n+i+2}{2}q^2\cdots\right], \qquad q=e^{-(i+1)/n},$$

is given by Laplace (*Théorie Analytique des Probabilités*, 2nd ed., Paris, 1814, p. 195) as an approximate value of $\dfrac{\Delta^n 0^i}{n^i}$ when n and i are very large numbers, and is applied immediately afterwards to the case where i is of the order $n\log n$. As remarked by Professor Tait, it is certainly not applicable to the case where i is of the order n; for taking $i=An$, where A is a given number however large, then q is indefinitely near to the very small value e^{-A}, but nevertheless the last term $-\frac{1}{2}(n+i+2)q^2$ by taking n sufficiently large, may be made as large as we please, and the value would thus come out negative. It is thus necessary that i should be at least of the order $n\log n$; but it may be of any higher order.

Writing for greater convenience $\dfrac{i+1}{n}=r$, $nq=re^{-1/n}\,r(1-X)$, if $X=1-e^{-1/n}$; and the formula becomes

$$\frac{\Delta^n 0^i}{n^i} = e^{-r(1-X)}\left[1+\frac{i+1-2n}{2n^2}\,e^{-r/n}\,r-\frac{n+i+2}{2n^2}\,e^{-r/n}\,r^2\cdots\right].$$

Here $X=\dfrac{1}{n}-\dfrac{1}{1\cdot 2}\dfrac{1}{n^2}+\dfrac{1}{1\cdot 2\cdot 3}\dfrac{1}{n^3}+\text{\&c.}$, and the exponential $e^{rX}=1+rX+\dfrac{r^2X^2}{1\cdot 2}+\cdots$ is thus also expansible in negative powers of n; the formula becomes

$$\frac{\Delta^n 0^i}{n^i} = e^{-r}\left(1+rX+\frac{r^2X^2}{1\cdot 2}+\cdots\right)\left[1+\frac{i+1-2n}{2n^2}\,e^{-r/n}\,r-\frac{n+i+2}{2n^2}\,e^{-r/n}\,r^2\cdots\right];$$

viz. putting for X its value,

$$\begin{aligned}
&= e^{-r}\Big\{1\\
&\quad + r\left(\frac{i+1-2n}{2n^2}e^{-r/n}+1-e^{-r/n}\right)\\
&\quad + r^2\left(\frac{-n-i-2}{2n^2}e^{-r/n}+\frac{i+1-2n}{2n^2}(1-e^{-r/n})e^{-r/n}+\tfrac{1}{2}(1-e^{-r/n})^2\right)\\
&\quad + \text{\&c.}\Big\};
\end{aligned}$$

or finally, expanding $e^{-r/n}$ and taking the whole result as far as $\dfrac{1}{n^2}$, the coefficient of r is

$$\left(-\frac{1}{n}+\frac{i+1}{n^2}\right)\left(1-\frac{1}{n}\right)+\frac{1}{n}-\frac{1}{2n^2}=\frac{1+\frac{1}{2}i}{n^2};$$

the coefficient of r^2 is

$$\left(-\frac{1}{n}+\frac{-2-i}{n^2}\right)\left(1-\frac{1}{n}\right)+\left(-\frac{1}{n}+\frac{4i}{n^2}\right)\frac{1}{n}+\frac{1}{2n^2}=-\frac{1}{n}-\frac{1}{2}-\tfrac{1}{2}i;$$

whence the formula becomes

$$\frac{\Delta^n 0^i}{n^i}=e^{-r}\left\{1+r\cdot\frac{1+\frac{1}{2}i}{n^2}+\frac{r^2}{1\cdot 2}\left(-\frac{1}{n}-\frac{-1-i}{n^2}\right)\cdots\right\}.$$

It seems to me that the correct result up to this order of approximation is

$$\frac{\Delta^n 0^i}{n^i} = e^{-r}\left\{1+r\cdot\frac{1+\frac{1}{2}i}{n^2}+\frac{r^2}{1\cdot 2}\left(-\frac{1}{n}+\frac{-1}{n^2}\right)\right\}.$$

My investigation is as follows: we have

$$\frac{\Delta^n 0^i}{n^i} = 1 - \frac{n}{1}\left(1-\frac{1}{n}\right)^i + \frac{n\cdot n-1}{1\cdot 2}\left(1-\frac{2}{n}\right)^i + \cdots,$$

the series being a finite one; but the number of terms is very large. But observe that, however large n is, we can take i so large that the second term $n\left(1-\frac{1}{n}\right)^i$ may be as small as we please; taking this term to be of moderate amount, say $= r_1$, the subsequent terms will be not very different from $\frac{r_1^2}{1\cdot 2}$, $\frac{r_1^3}{1\cdot 2\cdot 3}$, ..., and the approximate value is $1 - r_1 + \frac{r_1^2}{1\cdot 2} -$ &c., which is a convergent series having its sum $= e^{-r_1}$.

To work this properly out, I represent the successive terms by r_1, $\frac{r_2}{1\cdot 2}$, $\frac{r_3}{1\cdot 2\cdot 3}$, ..., so that the series is

$$= 1 - r_1 + \frac{r_2}{1\cdot 2} - \frac{r_3}{1\cdot 2\cdot 3} + \cdots.$$

Taking r a value at pleasure not very different from r_1, and multiplying by

$$(1+\nu)\,e^{-r},\quad e^{\nu} = e^{-r}\cdot\left(1 + r + \frac{r^2}{1\cdot 2} + \cdots\right),$$

the sum is

$$= e^{-r}\left\{1 + (r - r_1) + \frac{1}{1\cdot 2}(r^2 - 2rr_1 + r_2) + \frac{1}{1\cdot 2\cdot 3}(r^3 - 3r^2 r_1 + 3rr_2 - r_3) + \cdots\right\}.$$

Assuming now $r = ne^{-i/n}$, we have

$$r_1 = n\left(1-\frac{1}{n}\right)^i = ne^{i\log(1-1/n)} = r(1+X_1),$$

where $X_1 = e^{-\frac{i}{2}\frac{1}{n} - \frac{i}{3}\frac{1}{n^2} - \cdots} - 1$; and similarly

$$r_2 = n\cdot n{-}1\cdot\left(1-\frac{2}{n}\right)^i = n^2\left(1-\frac{1}{n}\right)e^{i\log(1-2/n)} = \left(1-\frac{1}{n}\right)r^2(1+X_2),$$

where $X_2 = e^{-\frac{2i}{2}\frac{1}{n} - \frac{2^2 i}{3}\frac{1}{n^2} - \cdots} - 1$; also

$$r_3 = n\cdot n{-}1\cdot\left(1-\frac{3}{n}\right)^i = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)r^3(1+X_3),$$

where $X_3 = e^{-\frac{3i}{2}\frac{1}{n} - \frac{3^2 i}{3}\frac{1}{n^2} - \cdots}$, and so on. It is now easy to calculate the successive terms $r - r_1$, $r^2 - 2rr_1 + r_2$, &c.; and it is to be observed that, in the parts independent of the X's, we have only terms divided by n, n^2, or higher powers of n: thus in $r^4 - 4r^3 r_1 + 6r^2 r_2 - 4rr_3 + r_4$, we have r^4 multiplied by

$$1 - 4 + 6\left(1-\frac{1}{n}\right) - 4\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) + \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right) = \frac{3}{n^2} - \frac{6}{n^3}.$$

We thus obtain the formula

$$\frac{\Delta^n 0^i}{n^i} = e^{-r}\Big\{1 + r(-1\,X_1) + \frac{r^2}{1\cdot 2}\big[-\tfrac{1}{n} - 2X_1 + (1-\tfrac{1}{n})X_2\big] + \frac{r^3}{1\cdot 2\cdot 3}\big[-\tfrac{2}{n^2} - 3X_1 + 3(1-\tfrac{1}{n})X_2 - (1-\tfrac{1}{n})(1-\tfrac{2}{n})X_3\big] + \cdots\Big\},$$

with the next term

$$+\frac{r^4}{1\cdot 2\cdot 3\cdot 4}\big[-\tfrac{3}{n^2} - \tfrac{6}{n^3} - 4X_1 + 6(1-\tfrac{1}{n})X_2 - 4(1-\tfrac{1}{n})(1-\tfrac{2}{n})X_3 + (1-\tfrac{1}{n})(1-\tfrac{2}{n})(1-\tfrac{3}{n})X_4\big] + \cdots,$$

where $r = ne^{-i/n}$ as above, and X_1, X_2, ... have the above-mentioned values, the exponentials being expanded in negative powers of n.

Writing

$$X_1 = -\tfrac{1}{2}\frac{i}{n^2}, \quad X_2 = -\frac{2i}{n^2},$$

we have

$$\frac{\Delta^n 0^i}{n^i} \ = \ e^{-r}\left\{1 + r\cdot\frac{1+\frac{1}{2}i}{n^2} + \frac{r^2}{2}\left(-\frac{1}{n} + \frac{-i}{n^2}\right)\right\},$$

which is the foregoing approximate value.

854. An Algebraical Transformation.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 58, 59.]

THE following algebraical transformation occurs in a paper by Hermite "On the theory of the Modular Equations," *Comptes Rendus*, t. XLVIII. (1859), p. 1100.

Writing $q = 1 - 2u^4$, $l = 1 - 2v^4$, then in the transformation of the fifth order, the modular equation was expressed by Jacobi in the form

$$\Omega \;=\; (q-l)^6 - 256(1-q^2)(1-l^2)\Big[16ql\big((9-q^2l)+9(45-ql)(q-l)\big)^2\Big] = 0;$$

and if we write herein $q = 1-2x$, $l = \dfrac{x+1}{x-1}$, or, what is the same thing, establish between q, l the relation $q = \dfrac{1+ql}{l-1}$, that is, between u, v the relation $v^4 = 1 \div (1-u^4)$, then the function Ω becomes

$$\Omega \;=\; \frac{64}{(1-x)^6}\Big[(x^2-x+1)^4 + 64(1-x)^2\,x^2\big\{(x^2-x+1)^3 + 2^4(x^2-x+1)^2 + 3^4(x^2-x+1)\big\}\Big];$$

or, what is the same thing, the equation $\Omega = 0$ gives for $\dfrac{x^2-x+1}{(x^2-x)^2}$ the values -2^4 and -3^4.

We, in fact, have

$$q - l = 3 + ql = -\frac{2(x^2-x+1)}{1-x}, \quad 1-q^2 = 4x(1-x), \quad 1-l^2 = -\frac{4x}{(1-x)^2},$$

and therefore

$$(1-q^2)(1-l^2) = -\frac{16x^2}{1-x}.$$

Hence

$$\Omega \;=\; \frac{64}{(1-x)^6}\Big[(x^2-x+1)^6 + 64(1-x)^2x^2\Big\{16ql\big((9-q^2l)+9(45-ql)(3-2ql)\big)^2\Big\}\Big]\;;$$

and, putting for a moment $ql = \theta - 3$, the term in $\{\quad\}$ is found to be

$$=\; 7l^4\theta + 34560\theta^3 - 6912;$$

viz. this is

$$= \frac{56(x^2-x+1)^6}{(1-x)^6} + \frac{6912(x^2-x+1)}{1-x} - 6912,$$

$$= \frac{8}{(1-x)^6}\big[7(x^2-x+1)^6 + 864(x-1)^2(x^2-x+1) + 864(x-1)^4\big],$$

$$= \frac{8}{(1-x)^6}\big[7(x^2-x+1)^6 + 864(x^2-x)^2\big].$$

Hence

$$\Omega \;=\; \frac{64}{(1-x)^6}\big[(x^2-x+1)^6 + 512(x^2-x)^2\big\{7(x^2-x+1)^6 + 864(x^2-x)^2\big\}\big]\,,$$

which is

$$= \frac{64}{(1-x)^6}\big[(x^2-x+1)^6 + 2^4(x^2-x)^2\big]\big[(x^2-x+1)^6 + 2^4\cdot 3^4(x^2-x)^2\big]\,.$$

855. Solution of $(a, b, c, d) = (a^2, b^2, c^2, d^2)$.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 59–61.]

IT is required to find four quantities (no one of them zero) which are in some order or other equal to their squares, say

$$(a, b, c, d) \;=\; (a^2, b^2, c^2, d^2).$$

Supposing that the required quantities (a, b, c, d) are the roots of the biquadratic equation

$$x^4 + px^3 + qx^2 + rx + s = 0,$$

(s not $= 0$), then the function

$$(x^4 + px^3 + qx^2 + rx + s)^2 \text{ must be } = x^8 + px^6 + qx^4 + rx^2 + s;$$

and we have thus the conditions

$$p^2 - 2q = p, \quad 2s + 2pr - q^2 = q, \quad 2qs - r^2 = r, \quad s^2 = s,$$

the last of which (since s is not $= 0$) gives $s = 1$, and the others then become

$$2q = p^2 + p, \quad 2(pr - 1) = q^2 - q, \quad 2q = r^2 + r;$$

viz. regarding p, q, r as the coordinates of a point in space, this is determined as the intersection of three quadric surfaces, and the number of solutions is thus $= 8$.

We, in fact, have $2q = p^2 + p = r^2 + r$; that is, $p^2 + p = r^2 + r$, or $(p - r)(p + r + 1) = 0$; hence $r = p$ or $r = -1 - p$.

First, if $r = p$: here $2q = p^2 + p$, $2(p^2 - 1) = q^2 - q$: the last equation multiplied by 4 gives

$$8(p^2 - 1) = (p^2 + p)(p^2 + p - 2),$$

that is, $p^4 + 2p^3 - 8 = 0$ or $(p - 1)(p^2 + 3p + 8) = 0$.

If $p^4 - 1 = 0$, then either $p = 1$, giving $q = 1$, $r = 1$, and hence the equation is

$$x^4 + x^3 + x^2 + x + 1 = 0,$$

or else $p = -1$, giving $q = 0$, $r = -1$, and hence the equation is

$$x^4 - x^3 + 1 = 0, \text{ that is, } (x - 1)(x^3 - x + 1) = 0.$$

If $p^2 + 2p - 8 = 0$, then either $p = 2$, giving $q = 3$, $r = 2$, and hence the equation is

$$x^4 + 2x^3 + 3x^2 + 2x + 1 = 0, \text{ that is, } (x^2 + x + 1)^2 = 0,$$

or else $p = -4$, giving $q = 6$, $r = -4$, and hence the equation is $x^4 + 6x^3 - 4x + 1 = 0$, that is, $(x - 1)^4 = 0$.

Secondly, if $r = -1 - p$; here

$$2q = p^2 + p, \quad 2(-p^2 - p - 1) = q^2 - q;$$

the last equation multiplied by 4 gives

$$8(-p^2 - p - 1) = (p^2 + p)(p^2 + p - 2),$$

that is,

$$p^4 + 2p^3 + 7p^2 + 6p + 8 = 0, \text{ or } (p^2 + p + 4)(p^2 + p + 2) = 0.$$

If $p^2 + p + 4 = 0$, then $p = \tfrac{1}{2}\{-1 \pm i\sqrt{15}\}$, whence

$$r = \tfrac{1}{2}\left\{-1 \mp i\sqrt{15}\right\}, \quad 2q = p^2 + p = -4, \text{ or } q = -2;$$

and the equation is

$$x^4 + \tfrac{1}{2}\Big\{-1 \pm i\sqrt{15}\Big\}x^3 - 2x^2 + \tfrac{1}{2}\Big\{-1 \mp i\sqrt{15}\Big\}x + 1 = 0.$$

If $p^2 + p + 2 = 0$, then $p = \frac{1}{2}\{-1 \pm i\sqrt{7}\}$, whence

$$r = \tfrac{1}{2}\Big\{-1 \mp i\sqrt{7}\Big\}, \quad 2q = p^2 + p = -2, \text{ or } q = -1;$$

and the equation is

$$x^4 + \tfrac{1}{2}\Big\{-1 \pm i\sqrt{7}\Big\}x^3 - x^2 + \tfrac{1}{2}\Big\{-1 \mp i\sqrt{7}\Big\}x - 1 = 0,$$

that is,

$$(x-1)\Big[x^3 + \tfrac{1}{2}\Big\{1 \pm i\sqrt{7}\Big\}x^2 + \tfrac{1}{2}\Big\{-1 \mp i\sqrt{7}\Big\}x - 1\Big] = 0.$$

We thus see that the eight equations are

$$\begin{array}{ll}
1 & (x-1)^4 = 0,\\
1 & (x^2+x+1)^2 = 0,\\
1 & (x-1)(x^3-x+1) = 0,\\
1 & x^4+x^3+x^2+x+1 = 0,\\
2 & (x-1)\Big[x^3 + \frac{1}{2}(1 \pm i\sqrt{7})\,x^2 + \frac{1}{2}(-1 \mp i\sqrt{7})\,x - 1\Big] = 0,\\
2 & x^4 + \frac{1}{2}(-1 \pm i\sqrt{15})\,x^3 - 2x^2 + \frac{1}{2}(-1 \mp i\sqrt{15})\,x + 1 = 0,
\end{array}$$

(totalling 8) and it hence appears that, writing γ, ϵ, θ to denote respectively an imaginary cube root, fifth root, and seventh root of unity, then the values of (a, b, c, d) are

$$\begin{array}{l}
1,\ 1,\ 1,\ 1;\\
\gamma,\ \gamma,\ \gamma^2,\ \gamma^2;\\
1,\ 1,\ \gamma,\ \gamma^2;\\
\epsilon,\ \epsilon^2,\ \epsilon^3,\ \epsilon^4;\\
\epsilon^2\gamma,\ \epsilon\gamma^2,\ \epsilon^4\gamma,\ \epsilon^3\gamma^2;\\
\epsilon^2\gamma,\ \epsilon\gamma^2,\ \epsilon^3\gamma^2,\ \epsilon^4\gamma;\\
1,\ \theta,\ \theta^2,\ \theta^4;\\
1,\ \theta^3,\ \theta^6,\ \theta^5;
\end{array}$$

viz. for each of these systems we have the required relation

$$(a, b, c, d) \ = \ (a^2, b^2, c^2, d^2).$$

It may be noticed that out of the eight equations we have the following three which are irreducible:

$$\begin{array}{l}
x^4+x^3+x^2+x+1 = 0,\\
x^4 + \frac{1}{2}(-1 + i\sqrt{15})\,x^3 - 2x^2 + \frac{1}{2}(-1 - i\sqrt{15})\,x + 1 = 0,\\
x^4 + \frac{1}{2}(-1 - i\sqrt{15})\,x^3 - 2x^2 + \frac{1}{2}(-1 + i\sqrt{15})\,x + 1 = 0.
\end{array}$$

Each of these is an Abelian equation, viz. the roots are of the form

$$\alpha,\ \theta(\alpha),\ \theta^2(\alpha),\ \theta^3(\alpha), \quad (= \alpha,\ \alpha^a,\ \alpha^{a^2},\ \alpha^{a^3}),$$

where $\theta^4(\alpha) = \alpha$, not identically but in virtue of the value of α, viz. we have $\alpha^{a^4} = \alpha^{16} = \alpha$, $\alpha^{15} = 1$ (in the first equation $a^5 = 1$, and therefore $a^{15} = 1$; in each of the other two, a^{15} is the lowest power which is $= 1$).

In the first equation, we have evidently

$$x^4 + x^3 + x^2 + x + 1$$

as the irreducible factor of $x^5 - 1$.

The second and third equations combined together give

$$\left(x^4 - \tfrac{1}{2}x^3 + \tfrac{1}{2}x - 2x^2 + \tfrac{15}{4}\right)\left(x^4 - x^3\right) = 0;$$

that is,

$$x^8 - x^7 + x^6 - x^4 + x^2 - x + 1 = 0,$$

where the left-hand side is the irreducible factor of $x^{15} - 1$.

856. Note on a Cubic Equation.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 62–64.]

CONSIDER the cubic equation

$$x^3 + 3cx + d = 0;$$

then effecting upon this the Tschirnhausen-Hermite transformation

$$y = xT_1 + (x^2 + 2c)T_2,$$

the resulting equation in y is

$$\begin{aligned} y^3 &+ 3y(cT_1^2 + dT_1T_2 - c^2T_2^2) \\ &+ dT_1^3 - 6c^2T_1^2T_2 - 3cdT_1T_2^2 - (d^2 + 2c^3)T_2^3 = 0, \end{aligned}$$

and this will be

$$y^3 + 3cy + d = 0,$$

if only

$$c = cT_1^2 + dT_1T_2 - c^2T_2^2, \quad d = dT_1^3 - 6c^2T_1^2T_2 - 3cdT_1T_2^2 - (d^2 + 2c^3)T_2^3,$$

equations which give

$$(d^2 + 4c^3) = (d^2 + 4c^3)({T_1}^3 + 3cT_1{T_2}^2 + d{T_2}^3),$$

viz. assuming that $d^2 + 4c^3$ not $= 0$, this is

$$1 \ = \ {T_1}^3 + 3cT_1{T_2}^2 + d{T_2}^3.$$

Hence the coefficients T_1, T_2 being such as to satisfy these relations, the equation in y is identical with the equation in x; or, what is the same thing, if α, β, γ are the roots of the equation in x, then we have between these roots the relations

$$\begin{aligned} \beta &= \alpha T_1 + (\alpha^2 + 2c)T_2, \\ \gamma &= \beta T_1 + (\beta^2 + 2c)T_2, \\ \alpha &= \gamma T_1 + (\gamma^2 + 2c)T_2, \end{aligned}$$

viz. the general cubic equation $x^3 + 3cx + d = 0$, adjoining thereto the radicals T_1, T_2, may be regarded as an Abelian equation.

In particular, if c, $d = -1$, 1, then we may write $T_1 = 0$, $T_2 = 1$; the cubic equation is here

$$x^3 - 3x + 1 = 0,$$

and the roots α, β, γ are such that $\beta = \alpha^2 - 2$, $\gamma = \beta^2 - 2$, $\alpha = \gamma^2 - 2$; in fact, taking θ a primitive ninth root of unity, $\theta^6 + \theta^3 + 1 = 0$; we have α, β, $\gamma = \theta + \theta^8$, $\theta^2 + \theta^7$, $\theta^4 + \theta^5$, values which satisfy $x^3 - 3x + 1 = 0$, and the relations in question.

The same question may be considered from a different point of view. Take the transforming equation to be

$$y = A + Bx + Cx^2,$$

then assuming that the values of y corresponding to the values $x = \alpha, \beta, \gamma$ are β, γ, α respectively, we have

$$\begin{aligned} \beta &= A + B\alpha + C\alpha^2, \\ \gamma &= A + B\beta + C\beta^2, \\ \alpha &= A + B\gamma + C\gamma^2, \end{aligned}$$

and the transforming equation thus is

$$\begin{vmatrix} y, & 1, & x, & x^2 \\ \beta, & 1, & \alpha, & \alpha^2 \\ \gamma, & 1, & \beta, & \beta^2 \\ \alpha, & 1, & \gamma, & \gamma^2 \end{vmatrix} = 0.$$

This may also be written

$$(\beta-\gamma)(\gamma-\alpha)(\alpha-\beta)\left\{y-\tfrac{1}{3}(\alpha+\beta+\gamma+x)\right\}$$
$$= \beta^2\gamma+\gamma^2\alpha+\alpha^2\beta-\tfrac{1}{3}(\beta^3+\gamma^3+\alpha^3+\beta^2\gamma+\gamma^2\alpha+\alpha^2\beta+\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2)$$
$$+x\left\{\beta\gamma+\gamma\alpha+\alpha\beta-\tfrac{1}{3}(\alpha^2+\beta^2+\gamma^2)\right\}.$$

We have

$$(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2 = -27\left(a^2d^2+4ac^3+4b^3d-3b^2c^2-6abcd\right) = -27\Delta,$$

or, say

$$(\beta-\gamma)(\gamma-\alpha)(\alpha-\beta) = \frac{3(\omega-\omega^2)}{a^2}\sqrt{(\Delta)},$$

if Δ be the discriminant, and ω an imaginary cube root of unity, $\{(\omega-\omega^2)^2=-3\}$.

The remaining functions of α, β, γ are of course expressible rationally in terms of the coefficients: we have

$$\begin{aligned}
\Sigma\beta^2\gamma &= \tfrac{1}{a^3}(-6bd+9c^2),\\
\Sigma\beta\gamma &= \tfrac{1}{a^3}(-3abd-18ac^2+27b^2c),\\
\Sigma\alpha^2 &= \tfrac{1}{a^3}(-3a^2d+7abc-9b^3),\\
\Sigma\beta\gamma &= \tfrac{1}{a^2}(3ad-9bc),\\
\Sigma\alpha\beta &= \tfrac{3c}{a},\\
\Sigma\alpha^2 &= \tfrac{1}{a^2}(9b^2-6ac),
\end{aligned}$$

and the final result is

$$\begin{aligned}
\tfrac{1}{2}(\omega-\omega^2)\sqrt{(\Delta)}\left\{2a(y+x)+3b\right\} = &-abd+4ac^2-3b^2c\\
&+x(-a^2d+7abc-6b^3)\\
&+x^2(2a^2c-2ab^2);
\end{aligned}$$

viz. we have thus an automorphic transformation of the equation

$$ax^3+3bx^2+3cx+d=0.$$

857. Analytical Geometrical Note on the Conic.

[From the *Messenger of Mathematics*, vol. xv. (1886), p. 192.]

TAKE (X, Y, Z) the coordinates of a point on the conic $yz + zx + xy = 0$, so that $YZ + ZX + XY = 0$; clearly (Y, Z, X) and (Z, X, Y) are the coordinates of two other points on the same conic; I say that the three points are the vertices of a triangle circumscribed about the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

In fact, the equation of one of the sides is

$$\begin{vmatrix} x, & y, & z \\ X, & Y, & Z \\ Y, & Z, & X \end{vmatrix} = 0,$$

say this is $AX + BY + CZ = 0$, where $A, B, C = XY - Z^2,\ YZ - X^2,\ XZ - Y^2$; and the condition in order that this side may touch the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0$$

is

$$BC + CA + AB = 0.$$

But we have

$$\begin{aligned} BC + CA + AB \ &= Y^2Z^2 + Z^2X^2 + X^2Y^2 - X(Y^3 + Z^3) - Y(Z^3 + X^3) - Z(X^3 + Y^3) \\ &\quad + XYZ + XYZ + XYZ \\ &= -(YZ + ZX + XY)\big(-X^3 - Y^3 - Z^3 + YZX + ZXY + XYZ\big)\,; \end{aligned}$$

and similarly for the other two sides. The point (X, Y, Z) is an arbitrary point on the conic $yz+zx+xy = 0$; and we thus see that we have a singly infinite series of triangles each inscribed in this conic and circumscribed about the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

858. Comparison of the Weierstrassian and Jacobian Elliptic Functions.

[From the *Messenger of Mathematics*, vol. xvi. (1887), pp. 129–132.]

THE Weierstrassian function $\sigma(u)$ corresponds of course with Jacobi's $H(u)$, but it is worth while to establish the actual formulas of transformation.

Writing, for a moment,

$$\omega = \omega_1 + i\omega_2, \quad \omega' = \omega_1' + i\omega_2',$$

it is convenient to assume $\omega_1\omega_2' - \omega_2\omega_1'$ positive; we then have

$$2(\eta\omega' - \eta'\omega) \ = \ \pi i;$$

in particular, this will be the case if $\omega = \omega_1$, $\omega' = i\omega_2$, where ω_1, ω_2 are each positive.

To reduce the periods into the Jacobian form, we may assume

$$\omega = \lambda K, \quad \omega' = \lambda i K',$$

(where observe that, if as usual k, K, K' are each real and positive, and if as above $\omega = \omega_1$, $\omega' = i\omega_2$, ω_1 and ω_2 positive, then also λ will be real and positive). We have

$$\frac{iK'}{K} \ = \ \frac{\omega'}{\omega}, \quad \text{or say} \quad q \ = \ e^{-\frac{\pi K'}{K}} = e^{\frac{i\pi\omega'}{\omega}},$$

which determines first q, and thence k, K, K' as functions of $\dfrac{\omega'}{\omega}$, and we then have

$$\lambda \ = \ \frac{\omega}{K} = \frac{-i\omega'}{K'},$$

either of which equations gives λ as a function of ω, ω'. Conversely, starting with k, λ, the original equations give the values of ω, ω'; those of η, η' will be determined presently.

The form of relation is at once seen to be

$$H(u) \ = \ Ae^{Bu^2}\sigma(\lambda u),$$

and observing that, for u small, we have $H(u) = \sqrt{\left(\dfrac{2kK}{\pi}\right)}u$ and $\sigma(\lambda u) = \lambda u$, we have $A = \dfrac{1}{\lambda}\sqrt{\left(\dfrac{2kK}{\pi}\right)}$; I first write down and afterwards verify the value of B, viz. this is $= -\frac{1}{2}\dfrac{\lambda\eta}{\omega}$; and the formula thus is

$$H(u) \ = \ \frac{1}{\lambda}\sqrt{\left(\frac{2kK}{\pi}\right)}\, e^{-\frac{1}{2}\frac{\lambda\eta}{\omega}u^2}\, \sigma(\lambda u).$$

In fact, for u writing successively $u + 2K$, and $u + 2iK'$, we obtain

$$\frac{H(u+2K)}{H(u)} \ = \ e^{-\frac{2\lambda\eta}{\omega}K(u+K)}\, \frac{\sigma(\lambda u + 2\omega)}{\sigma(\lambda u)},$$

$$\frac{H(u+2iK')}{H(u)} \ = \ e^{-\frac{2\lambda\eta}{\omega}iK'(u+iK')}\, \frac{\sigma(\lambda u + 2\omega')}{\sigma(\lambda u)},$$

which should be satisfied in virtue of

$$\frac{H(u+2K)}{H(u)} \ = \ -1, \quad \frac{\sigma(\lambda u + 2\omega)}{\sigma(\lambda u)} = -e^{2\eta(\lambda u+\omega)},$$

$$\frac{H(u+2iK')}{H(u)} \ = \ e^{-\frac{\pi}{K}(u+iK')}, \quad \frac{\sigma(\lambda u + 2\omega')}{\sigma(\lambda u)} = -e^{2\eta'(\lambda u+\omega')};$$

viz. we ought to have

$$0 \ = \ \frac{\lambda\eta}{\omega}\, 2K\,(u+K) + 2\eta(\lambda u + \omega)$$

$$-\frac{i\pi}{K}(u+iK') = \frac{\lambda\eta}{\omega}\,2iK'(u+iK') + 2\eta'(\lambda u+\omega').$$

The first of these is

$$0 \;=\; -\frac{\lambda K}{\omega}(u+K) + \Big(\lambda + \frac{\omega}{\lambda}\Big),$$

that is,

$$0 \;=\; (-1+1)(u+K);$$

and the second is

$$0 \;=\; \left(\frac{\lambda iK'}{\lambda K} - \frac{\lambda iK'}{\omega}\right)(u+iK') + \eta'\left(u+\frac{\omega'}{\lambda}\right),$$

viz. for iv' writing $2(\eta\omega'-\eta'\omega)$, this is

$$0 \;=\; \left(\frac{\eta\omega'-\eta'\omega}{\omega} - \frac{\eta\omega'}{\omega} + \eta'\right)(u+iK'),$$

and the two equations are thus each of them an identity.

The Weierstrassian function $\wp(u)$ is defined as

$$\wp(u) \;=\; -\frac{d^2}{du^2}\log\sigma(u),$$

or, what is the same thing, we have

$$\wp(u) \;=\; -\frac{d}{du}\frac{\sigma'(u)}{\sigma(u)}.$$

Hence

$$\wp(\lambda u) \;=\; -\frac{1}{\lambda}\frac{d}{du}\frac{\sigma'(\lambda u)}{\sigma(\lambda u)}.$$

But

$$\frac{H'(u)}{H(u)} \;=\; \frac{\lambda^2\eta\, u}{\omega} + \frac{\lambda\,\sigma'(\lambda u)}{\sigma(\lambda u)},$$

or

$$\frac{1}{\lambda}\frac{d}{du}\frac{H'(u)}{H(u)} \;=\; -\eta + \wp(\lambda u).$$

But from the equation $\sqrt{k}\,\mathrm{sn}\,u = \dfrac{H(u)}{\Theta(u)}$, we obtain

$$\frac{d}{du}\frac{H'(u)}{H(u)} \;=\; Z'(u) - \frac{1}{\mathrm{sn}^2 u} + k^2\,\mathrm{sn}^2 u, \quad = 1 - \frac{E}{K} - \frac{1}{\mathrm{sn}^2 u},$$

and consequently

$$\wp(\lambda u) \;=\; \frac{\eta}{\omega} - \frac{1}{\lambda^2}\left(1-\frac{E}{K}\right) + \frac{1}{\lambda^2\mathrm{sn}^2 u},$$

where, on the right-hand side, expanding in ascending powers of u, the constant term is

$$= -\frac{\eta}{\omega} - \frac{1}{\lambda^2}\left(1-\frac{E}{K}\right) + \frac{1}{\lambda^2}\cdot\frac{1}{3}(1+k^2).$$

But in the function $\wp(\lambda u)$ this constant term is $=0$, and we thus have

$$\frac{\eta}{\omega} \;=\; \frac{1}{\lambda^2}\left\{\tfrac{1}{3}(1+k^2) - 1 + \frac{E}{K}\right\};$$

and then, since

$$\frac{\eta}{\omega} - \frac{\eta'}{\omega'} \;=\; \frac{\frac{1}{2}\pi i}{\omega\omega'} = \frac{\frac{1}{2}\pi i}{\lambda^2 iKK'},$$

we have

$$\frac{\eta'}{\omega'} \;=\; -\frac{\frac{1}{2}\pi i}{\lambda^2 iKK'} + \frac{1}{\lambda^2}\left\{\tfrac{1}{3}(1+k^2) - 1 + \frac{E}{K}\right\};$$

or, as these equations may also be written,

$$\eta = \frac{K}{\lambda}\left\{\tfrac{1}{3}(1+k^2) - 1 + \frac{E}{K}\right\},$$
$$\eta' = \frac{\frac{1}{2}\pi i}{\lambda K} + \frac{2iK'}{\lambda}\left\{\tfrac{1}{3}(1+k^2) - 1 + \frac{E}{K}\right\},$$

which are the values of η, η'. And we then have

$$\wp(\lambda u) \;=\; \frac{1}{\lambda^2}\tfrac{1}{3}(1+k^2) + \frac{1}{\lambda^2 \mathrm{sn}^2 u},$$

the equation connecting $\wp(\lambda u)$ and $\mathrm{sn}\, u$.

859. On the Complex of Lines which meet a Unicursal Quartic Curve.

[From the *Proceedings of the London Mathematical Society*, vol. xvii. (1886), pp. 232–238.]

THE curve is taken to be that determined by the equations

$$x : y : z : w \;=\; 1 : \theta : \theta^2 : \theta^4,$$

viz. it is the common intersection of the quadric surface $\Theta = 0$, and the cubic surfaces $P = 0$, $Q = 0$, $R = 0$, where

$$\Theta = xw - yz, \quad P = x^2z - y^2w, \quad Q = xz^2 - y^2w, \quad R = xz - y^2.$$

Writing (a, b, c, f, g, h) as the six coordinates of a line, viz.

$$(a, b, c, f, g, h) \;=\; (\beta z - \gamma y,\ \gamma x - \alpha z,\ \alpha y - \beta x,\ \alpha w - \delta x,\ \beta w - \delta y,\ \gamma w - \delta z),$$

if $(\alpha, \beta, \gamma, \delta)$, (x, y, z, w) are the coordinates of any two points on the line; then, if the line meet the curve, we have

$$\begin{aligned} -\,h\theta + g\theta^2 + a\theta^4 &= 0,\\ -\,h + f\theta^2 + b\theta^4 &= 0,\\ g - f\theta + c\theta^4 &= 0,\\ -\,a - b\theta - c\theta^3 &= 0, \end{aligned}$$

from which four equations (equivalent, in virtue of the identity $af + bg + ch = 0$, to two independent equations), eliminating θ, we have the equation of the complex. The form may, of course, be modified at pleasure by means of the identity just referred to, but one form is

$$\Omega \;=\; a^4 - b^3h + bf^2g + cg^3 - acfh + 2c^2h^2 - 4a^2ch + af^3 - a^3f = 0,$$

as may be verified by substituting therein the values $a = \beta w - c\theta^3$, $g = f\theta - c\theta^2$, $h = f\theta^3 + b\theta^4$. The last-mentioned equation is thus the equation of the complex in question, in terms of the six coordinates (a, b, c, f, g, h).

If for the six coordinates we substitute their values, $\beta z - \gamma y$, &c., we obtain

$$\Omega \;=\; (x, y, z, w)^4\,(\alpha, \beta, \gamma, \delta)^4 = 0,$$

which, regarded as an equation in (x, y, z, w), is the equation of the cone, vertex $(\alpha, \beta, \gamma, \delta)$, passing through the quartic curve; this equation should evidently be satisfied if only Θ, P, Q, R are each $= 0$, viz. must be a linear function of (Θ, P, Q, R); and by symmetry, it must be also a linear function of (Θ', P', Q', R'), where

$$\Theta' = \alpha\delta - \beta\gamma, \quad P' = \alpha^2\gamma - \beta^3, \quad Q' = \alpha\gamma^2 - \beta^2\delta, \quad R' = \alpha\gamma - \beta^2,$$

viz. the form is $\Omega = (\Theta', P', Q', R')\,(\Theta, P, Q, R)$, an expression with coefficients which are of the first or second degree in (x, y, z, w) and also of the first or second degree in $(\alpha, \beta, \gamma, \delta)$.

To work this out, I first arrange in powers and products of (α, δ), (β, γ), expressing the quartic functions of (x, y, z, w) in terms of (Θ, P, Q, R), as follows:

$\Omega =$ **expansion table**

	a^4	$-b^3h$	$+bf^2g$	$+cg^3$	$-acfh$	$+2c^2h^2$	$-4a^2ch$	$+af^3$	$-a^3f$	sum	contribution
α^4										0	0
$\alpha^3\delta$		$-z^4$	$+yzw^2$							$-z^4+yzw^2$	$-zR$
$\alpha^2\delta^2$			$-2xyzw$	$-y^4$		$+2y^2z^2$				$-2xyzw+2y^2z^2$	$-2yz\Theta$
$\alpha\delta^3$			$+x^2yz$							$+x^2yz-y^4$	$+yP$
δ^4										0	0
$\alpha^3\beta$			$-zw^3$				$+zw^3$			0	0
$\alpha^2\beta\delta$			$+2xzw^2$		$+yz^2w$		$-3xzw^2$			$-xzw^2+yz^2w$	$-zw\Theta$
$\alpha\beta\delta^2$			$-x^2zw$	$+3y^3w$	$-xy^2w$	$-4xyz^2$	$+3x^2zw$			$+2x^2zw+3y^3w-3xy^2w-5xyz^2$	$+2xz\Theta-3yQ$
$\beta\delta^3$				$+xy^3$			$-x^2y$			$+xy^3-x^3z$	$-xP$
$\alpha^2\gamma$			$+z^3w$				$-yw^3$			$+z^3w-yw^3$	$+wR$
$\alpha^2\gamma\delta$			$+3xz^3$		$-yzw^2$		$-3xz^2w$	$+3xyw^2$		$+3xz^3+2xyw^2-5yzw^2$	$+2yw\Theta+3zQ$
$\alpha\gamma\delta^2$			$+2x^2zw$	$-x^2y$	$-xy^2w$		$-3x^2yw$	$+x^3y$		$-x^2yw+xyz^2$	$-xy\Theta$
$\gamma\delta^3$				$-x^3y$	$+x^2z^2$			$+x^3y$		0	0
$\alpha^2\beta^2$		$+xw^4$								$+xw^4-yzw^3$	$+w^2\Theta$
$\alpha^2\beta\gamma$		$-3xz^2w$	$+xw^3$		$-yzw^2$	$+2y^2zw$	$+4yz^3$			$-3xz^2w+3y^2zw-2yzw^2+4yz^3+3xz\Theta$	$-3wQ$
$\alpha^2\gamma^2$		$-2c^2w^2$		$-3y^3w^2$	$-xz^2w$	$+2xyzw$	$+8xyz^2$	$-3xz^3$		$-3y^3w^2-xz^2w+2xyzw+8xyz^2-3xz^3+10xyzw-9y^2z^2$	$-4z^2\Theta+3wQ$
$\alpha\beta^2\delta$		$-3xz^2$	$-2x^2w^2$		$+2xyzw$	$+8xyz^2$	$-9y^2z^2$			$-3xz^2-2x^2w^2+2xyzw+8xyz^2-3xz^3+10xyzw-9y^2z^2$	$-4y^3\Theta-3xQ$
$\alpha\beta\gamma\delta$			$+x^2w$	$-2xy^2w$	$+xy^2$		$-3xy^2w$	$+x^3w$		$-3xz^2w+2xy^2w+xz^2y+x^2y^2$	$+3xQ$
$\alpha\gamma^2\delta$		$+x^2w$	$-2xz^2$	$+x^3z$				$+x^3z$		$+x^3z-x^2y$	$+x^2\Theta$
$\beta\gamma\delta^2$				$+yw^3$	$+xzw^2$	$-4xyzw$	$-3yz^2w$			$+yw^3-z^2w-xzw\Theta$	$-wR$
$\gamma^2\delta^2$				$+xww^2$	$-xyzw$	$+8yz^2w$		$-3yz^2w$		$+xzw-yz^2w$	$+zw\Theta$
$\alpha\beta^3$			$+yw^3$		$+xzw^2$	$-4xyzw$	$-4y^2z$	$+xy^2w$	$-yz^3$	$+ywR\Theta-5y^2zw$	$-5yw\Theta$
$\alpha\beta^2\gamma$			$+xxw^3$	$-4y^2w^2$	$+8y^2zw$			$-5xyzw+5y^2zw$		$+3y^2zw-5xyz^2+5y^2zw$	$-3xz\Theta$
$\alpha\beta\gamma^2$			$+3x^2zw$	$-xy^2w$	$-4xyzw$	$+8xyz^2$	$-xy^2w$	$+x^2yw-xy^2z$		$+x^2yw-xy^2z$	$-5xz\Theta$
$\alpha\gamma^3$			$+3x^2zw$		$-xy^2w$	$-4xy^2z$		$+x^3w$	$-xy^3$	$+x^2zw-xy^2z$	$+xz\Theta$
$\beta^3\delta$			$+x^2z$		$-xw^3$				$+x^3z$	$+x^3z-xy^3$	$+xP$
$\beta^2\gamma\delta$			$+3x^2z$	$+3xyw^2$		$-x^2zw$	$-4x^2z^2$		$-z^4+xw^3$	$+z^4-xw^3$	$+zR-w^2\Theta$
$\beta\gamma^2\delta$					$+x^2yw$	$+2x^2w^2$	$-8xyzw$	$+6y^2z^2$		$+2x^2w^2-8xyzw+6y^2z^2$	$+(2xw-6yz)\Theta$
$\gamma^3\delta$			$+x^2z$		$+x^2yw$		$-4y^3z$	$+4xy^2z$		$-4y^3z+4xy^2z$	$+4y^2\Theta$
β^4	$+z^4$									$+z^4-xw^3$	$+zR-w^2\Theta$
$\beta^3\gamma$	$-4yz^3$						$+4xz^2w$			$+4xz^2w$	$+4z^2\Theta$
$\beta^2\gamma^2$	$+6y^2z^2$				$+2x^2w^2$	$-8xyzw$	$+6y^2z^2$			$+2x^2w^2-8xyzw+6y^2z^2$	$+(2xw-6yz)\Theta$
$\beta\gamma^3$	$-4y^3z$				$+4xy^2w$					$+4y^2w$	$+4y^2\Theta$
γ^4	$+y^4$	$-x^3w$								$+y^4-x^3w$	$-yP-x^2\Theta$

[Table reproduced from the scan: rows index the monomials of degree four in $(\alpha,\beta,\gamma,\delta)$, columns enumerate the monomials of Ω in the six line-coordinates (a,b,c,f,g,h); the penultimate column is the sum of the entries on that row, and the last column expresses that sum in the basis (Θ,P,Q,R).]

Collecting the terms multiplied by P, Q, R, Θ, respectively, we have

$$\begin{aligned}\Omega\ &= P\{y\alpha\delta^3 - x\beta\delta^3 + 3w\alpha y^3 + xy^3\delta - yy^3\}\\ &\quad + Q\{\Sigma y\alpha\beta\delta + 3z\alpha^2y^3\delta - 3w\alpha^2z^2\delta^2 + 3w\alpha\beta^2z^2\delta - z^2\delta^2y^2 + 3xy^3\alpha\gamma\}\\ &\quad + R\{xw - z^2\alpha^3\delta + 4(\alpha^3\delta - \alpha\beta^3) + 3w\alpha\beta y^3 - z\alpha\beta^3y - y^3\}\\ &\quad + \Theta\{w^2\alpha^2\beta^2 + 2\alpha^2\beta^2\gamma + 2xw + \alpha\beta\delta\cdot 2\,2\alpha\beta\delta + (4z + 8yz)\alpha\beta\gamma\delta + \ldots\}\,,\end{aligned}$$

which may be written as follows:

$$\begin{aligned}\Omega\ &= P\{y(\alpha\delta^3 - \beta^3) + x(\gamma^3 - \beta\delta^3)\} + P(3w\alpha y^3)\\ &\quad + Q\{3z(\beta^3 - \alpha\gamma^2y^3) + 3w(\alpha^2\delta - \alpha\gamma^2y) - 3y\delta(\alpha^2\beta\delta - \alpha\beta^2\gamma)\} + Q(3z\alpha\gamma^2y - 3z\alpha^2\beta^2y)\\ &\quad + R\{xw + 3z(\alpha^3\delta - \alpha\beta^3) + (\alpha^3\gamma - \alpha\beta^3)\} + R(-3z\alpha^3\gamma)\\ &\quad + \Theta\big[zw(-\alpha^2\beta\delta + \alpha\beta^2\gamma)\\ &\qquad + xz(-\alpha\beta^2\delta + \beta^2\gamma)\\ &\qquad + yw(-2\alpha\beta^2\delta + \alpha\beta^2\gamma)\\ &\qquad + xw\,2(-\alpha\beta\gamma\delta + \alpha\beta\gamma^2)\\ &\qquad + yz(-2\alpha^3\delta + 5\alpha^2\beta\delta - 6\alpha\beta^2\gamma)\\ &\qquad + x^2(\alpha\beta^3\delta - \alpha\beta^2\gamma^2)\\ &\qquad + y^2(-\alpha^4\delta + \beta^4)\\ &\qquad + z^2(-\alpha^3\delta + \beta^3\delta)\\ &\qquad + w^2(\alpha^2\beta^2 - \beta^3)\big]\,,\end{aligned}$$

in which all the terms contained in the { } admit of expression in terms of P', Q', R', Θ', the remaining six terms not included within { } may be written

$$\begin{aligned}&3wP\alpha(\gamma' - \beta\delta') + 3(wP - yQ)\alpha\beta\Theta' - 3w\alpha\gamma^2\beta\delta';\\ &3zR\beta(\beta' - \alpha'\gamma) + 3(-xR + zQ)\alpha\gamma^2 - 3\theta wxyz\gamma^2;\end{aligned}$$

which, observing that $wP - yQ = xz\Theta'$, and $-xR + zQ = yw\Theta$, are

$$\begin{aligned}&3wP\alpha\beta(\gamma' - \beta\delta') + 3xz\Theta\alpha\beta(\gamma\beta - \beta\delta) + 3(\beta^3 - \alpha\gamma^3)\\ &\qquad - 3zR\beta(\beta' - \alpha'\gamma)\\ &\qquad - (\alpha^2\gamma\delta - \alpha\beta\gamma^2)\Theta.\end{aligned}$$

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Papers 859 (concl.), 860, 861, 862, 863 (start)

Arthur Cayley

859 (concluded). On the Lines Which Meet a Unicursal Quartic Curve.

(Continuation of the determination of Ω; printed page 431.)

The expression thus becomes

$$\begin{aligned}
\Omega = P\cdot\quad & x(y\gamma^3 - y^3\gamma) && = \alpha\delta\Omega_1,\\
& + z(\alpha\delta^3 - \gamma^3\beta) && = y(-z\beta_1 + 2w\theta\Omega)\\
& + 3xw(\alpha\gamma - \beta\delta) && = 3\alpha\beta_1\\
& + Q\cdot -3z(\beta\delta - w\gamma^2) && = -2\alpha\delta\Omega_2\\
& \quad + 3y(\alpha\delta - x\gamma^2) && = 3\alpha\beta\delta_1\\
& + R\cdot -3x(\beta^3 - z\alpha^2\gamma) && = 3\alpha^2\delta_1\\
& \quad - z(\alpha\delta - \beta^2) && = (-\beta^4 + \alpha^3\delta)\\
& \quad + w(\alpha\delta^3 - \alpha\beta\gamma) && = -\alpha^2\gamma\delta\Omega\\
& \quad + zw(-\alpha^2\delta^2 - \beta^2\gamma^2) && = -2\alpha\beta\gamma\delta\Omega\\
& \quad + \tfrac{3}{2}\alpha\beta\delta - \tfrac{1}{2}\gamma^2 && = \alpha\beta\delta\Omega\\
& + S\cdot xw(\alpha\beta\gamma - \beta\delta^2) && = \alpha\beta\delta\Omega_2\\
& \quad - z(\alpha\delta^2 - \alpha\beta\gamma) && = -\alpha\beta\delta\Omega_1\\
& \quad + yz(-2\alpha^2\beta\delta^2 - \beta^4\gamma^2) && = -2\alpha\beta^2\delta\Omega\\
& \quad + \tfrac{3}{2}\alpha\beta\delta\gamma && = 3\alpha\beta^2\delta\Omega\\
& \quad - \tfrac{1}{2}\beta^4 + 6\beta^2\gamma && = -6\beta^3\Omega\\
& + w^2\cdot 2\alpha^2\delta^3 - 4\alpha\beta\delta && = -\alpha\beta^2\Omega\\
& \quad - (\alpha^3\beta\gamma - \beta^4) && = -\beta^3\delta\Omega
\end{aligned}$$

and we thus finally obtain

$$\begin{aligned}
\Omega\ &= PR\,(3\alpha\delta w - \gamma^2 y + S)\\
&\quad + RP\,(3\beta\gamma - \delta^2 z + \alpha w)\\
&\quad + PQ\cdot \tfrac{1}{3}\,S^2\,y\\
&\quad + Q\Omega\cdot -3\,(\alpha w + \cdots).
\end{aligned}$$

viz. this is the equation of the cone, vertex $(\alpha,\beta,\gamma,\delta)$, which passes through the quartic curve $x:y:z:w = 1:\theta:\theta^2:\theta^3$. As regards the symmetry of this expression, it is to be remarked that, changing (x,y,z,w) and $(\alpha,\beta,\gamma,\delta)$ into (w,z,y,x) and $(\delta,\gamma,\beta,\alpha)$ respectively, we change (Θ,P,Q,R) and (Θ',P',Q',R') into (Θ,R,Q,P) and (Θ',R',Q',P') respectively, and so leave Ω unaltered. Again, interchanging (x,y,z,w) and $(\alpha,\beta,\gamma,\delta)$, interchange (Θ,P,Q,R) and (Θ',P',Q',R'), and so leave Ω unaltered.

860. On Briot and Bouquet's Theory of the Differential Equation $F\left(u, \dfrac{du}{dz}\right) = 0.$

[From the *Proceedings of the London Mathematical Society*, vol. XVIII. (1887), pp. 314–324.]

I CONSIDER the theory of a differential equation of the first order as developed in Briot and Bouquet's *Théorie des fonctions doublement périodiques et en particulier des fonctions elliptiques* (§8, 1st ed., Paris, 1859), but I make some substantial variations in the mode of treatment.

I remark that, writing $u = x$, $\dfrac{du}{dz} = y$, I make the theory to depend altogether upon that of the curve $F(x, y) = 0$; viz. the form of my result is, in order that a differential equation of the first order $F\left(u, \dfrac{du}{dz}\right) = 0$ may have a monotropic integral, the curve $F(x, y) = 0$ must satisfy certain conditions. Briot and Bouquet give their result (Theorem iv., p. 296) under a somewhat different form, as follows; viz. changing the word "monodrome" into "monotrope," their theorem is:

"Pour qu'une équation différentielle du premier ordre de la forme

$$\left(\frac{dy}{dz}\right)^m + f_1(u)\left(\frac{dy}{dz}\right)^{m-1} + \cdots + f_m(u) = 0$$

admette une intégrale monotrope: 1° *les coefficients* $f_1(u)$, $f_2(u)$, …, $f_m(u)$ *doivent être des polynômes entiers en* u *et, au plus, le premier du second degré, le second du quatrième degré, …, le dernier du degré* $2m$*;* 2° *quand pour une certaine valeur de* u *l'équation a une racine multiple différente de zéro,* $\dfrac{dy}{dz}$ *doit rester monotrope par rapport à* u*;* 3° *quand pour une certaine valeur* u_1 *de* u*, l'équation a une racine multiple égale à zéro, le premier terme du développement de* $\dfrac{dy}{dz}$ *suivant les puissances croissantes de* $(u - u_1)^{1/n}$ *doit avoir l'exposant* $1 - \dfrac{1}{n}$*, si cet exposant est plus petit que l'unité;* 4° *enfin l'équation différentielle, que l'on déduit de la première en posant* $u = \dfrac{1}{v}$*, doit offrir pour* $v = 0$ *les mêmes caractères. Ces conditions sont suffisantes."*

I notice that this may be regarded as a statement referring to the curve

$$F(x, y) = y^m + f_1(x)\, y^{m-1} + \cdots + f_m(x) = 0,$$

but that in 2° a notion of monotropy is assumed, for y must be a monotropic function of x; and that 4° in effect introduces the new curve $F\left(\dfrac{1}{x}, y\right) = 0$.

1. We have between the variables u and z a differential equation of the first order $F\left(u, \dfrac{du}{dz}\right) = 0$, a rational and integral function of u and $\dfrac{du}{dz}$. The equation determines u as a function of z, and we wish to know whether u is a monotropic function of z. It will not be so if we have a tropical point, viz. a point $z = c$, such that, in the neighbourhood thereof, the value of u is given by a series $u - a = B(z - c)^{p/q} + \cdots$, in ascending powers of $z - c$, involving fractional powers of $z - c$. Conversely, if there is no tropical point, then u will be a monotropic function of z; but we must consider and exclude not only tropical points at a finite distance, but also the point at infinity, viz. there must not be any development $u = Bz^{p/q} + \cdots$, in descending powers, involving fractional powers of z.

2. A one-valued function (*eindeutige Function*, or *fonction uniforme* or *bien déterminée*) is monotropic; since there is only one value, the function must, after a circuit described by the point z, recover its original value. But, conversely, a monotropic function is one-valued, that is, no two- or more-valued function can be monotropic. For, suppose a two-valued function of z, having the values Z_1 and Z_2, is monotropic; there is by hypothesis no tropical point, that is, no point $z = c$, such that the function Z_1, after a circuit described by the point z, instead of recovering its original value, assumes the value Z_2; and the like as to Z_2. But, this being so, it is meaningless to regard Z_1 and Z_2 as two values of the same function; they must be

considered as two distinct and separate functions, each of them a one-valued monotropic function. And it thus appears that the two notions, monotropic and one-valued, are in fact equivalent.

3. In what precedes, u and z are regarded as complex variables geometrically represented by means of two real points in their own planes respectively. But we now write $\frac{du}{dz} = y$, $u = x$, whence $F(x, y) = 0$, regarding x, y as the coordinates, real or imaginary, of a point of the curve represented by this equation. And this curve has, moreover, to be considered from a non-projective point of view; we have to distinguish between, and separately consider, finite points (x and y each of them finite), and the points at infinity (x and y either or each of them infinite). And it is moreover necessary to distinguish between, and separately consider, points on the axis of x (or say axial points), and points off the axis of x (say non-axial points). Taking, for convenience, a to denote a finite value which may be $= 0$, b a finite value which is not $= 0$ (or say, a not $= \infty$, b not $= \infty$ or 0), we have thus the six cases, 1° (a, b); 2° $(a, 0)$; 3° (a, ∞); 4° (∞, b); 5° $(\infty, 0)$; 6° (∞, ∞). I regard also the axes of x and y as horizontal and vertical respectively, and speak of the tangent or element of the curve as horizontal, vertical, or inclined, accordingly.

4. In the neighbourhood of any given point of the curve, we have $y =$ a series of powers of x, the form of the series being different for different classes of points. I write $P(x)$ to denote a series (finite or infinite) $A+Bx+Cx^2+\cdots$, in ascending powers of x; it is to be throughout understood that the leading coefficient A is not $= 0$. Of course, $P(x-a)$, $P(x-a)^{1/n}$, $P(x^{-1})$, &c., will denote the like series $A+B(x-a)+\cdots$, $A+B(x-a)^{1/n}+\cdots$, $A+Bx^{-1}+\cdots$, &c. The coefficients after the leading coefficient A may any or all of them vanish; thus $P(x-a)^{1/n}$ extends to denote the series $P(x-a)$ of integer powers, but naturally the former notation is not used unless the series contains fractional powers. We can, by means of the symbol P, express for any given point of the curve the form of the expansion in the neighbourhood thereof; and, attaching to the point the expansion which belongs to it, we may, for instance, speak of a point $y = P(x-a)$, viz. if b is the constant term of the series, then this is a non-axial point (a, b), which is an ordinary (non-singular) point of the curve, and for which the element is not vertical; a like point with the element vertical would be a point $y = b + (x-a)^{\frac{1}{2}} P(x-a)^{\frac{1}{2}}$. Observe that, in the case of a multiple point, there is a separate expansion for each branch through the point, and in thus attaching an expansion to the point we regard the point as belonging to one of these branches; in dealing with a multiple point, it is necessary to consider separately all the different expansions, that is, all the branches through the point. It is clear that for a point (a, b), $(a, 0)$, or (a, ∞) the expression for y depends on $P(x-a)$, or $P(x-a)^{1/n}$; while, for a point (∞, b), $(\infty, 0)$, or (∞, ∞), it depends on $P(x^{-1})$ or $P(x^{-1/n})$; viz. in the two cases respectively, we have a series in ascending powers of $x-a$, or in descending powers of x.

5. In regard to any given point of the curve, substituting for y, x their values $\frac{du}{dz}$, u, and thence forming the reciprocal of $\frac{du}{dz}$, we obtain $\frac{dz}{du} =$ a series in u; viz. this is a series in ascending powers of $(u-a)$, or in descending powers of u. Integrating, we have $z - c =$ a series in u, which series may or may not contain a logarithmic term $\log u$ or $\log(u-a)$; and, reverting the series, we obtain $u =$ a series in $(z-c)$. This result, applicable to the neighbourhood of the point $u = a$, may contain only integer powers of $z-c$, and be accordingly of the form $u =$ a one-valued series in z; and we then say that the point on the curve is a "permissive" point. Or it may contain fractional powers of $z-c$, and be accordingly of the form $u =$ a more-than-one-valued series in z; and we then say that the point on the curve is a "prohibitive" point. The necessary and sufficient condition in order that u may be a monotropic function of z is: the points of the curve must be all of them permissive points; or, what is the same thing, there must not be any prohibitive point.

6. I form a complete table of permissive points, as follows:

$$\begin{array}{ll}
(a, b), & y = P(x-a), \\
(a, 0), & y = (x-a)P(x-a) \text{ or } (x-a)^{\frac{1}{n}} P(x-a)^{\frac{1}{n}}, \\
(a, \infty), & y = (x-a)^{-\frac{1}{2}} P(x-a)^{\frac{1}{2}}, \\
(\infty, b), & y = P(x^{-1}), \\
(\infty, 0),\ (\infty, \infty), & y = xP(x^{-1}) \text{ or } x^2 P(x^{-1}), \\
 & y = x^{1+\varepsilon/n} P(x^{-1/n})
\end{array}$$

(where n is a positive integer $= 2$ at least, $\varepsilon = 0$, 1, or -1,) viz. it is only a point (a, b), $(a, 0)$, (∞, b) or (∞, ∞) which may be permissive; a point (a, ∞) or $(\infty, 0)$ is prohibitive.

7. A point $y = P(x-a)$ is permissive. We have

$$\frac{du}{dz} = P(u-a), \qquad \frac{dz}{du} = \frac{1}{P(u-a)} = P(u-a),$$

$$z - c = (u-a)\,P(u-a),$$
$$u - a = (z-c)\,P(z-c),$$

the required result. The several steps will be readily understood; the reciprocal of a series $P(u-a)$ is a series of the like form $P(u-a)$; integrating this in regard to u as a series in $u-a$ (no constant being added on the right-hand side), we obtain a series $(u-a)\,P(u-a)$; and finally, $z-c$ being equal to this expression, we obtain, by reversion, a like expression $u-a = (z-c)\,P(z-c)$.

We might, for convenience, have written x, u, z in place of $x-a$, $u-a$, $z-c$, respectively. The proof would have run

$$y = P(x), \quad \frac{du}{dz} = P(u), \quad \frac{dz}{du} = P(u), \quad z = u\,P(u), \quad u = z\,P(z),$$

meaning $u-a = (z-c)\,P(z-c)$, as above; and, in what follows, this is done accordingly.

It is worth while to make an instance in which the integration can be performed: say we consider the point $(0,1)$ of the curve $y = \dfrac{1+x+x^2}{1+2x}$; we have

$$\frac{du}{dz} = \frac{1+u+u^2}{1+2u}, \qquad \frac{dz}{du} = \frac{1+2u}{1+u+u^2},$$

$$z = \log(1+u+u^2), \quad \text{or} \quad u+u^2 = e^z - 1,$$

giving a series $u = z\,P(z)$, or the point is a permissive point, as in the general case. We have here $u+u^2 = (e^z-1)$, viz. u is not a monotropic function of z; but this is by reason of a prohibitive point $(-\tfrac{1}{2}, \infty)$.

8. A point $y = (x-a)\,P(x-a)$ is permissive.

Writing as above u, z in place of $u-a$, $z-c$, we have

$$\frac{du}{dz} = u\,P(u), \qquad \frac{dz}{du} = \frac{1}{u}\,P(u), \quad \text{or say} \quad m\frac{dz}{du} = \frac{1}{u} + Pu,$$

whence

$$mz = \log u + u\,P(u),$$

that is,

$$u\,e^{u\,P(u)} = e^{mz}, \quad \text{or say} \quad u\,P(u) = e^{mz},$$

whence

$$u = e^{mz}\,P(e^{mz}), \quad \text{a one-valued series.}$$

I take a particular instance, the point $(0,0)$ for the curve $y = \dfrac{x - \frac{1}{3}x^3}{1-x^2}$; here

$$\frac{du}{dz} = \frac{u - \frac{1}{3}u^3}{1-u^2}, \qquad \frac{dz}{du} = \frac{1-u^2}{u - \frac{1}{3}u^3},$$

$$z = \log(u - \tfrac{1}{3}u^3), \quad \text{or} \quad u - \tfrac{1}{3}u^3 = e^z;$$

the point is permissive. The equation just obtained shows that u is not a monotropic function of z; but this is by reason of the prohibitive points $(\pm 1, \infty)$.

9. A point $y = (x-a)^2\,P(x-a)$ is permissive. Here

$$\frac{du}{dz} = u^2 P(u), \qquad \frac{dz}{du} = \frac{1}{u^2}P(u) = \frac{A}{u^2} + \frac{B}{u} + P(u),$$

whence

$$z = -\frac{A}{u} + B\log u + uP(u),$$

where the logarithmic term occurs or does not occur, according as B is not or is $= 0$. In the former case, we have

$$e^{z/B} = ue^{-\frac{A}{Bu}+uP(u)};$$

in the latter case,

$$z = -\frac{A}{u} + uP(u);$$

and in each case we can, by reversion, express u as a one-valued series in z. We may take as examples the two curves $y = \dfrac{x^2}{1-x}$ and $y = \dfrac{x^2}{1-x^2}$; in each case the point $(0,0)$ is a permissive point, but, by reason of the prohibitive points $(1,\infty)$ and $(\pm1,\infty)$, u is not a monotropic function of z.

10. As to the other forms of permissive points, the proofs depend upon those for the forms already considered. Thus for

$$y = (x-a)^{1+\varepsilon/n}P\big((x-a)^{1/n}\big), \quad \text{or} \quad \frac{du}{dz} = u^{1+\varepsilon/n}P(u^{1/n}),$$

writing $v = u^{1/n}$, we find

$$\frac{dv}{dz} = u^{(1+\varepsilon)/n}P(u^{1/n}) = v^{1+\varepsilon}P(v),$$

that is,

$$\frac{dv}{dz} = P(v), \quad vP(v), \quad \text{or} \quad v^2P(v);$$

in each case $v =$ one-valued series in z, and thence $u = v^n =$ one-valued series in z.

Similarly for $y = P(x^{-1})$, $y = x\,P(x^{-1})$, $y = x^2\,P(x^{-1})$, or say here

$$y = x^{1+\varepsilon}\,P(x^{-1});$$

$$\frac{du}{dz} = u^{1+\varepsilon}\,P(u^{-1}),$$

or, writing $v = u^{-1}$, we have

$$\frac{dv}{dz} = u^{\varepsilon-1}P(u^{-1}) = v^{1-\varepsilon}P(v);$$

viz. this is

$$\frac{dv}{dz} = P(v), \quad vP(v), \quad \text{and} \quad v^2P(v);$$

in each case $v =$ one-valued series in z, and thence $u = v^{-1} =$ one-valued series in z.

And finally, for

$$y = x^{1+\varepsilon/n}P\left(\frac{1}{x^{1/n}}\right);$$

here

$$\frac{du}{dz} = u^{1+\varepsilon/n}P\left(\frac{1}{u^{1/n}}\right),$$

or, writing $v = u^{-1/n}$, we have

$$\frac{dv}{dz} = u^{(\varepsilon-1)/n}P\left(\frac{1}{u^{1/n}}\right) = v^{1-\varepsilon}P(v),$$

that is,

$$\frac{dv}{dz} = P(v), \quad vP(v), \quad \text{or} \quad v^2P(v);$$

as before; $v =$ one-valued series in z, and thence $u = v^{-n} =$ one-valued series in z.

11. There is no difficulty in showing that every point of the curve, not being a permissive point as above, is a prohibitive point. Thus for a point (a, b), if the series for $y - b$ contain any fractional power, say, if we have $y = b + (x-a)^{1/n} + \cdots$, then

$$\frac{du}{dz} = b + u^{1/n}, \qquad \frac{dz}{du} = \frac{1}{b} - \frac{u^{1/n}}{b^2} + \cdots,$$

whence, reverting, $u = bz +$ series containing the fractional power $z^{1/n+1}$.

Again, a point $y = (x-a)^m\, P(x-a)$, $m = 3$ or any higher integer value, is prohibitive. Attending only to the leading term, we have $\dfrac{du}{dz} = u^m$, whence $\dfrac{dz}{du} = u^{-m}$ or u^{1-m}, whence, reverting, $z = u^{1-m}$, $u = z^{1/(1-m)}$, which is a fractional power; and this also is the case if m be a negative integer. And similarly, in other cases, the series for u will always contain fractional powers of z.

12. In order that u may be a monotropic function of z, the points of the curve must be all of them permissive, or, what is the same thing, there must be no prohibitive point; we have to consider the geometrical signification of this condition. If we attend first to the ordinary (or non-singular) points of the curve, a non-axial point (a, b) must be a point $y = P(x-a)$, viz. there must be no such point with a vertical element.

An axial point $(a, 0)$ may be $y = (x-a)\, P(x-a)$, viz. the element may be inclined; or the point may be $y = (x-a)^2\, P(x-a)$, viz. the element may be horizontal (but in this case there must be an ordinary or two-pointic contact only); or the point may be $y = (x-a)^{\frac{1}{2}}\, P(x-a)^{\frac{1}{2}}$, viz. the element may be vertical.

The point must not be (a, ∞), viz. there must be no asymptote parallel to (or coincident with) the axis of y.

The point may be (∞, b), $y = P(x^{-1})$, that is, there may be an ordinary, or osculating, asymptote parallel to the axis of x. But there is no point $(\infty, 0)$, that is, there must be no asymptote coincident with the axis of x.

There may be points (∞, ∞), $y = x\, P(x^{-1})$, that is, ordinary or osculating asymptotes inclined to the axes; and there may also be points (∞, ∞), $y = x^2\, P(x^{-1})$, that is, asymptotic parabolas of vertical axis.

13. We have, moreover, conditions as to the singular points; thus, every non-axial multiple point (a, b) must be a point with each branch of the form $y = P(x-a)$, that is, each branch must be an ordinary (non-cuspidal) branch, and must be non-vertical. The axial multiple or singular points $(a, 0)$ may have ordinary (non-cuspidal) branches $y = (x-a)\, P(x-a)$ and $y = (x-a)^2\, P(x-a)$, inclined or horizontal, or else $y = (x-a)^{\frac{1}{2}}\, P(x-a)^{\frac{1}{2}}$, vertical; and there may also be cuspidal branches $y = (x-a)\, P(x-a)^{1/n}$, $y = (x-a)^{1/n}$, and $y = (x-a)^{1+1/n}\, P(x-a)^{1/n}$, $P(x-a)^{1/n}$ inclined, horizontal, or vertical.

There must be no singular points (a, ∞); any such point is, in fact, excluded by the condition, no asymptote parallel to or coincident with the axis of y.

There may be multiple points (∞, b), but not $(\infty, 0)$; viz. as above, there may be asymptotes parallel to, but not coincident with, the axis of x; but in this case, the several branches must be each of them an ordinary branch $y = P(x^{-1})$.

Finally, there may be multiple or singular points (∞, ∞), but each branch must be either an ordinary branch $y = x\, P(x^{-1})$ or $y = x^2\, P(x^{-1})$, or a cuspidal branch.

14. The enumeration seems long and difficult. But for a given curve we have, in fact, only to see that there are no ordinary non-axial points (a, b) of vertical element, and to further examine the finite singular points (a, b) and $(a, 0)$, and also the several infinite branches as giving ordinary or else singular points (a, ∞), (∞, b), $(\infty, 0)$, or (∞, ∞); viz. it has to be seen for each infinite branch that the expansion is of the proper form.

15. Writing the equation of the curve in the form

$$f_0(x)\, y^m + f_1(x)\, y^{m-1} + \cdots + f_m(x) = 0,$$

where f_0, f_1, …, f_m denote rational and integral functions of x, then, if a be a root of the equation $f_0(x) = 0$, there will be on the curve a point (a, ∞) which is prohibitive; $f_0(x)$ is therefore a constant, or, taking it to be $= 1$, the form must be

$$y^m + f_1(x)\, y^{m-1} + \cdots + f_m(x) = 0,$$

where f_1, f_2, …, f_m denote rational and integral functions of x.

It is to be shown that the degrees of these functions are equal respectively to 2, 4, …, m at most. Supposing that the degrees are 2, 4, …, $2m$; then for the points at infinity we find an expansion $y = Ax^n + \cdots$, in descending powers of x, by substitution in an equation

$$y^m + a_1 x\, y^{m-1} + a_2 x^2\, y^{m-2} + \cdots + a_m x^m = 0;$$

whence $n = 2$, and A is determined by an equation of the order m; there are thus m branches $y = x^2\, P(x^{-1})$, corresponding to permissive points. If, however, any one of the y functions be of a higher degree, then it may be shown (but the formal proof is not easily given) that there will be at any rate a branch $y = Ax^n + \&c.$, where n has a value > 2, and we have thus a prohibitive point; hence the degrees must be as stated.

16. As an example of a differential equation with a monotropic solution, take

$$\left(\frac{du}{dz}\right)^2 = \left(\frac{du}{dz}\right) + 27u^2 - 4 = 0;$$

we have here the curve

$$y^3 = 3y + 27x^2 - 4 = 0,$$

which is a nodal cubic, as shown in the annexed figure. To put the node in evidence, the equation may be written in the form $(y+2)^2(y-1) + 27x^2 = 0$; viz. the node is the point $(0, -2)$, and the equation of the tangents is $(y+2)^2 - 27x^2 = 0$. The curve besides meets the line $x = 0$ in the point $y = 1$, where the tangent is horizontal, and it meets the axis $y = 0$ in the points $x = \pm\dfrac{2}{3\sqrt{3}}$, at each of which the tangent is vertical; these two points, as lying on the axis $y = 0$, are thus each of them permissive; and they are the only points where the tangent is vertical (in fact, the tangents from the point at infinity on the line $x = 0$ are the two lines $x = \pm\dfrac{2}{3\sqrt{3}}$, and the line infinity counting twice, as a tangent at an inflexion).

[Figure: nodal cubic curve $y^3 - 3y + 27x^2 - 4 = 0$, showing the loop at the node $(0, -2)$ with branches descending and the upper smooth arc through $(0, 1)$.]

For the points at infinity, we have $y = -3x^{\frac{2}{3}} - 1 - \frac{1}{3}x^{-\frac{2}{3}} + \&c.$, which is of the form $y = x^{\frac{2}{3}} P\left(\dfrac{1}{x^{1/3}}\right)$, included in $y = x^{1+\varepsilon/n}\, P(x^{-1/n})$, and the point is thus permissive; there is no prohibitive point, and the differential equation has a monotropic solution. This is, in fact, the rational solution $u = (z-c) - (z-c)^3$, or say $u = z - z^3$; the curve is thus given by the two equations $x = z - z^3$, $y = 1 - 3z^2$.

17. As a second example, take the differential equation

$$\left(\frac{du}{dz}\right)^3 - \left(\frac{du}{dz}\right)^2 + 27u^2 - 4 = 0.$$

We have here the curve

$$y^3 - y^2 + 4x^2 - 27x^4 = 0,$$

which is a trinodal quartic curve, as shown in the figure. There is a node at the origin with the tangents $y^2 - 4x^2 = 0$, and, writing the equation in the form

$$(3y+1)^2(3y-2)^2 - 8(27x^2 - 2)^2 = 0,$$

we have for the other two nodes $x = \pm\dfrac{1}{3\sqrt{3}}$, $y = \dfrac{2}{3}$. The lowest points are given by $x = \pm\dfrac{2}{3\sqrt{3}}$, $y = -\dfrac{1}{3}$; viz. the line $y = -\dfrac{1}{3}$ is a horizontal tangent. Writing $x = 0$, we have $y^3 - y^2 = 0$, that is, $y^2 = 0$, the node at the origin, and $y = 1$, the height of the loop; and, writing $y = 0$, we have $4x^2 - 27x^4 = 0$, that is, $x^2 = 0$, the node at the origin, and $x = \pm\dfrac{2}{3\sqrt{3}}$, the other two intersections with the axis of x. The tangents at these two points are vertical; as being on the axis, they are thus permissive points. And they are the only points with a vertical tangent; in fact, the point $(x = 0, y = \infty)$ is a point of the curve, with the line ∞ as an osculating tangent (4-pointic intersection); hence the tangents from the point $x = 0$, $y = \infty$ are the line ∞ counting four times, and the lines $x = \pm\dfrac{2}{3\sqrt{3}}$, touching at the points $\left(\pm\dfrac{2}{3\sqrt{3}}, -\dfrac{1}{3}\right)$, as above.

[Figure: trinodal quartic curve $y^3 - y^2 + 4x^2 - 27x^4 = 0$, showing the three nodes and the lower loops with horizontal tangent at $y = -\frac{1}{3}$.]

For the infinite branches, we have $y = 3^{\frac{2}{3}} x^{\frac{4}{3}} + \cdots$, which is of the right form $y = x^{1+\frac{1}{3}} P\left(\frac{1}{x^{1/3}}\right)$. We thus see that there is no prohibitive point; and the differential equation has a monotropic solution accordingly.

Writing z in place of $z - c$, the solution in fact is

$$u = \frac{e^z - e^{-z}}{e^z + e^{-z}} - \left(\frac{e^z - e^{-z}}{e^z + e^{-z}}\right)^3;$$

hence, putting $\theta = \dfrac{e^z - e^{-z}}{e^z + e^{-z}}$, the curve is given by the two equations

$$x = \theta - \theta^3, \qquad y = (1 - 3\theta^2)(1 - \theta^2) = 1 - 4\theta^2 + 3\theta^4;$$

the monotropic function u satisfies the differential equation.

18. $F\left(u, \dfrac{du}{dz}\right)$ must be either a rational function, a singly periodic function, or a doubly periodic function of z; say the forms are

$$u = P(z), \quad u = P(e^z),$$

and

$$u = P\{\mathrm{sn}(gz, k)\} + Q\{\mathrm{sn}(gz, k)\}\ \mathrm{cn}(gz, k)\ \mathrm{dn}(gz, k),$$

where P, Q denote rational functions. I do not at present consider the criteria (such as are given in Briot and Bouquet's Theorem v., p. 301) for determining by means of the curve which is the form of the integral. I remark, however, that in the first and second cases the curve is unicursal, while in the third case it is bicursal; or say that, according as the deficiency is 0 or 1, the integral is rational or simply periodic, or else it is doubly periodic. Moreover, the curve being unicursal, we can express the coordinates as equal to rational functions $P(\theta)$, $Q(\theta)$ of a parameter θ; and, being bicursal, we can express them as functions of the form

$$P(\theta)\sqrt{1 - \theta^2 \cdot 1 - k^2\theta^2} + P_1(\theta), \quad Q(\theta)\sqrt{1 - \theta^2 \cdot 1 - k^2\theta^2} + Q_1(\theta).$$

of the parameter θ; and supposing the coordinates (x, y), that is, u and $\dfrac{du}{dz}$, thus expressed, it should be easy to establish the relation of θ with z, e^z or $\mathrm{sn}(gz, k)$ in the three cases respectively.

861. Note on a Formula Relating to the Zero-Value of a Theta-Function.

[From *Crelle's Journal der Mathem.*, t. c. (1887), pp. 87, 88.]

I HAD some difficulty in verifying for the case of a single theta-function, a formula given in Herr Thomae's paper "Beitrag zur Theorie der ϑ-Functionen," *Crelle's Journal*, vol. LXXI. (1870), pp. 201–222. The formula in question (see p. 216) is given as follows:

$$(11) \qquad \vartheta(0,0,\ldots,0) = \sqrt{\frac{|A_\lambda^{(\lambda)}|}{(2\pi i)^p}}\,\sqrt[4]{\text{Discr.}(0,0,\ldots,0)\,\text{Discr.}'(0,0,\ldots,0)},$$

but the denominator factor should I think be $(\pi i)^p$ instead of $(2\pi i)^p$. Making this alteration, then in the case of a single theta-function, $p = 1$, and the function belongs to the radical

$$\sqrt{x-k_1\cdot x-k_2\cdot x-k_3\cdot x-k_4}$$

where

$$(k_1, k_2, k_3, k_4) = \left(-\tfrac{1}{k},\ -1,\ +1,\ +\tfrac{1}{k}\right).$$

The determinant $|A_\lambda^{(\lambda)}|$ is a single term $= A$, and the formula becomes

$$\vartheta(0) = \sqrt{\frac{A}{\pi i}}\,\sqrt[4]{(k_3-k_1)(k_4-k_2)},$$

where $k_3 - k_1$, $k_4 - k_2$ are each $= 1 + \dfrac{1}{k}$, and we have therefore

$$\vartheta(0) = \sqrt{\frac{A}{\pi i}\left(1+\frac{1}{k}\right)}.$$

A also denotes the integral

$$\int_{k_1}^{k_2} \frac{dx}{\sqrt{x-k_1\cdot x-k_2\cdot x-k_3\cdot x-k_4}} = \int_{-\frac{1}{k}}^{-1} \frac{k\,dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \int_1^{\frac{1}{k}} \frac{k\,dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = ikK',$$

and the formula thus is

$$\vartheta(0) = \sqrt{\frac{K'(1+k)}{\pi}}.$$

But observe that, in the theta-function as defined by the equation

$$\vartheta(x) = \Sigma e^{an^2+2mnx},$$

a is used to denote the value

$$a = a_1'B, \qquad = \frac{2\pi}{A}B,$$

where A is the above-mentioned integral, and B is the integral

$$B = \int_{k_2}^{k_3} \frac{dx}{\sqrt{x-k_1\cdot x-k_2\cdot x-k_3\cdot x-k_4}} = \int_{-1}^{1} \frac{k\,dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = 2kK,$$

which value must however be taken negatively, viz. we must write $B = -2kK$, and we then have

$$a = -\frac{2\pi K}{K'};$$

viz. writing as usual

$$q = e^{-\pi K'/K}, \qquad r = e^{-\pi K/K'},$$

the e^a of the theta-function is not q, but it is r^2; and the zero-value $\vartheta(0)$ is $= 1 + 2r^2 + 2r^8 + 2r^{18} + \cdots$. The equation thus is

$$1 + 2r^2 + 2r^8 + 2r^{18} + \cdots = \sqrt{\frac{K'(1+k)}{\pi}},$$

which is right; in fact, writing k' in place of k, and consequently K, q in place of K', r respectively, the equation becomes

$$1 + 2q^2 + 2q^8 + 2q^{18} + \cdots = \sqrt{\frac{K(1+k')}{\pi}};$$

we have

$$1 + 2q + 2q^4 + 2q^9 + \cdots = \sqrt{\frac{2K}{\pi}},$$

and changing q into q^2, then (*Fund. Nova*, p. 92) K is changed into $\dfrac{K(1+k')}{2}$, and we have the formula in question. As a verification for small values of q, observe that we have

$$\frac{2K}{\pi} = 1 + 4q + 4q^2, \qquad \frac{1+k'}{2} = 1 - 4q + 16q^2,$$

and thence

$$\frac{K(1+k')}{\pi} = 1 + 4q^2, \qquad \text{or} \qquad \sqrt{\frac{K(1+k')}{\pi}} = 1 + 2q^2.$$

Cambridge, 12 February, 1886.

862. Note on the Theory of Linear Differential Equations.

[From *Crelle's Journal der Mathem.*, t. c. (1887), pp. 286–295.]

1. I consider a linear differential equation

$$p_0 \frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

where p_0, p_1, ..., p_m are rational and integral functions of x, having no common factor: as usual, x is a complex magnitude represented by a point, and we consider the integrals belonging to a singular point $x = a$ of the differential equation. An integral may be a regular integral, or it may be what Thomé calls a normal elementary integral: the theory of these integrals (which I would rather call subregular integrals) requires, I think, further examination.

2. I retain x as the independent variable, but for shortness use t to denote $\frac{1}{x-a}$: and I take as the dependent variable z, $= \frac{1}{y}\frac{dy}{dx}$: we have then z determined from any value of z we derive a corresponding value $e^{\int z\,dx}$ of y.

3. To obtain the z-equation, using for shortness accents to denote differentiation in regard to x, we have $z = \frac{y'}{y}$, and thence $z' = \frac{y''}{y} - \left(\frac{y'}{y}\right)^2$, that is, $\frac{y''}{y} = z' + z^2$; also

$$\frac{y'''}{y} - \frac{y''\,y'}{y^2} = z'' + 2zz',$$

that is

$$\frac{y'''}{y} = z'' + 3zz' + z^3, \quad \text{or finally} \quad z'' + 3zz' + z^3 = \frac{y'''}{y},$$

and so on; viz. the values of $\frac{y'}{y}$, $\frac{y''}{y}$, $\frac{y'''}{y}$, ... are z, $z^2 + z'$, $z^3 + 3zz' + z''$, ...; $\frac{y^{(n)}}{y}$, generally for the first term is z^n and the last term is $z^{(n-1)}$. If, to fix the ideas, the y-equation is

$$\frac{d^3 y}{dx^3} + p_1 \frac{d^2 y}{dx^2} + p_2 \frac{dy}{dx} + p_3 y = 0,$$

then, dividing by y, the z-equation is

$$(z'' + 3zz' + z^3) + p_1(z' + z^2) + p_2\, z + p_3 = 0;$$

and similarly, from the y-equation of the order m, we derive a z-equation of the order $m-1$,

$$p_0\,[z^{(m-1)} + \cdots + z^m] + p_1[\cdots + z^{m-1}] + \cdots + p_{m-2}(z' + z^2) + p_{m-1}\, z + p_m = 0.$$

4. In this equation, each coefficient is in the first instance a rational and integral function of x, or, what is the same thing, of $x - a$; writing t to denote $\frac{1}{x-a}$, each coefficient is thus the sum of a finite number of negative integer powers of t; hence multiplying the whole equation by a proper positive power of t, the several coefficients become each of them the sum of a finite number of positive powers of t, or arranging in descending powers of t, say the general form is $p_r = \alpha_r t^\lambda + \beta_r t^{\lambda-1} + \cdots + \kappa_r$, where λ is a positive integer which may be $= 0$, and which has for each coefficient its proper value. We wish to satisfy the z-equation by a value $z =$ descending series in t, viz. the form is $z = At^\alpha + A't^{\alpha-1} + \cdots$, with powers which are integral or fractional, but where the number of positive powers is finite. The theory is almost identical with that of the solution in like form of the algebraical equation

$$p_0\, z^m + p_1\, z^{m-1} + \cdots + p_{m-1}\, z + p_m = 0,$$

or say the equation $U = 0$, where U is a rational and integral function of z and t, of the degree m as regards z.

5. Considering z and t as Cartesian coordinates, the equation in question, $U = 0$, represents a geometrical curve such for any given value of the abscissa t, the ordinate z has m values: the curve is of the order m at least, but it may be of a higher order $m + \kappa$; when this is so, there is at infinity on the axis $t = 0$, a κ-tuple point K, and thus any line parallel to the axis $t = 0$, intersects the curve in the point K counting as κ intersections, and in m other points, which are the points belonging to the m values of the ordinate z. Taking $s = 1$, and for z, t writing $\frac{z}{s}$, $\frac{t}{s}$ respectively, we have between the trilinear coordinates z, t, s an equation $U = 0$ of the order $m + \kappa$, where $s = 0$ is the equation of the line infinity: and the curve has a κ-tuple point at the point K for which $t = 0$, $s = 0$.

6. Starting from the equation $U = 0$, for an arbitrary value of t we have m values of z all of them different. These may be developed, each of them as a descending series in t, and the development is effected in the usual manner; viz. considering for the moment t as very large, then z is in general very large and it has a leading term At^α, which is determined by writing in the equation $z = At^\alpha$, and then finding α in such wise that there are two or more exponents equal to each other and greater (that is, nearer $+\infty$) than any other exponent: we have thus a highest power of t, the whole coefficient for which must vanish, viz. we obtain an equation of two or more terms giving for A a value or values not $= 0$; in the term At^α in question, α is in general positive, but it may be $= 0$, or be negative. The leading term At^α being found in this manner, the law for the exponents of the subsequent terms is frequently at once apparent; but, if necessary, we write $z = At^\alpha + A't^{\alpha-\alpha'} + \cdots$, and determine in like manner the exponent $\alpha - \alpha'$ and the coefficient A'. Proceeding in this manner until the law of the exponent becomes apparent, and then by the method of indeterminate coefficients, we finally arrive at a series

$$z = At^\alpha + A't^{\alpha-\alpha'} + A''t^{\alpha-\alpha''} + \cdots ,$$

in descending powers of t: the exponents are integral or fractional, but the number of positive exponents is always finite. There is either a single series or it may be two or more series; but the number of series is at most $= m$, and when it is less than m, then the coefficients in the several series or some of them will contain radicals, and by giving to each radical its different values, the system of series will determine m different values of z.

7. The curve meets the line infinity ($s = 0$) in the point K counting as κ intersections, and in m other points, some or all of which may coincide with K; the forms of the series depend on the configuration of the m points, or say on the relation of the curve to the line infinity. Thus suppose $\kappa = 0$, and further that the m points are all of them distinct, that is, let the curve be a curve of the order m meeting the line infinity in m distinct points, or, what is the same thing, having m asymptotes, no two of them parallel; we have in this case m series, each of the form

$$z = At + B + \frac{C}{t} + \frac{D}{t^2} + \cdots ,$$

where the several coefficients A have distinct values.

8. In the foregoing case A is determined by an equation of the order m, having m unequal roots; and taking for A any root at pleasure, the remaining coefficients B, C, ... are each of them linearly determined. If the equation has two equal roots, this may correspond to the case of two parallel asymptotes (the curve has here a node at infinity); the coefficients B, C, ... will in this case depend on a quadric radical, and by giving to this radical its two values, we obtain the two series

$$z = At + B + \frac{C}{t} + \cdots ,$$

corresponding to the two equal roots of the equation. But if instead of a node at infinity we have the line infinity a tangent to the curve, then instead of the two parallel asymptotes, we have an asymptotic parabola; the series assumes a new form, viz. it contains terms in $t^{\frac{1}{2}}$, $t^{-\frac{1}{2}}$, ...; the coefficients of the integer powers have determinate values, but those of the powers $t^{\frac{1}{2}}$, $t^{-\frac{1}{2}}$, ... contain as factor a quadric radical, and giving to this radical its two values, we have two series.

9. The like considerations apply in the case where the line infinity passes through higher multiple points of the curve, or has with it a contact or contacts other than a single ordinary contact: in every case,

reckoning as distinct series those obtained by attaching to each radical its different values, the number of distinct series will be $= m$ precisely.

10. Reverting now to the differential equation

$$p_0\,[z^{(m-1)} + \cdots + z^m] + p_1[\cdots + z^{m-1}] + \cdots + p_{m-2}(z' + z^2) + p_{m-1}\,z + p_m = 0,$$

it is to be observed that the very same process is applicable to the determination of z in the form of a descending series $z = At^\alpha + A't^{\alpha-\alpha'} + \cdots$; moreover, it frequently happens that the determination of the leading term At^α is made by means of the algebraical equation

$$p_0\,z^m + \cdots + p_{m-2}\,z^2 + p_{m-1}\,z + p_m = 0,$$

viz. this is so whenever the value of α is greater than 1: in fact, writing $z = At^\alpha$ $(\alpha > 1)$, then in the sets of terms $\{z^m + \cdots + z^{(m-1)}\}, \ldots, (z^3 + 3zz' + z''), (z^2 + z')$, the first terms z^m, z^{m-1}, $\ldots$, z^3, z^2 are each of them of a higher degree than any other term in the same brackets, so that attending only to the terms of highest degree the sets may be reduced to z^m, z^{m-1}, $\ldots$, z^3, z^2. But in the determination of the subsequent exponents and coefficients, the omitted terms or some of them would of course come into play: and it is at least not obvious that we can, for the forms of the series which satisfy the differential equation, employ geometrical considerations such as those which were used in regard to the series satisfying the algebraical equation.

11. Considering as above the coefficients p_0, p_1, $\ldots$, p_m as rational and integral functions of t, it is easy to determine the degrees of these coefficients in such wise that the exponent α of the leading term At^α shall have a given value. Suppose first, this given value is $\alpha = a$, a positive integer greater than 1: the degrees may be θ, $\theta + a$, $\theta + 2a$, $\ldots$, $\theta + ma$, ($\theta = 0$ or any positive integer at pleasure); and not only so, but if we write

$$p_0 = L_0\,t^\theta + \cdots\,, \quad p_1 = L_1\,t^{\theta+a} + \cdots\,, \quad \ldots, \quad p_m = L_m\,t^{\theta+ma} + \cdots\,,$$

then writing $z = At^a$, clearly the highest power in the equation is $t^{\theta+ma}$, and equating the coefficient hereof to zero, we have

$$L_0\,A^m + L_1\,A^{m-1} + \cdots + L_{m-1}\,A + L_m = 0,$$

an equation of the degree m for the determination of the coefficient of the leading term At^a $(a > 1)$. Assuming that the roots are all unequal, we have m series each of the form

$$z = At^a + Bt^{a-1} + \cdots + Mt + N + \frac{O}{t} + \cdots.$$

12. Suppose $a = 1$, that is, let the leading term be At; here the degrees may be θ, $\theta+1$, $\theta+2$, $\ldots$, $\theta+m$, ($\theta = 0$ or any positive integer as before), but we have a different form for the A-equation. In fact, here ($z = At$) in the sets of terms $\{z^m + \cdots + z^{(m-1)}\}, \ldots, (z^3 + 3zz' + z''), (z^2 + z')$, the terms in each brackets are of the same degree, viz. the degrees are m, $\ldots$, 3, 2; and not only so, but we have $z^2 + z' = (A^2 + A)\,t^2$, that is, $[A]_2\,t^2$; $z^3 + 3zz' + z'' = [A]_3\,t^3$, $\ldots$. Hence if

$$p_0 = L_0\,t^\theta + \cdots\,, \quad p_1 = L_1\,t^{\theta+1} + \cdots\,, \quad p_m = L_m\,t^{\theta+m} + \cdots\,,$$

the A-equation is

$$L_0\,[A]_m + L_1\,[A]_{m-1} + \cdots + L_{m-1}\,[A]_1 + L_m = 0,$$

an equation of the degree m as before; assuming that the roots are unequal, we have here m series each of the form

$$z = At + B + \frac{C}{t} + \frac{D}{t^2} + \cdots\,,$$

m being, as will presently appear, the case of regular integrals for the y-equation.

13. In either of the last-mentioned cases, the formula for the A-equation holds good if any two or more of the coefficients p_0, p_1, $\ldots$, p_m are of the degrees aforesaid, the others of them being of inferior degrees; we have only to put $= 0$ the L's which belong to the coefficients of inferior degrees. If neither L_0 nor L_m be $= 0$, the equation is still an equation of the order m, with m effective roots; but if any of the L coefficients

at the beginning of the equation vanish we thus introduce roots $= \infty$: and if any of the coefficients L at the end of the equation vanish, we thus introduce roots $= 0$: the number of effective roots is $m-$ the number of roots ∞ or 0; the case requires further consideration. If the equation has equal roots, then from what precedes, it is easy to infer that we may have distinct series containing as in the general case only integer powers of t, or we may have series beginning with At^a or At, but containing also fractional powers of t.

14. If the leading term is $z = At^{-a}$, a a value < 1, then in the sets of terms $\{z^m + \cdots + z^{(m-1)}\}, \ldots, (z^3 + 3zz' + z''), (z^2 + z')$ the terms $z^{(m-1)}, \ldots, z''$, z' are each of them of a higher degree than any other term in the same brackets, so that attending only to the terms of the highest degree the sets may be reduced to $z^{(m-1)}, \ldots, z''$, z', or we may consider the equation. In particular, this is the case for $z = At^{-a}$, a being a positive integer; in this case, the degrees of p_0, p_1, $\ldots$, p_m may be $\theta + m - 1$, $\theta + m$, $\theta + m + 1$, $\ldots$, $\theta + a$, and this being so, the leading coefficient A will be determined by a linear equation. But the case $\alpha = 0$ is, in fact, that of a non-singular point $x = a$: and I do not here further consider it.

15. Reverting to the case where the series is

$$z = At + B + \frac{C}{t} + \frac{D}{t^2} + \cdots ,$$

and substituting for t its value $\dfrac{1}{x-a}$, or for greater convenience $\dfrac{1}{x}$, this is

$$\frac{1}{y}\frac{dy}{dx} = \frac{A}{x} + B + Cx + \cdots ,$$

whence

$$\log y = A \log x + Bx + \tfrac{1}{2}Cx^2 + \cdots ,$$

or passing to the value of y, but expanding the exponential of $Bx + \frac{1}{2}Cx^2 + \cdots$, this is

$$y = x^A\,(1 + Bx + Cx^2 + \cdots),$$

viz. we have here a regular integral. By what precedes, it appears that, for the values $p_0 = L_0 x^{-\theta} + \cdots$, $p_1 = L_1 x^{-\theta-1} + \cdots$, $\ldots$, $p_m = L_m x^{-\theta-m} + \cdots$, the exponent A is determined by the equation

$$L_0\,[A]_m + L_1\,[A]_{m-1} + \cdots + L_{m-1}\,[A]_1 + L_m = 0,$$

which is the equation for that purpose considered by Fuchs, and is what I would call the indicial equation.

The logarithmic forms which belong to the case of equal roots may be deduced from the case of unequal roots, by supposing two or more of the roots to approach each other continuously and become ultimately equal.

16. Similarly, if the series be

$$z = \frac{F}{x^a} + \cdots + \frac{K}{x} + A + Bx + \tfrac{1}{2}Cx^2 + \cdots ,$$

(a being a positive integer $= 2$ at least), then writing as above $t = \dfrac{1}{x}$, this is

$$\frac{1}{y}\frac{dy}{dx} = \frac{F}{x^a} + \cdots + \frac{K}{x} + A,$$

whence, if for shortness $w = \dfrac{F}{a-1}\dfrac{1}{x^{a-1}} - \cdots - \dfrac{K}{x}$, we have

$$\log y = w + A \log x + Bx + \tfrac{1}{2}Cx^2 + \cdots ,$$

or passing to the value of y, but expanding as before the exponential of $Bw + \frac{1}{2}Cx^2 + \cdots$, this is

$$y = e^w\,x^A\,(1 + Bx + Cx^2 + \cdots),$$

a subregular integral. By what precedes, it appears that, if

$$p_0 = L_0\,x^{-\theta} + \cdots\,, \quad p_1 = L_1\,x^{-\theta-a} + \cdots\,, \quad \ldots, \quad p_m = L_m\,x^{-\theta-ma} + \cdots\,,$$

then the equation for the leading coefficient (previously represented by A) is

$$L_0\,F^m + L_1\,F^{m-1} + \cdots + L_{m-1}\,F + L_m = 0,$$

but I do not see any direct or simple way of obtaining the corresponding equation for the exponent A of this subregular integral.

17. To illustrate the case of a series with fractional powers, and also in order to work out a simple example with some completeness, I consider the differential equation of the second order $\dfrac{d^2y}{dx^2} + p_2\,y = 0$, giving the z-equation $z^2 + z' + p_2 = 0$; p_2 is a rational and integral function of $t\,(=\dfrac{1}{x})$, and we have two cases according as the order of the function is even or odd.

18. Suppose first that the order is even, and for convenience taking it to be $= 4$, and assuming the value of p_2 accordingly, I consider the equation

$$z^2 + z' = \alpha t^4 + \beta t^3 + \gamma t^2 + \delta t + \varepsilon.$$

Here the form is at once seen to be

$$z = At^2 + Bt + C + \frac{D}{t} + \frac{E}{t^2} + \cdots\,,$$

and observing that we have

$$z' = -t^2\frac{dz}{dt} = -2At^3 - Bt^2 + D + \frac{2E}{t} + \cdots\,,$$

we find

$$\begin{aligned} A^2 &= \alpha, \\ 2AB - 2A &= \beta, \\ 2AC + B^2 - B &= \gamma, \\ 2AD + 2BC + 0C &= \delta, \\ 2AE + 2BD + C^2 + D &= \varepsilon, \\ 2AF + 2BE + 2CD + 2E &= 0, \end{aligned}$$

equations which give

$$A = \sqrt{\alpha},$$

$$B = 1 + \frac{\beta}{2\sqrt{\alpha}},$$

$$C = \frac{\gamma}{2\sqrt{\alpha}} - \frac{\beta}{4\alpha} - \frac{\beta^2}{8\alpha\sqrt{\alpha}},$$

$$D = \frac{\delta}{2\sqrt{\alpha}} - \frac{\gamma}{2\alpha} + \frac{1}{4\alpha\sqrt{\alpha}}(\beta - \beta\gamma) + \frac{\beta^2}{4\alpha^2} + \frac{\beta^3}{16\alpha^2\sqrt{\alpha}},$$

$$E = \frac{\varepsilon}{2\sqrt{\alpha}} - \frac{3\delta}{4\alpha} + \frac{1}{4\alpha\sqrt{\alpha}}\left(3\gamma - \beta\delta - \tfrac{1}{2}\gamma^2\right) + \frac{1}{4\alpha^2}\left(-\tfrac{3}{2}\beta + 3\beta\gamma\right) + \frac{1}{16\alpha^2\sqrt{\alpha}}\left(-\tfrac{17}{2}\beta^2 + 3\beta^2\gamma\right) - \frac{\beta^3}{4\alpha^3} - \frac{5}{128}\frac{\beta^4}{\alpha^3\sqrt{\alpha}}, \quad \&c.$$

We then have

$$z = \frac{1}{y}\frac{dy}{dx} = \frac{A}{x^2} + \frac{B}{x} + C + Dx + Ex^2 + \cdots\,,$$

and consequently

$$\log y = -\frac{A}{x} + B\log x + Cx + \tfrac{1}{2}Dx^2 + \tfrac{1}{3}Ex^3 + \cdots\,,$$

$$y = e^{-A/x}\, x^B\, e^{Cx+\frac{1}{2}Dx^2+\frac{1}{3}Ex^3+\cdots},$$

where the second exponential is to be expanded in the form $1 + C'x + D'x^2 + \cdots$; there are of course two values, say y_1 and y_2, corresponding to the values $+\sqrt{\alpha}$ and $-\sqrt{\alpha}$ of the radical respectively.

19. Secondly, if the order be odd, and for convenience taking it to be $= 3$, and assuming the value of p_2 accordingly, I consider the equation

$$z^2 + z' = \beta t^3 + \gamma t^2 + \delta t + \varepsilon.$$

Here the series is at once seen to be

$$z = At^{3/2} + Bt + Ct^{1/2} + D + Et^{-1/2} + \cdots :$$

we have

$$z' = -\tfrac{3}{2}At^{5/2} - Bt^2 - \tfrac{1}{2}Ct^{3/2} + \tfrac{1}{2}Et^{1/2} + \cdots ,$$

and we thence derive the equations

$$\begin{aligned} A^2 &= \beta, \\ 2AB - \tfrac{3}{2}A &= 0, \\ 2AC + B^2 - B &= \gamma, \\ 2AD + 2BC - \tfrac{1}{2}C &= 0, \\ 2AE + 2BD + C^2 &= \delta, \\ 2AF + 2BE + 2CD + \tfrac{1}{2}E &= \varepsilon, \end{aligned}$$

(where in the next equation we have on the right-hand side ε, and in each of the subsequent equations we have on the right-hand side 0). We find

$$A = \sqrt{\beta}, \quad B = \tfrac{3}{4}, \quad C = \frac{1}{2\sqrt{\beta}}\left(\gamma + \tfrac{3}{16}\right), \quad D = -\frac{1}{4\beta}\left(\gamma + \tfrac{3}{16}\right), \quad E = \frac{1}{2\beta\sqrt{\beta}}\left(\delta + \tfrac{9}{32}\gamma - \tfrac{1}{4}\gamma^2 + \tfrac{63}{1024}\right), \quad \&c.,$$

where observe that the coefficients A, C, E, ... contain as a factor $\dfrac{1}{\sqrt{\beta}}$, but the coefficients B, D, ... are rational.

We have

$$z = \frac{1}{y}\frac{dy}{dx} = \frac{A}{x^{3/2}} + \frac{B}{x} + \frac{C}{x^{1/2}} + D + Ex^{1/2} + Fx + \cdots ,$$

giving

$$\log y = -\frac{2A}{x^{1/2}} + B\log x + 2Cx^{1/2} + Dx + \tfrac{2}{3}Ex^{3/2} + \tfrac{1}{2}Fx^2 + \cdots ;$$

and thence passing to the value of y, but expanding the exponential of $2Cx^{1/2} + Dx + \cdots$, we have say the integral

$$y_1 = e^{-2A/x^{1/2}}\, x^B\,(1 + C'x^{1/2} + D'x + E'x^{3/2} + \cdots),$$

and then also, changing the sign of $\sqrt{\beta}$, the integral

$$y_2 = e^{+2A/x^{1/2}}\, x^B\,(1 - C'x^{1/2} + D'x - E'x^{3/2} + \cdots).$$

Writing for a moment

$$Q = e^{-2A/x^{1/2}}, \qquad M = 1 + D'x + F'x^2 + \cdots , \qquad N = C' + E'x + G'x^2 + \cdots ,$$

the two integrals are $x^B\,Q\,(M + Nx^{1/2})$, and $x^B\,Q^{-1}\,(M - Nx^{1/2})$: these may be replaced by their sum and difference, viz. these are

$$\begin{aligned} y_1 + y_2 &= Mx^B\,(Q + Q^{-1}) + Nx^{B+1/2}\,(Q - Q^{-1}), \\ y_1 - y_2 &= Mx^B\,(Q - Q^{-1}) + Nx^{B+1/2}\,(Q + Q^{-1}), \end{aligned}$$

where $Q + Q^{-1}$ is a rational function of x, and $Q - Q^{-1}$ is $= x^{1/2}$ multiplied by a rational function of x; hence $y_1 + y_2$ is $= x^B$ multiplied by a rational function of x, but $y_1 - y_2$ is $= x^{B+\frac{1}{2}}$ multiplied by a rational function of x.

Cambridge, 28 April, 1886.

863. Note on the Theory of Linear Differential Equations.

[From *Crelle's Journal der Mathem.*, t. ci. (1887), pp. 209–213.]

The theorem v. given by Fuchs in the memoir "Zur Theorie der linearen Differentialgleichungen mit veränderlichen Coefficienten," *Crelle's Journal*, t. lxviii. (1868), pp. 354–385 (see p. 374) for the purpose of deciding whether the integrals belonging to a group of roots of the "determinirenden Fundamentalgleichung" (or as I call it, the Indicial equation) do or do not involve logarithms, may I think be exhibited in a clearer form.

Starting from the differential equation

$$P(y) := p_0 \frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

of the order m, then if X be any function of x not satisfying the differential equation, we can at once form a differential equation of the order $m+1$, satisfied by all the solutions of the differential equation, and having also the solution $y = X$; the required equation is in fact

$$d_x P(y) \cdot P(X) - P(y) \cdot d_x P(X) = 0.$$

This I call the augmented equation.

I recall that the equation $P(y) = 0$, considered by Fuchs, is an equation having for each singular point $x = a$, m regular integrals, viz. the coefficients $p_0, p_1, \ldots, p_m$, have the forms $q_0(x-a)^m$, $q_1(x-a)^{m-1}$, $\ldots$, q_m, where $q_0, q_1, \ldots, q_m$ are rational and integral functions of $x - a$, q_0 not vanishing for $x = a$, and the other functions $q_1, q_2, \ldots, q_m$ not in general vanishing for $x = a$. Writing $y = (x-a)^\theta$, we obtain

$$P(x-a)^\theta = q_0 \, I(\theta) \, (x-a)^\theta + \text{higher powers of } (x-a),$$

where $I(\theta)$, the coefficient of the lowest power of $(x - a)$, is a function of θ of the order m, which I call the indicial coefficient; and equating it to zero, we have $I(\theta) = 0$, the determinirende Fundamentalgleichung, or Indicial equation, being an equation of the order m. If the roots of this equation are such that no two of them are equal or differ only by an integer number, then we have m particular integrals each of them of the form

$$y = (x-a)^r + \text{higher powers of } (x-a),$$

where r is any root of the indicial equation: but if we have in the indicial equation a group of λ roots $r_1, r_2, \ldots, r_\lambda$, such that the difference of each two of them is either zero or an integer, then the integrals which correspond to these roots involve or may involve logarithms; in particular, if any two of the roots are equal, the integrals for the group will involve logarithms.

Consider now the differential equation $P(y) = 0$ in reference to the singular point $x = a$ as above, and writing $X = (x-a)^e f$ where e is in the first instance arbitrary, and f is a rational and integral function of $x - a$ not vanishing for $x = a$, we form the augmented equation which, observing that we have in general $P(X) = (x-a)^e \, Q$, Q a rational and integral function of $x - a$ not vanishing for $x = a$, and dividing the whole equation by $(x-a)^{e-1}$, may be written

$$d_x P(y) \cdot (x-a) Q - P(y) \left[e \, Q + (x-a) \, d_x Q \right] = 0,$$

an equation of the same form as the original equation (but of the order $m + 1$ instead of m), and having an indicial equation

$$(\theta - e) \, I(\theta) = 0.$$

In fact, writing as before $y = (x-a)^\theta$, we have in $d_x P(y) \cdot (x-a) \, Q$ the term of lowest order $\theta \, I(\theta) \, Q_0 \, (x-a)^{\theta-e}$, and in $P(y) \cdot e \, Q$ the term of lowest order $e \, I(\theta) \, Q_0 \, (x-a)^{\theta-e}$, whereas in $P(y) \, (x-a) \, d_x Q$ the term of lowest order is $(x-a)^{\theta-e+1}$; the indicial equation is thus as just found.

If however e be equal to a root of the indicial equation $I(\theta) = 0$, then instead of $P(X) = (x-a)^e \, Q$, we have $P(X) = (x-a)^\mu \, Q$, where the index μ is $= e +$ a positive integer, and where the value of the difference $\mu - e$ may depend upon the determination of the function f in the expression $(x-a)^e \, f$. The indicial equation for the augmented equation is in this case $(\theta - \mu) \, I(\theta) = 0$.

If the indicial equation $I(\theta) = 0$ of the given differential equation has a group of roots $r_1, r_2, \ldots, r_\lambda$, the difference of any two of these roots being zero or an integer, then taking $e =$ any one of these roots, the augmented equation will have a group of roots $(\mu, r_1, r_2, \ldots, r_\lambda)$.

If any two of the roots $r_1, r_2, \ldots, r_\lambda$ are equal, the group of integrals $u_1, u_2, \ldots, u_\lambda$ will involve logarithms: the question only arises when these roots are unequal, and taking them to be so, the theorem v. is in effect as follows: "If by taking $e =$ some one of the roots $r_1, r_2, \ldots, r_\lambda$, and by a proper determination of the function f we can make μ to be $=$ one of the same roots $r_1, r_2, \ldots, r_\lambda$, then the group of integrals $u_1, u_2, \ldots, u_\lambda$ will involve logarithms; but if μ cannot be made $=$ one of the roots $r_1, r_2, \ldots, r_\lambda$, then the group of integrals will be free from logarithms."

As an example, I consider the equation

$$\frac{d}{dx}\left[(1-x^2)\,\frac{d^2y}{dx^2}\right] - 2x\,\frac{dy}{dx} + n(n+1)\,y = 0.$$

This is Legendre's equation $(1-x^2)\,\dfrac{d^2y}{dx^2} - 2x\,\dfrac{dy}{dx} + n(n+1)\,y = 0$, with $\dfrac{1}{x}$ substituted for x, so that, instead of a singular point $x = \infty$, there may be a singular point $x = 0$. Attending to the singular point $x = 0$, we have

$$P(x)^\theta = \big[(\theta-n)(\theta-n-1) - 2\theta\big]\,x^\theta + \text{higher powers},$$

so that the indicial equation $I(\theta) = 0$ is $\theta^2 - \theta - n^2 - n = 0$, that is, $(\theta+n)(\theta-n-1) = 0$, or we have the roots $-n$, $n+1$, which differ by an integer, and thus form a group, if n be $=$ an integer, or be $=$ an integer $+\frac{1}{2}$; to fix the ideas, say that the roots are $-p$, $p+1$ or else $-p-\frac{1}{2}$, $p+\frac{1}{2}$, where p is a positive integer.

Writing for greater convenience $x^e f = x^e + F$, where F is a sum of powers of x higher than e, we find without difficulty

$$P\big(x^e f\big) = x^e\left\{(e+n)(e-n-1) - (e^2+e)x^2 + (x^2-x^4)x^{-e}F'' - (n^2+n)x^{-e}F\right\}$$

which, so long as e remains arbitrary, is of the form $x^e\,Q$, $Q = (e+n)(e-n-1) +$ powers of x; if however e be a root of the indicial equation, $=$ for instance, if $e = -n$, then the expression in brackets { } contains at any rate the factor x, so that the form is

$$P\big(x^{-n} f\big) = x^\mu\,Q,$$

where μ is $= -n+1$ at least; we can however, by a proper determination of the function f, make μ acquire a larger value.

For instance, suppose $-n$, $n+1 = -2$, 3, $\therefore n = 2$, and assume

$$f = x^e + x^{-2} + Bx^{-1} + Cx^0 + Dx^1 + Ex^2 + Fx^3 + Gx^4 + \cdots$$

To calculate $P\big(x^e f\big)$, we have

[Calculation table follows: the polynomial $P\big(x^e f\big)$ is set out in rows giving the coefficients of x^{e-2}, x^{e-1}, x^e, x^{e+1}, ..., x^{e+5}, etc., as a function of the indeterminates A, B, C, D, E, F, G, and the parameter $\mu = e+n$.]

Hence if B not $= 0$, we have $\mu = -1$; if $B = 0$, C not $= 0$, we have $\mu = 0$; if $B = 0$, $C = 0$, D not $= 0$, we have $\mu = 1$; if $B = 0$, $C = 0$, $D = 0$, E not $= 0$, we have $\mu = 2$; if $B = 0$, $C = 0$, $D = 0$, $E = 0$, F arbitrary, we can by giving proper values to the higher coefficients B, C, D, E, ... from this on make μ have any integer value. The values of $\mu = -1$, 0, 1, 2 being all of them less than the indicial roots $-p$, $p+1$, we thus see by the theorem that, E not $= 0$, then by a proper determination of the function f in the equation $X = x^e\,f$, we can bring μ up to the value p, and this being so, the indicial equation of the augmented equation is $\big(\theta-p\big)(\theta+p)(\theta-p-1) = 0$; that is, the group of integrals $u_1, u_2, \ldots, u_p$ which correspond to the roots $-p$, $p+1$ of the original equation will involve logarithms. The case $B = 0$, etc. is one which appears to me to require further consideration; but I do not here further consider it.

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Arthur Cayley

863 (concluded). Note on the Theory of Linear Differential Equations.

(Continuation of the example for $n = 2$; printed page 456.)

integer), do not involve logarithms: this is right, for the integrals are, in fact, the Legendrian functions of the first and second kinds P_p and Q_p, with only $\dfrac{1}{x}$ written therein instead of x.

Similarly, if for instance n, $n+1 = -\frac{1}{2}$, $\frac{3}{2}$, then, if $e = -n = -\frac{1}{2}$, assuming

$$x^e f \;=\; x^{-\frac{1}{2}} + Bx^{\frac{1}{2}} + Cx^{\frac{3}{2}} + Dx^{\frac{5}{2}} + Ex^{\frac{7}{2}} + \cdots\,,$$

we have

	$x^{-\frac{1}{2}}$	$x^{\frac{1}{2}}$	$x^{\frac{3}{2}}$	$x^{\frac{5}{2}}$
$\frac{1}{2}\cdot -\frac{1}{2}\,B$	$-\frac{1}{4}B$			
	$-\frac{3}{4}C$	$+\frac{3}{4}D$	$-\frac{7}{4}E$	
	$+1$	$-B$	$-3C$	
	$-\frac{1}{2}$	$-\frac{3}{4}B - \frac{3}{4}C$	$-\frac{3}{4}D - \frac{7}{4}E$	
$P(x^e f) =$	0	$-B$	$0\cdot C$	$3D$
		$+\frac{1}{2}$	$-\frac{3}{4}B$	$5E$

We have here B if not $= 0$, $\mu = \frac{1}{2}$; but if $B = 0$, then we cannot in any way make the coefficient of $x^{\frac{1}{2}}$ to vanish, and consequently $\mu = \frac{1}{2}$. With this last value of μ, the group $(\mu, -\frac{1}{2} + n, n-1)$, that is, $(\mu, -\frac{1}{2}, \frac{3}{2})$, becomes $(\frac{1}{2}, -\frac{1}{2}, \frac{3}{2})$ which contains two equal roots, and the conclusion from the theorem thus is that the integrals u_1, u_2 corresponding to the roots $-\frac{1}{2}$, $\frac{3}{2}$ involve logarithmic values. And similarly in general the integrals u_1, u_2, corresponding to the roots $-p + \frac{1}{2}$, $p + \frac{1}{2}$ (p any positive integer), involve logarithmic values: this also is right.

The examples exhibit the true character of the theorem, and show I think that it is a less remarkable one than would at first sight appear: in fact, in working them out, we really ascertain by an actual substitution whether the differential equation can be satisfied by series of powers only, without logarithms. Thus for $n = 2$ as above, it appears that the equation is satisfied by the series

$$y = x^{-2} + Bx^{-1} + Cx^0 + Dx^1 + Ex^2 + Fx^3 + Gx^4 + \cdots\,,$$

where $B = 0$, $C = -\frac{1}{2}$, $D = 0$, $E = 0$, $F = F$, $G = 0$, $H = -\frac{2}{7}F$, ..., that is, by

$$y = x^{-2} + A\bigl(x^0 - \tfrac{1}{2}\bigr) + F\bigl(x^3 + \tfrac{1}{5}\,\cdots\bigr);$$

in other words, that we have the two particular integrals $y = x^{-2} + \frac{1}{2}$, and $y = x^3 + \frac{1}{5}x\cdots$, belonging to the two roots -2, 3 respectively.

Similarly, when $n = \frac{1}{2}$, we cannot satisfy the equation by a series

$$y = x^{-\frac{1}{2}} + Bx^{\frac{1}{2}} + Cx^{\frac{3}{2}} + Dx^{\frac{5}{2}} + \cdots\,;$$

for in order to satisfy the equation, we must have $B = 0$, $C = \infty$, there is thus no series of powers $y = x^{-\frac{1}{2}} + Cx^{\frac{3}{2}} + \cdots$, corresponding to the root $-\frac{1}{2}$: but there is a series $y = x^{\frac{3}{2}} + kx^{\frac{5}{2}} + \cdots$ corresponding to the root $\frac{3}{2}$; and thus the integrals u_1, u_2, corresponding to these roots $-\frac{1}{2}$, $\frac{3}{2}$, involve logarithms.

Cambridge, 23 March 1887.

864. On Rudio's Inverse Centro-Surface.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXII. (1887), pp. 156–158.]

Dr F. Rudio, in an inaugural dissertation "Ueber diejenigen Flächen deren Krümmungsmittelpunkts-flächen confokale Flächen zweiten Grades sind," Berlin, 1880, and *Crelle's Journal*, t. XCV., p. 240, has determined the surfaces having for their centre-surface (i.e., the locus of centres of curvature) the aggregate of the confocal quadric surfaces

$$\frac{x^2}{a-\lambda}+\frac{y^2}{b-\lambda}+\frac{z^2}{c-\lambda}=1, \qquad \frac{x^2}{a-\mu}+\frac{y^2}{b-\mu}+\frac{z^2}{c-\mu}=1,$$

or, what is the same thing, the surfaces orthotomic to the common tangents of these two surfaces. He obtains, as the final result of an elegant analytical investigation, the following formulae:

$$x=\sqrt{(a-\lambda)}\sqrt{\frac{a-u\,.\,a-v}{a-b\,.\,a-c}},$$
$$y=\sqrt{(b-\lambda)}\sqrt{\frac{b-u\,.\,b-v}{b-c\,.\,b-a}},$$
$$z=\sqrt{(c-\lambda)}\sqrt{\frac{c-u\,.\,c-v}{c-a\,.\,c-b}},$$
$$U=\sqrt{\frac{a-u\,.\,b-u\,.\,c-u}{\lambda-u\,.\,\mu-u}},$$
$$V=\sqrt{\frac{a-v\,.\,b-v\,.\,c-v}{\lambda-v\,.\,\mu-v}},$$

$$\xi\,\frac{U-V}{x}=1-\frac{b-\lambda\,.\,c-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{a-\mu}{a-u\,.\,a-v}\,UV,$$
$$\eta\,\frac{U-V}{y}=1-\frac{c-\lambda\,.\,a-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{b-\mu}{b-u\,.\,b-v}\,UV,$$
$$\zeta\,\frac{U-V}{z}=1-\frac{a-\lambda\,.\,b-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{c-\mu}{c-u\,.\,c-v}\,UV,$$

(values which are such that $\xi^2+\eta^2+\zeta^2=1$),
And then

$$\rho=\tfrac{1}{2}\left(\int\frac{du}{U}+\int\frac{dv}{V}\right)+C;$$

$$x'=x+\rho\xi,\quad y'=y+\rho\eta,\quad z'=z+\rho\zeta,$$

viz. the equations give x, y, z, U, V, ξ, η, ζ, each of them as a function of two independent parameters u, v; ρ is a function of u, v and of the arbitrary constant C; hence, giving to C any assumed value, we have x', y', z' each of them a function of the two arbitrary parameters u, v; that is, x', y', z' are the coordinates of a point on a surface, one of a system of parallel surfaces (corresponding to the different values of C) which are the surfaces in question.

Observe that u, v are the elliptic coordinates of the point (x, y, z) on the first of the two confocal surfaces, and that ξ, η, ζ are the cosine-inclinations of one of the tangents from this point to the other confocal surface; so that, if ρ were left arbitrary, the equations $x'=x+\rho\xi$, $y'=y+\rho\eta$, $z'=z+\rho\zeta$ would be the equations of a common tangent of the two confocal surfaces; but ρ, as determined, is the distance of the point x, y, z from the point x', y', z' on the required surface. The expression for ρ involves hyper-elliptic integrals of the first species, which are the same as those which present themselves in the determination of the geodesic lines upon either of the two confocal surfaces.

I have, in quoting these remarkable results, written for greater simplicity a, b, c instead of the author's a^2, b^2, c^2.

Cambridge, Dec. 20, 1886.

865. On Multiple Algebra.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXII. (1887), pp. 270–308.]

1. I REPRODUCE a passage from my Presidential Address, British Association, Southport, 1883.

"Outside of ordinary mathematics we have some theories which must be referred to: algebraical, geometrical, logical. It is, as in many other cases, difficult to draw the line: we do in ordinary mathematics use symbols not denoting quantities, which we nevertheless combine in the way of addition and multiplication, $a+b$ and ab, and which may be such as not to obey the commutative law $ab = ba$; in particular, this is or may be so in regard to symbols of operation; and it could hardly be said that any development whatever of the theory of such symbols of operation did not belong to ordinary algebra. But I do separate from ordinary algebra the system of multiple algebra or linear associative algebra developed in the valuable memoir by the late Benjamin Peirce, "Linear Associative Algebra" (1870, reprinted 1881 in the *American Journal of Mathematics*, vol. IV., with notes and addenda by his son, C. S. Peirce): we here consider symbols A, B, &c., which are linear functions of a determinate number of letters or units i, j, k, l, &c., with coefficients which are ordinary analytical magnitudes real or imaginary, viz. the coefficients are in general of the form $x+iy$, where i is the before-mentioned imaginary, or $\surd(-1)$ of ordinary analysis. The letters i, j, k, &c., are such that every binary combination i^2, ij, ji, &c., (ij not in general $= ji$), is equal to a linear function of the letters, but under the restriction of satisfying the associative law; viz. for each combination of three letters $ij \cdot k$ is $= i \cdot jk$, so that there is a determinate and unique product of three or more letters; or, what is the same thing, the laws of combination of the units i, j, k, ... are defined by a multiplication table giving the values of i^2, ij, ji, &c.; the original units may be replaced by linear functions of these units, so as to give rise for the units finally adopted to a multiplication table of the most simple form; and it is very remarkable how frequently in these simplified forms we have nilpotent or idempotent symbols ($i^2 = 0$ or $= i$ as the case may be), and symbols i, j, such that $ij = ji = 0$; and consequently how simple are the forms of the multiplication tables which define the several systems respectively.

"I have spoken of this multiple algebra before referring to various geometrical theories of earlier date, because I consider it as the general analytical basis, and the true basis, of these theories. I do not realise to myself directly the notions of the addition or multiplication of two lines, areas, rotations, or other geometrical, kinematical, or mechanical entities; and I would formulate a general theory as follows: consider any such entity as determined by the proper number of parameters a, b, c, ... (for instance, in the case of a finite line given in magnitude and position, these might be the length, the coordinates of one end, and the direction-cosines of the line considered as drawn from this end); and represent it by or connect it with the linear function $ai+bj+ck+$&c., formed with these parameters as coefficients and with a given set of units i, j, k, &c. Conversely, any such linear function represents an entity of the kind in question. Two given entities are represented by two linear functions; the sum of these is a like function representing an entity of the same kind, which may be regarded as the sum of the two entities; and the product of them (taken in a determined order, when the order is material) is an entity of the same kind, which may be regarded as the product (in the same order) of the two entities. We thus establish by definition the notion of the sum of the two entities, and that of the product (in a determinate order, when the order is material) of the two entities. The value of the theory in regard to any kind of entity would of course depend on the choice of a system of units i, j, k, ..., with such laws of combination as would give a geometrical or kinematical or mechanical significance to the notions of the sum and product as thus defined.

"Among the geometrical theories referred to, we have a theory (that of Argand, Warren, and Peacock) of imaginaries in plane geometry; Sir W. R. Hamilton's very valuable and important theory of quaternions; the theories developed in Grassmann's *Ausdehnungslehre*, 1844 and 1862; Clifford's theory of biquaternions, and recent extensions of Grassmann's theory to non-Euclidian space by Mr Homersham Cox. These different theories have of course been developed, not in anywise from the point of view from which I have been considering them, but from the points of view of their several authors respectively."

2. The present paper is in a great measure the development of the views contained in the foregoing extract; but, instead of establishing *ab initio* a linear function $ai+bj+ck+\cdots$ as above, I deduce this, as will be seen from the notion of addition.

3. If x, y, ... denote ordinary (real or imaginary) analytical magnitudes, which (as such) are susceptible of addition and multiplication, and for each of these operations are commutative and associative, then we may consider a multiple symbol $(x, y, \ldots)$ involving any given number of letters, to fix the ideas say

(x, y), susceptible of addition and multiplication according to determinate laws

$$(x, y) + (x', y') = (P, Q), \qquad (x, y)(x', y') = (X, Y),$$

where P, Q, X, Y are given functions of x, y, x', y'. For greater simplicity the law of addition is taken to be

$$(x, y) + (x', y') = (x + x',\ y + y'),$$

so that as regards addition the multiple symbols are commutative and associative. But this is or is not the case for multiplication, according to the form of the given functions X, Y; for instance, if

$$(x, y)(x', y') = (xx' - yy',\ xy' + yx'),$$

then in regard to multiplication the symbols will be commutative and associative. But if

$$(x, y, z)(x', y', z') = (yz' - zy',\ zx' - xz',\ xy' - yx'),$$

then the symbols will be associative, but not commutative.

4. I remark here that we are in general concerned with symbols of a given multiplicity, double symbols (x, y), triple symbols (x, y, z), n-tuple symbols $(x_1, x_2, \ldots, x_n)$, as the case may be, and that as well the product as the sum is a symbol *ejusdem generis*, and consequently of the same multiplicity, with the component symbols; this is to be assumed throughout in the absence of an express statement to the contrary. It is, moreover, proper to narrow the notion of multiplication by restricting it to the case where the terms $(X, Y, \ldots)$ of the product are linear functions of the terms $(x, y, \ldots)$ and $(x', y', \ldots)$ of the component symbols respectively; any other form $(X, Y, \ldots)$ is better designated not as a product, but as a combination (or by some other name) of the component symbols $(x, y, \ldots)$ and $(x', y', \ldots)$.

5. I assume, moreover, that, if m be any ordinary analytical magnitude, this may be multiplied into a multiple symbol $(x, y, \ldots)$, according to the law

$$m(x, y, \ldots) = (mx, my, \ldots).$$

6. As a consequence of this last assumption and of the assumed law of addition, we have for instance

$$(x, y, z) = (x, 0, 0) + (0, y, 0) + (0, 0, z) = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1);$$

that is, using single letters i, j, k for the multiple symbols $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ respectively, we have

$$(x, y, z) = xi + yj + zk,$$

where the letters i, j, k, thus standing for determinate multiple symbols, may be termed "*extraordinaries.*" Each extraordinary may be multiplied into any ordinary symbol x, and is commutative therewith, $xi = ix$; moreover, each extraordinary may be multiplied into itself, or into another extraordinary, according to laws which are, in fact, determined by means of the assumed law of multiplication of the original multiple symbols; and, conversely, the law of multiplication of the extraordinaries determines that of the original multiple symbols; thus, if

$$(x, y)(x', y') = (xx' - yy',\ xy' + yx'),$$

then

$$(ix + jy)(ix' + jy') = i(xx' - yy') + j(xy' + yx'),$$

and also

$$(ix + jy)(ix' + jy') = i^2 xx' + ij\, xy' + ji\, yx' + j^2 yy',$$

which expressions will agree together if, and only if, $i^2 = i$, $ij = j$, $ji = j$, $j^2 = -i$, or, as these equations may be written,

$$\begin{array}{c|cc} & i & j \\ \hline i & i & j \\ j & j & -i \end{array}$$

And so in general we have a multiplication table giving each square or product as a homogeneous linear function of all or any of the extraordinaries.

7. And, conversely, from this multiplication table of the extraordinaries i, j, we have

$$(ix+jy)(ix'+jy') = i(xx'-yy') + j(xy'+yx'),$$

that is,

$$(x,y)(x',y') = (xx'-yy',\ xy'+yx'),$$

the originally assumed formula of multiplication.

In the example just given, we have $i^2 = i$, $ij = ji = j$, viz. the symbol i comports itself like unity, and may be put $= 1$; we have $(x,y) = x + jy$, with the multiplication table

$$\begin{array}{c|cc} & 1 & j \\ \hline 1 & 1 & j \\ j & j & -1 \end{array}$$

or simply the equation $j^2 = -1$; and then

$$(x+jy)(x'+jy') = xx' - yy' + j(xy'+yx'),$$

and it is convenient to regard 1 as an extraordinary, and speak of the system of extraordinaries 1, j.

8. The separate terms x, y, ..., whatever be their number, may always without loss of generality be arranged in a line; but it may be convenient to arrange them in a different form, for instance, in that of a square; we may have for instance symbols

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix}$$

with the laws of combination

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix} + \begin{vmatrix} x' & y' \\ z' & w' \end{vmatrix} = \begin{vmatrix} x+x' & y+y' \\ z+z' & w+w' \end{vmatrix},$$

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix} \cdot \begin{vmatrix} x' & y' \\ z' & w' \end{vmatrix} = \begin{vmatrix} xx'+yz' & xy'+yw' \\ zx'+wz' & zy'+ww' \end{vmatrix},$$

where observe that the multiplication is not commutative.

9. A multiple symbol may be connected with a geometrical or physical entity of any kind: viz. any such entity, depending on a number of parameters susceptible of analytical magnitude, to fix the ideas say on two parameters (x,y), may be connected with the multiple symbol (x,y). We cannot in general directly conceive the addition or multiplication of such entities, but the assumed laws of combination of the multiple symbols in effect serve as definitions of the operations. Thus

$$(x,y) + (x',y') = (x+x',\ y+y'); \quad (x,y)\cdot(x',y') = (xx'-yy',\ xy'+yx');$$

or the sum of the entities whose parameters are x, y and x', y' is by definition the entity whose parameters are $x+x'$, $y+y'$; and the product of the same entities is by definition the entity whose parameters are $xx'-yy'$, $xy'+yx'$. The entities are thus, as regards addition, commutative and associative; but, as regards multiplication they are or are not commutative, or associative, according to the assumed law of multiplication of the multiple symbols.

10. If, as above, the multiple symbol be represented by means of extraordinaries, $(x,y) = ix + jy$, then the extraordinaries i, j, qua multiple symbols $(1,0)$, $(0,1)$, are themselves special entities of the kind in question, their laws of combination (as such special entities) being included in the general laws for the combination of such entities; or, if we please, these general laws being derived from the assumed laws for the special entities. Thus, if $ix+jy$ represent the point whose coordinates are x, y, then i represents the point whose coordinates are $(1,0)$, and j the point whose coordinates are $(0,1)$.

11. It has been assumed that the multiple symbols combined together, and their sum and product, are symbols of one and the same multiplicity, say the types of combination are

$$(x,y,z) + (x',y',z') = (x+x',\ y+y',\ z+z'); \quad (x,y,z)(x',y',z') = (X,Y,Z);$$

if this were not so, if for instance we had

$$(x,y,z)(x',y',z') = \text{a double symbol } (X,Y),$$

then this given equation for the multiplication of two triple symbols would not serve as a definition for the multiplication of double symbols, or of a double and a triple symbol, and new definitions would be required. We might, however, have symbols of indefinite multiplicity $(x, y, z, w, \ldots)$, including within them all finite multiplicities, viz. (x, y) meaning $(x, y, 0, 0, \ldots)$, and so in other cases, these being combined into symbols of like indefinite multiplicity

$$(x, y, z, \ldots) + (x', y', z', \ldots) = (x + x',\ y + y',\ z + z',\ \ldots);$$

$$(x, y, z, \ldots)(x', y', z', \ldots) = (X, Y, Z, \ldots);$$

for instance, the law of multiplication might be

$$(x, y, z, \ldots)(x', y', z', \ldots) = (xx' + yx' + zx' + \cdots,\ xy' + yy' + \cdots,\ \ldots),$$

and we could hereby combine symbols of any finite multiplicities (the same or different) whatever.

12. Another peculiarity may be noticed: suppose that, in general,

$$(x, y, z, w) + (x', y', z', w') = (x + x',\ y + y',\ z + z',\ w + w');$$

$$(x, y, z, w)(x', y', z', w') = (X, Y, Z, W);$$

then, as a particular case hereof, we have the addition-equation

$$(x, y, 0, 0) + (x', y', 0, 0) = (x + x',\ y + y',\ 0,\ 0),$$

and it may very well be that for the same two symbols the multiplication-equation may take the form

$$(x, y, 0, 0)(x', y', 0, 0) = (0, 0, Z, W),$$

(Z, W each of them a function of x, y, x', y'); viz. here, in a different point of view, the component symbols $(x, y, 0, 0)$ and $(x', y', 0, 0)$ are double systems of a certain kind $\{x, y\}$, $\{x', y'\}$, and the product is a double system of a different kind $[Z, W]$, or it might have been $(0, Y, Z, W)$, a triple system $[Y, Z, W]$. Of course, all peculiarity disappears when we revert from the particular symbols $(x, y, 0, 0)$ to the general symbols (x, y, z, w), from which we may regard them as derived.

13. The peculiarity just referred to presents itself naturally when we regard the symbols as representing geometrical or physical entities; it may very well be that the product of two entities of a certain kind is taken to be an entity of a different kind (for instance, the product of two points to be a line), and that the analytical theory of the multiple symbol is constructed in order to such a relation; and it may further be that only the symbol $(x, y, 0, 0)$ or $\{x, y\}$ is interpreted with reference to an entity of the one kind, and only the symbol $(0, 0, Z, W)$, or $[Z, W]$ is interpreted by reference to an entity of the other kind, without any interpretation at all being given to the general symbol (x, y, z, w); thus the two forms $(x, y, 0, 0)$, and $(0, 0, Z, W)$, naturally present themselves as double symbols $\{x, y\}$ and $[Z, W]$ of different kinds.

14. In further illustration, suppose the symbols represented as linear functions of extraordinaries,

$$(x, y, z, w) = xe_1 + ye_2 + z\eta_1 + w\eta_2,$$

with a multiplication table for the four extraordinaries e_1, e_2, η_1, η_2. We then have $(x, y, 0, 0) = \{x, y\} = xe_1 + ye_2$; the e_1, e_2 have no proper multiplication table of their own, but the squares and products e_1^2, e_1e_2, e_2e_1, e_2^2 are each given as a linear function of η_1, η_2; hence we have

$$(xe_1 + ye_2)(x'e_1 + y'e_2) = Z\eta_1 + W\eta_2,$$

which is a symbol $(0, 0, Z, W)$, or $[Z, W]$, of a different kind; it may, in completion of the theory, be assumed that in like manner η_1, η_2 have no proper multiplication table of their own, but that the squares and products η_1^2, $\eta_1\eta_2$, $\eta_2\eta_1$, η_2^2 are each given as a linear function of e_1, e_2, leading to a relation

$$(z\eta_1 + w\eta_2)(z'\eta_1 + w'\eta_2) = Xe_1 + Ye_2,$$

which is a symbol $(X, Y, 0, 0)$ or $\{X, Y\}$ of the first kind; it may further happen that each of the several products $e\eta$, ηe is 0, in which case

$$(xe_1 + ye_2)(z\eta_1 + w\eta_2) = 0, \quad (z\eta_1 + w\eta_2)(xe_1 + ye_2) = 0.$$

For the complete theory, we require the multiplication table of the four extraordinaries e_1, e_2, η_1, η_2. The geometrical interpretation may be that $xe_1 + ye_2$ represents a point, $z\eta_1 + w\eta_2$ a line (with or without any geometrical interpretation of a symbol $xe_1 + ye_2 + z\eta_1 + w\eta_2$), the product of two points is a line, the product of two lines is a point; that of a line and point is $= 0$, (see *post* Grassmann, where however the system is a somewhat different one).

15. I do not propose in the present paper to consider the subject of multiple algebra from an analytical point of view; and I will make only a few remarks and give some references. The general theory of associative linear forms is treated in a very satisfactory manner in Peirce's Memoir (1870) above referred to, only it is assumed throughout *ab initio* that the forms are associative. In a Note "On Associative Imaginaries," *Johns Hopkins Circulars*, No. 15 (1882), [822], and more fully in the paper "On Double Algebra," *Proc. Lond. Math. Soc.*, t. XV. (1884), pp. 185–197, [814], starting from the assumed equations $x^2 = ax + by$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = gx + hy$, between the extraordinaries x, y, I considered under what conditions the algebra of these symbols was in fact associative. And in a paper "On the 8-square Imaginaries," *American Journal of Mathematics*, t. IV. (1881), pp. 293–296, [773], I showed that the extraordinaries 0, 1, 2, 3, 4, 5, 6, 7, connected with Euler's theorem of the 8 squares, are of necessity non-associative. I do not know that anything else has been done in regard to non-associative algebras: and it thus appears that the question has hardly been discussed at all. Matrices are associative: I shall have again to refer to them.

16. The object of the present paper is to discuss in detail, from the point of view explained in the extract from my British Association Address, the different theories in regard to geometrical or other entities. I do this under various headings.

The $i = \sqrt{(-1)}$ of Analysis and Analytical Geometry. Art. Nos. 17–26.

17. We may, of course, consider the i as an extraordinary. We have a double symbol (x, y), combining according to the laws

$$(x, y) + (x', y') = (x + x',\ y + y'),$$
$$(x, y)(x', y') = (xx' - yy',\ xy' + yx');$$

and then, introducing the extraordinaries k, i, and writing

$$(x, y) = kx + iy,$$

we have

$$(kx + iy)(kx' + iy') = k^2xx' + ki\,xy' + ik\,yx' + i^2yy';$$

whence

$$k^2 = k, \quad ki = ik = i, \quad i^2 = -k,$$

or the multiplication table is

$$\begin{array}{c|cc} & k & i \\ \hline k & k & i \\ i & i & -k \end{array}$$

But the conditions are satisfied by, and we accordingly assume, $k = 1$; and there then remains only the condition $i^2 = -1$. Compare herewith Sir W. R. Hamilton, "Theory of Conjugate Functions, or Algebraical Couples; with a Preliminary and Elementary Essay on Algebra as a Science of pure time" (*Trans. R. I. Acad.*, t. XVII., 1833–35). I refer to this paper in my Address, and remark upon it that I cannot appreciate the manner in which the author connects, with the notion of time, his algebraical couple or imaginary magnitude $a + bi$, $= a + b\sqrt{(-1)}$ as written in the memoir.

18. But we have, in fact, passed out of this view, and have come to regard $a + bi$ as an ordinary analytical magnitude; viz. in every case an ordinary symbol represents or may represent such a magnitude, and the magnitude (and as a particular case thereof, the symbol i) is commutable with the extraordinaries of any system of multiple algebra. And similarly in analytical geometry, without seeking for any real representation, we deal with imaginary points, lines, &c., that is, with points, lines, &c., depending on parameters of the foregoing form $a + bi$.

$\sqrt{(-1)}$ denotes Perpendicularity. Art. Nos. 19–26.

19. I give a list of the earlier works and memoirs, marking with an asterisk those which I have not seen.

WALLIS. *De Algebra Tractatus*, 1685 Anglice editus; 1693. Chapters 66, 67, 68, and 69 have respectively the titles: *De quadratis negativis eorumque radicibus dictis Imaginariis; Eorundem exemplificatio in Geometria; Effectiones geometricæ his accommodatæ; Aliæ quæ huc spectant constructiones geometricæ.* I quote a single sentence from Chap. 67: "Prout igitur cum æquationis quadraticæ radix prodit negativa, dicendum verbi gratia punctum B pro eo statu haberi non posse ut supponitur in AC exposita prorsum; posse tamen retrorsum ab A in eadem recta; hic vero (de radice quadrati negativi) dicendum, non haberi quidem posse punctum B ut erat suppositum in AC recta, vel ante vel retro: posse tamen (in eodem plano) supra rectam illam": viz. the distance represented by the square root of a negative quantity cannot be measured in the line, forwards or backwards; but can be measured (in the same plane) above the line or, as appears elsewhere, at right angles to the line, either in the plane, or in a plane at right angles thereto.

FORCENEX. "Réflexions sur les quantités imaginaires," *Mél. de Turin*, t. I. (1759), pp. 113–146. The author, after stating that there was no geometrical construction for imaginary quantities, proceeds "Cependant, pour conserver une certaine analogie avec les quantités négatives, un auteur dont nous avons un cours d'algèbre d'ailleurs fort estimable a prétendu les devoir prendre sur une ligne perpendiculaire à celle où l'on les avait supposées, si par exemple &c.," the example which he considers being as follows: On a given line AB to find a point P, such that the product of the distances AP, BP is $= \frac{1}{2}(AB)^2$. Taking $AB = 2a$ and $AP = x$, then $x(2a - x) = 2a^2$, that is, $(x - a)^2 = -a^2$, or $x = a + a\sqrt{(-1)}$, an imaginary value; the condition is, however, satisfied by a real point P off the line AB

$$(AM = MP = \tfrac{1}{2}\,AB = a), \quad \text{(fig. 1)},$$

and hence the author in question infers that the imaginary value $a\sqrt{(-1)}$ is represented by the line MP at right angles to AB. Forcenex objects to this: if the

[Figure 1: line AB with midpoint M, and point P directly above M at distance a, forming a right angle with AB.]

point P may be taken off the axis of x, then the condition $AP \cdot BP = 2a^2$ gives for the locus of P a certain curve; and there is no reason why the foregoing solution $x = a + a\sqrt{(-1)}$ should represent one point rather than another of this curve. As to this see *post* (Peacock).

*TRUEL, Henri Dominique, 1786.

*SUREMAIN-DE-MISSERY. *Théorie purement algébrique des quantités imaginaires*, Paris, 1801; it is referred to in the letter by Servois, *infra*.

20. BUÉE. "Mémoire sur les quantités imaginaires," *Phil. Trans.*, 1806, pp. 23–88.

I give an extract: "Du signe $\sqrt{(-1)}$. Je mets en titre, du signe $\sqrt{(-1)}$ et non de la quantité imaginaire ou de l'unité imaginaire, parce que $\sqrt{(-1)}$ est un signe particulier joint à l'unité réelle, non une quantité particulière: c'est un nouvel adjectif joint au substantif ordinaire et non un nouveau substantif.

"Mais que veut dire ce signe? il n'indique ni une addition ni une soustraction, ni une suppression ni une opposition par rapport aux signes + et −: une quantité accompagnée par $\sqrt{(-1)}$ n'est ni additive ni subtractive, ni égale à zéro. La qualité marquée par $\sqrt{(-1)}$ n'est opposée ni à celle qu'indique +, ni à celle marquée par −: qu'est-elle donc?

"Pour la découvrir supposons trois lignes égales AB, AC, AD (fig. 2) qui partent toutes du point A. Si je désigne la ligne AB par $+1$, la ligne AC sera -1, et la ligne AD qui est une moyenne proportionnelle entre $+1$ et -1, sera nécessairement $\sqrt{(-1 \cdot a)}$ ou plus simplement $\sqrt{(-1)}$. Ainsi $\sqrt{(-1)}$ est le signe de la PERPENDICULARITÉ: donc la propriété caractéristique est que tous les points de la perpendiculaire sont également éloignés de points placés à égale distance de part et d'autre de son pied. Le signe $\sqrt{(-1)}$ signifie tout cela et il est le seul qui l'exprime.

[Figure 2: three equal lines AB, AC, AD from point A, with C to the left of A on the horizontal through B, and D vertically above A.]

"Ce signe mis devant a (a signifiant une ligne ou une surface) veut donc dire *qu'il faut donner à a une situation perpendiculaire à celle qu'on lui donnerait si l'on avait simplement a ou −a.*"

And he, in fact, gives (Prob. vi.) the problem before referred to, and finds no difficulty in assigning to P a position off the line AB. I notice also that Buée considers a curve as having branches in a perpendicular plane; thus for the circle $x^2 + y^2 = a^2$ in the plane of the paper, putting $-x^2$ in place of x^2, there are branches, or, as he calls it, "une appendice," if $y^2 = a^2 + x^2$ (or say $y^2 = a^2 - z^2$), being a rectangular hyperbola, in a plane at right angles to the plane of the paper.

21. Other writings are:

ARGAND. *Essai sur une manière de représenter les quantités imaginaires.* Paris, 1806. Partly reproduced in Argand's memoir under the same title presently referred to.

FRANÇAIS, J. F. "Nouveaux principes de géométrie de position et interprétation géométrique des symboles imaginaires." *Gergonne Annales*, t. IV. 1813–14, pp. 61–72.

ARGAND. "Essai sur une manière, &c." *Ibid.* pp. 133–148.

FRANÇAIS, J. F. "Lettre au Rédacteur." *Ibid.* pp. 222–227.

SERVOIS. "Lettre au Rédacteur." *Ibid.* pp. 228–235.

This last is against the theory, with a running commentary of notes in favour by Gergonne.

FRANÇAIS, J. F. "Autre lettre." *Ibid.* pp. 364–366.

LACROIX. "Note transmise à M. Vecten"; it calls attention to Buée's memoir. *Ibid.* p. 367.

*MOUREY. *Vraie théorie des quantités négatives et des quantités prétendues imaginaires*, 1828, reprinted 1861.

22. WARREN, J. *A treatise on the geometrical representation of the square roots of negative quantities.* 8vo. Cambridge, 1828, pp. 1–154.

PEACOCK, Preface to *Algebra*, 1830, speaks of Warren's work as "distinguished for great originality and for extreme boldness in the use of definitions." There is no Preface or Introduction; the first Chapter is entitled Definitions, Addition, Subtraction, Proportion, Multiplication, Division, Fractions, and Raising of Powers. I quote certain articles as follows:

(1) All straight lines drawn in a given direction from a given point are represented in length and direction by algebraic quantities; and in the following treatise whenever the word quantity is used it is to be understood as signifying a line.

(2) DEF. The given point from which the straight lines are measured is called the origin.

(3) DEF. The sum of two quantities is the diagonal of the parallelogram whose sides are the two quantities.

(12) DEF. The first of four quantities is said to have to the second the same ratio which the third has to the fourth: when the first has in length to the second the same ratio which the third has in length to the fourth, according to Euclid's definition; and also the angle at which the fourth is inclined to the third is equal to the angle at which the second is inclined to the first, and is measured in the same direction.

(17) DEF. Unity is a positive quantity arbitrarily assumed, from a comparison of which the values of other quantities are obtained.

(18) DEF. If there be three quantities such that unity is to the first as the second is to the third, then the third is called the product which arises from the multiplication of the second by the first.

23. The signification of these definitions may be thus expressed. Let the line, drawn from an origin O to the point whose rectangular coordinates are x, y, be called the line $x + iy$; of course the line, length unity in the direction of the axis of x, is the line 1. Then, observing that, putting $x, y = r\cos\theta, r\sin\theta$, and $x', y' = r'\cos\theta', r'\sin\theta'$, we have $(x+iy)(x'+iy') = rr'\{\cos(\theta+\theta') + i\sin(\theta+\theta')\}$, the two definitions are:

The sum of the lines $x + iy$ and $x' + iy'$ is the line

$$x + x' + i(y + y');$$

and the product of the same lines is the line

$$(x + iy)(x' + iy');$$

viz. Warren in effect establishes by definition the notions of the sum and product of two lines, and that by connecting the line with a linear symbol $x + iy$.

It may be right to remark that the general notion of a proportion, used as above to define multiplication, is geometrically the more simple of the two, and that the condition for the proportion $OA : OB = OC : OD$

may be thus expressed: the lengths of the lines are proportional, and the inclination OA to OB is equal to the inclination OC to OD, these inclinations being in the same sense.

24. PEACOCK, G. *A Treatise on Algebra.* 8vo. Cambridge, 1830 (one Vol., pp. 5–38 and 1–685).

"Report on the Recent Progress and Present State of Certain Branches of Analysis." *B. A. Report*, (1833), pp. 185–352.

A Treatise on Algebra. Vol. I. Arithmetical Algebra; Vol. II. Symbolical Algebra and its Application to the Geometry of Position. 8vo. Cambridge, 1842 and 1845.

The statement of the theory is substantially identical with that of the earlier writers, thus *Algebra*, (1830), p. 362, No. 439, is: "We have shown on a former occasion that, if a designated a line in one direction, $-a$ must designate a line in the direction opposite to it, or making an angle with the former equal to two right angles or 180 degrees; but inasmuch as $a\{\sqrt{(-1)}\}^2 = -a$, it follows that the double affectation of the line a with the sign $\sqrt{(-1)}$ produces a result represented by $-a$, and is consequently equivalent to its transfer through 180°: it follows therefore that its single affectation with the sign $\sqrt{(-1)}$ is equivalent to its transfer through half that angle, or through 90°; or, in other words, if a represents a line, $a\sqrt{(-1)}$ will represent a line at right angles to it." But further, pp. 666, 667, the Author considers the before-mentioned problem (for convenience, I write $2a$ instead of a) in the form "to divide a line $2a$ into two such parts that the rectangle contained by them may be $= b^2$"; he takes x, y for the two parts, and in the case where $b^2 > a^2$ finds the two parts

$$x = a - \sqrt{(-1)}\sqrt{(b^2 - a^2)}, \quad y = a + \sqrt{(-1)}\sqrt{(b^2 - a^2)},$$

which he interprets as meaning the two parts are AC, CB, where C is a point off the line AB.

25. It seems to me that, except in Warren, the defect in the exposition of the theory is that it is not made clear, that the theory is in fact a theory of lines in a plane through a given point, with assumed definitions for the addition and for the multiplication of lines; thus in Peacock's problem to "divide a line a into two such parts that the product of them is $= b^2$," meaning the real "given a line OA, $= a$, and a line OB, $= b^2$ (a and b^2 real and positive, so that each of these lines is in the given direction Ox), to find lines OP, OQ such that their sum may be equal to the given line OA, and their product equal to the given line OB." To solve it, take the two lines to be $x + iy = z$, and $x' + iy' = z'$; then, working with z and z', we have $z + z' = 2a$, $zz' = b^2$, giving a solution which may be written in the two forms

$$\{z = a + \sqrt{(a^2 - b^2)}, \quad z' = a - \sqrt{(a^2 - b^2)}\},$$

and

$$\{z = a + i\sqrt{(b^2 - a^2)}, \quad z' = a - i\sqrt{(b^2 - a^2)}\},$$

viz. for $a > b$, the two lines lie each of them in the line Ox, while for $b > a$, they are inclined at equal angles on opposite sides of the line Ox. Or if we work with the real values x, y, x', y', then the same two equations give $x + x' = 2a$, $y + y' = 0$, $xx' - yy' = b^2$, $xy' + yx' = 0$; viz. writing $y' = -y$, we have $x + x' = 2a$, $xx' + y^2 = b^2$, $y(x' - x) = 0$, whence either $x' = x$ or else $y = 0$, and we have the same two solutions as before. Observe that the second condition is $zz' = b^2$, not (as Forcenex understood it) the product of the lengths OP, $OQ = b^2$; and his objection is thus answered. *[Discussion continues on the next page.]*

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Papers 865 (concl.), 866, 867, 868, 869 (start)

Arthur Cayley

865 (concluded). On Multiple Algebra.

(Continuation of §46; printed page 481.)

depending on the new extraordinaries e_1e_2, e_2e_3, e_3e_1. Grassmann further assumes

$$e_1^2 = 0, \quad e_2^2 = 0, \quad e_3^2 = 0,$$

$$e_2e_3 = -e_3e_2 = \eta_1, \quad e_3e_1 = -e_1e_3 = \eta_2, \quad e_1e_2 = -e_2e_1 = \eta_3;$$

he calls the foregoing combinations the complements (*Ergänzungen*) of e_1, e_2, e_3, denoting them by $/e_1$, $/e_2$, $/e_3$ respectively; it is better to use new letters, and I call them η_1, η_2, η_3 respectively; we thus have

$$e_2e_3 = -e_3e_2 = \eta_1, \quad e_3e_1 = -e_1e_3 = \eta_2, \quad e_1e_2 = -e_2e_1 = \eta_3.$$

But we require further assumptions for the combinations of the η's *inter se* and with the e's; the forms are

$$\begin{gathered}\eta_1e_1 = \eta_2e_2 = \eta_3e_3 = 1, \\ -\eta_2\eta_3 = \eta_3\eta_2 = e_1, \quad -\eta_3\eta_1 = \eta_1\eta_3 = e_2, \quad -\eta_1\eta_2 = \eta_2\eta_1 = e_3, \\ e_1\eta_1 = \eta_1e_1 = 1, \quad e_2\eta_2 = \eta_2e_2 = 1, \quad e_3\eta_3 = \eta_3e_3 = 0, \quad \&c.,\end{gathered}$$

(viz. each such combination is $= 0$); or, what is the same thing, we have for 1, e_1, e_2, e_3, η_1, η_2, η_3 the multiplication table

	1	e_1	e_2	e_3	η_1	η_2	η_3
1	1	e_1	e_2	e_3	η_1	η_2	η_3
e_1	e_1	0	η_3	$-\eta_2$	1	0	0
e_2	e_2	$-\eta_3$	0	η_1	0	1	0
e_3	e_3	η_2	$-\eta_1$	0	0	0	1
η_1	η_1	1	0	0	0	$-e_3$	e_2
η_2	η_2	0	1	0	e_3	0	$-e_1$
η_3	η_3	0	0	1	$-e_2$	e_1	0

It is proper to remark that in Grassmann's theory there is no interpretation for the general linear symbol

$$\omega + x_1e_1 + x_2e_2 + x_3e_3 + y_1\eta_1 + y_2\eta_2 + y_3\eta_3,$$

which presents itself in connexion with the seven extraordinaries $(1, e_1, e_2, e_3, \eta_1, \eta_2, \eta_3)$.

47. It is to be noticed that the symbols are not associative; assuming them to be so, we should have $e_2e_1 \cdot e_2e_3 = e_2 \cdot e_1^2 \cdot e_3 = 0$, which is inconsistent with $\eta_3\eta_1 = -e_2$; and so in other cases. But any *three* e's are associative, and hence a product

$$x \cdot y \cdot z \;=\; (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

[PDF p. 501 / printed p. 481]

is associative; in fact, if this be regarded as standing for $(x \cdot y)z$, the value is

$$= \big\{(x_2y_3 - x_3y_2)\,\eta_1 + (x_3y_1 - x_1y_3)\,\eta_2 + (x_1y_2 - x_2y_1)\,\eta_3\big\}(z_1e_1 + z_2e_2 + z_3e_3),$$

which is

$$= (x_2y_3 - x_3y_2)z_1 + (x_3y_1 - x_1y_3)z_2 + (x_1y_2 - x_2y_1)z_3 \;=\; \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix};$$

and similarly, regarding it as standing for $x(y \cdot z)$, the value is

$$= x_1(y_2z_3 - y_3z_2) + x_2(y_3z_1 - y_1z_3) + x_3(y_1z_2 - y_2z_1) \;=\; \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

as before. The product of the three points is thus a scalar, viz. it is equal to the area of the triangle formed by the three points divided by that of the triangle $A_1A_2A_3$.

48. But a product $x \cdot Y \cdot z$,

$$= (x_1e_1 + x_2e_2 + x_3e_3)(Y_1\eta_1 + Y_2\eta_2 + Y_3\eta_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

is not associative, and has thus no meaning until the grouping of the factors is determined. In fact, $(x \cdot Y)z$ will be

$$= (x_1Y_1 + x_2Y_2 + x_3Y_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

which denotes the point z_1 with a weight $= x_1Y_1 + x_2Y_2 + x_3Y_3$; whereas $x(Y \cdot z)$ will be

$$= (x_1e_1 + x_2e_2 + x_3e_3)(y_1Z_1 + y_2Z_2 + y_3Z_3),$$

which denotes the point x_1 with a weight $= y_1Z_1 + y_2Z_2 + y_3Z_3$.

Hence also a product $x \cdot y \cdot z \cdot w$,

$$= (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3)(z_1e_1 + z_2e_2 + z_3e_3)(w_1e_1 + w_2e_2 + w_3e_3),$$

of four factors is not associative, and has thus no meaning until the grouping of the factors is determined. Grassmann attributes to such a product the value $= 0$; but this is not a value corresponding to any grouping of the factors, and the equation $x \cdot y \cdot z \cdot w = 0$ can only be regarded as an independent assumption.

For the product of a point into itself, or say the square of a point, we have

$$x^2 = (x_1e_1 + x_2e_2 + x_3e_3)^2 = (x_1e_1 + x_2e_2 + x_3e_3)(x_1e_1 + x_2e_2 + x_3e_3) = 0,$$

viz. this is $=$ zero.

49. The product of two points x and y is defined to be the "Strecke," the finite line or stroke (xy), considered as drawn from x to y, say we have $x \cdot y = (xy)$; observe

that $y\cdot x = (yx)$: by what precedes $y\cdot x+x\cdot y = 0$, whence also $(yx)+(xy) = 0$, which is an equation in the addition of strokes. Hence, if as before, the points x, y are $x = x_1e_1+x_2e_2+x_3e_3$ and $y = y_1e_1+y_2e_2+y_3e_3$, then for the stroke (xy) we have

$$\begin{aligned}(xy) &= (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3),\\ &= (x_2y_3 - x_3y_2)\,\eta_1 + (x_3y_1 - x_1y_3)\,\eta_2 + (x_1y_2 - x_2y_1)\,\eta_3,\end{aligned}$$

which is, in fact,

$$= (A_1xy \cdot A_1 + A_2xy \cdot A_2 + A_3xy \cdot A_3) \div A_1A_2A_3.$$

if A_1xy, A_2xy, A_3xy, $A_1A_2A_3$ denote the areas of the triangles formed by the points (A_1, x, y), (A_2, x, y), (A_3, x, y), (A_1, A_2, A_3) respectively. These triangles remain unaltered if the stroke (xy) is slid along the indefinite line joining the original points x, y, the absolute distance xy being unaltered; that is, strokes of equal length upon the same line are regarded as identical. Hence also strokes on the same line may be added together by adding their lengths.

50. Consider two strokes (xy), (xz), (fig. 4), having a common point x, then completing the parallelogram, we prove that the sum $(xy) + (xz)$ is equal to the diagonal (xw).

[Figure 4: a parallelogram with vertex x at lower-left, y at lower-right, z at upper-left, w at upper-right; both diagonals drawn, illustrating $x + w = y + z$.]

We have, in fact,

$$x + w = y + z,$$

and thence

$$x(x + w) = x(y + z),$$

that is,

$$x^2 + x \cdot w = x \cdot y + x \cdot z,$$

or, since $x^2 = 0$ and $x \cdot y = (xy)$, &c., this is

$$(xw) = (xy) + (xz),$$

the required formula for the addition of the strokes.

51. The same thing appears thus: writing

$$\begin{aligned}x &= x_1e_1 + x_2e_2 + x_3e_3,\\ y &= y_1e_1 + y_2e_2 + y_3e_3,\\ z &= z_1e_1 + z_2e_2 + z_3e_3,\end{aligned}$$

we have

$$\begin{aligned}(xy) &= (x_2y_3 - x_3y_2)\,\eta_1 + (x_3y_1 - x_1y_3)\,\eta_2 + (x_1y_2 - x_2y_1)\,\eta_3,\\ (xz) &= (x_2z_3 - x_3z_2)\,\eta_1 + (x_3z_1 - x_1z_3)\,\eta_2 + (x_1z_2 - x_2z_1)\,\eta_3.\end{aligned}$$

Or, putting for a moment

$$m_1 = \tfrac{1}{2}(y_1 + z_1), \quad m_2 = \tfrac{1}{2}(y_2 + z_2), \quad m_3 = \tfrac{1}{2}(y_3 + z_3),$$

we have for the point m, midway between y and z, the equation $m = m_1e_1 + m_2e_2 + m_3e_3$; and then

$$\begin{aligned}(xy) + (xz) &= 2\{(x_2m_3 - x_3m_2)\,\eta_1 + (x_3m_1 - x_1m_3)\,\eta_2 + (x_1m_2 - x_2m_1)\,\eta_3\}\\ &= 2\,(xm), \ = xw,\end{aligned}$$

which is the theorem in question.

52. Recurring to the foregoing equation $x + w = y + z$, we deduce

$$xz - yw = zw - xy = (yzw) = (zwx) = (wxy) = (xyz),$$

if for shortness $(yzw) = yz + zw + wy$, and similarly for (zwx), (wxy), and (xyz). Observe here that $xz - yw$ and $zw - xy$ are each of them equal to the difference between two equal and parallel strokes; the strokes yz, zw, wy are the sides (in order) of a triangle; qua forces, the expressions in question represent each of them a couple.

53. We have thus arrived at the representation of a stroke

$$(xy) = (x_2y_3 - x_3y_2)\,\eta_1 + (x_3y_1 - x_1y_3)\,\eta_2 + (x_1y_2 - x_2y_1)\,\eta_3,$$

or say

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3,$$

where the meaning of the coordinates λ_1, λ_2, λ_3 is such that, if x, y are the two extremities of the stroke, then λ_1, λ_2, λ_3 are equal to the areas A_1xy, A_2xy, A_3xy, each divided by the area $A_1A_2A_3$. And we have also established the rule for the addition of strokes: it will be noticed that this has been done without the aid of any expression for the length of the stroke.

54. *Metrical relations.* For the expression of the distance of two points or length of a stroke, and of the inclinations of a stroke to the sides of the fundamental triangle, &c., we require the values of the angles and sides of this triangle $A_1A_2A_3$, say the angles are A_1, A_2, A_3, and the sides are $= R\sin A_1$, $R\sin A_2$, $R\sin A_3$ respectively; R is thus equal to the diameter of the circumscribed circle. Moreover, considering a stroke xy, if through x we draw lines $x\theta_1$, $x\theta_2$, $x\theta_3$ in the senses A_1A_1, A_1A_2, A_1A_3 respectively, then taking the angles $yx\theta_1$, $yx\theta_2$, $yx\theta_3$, each of them in the same sense, or say each right-handedly, we call these angles θ_1, θ_2, θ_3 respectively, and take also ρ, $= (xy)$ for the length of the stroke. This being so, and if the two extremities are

$$x = x_1e_1 + x_2e_2 + x_3e_3, \quad \text{and} \quad y = y_1e_1 + y_2e_2 + y_3e_3,$$

then we have, first,

$$\theta_1 - \theta_2 = \pi - A_3, \quad \theta_2 - \theta_3 = \pi - A_1, \quad \theta_3 - \theta_1 = \pi - A_2,$$

giving

$$A_1 + A_2 + A_3 = \pi,$$

and

$$\sin A_1 = \sin(\theta_2 - \theta_3), \quad \sin A_2 = \sin(\theta_3 - \theta_1), \quad \sin A_3 = \sin(\theta_1 - \theta_2).$$

And then

$$y_1 = x_1 + \frac{\rho}{R}\,\frac{\sin\theta_1}{\sin A_2\,\sin A_3},\quad y_2 = x_2 + \frac{\rho}{R}\,\frac{\sin\theta_2}{\sin A_3\,\sin A_1},\quad y_3 = x_3 + \frac{\rho}{R}\,\frac{\sin\theta_3}{\sin A_1\,\sin A_2}.$$

As regards these last equations, writing for a moment p_1, p_2, p_3 for the perpendicular distances of x, and q_1, q_2, q_3 for those of y from the sides of the triangle, then we have $q_1, q_2, q_3 = p_1+\rho\sin\theta_1,\ p_2+\rho\sin\theta_2,\ p_3+\rho\sin\theta_3$, and we thence easily derive the equations in question.

55. *Expression for length of a stroke.* We hence, by a somewhat complicated analytical process, find an expression for ρ in terms of the coordinates (x_1, x_2, x_3) and (y_1, y_2, y_3), which enter into it through the combinations

$$x_2y_3 - x_3y_2,\quad x_3y_1 - x_1y_3,\quad x_1y_2 - x_2y_1.$$

In fact, writing for shortness

$$\frac{x_1}{\sin A_1},\ \frac{x_2}{\sin A_2},\ \frac{x_3}{\sin A_3} = K_1,\ K_2,\ K_3,$$

we have

$$\begin{aligned}(x_2y_3 - x_3y_2)\sin A_1 &= \frac{\rho}{R}\,(K_2\sin\theta_3 - K_3\sin\theta_2),\\ (x_3y_1 - x_1y_3)\sin A_2 &= \frac{\rho}{R}\,(K_3\sin\theta_1 - K_1\sin\theta_3),\\ (x_1y_2 - x_2y_1)\sin A_3 &= \frac{\rho}{R}\,(K_1\sin\theta_2 - K_2\sin\theta_1);\end{aligned}$$

and hence also, if $(*\,|\,\xi,\eta,\zeta)^2$ denote the quadric function

$$(1,\ 1,\ 1,\ -\cos A_1,\ -\cos A_2,\ -\cos A_3\ |\ \xi,\eta,\zeta)^2,$$

we have

$$(*\,|\ (x_2y_3 - x_3y_2)\sin A_1, \ldots)^2 = \frac{\rho^2}{R^2}\,(*\,|\ K_2\sin\theta_3 - K_3\sin\theta_2,\ K_3\sin\theta_1 - K_1\sin\theta_3,\ K_1\sin\theta_2 - K_2\sin\theta_1)^2,$$

where the quadric function on the right-hand side is in fact

$$= (K_1\sin A_1 + K_2\sin A_2 + K_3\sin A_3)^2,$$

that is,

$$= (x_1 + x_2 + x_3)^2,\ = 1\,;$$

and we thus finally obtain

$$\rho^2 = R^2\big(*\,|\ (x_2y_3 - x_3y_2)\sin A_1,\ (x_3y_1 - x_1y_3)\sin A_2,\ (x_1y_2 - x_2y_1)\sin A_3\big)^2,$$

the required formula for the length of the stroke xy.

56. To effect the foregoing reduction of the quadric function

$$\big(*\,|\ K_2\sin\theta_3 - K_3\sin\theta_2,\ K_3\sin\theta_1 - K_1\sin\theta_3,\ K_1\sin\theta_2 - K_2\sin\theta_1\big)^2,$$

observe that the coefficient herein of K_1^2 is

$$= \sin^2\theta_2 + \sin^2\theta_3 + 2\cos A_1 \sin\theta_2 \sin\theta_3,$$

where $\theta_2 - \theta_3 = \pi - A_1$, and thence

$$\cos\theta_2 \cos\theta_3 + \sin\theta_2 \sin\theta_3 = -\cos A_1;$$

that is, we have

$$\cos A_1 + \sin\theta_2 \sin\theta_3 = -\cos\theta_2 \cos\theta_3,$$

and thence

$$1 - \sin^2\theta_2 - \sin^2\theta_3 + \sin^2\theta_2 \sin^2\theta_3 = \cos^2 A_1 + 2\cos A_1 \sin\theta_2 \sin\theta_3 + \sin^2\theta_2 \sin^2\theta_3;$$

whence the coefficient in question is $= \sin^2 A_1$. And finding in like manner the coefficients of K_2^2, K_3^2, K_2K_3, &c., the whole function is, as already mentioned,

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2, \text{ that is, } = 1.$$

57. *New representation of a stroke.* Representing the stroke in the form

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3,$$

we have only to replace $x_2y_3 - x_3y_2$, $x_3y_1 - x_1y_3$, $x_1y_2 - x_2y_1$ by their values $= \lambda_1, \lambda_2, \lambda_3$ respectively, viz. for the length ρ of the stroke, we have

$$\rho^2 = R^2\,(*\,|\ \lambda_1 \sin A_1,\ \lambda_2 \sin A_2,\ \lambda_3 \sin A_3)^2.$$

Say this equation is

$$\rho^2 = R^2 \cdot \Omega, \text{ then } \rho = R\sqrt{(\Omega)}, \text{ or } 1 = R\frac{\rho}{\sqrt{(\Omega)}},$$

and the stroke may be represented in the form

$$\lambda = \frac{\rho}{R\sqrt{(\Omega)}}\,(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3);$$

or, what is the same thing, if the absolute magnitudes of the coordinates $\lambda_1, \lambda_2, \lambda_3$ are such that

$$(*\,|\ \lambda_1 \sin A_1,\ \lambda_2 \sin A_2,\ \lambda_3 \sin A_3)^2 = 1,$$

then the expression is

$$\lambda = \frac{\rho}{R}\,(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

where ρ is the length of the stroke.

58. It has been seen that the coefficients of η_1, η_2, η_3 have the values A_1xy, A_2xy, A_3xy, each divided by $A_1A_2A_3$; or, what is the same thing, if p_1, p_2, p_3 are the lengths of the perpendiculars on the stroke from the points A_1, A_2, A_3 respectively, then these coefficients are $(\frac{1}{2}\rho p_1,\ \frac{1}{2}\rho p_2,\ \frac{1}{2}\rho p_3)$, each divided by $A_1A_2A_3$; or, what is the same thing, the perpendicular distances p_1, p_2, p_3 are

$$= \frac{2A_1A_2A_3}{R\sqrt{(\Omega)}}\,(\lambda_1,\ \lambda_2,\ \lambda_3)$$

respectively, where $A_1A_2A_3$ is the area of the fundamental triangle.

59. *Expression for the mutual inclination of two strokes.* In connexion with the stroke $\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3$, considering a stroke $\mu = \mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3$, suppose the inclinations hereof to the axes are ϕ_1, ϕ_2, ϕ_3, then

$$\phi_1 - \phi_2 = \pi - A_3, \quad \phi_2 - \phi_3 = \pi - A_1, \quad \phi_3 - \phi_1 = \pi - A_2,$$

and we have

$$\theta_1 - \phi_1 = \theta_2 - \phi_2 = \theta_3 - \phi_3,$$

viz. each of these equal angles is the inclination of the two strokes to each other, say this angle is $= \delta$. The expression for this angle is

$$\begin{aligned}\cos\delta = (*|\ &\lambda_1 \sin A_1,\ \lambda_2 \sin A_2,\ \lambda_3 \sin A_3\ |\ \mu_1 \sin A_1,\ \mu_2 \sin A_2,\ \mu_3 \sin A_3) \\ &\div \sqrt{(1,\ldots\ |\ \lambda_1 \sin A_1,\ \lambda_2 \sin A_2,\ \lambda_3 \sin A_3)^2} \cdot \sqrt{(1,\ldots\ |\ \mu_1 \sin A_1,\ \mu_2 \sin A_2,\ \mu_3 \sin A_3)^2}.\end{aligned}$$

In fact, considering (as we may do) the two strokes as proceeding from a common point $x = x_1e_1 + x_2e_2 + x_3e_3$, then the function in question is

$$= (*|\ K_2\sin\theta_3 - K_3\sin\theta_2,\ K_3\sin\theta_1 - K_1\sin\theta_3,\ K_1\sin\theta_2 - K_2\sin\theta_1\ |\ K_2\sin\phi_3 - K_3\sin\phi_2, \ldots),$$

which is

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos\delta,\ = \cos\delta.$$

60. In verification, observe that the whole coefficient of K_1^2 is

$$= \sin\theta_2 \sin\phi_3 + \sin\theta_3 \sin\phi_2 + \cos A_1 (\sin\theta_2 \sin\phi_3 + \sin\theta_3 \sin\phi_2),$$

where $\theta_2 - \theta_3 = \pi - A_1$, $\phi_2 - \phi_3 = \pi - A_1$. Hence

$$\sin\theta_3 = -\sin(\theta_2 - A_1), \quad \sin\phi_3 = -\sin(\phi_2 - A_1);$$

and the expression in question is

$$\begin{aligned}= &\sin(\theta_2 - A_1)\sin(\phi_2 - A_1) \\ &- \cos A_1 \sin(\theta_2 - A_1)\sin\phi_2 \\ &- \cos A_1 \sin(\phi_2 - A_1)\sin\theta_2 \\ &+ \sin\theta_2 \sin\phi_2,\end{aligned}$$

which, expanding the sines, and writing the last term in the form

$$= \sin\theta_2 \sin\phi_2 (\cos^2 A_1 + \sin^2 A_1)$$

is

$$\begin{aligned}= &\cos^2 A_1 (\sin\theta_2 \sin\phi_2 - \sin\theta_2 \sin\phi_2 - \sin\theta_2 \sin\phi_2 + \sin\theta_2 \sin\phi_2) \\ &+ \sin^2 A_1 (\cos\theta_2 \cos\phi_2 + \sin\theta_2 \sin\phi_2) \\ &+ \cos A_1 \sin A_1 (-\sin\theta_2 \cos\phi_2 - \sin\phi_2 \cos\theta_2 + \sin\phi_2 \cos\theta_2 + \sin\theta_2 \cos\phi_2), \\ = &\sin^2 A_1 \cos(\theta_2 - \phi_2),\ = \sin^2 A_1 \cos\delta,\end{aligned}$$

and reducing in like manner the coefficients of K_2^2, K_3^2, K_2K_3, &c., the whole expression becomes

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos\delta,\ = \cos\delta \text{ as above.}$$

61. The foregoing formulae agree with the theorem, No. 50, that the sum of the strokes

$$\lambda_1 = \frac{\rho}{R\sqrt{(\Omega)}}\,(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

and

$$\mu_1 = \frac{\sigma}{R\sqrt{(\Upsilon)}}\,(\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3),$$

is a stroke

$$= \frac{\tau}{R\sqrt{(\Pi)}}\,(\nu_1\eta_1 + \nu_2\eta_2 + \nu_3\eta_3),$$

which is the diagonal of the parallelogram constructed with the given strokes λ and μ; the proof is a little simplified by assuming (as is allowable) $\Omega = 1$ and $\Upsilon = 1$, in which case we have

$$\lambda + \mu = \frac{1}{R}\left[(\rho\lambda_1 + \sigma\mu_1)\,\eta_1 + (\rho\lambda_2 + \sigma\mu_2)\,\eta_2 + (\rho\lambda_3 + \sigma\mu_3)\,\eta_3\right],$$

from which the length and inclination may be calculated.

As already appearing, the product x, y, z of any three points is the scalar

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

which is equal to the area of the triangle xyz, divided by that of the triangle $A_1A_2A_3$. We have in like manner the product of a point x into a stroke λ, viz. this is

$$= (x_1\lambda_1 + x_2\lambda_2 + x_3\lambda_3)(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

which is $= x_1\lambda_1 + x_2\lambda_2 + x_3\lambda_3$, the two factors being in this case commutative; the value is equal to the area of the triangle formed by the point and the stroke, divided by the area $A_1A_2A_3$. Of course, if in the one case the three points are in a line, or if in the other the point is in the line of the stroke, then the product is $= 0$.

62. We have yet to consider the product of two strokes; say these are

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3, \quad \text{and} \quad \mu = \mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3,$$

then we have

$$\begin{aligned}\lambda\cdot\mu &= (\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3)(\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3)\\ &= -\left[(\lambda_2\mu_3 - \lambda_3\mu_2)\,e_1 + (\lambda_3\mu_1 - \lambda_1\mu_3)\,e_2 + (\lambda_1\mu_2 - \lambda_2\mu_1)\,e_3\right],\end{aligned}$$

which is a point regarded as having weight. If we take the two strokes to be

$$\begin{aligned}\lambda = (xy) &= (x_2y_3 - x_3y_2)\,\eta_1 + (x_3y_1 - x_1y_3)\,\eta_2 + (x_1y_2 - x_2y_1)\,\eta_3,\\ \text{and}\quad \mu = (xz) &= (x_2z_3 - x_3z_2)\,\eta_1 + (x_3z_1 - x_1z_3)\,\eta_2 + (x_1z_2 - x_2z_1)\,\eta_3,\end{aligned}$$

then the product is

$$\lambda\cdot\mu \;=\; -(x_1e_1+x_2e_2+x_3e_3)\begin{vmatrix} x_1 & x_2 & x_3\\ y_1 & y_2 & y_3\\ z_1 & z_2 & z_3\end{vmatrix},$$

viz. this is the common point x, with a weight = area of triangle $xyz\div$ area $A_1A_2A_3$, or say = the area of triangle formed by the two strokes $\div$ area $A_1A_2A_3$. The product is non-commutative: we, in fact, have $\lambda\cdot\mu=-\mu\cdot\lambda$.

63. The product of three strokes is associative; and it is a scalar. If the three strokes are the sides of a triangle LMN, then the value is $=(LMN)^2\div(A_1A_2A_3)^3$, where LMN and $A_1A_2A_3$ denote the areas of the triangle LMN and the triangle $A_1A_2A_3$ respectively; and if any stroke instead of being a side of the triangle LMN be a part only of this side, then the value is diminished proportionally; hence the value is $=\dfrac{\rho\sigma\tau}{MN\cdot NL\cdot LM}\left\{(LMN)^2\div(A_1A_2A_3)^3\right\}$, where ρ, σ, τ are the lengths of the three strokes, MN, NL, LM the lengths of the sides of the triangle LMN, $A_1A_2A_3$ are the areas as above.

64. It is to be remarked that the form

$$(*\,|\,\xi,\,\eta,\,\zeta)^2,\;=(1,\,1,\,1,\,-\cos A_1,\,-\cos A_2,\,-\cos A_3\;|\;\xi,\,\eta,\,\zeta)^2$$

is the product of two linear factors; it is

$$=(\xi-\eta\cos A_3-\zeta\cos A_2)^2+(\eta\sin A_3-\zeta\sin A_2)^2,$$

and thus

$$=\left(\xi-\eta e^{iA_3}-\zeta e^{-iA_2}\right)\left(\xi-\eta e^{-iA_3}-\zeta e^{iA_2}\right).$$

This corresponds to the theorem that the distance of two points is $=0$, when the line joining them passes through one of the circular points at infinity; the coordinates of one of these points may be written under the three equivalent forms

$$\begin{aligned} x_1:x_2:x_3 &= \sin A_1 \quad : -\sin A_2e^{-iA_3} \;:\; -\sin A_3e^{-iA_2},\\ &= -\sin A_1e^{-iA_3} \;:\; \sin A_2 \quad : -\sin A_3e^{iA_1},\\ &= -\sin A_1e^{iA_2} \;:\; -\sin A_2e^{iA_1} \;:\; \sin A_3,\end{aligned}$$

and those of the other are of course obtained therefrom by the change of i into $-i$.

*(To be continued)**

[* This paper remains incomplete; no continuation appears to have been prepared by Professor Cayley.]

866. Note on Kiepert's L-Equations, in the Transformation of Elliptic Functions.

[From the *Mathematische Annalen*, t. XXX. (1887), pp. 75–77.]

It appears, by comparison with Klein's paper "Ueber die Transformation u. s. w.," *Math. Annalen*, t. XIV. (1878), see p. 144, that Kiepert's L made use of in the Memoir "Ueber Theilung und Transformation der elliptischen Functionen," *Math. Annalen*, t. XXVI. (1886), pp. 369–454, is, in fact, the square of the multiplier, "für das durch $\sqrt[5]{\Delta}$ normirte Integral," viz. considering the general quartic function $(a,\ldots)(x,1)^4 = (a,b,c,d,e)(x,1)^4$ and the transformed function $(a_1,\ldots)(y,1)^4$, then we have

$$\frac{L\sqrt[5]{\Delta}\,dx}{\sqrt{(a,\ldots)(x,1)^4}} = \frac{\sqrt[5]{\Delta_1}\,dy}{\sqrt{(a_1,\ldots)(y,1)^4}},$$

where if

$$I = ae - 4bd + 3c^2,$$
$$J = ace - ad^2 - b^2e + 2bcd - c^3,$$

and similarly I_1, J_1, are the invariants of the two functions, then Δ, Δ_1, are the discriminants

$$\Delta = I^3 - 27J^2, \quad \Delta_1 = I_1^3 - 27J_1^2,$$

and the γ_1, γ_1 of Kiepert's equations are

$$\gamma_1 = I + \sqrt[3]{\Delta}, \quad \gamma_1 = J + \sqrt{\Delta},$$

whence

$$\gamma_1^3 - 27\gamma_1^2 = 1.$$

In particular, if the forms are

$$1 - x^5,\ 1 - k^2x^6, \text{ and } 1 - y^5 \cdot 1 - \lambda^2 y^6,$$

and if as usual $k = u^4$, $\lambda = v^4$, and M is the multiplier for the form

$$\frac{dx}{\sqrt{1-x^2\cdot 1-k^2x^2}} = \frac{M\,dy}{\sqrt{1-y^2\cdot 1-\lambda^2y^2}},$$

then we have

$$\begin{aligned}
I &= \tfrac{1}{12}\,(1+14u^8+u^{10}),\\
J &= \tfrac{1}{216}\,(1+u^8)\,(1-34u^8+u^{16}),\\
\Delta &= \tfrac{1}{16}\,u^8\,(1-u^8)^4, \quad \Delta_1 = \tfrac{1}{16}\,v^8\,(1-v^8)^4,\\
\gamma_1 &= \tfrac{1}{4}\,\sqrt[5]{2}\,\frac{1+14u^8+u^{16}}{u^4\,(1-u^8)^4}, \quad \gamma_1 = \tfrac{1}{4}\,\frac{(1+u^8)\,(1-34u^8+u^{16})}{u^4\,(1-u^8)^2}.
\end{aligned}$$

and thence

$$L^2 = \frac{v^8\,(1-v^8)^{\frac{4}{5}}}{u^8\,(1-u^8)^{\frac{4}{5}}}\,\frac{1}{M},$$

which last equation is the expression for L^2 in terms of the Jacobian symbols u, v, M.

As an easy verification in a particular case, suppose $n=5$. We have here

$$L^2 = \frac{v^8\,(1-v^8)^{\frac{4}{5}}}{u^8\,(1-u^8)^{\frac{4}{5}}}\cdot\frac{1}{M}, \quad M = \frac{v\,(1-uv^6)}{v-u^5}, \quad \left(= \frac{v+u^5}{5u\,(1+u^8v^8)}\right),$$

$$u^2 - v^2 + 5u^2v^2\,(u^2-v^2) + 4uv\,(1-u^4v^4) = 0,$$

$$\gamma_1 = \tfrac{1}{4}\,\sqrt[5]{2}\,\frac{1+14u^8+u^{16}}{u^4\,(1-u^8)^4};$$

and it should be possible, by eliminating u, v, M, to deduce hence the L-equation

$$L^6 + 10L^5 - 12\gamma_1 L^2 + 5 = 0. \quad \text{(Kiepert, p. 428.)}$$

It does not seem in any wise easy to do this in the case of an arbitrary modulus; but writing the modular equation in the form

$$(u^6-v^6)(u^6+6u^2v^2+v^2) + 4uv\,(1-u^4v^4) = 0,$$

we satisfy this by

$$uv - 1 = 0, \quad u^6 + 6u^2v^2 + v^2 = 0,$$

or say by

$$v = \frac{1}{u}, \quad u^8 + 6u^4 + 1 = 0,$$

and the equation may be verified for this particular modulus.

We have

$$1 + 14u^8 + u^{16} = 48u^8, \quad (1-u^8)^4 = 32u^8,$$

and consequently

$$\gamma_1 = \tfrac{1}{4}\,\sqrt[5]{2}\cdot\frac{48u^8}{u^4\,(32u^8)^{\frac{4}{5}}}, = \frac{1}{2\cdot 3}\cdot 2^{\frac{6}{5}}\,\frac{2^4\cdot 3u^8}{u^4\cdot 2^{\frac{12}{5}}\cdot u^{32}} = 1, \text{ (whence also } \gamma_1 = 0).$$

Moreover

$$M = \frac{\dfrac{1}{u} - \dfrac{u}{u^4}}{\dfrac{1}{u} - u^5} = -\frac{1}{u^2}\frac{1-u^2}{1-u^6} = \frac{-1}{u^2\,(1+u^2+u^4)},$$

and thence

$$L^2 = \frac{\dfrac{1}{u^8}\left(1-\dfrac{1}{u^8}\right)^{\frac{4}{5}}}{u^8\,(1-u^8)^{\frac{4}{5}}} \cdot \frac{1}{M} = \frac{(u^8-1)^{\frac{4}{5}}}{u^4\,(1-u^8)^{\frac{4}{5}}} \cdot \frac{1}{M} = -\frac{1}{u^2 M},$$

that is,

$$L^2 = 1 + u^2 + u^{-2}.$$

But we have

$$u^4 = -3 + 2\sqrt{2},$$

and thence

$$u^2 = i\,(1-\sqrt{2}), \quad u^{-2} = i\,(1+\sqrt{2}),$$

and

$$L^2 = 1 + 2i,$$

whence

$$L^{10} + 10L^5 - 12\gamma_1 L^2 + 5 = (117+44i) + 10\,(-11-2i) - 12\,(1+2i) + 5 = 0,$$

or the L-equation is satisfied.

Cambridge, 14 *March* 1887.

867. Note on the Jacobian Sextic Equation.

[From the *Mathematische Annalen*, t. XXX. (1887), pp. 78–84.]

In the Jacobian sextic equation

$$(z-a)^6 - 4a\,(z-a)^5 + 10\delta\,(z-a)^2 - 4\epsilon\,(z-a) + 5\delta^2 - 4a\epsilon = 0,$$

if A, B, C, D, E, F are the square roots (each with a determinate sign) of the roots z_1, z_2, z_3, z_4, z_5, z_6 of the equation, and if ϵ be an imaginary fifth root of unity, such that $\epsilon + \epsilon^2 - \epsilon^3 - \epsilon^4 = \sqrt{5}$, then we have between the square roots a system of three linear equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= \quad B + \epsilon C + \epsilon D + \epsilon^2 E + \epsilon^2 F, \\ 0 &= \quad B + \epsilon^2 C + \epsilon^2 D + \epsilon E + \epsilon^2 F, \end{aligned}$$

see Brioschi's *Funzioni ellittiche* (Milan, 1880), third appendix, p. 402.

The sextic equation is irreducible, and there is thus nothing to distinguish any one square root from any other square root. But in the foregoing system of equations, A is distinguished from the other square roots; hence the system must be one of 6 systems, wherein the place occupied by A is occupied by A, B, C, D, E, F respectively. Observe however that the letters being square roots, it may very well happen, and (as will appear) it does in fact happen, that the passage from the first to another system implies a change of sign in certain of the letters A, B, C, D, E, F.

The selection of one of the square roots as $= A$, does not determine which of the other square roots shall be $= B$, C, D, E, F respectively: and in fact each system of equations may be represented in 10 different forms; viz. by multiplying the second and third equations only by ϵ, ϵ^2, ϵ^3, ϵ^4 respectively, we obtain four new forms: we have thus five forms: and then in each of them transposing the second and third equations, we have five other forms: we have thus in all 10 forms.

Write $ABCDEF$ to denote the foregoing system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= \phantom{-A\sqrt{5} +{}} B + \epsilon C + \epsilon D + \epsilon^2 E + \epsilon^2 F, \\ 0 &= \phantom{-A\sqrt{5} +{}} B + \epsilon^2 C + \epsilon^2 D + \epsilon E + \epsilon^2 F; \end{aligned}$$

the 10 forms of the system are

$$\begin{array}{l|l} ABCDEF & ABFEDC \\ ACDEFB & ACBFED \\ ADEFBC & ADCBFE \\ AEFBCD & AEDCBF \\ AFBCDE & AFEDCB, \end{array}$$

where observe that the five forms in the same column are connected by cyclical substitutions on the last five letters; and that the forms in each line are connected by two inversions of the last four letters.

It is hardly necessary to remark that $ABFEDC$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + F + E + D + C, \\ 0 &= \phantom{-A\sqrt{5} +{}} B + \epsilon F + \epsilon E + \epsilon D + \epsilon C, \\ 0 &= \phantom{-A\sqrt{5} +{}} B + \epsilon^2 F + \epsilon^2 E + \epsilon D + \epsilon C, \end{aligned}$$

which are the same with the equations of the original form; and similarly that $ACDEFB$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + C + D + E + F + B, \\ 0 &= \phantom{-A\sqrt{5} +{}} C + \epsilon D + \epsilon E + \epsilon^2 F + \epsilon^2 B, \\ 0 &= \phantom{-A\sqrt{5} +{}} C + \epsilon^2 D + \epsilon^2 E + \epsilon F + \epsilon^2 B, \end{aligned}$$

where the first equation is the same with the first equation, and the second and third equations only differ by a factor from the second and third equations respectively of the original form. And similarly as regards the others of the 10 forms.

I say that we have a second system $BCAF\bar{D}\bar{E}$ (where $\bar{D}$, $\bar{E}$ mean $-D$, $-E$ respectively): in fact, this denotes the system of three equations

$$\begin{aligned} 0 &= -B\sqrt{5} + C + A + F - D - E, \\ 0 &= \phantom{-B\sqrt{5} +{}} C + \epsilon A + \epsilon F - \epsilon^2 D - \epsilon^2 E, \\ 0 &= \phantom{-B\sqrt{5} +{}} C + \epsilon^2 A + \epsilon^2 F - \epsilon D - \epsilon^2 E, \end{aligned}$$

which are deducible from the original three equations.

We have, in fact, the identities

$$\begin{aligned}
-\sqrt{5}\,[-B\sqrt{5}+C+A+F-D-E] &= 1\,(B+C-A\sqrt{5}+F+D+E)\\
&\quad +2\,(B+\epsilon C+\epsilon F+\epsilon^2 D+\epsilon^2 E)\\
&\quad +2\,(B+\epsilon^2 C+\epsilon^2 F+\epsilon D+\epsilon E),
\end{aligned}$$

$$\begin{aligned}
(1-\epsilon-\epsilon^2+\epsilon^4)\,[\ \ C+\epsilon A+\epsilon F-\epsilon^2 D-\epsilon^2 E]&\\
= 1\,(B+C-A\sqrt{5}+F+D+E)&\\
+(\epsilon+\epsilon^4)\,(B+\epsilon C+\epsilon F+\epsilon^2 D+\epsilon^2 E)&\\
+(\epsilon^2+\epsilon^3)\,(B+\epsilon^2 C+\epsilon^2 F+\epsilon D+\epsilon E),&
\end{aligned}$$

and

$$\begin{aligned}
(1+\epsilon-\epsilon^2-\epsilon^4)\,[\ \ C+\epsilon^2 A+\epsilon^2 F-\epsilon D-\epsilon E]&\\
= 1\,(B+C-A\sqrt{5}+F+D+E)&\\
+(\epsilon^2+\epsilon^3)\,(B+\epsilon C+\epsilon F+\epsilon^2 D+\epsilon^2 E)&\\
+(\epsilon+\epsilon^4)\,(B+\epsilon^2 C+\epsilon^2 F+\epsilon D+\epsilon E),&
\end{aligned}$$

all which identities are at once verified on recollecting that

$$1=-\epsilon-\epsilon^2-\epsilon^3-\epsilon^4 \quad \text{and} \quad \sqrt{5}=\epsilon-\epsilon^2-\epsilon^3+\epsilon^4.$$

We can now write down the 6 systems, each of them under its 10 forms; these, in fact, are

$ABCDEF$	$ABFEDC$
$ACDEFB$	$ACBFED$
$ADEFBC$	$ADCBFE$
$AEFBCD$	$AEDCBF$
$AFBCDE$	$AFEDCB$
$BCAF\bar{D}\bar{E}$	$BC\bar{E}\bar{D}FA$
$BAF\bar{D}\bar{E}C$	$BAC\bar{E}\bar{D}F$
$BF\bar{D}\bar{E}CA$	$BFAC\bar{E}\bar{D}$
$B\bar{D}\bar{E}CAF$	$B\bar{D}FAC\bar{E}$
$B\bar{E}CAF\bar{D}$	$B\bar{E}\bar{D}FAC$
$CDABF\bar{E}$	$CDF\bar{E}BA$
$CABF\bar{E}D$	$CAD\bar{F}EB$
$CBF\bar{E}DA$	$CB\bar{A}DP\bar{E}$
$C\bar{E}\bar{F}DAB$	$C\bar{E}DABF$
$CFDAB\bar{E}$	$CFEBAD$

(continuation of the table of 6 systems under 10 forms each:)

$DEAC\bar{F}\bar{B}$	$DE\bar{B}\bar{F}CA$
$DAC\bar{F}\bar{B}E$	$DCF\bar{B}EA$
$DC\bar{F}\bar{B}EA$	$DCAE\bar{B}\bar{F}$
$D\bar{F}\bar{B}EAC$	$D\bar{F}CAE\bar{B}$
$D\bar{B}EAC\bar{F}$	$D\bar{B}\bar{F}CAE$
$EFAD\bar{B}\bar{C}$	$EF\bar{C}\bar{B}DA$
$EAD\bar{B}\bar{C}F$	$EAF\bar{C}\bar{B}D$
$ED\bar{B}\bar{C}FA$	$ED\bar{A}F\bar{C}\bar{B}$
$E\bar{B}\bar{C}FAD$	$E\bar{B}DAF\bar{C}$
$E\bar{C}FAD\bar{B}$	$E\bar{C}\bar{B}DAF$
$FAE\bar{C}\bar{D}B$	$FA\bar{D}\bar{C}EB$
$FE\bar{C}\bar{D}BA$	$FE\bar{B}A\bar{D}\bar{C}$
$F\bar{C}\bar{D}BAE$	$F\bar{C}EBA\bar{D}$
$F\bar{D}BAE\bar{C}$	$F\bar{D}\bar{C}EBA$
$FBAE\bar{C}\bar{D}$	$FBDCEA$,

where, as before, the superscript line denotes a change of sign, $\bar{A} = -A$, &c.

As to the formation of this table, observe that we have $ABCDEF$ and $BCAF\bar{D}\bar{E}$; repeating the substitution, we have

$$\begin{gathered} ABCDEF \\ BCAF\bar{D}\bar{E} \\ CABF\bar{E}\bar{D} \\ \bar{A}BCDEF, \end{gathered}$$

viz. we have thus the $CABF\bar{E}\bar{D}$ which belongs to the third system; but nothing further, for the substitution is periodic of the third order, and the three forms are merely repeated. But if, upon

$$\begin{gathered} ABCDEF \\ BCAF\bar{D}\bar{E}, \end{gathered}$$

we operate successively with the substitutions which change the upper line into the three other forms of the first system

$$ADEFBC, \;\; AEFBCD, \;\; AFBCDE,$$

then the lower line is changed into the three forms

$$DEACF\bar{B}, \;\; EFADB\bar{C}, \;\; FBAED\bar{C},$$

which are forms belonging to the fourth, fifth, and sixth systems respectively. By way of verification, observe that, for instance repeating upon the second line the substitution

$$ABCDEF,$$

in place of

$$DEACF\bar{B},$$

we obtain

$$GFDABE,$$

which is one of the forms of the third system, assumed to have been previously found; and so in other instances.

Reverting to the three equations belonging to the form $ABCDEF$, by subtracting the third from the second equation, we obtain a linear relation between the four square roots C, D, E, F, viz. this is

$$0 = (\epsilon^2 - \epsilon^3)(C - F) + (\epsilon - \epsilon^4)(D - E)\,;$$

and the same equation is obtained by means of the form $ABFEDC$. We thus obtain from the thirty lines of either column of the preceding table, thirty such equations, but obviously the number of such equations should be fifteen, for there can be but one relation between any four square roots C, D, E, F; consequently each equation will be obtained twice; and, in fact, it is clear that the forms $ABCDEF$ and $BAFDEC$, and so in general any two forms which begin with the same pair of letters, give the same equation. But for greater symmetry, I write down the thirty equations in the order in which they are given by the left-hand column of the table: the equations are

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$C - F$	$D - E$
$D - B$	$E - F$
$E - C$	$F - B$
$F - D$	$B - C$
$B - E$	$C - D$
$A + E$	$F + D$
$F - C$	$-D + E$
$-D - A$	$-E - C$
$-E - F$	$C - A$
$C + D$	$A - F$
$A + F$	$B + E$
$B - D$	$-E + F$
$-E - A$	$-F - D$
$-F - B$	$D - A$
$D + E$	$A - B$

(continuation of the table of fifteen equations from the left-hand column:)

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$A + B$	$C + F$
$C - E$	$-F + B$
$-F - A$	$-B - E$
$-B - C$	$E - A$
$E + F$	$A - C$
$A + C$	$D + B$
$D - F$	$-B + C$
$-B - A$	$-C - F$
$-C - D$	$F - A$
$F + B$	$A - D$
$E - B$	$-C + D$
$-C - A$	$-D - B$
$-D - E$	$B - A$
$B + C$	$A - E$
$A + D$	$E + C$

where it is to be observed that, adding together the five equations given by any one of the systems, we obtain the identical result $0 = 0$.

I write down the 15 equations in a different order, in some cases changing the sign of the whole equation, as follows

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$A + C$	$B + D$
$B + C$	$A - E$
$A + B$	$C + F$
$B + E$	$A - B$
$B + F$	$A - D$
$A + F$	$B + E$
$A + D$	$C + E$
$C + D$	$A - F$
$E + F$	$A - C$
$A + E$	$D + F$
$B - E$	$C - D$
$F - D$	$B - C$
$E - C$	$F - B$
$D - B$	$E - F$
$C - F$	$D - E$

The first three of these equations, or writing $\lambda = \epsilon^2 - \epsilon^3$, $\mu = \epsilon - \epsilon^4$, say

$$\lambda\,(A+C) + \mu\,(B+D) = 0,$$
$$\lambda\,(B+C) + \mu\,(A-E) = 0,$$
$$\lambda\,(A+B) + \mu\,(C+F) = 0,$$

constitute the entire system of independent linear relations between the square roots A, B, C, D, E, F. The coefficients λ, μ are such that

$$\lambda^2 - \mu^2 = \lambda\mu\,(= \epsilon - \epsilon^2 - \epsilon^3 + \epsilon^4,\ = \sqrt{5}),$$

and it is hence easy to verify that the remaining twelve equations can be deduced from these by the elimination of one or two of the square roots A, B, C. For instance, to eliminate A from the first and second equations, multiplying by $-\mu$, λ and adding, we obtain

$$(-\lambda\mu + \lambda^2)\,C + (-\mu^2 + \lambda^2)\,B - \mu^2 D - \lambda\mu E = 0,$$

that is,

$$\mu^2 C + \lambda\mu B - \mu^2 D - \lambda\mu E = 0,$$

or finally

$$\lambda\,(B-E) + \mu\,(C-D) = 0,$$

which is one of the equations. And so again eliminating A from the first and third equations, we find

$$\lambda\,(B-C) + \mu\,(C+F-B-D) = 0,$$

that is,

$$(\lambda-\mu)(B-C) + \mu\,(F-D) = 0,$$

or multiplying by λ,

$$\mu^2\,(B-C) + \lambda\mu\,(F-D) = 0,$$

that is, finally

$$\lambda\,(F-D) + \mu\,(B-C) = 0,$$

which is one of the equations.

Cambridge, 21 *March* 1887.

868. On the Intersection of Curves.

[From the *Mathematische Annalen*, t. XXX. (1887), pp. 85–90.]

It is only recently that I have studied Bacharach's paper "Ueber den Cayley'schen Schnittpunktsatz," *Math. Ann.*, t. XXVI. (1885), pp. 275–299; his theorem in regard to the case where the δ points lie on a curve of the order $\gamma - 3$ is a very interesting and valuable one, but I consider it rather as an addition than as a correction to my original theorem; and I cannot by any means agree that the method by counting of constants is to be rejected as not trustworthy; on the contrary, it seems to me to be the proper foundation of the whole theory; it must of course be employed with due consideration of special cases. I reproduce the theorem in what appears to me the complete form.

Writing with Bacharach

$$r \rightleftarrows m, \quad r \rightleftarrows n, \quad r \rightleftarrows m+n-3, \quad \gamma = m+n-r, \quad \delta = \tfrac{1}{2}(\gamma-1)(\gamma-2),$$

and assuming $n \rightleftarrows m$, (these conditions and inequalities are to be attended to throughout the present paper), I consider two curves of the orders m, n respectively meeting in δ points B, and in $(mn - \delta)$ points A; and I state the theorem as follows:

1°. The $mn - \delta$ points A are in general such that a curve of the order r, which passes through $mn - \delta - 1$ of these points, does not of necessity pass through the remaining point; and in this case the general curve of the order r, which passes through the $mn - \delta$ points A, has for its form of equation

$$0 = \Omega_r + L_{r-m}\, P_m + M_{r-n}\, Q_n,$$

where $P_m = 0$, $Q_n = 0$ are the equations of the given curves, and L_{r-m}, M_{r-n} denote functions of the orders $r - m$, $r - n$ respectively; and it thus appears that the curve of the order r through the $mn - \delta$ points A passes also through the δ points B.

2°. If however the δ points B are on a curve of the order $\gamma - 3$, then the $mn - \delta$ points A are a system such that every curve of the order r passing through $mn - \delta - 1$ of these points passes through the remaining point; and in this case the general curve of the order r, which passes through the $mn - \delta$ points A, has for its form of equation

$$0 = \Omega_r + L_{r-m}\, P_m + M_{r-n}\, Q_n,$$

$\Omega_r = 0$ is a particular curve through the $mn - \delta$ points A, which does not go through any of the points B; and consequently the curve of the order r does not pass through any of the points B.

For the proof of the theorem I premise as follows:

A curve of the order r depends upon $\frac{1}{2}r\,(r+3)$ constants, or, to use a convenient expression, its Postulandum is $= \frac{1}{2}r\,(r+3)$: if the curve has to pass through a given point, this imposes a single relation upon the constants, or say the Postulation is $= 1$; similarly, if the curve has to pass through k given points, this imposes k relations, or say the Postulation is $= k$. The points may be however a special system, for instance, they may be such that every curve of the order r which passes through them passes through one other point; the Postulation is in this case $= k - 1$; and so in other cases. Assuming the Postulation of the k points to be $= k$, then the Postulandum of a curve of the order r through the k points is $= \frac{1}{2}r\,(r+3) - k$. I stop to remark that the Postulation has reference to the particular curve or other entity in question; thus in the case of a curve passing through k points, the Postulation for a curve of a certain order may be $= k$, and for a curve of a different order it may be less than k:

Considering now, as above, two given curves of the orders m and n intersecting in the $mn - \delta$ points A and the δ points B, then assuming that the $mn - \delta$ points are not a special system, viz. that their Postulation in regard to a curve of the order r is $= mn - \delta$, the Postulandum of a curve of the order r through the $mn - \delta$ points is

$$\begin{aligned} &= \tfrac{1}{2}\,r\,(r+3) - mn + \delta, \\ &= \tfrac{1}{2}\,(r-m+1)(r-m+2) + \tfrac{1}{2}\,(r-n+1)(r-n+2) - 1, \end{aligned}$$

viz. this is identically true when for δ we write its value

$$= \tfrac{1}{2}\,(m+n-r-1)(m+n-r-2).$$

But we have through the $mn - \delta$ points A, a curve

$$L_{r-m}\, P_m + M_{r-n}\, Q_n = 0$$

with the proper Postulandum: viz. L_{r-m} contains $\frac{1}{2}\,(r-m+1)(r-m+2)$ constants, M_{r-n} contains $\frac{1}{2}\,(r-n+1)(r-n+2)$ constants, and there is a diminution -1 for the constant which divides out; hence this is the general equation of the curve of the order r through the $mn - \delta$ points A; and the curve passes through the remaining δ points B.

In the case where the δ points B are on a curve of the order $\gamma-3$, (observe that this is a single condition imposed on the δ, $=\frac{1}{2}(\gamma-1)(\gamma-2)$ points, for a curve of the order $(\gamma-3)$ can be drawn through $\frac{1}{2}\gamma(\gamma-3)$ points), it is to be shown that the Postulation of the $mn-\delta$ points A is $=mn-\delta-1$; for, this being so, the Postulandum of the curve of the order r through the $(mn-\delta)$ points A is

$$=\tfrac{1}{2}r(r+3)-mn+\delta+1,$$

and the equation of the curve will be of the foregoing form, but it will be of the form

$$\Omega_r+L_{r-m}P_m+M_{r-n}Q_n=0,$$

$\Omega_r=0$ being a particular curve through the $mn-\delta$ points A, which does not pass through any of the points B. The proof depends on the theory of Residuation; which for the present purpose may be presented under the following form.

Let A, B, … denote systems of points upon a given Basis-curve, for instance the foregoing curve $P_m=0$, of the order m. And let $A\equiv ci$ denote that the points A are the complete intersection of the basis-curve by some other curve; (this implies that the number of the points is $=km$, a multiple of m, and the intersecting curve is then of course a curve of the order k). It is clear that, if $A\equiv ci$, and $B\equiv ci$, then also $A+B\equiv ci$. But conversely we have the theorem that, if $A+B\equiv ci$ and $A\equiv ci$, then also $B\equiv ci$. And we at once deduce the further theorem: if $A+B\equiv ci$, $B+C\equiv ci$, $C+D\equiv ci$, then also $A+D\equiv ci$. For the first and third relations give $A+B+C+D\equiv ci$, and the second relation then gives $A+D\equiv ci$.

[Diagram: a rectangle with vertices labelled A at top-left, B at top-right, D at bottom-left, C at bottom-right; the top side has length $mn-\delta$, divided as $mn-ab-\delta$ on the left portion and b on the right; the right side has length $\gamma-3$; the bottom side $m(\gamma-3)-b$; and the left side $m(\gamma-3)+\delta$.]

Starting now (see the diagram) with the δ points B on a curve of the order $\gamma-3$, suppose that we have through these points the basis-curve $P_m=0$ of the order m, and another given curve $Q_n=0$, of the order n; and let these besides meet in the $mn-\delta$ points A. Let the curve of the order $\gamma-3$ besides meet the basis-curve in the $m(\gamma-3)-\delta$ points C; and through these let there be drawn a curve of the order $m-3$, which besides meets the basis-curve in the $m(r-n)+\delta$ points D. We have here $A+B\equiv ci$, $B+C\equiv ci$, $C+D\equiv ci$; consequently $A+D\equiv ci$, that is, the $mn-\delta$ points A and the $m(r-n)+\delta$ points D lie on a curve of the order r. The curve of the order $m-3$ passes through the $m(\gamma-3)-\delta$ points C; its Postulandum is thus

$$=\tfrac{1}{2}m(m-3)-m(\gamma-3)+\delta,$$

which is

$$= \tfrac{1}{2}(r-n+1)(r-n+2).$$

In fact, substituting for γ, δ their values $= m+n-r$, and $\frac{1}{2}(m+n-r-1)(m+n-r-2)$ respectively, this equation is satisfied identically. The system of the $m(r-n)+\delta$ points D thus depends upon $\frac{1}{2}(r-n+1)(r-n+2)$ constants, or say the Postulandum of the points D is $= \frac{1}{2}(r-n+1)(r-n+2)$. It follows that the curve of the order r through the $(mn-\delta)$ points A and the $m(r-n)+\delta$ points D cannot have an equation of the form

$$L_{r-m}\,P_m + M_{r-n}\,Q_n = 0;$$

for the intersections of this curve with the basis-curve $P_m = 0$ are given by the equation $M_{r-n}\,Q_n = 0$, which contains only the

$$\tfrac{1}{2}(r-n+1)(r-n+2)-1$$

constants of M_{r-n} (one constant of course divides out, giving the diminution -1). The equation must have the more general form

$$\Omega_r + L_{r-m}\,P_m + M_{r-n}\,Q_n = 0;$$

and it thus appears that the Postulation of the $mn-\delta$ points A, instead of being $= mn-\delta$, must be $= mn-\delta-1$. This completes the proof.

I notice that, combining the last-mentioned identity

$$\tfrac{1}{2}m(m-3) - m(\gamma-3) + \delta = \tfrac{1}{2}(r-n+1)(r-n+2)$$

with the like identity

$$\tfrac{1}{2}n(n-3) - n(\gamma-3) + \delta = \tfrac{1}{2}(r-m+1)(r-m+2),$$

we obtain

$$\begin{aligned} &\tfrac{1}{2}m(m-3) + \tfrac{1}{2}n(n-3) - (m+n)(\gamma-3) + 2\delta \\ &= \tfrac{1}{2}(r-n+1)(r-n+2) + \tfrac{1}{2}(r-m+1)(r-m+2), \end{aligned}$$

and consequently, referring to a former result, the left-hand side should be

$$= \tfrac{1}{2}r(r+3) - mn + \delta + 1\,;$$

substituting for γ, δ their values, this is at once verified.

As appears by what precedes, Bacharach's special case is that in which the δ, $= \frac{1}{2}(\gamma-1)(\gamma-2)$ points B satisfy the single condition of lying on a curve of the order $\gamma-3$. We may have between the points B more than a single relation; in particular, the points B may be such as to include among themselves the complete intersection of two curves of the orders a, b respectively ($ab \leqq \delta$): this will be the case, if the given curves are of the form

$$\begin{aligned} P_m &= \lambda_a\,S_{m-a} - \mu_b\,R_{m-b}, \\ Q_n &= \lambda_a\,V_{n-a} - \mu_b\,U_{n-b}, \end{aligned}$$

it being understood, here and in what follows, that the values of a, b are such that the suffixes are none of them negative.

The two curves here intersect in the ab points $(\lambda_a = 0,\ \mu_b = 0)$, and in the $mn - ab$ points

$$\begin{vmatrix} \lambda_a & R_{m-b} & U_{n-b} \\ \mu_b & S_{m-a} & V_{n-a} \end{vmatrix} = 0,$$

say the $(mn - \delta)$ points A are $mn - ab - \Theta$ of the last-mentioned points, and the δ points B are the remaining Θ points together with the ab points. Here the general form of the curve of the order r passing through the $mn - ab$ points, and therefore through the $mn - \delta$ points A, is

$$\begin{vmatrix} L_{r-m-a+a+b} & M_{r-m} & N_{r-n} \\ \lambda_a & R_{m-b} & U_{n-b} \\ \mu_b & S_{m-a} & V_{n-a} \end{vmatrix} = 0,$$

where $L_{r-m-a+a+b}$, M_{r-m}, N_{r-n} are arbitrary functions of the orders indicated by the respective suffixes. The theory in regard to the number of constants is of course altogether different from that which belongs to the case of the general functions P_m, Q_n; and it is probable that much interesting theory would present itself in the consideration of particular cases.

Cambridge, 22 *March* 1887.

869. On the Transformation of Elliptic Functions.

[From the *American Journal of Mathematics*, vol. IX. (1887), pp. 193–224.]

The algebraical theory of the Transformation of Elliptic Functions was established by Jacobi in a remarkably simple and elegant form, but it has not hitherto been developed with much completeness or success. The cases $n = 3$ and $n = 5$ are worked out very completely in the *Fundamenta Nova* (1829); viz. considering the equation

$$\frac{M\,dy}{\sqrt{1-y^2\cdot 1-\lambda^2y^2}} = \frac{dx}{\sqrt{1-x^2\cdot 1-k^2x^2}},$$

($k = u^4$, $\lambda = v^4$; say this is the $Mk\lambda$- or Mun-form), Jacobi finds, in the two cases respectively, a modular equation between the fourth roots u, v, say the un-modular equation, and, as rational functions of u, v, the value of M and the values of the coefficients of the several powers of x in the numerator and denominator of the fraction which gives the value of y; but there is no attempt at a like development of the general case. I shall have occasion to speak of other researches by Jacobi, Brioschi and myself; but I will first mention that my original idea in the present memoir was to develop the following mode of treatment of the theory:

In place of the $Mk\lambda$-form, using the $\rho q\Omega$-form

$$\frac{dy}{\sqrt{1-2\beta y^2+y^4}} = \frac{\rho\,dx}{\sqrt{1-2\alpha x^2+x^4}}$$

(I write for greater convenience 2α, 2β in place of the α of Jacobi and Brioschi and the β of Brioschi), we can, by expanding each side in a series, integrating, and reverting the resulting series for y, obtain y in the form

$$y = \rho x\,(1 + \Pi_2 x^2 + \Pi_4 x^4 + \cdots).$$

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Paper 869 (continuation/conclusion of memoir, Art. Nos. 1–47)

Arthur Cayley

869 (continued). On the Transformation of Elliptic Functions.

(Continuation, beginning of memoir on printed page 506. The opening pages of paper 869 occupy PDF pp. 526–552. The introductory matter of the memoir—the announcement of the $p\alpha\beta$-form, the comparison with Jacobi's results, and the survey of the eight sections to come—fills the present page.)

where $\Pi_1, \Pi_2, \Pi_3, \ldots$ denote given functions of ρ, α, β. Taking n odd and $= 2s+1$, we assume for y an expression

$$y = \frac{x(A_1 + A_{s-1}x^2 + \cdots + A_1 x^{2s-2} + x^{2s})}{1 + A_1 x^2 + \cdots + A_{s-1}x^{2s-2} + A_s x^{2s}},$$

where the last coefficient A_s is at once seen to be $= \rho$. Comparing with the series-value $y = \rho x(1 + \Pi_1 x^2 + \Pi_2 x^4 + \cdots)$, we have an infinite series of equations. The first of these is, in fact, $A_s = \rho$; the next equations give linearly $A_{s-1}, A_{s-2}, \ldots, A_1$ in terms of the coefficients Π; that is, of ρ, α, β: the two which follow serve in effect to determine ρ, β as functions of α: and then, ρ and β having these values, all the remaining equations will be satisfied identically.

The process is an eminently practical one, so far as regards the determination of the coefficients $A_1, A_2, \ldots, A_{s-1}$ as functions of ρ, α, β; it is less so, and requires eliminations more or less complicated, as regards the determination of the relations between ρ, α, β. As to this, it may be remarked that the problem is not so much the determination of the equation between ρ and α (or say the $p\alpha$-multiplier equation, or simply the $p\alpha$-equation), and of the equation between β, α (or say the $\alpha\beta$-modular equation, or simply the $\alpha\beta$-equation), as it is to determine the complete system of relations between ρ, α, β; treating these as coordinates, we have what may be called the multiplier-modular-curve, or say the MM-curve, and the relations in question are those which determine this curve.

In the absence of special exceptions, it follows from general principles that the coefficients $A_1, A_2, \ldots, A_{s-1}$, qua rational functions of ρ, α, β, must also be rational functions of α, β or of α, ρ; and I think it may be assumed that this is the case; the method, however, affords but little assistance towards thus expressing them.

In connexion with the foregoing theory, I consider the solutions of the problem of transformation given by Jacobi's partial differential equation ("Suite de Notices sur les Fonctions elliptiques," *Crelle*, t. IV. (1829), pp. 185–193), and by what I call the Jacobi–Brioschi differential equations. The first and third of these were obtained by Jacobi in the memoir[1], "De functionibus ellipticis Commentatio," *Crelle*, t. IV. (1829), pp. 371–390 (see p. 376); but the second equation, which completes the system, was, I believe, first given by Brioschi in the second appendix to his translation of my *Elliptic Functions: Trattato elementare delle Funzioni ellittiche*: Milan, 1880. I had, strangely enough, overlooked the great importance of these equations. I shall have occasion also to refer to results, and further develop the theory contained in my memoir, "On the Transformation of Elliptic Functions," *Phil. Trans.*, t. CLXIV. (1874), pp. 397–456, [578], and the addition thereto, *Phil. Trans.*, t. CLXXXIX. (1878), pp. 419–424, [692].

I remark that, while I have only worked out the formulæ for the cases $n = 3$ and $n = 5$, and a few formulæ for the case $n = 7$, the memoir is intended to be a contribution to the general theory of the $p\alpha\beta$-transformation; I hope to be able to complete the theory for the case $n = 7$.

[1] *Ges. Werke*, bd. i., pp. 295–318; in particular, p. 303.

Comparison of the $Mk\lambda$- and $p\alpha\beta$-Forms. The Modular and Multiplier Equations. Art. Nos. 1 to 12.

1. The equation

$$\frac{M\,dy}{\sqrt{1-y^2}\cdot\sqrt{1-\lambda^2y^2}} = \frac{dx}{\sqrt{1-x^2}\cdot\sqrt{1-k^2x^2}}\,,$$

if we write therein

$$x = \frac{x}{\sqrt{k}}\,,\quad y = \frac{y}{\sqrt{\lambda}}\,,\quad M = \frac{M}{\rho}\,,$$

becomes

$$\frac{M\,dy}{\sqrt{1-(\lambda^{-1}+\lambda)y^2+y^4}} = \frac{dx}{\sqrt{1-(k^{-1}+k)x^2+x^4}}\,;$$

viz., this is

$$\frac{dy}{\sqrt{1-2\beta y^2+y^4}} = \frac{\rho\,dx}{\sqrt{1-2\alpha x^2+x^4}}\,,$$

if only

$$2\alpha = k^{-1}+k\,,\quad 2\beta = \lambda^{-1}+\lambda\,,\quad \rho = \frac{1}{M}\,.$$

2. We have a uv-modular equation, see ante my Transformation Memoir*, p. 450, this may be converted into a uw-modular equation; in particular, $n=3$, the equation is

$$y^4+6x^2y^2+x^4-4xy\,(4x^2y^2-3x^2-3y^2+4)=0,$$

where x, y denote u^2, v^2 respectively; say the equation is

$$F(x,\,y) \;=\; x^4+x^2(-16y^2+12y)+x^2\,(6y^2+x(12y^2-16y))+y^4=0.$$

From the equation $F(x,\,y)=0$, we derive

$$x^{-2}F(x,\,y)\,,\quad x^{-2}F(x,\,y^{-1})=0\,;$$

say this is

$$(Ax^2+Bx+C+Dx^{-1}+Ex^{-2})\,(A'x^2+Bx+C'+D'x^{-1}+E'x^{-2})=0,$$

viz. the equation is

$$AA'x^4+(AB'+A'B)x^3+\cdots+EE'x^{-4}=0\,;$$

where, by reason of the symmetry of $F(x,\,y)$, the coefficients AA', EE' of x^4, x^{-4}, those of x^3, x^{-3}, &c., have equal values; the form thus is

$$\mathfrak{A}\,(x^4+x^{-4})+\mathfrak{B}\,(x^3+x^{-3})+\mathfrak{C}\,(x^2+x^{-2})+\mathfrak{D}\,(x+x^{-1})+\mathfrak{E} \;=\; 0,$$

where $x+x^{-1}=2\alpha$,

$$x^2+x^{-2}=2\alpha\,,$$
$$x^3+x^{-3}=8\alpha^3-6\alpha\,,$$
$$x^4+x^{-4}=16\alpha^4-16\alpha^2+2.$$

[* This Collection, vol. XI., p. 170.]

The coefficients $\mathfrak{A}, \mathfrak{B}, \ldots$ are in like manner expressible as functions of $y + y^{-1} = 2\beta$; thus we have $\mathfrak{A} = 1$,

$$\mathfrak{B} \;=\; AB' + A'B \;=\; -16(y + y^{-1}) + 12(y^3 + y^{-3}) \;=\; -16(8\beta^3 - 6\beta) + 12 \cdot 2\beta;$$

or, finally, $\mathfrak{B} = -128\beta^3 + 120\beta$; and so for the other coefficients. The numerical coefficients contain, all of them, the factor 16; and, throwing this out, we obtain, for $n = 3$, the $\alpha\beta$-modular equation in the form

	α^4	α^3	α^2	α	1
β^4					+1
β^3	−64		+60		
β^2			−186		+192
β		+60		−64	
1	+1		+192		−192

$= 0$

where observe that the form is symmetrical as regards α, β; and, further, that the sums of the numerical coefficients in the lines or columns are the binomial coefficients 1, −6, +6, −4, +1. Observe, further, that the sums in the direction of the sinister diagonal are −64, −64, +320, −192; viz. dividing by −64, it thus appears that, writing $\beta = \alpha$, the equation becomes

$$\alpha^4 + \alpha^3 - 5\alpha^2 + 3 = 0\,;$$

that is, $(\alpha^2 - 1)^2(\alpha^2 + 3) = 0$.

Again, writing $\beta = -\alpha$, then dividing by 16, the equation becomes

$$4\alpha^4 - 19\alpha^2 + 28\alpha - 12 = 0\,;$$

that is, $(4\alpha - 3)(\alpha^3 - 2)^2 = 0$.

3. So also, for $n = 5$, we have the $\alpha\beta$-modular equation in the form

$$x^4 + 655x^3y^3 + 655x^3y^3 + y^4 - 640x^3y^3 - 640x^2y^2 + xy(-256 + 320x^2 + 320y^2 - 70x^4 - 660x^2y^2 - 70y^4) + 320x^4y^2 + 320x^2y^4 - 256x^3y^3 = 0$$

and in precisely the same manner, we obtain the $\alpha\beta$-modular equation; viz. casting out a factor 64, this is

	β^6	β^5	β^4	β^3	β^2	β	1
α^6							+1
α^5		−4096		+6400		−2310	
α^4			+60120		−172785		+103680
α^3		+6400		−133140		+126720	
α^2			−172785		+126720		−103680
α		−2310		+126720		−124416	
1	+1		+103680		−103680		+1

$= 0,$

where the form is symmetrical as regards α, β; the sums of the numerical coefficients in the lines or columns are 1, −6, +15, −20, +15, −6, +1. The sums in the direction of the sinister diagonal all divide by −4096; viz. throwing out this factor, we have, for $\beta = \alpha$, the equation

$$\alpha^{10} - 20\alpha^8 + 118\alpha^6 - 180\alpha^4 + 81\alpha^2 = 0\,;$$

that is, $\alpha^2(\alpha^2 - 1)^2(\alpha^2 - 9)^2 = 0$.

If $\beta = -\alpha$, the coefficients divide by 64; and throwing out this factor, the equation is

$$64\alpha^{10} + 8800\alpha^8 - 32474\alpha^6 + 36000\alpha^4 - 12296\alpha^2 = 0\,;$$

that is, $\alpha^2(\alpha^4 + 16\alpha^2 - 16)(8\alpha^2 - 9) = 0$.

4. We have a Mu-multiplier equation of the form $F\big(\frac{1}{M},\, 2\alpha^2 - 1\big) = 0$ (see Memoir*, pp. 420–422), but we cannot, by the preceding formulæ, deduce thence a $p\alpha$-multiplier equation; in fact, writing therein $\frac{1}{M} = \frac{\alpha^2\rho}{x^2}$, the resulting equation is

$$F\left(\frac{\alpha^2\rho}{x^2},\ 1 - 2x^2\right) = 0,$$

which is a $\rho\alpha$-multiplier equation only on the assumption that $1 - 2x^2$, α^2 and x^2 are therein regarded as given functions of α. But it is very

[* This Collection, vol. x., pp. 384, 385.]

remarkable that the $\rho\alpha$-equation, in fact, is $F(\rho,\,\alpha)=0$. To prove this, assume that the equation

$$\frac{dy}{\sqrt{1-2\beta y^2+y^4}}\;=\;\frac{\rho\,dx}{\sqrt{1-2\alpha x^2+x^4}}$$

has a $\rho\alpha$-multiplier equation $F(\rho,\,\alpha)=0$. Starting from the equation

$$\frac{M\,dy}{\sqrt{1-y^2}\cdot\sqrt{1-\lambda^2y^2}}\;=\;\frac{dx}{\sqrt{1-x^2}\cdot\sqrt{1-k^2x^2}},$$

we may, by effecting on each side a quadric transformation, convert this into

$$\frac{M^{-1}dy}{\sqrt{1-2(2\beta^2-1)y^2+y^4}}\;=\;\frac{dx}{\sqrt{1-2(2\alpha^2-1)x^2+x^4}}\,;$$

and this being so, we have, between M^{-1} and $2\alpha^2-1$, the relation

$$F\left(\frac{1}{M},\;2\alpha^2-1\right)=0;$$

or, conversely, if this be the form of the Mu-multiplier equation, then the $\rho\alpha$-multiplier equation is $F(\rho,\,\alpha)=0$.

5. The quadric transformations are

$$x=\frac{\sqrt{1-x^2}}{x\sqrt{1-k^2x^2}}\,,\qquad y=\frac{\sqrt{1-y^2}}{y\sqrt{1-\lambda^2y^2}}\,.$$

We have then only to show that

$$\frac{dx}{\sqrt{1-2(2\alpha^2-1)x^2+x^4}}\;=\;\frac{dx}{\sqrt{1-x^2}\cdot\sqrt{1-k^2x^2}}\,,$$

for then, in like manner,

$$\frac{dy}{\sqrt{1-2(2\beta^2-1)y^2+y^4}}\;=\;\frac{dy}{\sqrt{1-y^2}\cdot\sqrt{1-\lambda^2y^2}}\,,$$

and we pass from the assumed differential relation between x, y to the above-mentioned differential equation between x, y.

6. For the quadric transformation between x, x, write

$$\theta^k=k-ik'\,,\qquad \theta^{-k}=k+ik'\,,$$

$\left(\text{whence also } \theta=\dfrac{k-ik'}{k+ik'}\right)$, and therefore

$$\theta^k+\theta^{-k}=2k\,,\qquad \theta+\theta^{-1}=2k^2-2k'^2;$$

we have

$$1-\theta^kx^2=1-\theta\,\frac{1-x^2}{x^2(1-k^2x^2)}\;=\;\frac{1}{x^2(1-k^2x^2)}\big(\theta+(\theta+1)x^2-k^2x^4\big),$$
$$=\;\frac{-\theta}{x^2(1-k^2x^2)}\,(1-\theta^{-k}k^2x^2);$$

and similarly,

$$1-\theta^{-k}x^2 \;=\; \frac{-\theta^{-1}}{x^2(1-k^2x^2)}\,(1-\theta^k k^2x^2).$$

Consequently,

$$(1-\theta^k x^2)(1-\theta^{-k}x^2) \;=\; 1-2(2k^2-1)x^2+x^4 \;=\; \frac{1}{x^4(1-k^2x^2)^2}\,(1-2k^2x^2+k^2x^4)^2\,;$$

or say

$$\sqrt{1-2(2k^2-1)x^2+x^4} \;=\; \frac{1}{x^2(1-k^2x^2)}\,(1-2k^2x^2+k^2x^4).$$

Moreover,

$$dx \;=\; \frac{dx}{x^2(1-x^2)^{1/2}(1-k^2x^2)^{1/2}}\,(1-2k^2x^2+k^2x^4),$$

and thence the required equation

$$\frac{dx}{\sqrt{1-2(1-2k^2)x^2+x^4}} \;=\; \frac{dx}{\sqrt{1-x^2}\cdot\sqrt{1-k^2x^2}}\,;$$

this completes the proof.

7. Thus, referring to the Mu-equations given in the place referred to, we obtain the following $\rho\alpha$-multiplier equations. When $n=3$, we have

$$\rho^4-6\rho^2-8\alpha\rho-3=0.$$

This may be written in the forms

$$\begin{aligned} 8\alpha\rho &= \rho^4-6\rho^2-3,\\ 8(\alpha+1)\rho &= (\rho-1)^3(\rho+3),\\ 8(\alpha-1)\rho &= (\rho+1)^3(\rho-3). \end{aligned}$$

Next, for $n=5$, we have $\rho^6-10\rho^5+35\rho^4-60\rho^3+55\rho^2+(38-64\alpha^2)\rho+5=0$.
This may be written in the two forms

$$64\alpha\rho \;=\; (\rho^2-4\rho-1)(\rho^2-2\rho+5)^2$$

and

$$64(\alpha^2-1)\rho \;=\; (\rho-1)^5(\rho-5).$$

And, for $n=7$, we have

$$\rho^8-28\rho^6-112\alpha\rho^5-210\rho^4-224\alpha\rho^3+(-1484+1344\alpha^2)\rho^2+(-560\alpha+512\alpha^3)\rho+7=0.$$

8. The relation between ρ and β, or say the $\rho\beta$-multiplier equation, may be obtained by a known property of elliptic functions; viz. writing $\rho\sigma=\pm n$ (the sign is $-$ for $n=3$, $n=7$, or generally for any prime value $4p+3$: and it is $+$ for $n=5$ and generally for any prime value $4p+1$), then we have between σ, β the same relation as between ρ, α. Thus, if $n=3$, $\sigma=-\dfrac{3}{\rho}$, for ρ, α writing σ, β, the equation is

$$\sigma^4-6\sigma^2-8\beta\sigma-3=0\,;$$

or, as this may be written,

$$\rho^4+8\beta\rho^3+18\rho^2-27=0\,;$$

and so for the other cases; but it is perhaps more convenient to retain the σ; thus, if $n=5$, $\sigma=\dfrac{5}{\rho}$, we have

$$\sigma^6-10\sigma^5+35\sigma^4-60\sigma^3+55\sigma^2+(38-64\beta^2)\sigma+5=0.$$

9. We are hence able to express β as a rational function of ρ, α. We, in fact, have

$$\sqrt{\rho}\,8\beta \;=\; \frac{1}{\sqrt{\rho}}(\rho^2-4\rho-1)\sqrt{\rho^2-2\rho+5}\,, \qquad 8\beta \;=\; \frac{1}{\sqrt{\rho}}(\sigma^2-4\sigma-1)\sqrt{\sigma^2-2\sigma+5}\,,$$

(the signs must be opposite), and then for σ, substituting its value $=\frac{5}{\rho}$, and observing that $\sigma^2-2\sigma+5$ is thus $=\frac{5}{\rho^2}(\rho^2-2\rho+5)$, we find

$$\frac{\beta}{\alpha} \;=\; \frac{\rho^2+20\rho-25}{\rho^2(\rho^2-4\rho-1)},$$

which is the required formula.

Observe that, for $\rho=\sigma=\sqrt{5}$, the formulæ with the sign $-$, as above, give $\beta=-\alpha$, whereas with the sign $+$ they would have given $\beta=\alpha$. For the value in question, $\rho=\sqrt{5}$, the equation

$$64\alpha^2 \;=\; -\,(\rho^2-4\rho-1)^2(\rho^2-2\rho+5),$$

gives

$$64\alpha^2 \;=\; -\,16\,(1-\sqrt{5})^2(10-2\sqrt{5})\,;$$

that is,

$$\alpha^2 \;=\; -\tfrac{1}{4}\,(1-\sqrt{5})^2(5-\sqrt{5})\,,\; = \;(3-\sqrt{5})(\sqrt{5}-1)\,;$$

that is, $\alpha^2=-8+4\sqrt{5}$, or $\alpha^4+16\alpha^2-16=0$; it appears, *ante* No. 3, that this value belongs to the case $\beta=-\alpha$ and not to $\beta=\alpha$.

10. But there is another way of arriving at a formula containing β. Starting from Jacobi's equation

$$nM^k \;=\; \frac{\lambda\lambda'^2}{kk'^2}\cdot\frac{dk}{d\lambda},$$

and introducing for λ, λ', k, k', M their values in terms of u, v, we have

$$\frac{nv^k}{u^k\rho^k} \;=\; \frac{v^4(1-v^8)}{u^4(1-u^8)v^4\rho^4}\,\frac{dv}{du},$$

that is,

$$\frac{dv}{du} \;=\; \frac{\rho^2u^3(1-v^8)}{nv^3(1-u^8)}\,;$$

but, from the values of α, β, we find

$$\frac{dv}{du} \;=\; \frac{v^5(1-u^8)}{u^5(1-v^8)}\,\frac{d\beta}{d\alpha},$$

and, combining these results,

$$\frac{d\beta}{d\alpha} \;=\; \frac{\rho^2u^8}{nv^8}\cdot\frac{(1-v^8)^2}{(1-u^8)^2} \;=\; \frac{\rho^2}{n}\cdot\frac{(v^4-v^{-4})^2}{(u^4-u^{-4})^2}\,;$$

that is,

$$\frac{d\beta}{d\alpha} = \frac{\rho^2}{n} \cdot \frac{\beta^2-1}{\alpha^2-1}.$$

We have, consequently,

$$\frac{d\beta}{\beta^2-1} = \frac{\rho^2\, d\alpha}{n(\alpha^2-1)},$$

and therefore

$$\tfrac{1}{2}\log\frac{\beta-1}{\beta+1} = \frac{1}{n}\int \frac{\rho^2\, d\alpha}{\alpha^2-1},$$

where ρ^2 must be regarded as a function of α, or α of ρ; and from the form of the equation, it appears that the integral must be expressible as the logarithm of an algebraic function of ρ, α.

11. Thus, when $n=3$, we have

$$8\alpha = \rho^3 - 6\rho - \frac{3}{\rho};$$

$$8\frac{d\alpha}{d\rho} = 3\rho^2 - 6 + \frac{3}{\rho^2} = \frac{3}{\rho^2}(\rho^2-1)^2,$$

and thence easily

$$\begin{aligned}\tfrac{1}{2}\log\frac{\beta-1}{\beta+1} &= \int\frac{8\rho\, d\rho}{\rho^2-1\cdot\rho^2-9} = -\int\frac{d\rho}{\rho^2-1} + 9\int\frac{d\rho}{\rho^2-9}\\ &= -\tfrac{1}{2}\log\frac{\rho-1}{\rho+1} + \tfrac{3}{2}\log\frac{\rho-3}{\rho+3};\end{aligned}$$

that is,

$$\frac{\beta-1}{\beta+1} = \frac{(\rho-3)^2(\rho+1)}{(\rho+3)^2(\rho-1)},$$

as may be at once verified.

12. In the case $n=5$, I verify the equation under the form

$$\frac{d\beta}{\beta^2-1} = \frac{\rho^2}{5}\cdot\frac{d\alpha}{\alpha^2-1}.$$

From the equations

$$64(\alpha^2-1) = -(\rho-1)^5(\rho-5), \quad \text{and} \quad 8\alpha = \frac{1}{\sqrt{\rho}}(\rho^2-4\rho-1)\sqrt{\rho^2-2\rho+5},$$

we have

$$\frac{128\alpha\, d\alpha}{\alpha^2-1} = \frac{5\,(\rho^2-4\rho-1)\, d\rho}{\rho\,(\rho-1)(\rho-5)},$$

and thence

$$\frac{16\, d\alpha}{\alpha^2-1} = \frac{5\, d\rho}{(\rho-1)(\rho-5)\sqrt{\rho}\sqrt{\rho^2-2\rho+5}}.$$

Similarly, observing the $-$ sign of 8β, we have

$$\frac{16\, d\beta}{\beta^2-1} = \frac{-5\, d\sigma}{(\sigma-1)(\sigma-5)\sqrt{\sigma}\sqrt{\sigma^2-2\sigma+5}},$$

whence, substituting for σ its value $= \dfrac{5}{\rho}$, we have

$$\frac{16\,d\beta}{\beta^2-1} \;=\; \frac{\rho^2\,d\rho}{(\rho-1)(\rho-5)\sqrt{\rho}\sqrt{\rho^2-2\rho+5}} \;=\; \frac{\rho^2}{5}\cdot\frac{16\,d\alpha}{\alpha^2-1}\,,$$

which is right.

Connexion of the $Mk\lambda$- and $p\alpha\beta$-Theories. Order of Modular Equation. Art. Nos. 13 to 18.

13. In the Transformation Memoir [578], starting from the equation

$$\frac{1-y}{1+y} \;=\; \frac{1-x}{1+x}\cdot\frac{P-Qx}{P+Qx}\,,$$

I sought to determine the coefficients of P, Q by the consideration that the relation between x, y remains unaltered when x, y are changed into $\dfrac{1}{kx}$, $\dfrac{1}{\lambda y}$ respectively. This comes to saying that, when for x, y we write $\dfrac{1}{kx}$, $\dfrac{1}{\lambda y}$ respectively, the relation between x and y presents itself in the form

$$y \;=\; \frac{x(A_s + A_{s-1}x^2 + \cdots + A_1x^{2s})}{A_s + A_1x^2 + \cdots + A_{s-1}x^{2s}},$$

where $s = \frac{1}{2}(n-1)$, as before. For instance, when $n = 7$, $P = \alpha + 7\chi^2$, $Q = \beta + \delta\chi^2$.

If, solving for y, we then for x, y write $\dfrac{1}{kx}$, $\dfrac{1}{\lambda y}$, we find

$$\frac{1}{\lambda y} \;=\; \frac{x\big\{(\alpha^2+2\alpha\beta) + (2\alpha\gamma+\beta^2+2\alpha\delta+2\beta\gamma)\alpha\beta x^{-4} + (\gamma^2+2\beta\delta+2\gamma\delta)\alpha^2x^{-4} + (\delta^2+2\gamma\delta)\alpha^3x^{-6}\big\}}{\alpha^4 + \chi(2\alpha\gamma+\beta^2+2\beta\delta)x^{-2} + (\gamma^2+2\beta\delta+2\gamma\delta)x^{-4} + (\delta^2+2\gamma\delta)x^{-6} + \delta^2x^{-8}}\,;$$

and comparing this with

$$y \;=\; \frac{x(A_s + A_{s-1}x^2 + A_1x^4 + A_0x^6)}{A_s + A_1x^2 + A_{s-1}x^4 + A_sx^6}\,,$$

we have for each of the coefficients A two different expressions. Equating these and making a slight change of form, we obtain the relations between u, v, α, β, γ, δ used in the Memoir: thus,

$$A_s = \alpha^2 = v^4u^{-4}\Omega^2\,, \quad A_3 = v^4u^{-4}\,(\gamma^2+2\beta\delta+2\gamma\delta) = u^{-4}(2\alpha\gamma+\beta^2+2\alpha\beta), \text{ \&c.};$$

in the Memoir, $k\,(=v^2)$ is used instead of u, and $\Omega\,(=v^{u-2})$ instead of v, and the equations thus are

$$kv^2 \;=\; \Omega\delta^2,$$

$$\begin{aligned} k\,(2\gamma\gamma+2\alpha\beta+\beta^2) &= \Omega\,(\gamma^2+2\gamma\delta+2\beta\delta), \\ \gamma^2+2\beta\gamma+2\alpha\delta+2\beta\delta &= \Omega k\,(2\alpha\gamma+2\beta\gamma+2\alpha\delta+\beta^2), \\ \delta^2+2\gamma\delta &= \Omega k\,(\alpha^2+2\alpha\beta)\,; \end{aligned}$$

viz. these are the equations [*Coll. Math. Papers*, vol. IX., p. 119]. The idea in the present Memoir is that of considering the coefficients A in the stead of α, β, ...

14. We have here, and in general for any odd value of n, equations of the form

$$(\Omega =)\ \frac{U}{U'} = \frac{V}{V'} = \cdots ,$$

where U, V, ..., U', V', ... are quadric functions of the coefficients α, β, γ, ...; and these equations serve to establish between Ω and k a relation called the Ωk-modular equation, and which in regard to Ω is of the same degree as the uv-modular equation is in regard to v. Leaving out the equation $(\Omega =)$, we have

$$\begin{vmatrix} U, & V, & W, & \ldots \\ U', & V', & W', & \ldots \end{vmatrix} = 0\,;$$

and to each system of values of α, β, γ, δ, ... (or say of their ratios) given by these equations, there corresponds a single value of Ω; the number of values of Ω, or the degree in Ω, of the Ωk-equation is thus found as $= (n+1)\,2^{\frac{1}{2}(n-3)}$. This is far too high; for $n = 3, 5, 7, \ldots$, the degrees are 4, 12, 32, ...; those of the proper Ωk-equations are 4, 6, 8, ...

15. I showed, or endeavoured to show, that in the case $n = 5$, the extraneous factor was $(\Omega-1)^6$, $(\Omega-1 = 0$, the Ωk-modular equation belonging to $n = 1$, for which the transformation is the trivial one $y = x)$, and that in the case $n = 7$, the extraneous factor was $\{(\Omega, 1)^4\}^6$, $((\Omega, 1)^4 = 0$, the Ωk-modular equation for the case $n = 3)$; generally the extraneous factors seem to depend on the Ωk-functions for the values $n-4$, $n-8$, &c. The ground for this is that, in the assumed formula for any given value n, we may take P, Q to contain a common factor $1 \pm kx^2$ $\big($observe that, to a factor près, this is unaltered by the change x into $\frac{1}{kx}$, viz. it becomes $\frac{1}{k^2x^2}(1 \pm kx^2)$, a condition which is necessary$\big)$, and we thereby reduce the equation to

$$\frac{1-y}{1+y} \;=\; \frac{1-x}{1+x}\cdot\frac{P'-Q'x}{P'+Q'x},$$

in which equation the degrees of the numerator and the denominator are each diminished by 4, and the equation thus belongs to the value $n-4$.

16. I remark here that, in the case of n an odd prime, the degree of the modular equation is $= n+1$; but for any other odd value, the degree is

$$\sigma'(n) \;=\; n\left(1+\frac{1}{a}\right)\left(1+\frac{1}{b}\right)\cdots ,$$

where a, b, ... are all the unequal prime factors of n; thus, if $n = a^\alpha$, the degree is

$$a^\alpha\left(1+\frac{1}{a}\right) \;=\; a^{\alpha-1}(a+1).$$

In the case of a number $n = abc\ldots$, without any squared factor, the degree is

$$abc\cdots\left(1+\frac{1}{a}\right)\left(1+\frac{1}{b}\right)\left(1+\frac{1}{c}\right)\cdots \;=\; (a+1)(b+1)(c+1)\cdots ,$$

the sum of the factors of n. We have

$$\sigma'(n) \;=\; \text{coeff. } x^n \text{ in } \Sigma\phi\,(x^N),$$

where

$$\phi(x) \;=\; x + 3x^3 + 5x^5 + \cdots \;=\; \frac{x(1+x^2)}{(1-x^2)^2}\,,$$

and the summation extends to all odd values of N having no squared factor; thus,

$$\begin{aligned}
\phi(x) \;&=\; x + 3x^3 + 5x^5 + 7x^7 + 9x^9 + 11x^{11} + 13x^{13} + 15x^{15} + \cdots\,,\\
\phi(x^3) \;&=\; 1x^3 \qquad\qquad + 3x^9 \qquad\qquad + 5x^{15} + \cdots\,,\\
\phi(x^5) \;&=\; 1x^5 \qquad\qquad\qquad + 3x^{15} + \cdots\,,\\
\phi(x^7) \;&=\; 1x^7 + \cdots\,,\\
\phi(x^{11}) \;&=\; 1x^{11} + \cdots\,,\\
\phi(x^{13}) \;&=\; 1x^{13} + \cdots\,,\\
\Sigma\phi(x^N) \;&=\; x + 4x^3 + 6x^5 + 8x^7 + \cdots\,.
\end{aligned}$$

17. Supposing that the reduction is completely accounted for as above, then, to obtain numerical relations, the numbers 1, 4, 12, 32, …, $(n+1)\,2^{\frac{1}{2}(n-3)}$ have to be expressed linearly in terms of 1, 4, 6, 8, …, $\sigma'(n)$, viz. $(n+1)\,2^{\frac{1}{2}(n-3)}$ as a linear function of $\sigma'(n)$, $\sigma'(n-4)$, $\sigma'(n-8)$, …, and we have

$$\begin{gathered}
1 = 1,\\
4 = 4,\\
12 = 6 + 6\cdot 1,\\
32 = 8 + 6\cdot 4,\\
80 = 12 + 6\cdot 6 + 32\cdot 1,\\
192 = 12 + 6\cdot 8 + 33\cdot 4,\\
448 = 14 + 6\cdot 12 + 33\cdot 6 + 164\cdot 1,\\
1024 = 24 + 6\cdot 12 + 33\cdot 8 + 166\cdot 4,\\
2304 = 18 + 6\cdot 14 + 33\cdot 12 + 166\cdot 6 + 810\cdot 1,\\
5120 = 20 + 6\cdot 24 + 33\cdot 12 + 166\cdot 8 + 817\cdot 4,\\
11264 = 32 + 6\cdot 18 + 33\cdot 14 + 166\cdot 12 + 817\cdot 6 + 3768\cdot 1,\\
24576 = 24 + 6\cdot 20 + 33\cdot 24 + 166\cdot 12 + 817\cdot 8 + 3778\cdot 4,
\end{gathered}$$

but it is of course very doubtful whether these relations have any value in regard to the present theory.

18. In the same way that, by assuming a common factor, $1+kx^2$ in the values of P and Q, we pass from the case n to the case $n-4$, so, by assuming a common factor, $1+x^2$ in the numerator and denominator of the expression for y in terms of x and the coefficients B, we pass from the case n to the case $n-2$. Contrariwise, in the solutions given by the Jacobi–Brioschi differential equations and by the Jacobi partial differential equation, the solution for a given value of n does not thus contain the solution for an inferior value of n; see *post* Nos. 36 and 43.

I pass now to the theory before referred to.

The Development $y = \rho x(1 + \Pi_1 x^2 + \Pi_2 x^4 + \cdots)$. Art. Nos. 19 and 20.

19. Starting from the equation

$$\frac{dy}{\sqrt{1-2\beta y^2+y^4}} = \frac{\rho\, dx}{\sqrt{1-2\alpha x^2+x^4}},$$

and writing for shortness

$$R_1 = \tfrac{1}{2}\alpha, \quad R_2 = \tfrac{1}{4}\left(\tfrac{3}{2}\alpha^2 - \tfrac{1}{2}\right), \quad R_3 = \tfrac{1}{6}\left(\tfrac{5}{2}\alpha^3 - \tfrac{3}{2}\alpha\right), \quad R_4 = \tfrac{1}{8}\left(\tfrac{35}{8}\alpha^4 - \tfrac{15}{4}\alpha^2 + \tfrac{3}{8}\right),$$

$$S_1 = \tfrac{1}{2}\beta, \quad S_2 = \tfrac{1}{4}\left(\tfrac{3}{2}\beta^2 - \tfrac{1}{2}\right), \quad S_3 = \tfrac{1}{6}\left(\tfrac{5}{2}\beta^3 - \tfrac{3}{2}\beta\right), \quad S_4 = \tfrac{1}{8}\left(\tfrac{35}{8}\beta^4 - \tfrac{15}{4}\beta^2 + \tfrac{3}{8}\right),$$

(viz. save as to the exterior factors $\frac{1}{2}, \frac{1}{4}, \ldots, 3R_1, 5R_2, \ldots$ are the Legendrian functions of α, and $3S_1, 5S_2, \ldots$ the Legendrian functions of β), we have

$$dy\,(1 + 3S_1 y^2 + 5S_2 y^4 + \cdots) = \rho\, dx\,(1 + 3R_1 x^2 + 5R_2 x^4 + \cdots),$$

whence integrating, so that y may vanish with x, we have

$$y + S_1 y^3 + S_2 y^5 + \cdots = \rho\,(x + R_1 x^3 + R_2 x^5 + \cdots),$$

say this is

$$y = u.$$

20. We then have $y = ufy$, where $fy = S_1 + S_2 y^2 + \cdots$; and thence, expanding by Lagrange's theorem,

$$y = -fu + \frac{1}{2}\,(fu)^{2\prime} - \frac{1}{2\cdot 3}\,(f^3 u)'' + \frac{1}{2\cdot 3\cdot 4}\,(f^4 u)''' - \cdots,$$

we have

$$fu = S_1 u^4 + S_2 u^6 + S_3 u^8 + S_4 u^{10} + \cdots,$$

and thence

$$f^2 u = S_1^2 u^8 + 2S_1 S_2 u^{10} + (2S_1 S_3 + S_2^2) u^{12} + \cdots,$$
$$f^3 u = S_1^3 u^{12} + 3S_1^2 S_2 u^{14} + \cdots,$$
$$f^4 u = S_1^4 u^{16} + \cdots;$$

consequently,

$$\begin{gathered} y \;=\; u\,, \\ +u^3\,(-S_1)\,, \\ +u^5\,(-S_2+3S_1^3)\,, \\ +u^7\,(-S_3+8S_1S_2-12S_1^4)\,, \\ +u^9\,(-S_4+10S_1S_3+5S_2^2-55S_1^2S_2+55S_1^5)\,, \\ +\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\,, \\ +\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\,; \end{gathered}$$

and writing herein

$$\begin{gathered} u = \rho\,(x+R_1x^3+R_2x^5+\cdots), \\ u^3 = \rho^3\,[x^3+3R_1x^5+(3R_1^2+3R_2)x^7+\cdots], \\ u^5 = \rho^5\,[x^5+5R_1x^7+\cdots], \\ u^7 = \rho^7\,[x^7+\cdots], \\ \cdots\cdots\cdots\cdots\cdots\,, \end{gathered}$$

we have the required series

$$y \;=\; \rho x\,(1+\Pi_1x^2+\Pi_2x^4+\cdots),$$

where the values of the coefficients are

$$\Pi_1 \;=\; R_1+(-S_1)\,\rho^2,$$

$$\Pi_2 \;=\; R_2+(-S_1)\,3R_1\rho^2+(-S_2+3S_1^3)\,\rho^4,$$

$$\Pi_3 \;=\; R_3+(-S_1)\,(3R_1^2+3R_2)\,\rho^2+(3R_1R_2+R_2^2)\,\rho^2+(-S_2+3S_1^3)\,5R_1\rho^4+(-S_3+8S_1S_2-12S_1^4)\,\rho^6,$$

$$\Pi_4 \;=\; R_4+(-S_1)\,(3R_1R_2+6R_1R_3+R_2^2)\,\rho^2+(-S_2+3S_1^3)\,(3R_1^2+3R_2)\,\rho^4+(-S_3+8S_1S_2-12S_1^4)\,7R_1\rho^6+(-S_4+10S_1S_3+5S_2^2-55S_1^2S_2+55S_1^5)\,\rho^8,$$

and so on, as far as we please.

The Cubic Transformation, $n=3$. Art. Nos. 21 to 28.

21. We have here

$$\frac{\rho+x^2}{1+\rho x^2} \;=\; \rho\,(1+\Pi_1x^2+\Pi_2x^4+\cdots);$$

whence, developing the left-hand side and equating coefficients,

$$\rho\Pi_1 \;=\; -\rho^2+1\,, \quad \rho\Pi_2 \;=\; \rho^3-\rho\,, \quad \rho\Pi_3 \;=\; -\rho^4+\rho^2\,,\ldots$$

It will be convenient to write

$$\Theta_1 \;=\; \rho\,\Pi_1 + \rho^2 - 1\,, \quad = \; -S_1\rho^2 + \rho^2 + R_1\rho - 1\,,$$

$$\Theta_2 \;=\; \Pi_2 - \rho^2 + 1\,, \quad = \; (-S_2 + 3S_1^2)\rho^2 + (3R_1S_1 + 15R_2S_1^2)\rho + R_2 + 1\,,$$

$$\Theta_3 \;=\; \Pi_3 + \rho^2 - \rho\,, \quad = \; (-S_3 + 8S_1S_2 - 12S_1^4)\rho^3 + (-5R_1S_2 + 15R_1S_1^2)\rho^2 + (-3R_2S_1 - 3R_1^2S_1)\rho + R_3\,,$$

where observe the difference of form in the function Θ_1, and in the subsequent functions $\Theta_2, \Theta_3, \ldots$. In these last, a factor ρ is thrown out.

22. The two equations $\Theta_1 = 0$ and $\Theta_2 = 0$ serve to determine ρ, β in terms of α; the subsequent equations $\Theta_3 = 0$, $\Theta_4 = 0$, ... will then be, all of them, satisfied identically. This implies that $\Theta_3, \Theta_4, \ldots$ are each of them a linear function of Θ_1, Θ_2. The *à posteriori* verification and determination of the factors is by no means easy; I have effected it only for Θ_3; we have

$$7\Theta_3 \;=\; (\rho^3 - 27R_1)\Theta_1 - 10\rho + 25\Theta_2,$$

or, at full length,

$$7\left[(-S_3 + 8S_1S_2 - 12S_1^4)\rho^3 + \cdots\right] \;=\; (\rho^3 - 3S_1\rho^2 - 2\rho + 27R_1)\,(-S_1\rho^2 + \rho^2 + R_1\rho - 1)$$

in verifying which we must, of course, take account of the relations between the expressions R and those between the expressions S; we have

$$\alpha \;=\; 3R_1\,, \quad \text{and thence} \quad 10R_2 = 27R_1^2 - 1,$$

similarly,

$$10S_2 = 27S_1^2 - 1,$$

. . . .

Equating the coefficients of ρ^6, we have

$$-7\,S_3 + 56S_1S_2 - 84S_1^4 \;=\; \ldots$$

viz. multiplying by 2, this is

$$\cdots + 110S_1S_2 - 162S_1^4 + 2S_3 \ = \ 0\,;$$

or, finally, it is

$$(-135S_1^3 + 9S_1) + (297S_1^3 - 11S_1) - 162S_1^4 + 2S_3 \ = \ 0,$$

an identity, as it should be. The identity of the coefficients of ρ^5, ρ^4, ρ^3, ρ^2, ρ, 1 may be verified in like manner.

23. Considering α as known, the values of ρ and β are determined by the foregoing equations $\Theta_1 = 0$, $\Theta_2 = 0$; that is,

$$-S_1\rho^2 + \rho^2 + R_1\rho - 1 \ = \ 0,$$

$$(-S_2 + 3S_1^2)\rho^4 - (3R_1S_1 + 1)\rho^2 + R_2 + 1 \ = \ 0,$$

where, of course, the R's and S's are regarded as given functions of α and β respectively.

It is to be observed that the equations are satisfied by $\rho^2 = 1$, $\alpha = \beta$; viz. we have the transformation $y = \dfrac{x(1+x^2)}{1+x^2}$; that is, $y = x$, which is the transformation of the first order, $n = 1$. The two equations represent surfaces of the orders 4 and 6 respectively, and they have thus a complete intersection of the order 24. As part of this, we have, as just shown, each of the two lines $(\rho = 1, \alpha = \beta)$ and $(\rho = -1, \alpha = -\beta)$; but there is a more considerable reduction of order to be accounted for, the proper MM-curve being, as will appear, a unicursal curve of the order $= 6$.

24. Multiplying the second equation by $10\rho^2$, and for $10R_2$ and $10S_2$ writing their values $27R_1^2 - 1$ and $27S_1^2 - 1$ respectively, we have

$$(3S_1^2 + 1)\rho^6 - (30R_1S_1 + 10)\rho^4 + (27R_1^2 + 9)\rho^2 \ = \ 0\,;$$

and if herein we substitute for $S_1\rho^3$ its value from the first equation, $= \rho + R_1\rho^2 - 1$, we have

$$3\,(\rho^2 + R_1\rho - 1)^2 + \rho^6 - 30R_1\rho\,(\rho^2 + R_1\rho - 1) - 10\rho^4 + (27R_1^2 + 9)\rho^2 \ = \ 0\,;$$

that is,

$$\rho^6 - 7\rho^4 - 24R_1\rho^3 + 3\rho^2 + 24R_1\rho + 3 \ = \ 0\,;$$

viz. this is

$$(\rho^2 - 1)\,(\rho^4 - 6\rho^2 - 24R_1\rho - 3) \ = \ 0,$$

containing, as it ought to do, the factor $\rho^2 - 1$. Throwing this out, and repeating the first equation, we have

$$-S_1\rho^2 + \rho^2 + S_1\rho - 1 \ = \ 0,$$

$$\rho^4 - 6\rho^2 - 24R_1\rho - 3 \ = \ 0,$$

which two equations may be replaced by

$$\rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 \ = \ 0,$$

$$\rho^4 - 6\rho^2 - 24R_1\rho - 3 \ = \ 0,$$

which are the $\rho\beta$- and $\rho\alpha$-equations respectively. Recollecting that R_1 and S_1 denote $\frac{1}{2}\alpha$ and $\frac{1}{2}\beta$, they agree with the results obtained in No. 7. The $\alpha\beta$-modular equation is obtained by the elimination of ρ from these two equations, and may be at once written down in the form, Det. $= 0$, where the determinant is of the order 8, but contains S_1 and R_1, that is, β and α, each of them, in the fourth order only: the form is thus the same with that of the $\alpha\beta$-equation obtained in No. 2; but the identification would be a work of some labour.

25. The equations may be written

$$\begin{aligned} 24S_1\rho^3 &= \rho^4 + 18\rho^2 - 27, \\ 24R_1\rho &= \rho^4 - 6\rho^2 - 3, \end{aligned}$$

and, treating R_1, S_1, ρ as coordinates, it hence appears that the MM-curve is (as mentioned above) a unicursal curve of the order 6; in fact, we have R_1, S_1, each of them given as a rational function of ρ; and cutting the curve by an arbitrary plane $AR_1 + BS_1 + C\rho + D = 0$, the substitution of the values of R_1, S_1 in this equation gives for ρ an equation of the order 6.

26. The same conclusion may be obtained from the foregoing system of a cubic and a quartic equation in ρ. Considering R_1, S_1, ρ as coordinates, they represent, each of them, a surface of the order 4, and the complete intersection is of the order 16; but this is made up of a line in the plane infinity counting 10 times, and of the MM-curve, which is thus of the order $16 - 10 = 6$. In fact, introducing, for homogeneity, a fourth coordinate θ, the two equations are

$$\begin{aligned} -S_1\rho^3 + \rho^2\theta^2 + R_1\rho\theta^2 - \theta^4 &= 0, \\ \rho^4 - 6\rho^2\theta^2 - 24R_1\rho\theta^3 - 3\theta^4 &= 0, \end{aligned}$$

and the line $\rho = 0$, $\theta = 0$ is thus a triple line on each of these surfaces; viz. cutting them by an arbitrary plane, we have for the first surface an ordinary triple point, as shown by the continuous lines of the annexed figure, and for the second surface a triple point = cusp + two nodes, as shown by the dotted lines of the figure. There is, moreover, as shown in the figure, a contact of two branches, and the number of intersections is thus = 10.

[Figure: small schematic showing three concurrent line-pairs meeting at a single triple point, with continuous lines (first surface, ordinary triple point) and dotted lines (second surface, cusp + two nodes) crossing through it; a pair of branches indicates a tangency.]

27. If we assume $\rho\sigma = -3$, that is, $\rho = -\dfrac{3}{\sigma}$, and substitute this value in the equation for S_1, the two equations become

$$24S_1\sigma = \sigma^4 - 6\sigma^2 - 3,$$

viz. S_1 is the same function of $\sigma\ \bigl(= -\dfrac{3}{\rho}\bigr)$ that α is of ρ. This accords with the theorem in Elliptic Functions that a combination of two transformations leads to a multiplication.

28. We have

$$24(R_1+1)\rho \ = \ \rho^4-6\rho^2-8\rho-3,$$

or, what is the same thing,

$$24(R_1+1)\rho \ = \ (\rho-1)^2(\rho^2+2\rho+2);$$

and, in like manner,

$$24(R_1-1)\rho \ = \ \rho^4-6\rho^2-8\rho-3 \ = \ (\rho+1)^2(\rho^2-2\rho+2),$$

and, consequently,

$$24R_1\rho \ = \ \rho^4-6\rho^2-3,$$

with the like equations between S_1, α, ρ. It will be recollected that

$$R_1 \ = \ \tfrac{1}{2}\alpha\,, \quad \alpha \ = \ \tfrac{1}{2}\bigl(u^2+\frac{1}{u^2}\bigr)\,;$$

hence

$$(R_1+1) \ = \ \tfrac{1}{2}\Bigl(u+\frac{1}{u}\Bigr)^2, \quad (R_1-1) \ = \ \tfrac{1}{2}\Bigl(u-\frac{1}{u}\Bigr)^2.$$

The formulæ just obtained are useful for obtaining the uv-modular equation from the foregoing equations; or say

$$4\Bigl(v^4+\frac{1}{v^4}\Bigr)\sigma \ = \ \sigma^4-6\sigma^2-3,$$

where $\rho\sigma=-3$, and we have to eliminate ρ and σ; the elimination gives

$$\frac{u^2}{v^2}-\frac{v^2}{u^2}+2vu-\frac{2}{uv} \ = \ 0,$$

that is,

$$v^4+2v^3u^3-2vu^4-u^4 \ = \ 0.$$

The Quintic Transformation, $n=5$. Art. Nos. 29 to 32.

29. We have here

$$\frac{\rho+A_1x^2+x^4}{1+A_1x^2+\rho x^4} \ = \ \rho\,(1+\Pi_1x^2+\Pi_2x^4+\cdots),$$

and multiplying by $1+A_1x^2+\rho x^4$, we obtain an infinite series of equations, the first three of which are

$$A_1 \ = \ \rho\,\Pi_1+A_1\rho,$$

$$1 \ = \ \rho\,\Pi_2+A_1\rho\,\Pi_1,$$

$$0 \ = \ \rho\,\Pi_3+A_1\rho\,\Pi_2+\rho^2\Pi_1,$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

The first of these gives

$$A_1 \;=\; \frac{\rho\,\Pi_1}{-\rho+1}\,, \quad = \frac{\Theta_1-\rho^2+1}{-\rho+1}\,;$$

and the other two equations then determine the MM-curve. These being satisfied, the remaining equations will be satisfied identically. It is proper to introduce $\Theta_1, \Theta_2, \Theta_3$ into the equations instead of Π_1, Π_2, Π_3. We have first

$$1 \;=\; \rho\,\Pi_2 + \rho\,\Pi_1 \cdot \frac{\Theta_1-\rho^2+1}{-\rho+1} + \rho^2,$$

that is,

$$0 \;=\; (\Theta_1-\rho^2+1)\,\Theta_1 + (-\rho+1)\,(\rho\,\Pi_2+\rho^2-1)\,;$$

viz. this is

$$0 \;=\; (\Theta_1-\rho^2+1)\,\Theta_1 + (-\rho+1)\,\Theta_2 + \rho^2-1\,;$$

or, finally, this is

$$\rho\,(\rho-1)\,\Theta_1 - \Theta_1^2 - \Theta_2 + 2\Theta_1\,(\rho-1) \;=\; 0.$$

Secondly, we have

$$0 \;=\; \rho\,\Pi_3 + A_1\rho\,\Pi_2 + \rho^2\Pi_1,$$

that is,

$$0 \;=\; \Theta_3\,(-\rho+1) + \Theta_1^2 + \Theta_1\,(\rho^2-\rho) - \Theta_2\,(\rho^2-1) \;=\; 0,$$

viz. this is

$$\Theta_3\,(-\rho+1) + \Theta_1^2 + \Theta_1\,(\rho^2-\rho) - \Theta_2\,(\rho^2-1) \;=\; 0;$$

or, finally, it is

$$\Theta_3\,(-\rho+1) + \Theta_1^2 + \Theta_1\,(\rho^2-\rho) - \Theta_2\,(\rho^2-1) \;=\; 0.$$

30. We have thus the two equations

$$(\rho-\rho)\Theta_1 - \Theta_1^2 + 2\Theta_1\,(\rho-1) \;=\; 0,$$

$$\Theta_3\,(-\rho+1) + \Theta_1^2 + \Theta_1\,(\rho^2-\rho) - \Theta_2\,(\rho^2-1) \;=\; 0\,;$$

and recollecting that Θ_3 is of the form $L\,\Theta_1 + M\,\Theta_2$, we see that each of these equations is satisfied if only $\Theta_1 = 0$, $\Theta_2 = 0$ (the formulæ belonging to the cubic transformation). This ought to be the case, for the expression $\dfrac{x(\rho+A_1x^2+x^4)}{1+A_1x^2+\rho x^4}$, by writing $A_1 = \rho+1$, reduce the expression to $\dfrac{x(\rho+x^2)}{1+\rho x^2}$, which belongs to the cubic transformation (see *ante* No. 17). The equations may be written

$$\rho^2 \;=\; -\,(2\rho+2)\,\Theta_1 + \cdots\,,$$

31. The investigation may be presented in a slightly different form by introducing the functions Θ at an earlier stage; viz. writing

$$\frac{\rho + A_1x^2 + x^4}{1 + A_1x^2 + \rho x^4} \;=\; \rho\,(1 + \Pi_1 x^2 + \Pi_2 x^4 + \cdots),$$

we have

$$\frac{\rho + A_1x^2 + x^4}{1 + A_1x^2 + \rho x^4} \;=\; \rho + \frac{\Theta_1}{1 + A_1x^2 + \rho x^4}.$$

Transposing, reducing, and dividing by x^2, we have

$$\Theta_1\,(\rho - 1 + A_1\rho x^2 + \rho x^4) \;=\; \Theta_1\,(1 + A_1x^2 + \rho x^4),$$

whence clearly $\rho - 1 + A_1\,(-\rho + 1) = \Theta_1$, giving for A_1 the before-mentioned value; and we then have

$$1 + A_1x^2 + \rho x^4 \;=\; 1 + (\rho+1)x^2 + \rho x^4 - \frac{\Theta_1}{\rho - 1}\,x^2 \;=\; (1+x^2)(1+\rho x^2) - \frac{\Theta_1}{\rho-1}\,x^2.$$

The equation thus becomes

$$(1 - x^2)(1 + \rho x^2)\,\Theta_1 + \cdots$$

and expanding the left-hand side, first in the form

$$x_0 + \frac{\Theta_2 x^2}{(\rho-1)(1+x^2)} + \frac{\Theta_2^2 x^4}{(\rho-1)^2(1+x^2)^2(1+\rho x^2)} + \frac{\Theta_2^3 x^6}{(\rho-1)^3(1+x^2)^3(1+\rho x^2)^2} + \cdots,$$

and then each of these terms separately in powers of x, and comparing with $\Pi_1x^2 + \Pi_2x^4 + \Pi_3x^6 + \cdots$, we have the two equations in the last-mentioned form, and an infinite series of other equations, which will be satisfied identically.

32. The successive coefficients might be called $\Phi_2, \Phi_3, \ldots$; say

$$\Phi_2 \;=\; (\rho^2 - \rho)\,\Theta_1 - \Theta_1^2 + 2\,(\rho^2 - 1)(\Pi)_1,$$

$$\Phi_3 \;=\; (-\rho + 1)\,\Theta_3 + \Theta_1\Theta_2 + (2\Theta_1 + \Theta_1^2)\,\rho,$$

and similarly for $\Phi_4, \ldots$; and it would then be proper to show *à posteriori* that each of the equations $\Phi_4 = 0$, $\Phi_5 = 0, \ldots$ is satisfied identically in virtue of the two equations $\Phi_2 = 0$, $\Phi_3 = 0$, or, what is the same thing, that the functions $\Phi_4, \Phi_5, \ldots$ are each of them a linear function (with coefficients which are functions of ρ) of the two functions Φ_2 and Φ_3. I do not attempt to do this, nor even to discuss the MM-curve by means of the equations $\Phi_2 = 0$, $\Phi_3 = 0$; but I will obtain equivalent results, and complete the solution by means of the Jacobi–Brioschi equations, in effect reproducing the investigation contained in the third appendix of the *Funzioni ellittiche*.

The General Transformation, $n = 2s + 1$. Art. No. 33.

33. The equation here is

$$\frac{\rho + A_{s-1}x^2 + \cdots}{1 + A_1 x^2 + \cdots} \;=\; \rho\,(1 + \Pi_1 x^2 + \cdots).$$

The general theory is sufficiently illustrated by the preceding particular cases, and I wish at present only to notice the equation obtained by comparing the coefficients of x^2; viz. this is $A_{s-1} = \rho\, A_1\, \Pi_1$, or, substituting for Π_1 its value,

$$A_{s-1} - \rho A_1 \;=\; \tfrac{1}{2}\,(\alpha\rho - \beta\rho^3).$$

The Jacobi–Brioschi Equations. Art. Nos. 34 to 42.

34. These were obtained for the differential equation

$$\frac{dx}{\sqrt{ax^4 + bx^3 + cx^2 + dx + e}} \;=\; \frac{dy}{\sqrt{ay^4 + by^3 + cy^2 + dy + e}}\,,$$

viz. if this be satisfied by $y = U \div V$, where U, V are rational and integral functions of x of the degrees n and $n-1$ respectively, then, writing for shortness

$$\phi \;=\; ax^4 + bx^3 + cx^2 + dx + e,$$

and using accents to denote differentiation in regard to x, the numerator and denominator U, V satisfy the equations

$$\begin{gathered}(VV'' - V'^2)\,\phi + \tfrac{1}{2}\,VV'\,\phi' + aU^4 + \rho V^2 = 0,\\ -(VU'' - U'V')\,\phi - \tfrac{1}{2}\,(VU' + UV')\,\phi' + bU^3 + \rho UV = 0,\\ (UU'' - U'^2)\,\phi + \tfrac{1}{2}\,UU'\,\phi' + eU^2 + \rho U^2 = 0,\end{gathered}$$

where ρ is a function $= \alpha x^2 + bx + c$, with coefficients α, b, c, the values of which have to be determined. By way of verification, observe that, multiplying by U^2, UV, V^2, and adding, we obtain

$$\frac{V}{V}\,(\cdots\cdots) \;=$$

that is,

$$-\,(VU' - V'U)^2(\phi) + ax^4 + bx^3 + cx^2 + dx + e + ay^4 + by^3 + cy^2 + dy + e \;=\; 0,$$

the result obtained by substituting for y its value $= U \div V$ in the differential equation.

35. Considering the foregoing special form

$$\frac{dx}{\sqrt{1 - 2\alpha x^2 + x^4}} \;=\; \frac{dy}{\rho\sqrt{1 - 2\beta y^2 + y^4}}\,,$$

so that a, b, c, d, e have the values ρ^2, 0, $-2\beta\rho^2$, 0, ρ^2 and ϕ is $= 1 - 2\alpha x^2 + x^4$, the equations are

$$(VV'' - V'^2)\,\phi + \tfrac{1}{2}\,VV'\,\phi' + \rho^2 U^4 + \rho V^2 = 0,$$

$$\cdots\cdots\cdots = 0,$$

where, writing as before, $n = 2s+1$, and assuming that the last coefficient, $A_{(2s+1)/2}$ or A_s, is $= \rho$, we have

$$U = x\left(\rho + A_{s-1}x^2 + A_{s-2}x^4 + \cdots + A_1 x^{2s-2} + x^{2s}\right),$$
$$V = 1 + A_1 x^2 + A_2 x^4 + \cdots + A_{s-1}x^{2s-2} + \rho x^{2s},$$

and where, as is easily shown, ρ has the value $= -\left[2A_1 + (2s+1)\,x^2\right]$. In comparing with Brioschi, it will be recollected that 2α, 2β are written in place of his α, β.

36. The equations contain n, and they are not satisfied by values of U, V belonging to any inferior value of n; U, V may each of them be multiplied by any common constant factor at pleasure, but not by a common variable factor P; viz. it is assumed that the fraction $U \div V$ is in its least terms, and consequently that (save as to a constant factor) U, V are determinate functions. It is easy to verify that the equations (being verified by U, V) are not verified by PU, PV, but it is interesting to show *à priori* why this is so. The equations are obtained as follows. Consider the differential equation in the form $\dfrac{dx}{\sqrt{X}} = \dfrac{dy}{\sqrt{Y}}$, and suppose that an integral equation is given in the form $F = 0$ (F a rational and integral function of x, y); we thence deduce a relation $L\,dx + M\,dy = 0$ between the differentials, and this must agree with the given differential equation; that is, we have $L\sqrt{X} + M\sqrt{Y} = 0$, or, rationalizing, $L^2X - M^2Y = 0$; viz. this last equation must agree with the equation $F = 0$, or, what is the same thing, $L^2X - M^2Y$ must contain F as a factor; say we have

$$L^2X - M^2Y = F \cdot G,$$

where G is a function of x, y. In the particular case where the integral is of the form

$$y = U + V,$$

we have

$$F = Vy - U,$$

and we have therefore

$$L^2X - M^2Y = G\,(Vy - U);$$

and it is by means of this identity that the equations are obtained. But suppose that there is a common factor P, or that we have $y = PU \div PV$; then, if we write $F = PVy - PVU$, $= P\,(Vy - U)$, there is no necessity that $L^2X - M^2Y$ should contain as a factor this expression of F, and it will not in fact contain it; all that is necessary is that $L^2X - M^2Y$ shall contain the factor $Vy - U$; and thus the equations obtained for U, V do not apply to PU, PV. We might, of course, introduce an arbitrary constant factor Θ; contrast herewith the solution by means of the Jacobi partial differential equation, *post* No. 43, where Θ is not arbitrary but has a determinate value.

37. In virtue of the assumed forms of U, V, the first and the third equations give each of them the same relations between the coefficients A; and only one of these equations, say the first, need be attended to. It will be observed that this equation

does not contain β; it consequently serves to determine the coefficients A in terms of ρ, α, and to establish a relation between ρ, α, that is, the multiplier equation. We can from this, as will be explained, deduce the equation between ρ, β; the theory thus depends entirely upon the first equation; say this is

$$(VV'' - V'^2)(1 - 2\alpha x^2 + x^4) + \tfrac{1}{2}VV'(-4\alpha x + 4x^3) + \rho^2 U^4 + (2A_1 + (2s+1)x^2)V^2 = 0.$$

38. We have $V = 1 + A_1x^2 + A_2x^4 + \cdots$, but the equation contains the quadric functions VV'', V'^2, VV', and V^2; it is convenient to write

$$VV'' - V'^2 = K_1 + K_2x^2 + K_3x^4 + \cdots,$$

whence of course

$$VV' = 2L_1x + 4L_2x^3 + \cdots,$$

and we have

$K_1 =$	$K_2 =$	$K_3 =$	$K_4 =$	$K_5 =$	$K_6 =$	$\cdots$
$2A_1$	$12A_2$ $-2A_1^2$	$30A_3$ $-2A_1A_2$	$56A_4$ $+8A_1A_3$ $-4A_2^2$	$90A_5$ $+26A_1A_4$ $-4A_2A_3$	$132A_6$ $+52A_1A_5$ $+4A_2A_4$ $-6A_3^2$	

$L_1 =$	$L_2 =$	$L_3 =$	$L_4 =$	$L_5 =$	$\cdots$
1	$2A_1$ $+A_1^2$	$3A_2$ $+2A_1A_2$	$4A_3$ $+2A_1A_3$ $+A_2^2$	$5A_4$ $+2A_1A_4$	

The coefficients of U^4 are at once obtained; say we have $U^4 = A_0x^4 + A_1x^6 + A_2x^8 + \cdots$,

$A_0 =$	$A_1 =$	$A_2 =$	$\cdots$
ρ^2	$2\rho A_{s-1}$	$2\rho A_{s-2}$ $+A_{s-1}^2$	

Substituting in the equation and equating to zero the coefficients of the several powers of x, we find

$$K_1 - 2A_1L_1 = 0,$$

$$K_2 - 2A_1L_2 + (-2s-1)L_1 - 2\alpha\,(K_1 + L_1) + \rho^2 A_1 = 0,$$
$$K_3 + K_1 - 2A_1L_3 + (-2s+1)L_2 - 2\alpha\,(K_2 + 2L_2) + \rho^2 A_2 = 0,$$
$$K_4 + K_2 - 2A_1L_4 + (-2s+3)L_3 - 2\alpha\,(K_3 + 3L_3) + \rho^2 A_3 = 0,$$
$$K_5 + K_3 - 2A_1L_5 + (-2s+5)L_4 - 2\alpha\,(K_4 + 4L_4) + \rho^2 A_4 = 0,$$
$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

The number of equations is $= 2(s+1)$, for the equation contains terms in x^0, x^2, x^4, $\ldots$, x^{4s+2}; but the first equation, and also the last and last but one equations, are in fact identities; there remain thus $2(s+1)-3 = 2s-1$ equations; but these are equivalent to s independent equations, serving to determine the $s-1$ coefficients $A_1, A_2, \ldots, A_{s-1}$ and to determine the relation between ρ and α. In writing down the equations for a determinate value of s, the coefficients A_0, A_s must be taken to be $= 0$ and ρ respectively; and coefficients with a negative suffix or a suffix greater than s, must be taken to be each $= 0$.

39. Thus, $(n=3)$ $s=1$, we have the $2(s+1)$, $= 4$ equations:

$$2\rho - 2\rho\cdot 1 = 0,$$

$$-2\rho^2 - 2\rho\cdot 2\rho + (-3)\,1 - 2\alpha\,(-2\rho + 2\rho) + \rho^2\cdot\rho^2 = 0,$$
$$+2\rho - 2\rho\cdot\rho^2 + (-1)\,2\rho - 2\alpha\,(-2\rho + 2\rho^2) + \rho^2\cdot 2\rho = 0,$$
$$-2\rho - 2\rho\cdot 1 + (+1)\rho^2 - 2\alpha\,(0 + 3\cdot 0) + \rho^2\cdot 2\cdot 1 = 0,$$

where the first, third and fourth equations are each of them an identity; the second equation is $-4\rho^4 + 2\rho^4 - 8\alpha\rho + \rho^4 = 0$; viz. in accordance with what precedes, writing $\alpha = 3R_1$, this is the foregoing equation

$$\rho^4 - 6\rho^2 - 24R_1\rho - 3 = 0.$$

To complete the solution, we use the theorem in elliptic functions referred to *ante* (No. 8); viz. writing $\rho\sigma = -3$, then we have β the same function of σ that α is of ρ; whence

$$\sigma^4 - 6\sigma^2 - 24S_1\sigma - 3 = 0,$$

and we thus have two equations giving the MM-curve.

40. In the case, $(n=5)$ $s=2$, we have the $2(s+1)$, $= 6$ equations:

$$2A_1 - 2A_1\cdot 1 \;=\; 0,$$

$$12\rho - 2A_1^2 - 2A_1\,(2A_1) - 5\cdot 1 - 2\alpha\cdot 2A_1 + \rho^2\cdot\rho^2 \;=\; 0,$$
$$-4\rho + 12\rho - 2A_1^2 - 2A_1\cdot 2A_1\rho - 1\,(2\rho + A_1^2) - 2\alpha\,\{-2\rho + 3\cdot 2A_1\rho + \cdots\} + \rho^2\,(2\rho + A_1^2) \;=\; 0,$$
$$-2\rho A_1 - 2A_1\cdot\rho^2 - 2\alpha\,\{-4\rho + 1\cdot 2A_1\rho + 4\rho\} + \rho^2\cdot 2A_1 \;=\; 0,$$
$$-4\rho - 2A_2\cdot\, Q_2 + 3\cdot\rho^2 - 2\alpha\,\{\cdots + 5\cdot 0\} + \rho^2\cdot 1 \;=\; 0,$$
$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

where the first, fifth and sixth equations are each of them an identity. The remaining equations are

$$\begin{aligned}(\rho^2-2\rho+5)(\rho^2+2\rho-1)-6A_1^2-8A_1\alpha &= 0,\\ 2\rho^2A_1-6\rho A_1-32\rho\alpha-2A_1^3-\tfrac{4}{3}A_1 &= 0,\\ 2\rho\,(\rho^2-2\rho+5)+10\rho^2-4A_1^2\rho-8\alpha A_1\rho+3A_1\rho^2 &= 0.\end{aligned}$$

41. Writing the first and third of these in the forms

$$\begin{aligned}-6A_1^2-8A_1\alpha+(\rho^2-2\rho+5)(\rho^2+2\rho-1) &= 0,\\ A_1^2\,(\rho^2-4\rho+3)-8A_1\alpha\rho+(\rho^2-2\rho+5)\,2\rho &= 0,\end{aligned}$$

they determine A_1^2, $8A_1\alpha$ in terms of ρ; viz. we find

$$\begin{aligned}A_1^2 &= \tfrac{1}{2}\,(\rho^2-2\rho+5)\rho,\\ 8A_1\alpha &= (\rho^2-2\rho+5)(\rho^2-4\rho-1);\end{aligned}$$

and then, writing the second equation in the form

$$(\rho^3-3\rho)\,A_1-2\rho A_1^2-16\rho^2A_1\alpha-A_1^3 = 0,$$

and substituting these values of A_1^2 and $8A_1\alpha$, and omitting the factor $\rho^2-2\rho-5$, we have the identity

$$\rho\,(\rho^3-3\rho-2)-2\rho\,(\rho^2-4\rho-1)-\rho^2(\rho^2-2\rho+5) = 0\,;$$

viz. the second equation is then also satisfied.

Forming the square of $8A_1\alpha$, and for A_1^2 substituting its value, then omitting a factor $\rho^2-2\rho+5$, we find

$$\begin{aligned}64\rho\alpha^2 &= (\rho^2-2\rho+5)(\rho^2-4\rho-1)^2,\\ &= \rho^6-10\rho^5+35\rho^4-60\rho^3+55\rho^2+38\rho+5\,;\end{aligned}$$

or, as this may also be written,

$$64\rho\,(\alpha^2-1) = (\rho-1)^5(\rho-5),$$

and we then have also, as before,

$$64\rho\,(\beta^2-1) = (\sigma-1)^5(\sigma-5),$$

which two equations determine the MM-curve.

The coefficient A_1 is given by the foregoing equation for $8A_1\alpha$, say the value is

$$A_1 = \frac{1}{8\alpha}\,(\rho^2-2\rho+5)(\rho^2-4\rho-1).$$

The value $A_1=\dfrac{\rho\,\Pi_1}{-\rho+1}$, obtained in No. 29, on substituting for Π_1 its value, is

$$A_1 = \frac{\tfrac{1}{2}\,(\beta\rho^2-\alpha\rho)}{\rho-1},$$

and these two values are, in fact, equivalent in virtue of the value of β obtained in No. 9.

42. I consider the case $n = 7$, in order to show the form of the equations which have to be solved; these equations are

$$2A_1 - 2A_1 \cdot 1 \;=\; 0,$$

$$12A_2 - 2A_1^2 - 2A_1\,(2A_1) - 7\cdot 1 - 2\alpha\,(2A_1 + 1\cdot 2A_1) + \rho^2\cdot\rho^2 \;=\; 0,$$

$$30\rho - 2A_1A_3 + 2A_2 - 2A_1\,(2A_2 + A_1^2) - 5\cdot 2A_1 - 2\alpha\,(12A_2 - 2A_1^2 + 2\,(2A_2 + A_1^2)) + \rho^2\cdot 2\rho A_2 \;=\; 0,$$

$$8A_1\rho - 4A_2^2 + 12A_2 - 2A_1^2 - 2A_1\,(2\rho + 2A_1A_2) - 3\,(2A_2 + A_1^2) - 2\alpha\,(30\rho - 2A_2A_1 + 3\,(2\rho + 2A_1A_2)) + \rho^2\,(2\rho A_1 + A_2^2) \;=\; 0,$$

$$-6\rho^2 + 30\rho - 2A_1A_3 - 2A_1\,(2A_1\rho + A_2^2) - 1\,(2\rho + 2A_1A_2) - 2\alpha\,(8A_1\rho - 4A_2^2 + 4\,(2A_1\rho + A_2^2)) + \rho^2\,(2\rho + 2A_1A_2) \;=\; 0,$$

$$-6\rho^2 + 8A_1\rho - 4A_2^2 - 2A_1\,(2A_2\rho) + 1\,(2A_2\rho + A_1^2) - 2\alpha\,(-6\rho^2 + 5\,(2A_2\rho)) + \rho^2\,(2A_1\rho + A_2^2) \;=\; 0,$$

$$-6\rho^2 - 2A_1\cdot\rho^2 + 3\,(2A_2\rho) - 2\alpha\,(-6\rho^2 + 6\cdot\rho^2 + \rho^2\cdot 2A_1) + \rho^2\cdot 2A_1 \;=\; 0,$$

$$-6\rho^2 + 5\cdot\rho^2 - 2\alpha\,(0 + 7\cdot 0) + \rho^2\cdot 1 \;=\; 0\,;$$

viz. the first, seventh and eighth equations are satisfied identically, and there remain five equations connecting ρ, α, A_1, A_2.

These equations[2] should lead to the before-mentioned $\alpha\beta$-modular equation

$$\rho^8 - 28\rho^6 - 112\alpha\rho^5 - 210\rho^4 - 224\alpha\rho^3 + (-1484 + 1344\alpha^2)\rho^2 + (-560\alpha + 512\alpha^3)\rho + 7 = 0,$$

and to expressions for A_1, A_2 as rational functions of α, ρ: and they should be, all five of them, satisfied by these results; but I do not see how the results are to be worked out; there is, so far as appears, no clue to the discovery of the rational functions of α, ρ.

The Jacobi Partial Differential Equation. Art. Nos. 43 to 48.

43. Writing, as above, 2α in place of Jacobi's α, this is

$$(1 - 2\alpha x^2 + x^4)\,\frac{d^2y}{dx^2} + (n-1)\,(2\alpha x - 2x^3)\,\frac{dy}{dx} + n\,(n-1)\,x^2y - 4n\,(\alpha^2 - 1)\,x\,\frac{dx}{d\alpha} \;=\; 0,$$

satisfied by the numerator and denominator U, V, each of them taken with the same proper value of the coefficient A_1; or, what is the same thing, by the values

$$U \;=\; \Theta x\,(A_s + A_{s-1}x^2 + A_{s-2}x^4 + \cdots + A_1x^{2s-2} + x^{2s}),$$

$$V \;=\; \Theta\,(1 + A_1x^2 + A_2x^4 + \cdots + A_{s-1}x^{2s-2} + A_sx^{2s}),$$

[* See this volume, p. 535.]

[2][See this volume, p. 535.]

where $A_s = \rho$ as before: Θ has its proper value; viz. disregarding an arbitrary merely numerical factor which might of course be introduced, the value is

$$\Theta = \sqrt{\frac{1}{M}\cdot\frac{\lambda'}{k'}}, \quad = \sqrt{\frac{v^2\rho}{u^2}\cdot\frac{\sqrt{1-v^4}}{\sqrt{1-u^4}}}, \quad = \sqrt{\rho}\cdot\frac{\sqrt[4]{v^4-v^{-4}}}{\sqrt[4]{u^4-u^{-4}}},$$

or, what is the same thing,

$$\Theta = \sqrt{\rho}\cdot\sqrt[4]{\frac{\beta^2-1}{\alpha^2-1}}.$$

If for x we write Θx, then the equation becomes

$$(1-2\alpha x^2+x^4)\frac{d^2V}{dx^2}+(n-1)(2\alpha x-2x^3)\frac{dV}{dx}+(n-1)x^2V-4n(\alpha^2-1)\left(\frac{dV}{d\alpha}+\frac{1}{\Theta}\frac{d\Theta}{d\alpha}V\right) = 0,$$

satisfied by the foregoing values without the factor Θ; or, attending only to the denominator, say by the value

$$V' = 1+A_1x^2+A_2x^4+\cdots+A_{s-1}x^{2s-2}+\rho x^{2s}.$$

44. To calculate the value of $\frac{1}{\Theta}\frac{d\Theta}{d\alpha}$, we have

$$\frac{1}{\Theta}\frac{d\Theta}{d\alpha} = \frac{1}{\rho}\frac{d\rho}{d\alpha}+\frac{1}{2}\frac{d}{d\alpha}\log\frac{\beta^2-1}{\alpha^2-1};$$

but it has been seen (No. 10) that we have

$$\frac{d\beta}{d\alpha} = \frac{\rho^2}{n}\cdot\frac{\beta^2-1}{\alpha^2-1},$$

and the formula thus becomes

$$\frac{1}{\Theta}\frac{d\Theta}{d\alpha} = \frac{\frac{1}{2}\frac{d\rho}{d\alpha}(\beta\rho^2-n\alpha)}{\alpha^2-1}.$$

We have, as the first of the equations obtained by substituting in the partial differential equation,

$$2A_1-4n(\alpha^2-1)\frac{1}{\Theta}\frac{d\Theta}{d\alpha} = 0,$$

and we have hence the value of the first coefficient,

$$A_1 = n(\alpha^2-1)\frac{1}{\rho}\frac{d\rho}{d\alpha}+\tfrac{1}{2}(\beta\rho-n\alpha);$$

or we may, by means of this result, get rid of the term $\frac{1}{\Theta}\frac{d\Theta}{d\alpha}$ from the partial differential equation; viz. the equation may be written

$$(1-2\alpha x^2+x^4)\frac{d^2V}{dx^2}+(n-1)(2\alpha x-2x^3)\frac{dV}{dx}+\left[n(n-1)x^2-2A_1\right]\zeta-4n(\alpha^2-1)\frac{dV}{d\alpha} = 0.$$

Before going further, I remark that the last of the equations obtained by the substitution gives the coefficient A_{s-1}: but this is also given in terms of A_1 by the formula No. 33, $A_{s-1} - \rho A_1 = \frac{1}{2}(\alpha\rho - \beta\rho^3)$; combining the two formulae, we have

$$A_1 = \frac{1}{\rho}(\alpha^2-1)\frac{d\rho}{d\alpha} - \tfrac{1}{2}\alpha + \tfrac{1}{2}\beta\rho^2,$$

$$A_{s-1} = n(\alpha^2-1)\frac{d\rho}{d\alpha} + \left(-\tfrac{1}{2}\alpha + \tfrac{1}{2}\right)\rho + \tfrac{1}{2}\beta\rho^3.$$

45. In the case $n=3$, we have $A_{s-1} = A_1 = 1$, $A_s = \rho$; the two equations become

$$3(\alpha^2-1)\frac{d\rho}{d\alpha} - \tfrac{1}{2}\alpha\rho - \rho^2 + \tfrac{1}{2}\beta\rho^3 = 0,$$

$$3(\alpha^2-1)\frac{d\rho}{d\alpha} - 1 - \tfrac{1}{2}\alpha\rho + \tfrac{1}{2}\beta\rho^3 = 0,$$

each of which is easily verified.

I remark also that, in the same case, $(n=3)$, we have

$$\rho^4\,\frac{\beta^2-1}{\alpha^2-1} = \left(\frac{\rho^2-9}{\rho^2-1}\right)^2, \quad \text{and thence} \quad \Theta = \sqrt{\rho}\sqrt{\frac{\beta^2-1}{\alpha^2-1}} = \sqrt{\frac{\rho^2-9}{\rho^2-1}};$$

hence

$$\frac{1}{\Theta}\frac{d\Theta}{d\alpha} = \frac{4\rho}{(\rho^2-1)(\rho^2-9)};$$

and writing the equation $A_1 - 2n(\alpha^2-1)\dfrac{1}{\Theta}\dfrac{d\Theta}{d\alpha} = 0$ in the form

$$\rho - \alpha^2 n\,\frac{(\alpha^2-1)\dfrac{1}{\Theta}\dfrac{d\Theta}{d\rho}\dfrac{d\rho}{d\alpha}}{1} = 0,$$

we can verify this equation.

46. In the case $n=5$, we have for A_1 two equations, each ultimately giving the foregoing value

$$A_1 = \frac{1}{8\alpha}(\rho^2-2\rho+5)(\rho^2-4\rho-1).$$

Moreover, the equation $\Theta = \sqrt{\rho}\sqrt[4]{\dfrac{\beta^2-1}{\alpha^2-1}}$ gives, without difficulty, $\Theta = \frac{1}{\sqrt{\rho}}\cdot\dfrac{\rho-5}{\rho-1}$.

47. In the case $n=7$, the formulas give the two coefficients A_1, A_2; viz. we have

$$A_1 = \tfrac{1}{2}\,7(\alpha^2-1)\frac{d\rho}{d\alpha} - \tfrac{1}{2}\alpha + \tfrac{1}{2}\beta\rho^2,$$

$$A_2 = 7(\alpha^2-1)\frac{d\rho}{d\alpha} - \tfrac{1}{2}\alpha\rho + \tfrac{1}{2}\beta\rho^3,$$

where the value of $\dfrac{d\rho}{d\alpha}$ must of course be obtained from the before-mentioned $\rho\alpha$-equation (given in No. 7). I have not considered these results nor endeavoured to compare them with the results for this case, obtained in the Transformation Memoir and the addition thereto, [578, 692].